

MATH 345: Linear Algebra and Optimization

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Preface

This course is about linear algebra, together with the parts of multivariable calculus most relevant to *optimization problems*: given a quantity depending on multiple parameters, finding parameters that maximize or minimize that quantity. These tools are fundamental to data science, where one often builds models of data by choosing parameters that make a model best approximate a given data set. The course has one recurring theme: linear algebra studies linear maps; differential calculus studies nonlinear functions by replacing them locally with linear maps; optimization uses both to solve data problems.

The book is organized into seven units. Unit 1 introduces vectors, matrices, and linear maps. Unit 2 studies the anatomy of a linear map: systems, geometry, redundancy, rank, and forgotten directions. Unit 3 develops local linearization, directional derivatives, critical points, and gradient descent. Unit 4 develops orthogonality, projection, and least squares. Unit 5 develops eigenvalues, diagonalization, quadratic forms, Hessians, and least-squares applications. Unit 6 develops constrained optimization and Lagrange multipliers, then applies constrained quadratic forms to principal component analysis (PCA) and the singular value decomposition (SVD). Unit 7 uses abstract vector spaces and polynomial approximation to synthesize projection, least squares, Taylor approximation, and Gram-Schmidt.

Applications. The application threads are intentionally recurrent. Vectors, distance, and dot products introduce embeddings, nearest neighbors, and cosine similarity in Unit 1; dot products return through projection and least squares in Unit 4 and through eigenvector geometry and quadratic forms in Unit 5. Weighted linear combinations preview attention in Unit 1 and return through score-then-combine comparisons in Unit 4. Linear maps $\mathbf{x} \mapsto A\mathbf{x}$ begin in Unit 1 and return through Units 2 and 3. Systems $A\mathbf{x} = \mathbf{b}$, null spaces, rank, lines and planes, and local linearization connect solution sets, forgotten directions, tangent planes, and sensitivity across Units 2 and 3. Projection and least squares drive regression in Unit 4 and return in Units 5 and 7. Gradients and gradient descent organize first-order optimization in Unit 3; eigenvalues, quadratic forms, and Hessians organize second-order analysis in Unit 5; Lagrange multipliers, PCA, and SVD organize constraints and principal directions in Unit 6; polynomial inner products organize Taylor approximation and regression in Unit 7. The Transformer Wrap-Up Worksheet revisits token embeddings, attention scores, weighted averages, affine layers, nonlinear blocks, training loss, gradients, and low-rank updates. The course is not a course on large language models. The goal is to learn mathematical tools that are useful across many areas: data science, optimization, modeling, numerical computation, statistics, engineering, and machine learning.

How to use learning outcomes. Each unit begins with learning outcomes labeled U1-LO1, U1-LO2, and so on. Each outcome names a skill students should be able to use in a new problem. Toolbox skills are smaller calculations that support these outcomes. Activities, exercises, and the lab index identify the relevant outcomes using the same labels. For example, *Learning outcomes*. U4-LO4 identifies work on the Unit 4 least-squares residual outcome. Optional material is identified

separately.

Outcome recurrence. Dot products and cosine similarity first appear in U1-LO1 and U1-LO2, then return in Units 4 and 5 through projection, least squares, eigenvector geometry, and quadratic forms. Matrix-vector products and composition first appear in U1-LO4 and U1-LO6, then return in Units 2 and 3 through linear maps, matrix actions, and local linearization. Reachable outputs and null directions first appear in U2-LO3 and U2-LO6, then return in Units 4 and 6 through least squares, forgotten directions, rank, and the four fundamental subspaces. Gradients and gradient descent first appear in Unit 3. Eigenvalues, quadratic forms, and Hessian classifications appear in Unit 5. Projection and least squares first appear in U4-LO4 and U4-LO5, then return in Units 5, 6, and 7 through regression, normal equations, PCA reconstruction, and polynomial approximation. Symmetric quadratic forms from U5-LO4 return in U6-LO4 and U6-LO5 through constrained extrema, maximum stretch, and PCA. SVD first appears in U6-LO6 and U6-LO7, connecting orthonormal input and output directions with matrix stretch, rank, the four fundamental subspaces, and PCA scores. Polynomial inner products and best approximation first appear in U7-LO3 and U7-LO5 and close the final synthesis through Taylor approximation, continuous least squares, and sampled least squares.

Labs. The notebooks are linked supplements to the textbook. The lab sequence includes Lab U1 on vectors, similarity, attention, and matrix actions; Lab U2 on auditing a linear map; Lab U3 on Jacobian matrices and local linearization; Lab U4 on regression as projection; Lab U5 on Hessian eigenvalues, least squares, and fixed nonlinear features, with a short review of Unit 3 gradient descent; Lab U6 on constraints, PCA, and SVD; and Lab U7 on polynomial approximation. The Transformer Wrap-Up Lab is an extension attached to the Transformer Wrap-Up Worksheet appendix. A detailed lab index appears at the end of the book.

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For a summary of adapted source material, see Sources and Licenses.

Chapter 1

Vectors, matrices, and linear maps

Vectors can represent displacements in space, but in this course they will more often represent data or model parameters. A word, document, image, or user can be encoded as a vector. Matrices then act on vectors, and the matrix-vector product gives the first large family of functions we study: linear maps.

Big questions. How do vectors represent data, how do matrices act on vectors, and what is special about linear functions?

Learning outcomes. By the end of this unit, students should be able to:

- *U1-LO1.* Compute and interpret vector norms, dot products, distances, orthogonality, and cosine similarity.
- *U1-LO2.* Represent objects or data as vectors and explain what cosine similarity and Euclidean distance measure differently.
- *U1-LO3.* Perform basic matrix operations and check dimensions for matrix, vector, and matrix-product expressions.
- *U1-LO4.* Compute Ax and interpret it both as row dot products and as a linear combination of the columns of A .
- *U1-LO5.* Identify, construct, and interpret matrix representations of linear transformations.
- *U1-LO6.* Compute small matrix products and explain how multiplication represents composition of linear maps.
- *U1-LO7.* Distinguish linear maps from affine maps such as $\mathbf{x} \mapsto W\mathbf{x} + \mathbf{b}$.
- *U1-LO8.* Interpret simple code, tables, or diagrams involving cosine similarity, attention-style weighted averages, and applying one matrix to many inputs.

Toolbox skills. Compute vector norms, distances, dot products, cosine similarities, matrix sums, scalar multiples, transposes, matrix-vector products, small matrix products, and simple weighted averages.

1.1 Vectors

1.1.1 Vectors as lists of numbers

A real quantity is also called a *scalar*.

Definition 1.1.1 (Vector).

An n -vector is an ordered list of n scalars, often written in the form $\mathbf{v} = (v_1, v_2, \dots, v_n)$. The collection of all n -vectors is denoted $\mathbb{R}^n$.

We will also encounter vectors written as a row or column of numbers in square brackets, i.e.,

$$[v_1 \quad \cdots \quad v_n] \quad \text{and} \quad \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$$

It is customary to use bold letters for variables, such as $\mathbf{v}$ or $\mathbf{w}$, to represent vectors. We let $\mathbf{0}$ denote the vector $(0, \dots, 0)$ consisting only of zeros. Two vectors $\mathbf{v}$ and $\mathbf{w}$ are *equal* if they have the same number of entries and all the corresponding entries are equal. A point in n -dimensional space can also be identified with an ordered list of n numbers using Cartesian coordinates. When referring specifically to a point P , we write $P(x_1, x_2, \dots, x_n)$ to indicate its coordinates.

1.1.2 Vectors as displacements in space

Vectors can be used to represent displacements between points in space. The vector $\mathbf{v} = (v_1, v_2, v_3)$ represents the displacement from $P(x_1, y_1, z_1)$ to $Q(x_2, y_2, z_2)$, where $x_2 = x_1 + v_1$, $y_2 = y_1 + v_2$, and $z_2 = z_1 + v_3$. This vector is also written as $\overrightarrow{PQ}$. Because vectors often indicate displacements, they are drawn pictorially as arrows starting at a point, and ending where that point is displaced by the vector.

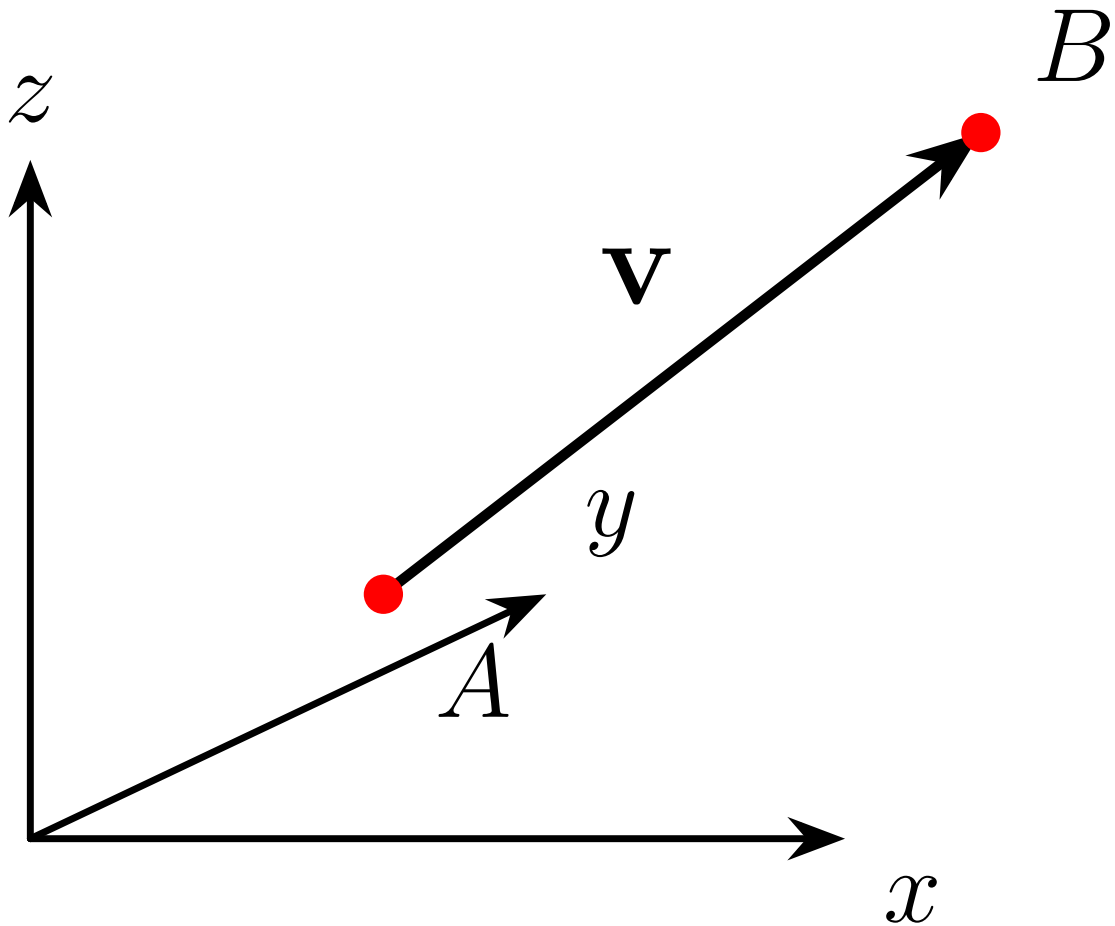


Figure 1.1. A vector $\mathbf{v}$ represented as an arrow between two points P and Q .

Activity: Vectors as displacements in space

What is the vector representing the displacement from the point $P(1, 2)$ to the point $Q(5, 7)$.

Solution.

If $\mathbf{v} = (v_1, v_2)$ was the displacement, then we would have $(1 + v_1, 2 + v_2) = (5, 7)$, i.e., so that $1 + v_1 = 5$ and $2 + v_2 = 7$. Solving these equations gives $v_1 = 4$ and $v_2 = 5$, so that the displacement vector is $\mathbf{v} = (4, 5)$.

Definition 1.1.2 (Length of a vector).

The *length* of a vector $\mathbf{v} = (v_1, \dots, v_n)$, also called its *Euclidean norm*, is the quantity

$$\|\mathbf{v}\| = \sqrt{v_1^2 + \dots + v_n^2}$$

Activity

Verify, using planar geometry, that the length of a 2-vector is the length of the line-segment of the arrow representing the vector.

Solution.

Let $\mathbf{v} = (x, y)$ be an arbitrary 2-vector. Consider a triangle with vertices $P(0, 0)$, $Q(x, 0)$, and $R(x, y)$ (see 1.2).

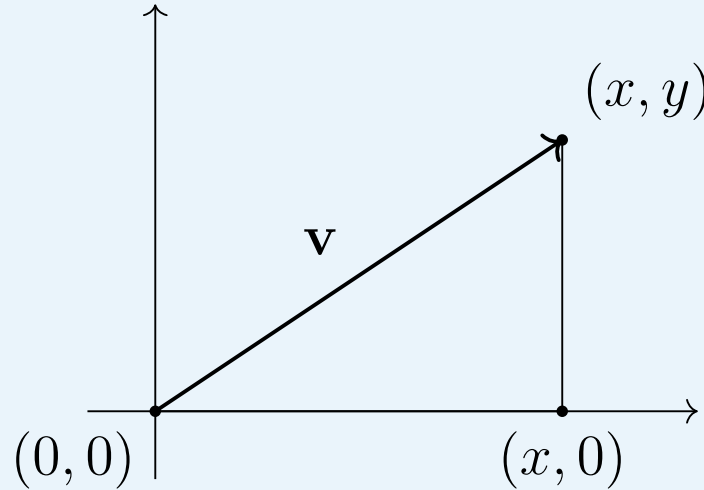


Figure 1.2. A triangle with vertices $P(0, 0)$, $Q(x, 0)$, and $R(x, y)$, drawn to determine the length of the line-segment upon which a 2-vector is based.

This triangle is right-angled (the two legs are horizontal and vertical), and the hypotenuse is the line-segment upon which the vector $\mathbf{v}$ is built. The length of the leg from $P(0, 0)$ to $Q(x, 0)$ is $|x|$, and the length of the leg from $Q(x, 0)$ to $R(x, y)$ is $|y|$. By Pythagoras' theorem, we conclude that the length of the hypotenuse is

$$\sqrt{|x|^2 + |y|^2} = \sqrt{x^2 + y^2} = \|\mathbf{v}\|$$

In a calculus course, you may have mostly seen a vector as a displacement, but in this course it can also be a row of data or a list of model parameters, among other important applications. The same operations below will later compare documents, tokens, and feature vectors.

1.1.3 Adding and scaling vectors

Two displacements can be combined: just apply one displacement after the other is applied. This naturally leads to the notion of *adding* two vectors.

Definition 1.1.3 (Adding vectors).

The sum of two n -vectors $\mathbf{v} = (v_1, \dots, v_n)$ and $\mathbf{w} = (w_1, \dots, w_n)$, denoted $\mathbf{v} + \mathbf{w}$, is the vector given by the tuple $(v_1 + w_1, \dots, v_n + w_n)$.

Remark

The sum of two vectors can be illustrated pictorially as in 1.3 by drawing the parallelogram with the vectors placed along two sides of the parallelogram.

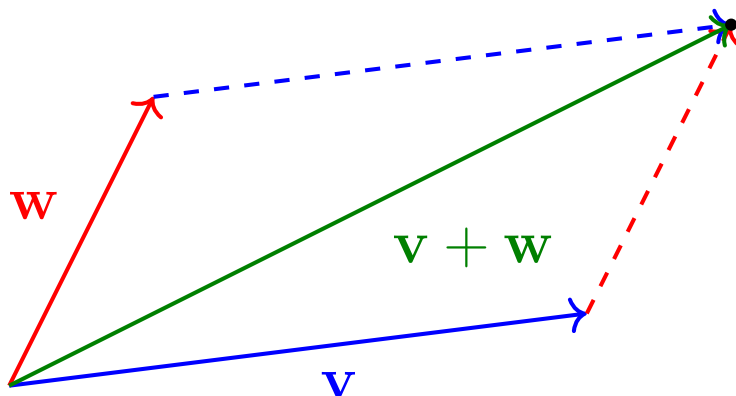


Figure 1.3. The sum of two vectors $\mathbf{v}$ and $\mathbf{w}$.

One can also *scale* a vector by a given scalar quantity.

Definition 1.1.4 (Scaling vectors).

If $\mathbf{v} = (v_1, \dots, v_n)$ is an n -vector and $t \in \mathbb{R}$ is a scalar, we let $t\mathbf{v}$ denote the vector $(tv_1, \dots, tv_n)$, i.e., multiplying each entry of the vector $\mathbf{v}$ by t .

Definition 1.1.5 (Linear combinations and convex combinations of vectors).

If $\mathbf{v}_1, \dots, \mathbf{v}_k$ are vectors in $\mathbb{R}^n$ and $c_1, \dots, c_k$ are scalars, then

$$c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k$$

is a *linear combination* of $\mathbf{v}_1, \dots, \mathbf{v}_k$. The scalars $c_1, \dots, c_k$ are the *coefficients* of the linear combination. If $c_i \geq 0$ for all i and $c_1 + \dots + c_k = 1$, then the linear combination is a *convex combination*.

A convex combination can be interpreted as a weighted average.

Activity: A weighted average

Let $\mathbf{v}_1 = (10, 0)$, $\mathbf{v}_2 = (0, 10)$, and $\mathbf{v}_3 = (10, 10)$. Let $\alpha_1 = 1/4$, $\alpha_2 = 1/4$, and $\alpha_3 = 1/2$. Compute $\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \alpha_3 \mathbf{v}_3$.

Solution.

We have

$$\begin{aligned}
\frac{1}{4}(10, 0) &= \left(\frac{5}{2}, 0\right) \\
\frac{1}{4}(0, 10) &= \left(0, \frac{5}{2}\right) \\
\frac{1}{2}(10, 10) &= (5, 5) \\
\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \alpha_3 \mathbf{v}_3 &= \left(\frac{15}{2}, \frac{15}{2}\right) \\
&= (7.5, 7.5)
\end{aligned}$$

. The coefficients are nonnegative and add to 1, so this is a convex combination.

Example: Subtracting vectors as a linear combination

Subtracting a vector means adding its negative:

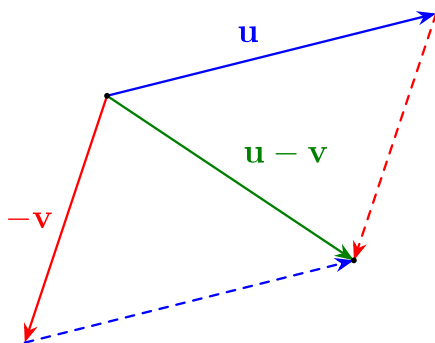
$$\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v}) = 1\mathbf{u} + (-1)\mathbf{v}.$$

Thus, $\mathbf{u} - \mathbf{v}$ is a linear combination of $\mathbf{u}$ and $\mathbf{v}$ with coefficients 1 and -1 . For example, if $\mathbf{u} = (4, 1)$ and $\mathbf{v} = (1, 3)$, then

$$\mathbf{u} - \mathbf{v} = (4, 1) + (-1, -3) = (3, -2).$$

Geometrically, $-\mathbf{v}$ has the same length as $\mathbf{v}$ but points in the opposite direction. We can therefore use the parallelogram construction from 1.3 with $\mathbf{u}$ and $-\mathbf{v}$, as shown on the left in 1.4. Equivalently, when $\mathbf{u}$ and $\mathbf{v}$ start at the same point, the arrow from the tip of $\mathbf{v}$ to the tip of $\mathbf{u}$ represents $\mathbf{u} - \mathbf{v}$. Here, it is the displacement from $P(1, 3)$ to $Q(4, 1)$, shown on the right.

(a) Add the negative vector



(b) Join the tips

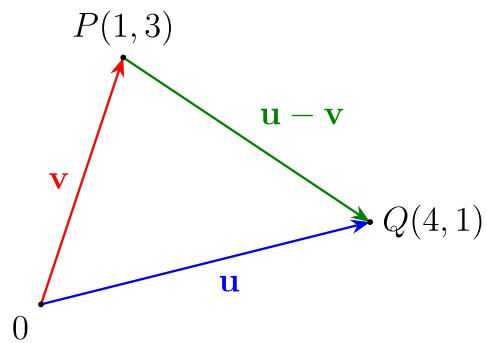


Figure 1.4. The difference $\mathbf{u} - \mathbf{v}$ as the sum $\mathbf{u} + (-\mathbf{v})$ (left) and as the displacement from the tip of $\mathbf{v}$ to the tip of $\mathbf{u}$ (right).

1.1.4 Dot products of vectors

Definition 1.1.6 (Dot product).

The *dot product* of two n -vectors $\mathbf{v} = (v_1, \dots, v_n)$ and $\mathbf{w} = (w_1, \dots, w_n)$ is the scalar quantity

$$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + \dots + v_n w_n$$

Geometrically, the dot product of two vectors is a quantity which is related to the *angle* between the two vectors $\mathbf{v}$ and $\mathbf{w}$. If we draw the two vectors as arrows with the same starting point, then they form an angle on the plane containing both vectors, and the angle θ is given below.

Theorem 1.1.7 (Angle between vectors).

For two nonzero vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$, the angle θ between the vectors is the quantity satisfying the equation

$$\cos \theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|}$$

The dot product can be thought of as a measure of the *similarity* between two vectors. Suppose θ is the angle between two vectors $\mathbf{v}$ and $\mathbf{w}$:

- If θ is close to zero, then the two vectors are close to pointing in the same direction. Since $\cos(0^\circ) = 1$, this occurs precisely when

$$\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|} \approx 1$$

- If θ is close to 180° , then the two vectors are close to pointing in opposite directions. Since $\cos(180^\circ) = -1$, this occurs precisely when

$$\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|} \approx -1$$

- If θ is close to 90° , the two vectors are close to pointing at right angles. Since $\cos(90^\circ) = 0$, this occurs precisely when

$$\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|} \approx 0$$

. When $\mathbf{v} \cdot \mathbf{w} = 0$, so that $\theta = 90^\circ$, we say the two vectors are perpendicular, or *orthogonal*.

The *cosine similarity* between the nonzero vectors $\mathbf{v}$ and $\mathbf{w}$ is denoted by

$$\text{cosim}(\mathbf{v}, \mathbf{w}) = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|}.$$

It ranges between -1 and 1 , and measures the degree to which the two vectors point in the same, or opposite, directions. This terminology is especially common in certain areas of data science.

Warning

Cosine similarity is undefined if either vector is the zero vector.

Activity

Calculate the cosine similarity between the two vectors $\mathbf{v} = (1, 2)$ and $\mathbf{w} = (\sqrt{3} + 2, 2\sqrt{3} - 1)$.

Solution.

We begin by calculating the lengths of the two vectors, i.e

$$\|\mathbf{v}\| = \sqrt{(1)^2 + (2)^2} = \sqrt{5}$$

and

$$\begin{aligned}\|\mathbf{w}\| &= \sqrt{(\sqrt{3} + 2)^2 + (2\sqrt{3} - 1)^2} \\ &= \sqrt{(3 + 4\sqrt{3} + 4) + (12 - 4\sqrt{3} + 1)} \\ &= \sqrt{20}\end{aligned}$$

. Next, we calculate the dot product

$$\begin{aligned}\mathbf{v} \cdot \mathbf{w} &= (1)(\sqrt{3} + 2) + (2)(2\sqrt{3} - 1) \\ &= \sqrt{3} + 2 + 4\sqrt{3} - 2 \\ &= 5\sqrt{3}\end{aligned}$$

. Therefore, the cosine similarity between the two vectors is

$$\begin{aligned}\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|} &= \frac{5\sqrt{3}}{\sqrt{5}\sqrt{20}} \\ &= \sqrt{3}/2\end{aligned}$$

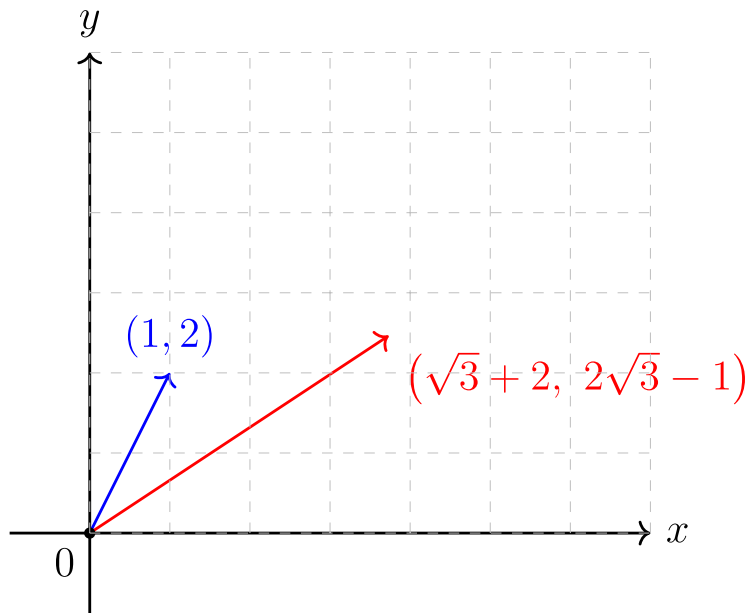


Figure 1.5. The vectors $\mathbf{v}$ and $\mathbf{w}$ from 1.1.4, drawn as arrows from the origin.

Activity

Calculate the angle between the two vectors

$$\mathbf{v} = \begin{bmatrix} 2 \\ 3 \\ 0 \\ 2 \end{bmatrix} \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

.

Solution.

We calculate that

$$\begin{aligned} \mathbf{v} \cdot \mathbf{w} &= (2)(1) + (3)(0) + (0)(1) + (2)(-1) \\ &= 2 + 0 + 0 - 2 = 0 \end{aligned}$$

. Thus if θ is the angle between the two vectors, then

$$\cos(\theta) = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|} = \frac{0}{\|\mathbf{v}\|\|\mathbf{w}\|} = 0$$

. Thus

$$\theta = \cos^{-1}(0) = 90^\circ$$

.

Note

Note that $\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}$ for any $\mathbf{v} \in \mathbb{R}^n$.

We will study the dot product in far more detail in 4 and 7.

Definition 1.1.8 (Distance between vectors).

The *Euclidean distance* between two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is

$$\text{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

As illustrated in 1.4, $\mathbf{u} - \mathbf{v}$ points from the tip of $\mathbf{v}$ to the tip of $\mathbf{u}$ when the two vectors start at the same point. Its Euclidean norm therefore measures how far apart those endpoints are. For the vectors in 1.3,

$$\text{dist}(\mathbf{u}, \mathbf{v}) = \|(3, -2)\| = \sqrt{3^2 + (-2)^2} = \sqrt{13}.$$

In contrast, cosine similarity compares direction after normalization.

1.1.5 Applications of vectors

Vectors can represent quantities other than displacements. The entries of a vector can record measurements, counts, samples, or features. The meaning of a vector depends on what its coordinates represent.

- **Object:** Color
Meaning of the entries: red, green, and blue intensities
- **Object:** Time series
Meaning of the entries: measurements at successive times
- **Object:** Portfolio
Meaning of the entries: amounts held in each asset
- **Object:** Image
Meaning of the entries: pixel intensities, listed in a fixed order
- **Object:** Document
Meaning of the entries: counts of selected words
- **Object:** Customer
Meaning of the entries: purchases of selected products
- **Object:** Object with features
Meaning of the entries: measured attributes such as size, price, weight, or rating

The same vector operations can have different interpretations. A sum can add purchases, add word counts, or add two time series. A scalar multiple can rescale an image, double a portfolio, or change units. A dot product can produce a score. A distance can compare two feature vectors. Cosine similarity compares direction after normalization.

Activity: Word-count vectors

Use the dictionary `linear, matrix, data`. A document vector records the number of times these words appear, in this order. The query

$$\mathbf{q} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

represents the phrase “linear data”. Consider three document vectors

$$\mathbf{D}_1 = \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix}, \quad \mathbf{D}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{D}_3 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}.$$

Compute the cosine similarities and Euclidean distances between $\mathbf{q}$ and each $\mathbf{D}_i$. Rank the documents from most to least similar by cosine similarity and from closest to farthest by Euclidean distance.

Solution.

First,

$$\begin{aligned} \text{cosim}(\mathbf{q}, \mathbf{D}_1) &= \frac{\mathbf{q} \cdot \mathbf{D}_1}{\|\mathbf{q}\| \|\mathbf{D}_1\|} \\ &= \frac{6}{\sqrt{2}\sqrt{18}} \\ &= 1. \end{aligned}$$

Next,

$$\begin{aligned} \text{cosim}(\mathbf{q}, \mathbf{D}_2) &= \frac{\mathbf{q} \cdot \mathbf{D}_2}{\|\mathbf{q}\| \|\mathbf{D}_2\|} \\ &= \frac{1}{\sqrt{2}\sqrt{2}} \\ &= \frac{1}{2}. \end{aligned}$$

Finally, $\text{cosim}(\mathbf{q}, \mathbf{D}_3) = 0$.

The vector $\mathbf{D}_1$ points in exactly the same direction as $\mathbf{q}$ because $\mathbf{D}_1 = 3\mathbf{q}$. The vector $\mathbf{D}_2$ shares the word “linear” with the query but also contains “matrix”. The vector $\mathbf{D}_3$ contains only “matrix”, so it is orthogonal to the query.

So the ranking by cosine similarity is

$$\mathbf{D}_1, \mathbf{D}_2, \mathbf{D}_3.$$

For the Euclidean distances,

$$\begin{aligned}
\text{dist}(\mathbf{q}, \mathbf{D}_1) &= \|\mathbf{q} - \mathbf{D}_1\| \\
&= \left\| \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix} \right\| \\
&= \left\| \begin{bmatrix} -2 \\ 0 \\ -2 \end{bmatrix} \right\| = \sqrt{8} = 2\sqrt{2}, \\
\text{dist}(\mathbf{q}, \mathbf{D}_2) &= \|\mathbf{q} - \mathbf{D}_2\| \\
&= \left\| \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\| \\
&= \left\| \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\| = \sqrt{2}, \\
\text{dist}(\mathbf{q}, \mathbf{D}_3) &= \|\mathbf{q} - \mathbf{D}_3\| \\
&= \left\| \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \right\| \\
&= \left\| \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \right\| = \sqrt{6}.
\end{aligned}$$

Since $\sqrt{2} < \sqrt{6} < 2\sqrt{2}$, the ranking from smallest to largest Euclidean distance is

$$\mathbf{D}_2, \mathbf{D}_3, \mathbf{D}_1.$$

In the previous activity, $\mathbf{D}_1$ is most similar to the query by cosine similarity but farthest from it by Euclidean distance; $\mathbf{D}_2$ is closest by Euclidean distance. The two measures select different documents because cosine similarity compares direction after normalization, while Euclidean distance also depends on vector length.

Activity: The same rankings in code

The following code repeats the cosine-similarity and Euclidean-distance rankings from 1.1.5. We compute the cosine similarities between the query and each document, then the Euclidean distances by taking the norm of each difference.

```
import numpy as np

D1 = np.array([3., 0., 3.]
```

```

D2 = np.array([1., 1., 0.])
D3 = np.array([0., 2., 0.])
q = np.array([1., 0., 1.])

score1 = (D1 @ q) / (np.linalg.norm(D1) * np.linalg.norm(q))
score2 = (D2 @ q) / (np.linalg.norm(D2) * np.linalg.norm(q))
score3 = (D3 @ q) / (np.linalg.norm(D3) * np.linalg.norm(q))

distance1 = np.linalg.norm(D1 - q)
distance2 = np.linalg.norm(D2 - q)
distance3 = np.linalg.norm(D3 - q)

(score1, score2, score3), (distance1, distance2, distance3)

```

Output (approximately):

```

((1.0, 0.5, 0.0),
 (2.8284271247461903, 1.4142135623730951, 2.449489742783178))

```

The first tuple gives the cosine similarities and the second gives the Euclidean distances, each in the order D_1, D_2, D_3 relative to q . Smaller distances mean closer documents.

1. Which score corresponds to D_2 ?
2. Where does the code compute $D_1 \cdot q$?
3. Which document is most similar to the query by cosine similarity?
4. Which line computes $\|D_2 - q\|$?
5. Rank the documents from closest to farthest from the query by Euclidean distance. How does this ranking compare with the cosine-similarity ranking?

Solution.

The score for D_2 is score2. The expression $D1 @ q$ computes $D_1 \cdot q$. Since score1 is largest, D_1 is most similar to the query.

The line `distance2 = np.linalg.norm(D2 - q)` computes $\|D_2 - q\|$. The distances are $2\sqrt{2}$, $\sqrt{2}$, and $\sqrt{6}$, respectively, so the ranking from closest to farthest is D_2 , then D_3 , then D_1 . By cosine similarity, the ranking is D_1 , then D_2 , then D_3 . Thus, D_1 is first by cosine similarity, while D_2 is first by Euclidean distance.

Example: Neural networks and transformers

A neural network is a function built from layers. A layer takes numbers as input, combines

them using weights, applies a rule, and sends output numbers to the next layer. In many neural networks, the input and output of a layer are vectors. The weights are adjusted from data during training.

A transformer is a neural-network architecture for sequences. In a language model, text is first broken into tokens. A token can be a word, part of a word, punctuation mark, or other text fragment. Each token position carries a vector.

A transformer layer produces a new vector at each token position. The token itself does not change; its vector becomes context-dependent. In the phrase “small red bird”, the new vector at the bird position can gather information from “small” and “red”. In a next-token model, the vector at this final position can then help predict what comes next.

For one output position, its *query* is used to decide which positions are relevant. Each token has a *key* used in that comparison and a *value* containing the information it can contribute. Queries and keys determine the weights; values are what get averaged. Every position has all three roles, but the next activity computes only the attention output at the bird position.

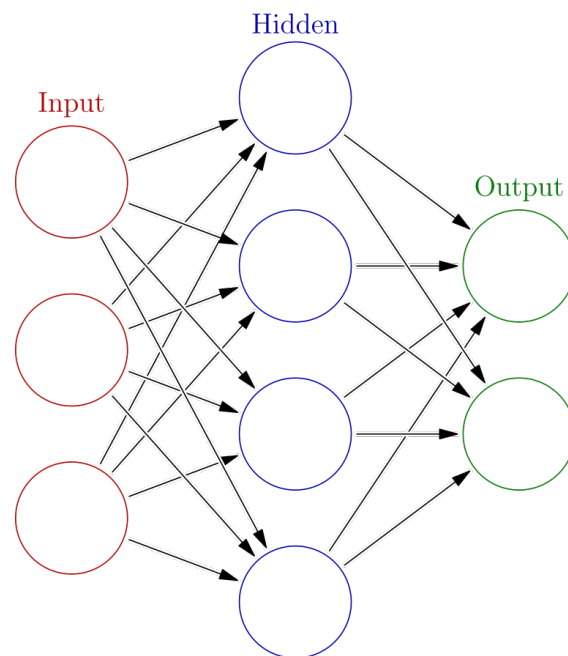


Figure 1.6. A neural network diagram with input, hidden, and output layers. Source: Wikipedia, Neural network (machine learning), using Colored neural network.svg by Glosser.ca, derivative of Artificial neural network.svg, licensed under CC BY-SA 3.0.

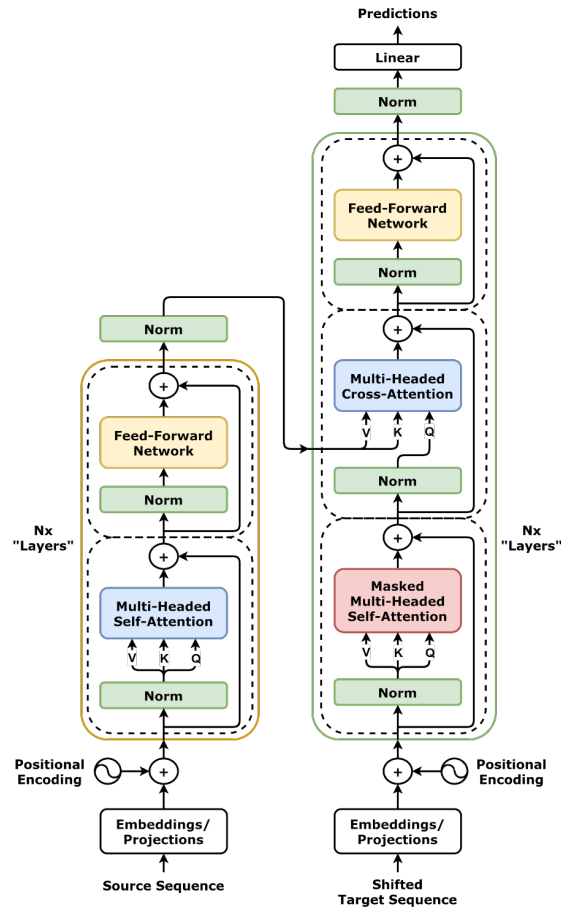


Figure 1.7. A transformer processes a sequence of token vectors. The activities below focus on the vector operations visible in the diagram: token embeddings, dot-product scores, weighted averages, and affine layers. Source: Wikipedia, Transformer (deep learning architecture), using Transformer, full architecture.png by dvigodoy, licensed under CC BY 4.0.

Activity: Gathering context for the bird position

Consider the token sequence

small, red, bird.

We compute the attention output at the position occupied by “bird”. The bird position is the destination; “small”, “red”, and “bird” itself are possible sources of information.

The query q_{bird} is compared with each key k_i to produce a relevance score. The scores are converted into weights, and those weights are used to average the value vectors:

query-key scores $\longrightarrow$ weights
 $\longrightarrow$ weighted average of values.

For this activity, the score for token i is

$$\text{score}_i = \mathbf{q}_{\text{bird}} \cdot \mathbf{k}_i.$$

Use

$$\mathbf{q}_{\text{bird}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{k}_{\text{small}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{k}_{\text{red}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{k}_{\text{bird}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

1. Compute the three scores

$$\mathbf{q}_{\text{bird}} \cdot \mathbf{k}_{\text{small}}, \quad \mathbf{q}_{\text{bird}} \cdot \mathbf{k}_{\text{red}}, \quad \mathbf{q}_{\text{bird}} \cdot \mathbf{k}_{\text{bird}}.$$

2. Normalize the positive scores by dividing each score by the sum of all three scores. Call the resulting weights

$$\alpha = \begin{bmatrix} \alpha_{\text{small}} \\ \alpha_{\text{red}} \\ \alpha_{\text{bird}} \end{bmatrix}.$$

3. The coordinates of the value vectors are abstract toy features. Use

$$\mathbf{v}_{\text{small}} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}, \quad \mathbf{v}_{\text{red}} = \begin{bmatrix} 0 \\ 4 \end{bmatrix}, \quad \mathbf{v}_{\text{bird}} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

to compute

$$\mathbf{h}_{\text{bird}} = \alpha_{\text{small}} \mathbf{v}_{\text{small}} + \alpha_{\text{red}} \mathbf{v}_{\text{red}} + \alpha_{\text{bird}} \mathbf{v}_{\text{bird}}.$$

4. Which token receives the largest weight? What total weight is assigned to “small” and “red” together? What does this say about the information gathered at the bird position?

Solution.

The scores are

$$\begin{aligned} \mathbf{q}_{\text{bird}} \cdot \mathbf{k}_{\text{small}} &= 1, \\ \mathbf{q}_{\text{bird}} \cdot \mathbf{k}_{\text{red}} &= 1, \\ \mathbf{q}_{\text{bird}} \cdot \mathbf{k}_{\text{bird}} &= 2. \end{aligned}$$

The sum of the scores is

$$1 + 1 + 2 = 4.$$

Thus

$$\alpha = \begin{bmatrix} 1/4 \\ 1/4 \\ 1/2 \end{bmatrix}.$$

The attention output at the bird position is the weighted average

$$\mathbf{h}_{\text{bird}} = \frac{1}{4} \begin{bmatrix} 4 \\ 0 \end{bmatrix} + \frac{1}{4} \begin{bmatrix} 0 \\ 4 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}.$$

The token “bird” receives the largest weight because its key vector has the largest dot product with $\mathbf{q}_{\text{bird}}$. The bird position assigns weight 1/2 to “bird” itself and total weight 1/2 to “small” and “red”. Thus the attention output combines information from the token itself with information from its context. Including the destination among the possible sources helps retain information about the token at that position. The coordinates of $\mathbf{h}_{\text{bird}}$ remain abstract toy features; the important interpretation is how the weights distribute across the three source positions. A full transformer layer combines this attention output with the current vector at the bird position and processes it further.

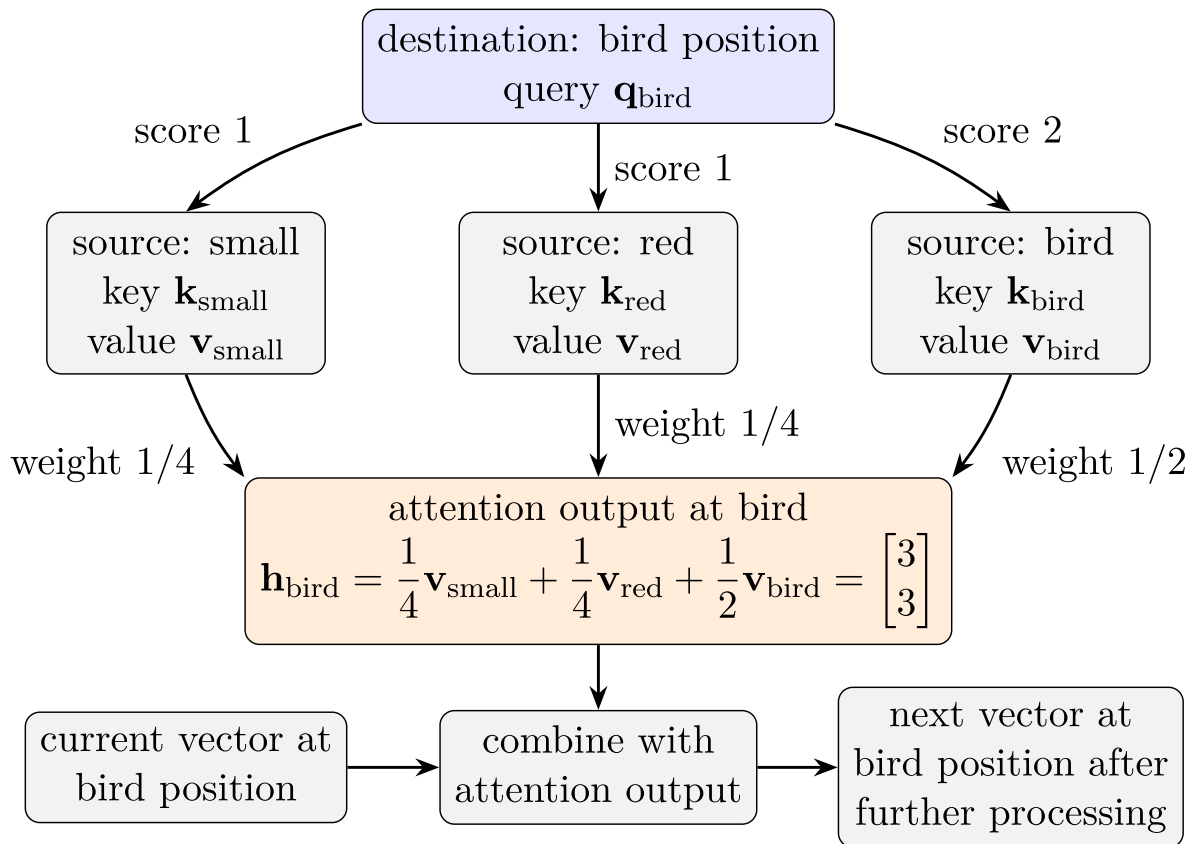


Figure 1.8. Attention at one destination position. The bird query produces scores 1, 1, 2 and weights 1/4, 1/4, 1/2 for the value vectors at “small”, “red”, and “bird”. Their weighted average is the attention output at the bird position. A full transformer layer performs an analogous calculation at every token

position and processes each attention output further.

Warning

This activity isolates one part of a transformer layer. It uses normalization by the sum of positive scores to keep the arithmetic simple. Real transformer attention usually uses a different rule to convert scores into weights, but it follows the same pattern:

$$\text{scores} \longrightarrow \text{weights} \longrightarrow \text{weighted average of values.}$$

A full layer then combines the attention output with the current token representation and processes the result further.

1.2 Matrices

Matrices can be viewed in two complementary ways: as arrays of data and as functions acting on vectors. We begin with basic matrix arithmetic, then use matrix-vector multiplication to describe linear maps.

1.2.1 Matrices as arrays of numbers

Definition 1.2.1 (Matrices).

An $m \times n$ matrix A is a rectangular array of $m \cdot n$ real numbers arranged in m (*horizontal*) rows and n (*vertical*) columns:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

The i -th row of A is

$$[a_{i1} \quad a_{i2} \quad \cdots \quad a_{in}],$$

and the j -th column of A is

$$\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}.$$

The number a_{ij} , which is in the i -th row and j -th column of A , is the (i, j) -entry of A , and we often write $A = [a_{ij}]$. We say A is an " m by n " matrix.

Activity

Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & -8 & -13 \end{bmatrix}$$

A is a 2×3 matrix. Compute the following:

Task

a_{11}

Solution.

1

Task

a_{12}

Solution.

2

Task

a_{21}

Solution.

5

Task

a_{23}

Solution.

−13

Task

a_{32}

Solution.

No such entry exists.

An important application of matrices (but far from the only one!) is to record data. The following table gives a few examples.

- **Object:** Monochrome image

Meaning of an entry: X_{ij} is the pixel value in row i and column j .

- **Object:** Rainfall data

Meaning of an entry: A_{ij} is the rainfall at location j on day i .

- **Object:** Asset returns
Meaning of an entry: R_{ij} is the return of asset i in period j .
- **Object:** Feature matrix
Meaning of an entry: X_{ij} is the value of feature j for entity i .

1.2.2 Operations on matrices

Definition 1.2.2 (Equality of matrices).

Two matrices A and B are *equal* if they have the same size and all the corresponding entries are equal.

Activity

Suppose

$$A = \begin{bmatrix} 3 & y \\ 4 & -7 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 6+x & -2 \\ 4 & -7 \end{bmatrix}$$

and $A = B$. Find x and y .

Solution.

Since $A = B$, all entries of A must equal the corresponding entries in B . So it must be true, comparing corresponding entries, that

$$3 = 6 + x$$

$$y = -2$$

Therefore, $x = -3$ and $y = -2$.

Definition 1.2.3 (Sums of matrices).

If $A = [a_{ij}]$ and $B = [b_{ij}]$ are both $m \times n$ matrices, then their *sum* $A + B$ is the matrix $C = [c_{ij}]$ where $c_{ij} = a_{ij} + b_{ij}$.

Activity

For the matrices

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & -8 & -13 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 3 & -4 \end{bmatrix}$$

calculate $A + B$.

Solution.

$$\begin{aligned}
 A + B &= \begin{bmatrix} 1+0 & 2+2 & 3+1 \\ 5+1 & -8+3 & -13+(-4) \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 4 & 4 \\ 6 & -5 & -17 \end{bmatrix}.
 \end{aligned}$$

Warning

For $A + B$ to be defined, A and B must be the same size. From now on, if we write $A + B$, assume that this is the case.

Definition 1.2.4 (Scalar multiples of matrices).

If $A = [a_{ij}]$ is an $m \times n$ matrix and r is a real number, then the *scalar multiple* of A by r , written rA , is the $m \times n$ matrix $C = [c_{ij}]$, where $c_{ij} = ra_{ij}$, that is, C is the matrix obtained by multiplying every entry of A by r .

Activity

For A as in 1.2.2, calculate $-2A$.

Solution.

$$-2A = \begin{bmatrix} -2 & -4 & -6 \\ -10 & 16 & 26 \end{bmatrix}$$

Note

The $m \times n$ matrix with zeros in every entry is called the *zero matrix* and is denoted by O . For every $m \times n$ matrix A , we have $A + O = A$. The *negative* of A is the matrix $-A = (-1)A$, and $A + (-A) = O$.

Theorem 1.2.5.

Let A , B , and C be $m \times n$ matrices.

1. $A + B = B + A$, i.e., matrix addition is *commutative*.
2. $A + (B + C) = (A + B) + C$, i.e., matrix addition is *associative*.
3. Let r and s be real numbers. Then:
 - (a) $r(sA) = (rs)A$.
 - (b) $(r + s)A = rA + sA$.
 - (c) $r(A + B) = rA + rB$.

Proof.

We prove Property 1 only, i.e., the commutativity of addition. Let $A = [a_{ij}]$ and $B = [b_{ij}]$. Then:

$$\begin{aligned} A + B &= [a_{ij} + b_{ij}] \\ &= [b_{ij} + a_{ij}] && \text{(since real numbers are commutative)} \\ &= B + A \end{aligned}$$

Definition 1.2.6.

If $A = [a_{ij}]$ is an $m \times n$ matrix, then the *transpose* of A , denoted $A^T = [a_{ij}^T]$, is the $n \times m$ matrix defined by

$$a_{ij}^T = a_{ji}$$

In other words, the transpose of A is obtained by *interchanging* the rows and the columns of A .

Activity

Compute the transpose for each of the given matrices:

Task

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & -8 & -13 \end{bmatrix}.$$

Solution.

$$A^T = \begin{bmatrix} 1 & 5 \\ 2 & -8 \\ 3 & -13 \end{bmatrix}$$

Task

$$B = \begin{bmatrix} 5 & 2 & 3 \\ 6 & 2 & 3 \\ -1 & -2 & 3 \end{bmatrix}$$

Solution.

$$B^T = \begin{bmatrix} 5 & 6 & -1 \\ 2 & 2 & -2 \\ 3 & 3 & 3 \end{bmatrix}$$

Task

$$C = \begin{bmatrix} 10 \\ 20 \\ 30 \end{bmatrix}.$$

Solution.

$$C^T = [10 \quad 20 \quad 30]$$

Note

Observe from the previous activity that, when transposed, a column vector becomes a row vector. And vice versa.

Definition 1.2.7 (Main diagonal).

If $A = [a_{ij}]$ is an $m \times n$ matrix, the elements $a_{11}, a_{22}, a_{33}, \dots$ are called the *main diagonal* of A . A matrix A is called *diagonal* if its only nonzero entries occur on its main diagonal.

Below are four matrices of various dimensions, with the main diagonal written in bold font.

$$\begin{bmatrix} \mathbf{a_{11}} & a_{12} & a_{13} \\ a_{21} & \mathbf{a_{22}} & a_{23} \\ a_{31} & a_{32} & \mathbf{a_{33}} \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{a_{11}} & a_{12} \\ a_{21} & \mathbf{a_{22}} \\ a_{31} & a_{32} \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{a_{11}} & a_{12} & a_{13} \\ a_{21} & \mathbf{a_{22}} & a_{23} \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{a_{11}} \\ a_{21} \end{bmatrix}$$

Forming the transpose of a matrix A can be viewed as *flipping* A about its main diagonal.

Theorem 1.2.8.

If r is a scalar and A and B are matrices of the appropriate sizes, then:

1. $(A^T)^T = A$.
2. $(A + B)^T = A^T + B^T$.
3. $(rA)^T = rA^T$.

Proof.

Proof of property 2:

Let $A = [a_{ij}]$ and $B = [b_{ij}]$. Then $A + B = [c_{ij}]$ where $c_{ij} = a_{ij} + b_{ij}$.

Then

$$\begin{aligned}
 (A + B)^T &= [c_{ij}^T] \\
 &= [c_{ji}] && \text{By definition of transpose} \\
 &= [a_{ji} + b_{ji}] && \text{Since } c_{ij} = a_{ij} + b_{ij} \\
 &= [a_{ji}] + [b_{ji}] && \text{By definition of matrix addition} \\
 &= [a_{ij}^T] + [b_{ij}^T] && \text{By definition of transpose} \\
 &= A^T + B^T
 \end{aligned}$$

Therefore, $(A + B)^T = A^T + B^T$.

Definition 1.2.9 (Symmetric matrices).

A matrix A with real entries is called *symmetric* if $A^T = A$.

Warning

The previous definition only makes sense if the matrix A is *square*, i.e., if it has the same number of rows and columns.

Activity

Determine whether each of the following matrices is symmetric or not symmetric:

Task

$$A = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 5 \\ 3 & -5 & 0 \end{bmatrix}$$

Solution.

Interchanging rows and columns gives

$$A^T = \begin{bmatrix} 0 & -2 & 3 \\ 2 & 0 & -5 \\ -3 & 5 & 0 \end{bmatrix}.$$

Since $A^T \neq A$, A is not symmetric.

Task

$$B = \begin{bmatrix} 3 & 5 & 2 \\ 5 & 1 & 4 \\ 2 & 4 & -1 \end{bmatrix}$$

Solution.

Interchanging rows and columns gives

$$B^T = \begin{bmatrix} 3 & 5 & 2 \\ 5 & 1 & 4 \\ 2 & 4 & -1 \end{bmatrix}.$$

B is symmetric since $B^T = B$.

Task

$$C = \begin{bmatrix} 1 & 2 & -3 \\ -2 & 0 & 5 \\ 3 & 5 & 0 \end{bmatrix}$$

Solution.

Interchanging rows and columns gives

$$C^T = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 0 & 5 \\ -3 & 5 & 0 \end{bmatrix}.$$

Since $C^T \neq C$, C is not symmetric.

Task

$$D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Solution.

Interchanging rows and columns gives

$$D^T = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Since $D^T = D$, D is symmetric.

Note: Shape habit

Before adding or transposing matrices, first identify their shapes. A 2×3 matrix has two rows and three columns. Matrix addition requires the same shape, while transposing a 2×3 matrix produces a 3×2 matrix.

1.3 Matrix-vector product and linear maps

1.3.1 Matrix-vector product

A vector can be written as a row or as a column. When a matrix acts on a vector in this course, we usually write the vector as a column:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

From now on, when we write $\mathbf{x} = (x_1, \dots, x_n)$, $\mathbf{x}$ is interpreted as a column vector. If A is an $m \times n$ matrix, then A has n columns. We write

$$A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n]$$

where each $\mathbf{a}_j$ is a column vector in $\mathbb{R}^m$.

Definition 1.3.1 (Matrix-vector product).

Let $A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n]$ be an $m \times n$ matrix and let $\mathbf{x} = (x_1, \dots, x_n)$ be a column vector in $\mathbb{R}^n$. The *matrix-vector product* is

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

Activity: Computing a matrix-vector product

Compute $A\mathbf{x}$ where

$$A = \begin{bmatrix} -1 & 4 & -5 \\ 3 & 1 & -2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}.$$

Solution.

Using columns,

$$\begin{aligned} A\mathbf{x} &= 2 \begin{bmatrix} -1 \\ 3 \end{bmatrix} - 3 \begin{bmatrix} 4 \\ 1 \end{bmatrix} + 4 \begin{bmatrix} -5 \\ -2 \end{bmatrix} \\ &= \begin{bmatrix} 2(-1) - 3(4) + 4(-5) \\ 2(3) - 3(1) + 4(-2) \end{bmatrix} \\ &= \begin{bmatrix} -34 \\ -5 \end{bmatrix}. \end{aligned}$$

Theorem 1.3.2 (Properties of matrix-vector products).

Let A and B be $m \times n$ matrices, let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, and let r be a scalar. Then

- $A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y}$,
- $A(r\mathbf{x}) = r(A\mathbf{x}) = (rA)\mathbf{x}$,

- $(A + B)\mathbf{x} = A\mathbf{x} + B\mathbf{x}$.

Proof.

We prove the first property. Write $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$, $\mathbf{x} = (x_1, \dots, x_n)$, and $\mathbf{y} = (y_1, \dots, y_n)$. By the definition above,

$$\begin{aligned} A(\mathbf{x} + \mathbf{y}) &= (x_1 + y_1)\mathbf{a}_1 + \cdots + (x_n + y_n)\mathbf{a}_n \\ &= A\mathbf{x} + A\mathbf{y}. \end{aligned}$$

Theorem 1.3.3 (Row dot-product view).

If A is an $m \times n$ matrix with rows $\text{row}_1(A), \dots, \text{row}_m(A)$ as n -vectors, then

$$A\mathbf{x} = \begin{bmatrix} \text{row}_1(A) \cdot \mathbf{x} \\ \text{row}_2(A) \cdot \mathbf{x} \\ \vdots \\ \text{row}_m(A) \cdot \mathbf{x} \end{bmatrix}.$$

Proof.

Write $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$, where $\mathbf{a}_j = (a_{1j}, \dots, a_{mj})$, and let $\mathbf{x} = (x_1, \dots, x_n)$. Since

$$A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n,$$

its i -th entry is

$$x_1a_{i1} + \cdots + x_na_{in} = \text{row}_i(A) \cdot \mathbf{x}.$$

Stacking these entries gives the stated formula.

Example: Computing the same product using rows

We revisit 1.3.1 using row dot products. Compute $A\mathbf{x}$ where

$$A = \begin{bmatrix} -1 & 4 & -5 \\ 3 & 1 & -2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}.$$

Solution.

Take the dot product of each row of A with $\mathbf{x}$:

$$\begin{aligned} A\mathbf{x} &= \begin{bmatrix} (-1, 4, -5) \cdot (2, -3, 4) \\ (3, 1, -2) \cdot (2, -3, 4) \end{bmatrix} \\ &= \begin{bmatrix} (-1)(2) + 4(-3) + (-5)(4) \\ 3(2) + 1(-3) + (-2)(4) \end{bmatrix} \\ &= \begin{bmatrix} -34 \\ -5 \end{bmatrix}. \end{aligned}$$

Each dot product gives one component of the same output vector obtained using columns.

Note: Shape habit for matrix-vector products

Before forming Ax , check that the number of columns of A equals the number of entries of x . If A is $m \times n$ and $x \in \mathbb{R}^n$, then $Ax \in \mathbb{R}^m$: its m entries come from the m row dot products above.

Note: Rows measure; columns contribute

A matrix-vector product has two complementary readings. In the row view, each row measures the input by taking a dot product with x . In the column view,

$$Ax = x_1 a_1 + \cdots + x_n a_n,$$

so the input coordinates tell how much each column contributes to the output.

Examples of matrix-vector products

The same product Ax can represent different operations, depending on what the rows and columns of A mean. The next activities show three common patterns: scoring, selecting, and measuring changes.

Activity: Scoring objects by features

Let the rows of X represent three objects with two features:

$$X = \begin{bmatrix} 3 & 1 \\ 1 & 4 \\ 2 & 2 \end{bmatrix}, \quad \mathbf{w} = (2, -1).$$

Compute $X\mathbf{w}$. Interpret the result.

Solution.

We have

$$X\mathbf{w} = \begin{bmatrix} 3 & 1 \\ 1 & 4 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 2 \end{bmatrix}.$$

Each output is a score:

$$2(\text{first feature}) - (\text{second feature}).$$

Activity: A selector matrix

Let

$$S = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \end{bmatrix}.$$

Compute $S\mathbf{x}$. What does S do?

Solution.

We have

$$\begin{aligned} S\mathbf{x} &= \begin{bmatrix} 1 \cdot 5 + 0 \cdot 6 + 0 \cdot 7 + 0 \cdot 8 \\ 0 \cdot 5 + 0 \cdot 6 + 1 \cdot 7 + 0 \cdot 8 \end{bmatrix} \\ &= \begin{bmatrix} 5 \\ 7 \end{bmatrix}. \end{aligned}$$

The matrix S selects the first and third entries of $\mathbf{x}$.

Activity: A difference matrix

Let

$$D = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 5 \\ 9 \end{bmatrix}.$$

Compute $D\mathbf{x}$. What does D measure?

Solution.

We have

$$\begin{aligned} D\mathbf{x} &= \begin{bmatrix} -1 \cdot 2 + 1 \cdot 5 + 0 \cdot 9 \\ 0 \cdot 2 + (-1) \cdot 5 + 1 \cdot 9 \end{bmatrix} \\ &= \begin{bmatrix} 3 \\ 4 \end{bmatrix}. \end{aligned}$$

The entries are consecutive differences:

$$5 - 2 = 3, \quad 9 - 5 = 4.$$

Thus D measures changes in a short time series.

Activity: Document ranking in matrix code

In 1.1.5, we ranked documents by cosine similarity one score at a time. The following code repeats that ranking using a data matrix whose rows are the document vectors.

```
import numpy as np

doc_names = np.array(["D1", "D2", "D3"])

X = np.array([
    [3., 0., 3.],
    [1., 1., 0.],
    [0., 2., 0.],
])

q = np.array([1., 0., 1.])

scores = (X @ q) / (
    np.linalg.norm(X, axis=1) * np.linalg.norm(q)
)
ranking = doc_names[np.argsort(scores)[:, :-1]]

scores, ranking
```

Output:

```
(array([1. , 0.5, 0. ]),
 array(['D1', 'D2', 'D3']))
```

1. What does the second row of X represent?
2. Which expression computes the three dot products $D_i \cdot q$?
3. What does `np.argsort(scores)[:, :-1]` do?

Solution.

The rows of X are D_1 , D_2 , and D_3 . The product $X @ q$ computes the three dot products with the query.

To understand `np.argsort(scores)[:, :-1]`, read it in two steps:

1. `np.argsort(scores)` returns the indices that put the scores in increasing order, rather than the sorted score values. NumPy indices start at 0. For scores `[1, 0.5, 0]`, it returns `[2, 1, 0]`: the smallest score is at index 2, the next at index 1, and the

largest at index 0.

2. `[::-1]` reverses that index array. A slice has the form `[start:stop:step]`; leaving the endpoints blank and using step `-1` traverses the whole array backward. Thus `[2, 1, 0]` becomes `[0, 1, 2]`, which lists the indices from largest score to smallest.

The assignment to `ranking` uses these indices to select entries from `doc_names`. We could write the same calculation in two lines:

```
order = np.argsort(scores)[::-1]
ranking = doc_names[order]
```

Here `order` is `[0, 1, 2]`, so `ranking` contains `doc_names[0]`, `doc_names[1]`, and `doc_names[2]`, in that order. Thus the document names are arranged from highest score to lowest. If the scores changed, the indices could select the names in a different order.

The displayed output `array(['D1', 'D2', 'D3'])` describes a one-dimensional NumPy array containing three text strings.

The scores agree with 1.1.5: $1, \frac{1}{2}, 0$.

Example: The attention activity in matrix form

In 1.1.5, we computed the scores

$$\mathbf{q} \cdot \mathbf{k}_{\text{small}}, \quad \mathbf{q} \cdot \mathbf{k}_{\text{red}}, \quad \mathbf{q} \cdot \mathbf{k}_{\text{bird}}$$

one at a time. Matrix-vector multiplication computes the same scores at once.

Put the key vectors as the rows of

$$K = \begin{bmatrix} \mathbf{k}_{\text{small}}^T \\ \mathbf{k}_{\text{red}}^T \\ \mathbf{k}_{\text{bird}}^T \end{bmatrix}.$$

Then

$$K\mathbf{q} = \begin{bmatrix} \mathbf{k}_{\text{small}} \cdot \mathbf{q} \\ \mathbf{k}_{\text{red}} \cdot \mathbf{q} \\ \mathbf{k}_{\text{bird}} \cdot \mathbf{q} \end{bmatrix}.$$

For the numbers in 1.1.5,

$$K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{q} = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

so

$$K\mathbf{q} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

Activity: The same attention calculation in code

The following code performs the calculation from the token attention activity.

```
import numpy as np

tokens = np.array(["small", "red", "bird"])

q = np.array([1., 1.])

K = np.array([
    [1., 0.],
    [0., 1.],
    [1., 1.],
])

V = np.array([
    [4., 0.],
    [0., 4.],
    [4., 4.],
])

s = K @ q
alpha = s / s.sum()
out = alpha @ V

s, alpha, out
```

Output:

```
(array([1., 1., 2.]),
 array([0.25, 0.25, 0.5 ]),
 array([3., 3.]))
```

1. Which line computes the token scores?
2. Which token receives the largest weight?
3. Which line forms the weighted average?
4. Why do the entries of `alpha` add to 1?

Solution.

The line $s = K @ q$ computes the three dot products. The token “bird” receives the largest weight. The line $out = alpha @ v$ forms the weighted average of the rows of V . The entries of $alpha$ add to 1 because the scores were divided by their sum.

1.3.2 Geometric matrix actions in $\mathbb{R}^2$ **Definition 1.3.4 (Identity matrix).**

The *identity matrix* I_n is the $n \times n$ matrix with 1s on the main diagonal and zeros elsewhere.

Fact

For every column vector $x \in \mathbb{R}^n$, $I_n x = x$.

Definition 1.3.5 (Standard basis).

The j -th *standard basis vector* e_j is the column vector with a 1 in position j and zeros elsewhere.

The term “basis” will be explained later in the course. For now, use the concrete description of e_j above.

Fact

If $A = [a_1 \ \cdots \ a_n]$, then $Ae_j = a_j$. Applying a matrix to a standard basis vector picks out a column.

In $\mathbb{R}^2$, a matrix $A = [a_1 \ a_2]$ sends e_1 to a_1 and e_2 to a_2 . Since every vector $x = (x_1, x_2)$ can be written as $x_1 e_1 + x_2 e_2$, we have

$$Ax = x_1 a_1 + x_2 a_2.$$

Thus a 2×2 matrix is geometrically determined by where it sends the two coordinate directions.

To understand a 2×2 matrix A , apply it to the four corners of the unit square. The image is usually a parallelogram. The two edges leaving the origin are the columns of A . We will refer to this as the *unit-square visualization*.

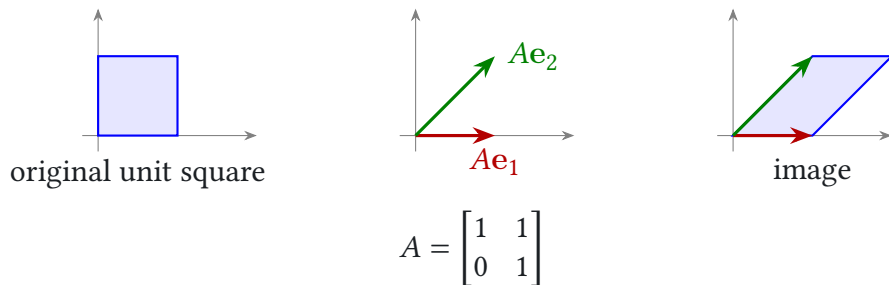
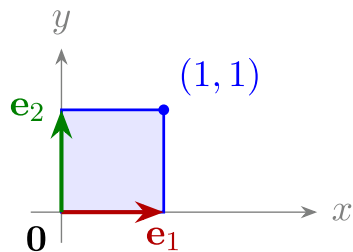
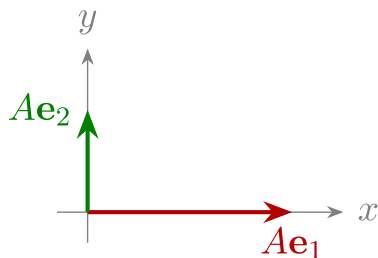


Figure 1.9. A 2×2 matrix sends the unit square to the parallelogram determined by its two columns. This is the geometric version of columns contribute.

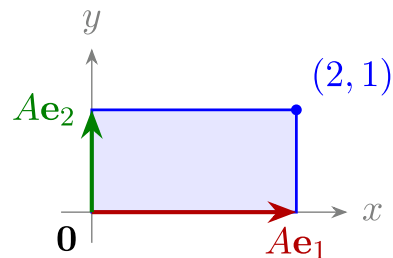
original unit square



columns



image



$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

Figure 1.10. A horizontal stretch by a factor of 2 sends the unit square to a rectangle twice as wide. The first column doubles the horizontal basis vector, while the second column leaves the vertical basis vector unchanged.

- **Matrix:** $\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$

$$Ae_1: \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$Ae_2: \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Name: horizontal stretch

Unit-square effect: The right edge moves farther right.

- **Matrix:** $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

$$Ae_1: \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$Ae_2: \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Name: projection onto the y-axis

Unit-square effect: Horizontal information is forgotten.

- **Matrix:** $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

$$Ae_1: \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$Ae_2: \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Name: reflection across the y-axis

Unit-square effect: The square flips left.

- **Matrix:** $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

$$Ae_1: \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$Ae_2: \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Name: reflection across the line $y = x$

Unit-square effect: The square stays in the same region, but the horizontal and vertical coordinate directions exchange.

• **Matrix:** $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$$Ae_1: \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$Ae_2: \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Name: horizontal shear

Unit-square effect: The bottom edge stays fixed; the top edge shifts right.

• **Matrix:** $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

$$Ae_1: \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$Ae_2: \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

Name: 90° counterclockwise rotation

Unit-square effect: The square rotates around the origin.

• **Matrix:** $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$$Ae_1: \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$Ae_2: \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Name: projection onto the x -axis

Unit-square effect: Vertical information is forgotten.

For more illustrations of these transformations, see Lab U1: Vectors, Similarity, Attention, and Matrix Actions.

Activity: Horizontal shear

Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Compute Ae_1 , Ae_2 , and Ax , where $x = (1, 1)$.

Solution.

We have

$$A\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad A\mathbf{e}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad A\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

The formula is $A(x, y) = (x + y, y)$. Points with $y = 0$ stay fixed, and points with $y = 1$ shift right by 1.

Note

For the shear matrix $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, the row view gives

$$A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} (1, 1) \cdot (x, y) \\ (0, 1) \cdot (x, y) \end{bmatrix} = \begin{bmatrix} x + y \\ y \end{bmatrix}.$$

The first row measures horizontal plus vertical, while the second row measures vertical only. The column view gives

$$A \begin{bmatrix} x \\ y \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

so the y -coordinate contributes both upward and rightward motion.

Activity: Reflection across the y -axis

Let $A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$. Compute $A\mathbf{e}_1$, $A\mathbf{e}_2$, $A\mathbf{x}$, and $A\mathbf{y}$, where $\mathbf{x} = (1, 1)$ and $\mathbf{y} = (0, 1)$.

Solution.

We get

$$\begin{aligned} A\mathbf{e}_1 &= \begin{bmatrix} -1 \\ 0 \end{bmatrix}, & A\mathbf{e}_2 &= \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \\ A\mathbf{x} &= \begin{bmatrix} -1 \\ 1 \end{bmatrix}, & A\mathbf{y} &= \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \end{aligned}$$

The first coordinate flips sign and the second coordinate is unchanged.

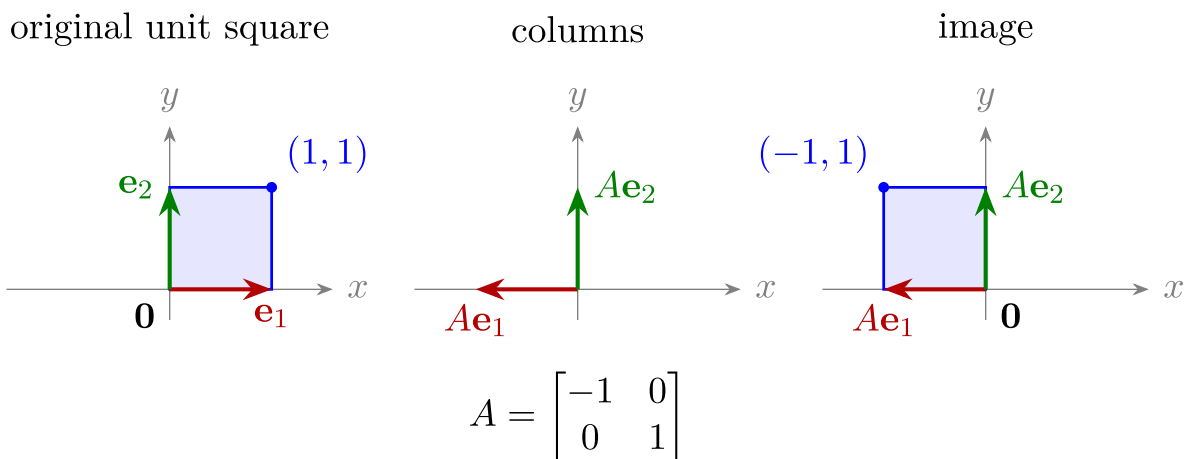


Figure 1.11. Reflection across the y -axis sends the unit square to its left-hand mirror image. The first column reverses the horizontal direction, while the second column leaves the vertical direction unchanged.

Activity: Reflection across the line $y=x$

Let

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

1. Compute Ae_1 and Ae_2 .
2. Compute $A \begin{bmatrix} x \\ y \end{bmatrix}$.
3. What happens to a point on the line $y = x$?

Solution.

We have

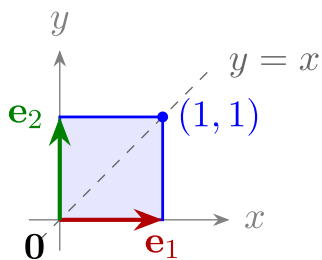
$$Ae_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = e_2, \quad Ae_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = e_1.$$

More generally, write the input in terms of the standard basis:

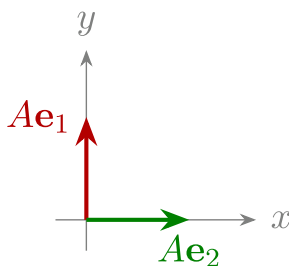
$$\begin{aligned} A \begin{bmatrix} x \\ y \end{bmatrix} &= A(xe_1 + ye_2) = xAe_1 + yAe_2 \\ &= xe_2 + ye_1 = \begin{bmatrix} y \\ x \end{bmatrix}. \end{aligned}$$

Thus the two coordinates are exchanged. If $x = y$, exchanging the coordinates does not move the point, so every point on the line $y = x$ stays fixed.

original unit square



columns



$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

image

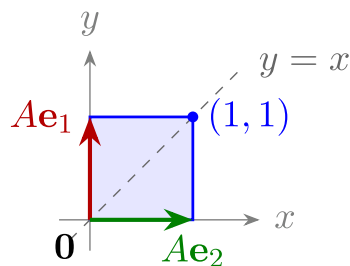


Figure 1.12. The unit square occupies the same region after reflection across $y = x$, even though its horizontal and vertical coordinate directions have exchanged, so an unlabeled image of the square can appear unchanged. The labeled red and green edges show the exchange. The corners $(0, 0)$ and $(1, 1)$ on the dashed mirror line stay fixed.

1.3.3 Linear and affine maps

A function assigns one output to each input. In this course, many functions have vectors as inputs and vectors as outputs:

$$F : \mathbb{R}^n \rightarrow \mathbb{R}^m.$$

The *domain* of F is its set of allowed inputs, $\mathbb{R}^n$, and its *co-domain* is the specified set containing its outputs, $\mathbb{R}^m$. A matrix gives one important source of such functions by the rule $\mathbf{x} \mapsto A\mathbf{x}$.

Definition 1.3.6 (Matrix map).

Given an $m \times n$ matrix A , the *matrix map* induced by A is the function $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by

$$T_A(\mathbf{x}) = A\mathbf{x}.$$

Definition 1.3.7 (Linear map).

A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is *linear* if

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= T(\mathbf{u}) + T(\mathbf{v}), \\ T(c\mathbf{v}) &= cT(\mathbf{v}) \end{aligned}$$

for all vectors $\mathbf{u}, \mathbf{v}$ and all scalars c .

Theorem 1.3.8 (Matrix maps are linear).

If A is an $m \times n$ matrix, then the function

$$T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad T_A(\mathbf{x}) = A\mathbf{x}$$

is linear.

Proof.

Matrix-vector multiplication distributes over vector addition and scalar multiplication:

$$A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}, \quad A(c\mathbf{v}) = cA\mathbf{v}.$$

These are exactly the two linearity rules.

Theorem 1.3.9 (Every linear map has a matrix).

Every linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is given by multiplication by a unique matrix. The columns of that matrix are $T(\mathbf{e}_1), \dots, T(\mathbf{e}_n)$.

Proof.

Write $\mathbf{x} = x_1\mathbf{e}_1 + \dots + x_n\mathbf{e}_n$ and let $A = [T(\mathbf{e}_1) \ \dots \ T(\mathbf{e}_n)]$. By linearity and the column view of matrix-vector multiplication,

$$T(\mathbf{x}) = x_1T(\mathbf{e}_1) + \dots + x_nT(\mathbf{e}_n) = A\mathbf{x}.$$

This matrix is unique because its j -th column must be $A\mathbf{e}_j = T(\mathbf{e}_j)$.

Activity: Images of basis vectors determine the matrix

Suppose a linear map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ satisfies $T(\mathbf{e}_1) = (2, 1)$ and $T(\mathbf{e}_2) = (-1, 3)$. Find the matrix A such that $T(\mathbf{x}) = A\mathbf{x}$.

Solution.

The columns are the images of the standard basis vectors, so

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}.$$

Warning

We use “linear map” and “linear transformation” interchangeably; “transformation” is often used when emphasizing geometry.

Definition 1.3.10 (Affine map).

A function of the form

$$\mathbf{x} \mapsto A\mathbf{x} + \mathbf{b}$$

is called *affine*. It is built from a linear map followed by a *translation*.

An affine map is linear only when $\mathbf{b} = \mathbf{0}$. A quick test is the zero vector: every linear map sends $\mathbf{0}$ to $\mathbf{0}$, but

$$A\mathbf{0} + \mathbf{b} = \mathbf{b}.$$

If $\mathbf{b} \neq \mathbf{0}$, the map is not linear.

Example: An affine layer

A layer in a neural network can take the form

$$\mathbf{x} \mapsto W\mathbf{x} + \mathbf{b}.$$

The matrix W mixes the input coordinates. The vector $\mathbf{b}$ shifts the result. Since

$$W\mathbf{0} + \mathbf{b} = \mathbf{b},$$

this map is affine. It is linear exactly when $\mathbf{b} = \mathbf{0}$.

1.4 Matrix multiplication and map composition

1.4.1 Matrix product formulas

Definition 1.4.1 (Matrix-matrix product).

Let A be an $m \times n$ matrix, let B be an $n \times k$ matrix, and write

$$B = [\mathbf{b}_1 \quad \mathbf{b}_2 \quad \cdots \quad \mathbf{b}_k]$$

where $\mathbf{b}_j$ is column j of B . The *product* AB is the $m \times k$ matrix

$$AB = [A\mathbf{b}_1 \quad A\mathbf{b}_2 \quad \cdots \quad A\mathbf{b}_k].$$

Activity: Computing a matrix product using columns

Let

$$A = \begin{bmatrix} 1 & -3 & 5 \\ 3 & 1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 1 \\ -2 & -6 \\ -1 & 0 \end{bmatrix}.$$

Compute AB .

Solution.

The columns of B are $\mathbf{b}_1 = (5, -2, -1)$ and $\mathbf{b}_2 = (1, -6, 0)$. Using the row dot-product rule in 1.3.3,

$$\begin{aligned} A\mathbf{b}_1 &= \begin{bmatrix} 1(5) + (-3)(-2) + 5(-1) \\ 3(5) + 1(-2) + 4(-1) \end{bmatrix} = \begin{bmatrix} 6 \\ 9 \end{bmatrix}, \\ A\mathbf{b}_2 &= \begin{bmatrix} 1(1) + (-3)(-6) + 5(0) \\ 3(1) + 1(-6) + 4(0) \end{bmatrix} = \begin{bmatrix} 19 \\ -3 \end{bmatrix}. \end{aligned}$$

Placing these columns side by side gives

$$AB = [A\mathbf{b}_1 \quad A\mathbf{b}_2] = \begin{bmatrix} 6 & 19 \\ 9 & -3 \end{bmatrix}.$$

Theorem 1.4.2.

If A is $m \times n$ and B is $n \times k$, then the (i, j) -entry of AB is the dot product of row i of A with column j of B :

$$(AB)_{ij} = \sum_{\ell=1}^n a_{i\ell} b_{\ell j}.$$

Proof.

By the definition of matrix multiplication, column j of AB is $A\mathbf{b}_j$, where $\mathbf{b}_j = (b_{1j}, \dots, b_{nj})$ is column j of B . The row dot-product rule for matrix-vector multiplication in 1.3.3 therefore gives

$$\begin{aligned} (AB)_{ij} &= (A\mathbf{b}_j)_i \\ &= \text{row}_i(A) \cdot \mathbf{b}_j \\ &= \sum_{\ell=1}^n a_{i\ell} b_{\ell j}. \end{aligned}$$

Note: Row-column interpretation

The sum above is the dot product of row i of A and column j of B . Viewing both as n -vectors,

$$(AB)_{ij} = (a_{i1}, \dots, a_{in}) \cdot (b_{1j}, \dots, b_{nj}).$$

Pair entries in corresponding positions, multiply, and add.

Activity: Computing a matrix entry

If

$$A = \begin{bmatrix} -3 & -4 & 1 \\ 2 & 4 & 0 \\ 1 & -4 & -5 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 2 & -4 \\ -4 & -3 & -1 \\ 4 & 3 & 1 \end{bmatrix},$$

what is the $(2, 3)$ -entry of AB ?

Solution.

Use row 2 of A and column 3 of B :

$$\begin{aligned}(2, 4, 0) \cdot (-4, -1, 1) &= 2(-4) + 4(-1) + 0(1) \\ &= -12.\end{aligned}$$

Theorem 1.4.3 (Properties of matrix multiplication).

Assume that A, B, C are matrices of appropriate sizes and r is a scalar. Then

1. $A(BC) = (AB)C$.
2. $(A + B)C = AC + BC$.
3. $C(A + B) = CA + CB$.
4. $A(rB) = r(AB) = (rA)B$.
5. $(AB)^T = B^T A^T$.

Proof.

We prove property 5. Let A be $m \times n$ and B be $n \times k$. Both $(AB)^T$ and $B^T A^T$ have shape $k \times m$. Transposition exchanges row and column indices, so the entry formula in 1.4.2 gives

$$\begin{aligned}((AB)^T)_{ij} &= (AB)_{ji} \\ &= \sum_{\ell=1}^n a_{j\ell} b_{\ell i} \\ &= \sum_{\ell=1}^n b_{\ell i} a_{j\ell} \\ &= \sum_{\ell=1}^n (B^T)_{i\ell} (A^T)_{\ell j} \\ &= (B^T A^T)_{ij}.\end{aligned}$$

We can reverse the two scalar factors in each summand because real-number multiplication is commutative. Every corresponding entry agrees, so $(AB)^T = B^T A^T$.

Note: Shape habit for matrix products

Before forming AB , check the inner dimensions. If A is $m \times n$ and B is $n \times k$, then the shared inner dimension n makes the product possible, and the outer dimensions give AB shape $m \times k$.

1.4.2 Matrix multiplication as composition

If $T_A(\mathbf{x}) = A\mathbf{x}$ and $T_B(\mathbf{x}) = B\mathbf{x}$, then applying B first and then A gives

$$T_A(T_B(\mathbf{x})) = A(B\mathbf{x}) = (AB)\mathbf{x}.$$

Matrix multiplication represents composition of matrix maps.

Activity: Composition shape check

Suppose A is 4×3 and B is 3×2 . What is the shape of AB , and what is the domain and codomain of the composed map $\mathbf{x} \mapsto A(B\mathbf{x})$?

Solution.

The product AB is 4×2 . The composed map takes inputs from $\mathbb{R}^2$ and outputs vectors in $\mathbb{R}^4$.

Activity: Applying one matrix to many points

Put the vertices of a shape in the columns of a matrix X . Then $Y = AX$ contains the transformed vertices.

```
import numpy as np

np.set_printoptions(precision=3, suppress=True)

# Columns are vertices of the unit square.
# The final column repeats the first vertex to close the shape.
X = np.array([
    [0., 1., 1., 0., 0.],
    [0., 0., 1., 1., 0.]
])

A = np.array([
    [1., 1.],
    [0., 1.]
])

Y = A @ X
Y
```

Output:

```
array([[0., 1., 2., 1., 0.],
       [0., 0., 1., 1., 0.]])
```

1. What are the columns of Y ?
2. Which transformation from the gallery is this?
3. Why does the bottom edge stay fixed?

Solution.

The columns of X are the vertices of the unit square:

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The final column repeats the first vertex to close the shape. The columns of Y are their images under the map $x \mapsto Ax$:

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

This matrix is a horizontal shear. It sends (x, y) to $(x + y, y)$. Points with $y = 0$ stay fixed.

Warning

In this geometric example, the columns of X are points, so $Y = AX$ applies the same matrix to every point. In many data tables, observations are stored as rows. For example, a document matrix may have one document per row. Always check what rows and columns represent before interpreting a product.

Warning: Later square-grid visualizations

The matrix computation $Y = AX$ applies one matrix to every column of X . For a 2×2 matrix, the images of the coordinate directions determine the whole grid. Later, in 3.1.1, the same columns-of-points convention is used for rules that are not matrix maps. Then four corners may not tell the whole story. A grid of input points gives more information.

1.4.3 A missing rule: AB need not equal BA

Multiplication of real numbers is commutative: $ab = ba$. Matrix multiplication is not commutative in general. For two square matrices A and B of the same size, we say that A and B *commute* when

$$AB = BA.$$

Some pairs commute and some do not.

Activity: Commute or not?

For each pair below:

1. compute both products;

2. determine whether $AB = BA$;

Pair A: two distinct diagonal transformations. Use the horizontal stretch and reflection across the y -axis:

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad F = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Pair B: a shared axis label is not enough. Use the same horizontal stretch and the horizontal shear:

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Pair C: reflection across the line $y = x$. Use the same horizontal stretch and

$$M = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Learning outcomes. U1-LO6.

Solution.

For Pair A, row-column dot products give

$$\begin{aligned} SF &= \begin{bmatrix} (2,0) \cdot (-1,0) & (2,0) \cdot (0,1) \\ (0,1) \cdot (-1,0) & (0,1) \cdot (0,1) \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & 1 \end{bmatrix}, \\ FS &= \begin{bmatrix} (-1,0) \cdot (2,0) & (-1,0) \cdot (0,1) \\ (0,1) \cdot (2,0) & (0,1) \cdot (0,1) \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & 1 \end{bmatrix}. \end{aligned}$$

The matrices commute. Flipping the horizontal coordinate and doubling it give the same result in either order.

For Pair B, row-column dot products give

$$\begin{aligned} SH &= \begin{bmatrix} (2,0) \cdot (1,0) & (2,0) \cdot (1,1) \\ (0,1) \cdot (1,0) & (0,1) \cdot (1,1) \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 0 & 1 \end{bmatrix}, \\ HS &= \begin{bmatrix} (1,1) \cdot (2,0) & (1,1) \cdot (0,1) \\ (0,1) \cdot (2,0) & (0,1) \cdot (0,1) \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}. \end{aligned}$$

The matrices do not commute. Both transformations involve the horizontal coordinate, but this does not suffice for the matrices to commute.

For Pair C, row-column dot products give

$$SM = \begin{bmatrix} (2,0) \cdot (0,1) & (2,0) \cdot (1,0) \\ (0,1) \cdot (0,1) & (0,1) \cdot (1,0) \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix},$$

$$MS = \begin{bmatrix} (0,1) \cdot (2,0) & (0,1) \cdot (0,1) \\ (1,0) \cdot (2,0) & (1,0) \cdot (0,1) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}.$$

The matrices do not commute. The reflection exchanges the coordinate directions. Stretching before that exchange is not the same as stretching afterward.

Activity: Two horizontal shears

Let

$$H = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad G = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

The matrix G is a stronger horizontal shear.

1. Justify the claim that applying H twice has the same effect as applying G once.
2. Compute HG and GH . Do the matrices commute?

Learning outcomes. U1-LO6.

Solution.

1. Applying H twice is represented by H^2 , and

$$H^2 = \begin{bmatrix} 1 \cdot 1 + 1 \cdot 0 & 1 \cdot 1 + 1 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 0 & 0 \cdot 1 + 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = G.$$

Thus $H(Hx) = Gx$ for every input x .

2. Computing both products directly, row-column dot products give

$$HG = \begin{bmatrix} (1,1) \cdot (1,0) & (1,1) \cdot (2,1) \\ (0,1) \cdot (1,0) & (0,1) \cdot (2,1) \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix},$$

$$GH = \begin{bmatrix} (1,2) \cdot (1,0) & (1,2) \cdot (1,1) \\ (0,1) \cdot (1,0) & (0,1) \cdot (1,1) \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}.$$

These matrices are distinct and non-diagonal, but they commute. The two horizontal shear amounts add in either order.

Part 1 also gives a shorter explanation: since $G = H^2$, associativity gives

$$HG = H(HH) = (HH)H = GH.$$

Both orders apply H three times, so the result of part 1 lets us conclude that the matrices commute without computing both products.

Note

Some structural patterns settle the question quickly. Two diagonal matrices of the same size commute. For example, if

$$A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}, \quad B = \begin{bmatrix} c & 0 \\ 0 & d \end{bmatrix},$$

then

$$AB = \begin{bmatrix} ac & 0 \\ 0 & bd \end{bmatrix},$$

$$BA = \begin{bmatrix} ca & 0 \\ 0 & db \end{bmatrix} = \begin{bmatrix} ac & 0 \\ 0 & bd \end{bmatrix} = AB.$$

The equality follows because multiplication of real numbers is commutative.

Every square matrix commutes with itself. The two horizontal shears in 1.4.3 are closely related to this fact: as the solution shows, $G = H^2$, so both HG and GH apply the same matrix H three times. Associativity gives

$$HG = H(HH) = (HH)H = GH.$$

In general, compute AB and BA and compare.

1.4.4 Many token queries and keys

Example: Many token queries at once

Suppose a sequence has L tokens. Store the token vectors as rows of a matrix

$$X = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_L^T \end{bmatrix} \in \mathbb{R}^{L \times d}.$$

A transformer layer forms three views of the same token vectors:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V.$$

The matrices W_Q, W_K, W_V are weight matrices. Their entries are adjusted during training. For now, treat them as ordinary matrices that assign each token three roles.

For destination position i , the vector $\mathbf{q}_i$ is its query: what that position uses to decide which source positions are relevant to its attention output.

For source position j , the vector $\mathbf{k}_j$ is its key: what is compared with the query from a destination position.

For source position j , the vector $\mathbf{v}_j$ is its value: the information that position can contribute after weights are chosen.

Write

$$Q = \begin{bmatrix} \mathbf{q}_1^T \\ \mathbf{q}_2^T \\ \vdots \\ \mathbf{q}_L^T \end{bmatrix}, \quad K = \begin{bmatrix} \mathbf{k}_1^T \\ \mathbf{k}_2^T \\ \vdots \\ \mathbf{k}_L^T \end{bmatrix}.$$

Then

$$K^T = [\mathbf{k}_1 \quad \mathbf{k}_2 \quad \cdots \quad \mathbf{k}_L].$$

So

$$QK^T = \begin{bmatrix} \mathbf{q}_1^T \\ \mathbf{q}_2^T \\ \vdots \\ \mathbf{q}_L^T \end{bmatrix} [\mathbf{k}_1 \quad \mathbf{k}_2 \quad \cdots \quad \mathbf{k}_L] = \begin{bmatrix} \mathbf{q}_1 \cdot \mathbf{k}_1 & \mathbf{q}_1 \cdot \mathbf{k}_2 & \cdots & \mathbf{q}_1 \cdot \mathbf{k}_L \\ \mathbf{q}_2 \cdot \mathbf{k}_1 & \mathbf{q}_2 \cdot \mathbf{k}_2 & \cdots & \mathbf{q}_2 \cdot \mathbf{k}_L \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{q}_L \cdot \mathbf{k}_1 & \mathbf{q}_L \cdot \mathbf{k}_2 & \cdots & \mathbf{q}_L \cdot \mathbf{k}_L \end{bmatrix}.$$

Thus the score matrix

$$S = QK^T$$

has entries

$$S_{ij} = \mathbf{q}_i \cdot \mathbf{k}_j.$$

Row i of S contains the scores used when computing the attention output at position i .

Activity: Many-token shape and entry check

Suppose

$$X \in \mathbb{R}^{5 \times 4}$$

and

$$W_Q, W_K, W_V \in \mathbb{R}^{4 \times 3}.$$

Let

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \quad S = QK^T.$$

1. What are the shapes of Q , K , and V ?

2. What is the shape of K^T ?
3. What is the shape of $S = QK^T$?
4. Suppose

$$\mathbf{q}_2 = \begin{bmatrix} q_{21} \\ q_{22} \\ q_{23} \end{bmatrix}, \quad \mathbf{k}_4 = \begin{bmatrix} k_{41} \\ k_{42} \\ k_{43} \end{bmatrix}.$$

Expand the entry S_{24} .

5. What does S_{24} compare?
6. What does row 2 of S contain?

Solution.

Since

$$X \in \mathbb{R}^{5 \times 4} \quad \text{and} \quad W_Q, W_K, W_V \in \mathbb{R}^{4 \times 3},$$

the matrices Q , K , and V all have shape

$$5 \times 3.$$

Thus

$$K^T \in \mathbb{R}^{3 \times 5}.$$

Therefore

$$S = QK^T$$

has shape

$$5 \times 5.$$

The $(2, 4)$ -entry comes from row 2 of Q and column 4 of K^T . But column 4 of K^T is the transpose of row 4 of K . Hence

$$\begin{aligned} S_{24} &= \begin{bmatrix} q_{21} & q_{22} & q_{23} \end{bmatrix} \begin{bmatrix} k_{41} \\ k_{42} \\ k_{43} \end{bmatrix} \\ &= q_{21}k_{41} + q_{22}k_{42} + q_{23}k_{43}. \end{aligned}$$

So

$$S_{24} = \mathbf{q}_2 \cdot \mathbf{k}_4.$$

This entry compares the query for token 2 with the key for token 4. Row 2 of S contains the scores comparing token 2's query with all five token keys.

A later weighting step converts each row of S into weights that add to 1. Those weights are then used to average the value vectors.

From attention scores to attention output

For this section, an attention calculation has three steps:

dot-product scores $\longrightarrow$ weights $\longrightarrow$ weighted averages.

The score S_{ij} compares the query at destination position i with the key at source position j . A larger score means source position j is more relevant to the attention output at destination position i .

The scores are not yet the output. For each token i , a weighting rule converts row i of S into numbers

$$\alpha_{i1}, \alpha_{i2}, \dots, \alpha_{iL}$$

that are nonnegative and add to 1. These are the attention weights for token i . They say how much token i uses each value vector.

Let

$$A_{\text{att}} = [\alpha_{ij}]$$

be the attention-weight matrix. If the value vectors are stored as rows,

$$V = \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_L^T \end{bmatrix},$$

then the attention output is

$$H = A_{\text{att}}V.$$

Row i of H is

$$\mathbf{h}_i^T = \alpha_{i1}\mathbf{v}_1^T + \alpha_{i2}\mathbf{v}_2^T + \dots + \alpha_{iL}\mathbf{v}_L^T.$$

Equivalently,

$$\mathbf{h}_i = \alpha_{i1}\mathbf{v}_1 + \alpha_{i2}\mathbf{v}_2 + \dots + \alpha_{iL}\mathbf{v}_L.$$

Thus H is the attention-output matrix, and each row of H is a weighted average of value vectors. These rows contribute to later token representations; a full transformer layer combines them with the current representations and processes them further.

Activity: Many-token attention output

Suppose

$$X \in \mathbb{R}^{4 \times 3}, \quad W_Q, W_K, W_V \in \mathbb{R}^{3 \times 2}.$$

Define

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \quad S = QK^T.$$

A weighting step converts S into an attention-weight matrix

$$A_{\text{att}} \in \mathbb{R}^{4 \times 4}.$$

The attention output is

$$H = A_{\text{att}}V.$$

1. What are the shapes of Q , K , and V ?
2. What is the shape of $S = QK^T$?
3. What does S_{ij} compare?
4. What does row i of A_{att} tell you?
5. What is the shape of H ?
6. Why is each row of H a weighted average of value vectors?
7. Now suppose row 2 of A_{att} is

$$\alpha_2^T = [0 \quad 1/2 \quad 1/4 \quad 1/4]$$

and

$$V = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 4 \\ 4 & 4 \end{bmatrix}.$$

Compute the attention-output vector $\mathbf{h}_2$.

Solution.

Since

$$X \in \mathbb{R}^{4 \times 3} \quad \text{and} \quad W_Q, W_K, W_V \in \mathbb{R}^{3 \times 2},$$

the matrices Q , K , and V are all 4×2 .

The score matrix is

$$S = QK^T.$$

Since Q is 4×2 and K^T is 2×4 , the matrix S is 4×4 . The entry S_{ij} compares the query for token i with the key for token j :

$$S_{ij} = \mathbf{q}_i \cdot \mathbf{k}_j.$$

Row i of A_{att} gives the weights token i uses when averaging the value vectors. Since A_{att} is 4×4 and V is 4×2 ,

$$H = A_{\text{att}}V$$

has shape 4×2 .

Row i of H is a weighted average because

$$\mathbf{h}_i = \alpha_{i1}\mathbf{v}_1 + \alpha_{i2}\mathbf{v}_2 + \alpha_{i3}\mathbf{v}_3 + \alpha_{i4}\mathbf{v}_4,$$

where the weights in row i of A_{att} are nonnegative and add to 1.

For row 2,

$$\mathbf{h}_2 = 0 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 2 \\ 0 \end{bmatrix} + \frac{1}{4} \begin{bmatrix} 0 \\ 4 \end{bmatrix} + \frac{1}{4} \begin{bmatrix} 4 \\ 4 \end{bmatrix}.$$

Thus the attention-output vector is

$$\mathbf{h}_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

The second token ignores value vector 1, uses value vector 2 with weight $1/2$, and uses value vectors 3 and 4 with weight $1/4$ each.

Warning

In real transformer attention, the weighting rule usually uses scaled and masked scores before forming weights. For Unit 1, the important facts are: rows of A_{att} are weights, the weights add to 1, and $A_{\text{att}}V$ forms weighted averages of the rows of V .

Activity: The same shape check in code

The following code performs a many-token shape check similar to the previous activity.

```
import numpy as np
```

```
X_tokens = np.ones((5, 4))
WQ = np.ones((4, 3))
WK = np.ones((4, 3))
WV = np.ones((4, 3))

Q = X_tokens @ WQ
K_tokens = X_tokens @ WK
V_tokens = X_tokens @ WV
scores_many = Q @ K_tokens.T

Q.shape, K_tokens.shape, V_tokens.shape, scores_many.shape
```

Output:

```
((5, 3), (5, 3), (5, 3), (5, 5))
```

1. Which line creates all query-key scores at once?
2. Why is `scores_many.shape` equal to `(5, 5)`?

Solution.

The line `scores_many = Q @ K_tokens.T` creates all query-key scores at once. Since `Q` has shape `(5, 3)` and `K_tokens.T` has shape `(3, 5)`, the score matrix has shape `(5, 5)`.

1.5 Coding recap

1.5.1 Linked notebook

Run Lab U1: Vectors, similarity, attention, and matrix actions. The main sequence practices vector applications, document similarity, token attention, matrix-vector products, geometric matrix actions, applying one matrix to many points, composition, affine maps, and short code outputs. The final part contains review questions and a short forward look to square-grid visualizations.

For a quick reference on arrays, matrix products, norms, rankings, and slicing, see the programming appendix sections A.2, A.4, A.5, A.6, and A.3.

1.5.2 Review activities

Activity: Reading a ranking

Continuing the cosine-similarity ranking from 1.1.5, the code is

```
scores = np.array([1.0, 0.5, 0.0])
doc_names = np.array(["D1", "D2", "D3"])
ranking = doc_names[np.argsort(scores)[::-1]]
ranking
```

1. What is the value of `ranking`?
2. What is being ranked?
3. Is this ranking based on distance or direction?

Solution.

The value is `["D1", "D2", "D3"]`. The documents are being ranked by cosine similarity with the query. This is a direction-based ranking because cosine similarity compares direction after normalization.

Activity: Reading an attention computation

In 1.3.1, the code was

```
s = K @ q
alpha = s / s.sum()
out = alpha @ V
```

1. What does `s` contain?
2. Why do the entries of `alpha` add to 1?
3. What does `out` represent?

Solution.

The vector `s` contains dot-product scores. The vector `alpha` contains weights because the scores were divided by their sum. The vector `out` is a weighted average of the value vectors stored in the rows of `V`.

Activity: Reading a geometric computation

In 1.4.2, the code was

```
X = np.array([
    [0, 1, 1, 0, 0],
    [0, 0, 1, 1, 0],
], dtype=float)

A = np.array([
    [1, 1],
    [0, 1],
], dtype=float)

Y = A @ X
Y
```

1. What are the columns of X ?
2. What are the columns of Y ?
3. Which transformation from the gallery is this?

Solution.

The columns of X are vertices of the unit square. The columns of Y are their images under A . This is a horizontal shear because A sends (x, y) to $(x + y, y)$.

Activity: Reading composition order

```
S = np.array([
    [2., 0.],
    [0., 1.],
])

R = np.array([
    [0., -1.],
    [1., 0.],
])

R @ S, S @ R
```

1. Which product applies S first and then R ?

2. Are the two products equal?
3. What does the difference mean geometrically?

Solution.

The product $R @ S$ applies S first and then R , because the rightmost matrix acts first. The two products are not equal:

$$RS = \begin{bmatrix} 0 & -1 \\ 2 & 0 \end{bmatrix}, \quad SR = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix}.$$

Rotating after a horizontal stretch is not the same as stretching after a rotation.

Activity: Reading an affine rule

```
W = np.array([
    [1., 1.],
    [0., 2.],
])

b = np.array([3., -1.])

def f(x):
    return W @ x + b
```

Is f linear? Justify your answer using the zero vector.

Solution.

No. A linear map must send 0 to 0 . But $f(0) = W0 + b = b$, and b is not the zero vector. The rule is affine, but not linear.

1.6 Exercises

Learning outcomes. The labels below identify the learning outcomes for each exercise group. Individual problems may also involve earlier outcomes.

1.6.1 Matrices

Learning outcomes. U1-LO3.

- Nicholson 2.1.3(f)

- Nicholson 2.1.8(b)
- Nicholson 2.1.15(b)

1.6.2 Matrix-vector multiplication

Learning outcomes. U1-LO4.

- Nicholson 2.2.2(b)
- Nicholson 2.2.3(b)
- Nicholson 2.2.11(d)

1.6.3 Linear and affine maps

Learning outcomes. U1-LO5, U1-LO7.

- Nicholson 2.6.2(b)
- Nicholson 2.6.13(b)

1.6.4 Matrix multiplication

Learning outcomes. U1-LO6.

- Nicholson 2.3.1(b)
- Nicholson 2.3.3(b)
- Nicholson 2.3.5(b)
- Nicholson 2.3.16(b)

1.6.5 Additional exercises

These exercises connect the main Unit 1 ideas: vectors as data, cosine similarity, weighted averages, matrix-vector products, geometric matrix actions, composition, affine maps, and code interpretation. Solutions are collected in Appendix A.1.

Additional exercise: Word-count vectors and cosine similarity

Use the dictionary `linear`, `matrix`, `data`. Let

$$\mathbf{q} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix},$$

$$\mathbf{D}_1 = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{D}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{D}_3 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}.$$

1. Compute the cosine similarity between $\mathbf{q}$ and each document vector.
2. Rank the documents by cosine similarity.
3. Explain why $\mathbf{D}_1$ has cosine similarity 1 with $\mathbf{q}$.
4. Which document is orthogonal to $\mathbf{q}$?
5. What does that orthogonality mean in terms of the dictionary words?
6. Compute the Euclidean distance from $\mathbf{q}$ to each $\mathbf{D}_i$. Does the distance ranking match the cosine ranking?

Learning outcomes. U1-LO1, U1-LO2.

Additional exercise: Weighted average of value vectors

Let

$$\mathbf{v}_1 = \begin{bmatrix} 10 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 10 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 10 \\ 10 \end{bmatrix},$$

and let

$$\alpha = \begin{bmatrix} \frac{1}{4} \\ \frac{1}{1} \\ \frac{1}{4} \\ \frac{1}{2} \end{bmatrix}.$$

1. Compute $\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \alpha_3 \mathbf{v}_3$.
2. Explain why this is a weighted average.
3. Which value vector receives the largest weight?

Learning outcomes. U1-LO8.

Additional exercise: Attention bridge: hand scores versus Kq

Use the token attention activity with tokens “small”, “red”, “bird”. Let

$$\mathbf{q} = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

$$\mathbf{k}_{\text{small}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{k}_{\text{red}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{k}_{\text{bird}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

1. Compute the three dot products

$$\mathbf{q} \cdot \mathbf{k}_{\text{small}}, \quad \mathbf{q} \cdot \mathbf{k}_{\text{red}}, \quad \mathbf{q} \cdot \mathbf{k}_{\text{bird}}.$$

2. Put the key vectors as rows of the matrix

$$K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

Compute $K\mathbf{q}$.

3. Explain why the entries of $K\mathbf{q}$ are the same scores from part (a).
4. Normalize the scores by dividing by their sum. What is the weight vector α ?

Learning outcomes. U1-LO1, U1-LO4, U1-LO8.

Additional exercise: Rows measure; columns contribute

Let

$$A = \begin{bmatrix} -1 & 4 & -5 \\ 3 & 1 & -2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}.$$

1. Compute $A\mathbf{x}$ using row dot products.
2. Compute $A\mathbf{x}$ as a linear combination of the columns of A .
3. In this example, what does the first row of A measure?
4. In this example, what do the coordinates $2, -3, 4$ tell you in the column-combination view?

Learning outcomes. U1-LO4.

Additional exercise: The unit-square visualization

Let

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}.$$

1. Compute $A\mathbf{e}_1$ and $A\mathbf{e}_2$.
2. Compute $A \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.
3. Apply A to the four corners of the unit square:

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

4. Sketch the image of the unit square.
5. Which two transformed edges leaving the origin are the columns of A ?

Learning outcomes. U1-LO5, U1-LO8.

Additional exercise: Transformation matching

Match each matrix with the correct geometric description.

$$A_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix},$$
$$A_4 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad A_5 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad A_6 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

1. Reflection across the y -axis.
2. Projection onto the x -axis.
3. Horizontal stretch.
4. Horizontal shear.
5. 90° counterclockwise rotation.
6. Reflection across the line $y = x$.

For each match, briefly justify your answer by describing what happens to $\mathbf{e}_1$ and $\mathbf{e}_2$.

Learning outcomes. U1-LO5.

Additional exercise: Composition order: RS versus SR

Let

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

1. Compute RS .
2. Compute SR .
3. Are RS and SR equal?
4. Which product represents “apply S first, then apply R ”?
5. Explain geometrically why the two orders produce different results.

Learning outcomes. U1-LO6.

Additional exercise: Affine but not linear

Consider the transformation

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + 1 \\ y \end{bmatrix}.$$

1. Compute $T(0, 0)$.
2. Explain why T cannot be written in the form $T(\mathbf{x}) = A\mathbf{x}$ for any matrix A .
3. Write T in the affine form

$$T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}.$$

4. Is T linear, affine but not linear, or neither?

Learning outcomes. U1-LO7.

Additional exercise: Code interpretation: sorting similarity scores

Consider the code:

```
import numpy as np

doc_names = np.array(["D1", "D2", "D3"])
scores = np.array([1.0, 0.5, 0.0])
```

```
ranking = doc_names[np.argsort(scores)[::-1]]
ranking
```

1. What is the value of `ranking`?
2. What does `np.argsort(scores)` return?
3. What does `[::-1]` do?
4. What is being ranked?
5. Is this ranking based on direction, distance, or alphabetical order?

Learning outcomes. U1-LO2, U1-LO8.

Additional exercise: Transformer shape check

Suppose a sequence of token vectors is stored in a matrix

$$X \in \mathbb{R}^{6 \times 5}.$$

Let

$$W_Q, W_K, W_V \in \mathbb{R}^{5 \times 2},$$

and define

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \quad S = QK^T.$$

A later weighting step produces an attention-weight matrix A_{att} , and the attention output is

$$H = A_{\text{att}}V.$$

1. What are the shapes of Q , K , and V ?
2. What is the shape of K^T ?
3. What is the shape of $S = QK^T$?
4. What is the shape of A_{att} ?
5. What is the shape of H ?
6. What does the (i, j) -entry of S compute?
7. Which product forms the weighted averages of value vectors?

8. In this convention, are token vectors stored as rows or columns of X ?

Learning outcomes. U1-LO3, U1-LO6, U1-LO8.

1.7 Unit 1 highlights

1.7.1 Mathematical quick reference

Vectors and linear combinations

Vector. An ordered list of scalars, $\mathbf{v} = (v_1, \dots, v_n) \in \mathbb{R}^n$. Vectors are equal when their dimensions and corresponding entries agree. The zero vector has every entry zero.

Operations. For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and $c \in \mathbb{R}$,

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, \dots, u_n + v_n), \quad c\mathbf{v} = (cv_1, \dots, cv_n).$$

Displacement. From a point P to a point Q , the displacement vector is $\overrightarrow{PQ} = Q - P$.

Linear combination. $c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k$. It is a **convex combination** when every $c_i \geq 0$ and $\sum_i c_i = 1$.

Norm, dot product, and angle

Norm and dot product. For vectors of the same dimension,

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i v_i, \quad \|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}, \quad \text{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

Unit vector. A vector of norm 1. Normalize a nonzero vector by $\mathbf{v}/\|\mathbf{v}\|$.

Angle and cosine similarity. For nonzero $\mathbf{u}, \mathbf{v}$ and their angle $\theta \in [0, \pi]$,

$$\text{cosim}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|} = \cos \theta \in [-1, 1].$$

Angle and cosine similarity are undefined if either vector is zero.

Orthogonality. $\mathbf{u} \perp \mathbf{v}$ means $\mathbf{u} \cdot \mathbf{v} = 0$; for nonzero vectors, this is equivalent to a right angle.

Matrix operations and dimensions

Matrix. $A = [a_{ij}] \in \mathbb{R}^{m \times n}$ has m rows and n columns. Addition requires equal sizes and is entrywise, as is scalar multiplication. A square matrix has $m = n$.

Transpose. $A^T \in \mathbb{R}^{n \times m}$ has entries $(A^T)_{ij} = a_{ji}$. A square matrix is symmetric when $A^T = A$.

Product. If $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$, then

$$AB \in \mathbb{R}^{m \times p}, \quad (AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

Identities. Whenever dimensions permit,

$$A(BC) = (AB)C, \quad A(B + C) = AB + AC, \quad (A + B)C = AC + BC,$$

$$(AB)^T = B^T A^T, \quad (A^T)^T = A, \quad A I_n = I_m A = A.$$

In general $AB \neq BA$; matrices **commute** when both products exist and $AB = BA$.

Matrix-vector products and linear maps

Two descriptions of a product. If $A = [a_1 \ \cdots \ a_n]$ and $\mathbf{x} \in \mathbb{R}^n$, then

$$A\mathbf{x} = x_1 \mathbf{a}_1 + \cdots + x_n \mathbf{a}_n, \quad (A\mathbf{x})_i = \sum_{j=1}^n a_{ij} x_j.$$

Columns contribute scaled vectors; each output entry is a row dot product.

Linear map. $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear when

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}), \quad T(c\mathbf{v}) = cT(\mathbf{v}).$$

In particular, $T(\mathbf{0}) = \mathbf{0}$. Its matrix is

$$A = [T(\mathbf{e}_1) \ \cdots \ T(\mathbf{e}_n)], \quad T(\mathbf{x}) = A\mathbf{x}.$$

Affine map. $F(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ is linear exactly when $\mathbf{b} = \mathbf{0}$.

Composition. Applying B first and A second gives $\mathbf{x} \mapsto AB\mathbf{x}$. Check the inner dimensions before multiplying.

Geometric matrix actions

Read the columns. In the plane, the columns of A are the images of $\mathbf{e}_1$ and $\mathbf{e}_2$; they determine the image of the unit square.

Scaling and counterclockwise rotation.

$$D = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}, \quad R_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$$

Reflections and horizontal shear. Reflection across the y -axis, reflection across $y = x$, and the shear $(x, y) \mapsto (x + ky, y)$ have matrices

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}.$$

1.7.2 Common mistakes

Confusing a dot product with a distance; using cosine similarity with a zero vector; omitting normalization; mixing up rows and columns; ignoring product dimensions; reversing composition order; assuming $AB = BA$; assuming that non-diagonal matrices cannot commute; or calling an affine map linear when its constant term is nonzero.

1.7.3 Connections

Unit 2 uses matrix equations to describe solution sets, column spaces, and null spaces. Unit 3 uses affine maps for local approximation and dot products for directional derivatives. Unit 4 develops dot products into orthogonality and projection. Unit 5 studies eigenvectors and quadratic forms of matrix maps.

Chapter 2

Anatomy of a linear map

A matrix acts on inputs by $x \mapsto Ax$. In applications, Ax might represent measurements from features, predictions from parameters, or scores from a model layer. This unit studies its anatomy through the equations $Ax = b$ (including the important special case where $b = 0$): which outputs are reachable, which directions are forgotten, when inputs are non-unique, and how subspaces, independence, rank, nullity, and inverses organize the answers.

Big question. Given a linear map, how do systems of equations reveal what it produces, what it forgets, and when it can be reversed?

Learning outcomes. By the end of this unit, students should be able to:

- *U2-LO1.* Translate among geometric descriptions, systems of equations, augmented matrices, and matrix equations $Ax = b$.
- *U2-LO2.* Use row reduction to solve linear systems, classify solution sets as empty, unique, or infinite, and describe solutions parametrically.
- *U2-LO3.* Compute and interpret $\text{im}(A)$, $\text{null}(A)$, rank, and nullity as reachable outputs, forgotten directions, transmitted dimensions, and non-unique inputs.
- *U2-LO4.* Use span, subspace, linear independence, basis, coordinates, and dimension to describe subspaces efficiently.
- *U2-LO5.* Determine whether a square matrix is invertible using row reduction, rank, null space, independence, inverses, and determinant criteria; compute inverses when appropriate.
- *U2-LO6.* Identify redundant columns or features and explain how null-space directions create different inputs or parameter vectors with the same output.
- *U2-LO7.* Compute determinants using cofactor expansion and determinant properties, and use determinants to test invertibility.

Toolbox skills. Set up augmented matrices, perform elementary row operations, find REF and RREF, solve homogeneous and nonhomogeneous systems, find basic solutions, compute bases for null spaces and column spaces, compute rank and nullity, test independence, compute inverses, and use determinant tests for invertibility.

2.1 Understanding a linear map: a preview

In Unit 1, a matrix defined a linear map

$$\mathbf{x} \mapsto A\mathbf{x}.$$

This unit answers the following questions about the behavior of this map. The second column previews the concepts introduced later in the unit for this purpose.

- **Question:** When do different inputs give the same output?

Concept/Method: $A(\mathbf{x} + \mathbf{z}) = A\mathbf{x}$

- **Question:** What input changes does A forget?

Concept/Method: $\text{null}(A)$

- **Question:** When can a desired output be hit?

Concept/Method: $A\mathbf{x} = \mathbf{b}$

- **Question:** What outputs can A produce?

Concept/Method: $\text{im}(A)$

- **Question:** Can the map be reversed?

Concept/Method: A^{-1} , when it exists

We give several preliminary examples.

Activity: A map that forgets height

Let

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

First compute $P\mathbf{x}_1$, $P\mathbf{x}_2$, and $P\mathbf{x}_3$, where

$$\mathbf{x}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}.$$

1. Which input coordinate changed?
2. Which part of the input did the output remember?
3. Which part of the input did the output forget?
4. Suppose the desired output is $(2, 1)$. What simple equations must the input coordinates satisfy?
5. Following up on the previous question, how many choices of the third coordinate work?

Solution.

We compute

$$P \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1x + 0y + 0z \\ 0x + 1y + 0z \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

So the outputs are

$$P\mathbf{x}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad P\mathbf{x}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad P\mathbf{x}_3 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

The third coordinate changed, but the output did not. The map remembers the first two coordinates and forgets height.

To get the desired output $(2, 1)$, an input $\mathbf{x} = (x, y, z)$ must satisfy

$$x = 2, \quad y = 1.$$

There is no equation restricting z , so many (in fact, infinitely many!) inputs work:

$$\mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ z \end{bmatrix}.$$

Activity: The same height loss in code

The following code repeats the same experiment.

```
import numpy as np

P = np.array([[1, 0, 0],
              [0, 1, 0]])

x1 = np.array([4, -2, 0])
x2 = np.array([4, -2, 6])
x3 = np.array([4, -2, -5])

P @ x1, P @ x2, P @ x3
```

Predict the three code outputs.

Solution.

The three outputs are $[4, -2]$, $[4, -2]$, and $[4, -2]$. The third coordinate changes across the inputs, but that change does not appear in the output.

Activity: Can this map produce the requested output?

Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

First compute $A\mathbf{x}_1$, $A\mathbf{x}_2$, and $A\mathbf{x}_3$, where

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

1. Can this map produce the output $(1, 2, 3)$? If yes, give an input.
2. What rule do the outputs above all obey?
3. For the desired output $(1, 2, 4)$, what equations would the unknown input $\mathbf{x} = (x, y)$ have to satisfy?
4. Following up on the previous question, can all of those equations be true at the same time?

Solution.

We compute

$$A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1x + 0y \\ 0x + 1y \\ 1x + 1y \end{bmatrix} = \begin{bmatrix} x \\ y \\ x + y \end{bmatrix}.$$

So the outputs are

$$A\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad A\mathbf{x}_2 = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}, \quad A\mathbf{x}_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The output $(1, 2, 3)$ is produced by the input $\mathbf{x}_1$. The outputs above all obey the rule: third coordinate equals first coordinate plus second coordinate.

For the desired output $(1, 2, 4)$, the unknown input $\mathbf{x} = (x, y)$ would have to satisfy

$$x = 1, \quad y = 2, \quad x + y = 4.$$

The first two equations force $x + y = 3$, not 4. So these equations cannot all be true at the same time, and the output $(1, 2, 4)$ cannot be produced.

Activity: Two inputs, one output

Let

$$F = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

First compute $F\mathbf{u}$ and $F\mathbf{v}$, where

$$\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}.$$

1. Are the inputs the same?
2. Are the outputs the same?
3. Specify $\mathbf{d}$ such that $\mathbf{v} = \mathbf{u} + \mathbf{d}$.
4. What equations does $\mathbf{d} = (d_1, d_2, d_3)$ satisfy if $F\mathbf{d} = \mathbf{0}$?

Solution.

The inputs are different:

$$\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}.$$

But the outputs are the same:

$$F\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad F\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

The change from $\mathbf{u}$ to $\mathbf{v}$ is

$$\mathbf{d} = \mathbf{v} - \mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

Then

$$F\mathbf{d} = \begin{bmatrix} 1d_1 + 1d_2 + 0d_3 \\ 0d_1 + 1d_2 + 1d_3 \end{bmatrix} = \begin{bmatrix} d_1 + d_2 \\ d_2 + d_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

So the change $(1, -1, 1)$ satisfies both equations.

Activity: Same output in code

The following code compares two more inputs for the same map.

```
F = np.array([[1, 1, 0],
              [0, 1, 1]])

u = np.array([0, 1, 2])
v = np.array([1, 0, 3])
d = v - u

F @ u, F @ v, d, F @ d
```

Predict the four code outputs.

Solution.

The four outputs are $[1, 3]$, $[1, 3]$, $[1, -1, 1]$, and $[0, 0]$. Thus $F @ u$ and $F @ v$ are the same, and $F @ d$ is $[0, 0]$.

The equation $F @ d = [0, 0]$ represents

$$d_1 + d_2 = 0, \quad d_2 + d_3 = 0.$$

The output does not always determine the input uniquely.

Activity: Output scores and a hidden direction

A final model layer can convert a hidden vector into output scores by

$$\ell = W\mathbf{h} + \mathbf{b}.$$

Let

$$W = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 0.5 \\ -0.5 \end{bmatrix}, \quad \mathbf{h} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{d} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

1. Compute $W\mathbf{h} + \mathbf{b}$.
2. Compute $W\mathbf{d}$.
3. Use part 2 to compute $W(\mathbf{h} + \mathbf{d}) + \mathbf{b}$.
4. Are $\mathbf{h}$ and $\mathbf{h} + \mathbf{d}$ the same hidden vector?
5. Are their output-score vectors the same?
6. Did the bias vector $\mathbf{b}$ affect which hidden-vector change was invisible?

Solution.

First,

$$W\mathbf{h} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad W\mathbf{h} + \mathbf{b} = \begin{bmatrix} 2.5 \\ 0.5 \end{bmatrix}.$$

Next,

$$W\mathbf{d} = \begin{bmatrix} 1 - 1 \\ -1 + 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Therefore,

$$\begin{aligned} W(\mathbf{h} + \mathbf{d}) + \mathbf{b} &= W\mathbf{h} + W\mathbf{d} + \mathbf{b} \\ &= W\mathbf{h} + \mathbf{b} \\ &= \begin{bmatrix} 2.5 \\ 0.5 \end{bmatrix}. \end{aligned}$$

The hidden vectors $\mathbf{h}$ and $\mathbf{h} + \mathbf{d}$ are different, but they produce the same output scores. The change $\mathbf{d}$ is invisible to W because $W\mathbf{d} = \mathbf{0}$.

The bias vector shifts the output scores, but it does not affect which hidden-vector changes are invisible.

Here is the same issue in a data-matrix setting.

Activity: Two coefficient choices, same predictions

A data matrix stores three measured features for four observations:

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}.$$

Observe that the third feature is the sum of the first two features in every row. Consider the two coefficient vectors

$$\mathbf{c} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{c}' = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

In this example, the linear map takes a coefficient vector $\mathbf{c}$ as input and returns the prediction vector $X\mathbf{c}$.

1. Compute $X\mathbf{c}$.
2. Compute $X\mathbf{c}'$.
3. Are the coefficient vectors the same?

4. Are the predictions the same?
5. Let $\mathbf{d} = \mathbf{c} - \mathbf{c}'$. What equations does $X\mathbf{d} = \mathbf{0}$ represent?

Solution.

We have

$$X\mathbf{c} = \begin{bmatrix} 4 \\ 8 \\ 1 \\ 1 \end{bmatrix}, \quad X\mathbf{c}' = \begin{bmatrix} 4 \\ 8 \\ 1 \\ 1 \end{bmatrix}.$$

The coefficient vectors are different, but the predictions are the same.
The difference is

$$\mathbf{d} = \mathbf{c} - \mathbf{c}' = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

The equation $X\mathbf{d} = \mathbf{0}$, with $\mathbf{d} = (d_1, d_2, d_3)$, represents

$$\begin{aligned} d_1 + 2d_2 + 3d_3 &= 0, \\ 2d_1 + 4d_2 + 6d_3 &= 0, \\ d_2 + d_3 &= 0, \\ d_1 - d_2 &= 0. \end{aligned}$$

The vector $(1, 1, -1)$ satisfies these equations. This happens because the third feature repeats information already contained in the first two features: feature 3 equals feature 1 plus feature 2 in every row.

Each preview led to the same kind of question: can certain linear equations in the unknown input coordinates be solved? The next section gives this question a formal name and a systematic method.

2.2 Linear systems

The preview examples had different stories: reachable outputs, invisible changes, redundant features. Algebraically, they all ask the same question: can a list of linear equations in the unknown input coordinates be solved?

2.2.1 Definitions and examples

Definition 2.2.1 (Linear equations).

An equation of the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where $a_1, \dots, a_n, b$ are scalars, and $x_1, \dots, x_n$ are variables is called a *linear equation*. The real quantities $a_1, \dots, a_n$ are called *coefficients*, b is called a *constant*, while the variables $x_1, \dots, x_n$ are called *unknowns*.

Definition 2.2.2 (Systems of linear equations).

A *system of linear equations* or *linear system* is a set of m linear equations each in n unknowns. A linear system has the form

$$\begin{array}{ccccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n & = & b_2 \\ \vdots & & \vdots & & & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \cdots & + & a_{mn}x_n & = & b_m \end{array} \quad (2.1)$$

where $a_{11}, \dots, a_{mn}$ and $b_1, \dots, b_m$ are constants.

A *solution* to the linear system 2.1 is a sequence of n numbers $s_1, \dots, s_n$ so that *each* equation in 2.1 is satisfied when the substitutions $x_1 = s_1, x_2 = s_2, \dots, x_n = s_n$ are made.

- If a linear system has *no* solution, it is called *inconsistent*.
- If a linear system has at least one solution, it is called *consistent*. Note that an equation with infinitely many solutions is still consistent!

If two linear systems have exactly the same solutions, they are called *equivalent*.

Warning: Variable names

We use x, y, z in geometric examples and $x_1, \dots, x_n$ for general systems.

Activity: Nonlinear equations

What are some equations that are not linear equations?

Solution.

Here are four examples:

- $x^2 + y = 1$ is nonlinear because of the squared term x^2 .
- $xy = 2$ is nonlinear because of the product of variables xy .
- $\sin(x) + y = 0$ is nonlinear because of the sine function $\sin$.
- $\sqrt{x} + y = 1$ is nonlinear because of the square root $\sqrt{x}$.

In 2.1, the question “can this output be produced?” became a question about whether a system can be solved.

Activity: Revisiting a requested output as equations

Return to 2.1:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

Write $A\mathbf{x} = \mathbf{b}$ for the two targets

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

What equations do these two targets produce? Which target is reachable?

Solution.

For $\mathbf{b}_1$, the equations are

$$x_1 = 1, \quad x_2 = 2, \quad x_1 + x_2 = 3.$$

These equations can all hold at the same time, so $\mathbf{b}_1$ is reachable.

For $\mathbf{b}_2$, the equations are

$$x_1 = 1, \quad x_2 = 2, \quad x_1 + x_2 = 4.$$

The first two equations force $x_1 + x_2 = 3$, not 4. So $\mathbf{b}_2$ is not reachable.

Before introducing a general algorithm, we look at a few more small systems where the equations can be analyzed directly.

Activity: Forest animals

In a Wisconsin forest, there are robins and badgers. Together they have 18 heads and 56 legs. How many robins and badgers are in the forest?

Solution.

Let x be the number of robins, and y the number of badgers.

Heads $x + y = 18$

Legs $2x + 4y = 56$

Rearranging the first equation, we find $y = 18 - x$. Substituting this into the second equation:

$$\begin{aligned}
 2x + 4(18 - x) &= 56 \\
 2x + 72 - 4x &= 56 \\
 -2x &= -16 \\
 x &= 8
 \end{aligned}$$

Since $y = 18 - x$ and $x = 8$, $y = 10$.
 So there are 8 robins in the forest, and 10 badgers in the forest.

Activity: Identifying constants and unknowns

Identify the constants and unknowns in each linear equation in 2.2.1, i.e., the two linear equations $x + y = 18$ and $2x + 4y = 56$.

Solution.

- **Equation:** $x + y = 18$
Constants: 1, 1, 18
Unknowns: x, y
- **Equation:** $2x + 4y = 56$
Constants: 2, 4, 56
Unknowns: x, y

Activity: Another forest problem

In another Wisconsin forest, there are also robins and badgers. Together they have 18 heads, 36 eyes, and 56 legs. How many robins and badgers are in the forest? Can we use any new terminology to describe the resulting system of linear equations that we use to determine the number of robins and badgers?

Solution.

Let x be the number of robins, and y the number of badgers:

Heads $x + y = 18$

Eyes $2x + 2y = 36$

Legs $2x + 4y = 56$

Note that the second equation gives $x + y = 18$ after dividing each side by 2, which is the same as the first equation.

From the first equation: $y = 18 - x$.

Substitute this into the third equation:

$$2x + 4(18 - x) = 56$$

$$2x + 72 - 4x = 56$$

$$-2x = -16$$

$$x = 8$$

Since $y = 18 - x$ and $x = 8$, $y = 10$.

So there are 8 robins and 10 badgers in the forest. The system of three linear equations in two unknowns that we have solved in this problem is *consistent*.

Activity: Forest analysis

In a third Wisconsin forest, there are deer and badgers. Together they have 18 heads and 70 legs. How many deer and badgers are in the forest? Can we use any new terminology to describe this system?

Solution.

Let x be the number of deer, and y the number of badgers.

Heads $x + y = 18$

Legs $4x + 4y = 70$

From the second equation:

$$4(x + y) = 70$$

$$x + y = 70/4 = 17.5$$

This system is inconsistent, since the first equation gives $x + y = 18$ but the second gives $x + y = 17.5$. Try substituting the first equation into the second one.

Geometric examples

Recall that a line in the plane which passes through a point $P_0(x_0, y_0)$, and has slope m , can be written in *point-slope form*, as a set of points $P(x, y)$ which solve the equation

$$y - y_0 = m(x - x_0).$$

Activity: The geometric point of view

Consider the following three systems, inspired by our forest examples. Graph the lines defined by these equations and check how this reflects the number of solutions in each case.

Task

System 1

$$\begin{aligned}x + y &= 18 \\ 2x + 4y &= 56\end{aligned}$$

Task

System 2

$$\begin{aligned}x + y &= 18 \\ 4x + 4y &= 72\end{aligned}$$

Task

System 3

$$\begin{aligned}x + y &= 18 \\ 4x + 4y &= 56\end{aligned}$$

Solution.

System 1 has one intersection point: subtracting $x + y = 18$ from $x + 2y = 28$ gives $y = 10$, so $x = 8$. System 2 is the same line twice, since $4x + 4y = 72$ is equivalent to $x + y = 18$, so it has infinitely many solutions. System 3 gives $x + y = 18$ and $x + y = 14$, so the two lines are parallel and the system has no solution.

The same word “solution” also has a geometric meaning. A solution set can be a point, a line, a plane, empty, or higher-dimensional.

To obtain an equation which describes a plane in three-dimensional space, we take a different approach to the point-slope form equation for lines in two-dimensional space. Instead of defining the equation in terms of the direction in which the plane travels, we instead describe the plane in terms of the one direction in which the plane *does not travel*, i.e., the direction at a right angle to the plane.

Definition 2.2.3 (Normal vector equation for planes).

The plane passing through $P_0(x_0, y_0, z_0)$ with normal vector $\mathbf{n}$ is the set of all points $P(x, y, z)$ such that $\mathbf{x} - \mathbf{x}_0$ is orthogonal to $\mathbf{n}$. That is,

$$\mathbf{n} \cdot (\mathbf{x} - \mathbf{x}_0) = 0,$$

or in scalar form, if $\mathbf{n} = \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix}$ and $\mathbf{x}_0 = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix}$,

$$n_1(x - x_0) + n_2(y - y_0) + n_3(z - z_0) = 0.$$

In general form, this equation is written as

$$ax + by + cz = d,$$

where $a = n_1$, $b = n_2$, $c = n_3$, and

$$d = n_1x_0 + n_2y_0 + n_3z_0.$$

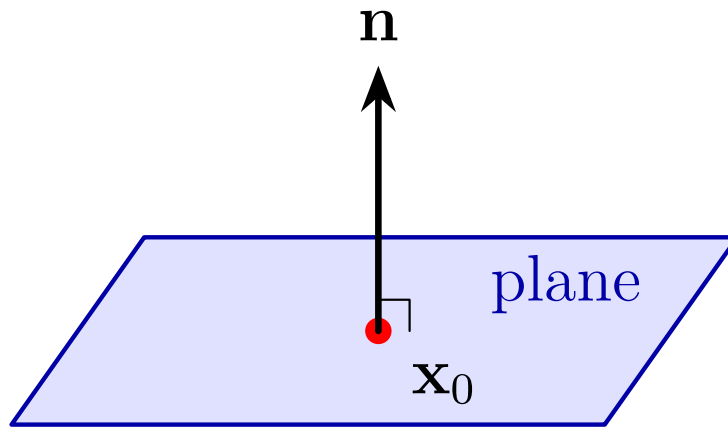


Figure 2.1. A plane passing through a point $\mathbf{x}_0$ with normal vector $\mathbf{n}$.

Activity

Find the equation of the plane passing through the point $P(1, 0, 3)$, normal to the vector

$$\mathbf{n} = \begin{bmatrix} 2 \\ 6 \\ 4 \end{bmatrix}.$$

Solution.

The equation describing the plane is

$$\mathbf{n} \cdot (\mathbf{x} - \mathbf{x}_0) = 0,$$

which expands as

$$2(x - 1) + 6y + 4(z - 3) = 0$$

and simplifies to

$$2x + 6y + 4z = 14.$$

Definition 2.2.4 (Parametric equation for a line).

The line through $P_0(x_0, y_0, z_0)$ with direction vector $\mathbf{d}$ is the set of points $P(x, y, z)$ of the form

$$\mathbf{x} = \mathbf{x}_0 + t\mathbf{d},$$

where $\mathbf{x}_0$ is the vector corresponding to P_0 . In coordinates, if $\mathbf{d} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$, this says

$$(x, y, z) = (x_0 + td_1, y_0 + td_2, z_0 + td_3).$$

If d_1, d_2, d_3 are all nonzero, we can also write the line *in symmetric form* as the set of points $P(x, y, z)$ such that

$$\frac{x - x_0}{d_1} = \frac{y - y_0}{d_2} = \frac{z - z_0}{d_3}.$$

All three values in this equation are equal to the scalar t which occurs in the first equations.

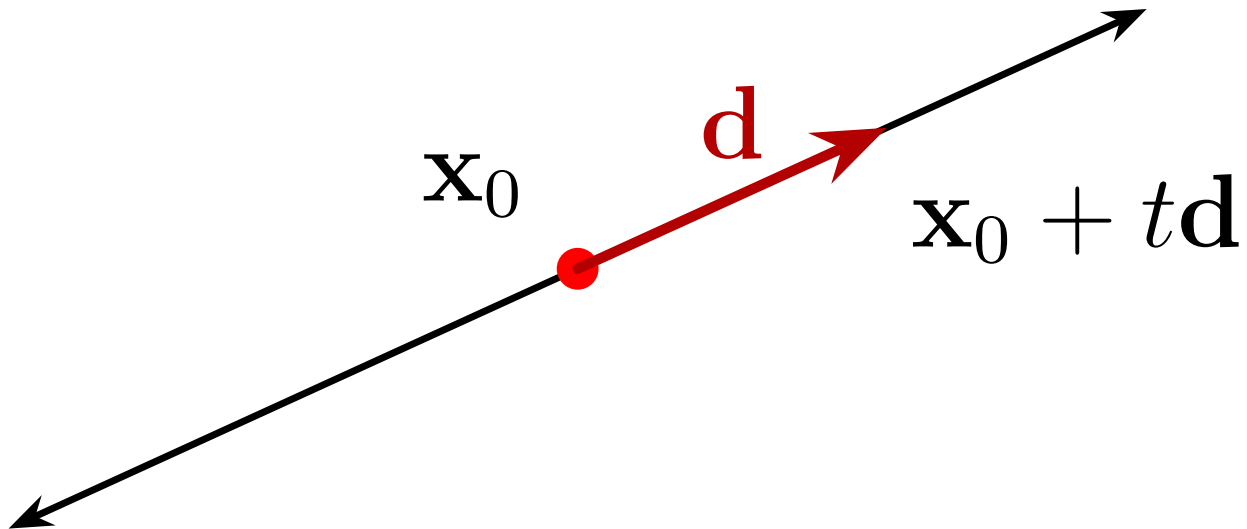


Figure 2.2. A line defined by a direction vector $\mathbf{d}$ and a point P_0 , with corresponding vector $\mathbf{x}_0$.

Activity

Find a parametric equation for the line passing through the two points $(3, 1, 0)$ and $(-3, 1, 4)$.

Solution.

We may choose $\mathbf{d}$ to be equal to

$$\begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} -3 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ -4 \end{bmatrix}.$$

If we let $\mathbf{x}_0 = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$, then the parametric equation is

$$\mathbf{x} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 6 \\ 0 \\ -4 \end{bmatrix}.$$

If we let $\mathbf{x}_0 = \begin{bmatrix} -3 \\ 1 \\ 4 \end{bmatrix}$, then we obtain a different parameterization, namely

$$\mathbf{x} = \begin{bmatrix} -3 \\ 1 \\ 4 \end{bmatrix} + t \begin{bmatrix} 6 \\ 0 \\ -4 \end{bmatrix}.$$

Both parameterizations define the same line. Because the second component of the direction vector is zero, a symmetric-equation form would need a separate caveat; the parametric form above is the clean description of the line.

Activity: A plane perpendicular to a line

Find the scalar equation of the plane through $P = (4, 2, 1)$ and perpendicular to the line

$$x = 1 + 5t, \quad y = -1 + 4t, \quad z = 2 + 3t.$$

Solution.

The direction vector of the line is

$$\mathbf{d} = \begin{bmatrix} 5 \\ 4 \\ 3 \end{bmatrix}.$$

Because the plane is perpendicular to the line, $\mathbf{d}$ is a normal vector for the plane. Therefore

$$5(x - 4) + 4(y - 2) + 3(z - 1) = 0.$$

Shapes in three-dimensional space described as the zeros of a single equation are in most cases *two-dimensional*, so lines must be described as points which satisfy *two* different equations. If a line in three-dimensional space is equal to the intersection of two planes passing through a point $\mathbf{x}_0$ (see 2.3), with normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$, then the line is precisely the set of points $\mathbf{x}$ which satisfy the two equations

$$\mathbf{n}_1 \cdot (\mathbf{x} - \mathbf{x}_0) = 0$$

and

$$\mathbf{n}_2 \cdot (\mathbf{x} - \mathbf{x}_0) = 0.$$

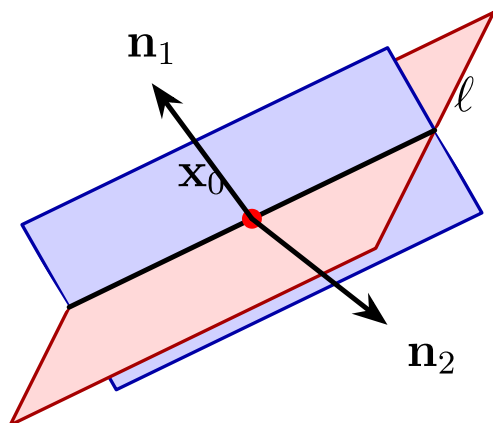


Figure 2.3. A line defined as the intersection of two planes, with normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$, passing through a common point $\mathbf{x}_0$.

Example: A line of intersection from two planes

Find a parametric equation for the line of intersection of

$$x + y + z = 1, \quad -x + 2y - 3z = -1.$$

First set $z = 0$. Then

$$x + y = 1, \quad -x + 2y = -1.$$

Solving gives $x = 1$ and $y = 0$, so $P(1, 0, 0)$ is on the line.

Next set $z = 1$. Then

$$x + y = 0, \quad -x + 2y = 2.$$

Solving gives $x = -\frac{2}{3}$ and $y = \frac{2}{3}$, so $Q(-\frac{2}{3}, \frac{2}{3}, 1)$ is also on the line.

A direction vector is

$$\mathbf{v} = \overrightarrow{PQ} = \begin{bmatrix} -\frac{5}{3} \\ \frac{2}{3} \\ 1 \end{bmatrix}.$$

Thus one parametrization is

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -\frac{5}{3} \\ \frac{2}{3} \\ 1 \end{bmatrix}.$$

This example shows how finding the intersection of planes leads naturally to solving systems of linear equations.

Warning: Parallel planes

The method above assumes the two planes intersect in a line. If the normal vectors are parallel, then the planes are either identical or have no intersection.

2.2.2 Row operations

All of these examples have the same form. The coefficients form a matrix A , the unknowns form a vector $\mathbf{x}$, and the constants form a vector $\mathbf{b}$. The system is $A\mathbf{x} = \mathbf{b}$.

A linear system of m equations in n unknowns can be written as one matrix equation

$$A\mathbf{x} = \mathbf{b}.$$

The system is consistent exactly when $\mathbf{b}$ is a linear combination of the columns of A .

Activity: Matrix equation for a linear system

For

$$\begin{array}{rrcrcl} x_1 & + & 2x_2 & + & 3x_3 & = & 6 \\ 2x_1 & - & 3x_2 & + & 2x_3 & = & 14 \\ 3x_1 & + & x_2 & - & x_3 & = & -2, \end{array}$$

write the system as $A\mathbf{x} = \mathbf{b}$.

Solution.

The equation is

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & -3 & 2 \\ 3 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 14 \\ -2 \end{bmatrix}.$$

Definition 2.2.5 (Matrices corresponding to linear equations).

The *coefficient matrix* of a linear system of the form 2.1 is the $m \times n$ matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

The *constant vector* is

$$\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

We can adjoin $\mathbf{b}$ to matrix A to create the *augmented matrix* representing our linear system:

$$[A \mid \mathbf{b}] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right]$$

Note

For the system $A\mathbf{x} = \mathbf{b}$, there are three related objects.

The matrix A is the coefficient matrix. The vector $\mathbf{b}$ is the right-hand side. The augmented matrix is

$$[A \mid \mathbf{b}].$$

When solving a system, row-reduce the augmented matrix $[A \mid \mathbf{b}]$. Only columns to the left of the vertical line correspond to variables. The column to the right of the vertical line is not a variable column.

We will build a method of efficiently solving a large system of linear equations by applying one of several elementary operations to the system, which do not change the solutions to the system, with the hope of eventually simplifying the system to one in which the set of solutions is obvious.

Definition 2.2.6 (Elementary operations).

The following operations, called *elementary operations*, can be performed on systems of linear equations, and produce *equivalent* systems.

1. *Interchanging* the i -th and j -th equations.
2. *Multiplying* an equation by a nonzero constant.
3. *Replacing* the i -th equation by c times the j -th equation *plus* the i -th equation, where $i \neq j$, for some real quantity c .

We can track these operations more easily using an array/matrix to record the coefficients and constants.

Activity: Elementary operations with matrices

Consider the linear system

This system is reduced to a simpler system using elementary operations. Find the augmented matrix for the linear systems, and describe the operation that is performed to obtain this matrix from the previous one.

The initial system and augmented matrix is

Interchange rows 1 and 2:

Divide first row by 2:

Replace row 3 by row 3 – 3(row 1):

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Replace row 3 by row 3 $- 2$ (row 2):

$$\begin{array}{rcrcrcrcrcl} x & & & + & 3z & = & 2 \\ & y & & - & z & = & 4 \\ & & & & -5z & = & -15 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 3 & 2 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & -5 & -15 \end{array} \right]$$

Multiply row 3 by $-1/5$:

$$\begin{array}{rcrcrcrcrcl} x & & & + & 3z & = & 2 \\ & y & & - & z & = & 4 \\ & & & & z & = & 3 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 3 & 2 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Replace row 2 by row 2 + row 3:

$$\begin{array}{rcrcrcrcrcl} x & & & + & 3z & = & 2 \\ & y & & & & = & 7 \\ & & & & z & = & 3 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 3 & 2 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Replace row 1 by row 1 $- 3$ (row 3):

$$\begin{array}{rcrcrcrcrcl} x & & & & & = & -7 \\ & y & & & & = & 7 \\ & & & & z & = & 3 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -7 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Notice that when we apply elementary operations to a system of linear equations, the corresponding rows of their corresponding matrices also change in an analogous way.

Definition 2.2.7 (Elementary row operations).

An *elementary row operation* on a matrix M is any one of the following operations:

1. *Interchange* two rows.
2. *Multiply* a row by a nonzero scalar.

$$\begin{array}{rclcl} x & + & 2(8) & + & (-3) & = & 4 \\ x & & & & & = & -9 \end{array}$$

Solution: $x = -9$, $y = 8$, and $z = -3$.

Task

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \end{array} \right]$$

Solution.

Rewrite as:

$$\begin{array}{rclcl} x & & & = & -5 \\ & y & & = & 2 \\ & & z & = & 4 \end{array}$$

Solution: $x = -5$, $y = 2$, $z = 4$.

Definition 2.2.10 (Row echelon form and reduced row echelon form).

An $m \times n$ matrix M is in *row echelon form* (REF) if it satisfies the following properties:

1. All zero rows, if there are any, appear at the *bottom* of the matrix.
2. The *leftmost nonzero entry* in any nonzero row is a 1. This entry is called the *leading one* of its row.
3. For each nonzero row, the leading one appears to the *right and below* any leading ones in *preceding* rows.

We say M is in *reduced row echelon form* (RREF) if M is in REF and also satisfies:

- all columns containing leading ones contain no other non-zero entries.

A matrix in row echelon form appears as a staircase (or *echelon*) pattern of leading ones descending from the upper left corner of the matrix. 2.4 gives a schematic of a matrix in row echelon form, where the asterisks denote arbitrary constants.

$$\left[\begin{array}{ccccccc} 0 & 1 & * & * & * & * & * \\ 0 & 0 & 0 & 1 & * & * & * \\ 0 & 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Figure 2.4. A schematic of a matrix in row echelon form.

Activity: Identifying REF and RREF matrices

For each of the following matrices, determine whether it is in row echelon form, reduced row echelon form, both, or neither.

Task

$$A = \begin{bmatrix} 1 & 3 & 0 & 0 & 1 \\ 0 & 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution.

Both REF and RREF (staircase pattern, 0s in columns with leading ones).

Task

$$B = \begin{bmatrix} 1 & 4 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

Solution.

Not REF (zero row is not a bottom row).

Task

$$C = \begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & 1 & -1 & 5 \\ 0 & 0 & 1 & 10 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution.

REF but not RREF (has nonzero entries above leading ones).

Task

$$D = \begin{bmatrix} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 0 & 1 & 6 \end{bmatrix}$$

Solution.

Both REF and RREF (has zeros above and below leading ones).

Task

$$E = \begin{bmatrix} 1 & 0 & 4 & 3 \\ 0 & 1 & 2 & 2 \\ 0 & 1 & -5 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution.

Not REF (third row has a leading one in same column as second row).

Definition 2.2.11 (Pivot entries, positions, and columns).

For a matrix M in REF, each leading one is a *pivot entry*. Its location is a *pivot position*. A column containing a pivot position is a *pivot column*.

Note

For an augmented matrix $[A \mid \mathbf{b}]$, only columns to the left of the vertical line are variable columns. A pivot to the right of the vertical line means the system is inconsistent.

Activity: Reading row reduction in code

The following code row reduces the augmented matrix for the two planes

$$x + y + z = 1, \quad -x + 2y - 3z = -1.$$

```
import sympy as sp

M = sp.Matrix([
    [1, 1, 1, 1],
    [-1, 2, -3, -1],
])

M.rref()
```

Output:

```
(Matrix([
[1, 0, 5/3, 1],
[0, 1, -2/3, 0]]), (0, 1))
```

The output has the form

```
(reduced matrix, pivot columns)
```

SymPy numbers columns starting at 0. Thus $(0, 1)$ means the first and second columns. Read M as an augmented matrix $[A \mid \mathbf{b}]$: the first three columns are coefficient columns, and the last column is the right-hand side.

1. Which variable columns are pivot columns?
2. Which variable column is not a pivot column?
3. Let $z = t$. Use the reduced matrix to write x and y in terms of t .
4. Write the solution set in parametric vector form.
5. What kind of solution set is this?
6. How does this connect to the line of intersection of two planes?

Solution.

SymPy reports pivot columns $(0, 1)$, meaning mathematical columns 1 and 2. These are pivot columns to the left of the vertical line, so x and y are basic variables. The variable z is free.

The reduced matrix represents

$$\begin{aligned}x + \frac{5}{3}z &= 1, \\ y - \frac{2}{3}z &= 0.\end{aligned}$$

Let $z = t$. Then

$$\begin{aligned}x &= 1 - \frac{5}{3}t, \\ y &= \frac{2}{3}t, \\ z &= t.\end{aligned}$$

So the solution set is

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -\frac{5}{3} \\ \frac{2}{3} \\ 1 \end{bmatrix}.$$

This is a line. It is the line of intersection of the two planes.

Theorem 2.2.12.

Every matrix M is *row equivalent* to a *unique* matrix in RREF.

The Gaussian Algorithm gives a procedure to find a row-echelon matrix which is row equivalent to any given matrix. The Heuristic is to move from top to bottom and outside in until you obtain a matrix in row-echelon form.

Algorithm: The Gaussian Elimination Algorithm

Given a matrix, this algorithm will compute a matrix in row echelon form which is row-equivalent to the original matrix.

1. If the matrix consists entirely of zeros, stop — it is already in row-echelon form.
2. Otherwise, find the left-most column containing a nonzero entry a , and move the row containing that entry to the top position.
3. Now multiply the new top row by $1/a$ so that the row has a leading one.
4. By subtracting multiples of that row from rows below it, make each entry below this leading one equal to zero.

This completes the first row. Now Repeat Steps 1-4 on the matrix consisting of all other rows. The process stops when either no rows remain, or the remaining rows consist entirely of zeros.

Activity: Finding row echelon forms

Find a row echelon form of the given matrix A .

$$A = \begin{bmatrix} 0 & 0 & 1 & 2 \\ 3 & 3 & 6 & -9 \\ 2 & 3 & 0 & -2 \end{bmatrix}$$

Solution.

First nonzero column is column 1. Interchange rows 1 and 2:

$$\begin{bmatrix} 3 & 3 & 6 & -9 \\ 0 & 0 & 1 & 2 \\ 2 & 3 & 0 & -2 \end{bmatrix}$$

Multiply first row by 1/3:

$$\begin{bmatrix} 1 & 1 & 2 & -3 \\ 0 & 0 & 1 & 2 \\ 2 & 3 & 0 & -2 \end{bmatrix}$$

Subtract 2 times first row from Row 3:

$$\begin{bmatrix} 1 & 1 & 2 & -3 \\ 0 & 0 & 1 & 2 \\ 0 & 1 & -4 & 4 \end{bmatrix}$$

Interchange rows 2 and 3:

$$\begin{bmatrix} 1 & 1 & 2 & -3 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

The matrix is now in REF with leading ones in columns 1, 2, and 3.

Since we can solve systems of linear equations whose augmented matrices are in row echelon form using back substitution as in 2.2.3, 2.4 gives a practical method of solving any system of linear equations.

Activity: Solving a linear system using row echelon form

Solve the linear system

$$\begin{array}{rcrcrcrcrcl} x & + & 2y & + & 3z & = & 9 \\ 2x & - & y & + & z & = & 8 \\ 3x & & & - & z & = & 3 \end{array}$$

by using 2.4 to find a matrix in row echelon form which is row equivalent to the augmented matrix $[A \mid \mathbf{b}]$.

The Elimination Scratchpad

Solution.

Augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 9 \\ 2 & -1 & 1 & 8 \\ 3 & 0 & -1 & 3 \end{array} \right]$$

Subtract 2 times Row 1 from Row 2 and subtract 3 times Row 1 from Row 3:

Multiply Row 2 by $-1/5$:

Add 6 times Row 2 to Row 3:

Multiply Row 3 by $-1/4$:

This matrix is the augmented matrix of the system of linear equations

By 2.2.9, we know that this system is equivalent to the original system of equations specified by the problem. We now apply back substitution to solve the system:

Solving for z Row 3 tells us $z = 3$.

Solving for y Row 2 gives $y + z = 2$, and since $z = 3$, we calculate that $y = -1$.

Solving for x Row 1 gives $x + 2y + 3z = 9$, and since $y = -1$ and $z = 3$, we calculate that $x = 2$.

So the unique solution is given by $(x, y, z) = (2, -1, 3)$.

In 2.4, every variable column has a pivot. Equivalently, the coefficient matrix can be row-reduced all the way to I_3 . This is why the system has a unique solution.

If the augmented matrix of a linear system is a matrix in row echelon form, we say the linear system is in *echelon form*. The variables corresponding to variable columns with leading entries are called *leading variables* (or *basic variables*), while the other variables are called *free variables*.

Back substitution allows us to write a general solution to a linear system by writing the basic variables in terms of the free variables.

Activity: Solving a system in REF

Let

$$[C \mid d] = \left[\begin{array}{ccccc|c} 1 & 2 & 3 & 4 & 5 & 6 \\ 0 & 1 & 2 & 3 & -1 & 7 \\ 0 & 0 & 1 & 2 & 3 & 7 \\ 0 & 0 & 0 & 1 & 2 & 9 \end{array} \right]$$

be the augmented matrix of a linear system. Solve the system.

Solution.

The matrix is already in row echelon form, and so we need only back substitute to solve the basic variables. Let x_1, x_2, x_3, x_4, x_5 be the variables. Then the linear system with the corresponding augmented matrix is given by

$$\begin{aligned} x_1 + 2x_2 + 3x_3 + 4x_4 + 5x_5 &= 6 \\ x_2 + 2x_3 + 3x_4 - x_5 &= 7 \\ x_3 + 2x_4 + 3x_5 &= 7 \\ x_4 + 2x_5 &= 9 \end{aligned}$$

The variables x_1, x_2, x_3 , and x_4 are the basic variables, and x_5 is a free variable. Write the free variable as a parameter t , i.e., write

$$x_5 = t$$

. The last row states that $x_4 + 2x_5 = 9$, which, given that $x_5 = t$, can be rearranged to read that

$$x_4 = 9 - 2t$$

. The third row states that $x_3 + 2x_4 + 3x_5 = 7$. Substituting $x_4 = 9 - 2t$ and $x_5 = t$ into this equation and rearranging, we find that

$$\begin{aligned} x_3 + 2(9 - 2t) + 3(t) &= 7 \\ x_3 + 18 - 4t + 3t &= 7 \\ x_3 + 18 - t &= 7 \\ x_3 &= -11 + t \end{aligned}$$

The second row states that $x_2 + 2x_3 + 3x_4 - x_5 = 7$. Substituting $x_3 = -11 + t$, $x_4 = 9 - 2t$, and $x_5 = t$ into this equation, and rearranging, we find that

$$\begin{aligned}
 x_2 + 2(-11 + t) + 3(9 - 2t) - (t) &= 7 \\
 x_2 - 22 + 2t + 27 - 6t - t &= 7 \\
 x_2 + 5 - 5t &= 7 \\
 x_2 &= 2 + 5t
 \end{aligned}$$

Finally, the first row says that $x_1 + 2x_2 + 3x_3 + 4x_4 + 5x_5 = 6$. Substituting $x_2 = 2 + 5t$, $x_3 = -11 + t$, $x_4 = 9 - 2t$, and $x_5 = t$ into this equation and rearranging, we find that

$$\begin{aligned}
 x_1 + 2(2 + 5t) + 3(-11 + t) + 4(9 - 2t) + 5(t) &= 6 \\
 x_1 + 4 + 10t - 33 + 3t + 36 - 8t + 5t &= 6 \\
 x_1 + 7 + 10t &= 6 \\
 x_1 &= -1 - 10t
 \end{aligned}$$

We now have our solution. All solutions to the linear equation can be written as $x_1 = -1 - 10t$, $x_2 = 2 + 5t$, $x_3 = -11 + t$, $x_4 = 9 - 2t$, and $x_5 = t$, where t is any real number.

Activity: Analyzing an inconsistent system

Consider a linear system with augmented matrix

$$[C \mid \mathbf{d}] = \left[\begin{array}{cccc|c} 1 & 2 & -3 & 4 & 5 \\ 0 & 1 & 2 & 1 & -6 \\ 0 & 0 & 0 & 0 & 1 \end{array} \right]$$

. Solve the system.

Solution.

The last row has the form

$$[0 \ 0 \ 0 \ 0 \mid 1].$$

It represents the equation $0 = 1$, which is impossible. Therefore the system is inconsistent.

Algorithm: Solving $Ax=b$ by Gaussian elimination

1. Write the augmented matrix $[A \mid \mathbf{b}]$.
2. Row-reduce $[A \mid \mathbf{b}]$ to a row echelon form $[C \mid \mathbf{d}]$.
3. If a row has the form

$$[0 \ 0 \ \cdots \ 0 \mid c], \quad c \neq 0,$$

then the system is inconsistent.

4. Otherwise, pivot columns to the left of the vertical line give basic variables.
5. Non-pivot columns to the left of the vertical line give free variables.
6. Use back substitution to write the basic variables in terms of the free variables.

Row echelon matrices are not unique for a given starting matrix. For example,

$$\begin{bmatrix} 1 & 3 & 9 \\ 0 & 1 & -6 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 9 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

are row equivalent row echelon matrices, but they are not equal. However, the number of leading ones will be the same in each row equivalent row echelon matrix.

For now, rank is a row-reduction count: the number of pivots. Later we will reinterpret it as the number of independent directions transmitted by a matrix.

Definition 2.2.14 (Rank of a matrix).

The *rank* of a matrix M is the number of pivots in any row echelon form row equivalent to M .

Activity: Finding the rank of a matrix

Find the rank of

$$C = \begin{bmatrix} 1 & 1 & 2 & 7 \\ 0 & 1 & 0 & 3 \\ 2 & 1 & 4 & 11 \end{bmatrix}$$

Solution.

We convert the matrix to row echelon form using 2.4. Subtract 2 times Row 1 from Row 3:

$$\begin{bmatrix} 1 & 1 & 2 & 7 \\ 0 & 1 & 0 & 3 \\ 0 & -1 & 0 & -3 \end{bmatrix}$$

Add Row 2 to Row 3:

$$\begin{bmatrix} 1 & 1 & 2 & 7 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The matrix is now in REF with 2 leading ones, and so the rank of C is 2.

Activity: Finding the rank of a matrix (continued)

Find the rank of

$$D = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 3 \\ 2 & 1 & 2 & -1 \end{bmatrix}$$

Solution.

Subtract 2 times Row 1 from Row 3:

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 1 & 0 & -5 \end{bmatrix}$$

Subtract Row 2 from Row 3:

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & -8 \end{bmatrix}$$

Multiply Row 3 by $(-1/8)$:

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix is now in REF with 3 leading ones, and so the rank of D is 3.

Theorem 2.2.15.

Suppose $Ax = b$ is a consistent system with n variables. Let r be the number of pivot columns to the left of the vertical line in a row echelon form of $[A \mid b]$. Then the solution set is described using exactly $n - r$ parameters.

In particular:

- if $r = n$, the system has a unique solution;
- if $r < n$, the system has infinitely many solutions.

The number r is the rank of the coefficient matrix A .

See 2.2.15. Together with the inconsistency test, it gives a method for finding the number of solutions to any given linear system.

Note: The Number of Solutions to a Linear System

Any linear system belongs to exactly one of three categories.

- *No solution.* This occurs when a row echelon form of $[A \mid \mathbf{b}]$ has a row

$$[0 \ 0 \ \cdots \ 0 \mid c], \quad c \neq 0.$$

- *A unique solution.* This occurs when the system is consistent and every variable column, meaning every column to the left of the vertical line, is a pivot column.
- *Infinitely many solutions.* This occurs when the system is consistent and at least one variable column, meaning a column to the left of the vertical line, is not a pivot column.

Note: Row-operation shorthand

We often record row operations using the symbol R_i for row i . The notation $R_i \rightarrow R_i + cR_j$ means to replace row i by row i plus c times row j . All rows not named on the left side stay the same.

- $R_3 \rightarrow R_3 - 2R_1$ means replace row 3 by row 3 minus twice row 1.
- $R_2 \rightarrow \frac{1}{5}R_2$ means multiply row 2 by $1/5$.
- $R_1 \leftrightarrow R_2$ means interchange rows 1 and 2.

Activity: Analyzing solution sets using rank

If

$$C = \left[\begin{array}{ccc|c} 1 & 1 & 2 & 7 \\ 0 & 1 & 0 & 3 \\ 2 & 1 & 4 & 11 \end{array} \right]$$

represents the augmented matrix for a system of 3 equations and 3 variables, what can we say about the solution set of the system?

Solution.

We apply the Gaussian elimination algorithm:

$$R_3 \rightarrow R_3 - 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 7 \\ 0 & 1 & 0 & 3 \\ 0 & -1 & 0 & -3 \end{array} \right]$$

$$R_3 \rightarrow R_3 + R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 7 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

There are two pivot columns to the left of the vertical line and no pivot to the right of the vertical line. The system is consistent. Since there are three variable columns, there is one free variable. The system has infinitely many solutions, parameterized by the free variable.

Activity: Analyzing solution sets using rank (continued)

If

$$D = \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 3 \\ 2 & 1 & 2 & -1 \end{array} \right]$$

represents the augmented matrix for a system of 3 equations and 3 variables, what can we say about the solution set of the system?

Solution.

We start by applying Gaussian elimination to convert the matrix to row echelon form:

$$R_3 \rightarrow R_3 - 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 1 & 0 & -5 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & -8 \end{array} \right]$$

$$R_3 \rightarrow (-1/8)R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

The last row has the form

$$[0 \ 0 \ 0 \ | \ 1].$$

This is a pivot to the right of the vertical line, so the system is inconsistent; there are no solutions to the system of linear equations.

2.2.4 Homogeneous systems and basic solutions

The preview also asked what changes a map forgets. That is the special system $Ax = 0$.

Definition 2.2.16 (Homogeneous systems).

If a linear system is of the form

$$\begin{array}{cccccc} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & 0 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n & = & 0 \\ \vdots & & \vdots & & & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \cdots & + & a_{mn}x_n & = & 0 \end{array} \quad (2.2)$$

it is called a *homogeneous system*.

- Note that $x_1 = x_2 = \cdots = x_n = 0$ is always a solution to a homogeneous system. It is called the *trivial* solution.
- A solution to a homogeneous system where *not* all x_i are 0 is called a *nontrivial* solution.

Activity: Solving a homogeneous system

Solve the homogeneous system

$$\begin{array}{l} 2x_1 + 4x_3 + 6x_4 = 0 \\ x_1 + 2x_2 + 7x_4 = 0 \end{array}$$

The Elimination Scratchpad

Solution.

Start by writing out the augmented matrix corresponding to the homogeneous system:

$$\left[\begin{array}{cccc|c} 2 & 0 & 4 & 6 & 0 \\ 1 & 2 & 0 & 7 & 0 \end{array} \right]$$

We next use Gaussian elimination (recall 2.4) to obtain a matrix in row echelon form which is row equivalent to the augmented matrix:

$$R_1 \rightarrow (1/2)R_1$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 2 & 3 & 0 \\ 1 & 2 & 0 & 7 & 0 \end{array} \right]$$

$$R_2 \rightarrow R_2 - R_1$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 2 & 3 & 0 \\ 0 & 2 & -2 & 4 & 0 \end{array} \right]$$

$$R_2 \rightarrow (1/2)R_2$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 2 & 3 & 0 \\ 0 & 1 & -1 & 2 & 0 \end{array} \right]$$

If we convert this augmented matrix back to a system of equations, we can (recalling 2.2.13) write the leading variables of the matrix (in this case, x_1 and x_2) in terms of the free variables (x_3 and x_4):

$$x_1 = -2x_3 - 3x_4$$

$$x_2 = x_3 - 2x_4$$

So if we let $x_3 = s$ and $x_4 = t$, all solutions to the original linear system are of the form $x_1 = -2s - 3t$, $x_2 = s - 2t$, $x_3 = s$, and $x_4 = t$, where s and t are arbitrary real quantities.

Activity: A plane through three points without cross products

Find an equation for the plane containing

$$P(1, 1, -2), \quad Q(0, 2, 1), \quad R(-1, -1, 0),$$

(*without* using the cross-product, if you happen to know what that is).

Solution.

First find two displacement vectors in the plane:

$$\mathbf{u} = \overrightarrow{PQ} = \begin{bmatrix} -1 \\ 1 \\ 3 \end{bmatrix}, \quad \mathbf{v} = \overrightarrow{PR} = \begin{bmatrix} -2 \\ -2 \\ 2 \end{bmatrix}.$$

Let $\mathbf{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ be a normal vector. Then $\mathbf{n}$ must be perpendicular to both displacement vectors:

$$-a + b + 3c = 0, \quad -2a - 2b + 2c = 0.$$

The Elimination Scratchpad

Choose $c = 1$. The first equation gives $b = a - 3$. Substituting into the second gives

$$-2a - 2(a - 3) + 2 = 0,$$

so $a = 2$, and then $b = -1$. One normal vector is

$$\mathbf{n} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}.$$

Using the point $P(1, 1, -2)$, the plane equation is

$$2(x - 1) - (y - 1) + (z + 2) = 0,$$

or

$$2x - y + z = -1.$$

Note: Normal vectors from homogeneous systems

The plane-through-three-points activity found a normal vector by solving a homogeneous system. Basic solutions below make this procedure systematic.

Note: Existence of solutions to homogeneous equations

A homogeneous system of m linear equations in n unknowns *always* has a nontrivial solution if $n > m$, that is if the number of unknowns exceeds the number of equations.

Activity: Thinking about homogeneous systems

Does a homogeneous system *need* to have more variables than equations in order to have infinitely many solutions?

Solution.

No. Such a system has infinitely many solutions if the rank of the coefficient matrix for the system is smaller than n .

The solution set of a homogeneous system can be written in vector form.

Activity: Solutions as linear combinations

Rewrite the general solution to 2.2.4 as an expression involving a linear combination of certain column vectors.

The Elimination Scratchpad

Solution.

Solution set:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ -2 \\ 0 \\ 1 \end{bmatrix}$$

Fact

If we consider solutions to a homogeneous system as vectors, then any linear combination of solutions to a homogeneous system is again a solution to the homogeneous system.

Activity: Finding basic solutions

Consider the homogeneous system whose augmented matrix, when put in RREF, is

$$[A \mid 0] = \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & -2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Solve the system, and write the solutions as a linear combination of particular solutions.

Solution.

Interpret the RREF matrix as a system of linear equations:

$$x_1 - 2x_5 = 0$$

$$x_3 = 0$$

$$x_4 + 5x_5 = 0$$

Next, solve for pivot variables:

$$x_1 = 2x_5$$

$$x_3 = 0$$

$$x_4 = -5x_5$$

Express free variables as parameters:
Let $x_2 = s$ and $x_5 = t$. Then

$$\begin{aligned}x_1 &= 2t \\x_2 &= s \\x_3 &= 0 \\x_4 &= -5t \\x_5 &= t\end{aligned}$$

Now write the solutions in vector form:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = s \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 2 \\ 0 \\ 0 \\ -5 \\ 1 \end{bmatrix}$$

Definition 2.2.17 (Basic solutions to homogeneous linear equations).

The Gaussian elimination algorithm produces, for each free variable, a solution to the homogeneous equation $A\mathbf{x} = \mathbf{0}$, by setting that free variable equal to one, and all other free variables equal to 0. These solutions are called *basic solutions*, one for every parameter. Every solution can be written uniquely as a linear combination of the basic solutions.

Theorem 2.2.18.

Let A be a rank r matrix with m rows and n columns. Consider the homogeneous system in n variables with A as the coefficient matrix. Then:

1. The system has exactly $n - r$ basic solutions, one for each parameter. Note that $n - r$ is also the number of *free* variables of the linear system when the augmented matrix $[A \mid \mathbf{0}]$ is reduced to row echelon form (recall 2.2.13).
2. Every solution is a linear combination of these basic solutions.

For a homogeneous system, row-reducing $[A \mid \mathbf{0}]$ is equivalent to row-reducing A while remembering that the right-hand side stays zero. The next section gives the solution set of $A\mathbf{x} = \mathbf{0}$ a name, and from that point on we work with A alone.

Homogeneous solution sets are closed under linear combinations. 2.3 gives a name to sets with this behavior: subspaces.

2.3 Subspaces

2.3.1 Definition of subspaces

The outputs a matrix can “reach” form a subspace. The directions a matrix “forgets” also form a subspace. The definitions below make this precise. In particular, the closure rules are not arbitrary. If two output vectors are reachable, then any linear combination of them is reachable. If two input changes are forgotten by a matrix, then any linear combination of those changes is also forgotten.

We are already familiar with geometric spaces living inside bigger ones, like lines lying in the plane, or planes lying in 3-space. We are now going to formalize this notion algebraically. Recall the definition of the set $\mathbb{R}^n$ in 1.1.1.

Definition 2.3.1 (Subspaces).

A set U of vectors in $\mathbb{R}^n$ is called a *subspace* of $\mathbb{R}^n$ if it satisfies the following properties:

- The zero vector (written as $\mathbf{0}$) is in U .
- If $\mathbf{x}$ is in U and $\mathbf{y}$ is in U , then $\mathbf{x} + \mathbf{y}$ is in U . This means U is *closed under addition*.
- If $\mathbf{x}$ is in U , then $a\mathbf{x}$ is in U for every scalar a . This means U is *closed under scalar multiplication*.

Example

The whole set $\mathbb{R}^n$ is a subspace of itself.

Example

The set containing only the zero vector, i.e., the set $\{\mathbf{0}\}$, is a subspace of $\mathbb{R}^n$ (called the *trivial subspace* or *zero subspace*).

Activity

Verify that $\{\mathbf{0}\}$ is a subspace of $\mathbb{R}^n$.

Solution.

To verify that $\{\mathbf{0}\}$ is a subspace, we must verify three properties:

Contains the zero vector:

The zero $\mathbf{0}$ is evidently an element of $\{\mathbf{0}\}$.

Closure under addition:

If $\mathbf{x}, \mathbf{y} \in \{\mathbf{0}\}$, then $\mathbf{x} = \mathbf{0}$ and $\mathbf{y} = \mathbf{0}$, so $\mathbf{x} + \mathbf{y} = \mathbf{0}$, which is an element of $\{\mathbf{0}\}$.

Closure under scalar multiplication:

If $\mathbf{x} \in \{\mathbf{0}\}$ and a is a scalar, then $\mathbf{x} = \mathbf{0}$, and $a\mathbf{x} = a\mathbf{0} = \mathbf{0}$ is in $\{\mathbf{0}\}$.

Since $\{\mathbf{0}\}$ satisfies all three of the necessary conditions, it is a subspace.

Definition 2.3.2 (Proper Subspaces).

Any subspace of $\mathbb{R}^n$ other than $\mathbb{R}^n$ or $\{0\}$ is a *proper subspace* of $\mathbb{R}^n$.

Example

Planes and lines through the origin in $\mathbb{R}^3$ are all subspaces of $\mathbb{R}^3$.

Activity

Task

Verify that planes through the origin in $\mathbb{R}^3$ are all proper subspaces of $\mathbb{R}^3$.

Solution.

Let R be a plane through the origin. Then there exist scalars a, b, c , not all zero, such that R is the set of points (x, y, z) satisfying the equation $ax + by + cz = 0$. The vector

$$\mathbf{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

is a normal vector for the plane, and we can write R as the set of vectors $\mathbf{v} \in \mathbb{R}^3$ such that $\mathbf{n} \cdot \mathbf{v} = 0$.

Let us verify that R satisfies the conditions to be a subspace:

Contains the zero vector $\mathbf{0} \in R$ because $\mathbf{n} \cdot \mathbf{0} = 0$.

Closure under addition

If $\mathbf{v} \in R$ and $\mathbf{w} \in R$ then to verify $\mathbf{v} + \mathbf{w} \in R$ we must argue that $\mathbf{n} \cdot (\mathbf{v} + \mathbf{w}) = 0$. Indeed,

$$\mathbf{n} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{n} \cdot \mathbf{v} + \mathbf{n} \cdot \mathbf{w} = 0$$

.

Closure under scalar multiplication

If $\mathbf{v} \in R$, and a is a scalar, to verify $a\mathbf{v} \in R$ we must argue that $\mathbf{n} \cdot (a\mathbf{v}) = 0$. Indeed,

$$\mathbf{n} \cdot (a\mathbf{v}) = a(\mathbf{n} \cdot \mathbf{v}) = a(0) = 0$$

.

We have verified the three properties required for R to be a subspace.

Task

Verify that lines through the origin in $\mathbb{R}^3$ are subspaces of $\mathbb{R}^3$.

Solution.

Let L be a line through the origin in $\mathbb{R}^3$. If L has direction vector $\mathbf{d}$, then $L = \{t\mathbf{d} : t \in \mathbb{R}\}$, i.e., the line L consists of all scalar multiples of $\mathbf{d}$.

Let us verify that L satisfies the conditions to be a subspace:

Contains the zero vector $\mathbf{0} \in L$ because $\mathbf{0} = 0\mathbf{d}$.

Closure under addition

Suppose $\mathbf{v} \in L$ and $\mathbf{w} \in L$. Then $\mathbf{v}$ and $\mathbf{w}$ are both scalar multiples of $\mathbf{d}$, i.e., $\mathbf{v} = t_1\mathbf{d}$ and $\mathbf{w} = t_2\mathbf{d}$. Then

$$\mathbf{v} + \mathbf{w} = t_1\mathbf{d} + t_2\mathbf{d} = (t_1 + t_2)\mathbf{d}$$

. Thus $\mathbf{v} + \mathbf{w}$ is a scalar multiple of $\mathbf{d}$, and so $\mathbf{v} + \mathbf{w} \in L$.

Closure under scalar multiplication

If $\mathbf{v} \in L$, and a is a scalar, then we may write $\mathbf{v} = t\mathbf{d}$, and then

$$a\mathbf{v} = a(t\mathbf{d}) = (at)\mathbf{d}$$

. So $a\mathbf{v}$ is a scalar multiple of $\mathbf{d}$, and thus $a\mathbf{v} \in L$.

We have verified the three properties required for L to be a subspace.

Note

Lines and planes which do not pass through the origin in $\mathbb{R}^3$ are *not* subspaces of $\mathbb{R}^3$. Nonetheless, if Π is a plane, and we consider the set of displacement vectors

$$\Delta\Pi = \{\mathbf{v}_1 - \mathbf{v}_0 : \mathbf{v}_0, \mathbf{v}_1 \in \Pi\}$$

, then $\Delta\Pi$ forms a plane passing through the origin, and so is a subspace of $\mathbb{R}^3$. A similar result holds for sets of displacement vectors given by a line not passing through the origin.

Activity

Show that the set

$$U = \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} : x_1 x_2 = 0 \right\}$$

is *not* a subspace of $\mathbb{R}^2$.

Solution.

It suffices to show *one* of the conditions required to be a subspace does not hold. U *does* contain the zero vector, and *is* closed under scalar multiplication. But U *is not* closed under addition. Define

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

. Then $\mathbf{x} \in U$ and $\mathbf{y} \in U$. But

$$\mathbf{x} + \mathbf{y} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

is not in U , since $(1)(1) \neq 0$. Since U fails to be closed under addition, it is not a subspace of $\mathbb{R}^2$.

Activity

Let W be the set of all vectors in $\mathbb{R}^3$ of the form

$$\begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix}$$

, where $r, s \in \mathbb{R}$. Is W a proper subspace of $\mathbb{R}^3$?

Solution.

We check whether the required properties hold in order to be a subspace:

Contains the zero vector:

The zero vector is in W , since

$$\mathbf{0} = \begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix}$$

when $r = 0$ and $s = 0$.

Closed under addition:

If

$$\mathbf{v} = \begin{bmatrix} r_1 \\ s_1 \\ r_1 + 2s_1 \end{bmatrix} \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} r_2 \\ s_2 \\ r_2 + 2s_2 \end{bmatrix}$$

Then

$$\begin{aligned} \mathbf{v} + \mathbf{w} &= \begin{bmatrix} r_1 + r_2 \\ s_1 + s_2 \\ r_1 + 2s_1 + r_2 + 2s_2 \end{bmatrix} \\ &= \begin{bmatrix} r_1 + r_2 \\ s_1 + s_2 \\ (r_1 + r_2) + 2(s_1 + s_2) \end{bmatrix} \\ &= \begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix} \end{aligned}$$

, where $r = r_1 + r_2$ and $s = s_1 + s_2$. Thus $\mathbf{v} + \mathbf{w} \in W$.

Closed under scalar multiplication:

If

$$\mathbf{v} = \begin{bmatrix} r_1 \\ s_1 \\ r_1 + 2s_1 \end{bmatrix}$$

, and c is a scalar, then

$$c\mathbf{v} = \begin{bmatrix} cr_1 \\ cs_1 \\ c(r_1 + 2s_1) \end{bmatrix} = \begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix}$$

, with $r = cr_1$ and $s = cs_1$. Thus $c\mathbf{v} \in W$.

Thus W is a subspace of $\mathbb{R}^3$.

To check that W is a proper subspace of $\mathbb{R}^3$, we must show that $W \neq \mathbb{R}^3$ and $W \neq \{\mathbf{0}\}$. The vector

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

is not in W , since if it were of the form

$$\begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix}$$

, then $r = 1$ and $s = 1$, and the third entry would have to be 3. Thus W does not contain every vector, and thus cannot be equal to $\mathbb{R}^3$. But it also contains some non-zero vectors, such as

$$\begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$$

, and so $W \neq \{\mathbf{0}\}$.

Thus W is a proper subspace of $\mathbb{R}^3$.

Note

To show a set *is* a subspace, you tend to need to parameterize all elements of the set in order to show they all satisfy some property (closure under addition or scalar multiplication), and so your argument should use variables. To show a set *is not* a subspace, you need to find one or more counterexamples that show that some property does not hold for all elements; a single explicit counterexample usually suffices.

2.3.2 Null space and image space

The following important examples are representative of two canonical ways of describing a subspace.

From here on, homogeneous equations play a special role. To solve $A\mathbf{x} = \mathbf{0}$ by row reduction, we could write the augmented matrix $[A \mid \mathbf{0}]$. Since the right-hand side stays zero throughout the row operations, the coefficient matrix A carries the essential information.

Definition 2.3.3 (Null Space and Image Space).

Let A be an $m \times n$ matrix. The *null space* of A , denoted $\text{null}(A)$, is the set of vectors $\mathbf{x}$ in $\mathbb{R}^n$ which are solutions to the equation $A\mathbf{x} = \mathbf{0}$, i.e.,

$$\text{null}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$$

. The *image space* of A , denoted $\text{im}(A)$, is the set of vectors $\mathbf{y}$ in $\mathbb{R}^m$ such that $A\mathbf{x} = \mathbf{y}$ has a solution, i.e.,

$$\text{im}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\}$$

Note

The null space lives in the input space. It records input directions that A sends to $\mathbf{0}$. The image space lives in the output space. It records outputs that A can reach.

- **Object:** $\text{null}(A)$
Lives in: $\mathbb{R}^n$
How to read it: input directions A forgets
- **Object:** $\text{im}(A)$
Lives in: $\mathbb{R}^m$
How to read it: outputs A can reach

Activity: Revisiting the height map with subspace language

Return to 2.1. Let

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Describe the output subspace and the subspace of input directions that P forgets.

Solution.

For

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad P\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

As x and y vary freely, every vector in $\mathbb{R}^2$ occurs as an output. Therefore

$$\text{im}(P) = \mathbb{R}^2.$$

The directions that P forgets are the solutions to $P\mathbf{x} = \mathbf{0}$. This requires

$$x = 0, \quad y = 0,$$

while z is free. Thus

$$\text{null}(P) = \left\{ \begin{bmatrix} 0 \\ 0 \\ t \end{bmatrix} : t \in \mathbb{R} \right\}.$$

This subspace records the height direction.

Theorem 2.3.4.

For any $m \times n$ matrix, the null space of A is a subspace of $\mathbb{R}^n$, and the image space of A is a subspace of $\mathbb{R}^m$.

Proof.

We start by showing the null space of A is a subspace. Recall the properties discussed in 1.3.2:

Contains the zero vector $\mathbf{0} \in \text{null}(A)$ because $A\mathbf{0} = \mathbf{0}$.

Closure under addition

Suppose $\mathbf{v} \in \text{null}(A)$ and $\mathbf{w} \in \text{null}(A)$. Then $A\mathbf{v} = \mathbf{0}$ and $A\mathbf{w} = \mathbf{0}$. So

$$A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w} = \mathbf{0} + \mathbf{0} = \mathbf{0}$$

. Thus $\mathbf{v} + \mathbf{w} \in \text{null}(A)$.

Closure under scalar multiplication

If $\mathbf{v} \in \text{null}(A)$, and a is a scalar, then

$$A(a\mathbf{v}) = a(A\mathbf{v}) = a\mathbf{0} = \mathbf{0}$$

. Thus $a\mathbf{v} \in \text{null}(A)$.

So $\text{null}(A)$ is a subspace of $\mathbb{R}^n$.

Next, we show that $\text{im}(A)$ is a subspace.

Contains the zero vector $\mathbf{0} \in \text{im}(A)$ because $A\mathbf{0} = \mathbf{0}$.

Closure under addition

Suppose $\mathbf{v} \in \text{im}(A)$ and $\mathbf{w} \in \text{im}(A)$. Then the equations $A\mathbf{x} = \mathbf{v}$ and $A\mathbf{x} = \mathbf{w}$ both have solutions, i.e., there exists $\mathbf{x}_1$ and $\mathbf{x}_2$ such that $A\mathbf{x}_1 = \mathbf{v}$ and $A\mathbf{x}_2 = \mathbf{w}$. Then

$$A(\mathbf{x}_1 + \mathbf{x}_2) = A\mathbf{x}_1 + A\mathbf{x}_2 = \mathbf{v} + \mathbf{w}$$

, so the equation $A\mathbf{x} = \mathbf{v} + \mathbf{w}$ has a solution, and so $\mathbf{v} + \mathbf{w} \in \text{im}(A)$.

Closure under scalar multiplication

If $\mathbf{v} \in \text{im}(A)$, and a is a scalar, then we may write $\mathbf{v} = A\mathbf{x}_1$ for some $\mathbf{x}_1$. But then

$$A(a\mathbf{x}_1) = a(A\mathbf{x}_1) = a\mathbf{v}$$

, so the equation $A\mathbf{x} = a\mathbf{v}$ has a solution, so that $a\mathbf{v} \in \text{im}(A)$.

Thus $\text{im}(A)$ is a subspace of $\mathbb{R}^m$.

Warning

The solution set of $A\mathbf{x} = \mathbf{0}$ is a subspace. The solution set of $A\mathbf{x} = \mathbf{b}$ for $\mathbf{b} \neq \mathbf{0}$ is usually not a subspace; when it is nonempty, it is a shifted copy of $\text{null}(A)$.

2.3.3 Spanning Sets

Even though subspaces of $\mathbb{R}^n$ can have infinitely many vectors, it is possible to represent them with a finite amount of data. Recall that a *linear combination* is a sum of scalar multiples of vectors.

Activity

Show that there are two vectors (describe them explicitly), so that every vector in the set W discussed in 2.3.1 can be written as the linear combination of those two vectors.

Solution.

We may write

$$\begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix} = r \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + s \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

. Thus every vector in W can be written as a linear combination of

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

Definition 2.3.5 (The Span of a Set of Vectors).

The set of all linear combinations of a set of vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ is called the *span* of the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$, and is denoted

$$\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\} = \{t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k : t_1, \dots, t_k \in \mathbb{R}\}$$

. If $V = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$, we say that V is *spanned* by the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$, and that the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ *span* the space V .

For an image space, spanning vectors come from the columns of the matrix. If

$$A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \dots \ \mathbf{a}_n]$$

and

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$$

then

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \dots + x_n\mathbf{a}_n.$$

Thus every output $A\mathbf{x}$ is a linear combination of the columns of A .

Theorem 2.3.6.

Let $U = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ be a set in $\mathbb{R}^n$. Then

- U is a *subspace* of $\mathbb{R}^n$ containing each of the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$.
- If W is a subspace of $\mathbb{R}^n$ and each of the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ is in W , then U is a *subset* of W , i.e., $U \subseteq W$.

Proof.

Define an $n \times k$ matrix A , with the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ as columns. Then $U = \text{im}(A)$, and is thus a subspace of $\mathbb{R}^n$. Since $\mathbf{x}_i = A\mathbf{e}_i$ for $1 \leq i \leq k$, the set U contains each of the vectors $\mathbf{x}_i$.

Conversely, suppose W is a subspace of $\mathbb{R}^n$ and $\mathbf{x}_1, \dots, \mathbf{x}_k \in W$. Since W is closed under addition and scalar multiplication, all of the linear combinations of $\mathbf{x}_1, \dots, \mathbf{x}_k$ are elements of W , and so $U \subseteq W$.

Example

Recall the standard basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of $\mathbb{R}^n$ from 1.3.5. Note that if

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

, then

$$\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n$$

. This means that $\mathbb{R}^n = \text{span}\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$.

Activity

In $\mathbb{R}^3$, consider the two vectors

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$$

. Determine if the vector

$$\mathbf{w} = \begin{bmatrix} 1 \\ 5 \\ -7 \end{bmatrix}$$

belongs to $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$.

Solution.

We need to determine if $\mathbf{w}$ is a linear combination of the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$, i.e., if it is possible to find scalars a and b such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2$. Comparing entries of the vectors in each side of the equation, we obtain a system of equations in a and b , i.e.,

$$2a + b = 1$$

$$a - b = 5$$

$$a + 3b = -7$$

The Elimination Scratchpad

It now suffices to determine if this system has a solution. We could use the methods we developed in 2 to solve this system, but since there are only two variables we can get by more elementary methods.

The only values of a and b that satisfy the first pair of equations are $a = 2$ and $b = -3$. And these values *also* satisfy the third equation. Thus $\mathbf{w} = 2\mathbf{v}_1 - 3\mathbf{v}_2$, and so $\mathbf{w}$ *does* belong to the span of $\mathbf{v}_1$ and $\mathbf{v}_2$.

Activity

Find a spanning set for the null space of the matrix

$$A = \begin{bmatrix} 1 & -2 & -4 & 3 \\ 0 & -1 & -3 & 2 \\ 2 & 1 & 7 & -4 \end{bmatrix}$$

The Elimination Scratchpad

Solution.

To solve this problem, we begin by finding the basic solutions to the linear system $A\mathbf{x} = 0$, where

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

. Using the Gaussian elimination algorithm, we row reduce:

$$\begin{aligned}
& \begin{bmatrix} 1 & -2 & -4 & 3 \\ 0 & -1 & -3 & 2 \\ 2 & 1 & 7 & -4 \end{bmatrix} \\
& \rightarrow \begin{bmatrix} 1 & -2 & -4 & 3 \\ 0 & -1 & -3 & 2 \\ 0 & 5 & 15 & -10 \end{bmatrix} \\
& \rightarrow \begin{bmatrix} 1 & -2 & -4 & 3 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\
& \rightarrow \begin{bmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}
\end{aligned}$$

The free variables are x_3 and x_4 . Setting $x_3 = s$ and $x_4 = t$, we obtain that $A\mathbf{x} = 0$ when $x_1 = -2s + t$ and $x_2 = -3s + 2t$. Thus a general solution to this equation is given by

$$\begin{bmatrix} -2s + t \\ -3s + 2t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} -2 \\ -3 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

. Thus the null space is spanned by

$$\mathbf{v}_1 = \begin{bmatrix} -2 \\ -3 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

.

Activity: A spanning set for an image space

Find a spanning set for the image space of the matrix

$$A = \begin{bmatrix} 1 & -2 & -4 & 3 \\ 0 & -1 & -3 & 2 \\ 2 & 1 & 7 & -4 \end{bmatrix}$$

.

Solution.

Let

$$\mathbf{x} = \begin{bmatrix} t_1 \\ t_2 \\ t_3 \\ t_4 \end{bmatrix}.$$

Then

$$A\mathbf{x} = t_1 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + t_2 \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix} + t_3 \begin{bmatrix} -4 \\ -3 \\ 7 \end{bmatrix} + t_4 \begin{bmatrix} 3 \\ 2 \\ -4 \end{bmatrix}.$$

Therefore the image space is spanned by the columns of A :

$$\text{im}(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} -4 \\ -3 \\ 7 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ -4 \end{bmatrix} \right\}.$$

This is a spanning set. It is not claiming to be minimal.

To summarize: basic solutions of $A\mathbf{x} = \mathbf{0}$ form a spanning set for $\text{null}(A)$, and the columns of A form a spanning set for $\text{im}(A)$.

The next section studies when a spanning set has redundant vectors and how to remove them.

2.4 Independence, bases, and rank

2.4.1 Linear independence

Activity

In 2.3.1 we showed that the set

$$W = \left\{ \begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix} : r, s \in \mathbb{R} \right\}$$

is a subspace. In 2.3.3, we showed that the set

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}$$

is a spanning set for W . The set

$$T = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} \right\}$$

also spans W . But why is S ‘better’?

Solution.

It uses fewer vectors to span the same space. Which vectors in T are redundant?

$$\begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

So T spans W , but it contains repeated information.

A subspace U has many different spanning sets. We use the concept of *linear independence* to determine when sets of spanning vectors are the ‘most efficient’. In data terms, independence asks whether one feature direction is already built from the others.

Definition 2.4.1 (Linear Independence).

A set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ in $\mathbb{R}^n$ is said to be *linearly independent*, or simply *independent* if the only solution to the equation $t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k = \mathbf{0}$ is trivial, i.e., $t_1 = \dots = t_k = 0$.

A set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ in $\mathbb{R}^n$ is *linearly dependent* if it is *not* linearly independent, i.e., if there exists scalars $t_1, \dots, t_k$, not all zero, such that $t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k = \mathbf{0}$.

Activity

Determine whether the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 4 \\ -2 \end{bmatrix}$$

are linearly independent.

Solution.

We need to determine the solutions to the equation $t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + t_3\mathbf{v}_3 = \mathbf{0}$. Expanding out this equation gives the system

$$3t_1 + t_2 - t_3 = 0$$

$$2t_1 + 3t_2 + 4t_3 = 0$$

$$2t_1 + 0t_2 - 2t_3 = 0$$

The Elimination Scratchpad

We solve this homogeneous system of linear equations using the Gaussian elimination algorithm. First we create the augmented matrix

$$\left[\begin{array}{ccc|c} 3 & 1 & -1 & 0 \\ 2 & 3 & 4 & 0 \\ 2 & 0 & -2 & 0 \end{array} \right]$$

We reduce the matrix using row operations to row echelon form:

$$\begin{aligned} & \left[\begin{array}{ccc|c} 3 & 1 & -1 & 0 \\ 2 & 3 & 4 & 0 \\ 2 & 0 & -2 & 0 \end{array} \right] \\ & \rightarrow \left[\begin{array}{ccc|c} 3 & 1 & -1 & 0 \\ 0 & \frac{7}{3} & \frac{14}{3} & 0 \\ 0 & -\frac{2}{3} & -\frac{4}{3} & 0 \end{array} \right] \\ & \rightarrow \left[\begin{array}{ccc|c} 3 & 1 & -1 & 0 \\ 0 & \frac{7}{3} & \frac{14}{3} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{aligned}$$

There is one free variable corresponding to this matrix, which implies that there is one basic solution to the equation, and thus the system has a nontrivial solution. Thus the vectors are *not* linearly independent.

The last example demonstrates an algorithm to verify whether a given set of vectors is linearly independent.

Algorithm: A Test of Independence

To verify if a set $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ are linearly independent:

1. Consider the system of linear equations $t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k = \mathbf{0}$.
2. Solve the resulting linear homogeneous system.
3. If there is a nontrivial solution to the equation, then the vectors are linearly dependent. Otherwise, the vectors are linearly independent.

In matrix form: put the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ as the columns of a matrix M . Then $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is independent exactly when $M\mathbf{c} = \mathbf{0}$ has only the trivial solution, equivalently when every column of M is a pivot column.

Theorem 2.4.2.

If $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is a linearly independent set of vectors in $\mathbb{R}^n$, then every vector in their span $\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ has a unique representation as a linear combination of the $\mathbf{x}_i$.

Proof.

Let $\mathbf{x}$ be an element of $\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$. If the equation

$$\mathbf{x} = a_1\mathbf{x}_1 + \dots + a_k\mathbf{x}_k$$

had another solution, i.e.,

$$\mathbf{x} = b_1\mathbf{x}_1 + \dots + b_k\mathbf{x}_k$$

, then subtracting one equation from the other gives that

$$(a_1 - b_1)\mathbf{x}_1 + \dots + (a_k - b_k)\mathbf{x}_k = \mathbf{0}$$

. Thus the coefficients $a_1 - b_1, \dots, a_k - b_k$ give a solution to the equation

$$t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k = \mathbf{0}$$

. But the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ are linearly independent, and so this equation has only a trivial solution. Thus $a_1 - b_1, \dots, a_k - b_k = 0$, i.e., $a_1 = b_1, \dots, a_k = b_k$, proving that the representation of $\mathbf{x}$ as a linear combination is unique.

Activity

Determine whether the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ -3 \\ -1 \\ -1 \end{bmatrix}$$

are linearly independent.

Solution.

We must check if the equation

$$t_1 \mathbf{v}_1 + t_2 \mathbf{v}_2 + t_3 \mathbf{v}_3 = \mathbf{0}$$

has nontrivial solutions. Expanding out this equation gives the system

$$\begin{aligned}t_1 + t_2 &= 0 \\2t_1 - 2t_2 - 3t_3 &= 0 \\-t_3 &= 0 \\t_1 - t_3 &= 0\end{aligned}$$

The Elimination Scratchpad

We could use row-reduction here, though in this case a more ad-hoc approach is more efficient. The third equation gives $t_3 = 0$, and once this is deduced the fourth equation gives $t_1 = 0$. Then the first equation gives $t_2 = 0$. Thus the only solution to the equation is the trivial solution, and so the vectors are linearly independent.

Activity

Is the set $S = \{\mathbf{0}\}$ linearly independent?

Hint

Does the equation $t\mathbf{0} = \mathbf{0}$ have a nontrivial solution?

Fact

The nonzero rows of a matrix in row echelon form are linearly independent.

2.4.2 Basis and dimension

In geometry, we often describe lines as being ‘1-dimensional’, and planes as being ‘2-dimensional’. We can describe the notion of dimension more precisely using linear algebra.

Theorem 2.4.3 (A Fundamental Theorem).

Let U be a subspace of $\mathbb{R}^n$. If U is spanned by m vectors, and if U contains k linearly independent vectors, then $k \leq m$.

Activity

Is it possible to have 4 linearly independent vectors in $\mathbb{R}^3$?

Solution.

It is not possible. $\mathbb{R}^3$ is spanned by 3 vectors (for example, the standard basis vectors), so by 2.4.3, any linearly independent subset of $\mathbb{R}^3$ must contain at most 3 vectors.

Definition 2.4.4 (Bases of Vector Spaces).

If U is a subspace of $\mathbb{R}^n$, a set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ in U is called a *basis* of U if it satisfies the following two conditions:

- $\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ is *linearly independent*.
- $U = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$.

Theorem 2.4.5 (The Invariance Theorem).

If $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ and $\{\mathbf{w}_1, \dots, \mathbf{w}_k\}$ are both *bases* for a subspace U of $\mathbb{R}^n$, then $m = k$.

2.4.5 says that, even though a subspace has many different bases, all bases must have the *same number* of vectors.

Definition 2.4.6 (Dimensions of Vector Spaces).

The *dimension* of a non-zero subspace U of $\mathbb{R}^n$ is the *number* of vectors in a *basis* for U . We often write $\dim(U)$ for the dimension of U , and if $\dim(U) = d$, we say U is *d-dimensional*. By convention, the dimension of the trivial subspace $\{\mathbf{0}\}$ is *zero*.

- **Term:** spanning set
Meaning: enough directions to generate the whole subspace
- **Term:** independent set
Meaning: no direction is redundant
- **Term:** basis
Meaning: just enough directions: spanning and independent
- **Term:** dimension
Meaning: number of directions in any basis

Activity

Task

What is the dimension of $\mathbb{R}^3$?

Solution.

The standard basis for $\mathbb{R}^3$ is a basis, i.e., the set of three vectors $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ defined in 1.3.5 is linearly independent and spans $\mathbb{R}^3$. Thus $\mathbb{R}^3$ has dimension 3.

Task

What is the dimension of $\mathbb{R}^4$?

Solution.

The standard basis $\mathbf{e}_1, \dots, \mathbf{e}_4$ for $\mathbb{R}^4$ is a basis, and is a set of four vectors, so $\dim(\mathbb{R}^4) = 4$.

Task

What is the dimension of $\mathbb{R}^n$?

Solution.

The standard basis is a basis for $\mathbb{R}^n$, so $\dim(\mathbb{R}^n) = n$.

Activity

Find the dimension of the space

$$W = \left\{ \begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix} : r, s \in \mathbb{R} \right\}$$

Solution.

The set

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}$$

is a basis for W since S is linearly independent (neither vector in S is the scalar multiple of the other). In the solution to 2.3.3, we showed that these vectors span W , so S is a basis for W . Since S contains two vectors, $\dim(W) = 2$.

Fact

Let U be a non-zero subspace of $\mathbb{R}^n$. Then

- U has a basis and $\dim(U) \leq n$.
- Any independent set in U can be enlarged (by adding some subset of vectors in any fixed basis) to obtain a basis of U .
- Any spanning set for U contains a subset of vectors which form a basis for U .

Theorem 2.4.7.

Let U be a subspace of $\mathbb{R}^n$ where $\dim(U) = m$ and let $B = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ be a set of m vectors in U . Then B is linearly independent if and only if B spans U .

Proof.

Suppose B is linearly independent and $|B| = m = \dim(U)$. By 2.4.3, one cannot add any new vectors to the set B and remain linearly independent. But 2.4.2 says that if B is not already a spanning set for U , one can add new vectors to B from U , and remain linearly independent. Thus B must be a basis.

Conversely, if B spans U , then by 2.4.2 we can choose a subset of B which is a basis for U . But any proper subset of B contains fewer than m vectors, and thus by 2.4.5, cannot be a basis for U . Thus B is the only subset of itself that can be a basis, and because B must contain a basis, the set B must itself be a basis.

Theorem 2.4.8.

Let U and W be subspaces of $\mathbb{R}^n$ with $U \subseteq W$. Then:

- $\dim(U) \leq \dim(W)$.
- If $\dim(U) = \dim(W)$, then $U = W$.

Activity

If $\mathbf{v}$ and $\mathbf{w}$ are nonzero vectors in $\mathbb{R}^3$, show that $\{\mathbf{v}, \mathbf{w}\}$ is dependent if and only if $\mathbf{v}$ and $\mathbf{w}$ are parallel, i.e., if one vector is a scalar multiple of the other.

Solution.

If $\mathbf{v}$ and $\mathbf{w}$ are parallel, then one is a scalar multiple of the other, say $\mathbf{v} = a\mathbf{w}$ for some scalar a . Then $1\mathbf{v} - a\mathbf{w} = \mathbf{0}$, which is a nontrivial linear combination of $\mathbf{v}$ and $\mathbf{w}$, so $\{\mathbf{v}, \mathbf{w}\}$ is dependent.

Conversely, if $\{\mathbf{v}, \mathbf{w}\}$ is dependent, then $s\mathbf{v} + t\mathbf{w} = \mathbf{0}$ for some scalars s and t , where $s \neq 0$ or $t \neq 0$. If, say, $s \neq 0$, then $\mathbf{v} = -t/s\mathbf{w}$, so $\mathbf{v}$ and $\mathbf{w}$ are parallel. A similar argument works if $t \neq 0$.

Activity

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are non-zero vectors in $\mathbb{R}^n$, where $\{\mathbf{v}, \mathbf{w}\}$ is independent. Show that $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is independent if and only if $\mathbf{u}$ is not in the subspace $M = \text{span}\{\mathbf{v}, \mathbf{w}\}$. See 2.5 and 2.6.

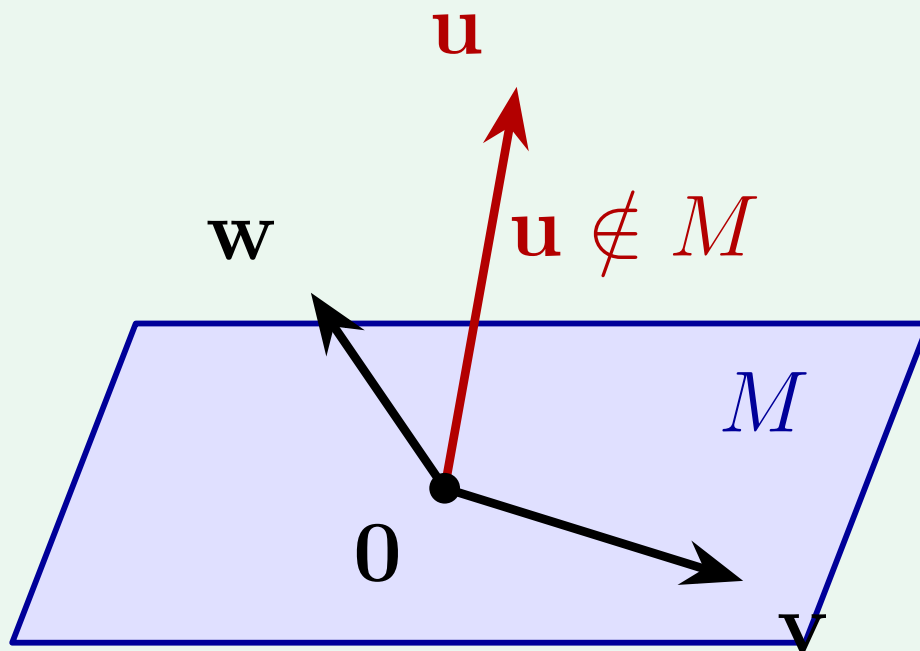


Figure 2.5. $\{u, v, w\}$ independent.

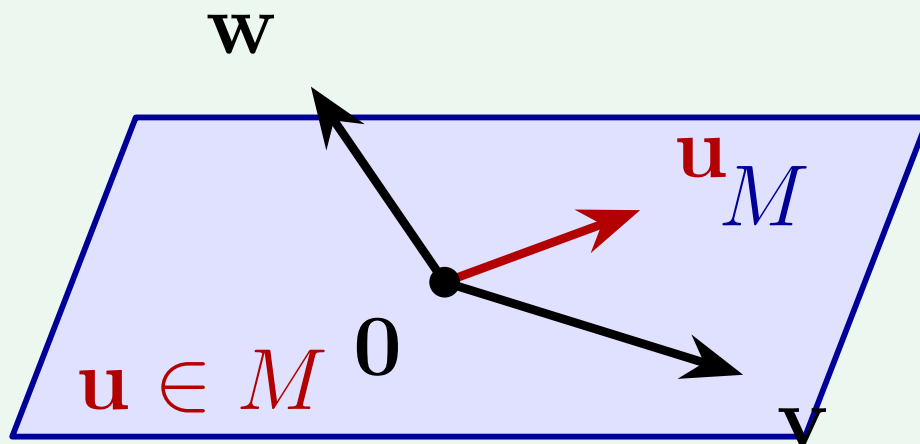


Figure 2.6. $\{u, v, w\}$ dependent.

Solution.

If $\{u, v, w\}$ is independent, suppose that u is in $M = \text{span}\{v, w\}$, say, that $u = av + bw$ for some scalars a and b . Then $1u - av - bw = 0$. Thus a nontrivial linear combination of u, v , and w is equal to zero, contradicting the independence of $\{u, v, w\}$.

On the other hand, suppose u is not in M . We need to show that $\{u, v, w\}$ is independent. If $ru + sv + tw = 0$, where r, s , and t are not all zero, then $r \neq 0$, for otherwise $sv + tw = 0$, which contradicts the independence of $\{v, w\}$. But then $u = -s/rv - t/rw$, which contradicts that u is not in M .

Note

More generally, if $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ are a linearly independent set of vectors in $\mathbb{R}^n$, and $\mathbf{x}_{k+1}$ is another vector in $\mathbb{R}^n$, then $\{\mathbf{x}_1, \dots, \mathbf{x}_{k+1}\}$ is a linearly independent set if and only if $\mathbf{x}_{k+1}$ is not contained in the span of the set $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$.

Activity

Find a basis for $\mathbb{R}^4$ that contains the two vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 3 \\ 3 \end{bmatrix}$$

Solution.

Since $\mathbb{R}^4$ is four dimensional, we must find two vectors $\mathbf{w}_1$ and $\mathbf{w}_2$ such that the vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{w}_1, \mathbf{w}_2\}$ is linearly independent. 2.4.2 states that we can choose these vectors from any basis, so we might as well choose them from the standard basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4\}$.

One strategy is to choose one vector $\mathbf{e}_i$ not contained in the span of the set $\{\mathbf{v}_1, \mathbf{v}_2\}$, and then choose a vector $\mathbf{e}_j$ not contained in the span of the set $\{\mathbf{e}_i, \mathbf{v}_1, \mathbf{v}_2\}$. By 2.4.2, the set $\{\mathbf{e}_j, \mathbf{e}_i, \mathbf{v}_1, \mathbf{v}_2\}$ is then linearly independent.

The vector $\mathbf{e}_2$ is not contained in the span of the set $\{\mathbf{v}_1, \mathbf{v}_2\}$, since the second entry in any linear combination of the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ is equal to zero, and thus the vector cannot be equal to $\mathbf{e}_2$.

We also claim that the vector $\mathbf{e}_1$ is not contained in the span of the set $\{\mathbf{e}_2, \mathbf{v}_1, \mathbf{v}_2\}$. If

$$\mathbf{e}_1 = a\mathbf{e}_2 + b\mathbf{v}_1 + c\mathbf{v}_2$$

, then checking each entry, we get the system of linear equations

$$1 = b$$

$$0 = a$$

$$0 = 2b + 3c$$

$$0 = 3c$$

The Elimination Scratchpad

The first, third, and fourth equations immediately contradict one another, so such a system cannot be solved.

Thus $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{v}_1, \mathbf{v}_2\}$ is a basis for $\mathbb{R}^4$.

2.4.3 Bases and coordinate systems

One way to think of a basis is as providing a coordinate system for a subspace. The uniquely determined coefficients in the linear combination of a vector in the subspace are the coordinates.

Definition 2.4.9 (Ordered Bases of a Subspace).

Let $S = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis of the subspace U . We call the tuple $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ an *ordered basis* of U .

As an example, $(\mathbf{e}_1, \mathbf{e}_2)$ and $(\mathbf{e}_2, \mathbf{e}_1)$ are two different ordered bases of $\mathbb{R}^2$. The two sets $\{\mathbf{e}_1, \mathbf{e}_2\}$ and $\{\mathbf{e}_2, \mathbf{e}_1\}$, and so describe the same bases of $\mathbb{R}^2$.

Note

Many Linear Algebra textbooks, including *Linear Algebra With Applications*, *Nicholson* write ordered bases using set notation $\{\dots\}$, despite the order in which the basis elements appear being important. We use tuple notation $(\dots)$ to emphasize the importance of the order.

Given an ordered basis B for an m -dimensional subspace U of $\mathbb{R}^n$, and any vector $\mathbf{v} \in U$, we can describe its ‘coordinates’ with respect to the basis B , identifying $\mathbf{v}$ with a vector in $\mathbb{R}^m$.

Definition 2.4.10 (Coordinate Vectors With Respect to a Basis).

Let U be an m -dimensional subspace of $\mathbb{R}^n$, and let $B = (\mathbf{b}_1, \dots, \mathbf{b}_m)$ be an ordered basis for U . Given any vector $\mathbf{v} \in U$, there exists unique scalars $t_1, \dots, t_m$ such that

$$\mathbf{v} = t_1 \mathbf{b}_1 + \dots + t_m \mathbf{b}_m$$

and we define the *coordinate vector* of a vector $\mathbf{v} \in U$ with respect to B to be the m -vector

$$C_B(\mathbf{v}) = \begin{bmatrix} t_1 \\ \vdots \\ t_m \end{bmatrix}$$

. See 2.7 for an illustration of this concept, taken from *Stanford’s MATH 51 Textbook*.

Note

If $B = (\mathbf{b}_1, \dots, \mathbf{b}_m)$ is an ordered basis for a subspace U , then $C_B(\mathbf{b}_i) = \mathbf{e}_i$, where $\mathbf{e}_i$ is the i th standard basis vector in $\mathbb{R}^m$.

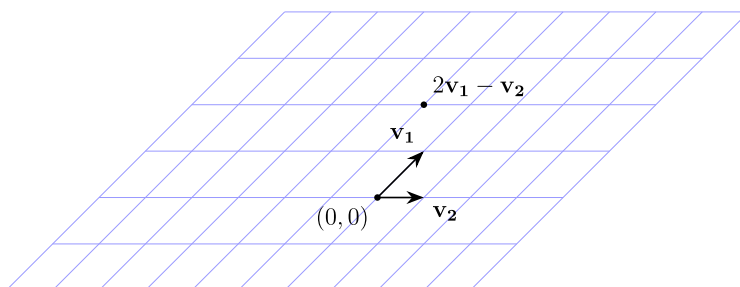


Figure 2.7. A parallelogram grid constructed from a basis $B = (\mathbf{v}_1, \mathbf{v}_2)$ for $\mathbb{R}^2$. The points lying at the intersection of lines on the grid are precisely those vectors $\mathbf{v} \in \mathbb{R}^2$ where the entries of the vector $C_B(\mathbf{v})$ are integers.

Activity

Let $B = (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ be an ordered basis for $\mathbb{R}^3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

. What is $C_B(\mathbf{v})$ if

$$\mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ -5 \end{bmatrix}$$

Solution.

We need to find scalars c_1, c_2 , and c_3 such that

$$\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3$$

. Comparing entries of the vectors gives a system of three equations, i.e.,

$$\begin{aligned} c_1 + c_2 + 2c_3 &= 1 \\ c_2 &= 1 \\ c_3 &= -5 \end{aligned}$$

The Elimination Scratchpad

Using back substitution, we find that $c_1 = 10$, $c_2 = 1$, and $c_3 = -5$ are the unique solution to this system. Thus

$$C_B(\mathbf{v}) = \begin{bmatrix} 10 \\ 1 \\ -5 \end{bmatrix}$$

2.4.4 Rank of a matrix

We are now going to connect the concept of dimension with the notion of rank. In this section, we will often realize vectors in $\mathbb{R}^n$ as rows rather than columns. The notion of span, linear independence, and basis are defined analogously as how they are defined for column vectors.

Definition 2.4.11 (Column and Row Space).

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

be an $m \times n$ matrix.

- The columns of A , considered as vectors in $\mathbb{R}^m$ span a *subspace* of $\mathbb{R}^m$ called the *column space* of A , denoted $\text{col}(A)$.
- The rows of A , considered as vectors in $\mathbb{R}^n$, span a *subspace* of $\mathbb{R}^n$ called the *row space* of A , denoted $\text{row}(A)$.

Note

The column space is the image space from 2.3 written with a column-based name:

$$\text{col}(A) = \text{im}(A).$$

The notation $\text{im}(A)$ emphasizes outputs of the map $\mathbf{x} \mapsto A\mathbf{x}$. The notation $\text{col}(A)$ emphasizes the columns that span those outputs.

Theorem 2.4.12.

Let A and B be two row equivalent $m \times n$ matrices. Then $\text{row}(A) = \text{row}(B)$.

Proof.

Each time we apply a row operation to a matrix, we replace one or more of its rows with a linear combination of other rows. Thus the row space of a matrix A' obtained by applying a row operation to a matrix A must be contained in the row space of A . But applying the argument in reverse (since every row operation is reversible), the row space of A must be contained in the row space of A' . Thus the two row spaces are equal. But if the row space is not changed after applying a single row operation, it is not changed after applying an arbitrary number of row operations, and so the result follows.

Theorem 2.4.13.

If R is a matrix in row echelon form (recall 2.2.10) then the non-zero rows of R are a basis of $\text{row}(R)$.

Proof.

The nonzero rows of R are linearly independent by 2.4.1, and span $\text{row}(R)$ by definition. They thus by definition form a basis for $\text{row}(R)$.

Remark

In 2.2.14, we defined the *rank* of a matrix as the number of leading 1s (i.e., the number of pivots) of any matrix R in row echelon form which is row-equivalent to A . By 2.4.13, $\text{rank}(A) = \dim(\text{row}(A))$, which justifies that the rank of A is independent of the choice of matrix R .

Activity

Find a basis for $V = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \quad \mathbf{v}_4 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

Solution.

The subspace V is the row space of the matrix

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

The Elimination Scratchpad The Scratchpad includes an extra zero column on the right. After reaching row echelon form, omit that column and use the nonzero rows as basis vectors for V .

If we reduce this matrix to row echelon form, we thus find a basis for V :

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0.5 & 0 \\ 0 & 1.5 & -3 \\ 0 & 1 & -2 \\ 0 & 0.5 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0.5 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

. Thus

$$\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \right\}$$

is a basis for V .

Theorem 2.4.14 (Rank Theorem).

Let A denote any $m \times n$ matrix of rank r . Then

$$\dim(\text{col}(A)) = \dim(\text{row}(A)) = r$$

. Moreover, if A is row-equivalent to a matrix R in row echelon form, then

- The r leading rows of R are a basis for $\text{row}(A)$.
- If the pivots of R lie in columns $j_1, j_2, \dots, j_r$, then the columns $j_1, j_2, \dots, j_r$ of A are a basis for $\text{col}(A)$.

Warning: Pivot columns come from the original matrix

Row reduction identifies pivot positions, but row operations do not preserve the original columns of A . When finding a basis for $\text{col}(A)$, use the pivot columns of the original matrix A , not the columns of the row-reduced matrix.

- **Goal:** basis for $\text{row}(A)$
Row reduction tells you: nonzero rows of REF/RREF
Use these vectors: rows of the reduced matrix
- **Goal:** basis for $\text{col}(A)$
Row reduction tells you: pivot column positions
Use these vectors: corresponding columns of original A
- **Goal:** basis for $\text{null}(A)$
Row reduction tells you: free variables
Use these vectors: basic solution vectors in $\mathbb{R}^n$
- **Goal:** rank
Row reduction tells you: number of pivots
Use these vectors: a number
- **Goal:** nullity
Row reduction tells you: number of free variables
Use these vectors: a number

Fact

It follows from 2.4.14 that for any matrix A , $\text{rank}(A) = \text{rank}(A^T)$.

Activity

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to 2.4.4.

Task

What is $\dim(\text{row}(A))$?

Solution.

The dimension of the row space of A is equal to the rank of A , which is 2 (the number of non-zero rows in the row echelon form for A , as computed in the solution to 2.4.4).

Task

Find a basis for the column space of A .

Solution.

The leading 1s in the row echelon form for A computed in 2.4.4 occur in the 1st and 2nd columns, so the 1st and 2nd columns of A form a basis for the column space, i.e.,

$$\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \\ 1 \end{bmatrix} \right\}$$

is a basis for $\text{col}(A)$.

Task

What is $\dim(\text{col}(A))$?

Solution.

The dimension of the column space of A is two, since we constructed a basis with two elements in the previous task. Alternatively, the dimension of the column space is equal to the rank of A , which we also computed to be two.

Task

Find a basis for the nullspace of A .

The Elimination Scratchpad

Solution.

To find a basis for the solutions to the equation $Ax = 0$, we use the row-echelon form computed in 2.4.4 and find basic solutions as in 2.2.17. The row-echelon form gives the equations $x_1 + \frac{1}{2}x_2 = 0$ and $x_2 - 2x_3 = 0$. Setting the free variable $x_3 = 1$, we obtain $x_2 = 2$ and $x_1 = -1$, so the basic solution is

$$\begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

. Thus $\left\{ \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \right\}$ is a basis for the nullspace of A .

Definition 2.4.15.

The *nullity* of a matrix A is equal to $\dim(\text{null}(A))$. It is denoted $\text{nullity}(A)$.

Rank counts independent directions transmitted by a matrix. Nullity counts independent input directions forgotten by a matrix. The row-reduction definition of rank now connects to dimension.

Theorem 2.4.16 (Rank-Nullity Theorem).

Let A denote an $m \times n$ matrix of rank r . Then

- The $n - r$ basic solutions to the equation $A\mathbf{x} = \mathbf{0}$ provided by the Gaussian algorithm are a basis for $\text{null}(A)$, so $\dim(\text{null}(A)) = n - r$.
- $\dim(\text{im}(A)) = r$.
- $\dim(\text{im}(A)) + \dim(\text{null}(A)) = n$.
- $\text{rank}(A) + \text{nullity}(A) = n$.

Note

input dimensions = transmitted dimensions + forgotten dimensions

$$n = \text{rank}(A) + \text{nullity}(A)$$

We can now state one of the preview patterns from Section 2.1, introduced in 2.1, in the language of null spaces.

Activity: Same output from a null-space direction

Suppose $A\mathbf{x} = \mathbf{y}$ and $A\mathbf{z} = \mathbf{0}$. Compute $A(\mathbf{x} + t\mathbf{z})$, where $t \in \mathbb{R}$. What does this say about uniqueness?

Solution.

$$\begin{aligned} A(\mathbf{x} + t\mathbf{z}) &= A\mathbf{x} + tA\mathbf{z} \\ &= \mathbf{y} + t\mathbf{0} \\ &= \mathbf{y}. \end{aligned}$$

So every vector $\mathbf{x} + t\mathbf{z}$ gives the same output. A nonzero vector in the null space creates a whole line of inputs with the same output.

Activity: Revisiting the height map with rank and nullity

Return again to 2.1. Let

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Find the column space, null space, rank, and nullity of P . How does rank-nullity describe the height direction that P forgets?

Solution.

The column space is

$$\text{col}(P) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} = \mathbb{R}^2,$$

so $\text{rank}(P) = 2$.

The null space is

$$\text{null}(P) = \text{span} \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\},$$

so $\text{nullity}(P) = 1$. Since P has three input coordinates,

$$\text{rank}(P) + \text{nullity}(P) = 2 + 1 = 3.$$

Rank-nullity says that the two output directions and the one forgotten height direction account for all three input directions.

Activity

Consider the matrix

$$B = \begin{bmatrix} 3 & -1 & 2 & 0 \\ -6 & 2 & -2 & 6 \end{bmatrix}$$

. You may use the fact that the reduced row echelon form of B is the matrix

$$\begin{bmatrix} 1 & -1/3 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{bmatrix} \quad (2.3)$$

.

Task

Find a basis for $\text{im}(B)$.

Solution.

We use the fact that $\text{im}(B) = \text{col}(B)$. Using 2.4.14, since the RREF in 2.3 has pivots in the 1st and 3rd columns, the column space of B , and thus the image of B , has a basis given by the 1st and 3rd columns of B , i.e., we have a basis

$$\left\{ \begin{bmatrix} 3 \\ -6 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \end{bmatrix} \right\}$$

.

Task

Find a basis for $\text{null}(B)$.

The Elimination Scratchpad

Solution.

The 2nd and 4th columns of the matrix in 2.3 do not contain pivots, so x_2 and x_4 correspond to free variables in the equation, and solutions to $Ax = 0$ are specified by the two equations $x_1 = x_2/3 + 2x_4$ and $x_3 = -3x_4$. The two basis solutions give a basis for the nullspace, and thus we have a basis of the form

$$\left\{ \begin{bmatrix} 1/3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -3 \\ 1 \end{bmatrix} \right\}$$

2.4.5 Redundant features in a data matrix

A data matrix can have redundant columns. This means one feature column is a linear combination of other feature columns.

Rows below represent houses. The three features are

first-level area, second-level area, total area,
measured in hundreds of square feet:

$$X = \begin{bmatrix} 9 & 7 & 16 \\ 11 & 9 & 20 \\ 14 & 0 & 14 \\ 8 & 8 & 16 \end{bmatrix}.$$

Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ be the columns of X . The third column satisfies

$$\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2,$$

so the third feature carries no new independent direction. The dependence relation is

$$\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0},$$

or equivalently

$$X \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \mathbf{0}.$$

The null-space direction

$$\mathbf{z} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

says that coefficient vectors differing by a multiple of $\mathbf{z}$ give the same prediction vector.

Activity: Redundant square-footage features

Use the data matrix

$$X = \begin{bmatrix} 9 & 7 & 16 \\ 11 & 9 & 20 \\ 14 & 0 & 14 \\ 8 & 8 & 16 \end{bmatrix}, \quad \mathbf{z} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

1. Compute $X\mathbf{z}$.
2. What does this say about the three feature columns?
3. Compare the coefficient vectors

$$\mathbf{c} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{c}' = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

Compute $X\mathbf{c}$ and $X\mathbf{c}'$.

4. The two coefficient vectors tell different stories. Why do they make the same predictions?
5. What warning does this give about asking which feature is most important?

Solution.

First,

$$X\mathbf{z} = X \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \mathbf{0}.$$

This means the columns are dependent. The total-area column is the sum of the first-level and second-level columns.

Now,

$$X\mathbf{c} = \begin{bmatrix} 34 \\ 42 \\ 42 \\ 32 \end{bmatrix}, \quad X\mathbf{c}' = \begin{bmatrix} 34 \\ 42 \\ 42 \\ 32 \end{bmatrix}.$$

The coefficient vectors are different, but they differ by the null-space direction:

$$\mathbf{c} - \mathbf{c}' = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \mathbf{z}.$$

Since $Xz = 0$,

$$Xc = X(c' + z) = Xc' + Xz = Xc'.$$

The predictions are the same, but the coefficient interpretations are different. One coefficient vector puts weight on first-level and second-level area separately. The other puts weight on first-level area and total area. With redundant features, individual coefficient importance is not uniquely determined by the data matrix.

Note: Common exact feature redundancies

Common exact feature redundancies include:

- first-level area, second-level area, and total area;
- minutes and seconds, where $\text{seconds} = 60 \cdot \text{minutes}$;
- height in inches and height in centimeters, where $\text{centimeters} = 2.54 \cdot \text{inches}$;
- subtotal, tax, tip, and total bill, where $\text{total} = \text{subtotal} + \text{tax} + \text{tip}$;
- exam parts and exam total, where $\text{total} = \text{part 1} + \text{part 2} + \text{part 3}$.

These are linear redundancies. Not every relationship between features is linear. For example, rectangle area is width times height, not a linear combination of width and height.

Activity: Which features are linearly redundant?

For each feature list, decide whether one feature is a linear combination of the others.

1. first-level area, second-level area, total area;
2. height in inches, height in centimeters;
3. width, height, area of a rectangle;
4. quiz 1 score, quiz 2 score, quiz total;
5. bedrooms, bathrooms, selling price.

Solution.

Items 1, 2, and 4 contain exact linear redundancies:

$$\text{total area} = \text{first-level area} + \text{second-level area},$$

$$\text{centimeters} = 2.54(\text{inches}),$$

and

$$\text{quiz total} = \text{quiz 1} + \text{quiz 2}.$$

Item 3 is related but not linearly redundant, because

$$\text{area} = (\text{width})(\text{height})$$

is not a linear combination of width and height.

Item 5 may contain useful statistical relationships, but there is no exact linear redundancy from the feature names alone.

Activity: Auditing redundant features in code

The matrix below stores the house features from the previous activity.

```
import numpy as np

X = np.array([
    [9, 7, 16],
    [11, 9, 20],
    [14, 0, 14],
    [8, 8, 16],
])

z = np.array([1, 1, -1])
c = np.array([3, 1, 0])
c_alt = np.array([2, 0, 1])

rank = int(np.linalg.matrix_rank(X))

X @ z, rank, X @ c, X @ c_alt
```

Here `c_alt` stores the vector $\mathbf{c}'$.

Output:

```
(array([0, 0, 0, 0]), 2, array([34, 42, 42, 32]), array([34, 42, 42, 32]))
```

1. What does the zero output mean?
2. What does the rank say about the three feature columns?

3. Why do $X\mathbf{c}$ and $X\mathbf{c}'$ agree?
4. What does this say about interpreting coefficients when features are redundant?

Solution.

The zero output means that $\mathbf{z}$ is a null-space direction. The rank is 2, so the three feature columns contain only two independent directions.

The vectors $\mathbf{c}$ and $\mathbf{c}'$ differ by $\mathbf{z}$, and $X\mathbf{z} = \mathbf{0}$. Therefore they give the same prediction vector. When features are redundant, different coefficient vectors can make the same predictions. Individual coefficients may not have a unique interpretation.

Activity: A difference matrix forgets level

Let

$$D = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

Compute $D \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$. What input direction is forgotten?

Solution.

We get

$$D \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \mathbf{0}.$$

The matrix records consecutive differences. Adding the same constant to every entry does not change the differences, so the constant direction $(1, 1, 1, 1)$ is forgotten.

Activity: Reading the difference matrix in code

```
D = np.array([
    [-1, 1, 0, 0],
    [0, -1, 1, 0],
    [0, 0, -1, 1],
])
```

```
x = np.array([2, 5, 9, 10])
ones = np.ones(4, dtype=int)

D @ x, D @ (x + 10*ones), D @ ones
```

Output:

```
(array([3, 4, 1]), array([3, 4, 1]), array([0, 0, 0]))
```

1. What does $D @ x$ measure?
2. Why do the first two outputs agree?
3. What does $D @ \text{ones}$ say about the map?

Solution.

The product $D @ x$ measures consecutive differences:

$$5 - 2 = 3, \quad 9 - 5 = 4, \quad 10 - 9 = 1.$$

Adding 10 to every entry changes the level but not the differences, so the first two outputs agree.

The output $D @ \text{ones}$ is $\mathbf{0}$, so the constant direction is a null-space direction. The difference matrix forgets level.

Redundant features create null-space directions. If $X\mathbf{z} = \mathbf{0}$, then

$$X(\mathbf{c} + t\mathbf{z}) = X\mathbf{c}$$

for every scalar t . Thus many coefficient vectors can produce the same prediction vector. Rank counts how many independent feature directions remain; nullity counts how many independent coefficient changes are invisible to the data matrix.

2.5 Inverses and determinants

2.5.1 Matrix invertibility and linear systems

In this section we finish the square-matrix version of the Unit 2 story. A square matrix is invertible exactly when it “does not forget any nonzero input direction” and “every output is reached once”. The inverse matrix gives the inverse map when it exists. The determinant, a fundamental quantity in linear algebra that we only briefly discuss in this course, gives a single-number test for invertibility.

Definition 2.5.1 (Invertibility of Matrices).

An $n \times n$ matrix A is called *nonsingular*, or *invertible*, if there exists an $n \times n$ matrix B such that $AB = BA = I_n$. Such a B is called the *inverse* of A . If no such B exists, A is called *singular* or *noninvertible*.

Note

The inverse of a matrix mimics the reciprocal of a real number.

Activity

Let

$$A = \begin{bmatrix} 1 & 3 \\ 1 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 4 & -3 \\ -1 & 1 \end{bmatrix}$$

. Compute both AB and BA , and make a conclusion using the language of inverses.

Solution.

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2 \quad \text{and} \quad BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

Therefore, B is the inverse of A .

Fact

The inverse of a matrix, if it exists, is unique. Therefore, we can write A^{-1} for the inverse of A .

Fact

For $n \times n$ matrices A and B , if $AB = I_n$, then $BA = I_n$.

Activity

Does $A = \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$ have an inverse?

Solution.

Let's suppose A had an inverse $B = \begin{bmatrix} x & y \\ z & w \end{bmatrix}$. Then:

$$AB = I_2$$

$$\begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

This gives us the system of linear equations in four unknowns:

$$\begin{array}{ccccccc} 2x & & + & z & & = & 1 \\ & 2y & & & + & w & = 0 \\ 0x & & + & 0z & & = & 0 \\ & 0y & & & + & 0w & = 1 \end{array}$$

The last equation states that $0 = 1$, which is impossible.

Therefore, A cannot have an inverse. This means A is singular.

Theorem 2.5.2.

If both A and B are nonsingular $n \times n$ matrices, then the matrix AB is nonsingular and its inverse is $(AB)^{-1} = B^{-1}A^{-1}$.

Proof.

It suffices to calculate that $(AB)(B^{-1}A^{-1}) = I_n$ and $(B^{-1}A^{-1})(AB) = I_n$. Firstly,

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_nA^{-1} = AA^{-1} = I_n$$

. Similarly,

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}I_nB = B^{-1}B = I_n$$

.

Fact: Follow Up Facts About Invertibility

- If $A_1, A_2, \dots, A_k$ are $n \times n$ invertible / nonsingular matrices, then $A_1A_2 \cdots A_k$ is invertible / nonsingular and

$$(A_1A_2 \cdots A_k)^{-1} = A_k^{-1} \cdots A_2^{-1}A_1^{-1}$$

.

- If A is invertible/nonsingular, then A^{-1} is invertible/nonsingular and

$$(A^{-1})^{-1} = A$$

.

- If A is invertible/nonsingular, then A^T is invertible/nonsingular and

$$(A^{-1})^T = (A^T)^{-1}$$

Let $T = T_A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a matrix transformation induced by the $n \times n$ matrix A . What does invertibility imply about the geometry of the transformation T_A ?

Consider $T' = T_{A^{-1}}$. What happens when we compose them?

Activity

Compute the composition of T and T' in both orders.

Solution.

Let's examine the composition of T and T' :

$$\begin{aligned} T \circ T'(\mathbf{x}) &= T_A(T_{A^{-1}}(\mathbf{x})) \\ &= T_A(A^{-1}\mathbf{x}) \\ &= A(A^{-1}\mathbf{x}) \\ &= (AA^{-1})\mathbf{x} \\ &= I_n\mathbf{x} \\ &= \mathbf{x} \end{aligned}$$

Similarly, for the reverse composition:

$$\begin{aligned} T' \circ T(\mathbf{x}) &= T_{A^{-1}}(T_A(\mathbf{x})) \\ &= T_{A^{-1}}(A\mathbf{x}) \\ &= A^{-1}(A\mathbf{x}) \\ &= (A^{-1}A)\mathbf{x} \\ &= I_n\mathbf{x} \\ &= \mathbf{x} \end{aligned}$$

Definition 2.5.3.

Given a matrix transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$, if there exists a transformation $S : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $S(T(\mathbf{x})) = T(S(\mathbf{x})) = \mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$, we say that T has an inverse.

Theorem 2.5.4.

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ denote the matrix transformation induced by an $n \times n$ matrix A . Then:

1. A is invertible if and only if T has an inverse.
2. In this case, T has exactly one inverse (which we denote as T^{-1}), and $T^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the transformation induced by the matrix A^{-1} .

In other words, $(T_A)^{-1} = T_{A^{-1}}$.

T and T^{-1} ‘reverse’ or ‘undo’ the action of each other.

Activity

Consider $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by reflection across the x -axis. Find the matrix representing this transformation, and find the inverse.

Solution.

T is given by matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Since $A^2 = I_2$, A is its own inverse, so $T^{-1} = T$.

Theorem 2.5.5.

The 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is invertible if and only if $ad - bc \neq 0$, in which case

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

We define the *determinant* of the matrix A to be the quantity $\det(A) = ad - bc$. Thus the matrix A is invertible if and only if its determinant is nonzero.

For now, $\det(A) = ad - bc$ is only the determinant of a 2×2 matrix. We will define determinants for larger square matrices later in this section.

Activity: Reading determinant checks in code

The following code computes determinants of two 2×2 matrices.

```
import numpy as np

A = np.array([
    [1., 1.],
    [0., 1.],
])

B = np.array([
```

```

    [1., 0.],
    [0., 0.],
])

detA = float(np.linalg.det(A))
detB = float(np.linalg.det(B))

detA, detB

```

Output:

```
(1.0, 0.0)
```

1. Which matrix is invertible?
2. Which matrix forgets a nonzero input direction?
3. How does the determinant tell you?

Solution.

The output says

$$\det(A) = 1, \quad \det(B) = 0.$$

Matrix A is invertible because its determinant is nonzero. Matrix B is singular because its determinant is zero.

The matrix B forgets the nonzero input direction

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

since

$$B \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

For a 2×2 matrix, a nonzero determinant means the matrix is invertible. A zero determinant means the matrix is singular and forgets some nonzero input direction.

Activity

Find the inverse of each matrix, if possible.

Task

$$A = \begin{bmatrix} -2 & -3 \\ 4 & 6 \end{bmatrix}$$

Solution.

$$\det(A) = (-2)(6) - (-3)(4) = -12 - (-12) = 0$$

, so A is not invertible.

Task

$$B = \begin{bmatrix} 1 & 6 \\ 2 & 3 \end{bmatrix}$$

Solution.

$$\det(B) = (1)(3) - (6)(2) = 3 - 12 = -9 \neq 0$$

, so

$$B^{-1} = \frac{1}{-9} \begin{bmatrix} 3 & -6 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -1/3 & 2/3 \\ 2/9 & -1/9 \end{bmatrix}$$

.

If A is an $n \times n$ matrix, then the linear system $A\mathbf{x} = \mathbf{b}$ is a system of n equations in n unknowns. Suppose A is nonsingular. How can we use A^{-1} to solve the system $A\mathbf{x} = \mathbf{b}$?

Activity

Solve the system using the inverse of A .

Solution.

Multiply both sides of $A\mathbf{x} = \mathbf{b}$ by A^{-1} on the left:

$$\begin{aligned}
 A\mathbf{x} &= \mathbf{b} \\
 A^{-1}(A\mathbf{x}) &= A^{-1}\mathbf{b} \\
 (A^{-1}A)\mathbf{x} &= A^{-1}\mathbf{b} \\
 I_n\mathbf{x} &= A^{-1}\mathbf{b} \\
 \mathbf{x} &= A^{-1}\mathbf{b}
 \end{aligned}$$

Therefore, when A is nonsingular, $\mathbf{x} = A^{-1}\mathbf{b}$ is the solution to $A\mathbf{x} = \mathbf{b}$.

Consequences:

- When A^{-1} exists, then $A\mathbf{x} = \mathbf{b}$ has a unique solution.
- If A is invertible/nonsingular, then the *only* solution to the homogeneous system $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$.
- Put differently, an invertible square matrix forgets no nonzero input direction.

Activity

Suppose that

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

. Use this information to calculate the following:

Task

$$A\mathbf{x} = \mathbf{b}$$

Solution.

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} -6 \\ 5 \end{bmatrix}$$

The Elimination Scratchpad

Task

$$A\mathbf{x} = \mathbf{c}$$

Solution.

$$\mathbf{x} = A^{-1}\mathbf{c} = \begin{bmatrix} -4 \\ \frac{7}{2} \end{bmatrix}$$

The Elimination Scratchpad

Task

$$A\mathbf{x} = \mathbf{0}$$

Solution.

$$\mathbf{x} = A^{-1}\mathbf{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The Elimination Scratchpad

Activity

Consider the transformation $P : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$P\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}\right) = \begin{bmatrix} a_1 \\ 0 \end{bmatrix}$$

.

Task

Is P a matrix transformation? If so, find the matrix A corresponding to P .

Solution.

Yes, P is a matrix transformation. It can be represented by the matrix

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

To verify:

$$P\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} a_1 \\ 0 \end{bmatrix}$$

Task

Is P an invertible transformation?

Solution.

No, P is not invertible. Note that $\det(A) = 0$, and so A is not an invertible matrix, and so P cannot be an invertible transformation.

Calculating inverses

Consider vectors $\mathbf{c}_1, \dots, \mathbf{c}_n$ in $\mathbb{R}^m$, and let A be the $m \times n$ matrix whose columns are the vectors $\mathbf{c}_1, \dots, \mathbf{c}_n$.

Note that $\{\mathbf{c}_1, \dots, \mathbf{c}_n\}$ are linearly independent precisely when the equation $A\mathbf{x} = \mathbf{0}$ has *only* the *trivial* solution.

Note that additionally, $\text{span}\{\mathbf{c}_1, \dots, \mathbf{c}_n\} = \mathbb{R}^m$ exactly when $A\mathbf{x} = \mathbf{b}$ has a solution for all $\mathbf{b}$ in $\mathbb{R}^m$.

We use these facts to extend our invertibility criteria for matrices.

Fact: Invertibility Criteria

Let A be a $n \times n$ matrix. Then the following are equivalent:

- A is invertible.
- The columns of A are linearly independent.
- The columns of A span $\mathbb{R}^n$.
- The rows of A are linearly independent.
- The rows of A span $\mathbb{R}^n$.

Activity

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 3 \\ 0 \\ -5 \end{bmatrix}$$

. Is S linearly dependent?

Solution.

Put the vectors in the columns of a matrix:

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & 0 \\ 2 & 2 & -5 \end{bmatrix}$$

The Elimination Scratchpad Use c_1, c_2, c_3 for the coefficients in $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{0}$.
Row reduction gives

$$\begin{aligned} \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & 0 \\ 2 & 2 & -5 \end{bmatrix} &\sim \begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & -6 \\ 0 & 2 & -11 \end{bmatrix} \\ &\sim \begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & -6 \\ 0 & 0 & -23 \end{bmatrix} \end{aligned}$$

. This echelon form has a pivot in every column. Therefore the equation $Ac = 0$ has only the trivial solution, so no nontrivial linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ gives zero. Thus S is linearly independent, not linearly dependent.

For a given $n \times n$ matrix A , if we are looking for an inverse for A (call this matrix X), we must have $AX = I_n$. We are solving n linear systems simultaneously:

Activity

Describe the system of linear equations for finding the inverse.

Solution.

We need to solve these systems:

$$A \begin{bmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{n1} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad A \begin{bmatrix} x_{12} \\ x_{22} \\ \vdots \\ x_{n2} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \dots, \quad A \begin{bmatrix} x_{1n} \\ x_{2n} \\ \vdots \\ x_{nn} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

Since, to solve each system, we wish to reduce the matrix A down to its RREF each time, we can combine all the systems into one big augmented matrix:

Activity

Write the augmented matrix for finding the inverse.

Solution.

$$[A \mid I_n] = \left[\begin{array}{cccc|cccc} a_{11} & a_{12} & \cdots & a_{1n} & 1 & 0 & \cdots & 0 \\ a_{21} & a_{22} & \cdots & a_{2n} & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & 0 & 0 & \cdots & 1 \end{array} \right]$$

It follows from 2.2.9 that if $[A \mid I_n]$ is row reducible to $[B \mid C]$, then $BX = C$. In particular, if we can row reduce $[A \mid I_n]$ to $[I_n \mid C]$, then it follows that $X = C$. Gaussian elimination (2.4) will reduce a given matrix to this form, provided that an inverse exists.

Algorithm: Algorithm for Computing A^{-1}

1. Set up the augmented matrix $[A \mid I_n]$.
2. Apply elementary row operations to $[A \mid I_n]$ to reduce it to $[I_n \mid A^{-1}]$.

If A is an $n \times n$ matrix, either A can be reduced to I_n by elementary row operations or it cannot. In the first case, the algorithm produces A^{-1} ; in the second case, A^{-1} does not exist.

Activity

Find A^{-1} where $A = \begin{bmatrix} 1 & 1 & 4 \\ 2 & 3 & 2 \\ 0 & 0 & 1 \end{bmatrix}$.

Solution.

Start with $[A \mid I_3]$:

$$[A \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & 1 & 4 & 1 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

Perform row operations:

$$R_2 \rightarrow R_2 - 2R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 4 & 1 & 0 & 0 \\ 0 & 1 & -6 & -2 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$R_1 \rightarrow R_1 - R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 10 & 3 & -1 & 0 \\ 0 & 1 & -6 & -2 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$R_1 \rightarrow R_1 - 10R_3$$

$$R_2 \rightarrow R_2 + 6R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -1 & -10 \\ 0 & 1 & 0 & -2 & 1 & 6 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

Therefore:

$$A^{-1} = \begin{bmatrix} 3 & -1 & -10 \\ -2 & 1 & 6 \\ 0 & 0 & 1 \end{bmatrix}$$

Theorem 2.5.6 (Box of Facts/Invertibility Criteria).

Expanding on 2.5.1, we can add row-reduction and solution-set criteria to the same list. Let A be an $n \times n$ matrix. The following are equivalent:

1. A is nonsingular/invertible.
2. The columns of A are linearly independent.
3. The columns of A span $\mathbb{R}^n$.
4. The rows of A are linearly independent.
5. The rows of A span $\mathbb{R}^n$.
6. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
7. A is row equivalent to I_n .
8. The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n \times 1$ matrix $\mathbf{b}$.
9. The rank of A is n .

2.5.2 General determinants

In this section we consider square $n \times n$ matrices. Our *goal* is to find a number, called the *determinant* that we can calculate for any matrix, which tells us whether or not the matrix is invertible.

Let's start with the 2×2 case. Recall that in 2.5.5 we defined the determinant of a 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

to be $\det(A) = ad - bc$. It has the required properties of the determinant that we will now define for more general matrices.

Activity

Find the determinant of the following matrices:

Task

$$A = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$$

Solution.

$$\det(A) = 3(2) - 1(4) = 6 - 4 = 2$$

.

Task

$$B = \begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix}$$

Solution.

$$\det(B) = 3(4) - 2(6) = 12 - 12 = 0$$

We use the 2×2 determinant to define the 3×3 determinant. The idea is recursive: a 3×3 determinant is built from 2×2 determinants. Later, an $n \times n$ determinant will be built from $(n - 1) \times (n - 1)$ determinants.

Definition 2.5.7 (Minors of a 3×3 matrix).

Let A be a 3×3 matrix. The (i, j) -th minor of A , denoted

$$\text{minor}(A)_{ij},$$

is the determinant of the 2×2 matrix obtained from A by deleting row i and column j .

Activity

Suppose

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

Task

Calculate $\text{minor}(A)_{11}$.

Solution.

$$\text{minor}(A)_{11} = \det \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} = 3(1) - 2(2) = -1$$

Task

Calculate $\text{minor}(A)_{23}$.

Solution.

$$\text{minor}(A)_{23} = \det \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} = 1(2) - 2(3) = -4$$

Definition 2.5.8 (Cofactors of a 3×3 matrix).

Let A be a 3×3 matrix. The (i, j) -cofactor of A , denoted

$$\text{cof}(A)_{ij},$$

is

$$\text{cof}(A)_{ij} = (-1)^{i+j} \text{minor}(A)_{ij}.$$

Activity

Define

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

Task

Compute $\text{cof}(A)_{11}$.

Solution.

$$\text{cof}(A)_{11} = (-1)^{1+1} \text{minor}(A)_{11} = (1)(-1) = -1$$

.

Task

Compute $\text{cof}(A)_{23}$.

Solution.

$$\text{cof}(A)_{23} = (-1)^{2+3} \text{minor}(A)_{23} = (-1)(-4) = 4$$

.

Definition 2.5.9 (Determinant of a 3×3 matrix).

Let $A = [a_{ij}]$ be a 3×3 matrix. The *determinant* of A , denoted $\det(A)$, is defined by expansion along the first row:

$$\det(A) = a_{11} \text{cof}(A)_{11} + a_{12} \text{cof}(A)_{12} + a_{13} \text{cof}(A)_{13}.$$

This process is called *cofactor expansion*.

Fact

For a 3×3 matrix, cofactor expansion along any row or any column gives the same determinant.

Activity

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

.

Task

Find the determinant of A by expanding along row 1.

Solution.

To calculate the determinant of A by expanding along row 1, we write

$$\det(A) = 1 \operatorname{cof}(A)_{11} + 2 \operatorname{cof}(A)_{12} + 3 \operatorname{cof}(A)_{13}$$

. We begin by calculating the cofactors:

$$\operatorname{cof}(A)_{11} = (-1)^{1+1} \det \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} = (1)(3 \cdot 1 - 2 \cdot 2) = -1$$

$$\operatorname{cof}(A)_{12} = (-1)^{1+2} \det \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix} = (-1)(4 \cdot 1 - 2 \cdot 3) = 2$$

$$\operatorname{cof}(A)_{13} = (-1)^{1+3} \det \begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix} = (1)(8 - 9) = -1$$

Therefore

$$\det(A) = 1(-1) + 2(2) + 3(-1) = -1 + 4 - 3 = 0$$

.

Task

Find the determinant of A by expanding along column 2.

Solution.

To calculate the determinant of A by expanding along column 2, we write

$$\det(A) = 2 \operatorname{cof}(A)_{12} + 3 \operatorname{cof}(A)_{22} + 2 \operatorname{cof}(A)_{32}$$

We calculate the cofactors:

$$\operatorname{cof}(A)_{12} = (-1)^{1+2} \det \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix} = (-1)(4 \cdot 1 - 2 \cdot 3) = 2$$

$$\operatorname{cof}(A)_{22} = (-1)^{2+2} \det \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} = (1)(1 \cdot 1 - 3 \cdot 3) = -8$$

$$\text{cof}(A)_{32} = (-1)^{3+2} \det \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix} = (-1)(2 - 12) = 10$$

. Therefore

$$\det(A) = 2(2) + 3(-8) + 2(10) = 4 - 24 + 20 = 0$$

.

Note

To track the sign $(-1)^{i+j}$ when computing cofactors, it may be helpful to view the matrix as a checkerboard as below:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

The sign in the i th row and j th position will be the sign of $(-1)^{i+j}$.

Note

Since any row or column gives the same determinant, choose a row or column with many zeros when possible.

Activity

Compute the determinant of the following matrices:

Task

$$A = \begin{bmatrix} 3 & -5 & 1 \\ 0 & 1 & 0 \\ -2 & -4 & 7 \end{bmatrix}$$

Solution.

The second row of A has two zeros, and so expanding along this row will likely result in the simplest computation of the determinant. We write

$$\begin{aligned} \det(A) &= 0 \cdot \text{cof}(A)_{21} + 1 \cdot \text{cof}(A)_{22} + 0 \cdot \text{cof}(A)_{23} \\ &= \text{cof}(A)_{22} \\ &= (-1)^{2+2} \det \begin{bmatrix} 3 & 1 \\ -2 & 7 \end{bmatrix} \\ &= (1)(3 \cdot 7 - 1 \cdot (-2)) \\ &= 21 + 2 = 23 \end{aligned}$$

.

Task

$$B = \begin{bmatrix} 0 & \pi & 200 \\ 0 & 0.03 & 4 \\ 0 & 19 & -3584 \end{bmatrix}$$

Solution.

If we expand the determinant of B along its first column, we find that

$$\det(B) = 0 \cdot \text{cof}(B)_{11} + 0 \cdot \text{cof}(B)_{21} + 0 \cdot \text{cof}(B)_{31} = 0$$

Note

Whenever a row or column of a matrix only has zeros as entries, expanding along that row shows that the determinant of the matrix is always zero, so no cofactor computations are required.

Now that 3×3 determinants have been defined, we can define larger determinants recursively. For a 4×4 matrix, each minor determinant is a 3×3 determinant. For a 5×5 matrix, each minor determinant is a 4×4 determinant, and so on.

Definition 2.5.10 (Determinant of a general square matrix).

Let $A = [a_{ij}]$ be an $n \times n$ matrix with $n \geq 4$. Assume determinants have already been defined for $(n-1) \times (n-1)$ matrices.

The (i, j) -th minor of A , denoted

$$\text{minor}(A)_{ij},$$

is the determinant of the $(n-1) \times (n-1)$ matrix obtained from A by deleting row i and column j . The (i, j) -cofactor of A is

$$\text{cof}(A)_{ij} = (-1)^{i+j} \text{minor}(A)_{ij}.$$

The determinant of A is defined by expansion along the first row:

$$\det(A) = \sum_{j=1}^n a_{1j} \text{cof}(A)_{1j}.$$

Fact

For an $n \times n$ matrix, cofactor expansion along any row or any column gives the same determinant:

$$\det(A) = \sum_{j=1}^n a_{ij} \text{cof}(A)_{ij}$$

for any fixed row i , and

$$\det(A) = \sum_{i=1}^n a_{ij} \operatorname{cof}(A)_{ij}$$

for any fixed column j .

Activity

Calculate the determinant of the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 4 & 2 & 3 \\ 1 & 3 & 4 & 5 \\ 3 & 4 & 3 & 2 \end{bmatrix}$$

Solution.

Expand along column 3:

$$\det(A) = 3 \operatorname{cof}(A)_{13} + 2 \operatorname{cof}(A)_{23} + 4 \operatorname{cof}(A)_{33} + 3 \operatorname{cof}(A)_{43}.$$

The signs in column 3 are

$$+, -, +, -.$$

So

$$\begin{aligned} \det(A) &= 3 \det \begin{bmatrix} 5 & 4 & 3 \\ 1 & 3 & 5 \\ 3 & 4 & 2 \end{bmatrix} \\ &\quad - 2 \det \begin{bmatrix} 1 & 2 & 4 \\ 1 & 3 & 5 \\ 3 & 4 & 2 \end{bmatrix} \\ &\quad + 4 \det \begin{bmatrix} 1 & 2 & 4 \\ 5 & 4 & 3 \\ 3 & 4 & 2 \end{bmatrix} \\ &\quad - 3 \det \begin{bmatrix} 1 & 2 & 4 \\ 5 & 4 & 3 \\ 1 & 3 & 5 \end{bmatrix}. \end{aligned}$$

Using the 3×3 determinant method,

$$\det \begin{bmatrix} 5 & 4 & 3 \\ 1 & 3 & 5 \\ 3 & 4 & 2 \end{bmatrix} = -33,$$

$$\det \begin{bmatrix} 1 & 2 & 4 \\ 1 & 3 & 5 \\ 3 & 4 & 2 \end{bmatrix} = -8,$$

$$\det \begin{bmatrix} 1 & 2 & 4 \\ 5 & 4 & 3 \\ 3 & 4 & 2 \end{bmatrix} = 26,$$

and

$$\det \begin{bmatrix} 1 & 2 & 4 \\ 5 & 4 & 3 \\ 1 & 3 & 5 \end{bmatrix} = 11.$$

Therefore

$$\begin{aligned} \det(A) &= 3(-33) - 2(-8) + 4(26) - 3(11) \\ &= -99 + 16 + 104 - 33 \\ &= -12. \end{aligned}$$

Activity

Compute $\det(A)$, where

$$A = \begin{bmatrix} 3 & 2 & 3 & 4 \\ 0 & 4 & 2 & 3 \\ 0 & 0 & 2 & 5 \\ 0 & 0 & 0 & -1 \end{bmatrix}.$$

Solution.

Expanding in cofactors along column 1 (because most of the entries of that column are zeros), we find

$$\begin{aligned} \det(A) &= 3 \cdot \text{cof}(A)_{11} + 0 \cdot \text{cof}(A)_{21} + 0 \cdot \text{cof}(A)_{31} + 0 \cdot \text{cof}(A)_{41} \\ &= 3(-1)^{1+1} \det \begin{bmatrix} 4 & 2 & 3 \\ 0 & 2 & 5 \\ 0 & 0 & -1 \end{bmatrix} = 3 \det \begin{bmatrix} 4 & 2 & 3 \\ 0 & 2 & 5 \\ 0 & 0 & -1 \end{bmatrix} \end{aligned}$$

Expanding along the first column again, we find that

$$\det \begin{bmatrix} 4 & 2 & 3 \\ 0 & 2 & 5 \\ 0 & 0 & -1 \end{bmatrix} = 4 \det \begin{bmatrix} 2 & 5 \\ 0 & -1 \end{bmatrix} \\ = 4((2)(-1) - (5)(0)) = -8$$

. So $\det(A) = (3)(-8) = -24$.

Recall 1.2.7.

Definition 2.5.11 (Triangular and Diagonal Matrices).

An $n \times n$ matrix $A = [a_{ij}]$ is called *upper triangular* if $a_{ij} = 0$ for $i > j$, and *lower triangular* if $a_{ij} = 0$ for $i < j$. A matrix is *diagonal* if it is both upper triangular and lower triangular, i.e., if $a_{ij} = 0$ for $i \neq j$. We denote the $n \times n$ diagonal matrix with entries $\lambda_1, \dots, \lambda_n$ on the diagonal by $\text{diag}(\lambda_1, \dots, \lambda_n)$.

Theorem 2.5.12.

If $A = [a_{ij}]$ is a upper or lower triangular matrix, then $\det(A) = a_{11} \cdots a_{nn}$. That is, the determinant of a triangular matrix is the product of entries in the *main diagonal*.

Proof.

A generalization of the argument used to calculate 2.5.2, i.e., expanding the determinant in cofactors along the first column iteratively.

Theorem 2.5.13.

If a row or column of a matrix A consists entirely of zeros, then $\det(A) = 0$.

Proof.

A generalization of the argument used to calculate the determinant in 2.5.2.

Theorem 2.5.14.

If a matrix B is obtained from a matrix A by swapping two rows or two columns, then $\det(B) = -\det(A)$.

Activity

Check directly that 2.5.14 is true when A and B are 2×2 matrices.

Proof.

If

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

and

$$B = \begin{bmatrix} c & d \\ a & b \end{bmatrix}$$

, then $\det(A) = ad - bc$ and $\det(B) = cb - ad = -(ad - bc) = -\det(A)$.

Theorem 2.5.15.

If two rows (or columns) of A are equal to one another, then $\det(A) = 0$.

Proof.

Let B be the matrix obtained by swapping the two identical rows / columns of A . Then $A = B$, and 2.5.14 implies that $\det(A) = -\det(B) = -\det(A)$. Therefore $2\det(A) = 0$, and thus that $\det(A) = 0$.

Theorem 2.5.16.

If B is obtained from A by multiplying each entry in a row or column of A by a quantity k , then $\det(B) = k\det(A)$.

Proof.

Suppose that B is obtained from A by multiplying each entry of row i by k . Consider a cofactor expansion along row i of B . Then

$$\begin{aligned} \det(B) &= ka_{i1}\text{cof}(B)_{i1} + ka_{i2}\text{cof}(B)_{i2} + \cdots + ka_{in}\text{cof}(B)_{in} \\ &= k[a_{i1}\text{cof}(A)_{i1} + \cdots + a_{in}\text{cof}(A)_{in}] \\ &= k\det(A) \end{aligned}$$

.

Fact

If A is a $n \times n$ matrix, then $\det(kA) = k^n \det(A)$.

Theorem 2.5.17.

If B is obtained by adding k times the i th row of A to the j th row, for $i \neq j$, which is a row operation which we denote by $R_j \rightarrow R_j + kR_i$, then $\det(B) = \det(A)$.

We can check 2.5.17 in the 2×2 case. If

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

and

$$B = \begin{bmatrix} a & b \\ c + ka & d + kb \end{bmatrix}$$

, then

$$\det(A) = ad - bc$$

and

$$\begin{aligned}\det(B) &= a(d + kb) - b(c + ka) \\ &= ad + kab - bc - kab \\ &= ad - bc = \det(A)\end{aligned}$$

Theorem 2.5.18 (How row operations affect the determinant).

Here is a summary of how the elementary row operations, when applied to a matrix, affect the determinant:

- $R_i \rightarrow kR_i$ multiplies the determinant by k .
- $R_i \leftrightarrow R_j$ multiplies the determinant by -1 .
- $R_j \rightarrow R_j + kR_i$ leaves the determinant unchanged.

Activity

Suppose a matrix B is obtained from a matrix A by applying the following row operations:

- $R_i \leftrightarrow R_j$
- $R_4 \rightarrow 4R_4$
- $R_3 \rightarrow R_3 + 3R_2$

If $\det(A) = 8$, what is the determinant of B .

Solution.

- $R_i \leftrightarrow R_j$ multiplies the determinant by -1 .
- $R_4 \rightarrow 4R_4$ multiplies the determinant by 4 .
- $R_3 \rightarrow R_3 + 3R_2$ does not change the determinant.

Therefore $\det(B) = (-1)(4)\det(A) = -32$.

Activity

Compute the determinant of the matrix

$$B = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 3 & 3 \\ 0 & 15 & 12 \end{bmatrix}$$

Solution.

We use the elementary row operations to make the matrix B triangular:

$$R_2 \rightarrow R_2 + R_1$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 5 & 4 \\ 0 & 15 & 12 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_2$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 5 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

None of the row operations we have performed change the determinant of the matrix, and the final matrix has determinant zero, so $\det(B) = 0$.

Theorem 2.5.19.

If A and B are $n \times n$ matrices, then $\det(AB) = \det(A)\det(B)$.

Activity

Let

$$A = \begin{bmatrix} 3 & 1 \\ -2 & 7 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix}$$

Task

Compute the determinant of A , B , and C .

Solution.

We use the formula for 2×2 matrices: First, that

$$\det(A) = (3)(7) - (1)(-2) = 23$$

, then that

$$\det(B) = (1)(4) - (2)(2) = 0$$

, and that

$$\det(C) = (1)(2) - (4)(0) = 2$$

.

Task

Compute the determinant of AC and BC .

Solution.

Using 2.5.19, we calculate that

$$\det(AC) = \det(A) \det(C) = (23)(2) = 46$$

, and that

$$\det(BC) = \det(B) \det(C) = (0)(2) = 0$$

. Alternatively, we can calculate the products of the matrices, i.e., that

$$AC = \begin{bmatrix} 3 & 14 \\ -2 & 6 \end{bmatrix}$$

and that

$$BC = \begin{bmatrix} 1 & 8 \\ 2 & 16 \end{bmatrix}$$

, and calculate the determinants directly.

Theorem 2.5.20.

If A is a $n \times n$ matrix, then A is *nonsingular* if and only if $\det(A) \neq 0$. In this case $\det(A^{-1}) = 1/\det(A)$.

Proof.

We prove only one direction of the theorem, that if A is nonsingular then $\det(A) \neq 0$ and $\det(A^{-1}) = 1/\det(A)$. Suppose that A is non-singular. Then $AA^{-1} = I$. Taking determinants of both sides, and using that I is diagonal, so that $\det(I) = 1$, we see that

$$1 = \det(I) = \det(AA^{-1}) = \det(A) \det(A^{-1})$$

. Thus $\det(A) \neq 0$ (since otherwise the right hand side of this equation would be zero), and rearranging gives $\det(A^{-1}) = 1/\det(A)$.

Theorem 2.5.21.

If A is a square matrix, then $\det(A) = \det(A^T)$.

Definition 2.5.22 (Orthogonal Matrices).

A square matrix A is *orthogonal* if $A^T = A^{-1}$.

Activity

If A is an orthogonal matrix, what are the possible values of $\det(A)$?

Solution.

If A is orthogonal, then applying the determinant to both sides of the equation $I = AA^T$ gives

$$1 = \det(I) = \det(AA^T) = \det(A)\det(A^T) = \det(A)^2$$

. Taking square roots gives that $\det(A) = \pm 1$.

2.6 Coding recap

2.6.1 Linked notebook

Run Lab U2: Understanding a Linear Map. The main sequence practices projection, reachable outputs, row reduction, plane intersections, difference matrices, rank, null vectors, redundant features, determinants, and inverse checks.

For a quick reference on arrays, matrix products, numerical rank, determinants, solves, row reduction, and null spaces, see the programming appendix sections A.2, A.4, A.10, and A.11.

2.6.2 Review activities

Activity: A sensor that drops height

A sensor records only horizontal position. It keeps the first two coordinates and drops height.

```
import numpy as np

P = np.array([
    [1, 0, 0],
    [0, 1, 0],
])

x = np.array([2, 1, 5])
z = np.array([0, 0, 1])

P @ x, P @ (x + 6*z), P @ z
```

Output:

```
(array([2, 1]), array([2, 1]), array([0, 0]))
```

1. What does P do to an input vector?
2. Why do $P\mathbf{x}$ and $P(\mathbf{x} + 6\mathbf{z})$ agree?
3. What does $P\mathbf{z} = \mathbf{0}$ say about the direction $\mathbf{z}$?
4. Why can the original input not be recovered uniquely from the output?

Solution.

The matrix P keeps the first two coordinates and forgets the third.

The vectors

$$\mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix} \quad \text{and} \quad \mathbf{x} + 6\mathbf{z} = \begin{bmatrix} 2 \\ 1 \\ 11 \end{bmatrix}$$

have different heights, but the same first two coordinates. Therefore their outputs agree. Since

$$P\mathbf{z} = \mathbf{0},$$

the direction

$$\mathbf{z} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

is a forgotten direction. Inputs that differ by a multiple of $\mathbf{z}$ have the same output.

Activity: Reading a null-space basis in code

The following code asks SymPy for a basis of a null space.

```
import sympy as sp

A = sp.Matrix([
    [1, 2, 3],
    [2, 4, 6],
    [0, 1, 1],
])
```

```
A.nullspace()
```

Output:

```
[Matrix([
[-1],
[-1],
[ 1]])]
```

1. What kind of object does `nullspace()` return?
2. Is the displayed vector an input direction or an output direction?
3. Verify from the code output that $\mathbf{z} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$ is in $\text{null}(A)$.
4. If $A\mathbf{x} = \mathbf{y}$, what is $A(\mathbf{x} + t\mathbf{z})$?

Solution.

The command returns a list of basis vectors for $\text{null}(A)$.
The vector

$$\mathbf{z} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

is an input direction. It satisfies

$$A\mathbf{z} = \mathbf{0}.$$

If $A\mathbf{x} = \mathbf{y}$, then

$$A(\mathbf{x} + t\mathbf{z}) = A\mathbf{x} + tA\mathbf{z} = \mathbf{y} + t\mathbf{0} = \mathbf{y}.$$

Thus every input of the form $\mathbf{x} + t\mathbf{z}$ gives the same output.

Activity: Reading an augmented rref output

The code row-reduces the augmented matrix for trying to solve

$$Ax = b, \quad A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

```
import sympy as sp

# Augmented matrix [A | b]
M = sp.Matrix([
    [1, 0, 1],
    [0, 1, 2],
    [1, 1, 4],
])

M.rref()
```

Output:

```
(Matrix([
[1, 0, 0],
[0, 1, 0],
[0, 0, 1]]), (0, 1, 2))
```

Read the displayed matrix as

$$\left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right].$$

1. Which column is to the right of the vertical line?
2. What does the last row represent?
3. Is b reachable as an output of A ?
4. What does the pivot to the right of the vertical line mean?

Solution.

The third column is to the right of the vertical line. It is the right-hand side column, not a variable column.

The last row is

$$[0 \ 0 \ | \ 1],$$

which represents the equation

$$0 = 1.$$

That equation is impossible, so the system is inconsistent. Therefore $\mathbf{b}$ is not reachable as an output of A .

A pivot to the right of the vertical line means the system has no solution.

Activity: Redundant square-footage features in code

Rows represent houses. The columns are first-level area, second-level area, and total area, measured in hundreds of square feet.

```
import numpy as np

X = np.array([
    [9, 7, 16],
    [11, 9, 20],
    [14, 0, 14],
    [8, 8, 16],
])

z = np.array([1, 1, -1])
c = np.array([3, 1, 0])
c_alt = np.array([2, 0, 1])

rank = int(np.linalg.matrix_rank(X))

X @ z, rank, X @ c, X @ c_alt
```

Output:

```
(array([0, 0, 0, 0]), 2, array([34, 42, 42, 32]), array([34, 42, 42, 32]))
```

1. What does $X\mathbf{z} = \mathbf{0}$ say about the three feature columns?
2. What does $\text{rank}(X) = 2$ say about the three features?
3. Why do $X\mathbf{c}$ and $X\mathbf{c}_{\text{alt}}$ agree?
4. What warning does this give about interpreting individual coefficients?

Solution.

The equation

$$X \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = 0$$

means that

$$\text{first-level area} + \text{second-level area} - \text{total area} = 0$$

for every row. Equivalently,

$$\text{total area} = \text{first-level area} + \text{second-level area}.$$

The three columns contain only two independent feature directions, so $\text{rank}(X) = 2$.

The coefficient vectors differ by the null-space direction:

$$\mathbf{c} - \mathbf{c}_{\text{alt}} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \mathbf{z}.$$

Since $X\mathbf{z} = \mathbf{0}$,

$$X\mathbf{c} = X(\mathbf{c}_{\text{alt}} + \mathbf{z}) = X\mathbf{c}_{\text{alt}} + X\mathbf{z} = X\mathbf{c}_{\text{alt}}.$$

The predictions agree, but the coefficient stories are different. With redundant features, individual coefficient importance may not be uniquely determined.

Activity: Debugging a column-space basis from code

A student computes an rref and pivot columns.

```
import sympy as sp

A = sp.Matrix([
    [1, 2, 3],
    [2, 4, 6],
    [0, 1, 1],
])

R, pivots = A.rref()
R, pivots
```

Output:

```
(Matrix([
```

```
[1, 0, 1],  
[0, 1, 1],  
[0, 0, 0]]), (0, 1))
```

The student says:

“Columns 1 and 2 of R are a basis for $\text{col}(A)$.”

1. What do the pivot indices $(0, 1)$ mean in mathematical column numbering?
2. What is wrong with using columns of R as a basis for $\text{col}(A)$?
3. Which columns should be used instead?
4. Write a basis for $\text{col}(A)$.

Solution.

The pivot indices $(0, 1)$ mean mathematical columns 1 and 2.

The mistake is that row operations do not preserve the original columns of A . Row reduction identifies pivot positions, but a basis for $\text{col}(A)$ must use columns of the original matrix A . Therefore we use columns 1 and 2 of A :

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix} \right\}.$$

These vectors form a basis for $\text{col}(A)$.

Activity: Determinant and inverse diagnostics in code

The following code checks determinant and rank for two square matrices.

```
import numpy as np  
  
A = np.array([  
    [1., 1.],  
    [0., 1.],  
])  
  
B = np.array([  
    [1., 0.],  
    [0., 0.],  
])
```

```

detA = float(np.linalg.det(A))
detB = float(np.linalg.det(B))

rankA = int(np.linalg.matrix_rank(A))
rankB = int(np.linalg.matrix_rank(B))

detA, rankA, detB, rankB

```

Output:

```
(1.0, 2, 0.0, 1)
```

1. Which matrix is invertible?
2. Which matrix forgets a nonzero input direction?
3. What do determinant and rank each say?
4. Should $Bx = b$ have a unique solution for every $b \in \mathbb{R}^2$?

Solution.

Matrix A is invertible because

$$\det(A) = 1 \neq 0$$

and

$$\text{rank}(A) = 2.$$

Matrix B is singular because

$$\det(B) = 0$$

and

$$\text{rank}(B) = 1 < 2.$$

The matrix B forgets the nonzero direction

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

since

$$B \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The system $B\mathbf{x} = \mathbf{b}$ should not be expected to have a unique solution for every $\mathbf{b}$. The map is not invertible.

2.7 Exercises

Learning outcomes. The labels below identify the learning outcomes for each exercise group. Individual problems may also involve earlier outcomes.

2.7.1 Systems, geometry, and row reduction

Learning outcomes. U2-LO1, U2-LO2.

- Nicholson 1.1.19
- Nicholson 1.2.1(b,d)
- Nicholson 1.2.3(b)
- Nicholson 1.2.5(b,d)
- Nicholson 1.2.8(b)
- Nicholson 1.2.10(b,d)
- Nicholson 1.2.18
- Nicholson 1.3.3(b)
- Nicholson 1.3.5(b)
- Nicholson 1.3.7(b,d)

2.7.2 Subspaces

Learning outcomes. U2-LO3, U2-LO4.

- Nicholson 5.1.1(d,f)
- Nicholson 5.1.2(b,d)

2.7.3 Independence, bases, and rank

Learning outcomes. U2-LO4.

- Nicholson 5.2.3(b)
- Nicholson 5.2.5(b)
- Nicholson 5.2.6(f)
- Nicholson 5.2.16(b)

2.7.4 Rank and nullity

Learning outcomes. U2-LO3, U2-LO4, U2-LO6.

- Nicholson 5.4.1(b)
- Nicholson 5.4.2(b)
- Nicholson 5.4.7(b)

2.7.5 Inverses and determinants

Learning outcomes. U2-LO5, U2-LO7.

- Nicholson 2.4.2(b)
- Nicholson 2.4.3(b)
- Nicholson 2.4.5(h)
- Nicholson 3.2.3(b)
- Nicholson 3.2.4(b)
- Nicholson 3.2.6(b)
- Nicholson 3.1.1(h)
- Nicholson 3.1.5(b)
- Nicholson 3.1.7(b)
- Nicholson 3.1.16(b)

2.7.6 Additional exercises

These exercises connect the main Unit 2 ideas through short applied computations: reachable outputs, forgotten directions, row reduction, subspaces, rank, nullity, redundant features, inverses, determinants, and code interpretation. Solutions are collected in Appendix A.2.

Additional exercise: Projection and forgotten height in code

A sensor records only horizontal position. It keeps the first two coordinates and drops height.

```
import numpy as np

P = np.array([
    [1, 0, 0],
    [0, 1, 0],
])

x = np.array([2, 1, 5])
z = np.array([0, 0, 1])

rank = int(np.linalg.matrix_rank(P))
nullity = P.shape[1] - rank

P @ x, P @ (x + 6*z), P @ z, rank, nullity
```

Output:

```
(array([2, 1]), array([2, 1]), array([0, 0]), 2, 1)
```

1. What does P do to an input vector?
2. Why do Px and $P(x + 6z)$ agree?
3. What does $Pz = \mathbf{0}$ say about the direction z ?
4. Interpret the rank and nullity.
5. Why can the original input not be recovered uniquely from the output?

Learning outcomes. U2-LO3, U2-LO6.

Additional exercise: Reachable outputs from augmented rref

Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{b}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

The following code row-reduces the augmented matrices for

$$Ax = b_1 \quad \text{and} \quad Ax = b_2.$$

```
import sympy as sp

M1 = sp.Matrix([
    [1, 0, 1],
    [0, 1, 2],
    [1, 1, 3],
])

M2 = sp.Matrix([
    [1, 0, 1],
    [0, 1, 2],
    [1, 1, 4],
])

M1.rref(), M2.rref()
```

Output:

```
((Matrix([
    [1, 0, 1],
    [0, 1, 2],
    [0, 0, 0]]), (0, 1)),
 (Matrix([
    [1, 0, 0],
    [0, 1, 0],
    [0, 0, 1]]), (0, 1, 2)))
```

Read the two rref matrices as

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right], \quad \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array} \right].$$

1. Which target vector is reachable?
2. Which target vector is not reachable?
3. What row shows inconsistency?
4. What does a pivot to the right of the vertical line mean?
5. For the reachable target, give one input x .

Learning outcomes. U2-LO1, U2-LO2, U2-LO3.

Additional exercise: Line of intersection from an rref output

Consider the two planes

$$x + y + z = 1, \quad -x + 2y - 3z = -1.$$

The following code row-reduces the augmented matrix for the system.

```
import sympy as sp

M = sp.Matrix([
    [1, 1, 1, 1],
    [-1, 2, -3, -1],
])

M.rref()
```

Output:

```
(Matrix([
    [1, 0, 5/3, 1],
    [0, 1, -2/3, 0]]), (0, 1))
```

Read the rref matrix as

$$\left[\begin{array}{ccc|c} 1 & 0 & 5/3 & 1 \\ 0 & 1 & -2/3 & 0 \end{array} \right].$$

1. Which variable is free?
2. Write x and y in terms of z .
3. Let $z = t$. Write the solution set in parametric vector form.
4. Why is the solution set a line?
5. How does this connect to the intersection of two planes?

Learning outcomes. U2-LO1, U2-LO2.

Additional exercise: Plane through three points from a null-space computation

Let

$$P = (1, 1, -2), \quad Q = (0, 2, 1), \quad R = (-1, -1, 0).$$

Two displacement vectors in the plane are

$$\overrightarrow{PQ} = \begin{bmatrix} -1 \\ 1 \\ 3 \end{bmatrix}, \quad \overrightarrow{PR} = \begin{bmatrix} -2 \\ -2 \\ 2 \end{bmatrix}.$$

A normal vector $\mathbf{n}$ must be perpendicular to both displacement vectors. The following code solves for such directions.

```
import sympy as sp

M = sp.Matrix([
    [-1, 1, 3],
    [-2, -2, 2],
])

M.nullspace()
```

Output:

```
[Matrix([
[ 2],
[-1],
[ 1]])]
```

1. What do the rows of M represent?
2. Why does a vector in $\text{null}(M)$ give a normal vector to the plane?
3. Use the output to choose a normal vector $\mathbf{n}$.
4. Write an equation of the plane.

Learning outcomes. U2-LO1, U2-LO2, U2-LO3.

Additional exercise: Reading a null-space basis in code

The following code asks SymPy for a basis of a null space.

```
import sympy as sp

A = sp.Matrix([
    [1, 2, 3],
    [2, 4, 6],
])
```

```
[0, 1, 1],
])

A.nullspace()
```

Output:

```
[Matrix([
[-1],
[-1],
[ 1]])]
```

Let

$$\mathbf{z} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}.$$

1. Is $\mathbf{z}$ an input direction or an output direction?
2. Verify that $A\mathbf{z} = \mathbf{0}$.
3. If $A\mathbf{x} = \mathbf{y}$, compute $A(\mathbf{x} + t\mathbf{z})$.
4. What does this say about uniqueness of inputs?

Learning outcomes. U2-LO3, U2-LO6.

Additional exercise: Redundant square-footage features in code

Rows represent houses. The columns are first-level area, second-level area, and total area, measured in hundreds of square feet.

```
import numpy as np

X = np.array([
    [9, 7, 16],
    [11, 9, 20],
    [14, 0, 14],
    [8, 8, 16],
])

z = np.array([1, 1, -1])
c = np.array([3, 1, 0])
```

```
c_alt = np.array([2, 0, 1])

rank = int(np.linalg.matrix_rank(X))

X @ z, rank, X @ c, X @ c_alt
```

Output:

```
(array([0, 0, 0, 0]), 2, array([34, 42, 42, 32]), array([34, 42, 42, 32]))
```

1. What feature relation does $Xz = 0$ show?
2. What does $\text{rank}(X) = 2$ say about the three feature columns?
3. Why do Xc and Xc_{alt} agree?
4. What warning does this give about interpreting individual coefficients?
5. In this example, why is it not clear whether first-level area, second-level area, or total area is the “most important” feature?

Learning outcomes. U2-LO3, U2-LO6.

Additional exercise: Difference matrix: levels versus changes

The following matrix computes consecutive differences in a short time series.

```
import numpy as np

D = np.array([
    [-1, 1, 0, 0],
    [0, -1, 1, 0],
    [0, 0, -1, 1],
])

x = np.array([2, 5, 9, 10])
ones = np.ones(4, dtype=int)

D @ x, D @ (x + 10*ones), D @ ones, int(np.linalg.matrix_rank(D))
```

Output:

```
(array([3, 4, 1]), array([3, 4, 1]), array([0, 0, 0]), 3)
```

1. What does $D\mathbf{x}$ measure?
2. Why do $D\mathbf{x}$ and $D(\mathbf{x} + 10\mathbf{1})$ agree?
3. What input direction does D forget?
4. What does the rank say about the three output differences?
5. What is the nullity of D ?

Learning outcomes. U2-LO3, U2-LO6.

Additional exercise: Debugging a column-space basis from code

A student computes an rref and pivot columns.

```
import sympy as sp

A = sp.Matrix([
    [1, 2, 3],
    [2, 4, 6],
    [0, 1, 1],
])

R, pivots = A.rref()
R, pivots
```

Output:

```
(Matrix([
    [1, 0, 1],
    [0, 1, 1],
    [0, 0, 0]]), (0, 1))
```

The student says:

“Columns 1 and 2 of R are a basis for $\text{col}(A)$.”

1. What do the pivot indices $(0, 1)$ mean in mathematical column numbering?
2. What is wrong with using columns of R as a basis for $\text{col}(A)$?
3. Which matrix should supply the basis vectors?
4. Write a basis for $\text{col}(A)$.

Learning outcomes. U2-LO3, U2-LO4.

Additional exercise: Invertibility gallery in code

The following code checks determinant and rank for five 2×2 matrices from the Unit 1 transformation gallery.

```
import numpy as np

mats = [
    np.array([[0., 0.], [0., 1.]]),
    np.array([[-1., 0.], [0., 1.]]),
    np.array([[1., 1.], [0., 1.]]),
    np.array([[0., -1.], [1., 0.]]),
    np.array([[1., 0.], [0., 0.]]),
]

[(float(np.linalg.det(A)), int(np.linalg.matrix_rank(A))) for A in mats]
```

Output:

```
[(0.0, 1), (-1.0, 2), (1.0, 2), (1.0, 2), (0.0, 1)]
```

1. Which matrices are invertible?
2. Which matrices forget a nonzero input direction?
3. How does determinant detect singularity here?
4. How does rank detect singularity here?
5. For each singular matrix, name one direction that is forgotten.

Learning outcomes. U2-LO5, U2-LO7.

Additional exercise: Reading an inverse-computation output

The following code row-reduces $[A \mid I_3]$.

```
import sympy as sp

A = sp.Matrix([
    [1, 1, 4],
    [2, 3, 2],
    [0, 0, 1],
])
```

```
M = A.row_join(sp.eye(3))
M.rref()
```

Output:

```
(Matrix([
[1, 0, 0, 3, -1, -10],
[0, 1, 0, -2, 1, 6],
[0, 0, 1, 0, 0, 1]]), (0, 1, 2))
```

Read the rref matrix as

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -1 & -10 \\ 0 & 1 & 0 & -2 & 1 & 6 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right].$$

1. What does the left side of the vertical line show?
2. Is A invertible?
3. Read off A^{-1} .
4. Why does row-reducing $[A \mid I_3]$ give the inverse?

Learning outcomes. U2-LO5, U2-LO7.

2.8 Unit 2 highlights

2.8.1 Mathematical quick reference

Systems and row reduction

Matrix equation. For $A \in \mathbb{R}^{m \times n}$, solving $Ax = b$ means finding $x \in \mathbb{R}^n$ for a specified $b \in \mathbb{R}^m$. Use the augmented matrix $[A \mid b]$.

Elementary row operations. Interchange rows, multiply a row by a nonzero scalar, or add a multiple of one row to another. These preserve the solution set.

Echelon forms. In REF, zero rows are last, each successive pivot is farther right, and entries below pivots are zero. In RREF, each pivot is 1 and is the only nonzero entry in its column.

Classify and solve. A row $[0 \cdots 0 \mid d]$ with $d \neq 0$ means no solution. Otherwise, pivot variables are determined by the free variables: no free variables gives one solution; at least one free variable gives infinitely many solutions. Back-substitute from REF or read the equations from RREF.

General solution. If x_p is one solution, then

$$\{\mathbf{x} : A\mathbf{x} = \mathbf{b}\} = \mathbf{x}_p + \text{null}(A).$$

For a homogeneous system, set one free variable to 1 and the others to 0 in turn to obtain a basis of basic solutions.

Lines, planes, and subspaces

Line. Through $\mathbf{p}$ with direction $\mathbf{v} \neq \mathbf{0}$:

$$\mathbf{x} = \mathbf{p} + t\mathbf{v}, \quad t \in \mathbb{R}.$$

Plane in three dimensions. Through $\mathbf{p}$ with normal $\mathbf{n} \neq \mathbf{0}$: $\mathbf{n} \cdot (\mathbf{x} - \mathbf{p}) = 0$. Equivalently, $\mathbf{x} = \mathbf{p} + s\mathbf{u} + t\mathbf{v}$ for two independent directions in the plane.

Subspace test. A subset of $\mathbb{R}^n$ is a subspace if it contains $\mathbf{0}$ and is closed under addition and scalar multiplication. Homogeneous solution sets are subspaces.

Span. $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is the set of all linear combinations of the listed vectors. To test membership, solve $[\mathbf{v}_1 \ \dots \ \mathbf{v}_k]\mathbf{c} = \mathbf{b}$.

Independence, bases, and coordinates

Linear independence. The list $\mathbf{v}_1, \dots, \mathbf{v}_k$ is independent exactly when

$$c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k = \mathbf{0} \implies c_1 = \dots = c_k = 0.$$

Test this by row-reducing the matrix with these vectors as columns.

Basis and dimension. A basis is an independent spanning list. Every basis of a finite-dimensional subspace has the same number of vectors, its dimension. The empty list is a basis of $\{\mathbf{0}\}$, which has dimension 0.

Coordinates. For the ordered basis $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_k)$,

$$\mathbf{x} = c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k, \quad [\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_k \end{bmatrix}.$$

Solve for the unique coefficient vector; changing the order of the basis changes the coordinates.

Fundamental spaces, rank, and nullity

Column space and null space.

$$\text{im}(A) = \text{col}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\}, \quad \text{null}(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}.$$

The first is a subspace of $\mathbb{R}^m$; the second is a subspace of $\mathbb{R}^n$.

Basis calculations. Locate pivot columns using RREF, then take those columns from the original A for a column-space basis. The nonzero rows of an echelon form give a row-space basis. Basic solutions of $A\mathbf{x} = \mathbf{0}$ give a null-space basis.

Rank-nullity.

$$\text{rank}(A) = \dim \text{col}(A) = \text{number of pivots}, \quad \text{nullity}(A) = n - \text{rank}(A).$$

Existence and uniqueness. $A\mathbf{x} = \mathbf{b}$ is consistent exactly when $\mathbf{b} \in \text{col}(A)$. For a consistent system, the solution is unique exactly when $\text{null}(A) = \{\mathbf{0}\}$.

Inverses and determinants

Invertibility. For a square $n \times n$ matrix A , the following are equivalent: A is invertible; $\text{rref}(A) = I_n$; $\text{rank}(A) = n$; $\text{null}(A) = \{\mathbf{0}\}$; its columns form a basis of $\mathbb{R}^n$; and $\det(A) \neq 0$.

Compute an inverse. Row-reduce $[A \mid I_n]$ to $[I_n \mid A^{-1}]$. Then $A\mathbf{x} = \mathbf{b}$ has solution $\mathbf{x} = A^{-1}\mathbf{b}$.

Two-by-two formula. When $ad - bc \neq 0$,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Cofactor expansion. If M_{ij} deletes row i and column j , put $C_{ij} = (-1)^{i+j} \det(M_{ij})$. Expand along any row or column:

$$\det(A) = \sum_j a_{ij} C_{ij} = \sum_i a_{ij} C_{ij},$$

with the row index fixed in the first sum and the column index fixed in the second.

Determinant rules. Swapping rows changes the sign; scaling a row by c scales the determinant by c ; adding a multiple of another row leaves it unchanged. For a triangular matrix, multiply the diagonal entries. For square matrices,

$$\det(AB) = \det(A) \det(B), \quad \det(A^T) = \det(A).$$

If A, B are invertible, $(AB)^{-1} = B^{-1}A^{-1}$ and $\det(A^{-1}) = 1/\det(A)$.

2.8.2 Common mistakes

Counting the augmented column as a variable column; missing an inconsistent row; omitting free parameters; using columns of RREF instead of the original matrix for a column-space basis; confusing row space and column space; assuming a spanning list is a basis; calling a translated affine set a subspace without checking the origin; or applying square-matrix inverse and determinant tests to a rectangular matrix.

2.8.3 Connections

Unit 1 supplies matrix operations and linear maps. Unit 3 reuses normal vectors for tangent planes and null spaces for local linear maps. Unit 4 uses column spaces to formulate projection and least squares. Unit 6 recovers rank and the fundamental subspaces from singular vectors.

Chapter 3

Beyond linear maps: local linearization and first-order optimization

Local linearization and first-order optimization

Unit 1 and Unit 2 focused on linear maps $\mathbf{x} \mapsto A\mathbf{x}$. Most functions are not linear. This unit studies general scalar-valued and vector-valued functions, then uses derivatives and Jacobian matrices to recover linear maps that describe nonlinear functions near one point. For scalar-valued functions, the same local model also identifies directional change, critical points, and descent steps.

$$F(\mathbf{a} + \mathbf{h}) \approx F(\mathbf{a}) + J_F(\mathbf{a})\mathbf{h}.$$

Big questions. If a function is not linear, what linear map best describes it near one point? How can that local model guide a step that decreases a scalar-valued function?

Learning outcomes. By the end of this unit, students should be able to:

- *U3-LO1.* Classify scalar-valued and vector-valued functions, write component functions, and describe domains, ranges, graphs, and level curves.
- *U3-LO2.* Compute limits, continuity checks, derivatives of vector-valued functions, and partial derivatives of multivariable functions as needed.
- *U3-LO3.* Compute and interpret partial derivatives, gradients, Hessians, and Jacobian matrices.
- *U3-LO4.* Construct tangent planes and local linear approximations for scalar-valued functions.
- *U3-LO5.* Use a Jacobian matrix as a local linear map for vector-valued functions: $F(\mathbf{a} + \mathbf{h}) \approx F(\mathbf{a}) + J_F(\mathbf{a})\mathbf{h}$.
- *U3-LO6.* Apply the chain rule in component form and as matrix multiplication of Jacobian matrices.
- *U3-LO7.* Interpret simple composed nonlinear systems, such as a tiny neural-network forward pass, by identifying affine pieces, nonlinear pieces, and the local linear approximation.
- *U3-LO8.* Compute and interpret directional derivatives, and use the gradient to identify directions of greatest increase and decrease.

- *U3-LO9.* Identify interior critical points as candidates for local extrema, take gradient-descent steps, and diagnose the effect of the learning rate from iterates, code, or loss tables.

Toolbox skills. Write component functions, compute partial derivatives and higher partial derivatives, compute gradients and Hessians, compute Jacobian matrices, write tangent plane equations, evaluate local linear approximations, check Jacobian matrix shapes, multiply Jacobian matrices in the chain rule, compare actual changes with linear predictions, find critical points, write critical-point equations, take gradient-descent steps, read a gradient-descent update in code, and compare learning rates from a loss table.

3.1 General functions and domains

3.1.1 Function types and component functions

Unit 1 and Unit 2 studied maps of the form $\mathbf{x} \mapsto A\mathbf{x}$. We now allow more general rules. The output may be a scalar, a vector, or several scalar component functions collected together.

Definition 3.1.1.

A *vector-valued map* is usually written $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$. A *scalar-valued function* is usually written $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Any vector-valued map $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ has m *component functions*, or *coordinate functions*, the scalar-valued functions $F_1, \dots, F_m : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$F(\mathbf{x}) = \begin{bmatrix} F_1(\mathbf{x}) \\ \vdots \\ F_m(\mathbf{x}) \end{bmatrix}$$

. More generally, vector-valued and scalar-valued functions may only be defined on a subset of vectors in $\mathbb{R}^n$, rather than defined everywhere, so their *domain* may be a proper subset of $\mathbb{R}^n$.

In other words, a scalar-valued function takes in an input and outputs *scalars*, and a vector-valued function takes in an input and outputs *vectors* (recall 1.1.1).

Activity

If $\mathbf{x}$ is a point in $\mathbb{R}^n$, recall the length of $\mathbf{x}$, denoted $\|\mathbf{x}\|$, from 1.1.2. The length of a vector defines a scalar-valued function from $\mathbb{R}^n$ to $\mathbb{R}$. Find two different vectors with the same length. What does this say about whether the output determines the input?

Solution.

There are many different choices one could make here. For instance, the vector $(3, 4)$ and the vector $(5, 0)$ both have the same length. Other examples are the vectors $(1, 2)$ and $(-1, 2)$.

Activity

Let A be an $m \times n$ matrix. The linear map (recall 1.3.7) $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by $T(\mathbf{x}) = T_A(\mathbf{x}) = A\mathbf{x}$ is a vector-valued map. Each component function is a row dot product.

Task

If $\mathbf{a}_1, \dots, \mathbf{a}_m$ are the rows of A , and $T_1, \dots, T_m$ are the component functions of T , then for each j , express the function T_j in terms of the row $\mathbf{a}_j$.

Solution.

Recalling the definition of matrix vector multiplication from 1.3.1, we can write

$$T_j(\mathbf{x}) = a_{j1}x_1 + \dots + a_{jn}x_n = \mathbf{a}_j \cdot \mathbf{x}$$

Task

Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{bmatrix}$$

, which defines a linear map $T_A : \mathbb{R}^3 \rightarrow \mathbb{R}^2$. Compute its component functions for a general input $\mathbf{x}$, and its output for the specific input $\mathbf{x} = (3, -1, 2)$.

Solution.

The component functions are:

$$T_1(\mathbf{x}) = (1, 2, -1) \cdot \mathbf{x} = x_1 + 2x_2 - x_3$$

and

$$T_2(\mathbf{x}) = (0, 3, 4) \cdot \mathbf{x} = 3x_2 + 4x_3$$

For an input $\mathbf{x} = (3, -1, 2)$, we compute that

$$T(\mathbf{x}) = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 5 \end{bmatrix}$$

Activity: Corners are not enough

Unit 1 used the unit-square visualization for 2×2 matrices. For nonlinear maps, the corners of a square do not tell the whole story.

Let $T, F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y \\ y \end{bmatrix}, \quad F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y^2 \\ y \end{bmatrix}.$$

1. Compute the images of the four corners of the unit square under T .

2. Compute the images of the four corners of the unit square under F .
3. Compute the images of $\begin{bmatrix} 0 \\ 1/2 \end{bmatrix}$ under both maps.
4. What does F do to a point $\begin{bmatrix} 0 \\ y \end{bmatrix}$ on the left edge of the unit square?

Solution.

The four corners are

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

For T ,

$$\begin{aligned} T\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right) &= \begin{bmatrix} 0 \\ 0 \end{bmatrix}, & T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) &= \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \\ T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) &= \begin{bmatrix} 2 \\ 1 \end{bmatrix}, & T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) &= \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \end{aligned}$$

For F , the four corner images are the same:

$$\begin{aligned} F\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right) &= \begin{bmatrix} 0 \\ 0 \end{bmatrix}, & F\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) &= \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \\ F\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) &= \begin{bmatrix} 2 \\ 1 \end{bmatrix}, & F\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) &= \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \end{aligned}$$

For the midpoint,

$$T\left(\begin{bmatrix} 0 \\ 1/2 \end{bmatrix}\right) = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}, \quad F\left(\begin{bmatrix} 0 \\ 1/2 \end{bmatrix}\right) = \begin{bmatrix} 1/4 \\ 1/2 \end{bmatrix}.$$

On the left edge,

$$F\left(\begin{bmatrix} 0 \\ y \end{bmatrix}\right) = \begin{bmatrix} y^2 \\ y \end{bmatrix}.$$

So F bends the left edge into a curve. A grid gives more information than the corners alone.

Note

A matrix sends grid lines to grid lines. A nonlinear map can bend grid lines. The square-grid visualization samples points and grid lines, not only the corners.

Activity: Reading a square-grid computation

The following code applies the nonlinear map

$$F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y^2 \\ y \end{bmatrix}$$

to several input points stored as columns.

```
import numpy as np

def F(P):
    x = P[0, :]
    y = P[1, :]
    return np.vstack([x + y**2, y])

P = np.array([
    [0.0, 1.0, 1.0, 0.0, 0.0],
    [0.0, 0.0, 1.0, 1.0, 0.5],
])

F(P)
```

Output:

```
array([[0. , 1. , 2. , 1. , 0.25],
       [0. , 0. , 1. , 1. , 0.5 ]])
```

1. What points are stored in the first four columns of P ?
2. What point is stored in the last column of P ?
3. What output corresponds to the last column?
4. Why do the first four columns not detect the difference between F and the shear $T(x, y) = (x + y, y)$?
5. Why is a grid more informative than the four corners?

Solution.

The first four columns are the corners

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The last column is

$$\begin{bmatrix} 0 \\ 1/2 \end{bmatrix}.$$

Its output is

$$F\left(\begin{bmatrix} 0 \\ 1/2 \end{bmatrix}\right) = \begin{bmatrix} 1/4 \\ 1/2 \end{bmatrix}.$$

The four corners have the same images under F and under $T(x, y) = (x + y, y)$. The midpoint does not. A grid includes edge and interior points, so it can show bending that the four corners miss.

Note: Reading square-grid diagrams

A square-grid diagram keeps track of input points and output points at the same time. The color identifies the original input point. The grid lines show how nearby input points move.

For a matrix map, grid lines stay straight and the whole grid is controlled by one matrix. For a smooth nonlinear map, grid lines may bend.

The purpose of the diagram is not to replace computation. It shows what the computation is doing.

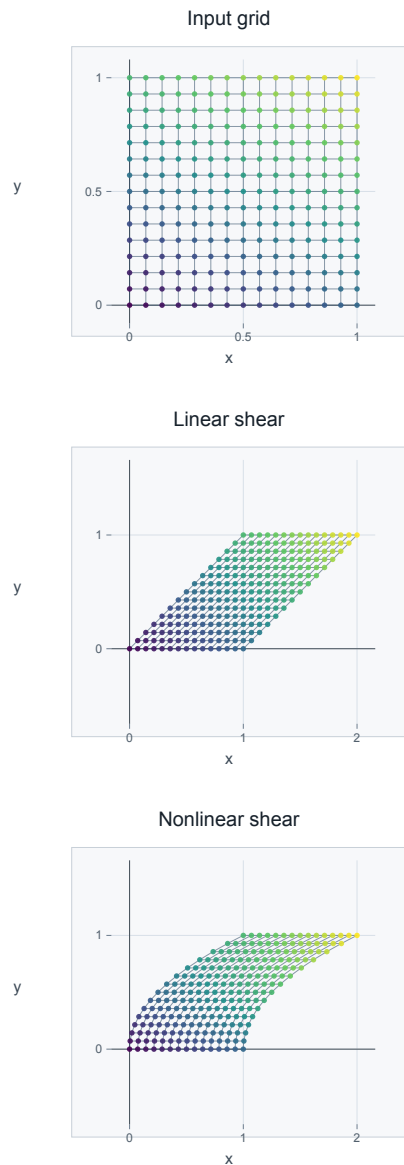


Figure 3.1. Square-grid visualization for $T(x, y) = (x + y, y)$ and $F(x, y) = (x + y^2, y)$. The color identifies the original input point. The two maps agree on the four corners, but the nonlinear map bends grid lines.

The notation $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ includes several cases. If $m = 1$, the output is scalar-valued. If $n = 1$ and $m > 1$, the map traces a curve. If $F(\mathbf{x}) = A\mathbf{x}$, the map is linear. If $F(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$, the map is affine.

3.1.2 Domains, ranges, graphs, and level curves

Domain and range

We encountered the domain of a function in Linear and affine maps. We now give a more formal definition and introduce the range.

Definition 3.1.2 (Domain and range).

The *domain* of a function is the set of all inputs for that function, and the *range* is the set of all possible outputs. More formally, if a function f takes inputs from a set X and outputs elements in a set Y , then the domain of f is the set X , and the range is the set of all $y \in Y$ for which $y = f(x)$ for some $x \in X$, which we might write in set notation as $\{y \in Y : y = f(x) \text{ for some } x \in X\}$.

In calculus, we often define the domain of a function f *implicitly*, by defining f in terms of an expression, and then letting the domain of f be the set of all values which, when substituted into that expression, lead to a well-defined quantity. For instance, we might discuss ‘the function’ $f(x, y) = x/y$, by which we mean the function with domain $\{(x, y) \in \mathbb{R}^2 : y \neq 0\}$ and range $\mathbb{R}$.

The range contains only outputs actually attained by the function, while the co-domain, defined in Linear and affine maps, is the specified output set and may be larger than the range.

For scalar-valued functions of two variables, domains and level curves are subsets of the plane. These pictures help us see where a function is defined and how its outputs change.

Activity

Sketch the domains of the following functions:

Task

$$f(x, y) = \ln(y - x^2) + \sqrt{x}.$$

Solution.

The expression that defines the function f consists of the sum of two terms, and so to find the domain of f , we must identify where both terms are well defined.

The term $\ln(y - x^2)$ is well defined when the input to $\ln$ is positive, i.e., when $y - x^2 > 0$. We may rearrange this inequality to read $y > x^2$.

The term $\sqrt{x}$ is well-defined when the input to $\sqrt{\cdot}$ is non-negative, i.e., when $x \geq 0$.

So f is well-defined when both $y > x^2$ and when $x \geq 0$, and so the domain of f is the set of all points (x, y) such that $x \geq 0$ and $y > x^2$. Using set notation, we can also write the domain as the set

$$\{(x, y) \in \mathbb{R}^2 : x \geq 0 \text{ and } y > x^2\}$$

. The set is the region in the plane bounded from below by the parabola $y = x^2$, and on the left by the y -axis, as depicted in 3.2.

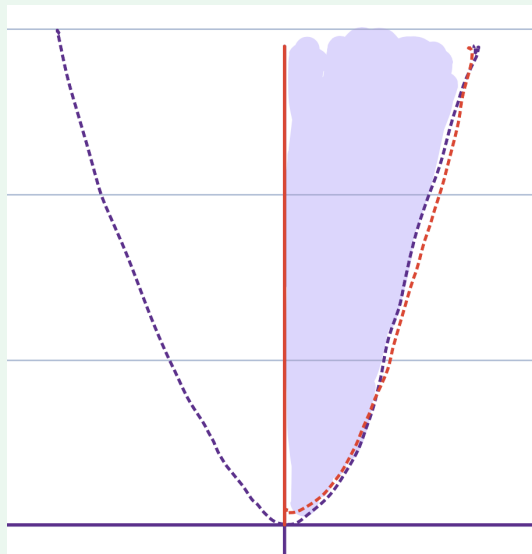


Figure 3.2. Domain of $f(x, y) = \ln(y - x^2) + \sqrt{x}$. Figure by Melissa Lindsey.

Task

$$f(x, y) = e^x \sin y$$

Solution.

The expression defining the function f consists of the product of e^x and $\sin y$, and so f is well-defined precisely when e^x and $\sin y$ are both well-defined. Both expressions are well-defined for all inputs. And so the domain of f is the set $\mathbb{R}^2$.

Task

$$f(x, y) = \frac{\sqrt{x}}{\sqrt{y}}$$

Solution.

The expression defining f is a fraction with numerator $\sqrt{x}$ and denominator $\sqrt{y}$. The expression is therefore well-defined when both the numerator and denominator are well defined, *and* when the denominator is non-zero.

The numerator $\sqrt{x}$ is well-defined when $x \geq 0$. The denominator $\sqrt{y}$ is well-defined when $y \geq 0$, and is non-zero when $y > 0$.

So the domain of f is the set of all $(x, y) \in \mathbb{R}^2$ with $x \geq 0$ and $y > 0$, or in set notation, the set

$$\{(x, y) \in \mathbb{R}^2 : x \geq 0 \text{ and } y > 0\}$$

. This set is the first quadrant in the xy -plane, including the positive y -axis, but excluding the origin and the positive x -axis. See 3.3.

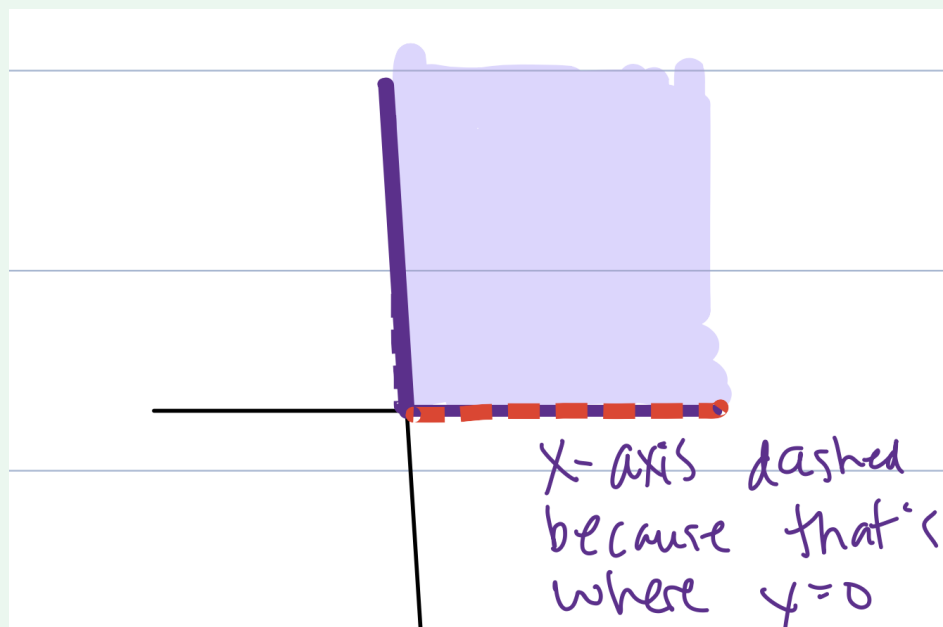


Figure 3.3. Domain of $f(x, y) = \frac{\sqrt{x}}{\sqrt{y}}$. Figure by Melissa Lindsey.

Graphs and level curves

Definition 3.1.3 (The Graph of a Function).

The *graph* of a function f is the set of all pairs $(x, f(x))$, with x in the domain of f .

When f is a scalar-function of two variables, the graph of f is a set of triples $(x, y, f(x, y))$, and thus a subset of $\mathbb{R}^3$, and can therefore be visualized as a surface. See 3.4 for a picture of the graph of the function $f(x, y) = xe^{-x^2-y^2}$.

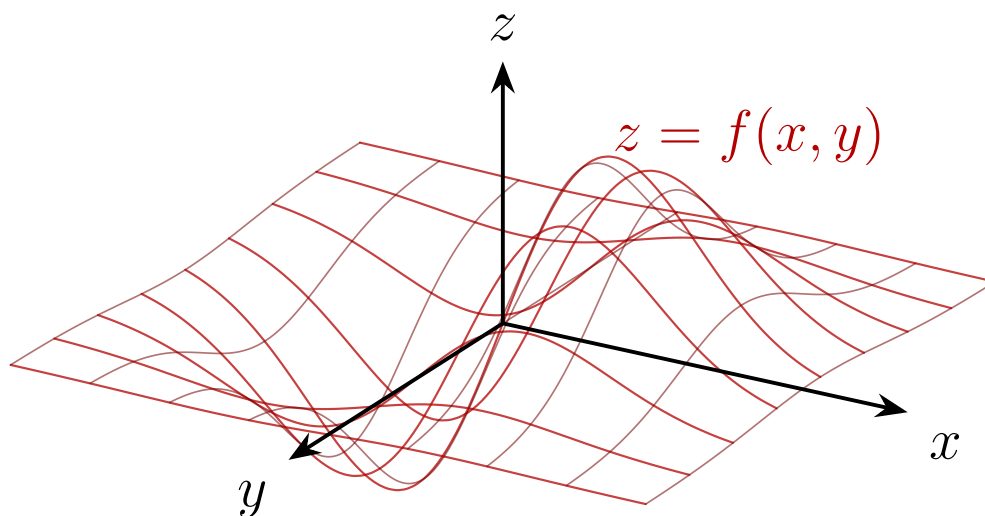


Figure 3.4. A graph of the function $f(x, y) = xe^{-x^2-y^2}$.

Definition 3.1.4 (Level Curves and Contour Plots).

A *level curve* of a scalar function f of two variables is the curve consisting of pairs (x, y) satisfying the equation $f(x, y) = c$, for some $c \in \mathbb{R}$. A *contour plot* is a sketch in $\mathbb{R}^2$ depicting level curves of a function f for many different values of c .

The different values of c which are chosen to be depicted in a contour plot are often (but not always) obtained by varying c by a common difference, obtaining ‘consecutive’ level sets.

Activity

Sketch a contour plot for the function

$$f(x, y) = e^{x^2+y^2}$$

by sketching the level curves $f(x, y) = c$ with $c = 1, c = e, c = e^4$, and $c = e^9$.

Solution.

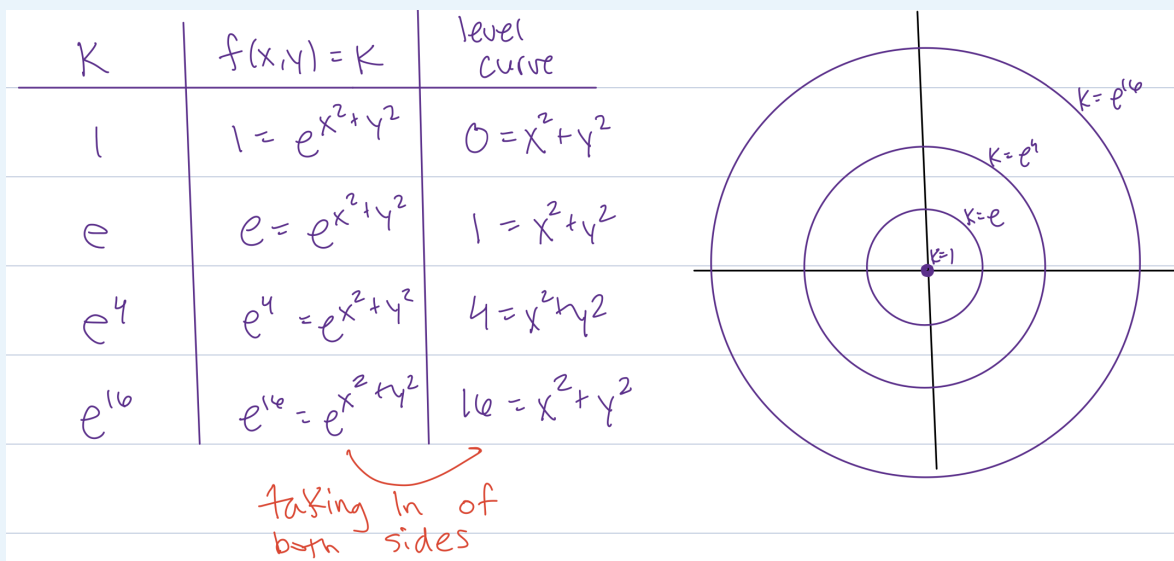


Figure 3.5. Level curves of $f(x, y) = e^{x^2+y^2}$. Figure by Melissa Lindsey.

3.2 Balls, limits, and continuity

3.2.1 Limits and continuity of scalar-valued functions

Balls, limits, and continuity

This short section supplies local language used in derivative and optimization statements: balls, limits, continuity, interior points, boundary points, closed sets, and bounded sets.

Recall the definition of a limit from single-variable calculus. For a function $f : \mathbb{R} \rightarrow \mathbb{R}$ and $a, b \in \mathbb{R}$, the equation

$$\lim_{x \rightarrow a} f(x) = b$$

holds if for each $\varepsilon > 0$, there is $\delta > 0$ so that if $0 < |x - a| < \delta$, then $|f(x) - b| < \varepsilon$. We can interpret this statement geometrically as follows: for any open interval J centered at b , there is an open interval I centered at a so that f maps points in I , excluding a itself, into J .

In $\mathbb{R}^2$, a ball is a disk.

Definition 3.2.1 (Disks in $\mathbb{R}^2$).

Consider a point $(a, b) \in \mathbb{R}^2$. The δ -disk centered at point (a, b) is the region of $\mathbb{R}^2$ bounded by the circle of radius δ centered at (a, b) , i.e., the set

$$\{(x, y) \in \mathbb{R}^2 : \sqrt{(x - a)^2 + (y - b)^2} < \delta\}$$

See 3.6 for a depiction of a δ disk.

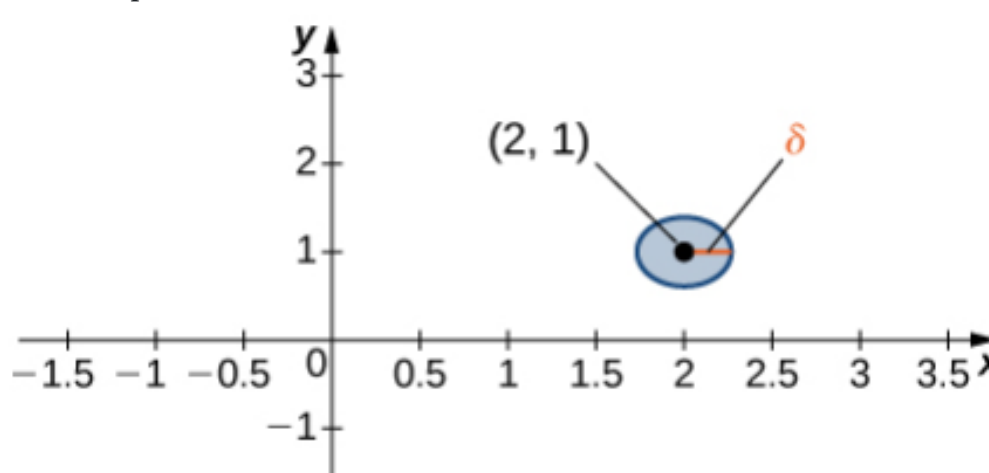


Figure 3.6. A δ -disk centered around the point $(2, 1)$. Figure 4.14 from Edwin “Jed” Herman and Gilbert Strang, *Calculus Volume 3*, OpenStax, © 2018 Rice University, licensed under CC BY-NC-SA 4.0; source: OpenStax Figure 4.14.

We now define the limit of a scalar-valued function f of two variables. For this definition, the equation

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$$

, will then hold precisely when, for each open interval I centered at L , there is a δ disk centered at (a, b) so that all points in this disk which are also contained in the domain of f , excluding the point (a, b) itself, are mapped by f into I .

Definition 3.2.2 (Limits of Scalar-Valued Functions of Two Variables).

Consider a scalar-valued function $f(x, y)$ of two variables. The *limit* of f as (x, y) approaches (a, b) is equal to L , written

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$$

, if for each $\varepsilon > 0$, there is $\delta > 0$ so that for any points in the domain of f with

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta$$

, the inequality $|f(x, y) - L| < \varepsilon$ holds.

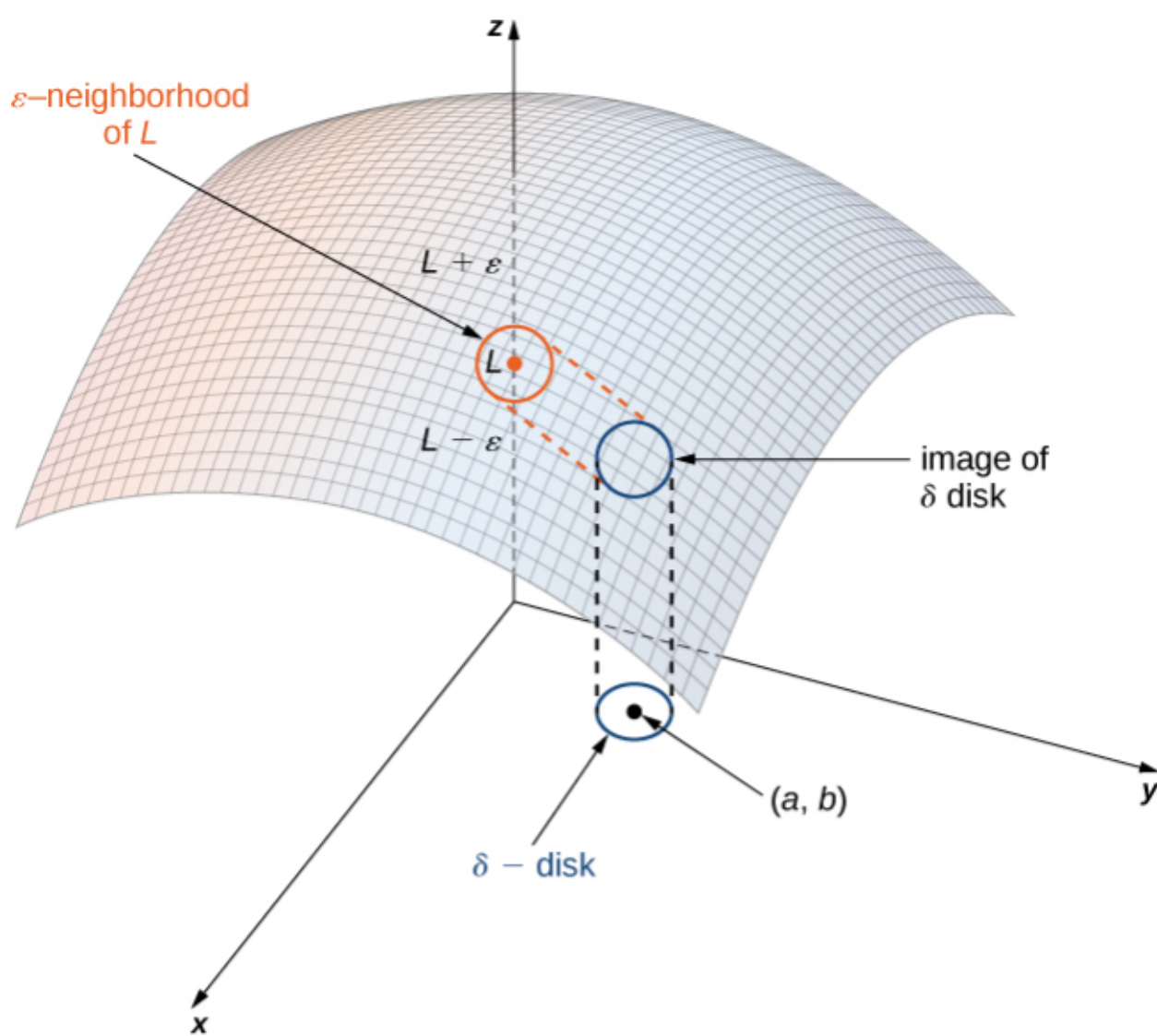


Figure 3.7. $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$ holds when $f(x, y)$ is within ε of L whenever (x, y) is within δ of (a, b) . Figure 4.15 from Edwin “Jed” Herman and Gilbert Strang, *Calculus Volume 3*, OpenStax, © 2018 Rice University, licensed under CC BY-NC-SA 4.0; source: OpenStax Figure 4.15.

Using vector-notation (recalling 1.1.2), we can write a δ -disk centered at a point $\mathbf{x}_0 \in \mathbb{R}^2$ as the set of points $\mathbf{x} \in \mathbb{R}^2$ so that

$$\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x} - \mathbf{x}_0\| < \delta\}$$

. This motivates the higher dimensional variant of a δ disk: a δ -ball.

Definition 3.2.3 (Balls in $\mathbb{R}^n$).

For $\mathbf{x}_0 \in \mathbb{R}^n$, the δ -ball centered at $\mathbf{x}_0$ is the set

$$\{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{x}_0\| < \delta\}$$

In $\mathbb{R}^3$ a δ -ball centered at a point (a, b, c) is precisely the set of points satisfying the equation

$$(x - a)^2 + (y - b)^2 + (z - c)^2 < \delta^2$$

. It is the region in $\mathbb{R}^3$ bounded by a sphere of radius δ .

By replacing disks with balls, we obtain the definition of a limit of a general scalar-valued function.

Definition 3.2.4 (The Limit of a Scalar-Valued Function).

Consider a scalar-valued function f . The *limit* of f as $\mathbf{x}$ approaches $\mathbf{x}_0$ is equal to L , written

$$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0} f(\mathbf{x}) = L$$

, if for each $\varepsilon > 0$, there is $\delta > 0$ so that for any points in the domain of f with $0 < \|\mathbf{x} - \mathbf{x}_0\| < \delta$, the inequality $|f(\mathbf{x}) - L| < \varepsilon$ holds.

Remark

In order for the limit of f , when it exists, to be unique, the point $\mathbf{x}_0$ must have the property that the domain of f contains a point other than $\mathbf{x}_0$ in any δ -ball centered at $\mathbf{x}_0$. This is only a small issue, since every point in the domain of any of the functions that naturally occur in calculus have this property.

Recall from the calculus of a single variable that a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous at a point a when $\lim_{x \rightarrow a} f(x) = f(a)$, i.e., when the limit on the left hand side exists and is equal to the right hand side. The continuity of functions of several variables is defined in exactly the same way.

Definition 3.2.5.

A scalar-valued function f is *continuous* at a point $\mathbf{x}_0$ in its domain if the limit of f as $\mathbf{x} \rightarrow \mathbf{x}_0$ exists, and

$$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0} f(\mathbf{x}) = f(\mathbf{x}_0)$$

Interior, boundary, and closed sets

This language lets us distinguish points inside a set from points on its boundary.

Definition 3.2.6 (Interior, boundary, and closed sets).

Let S be a subset of $\mathbb{R}^n$. Then a point $\mathbf{x}_0 \in S$ is an *interior* point if there is a δ -ball centered at $\mathbf{x}_0$ completely contained in S . A point $\mathbf{x}_0 \in \mathbb{R}^n$ is a *boundary point* if every ball centered at $\mathbf{x}_0$ contains at least one point of S and at least one point outside S . The set of all boundary points of a set S is called the *boundary* of the set S .

Consider the figure for an illustration of the example of an interior point and boundary point.

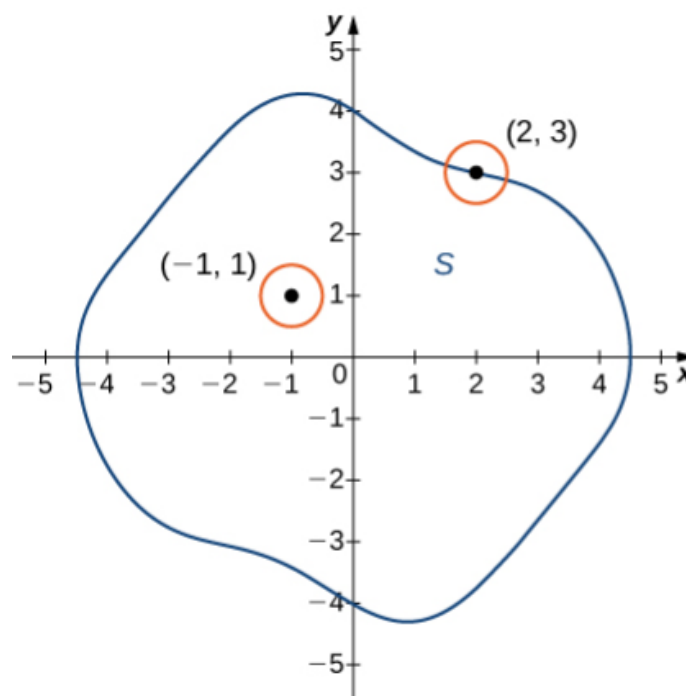


Figure 3.8. In the set S shown, $(-1, 1)$ is an interior point and $(2, 3)$ is a boundary point. Figure 4.17 from Edwin “Jed” Herman and Gilbert Strang, *Calculus Volume 3*, OpenStax, © 2018 Rice University, licensed under CC BY-NC-SA 4.0; source: OpenStax Figure 4.17.

Remark

Note that a boundary point of a set S *need not be contained in the set S* . For example, the δ -disk defined in this section does not contain its circular boundary. A set that contains its boundary is called *closed*.

Definition 3.2.7 (Closed sets).

A set $S \subset \mathbb{R}^n$ is called *closed* if it contains its boundary. A set $S \subset \mathbb{R}^n$ is *bounded* if it is contained in some ball centered at the origin, or equivalently if all points of S lie within some fixed distance of the origin.

3.3 One-parameter maps and changes

3.3.1 Vector-valued functions of one variable

Basic definitions

A map $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^n$ takes one scalar parameter and outputs a vector. It traces a curve.

Unit 2 lines are the special case $\mathbf{r}(t) = \mathbf{x}_0 + t\mathbf{d}$. Indeed, we already encountered parameterizations of curves when discussing parametric equations for straight lines, as in 2.2.4.

Activity

The line L_1 passes through point $(3, -2, 5)$ and is parallel to the vector $\mathbf{v} = (2, -3, 4)$.

Task

Find a parameterization of the line L_1 .

Solution.

The line passes through $(3, -2, 5)$ and has direction vector $(2, -3, 4)$, and so it is parameterized by the function

$$\mathbf{r}(t) = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix} + t \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 + 2t \\ -2 - 3t \\ 5 + 4t \end{bmatrix}$$

Task

Another line L_2 passes through the point $(1, 3, -1)$ and is perpendicular to L_1 . If L_2 also lies in the plane $z = -1$, find a parameterization of L_2 .

Solution.

Since L_2 lies in the plane $z = -1$, the z -coordinates of all the points of L_2 are the same, so any direction vector for L_2 must be of the form $\mathbf{d} = (a, b, 0)$ for some scalars a and b . Since L_2 is perpendicular to L_1 , direction vectors for L_2 must be perpendicular to direction vectors of L_1 , and since L_1 has $(2, -3, 4)$ as a direction vector, we find that

$$\begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ 0 \end{bmatrix} = 2a - 3b = 0$$

. Thus $a = (3/2)b$.

Any vector solving this equation gives a direction vector of L_2 , and so (to obtain integer values of a and b) we can pick $b = 2$, and thus $a = 3$. So $\mathbf{d} = (3, 2, 0)$ is a direction vector of L_2 , and since L_2 passes through $(1, 3, -1)$, we obtain a parameterization of L_2 of the form

$$\mathbf{r}(t) = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} + t \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 + 3t \\ 3 + 2t \\ -1 \end{bmatrix}$$

The parameterizations of straight lines given above are given by vector-valued functions whose components are linear functions. But we can also parameterize curves using nonlinear functions.

Example

As we vary t over its allowed values, the vector-valued function $\mathbf{r}(t)$ traces a curve in $\mathbb{R}^n$. We will not graph functions by hand in this course. For example, consider

$$\mathbf{r}(t) = 4 \cos t \mathbf{i} + 4 \sin t \mathbf{j} + t \mathbf{k}, \quad 0 \leq t \leq 4\pi$$

. Then the curve it traces is illustrated below.

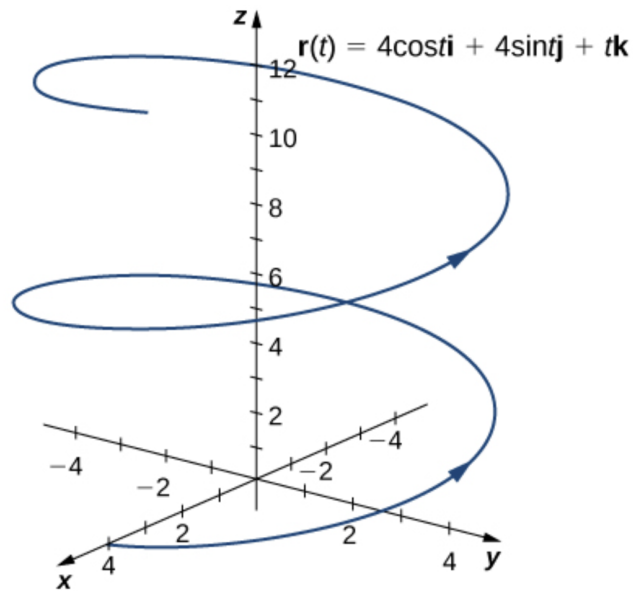


Figure 3.9. The helix traced by the vector-valued function $\mathbf{r}$. Figure 3.4 from Edwin “Jed” Herman and Gilbert Strang, *Calculus Volume 3*, OpenStax, © 2018 Rice University, licensed under CC BY-NC-SA 4.0; source: OpenStax Figure 3.4.

Activity

Recall the vectors $\mathbf{i}$ and $\mathbf{j}$ in $\mathbb{R}^2$ from 1.3.5. Consider the vector-valued function $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^3$ defined by the equation

$$\mathbf{r}(t) = (t^2 - 3t)\mathbf{i} + (4t + 1)\mathbf{j}$$

, i.e., the equation

$$\mathbf{r}(t) = \begin{bmatrix} t^2 - 3t \\ 4t + 1 \end{bmatrix}$$

.

Task

Evaluate $\mathbf{r}(0)$.

Solution.

We substitute 0 for t in the equation above, obtaining that

$$\begin{aligned} \mathbf{r}(0) &= ((0)^2 - 3(0))\mathbf{i} + (4(0) + 1)\mathbf{j} \\ &= 0\mathbf{i} + 1\mathbf{j} \\ &= \mathbf{j} \end{aligned}$$

. We can also rewrite the equation $\mathbf{r}(0) = \mathbf{j}$ as $\mathbf{r}(0) = (0, 1)$.

Task

Evaluate $\mathbf{r}(1)$.

Solution.

We calculate that

$$\begin{aligned}\mathbf{r}(1) &= ((1)^2 - 3(1))\mathbf{i} + (4(1) + 1)\mathbf{j} \\ &= (1 - 3)\mathbf{i} + (4 + 1)\mathbf{j} \\ &= -2\mathbf{i} + 5\mathbf{j}\end{aligned}$$

. Alternatively, we can write $\mathbf{r}(1) = (-2, 5)$.

Task

Evaluate $\mathbf{r}(-4)$.

Solution.

We calculate that

$$\begin{aligned}\mathbf{r}(-4) &= (16 + 12)\mathbf{i} + (-16 + 1)\mathbf{j} \\ &= 28\mathbf{i} - 15\mathbf{j}\end{aligned}$$

. So $\mathbf{r}(-4) = (28, -15)$.

We use parameterizations mainly to describe position, velocity, and tangent direction along a curve.

Limits and continuity

Definition 3.3.1 (Limit of a Vector-Valued Function).

A vector-valued function $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^n$ approaches the *limit* $\mathbf{v} \in \mathbb{R}^n$ as t approaches a , written

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{v}$$

, provided

$$\lim_{t \rightarrow a} \|\mathbf{r}(t) - \mathbf{v}\| = 0$$

. This is the limit of a scalar-valued function, and can thus be evaluated using the methods of single-variable calculus.

In practice, calculating and simplifying the expression $\|\mathbf{r}(t) - \mathbf{v}\|$ in order to evaluate the limit

$$\lim_{t \rightarrow a} \|\mathbf{r}(t) - \mathbf{v}\| = 0$$

can be quite cumbersome. Instead, the following theorem allows us to evaluate limits much more easily.

Theorem 3.3.2 (Limit of a Vector-Valued Function).

Let $\mathbf{r}(t) : \mathbb{R} \rightarrow \mathbb{R}^n$ be a function with component functions $r_1, \dots, r_n : \mathbb{R} \rightarrow \mathbb{R}$. Then if

$$\lim_{t \rightarrow a} r_1(t) = v_1 \quad \dots \quad \lim_{t \rightarrow a} r_n(t) = v_n$$

, and we define the vector $\mathbf{v} = (v_1, \dots, v_n)$, then

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{v}$$

. Equivalently,

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \begin{bmatrix} \lim_{t \rightarrow a} r_1(t) \\ \vdots \\ \lim_{t \rightarrow a} r_n(t) \end{bmatrix}$$

Activity

Calculate $\lim_{t \rightarrow 3} \mathbf{r}(t)$ for the given vector-valued functions $\mathbf{r}(t)$.

Task

The function $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^2$ defined by

$$\mathbf{r}(t) = (t^2 - 3t + 4)\mathbf{i} + (4t + 3)\mathbf{j}$$

.

Solution.

The component functions of the function $\mathbf{r}$ are $r_1(t) = t^2 - 3t + 4$ and $r_2(t) = 4t + 3$. We calculate that

$$\lim_{t \rightarrow 3} r_1(t) = \lim_{t \rightarrow 3} t^2 - 3t + 4 = (3)^2 - 3(3) + 4 = 4$$

and

$$\lim_{t \rightarrow 3} r_2(t) = \lim_{t \rightarrow 3} 4t + 3 = 4(3) + 3 = 15$$

. Thus by 3.3.2,

$$\lim_{t \rightarrow 3} \mathbf{r}(t) = \begin{bmatrix} 4 \\ 15 \end{bmatrix}$$

.

Task

The function $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^3$ defined by

$$\mathbf{r}(t) = \left(\frac{2t-4}{t+1} \right) \mathbf{i} + \left(\frac{t^2+1}{2} \right) \mathbf{j} + (4t-3) \mathbf{k}$$

Solution.

We calculate that

$$\begin{aligned} \lim_{t \rightarrow 3} \mathbf{r}(t) &= \lim_{t \rightarrow 3} \left[\frac{2t-4}{t+1} \mathbf{i} + \frac{t^2+1}{2} \mathbf{j} + (4t-3) \mathbf{k} \right] \\ &= \left[\lim_{t \rightarrow 3} \frac{2t-4}{t+1} \right] \mathbf{i} + \left[\lim_{t \rightarrow 3} \frac{t^2+1}{2} \right] \mathbf{j} + \left[\lim_{t \rightarrow 3} (4t-3) \right] \mathbf{k} \\ &= \left[\frac{2(3)-4}{3+1} \right] \mathbf{i} + \left[\frac{3^2+1}{2} \right] \mathbf{j} + [(4 \cdot 3 - 3)] \mathbf{k} \\ &= \left[\frac{6-4}{4} \right] \mathbf{i} + \left[\frac{9+1}{2} \right] \mathbf{j} + [12-3] \mathbf{k} \\ &= \frac{1}{2} \mathbf{i} + \frac{10}{2} \mathbf{j} + 9 \mathbf{k} \end{aligned}$$

Definition 3.3.3 (Continuity of Vector-Valued Functions).

A vector-valued function $\mathbf{r}(t)$ is *continuous* at point $t = a$ if

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{r}(a)$$

That is, a vector-valued function is continuous at point $t = a$ when the limit of the function as t approaches a equals the value of the function at a .

Theorem 3.3.4.

A vector-valued function $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^n$ with component functions $r_1, \dots, r_n$ is continuous at $t = a$ if and only if each of the component functions $r_1, \dots, r_n$ is continuous at $t = a$.

Derivatives

Now we define the derivative of a vector-valued function from $\mathbb{R}$ to $\mathbb{R}^n$.

Definition 3.3.5 (Derivative of a Vector-Valued Function).

The *derivative* of a function $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^n$ is

$$\frac{d\mathbf{r}}{dt}(t) = \mathbf{r}'(t) = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t},$$

provided the limit exists. If $\mathbf{r}'(t)$ exists, then we say $\mathbf{r}$ is *differentiable* at t . If $\mathbf{r}'(t)$ exists for all t in an open interval (a, b) , then we say $\mathbf{r}$ is *differentiable* on (a, b) . The vector $\mathbf{r}'(t)$ is called the *tangent vector* of the curve described by the function $\mathbf{r}$ at the point $\mathbf{r}(t)$.

Remark

The derivative $\mathbf{r}' : \mathbb{R} \rightarrow \mathbb{R}^n$ of a differentiable function $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^n$ is *also* a vector-valued function.

As in single variable calculus, we will often have much more efficient methods of calculating the derivative of a function $\mathbf{r}$. But you should be able to work directly from the definition in simple examples.

Activity

Use the definition to calculate the derivative of the function

$$\mathbf{r}(t) = (2t^2 + 3)\mathbf{i} + (5t - 6)\mathbf{j}$$

.

Solution.

The derivative of the function $\mathbf{r}(t)$ is given by the expression

$$\lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}$$

. As before, we use 3.3.2 to evaluate the limit. The two components of the expression

$$\frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}$$

are

$$\frac{(2(t + \Delta t)^2 + 3) - (2t^2 + 3)}{\Delta t}$$

and

$$\frac{(5(t + \Delta t) - 6) - (5t - 6)}{\Delta t}$$

, and it suffices to calculate the limits as $\Delta t \rightarrow 0$ for each component. For $\Delta t \neq 0$, expanding and canceling out like factors gives that

$$\frac{(2(t + \Delta t)^2 + 3) - (2t^2 + 3)}{\Delta t} = 4t + 2\Delta t$$

, and so

$$\lim_{\Delta t \rightarrow 0} \frac{(2(t + \Delta t)^2 + 3) - (2t^2 + 3)}{\Delta t} = \lim_{\Delta t \rightarrow 0} 4t + 2\Delta t = 4t$$

. Similarly, for $\Delta t \neq 0$ we can write

$$\frac{(5(t + \Delta t) - 6) - (5t - 6)}{\Delta t} = 5$$

, and so

$$\lim_{\Delta t \rightarrow 0} \frac{(5(t + \Delta t) - 6) - (5t - 6)}{\Delta t} = \lim_{\Delta t \rightarrow 0} 5 = 5$$

. We thus conclude that

$$\mathbf{r}'(t) = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} = \begin{bmatrix} 4t \\ 5 \end{bmatrix}$$

Since 3.3.2 tells us taking limits of functions from $\mathbb{R}$ to $\mathbb{R}^n$ is equivalent to taking the limits of each component function, taking the derivative of such a function is equivalent to taking the derivative of each component function.

Theorem 3.3.6.

If $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^n$ has components $r_1, \dots, r_n : \mathbb{R} \rightarrow \mathbb{R}$, then $\mathbf{r}$ is differentiable at t if and only if each component function $r_1, \dots, r_n$ is differentiable at t , and

$$\mathbf{r}'(t) = \begin{bmatrix} r_1'(t) \\ \vdots \\ r_n'(t) \end{bmatrix}$$

Because of 3.3.6, we can apply all the techniques from single-variable calculus to differentiate functions from $\mathbb{R}$ to $\mathbb{R}^n$.

Activity

Find $\mathbf{r}'(t)$, where

$$\mathbf{r}(t) = e^{2t} \mathbf{i} + \ln(1 + t) \mathbf{j} - (\cos t) \mathbf{k}$$

Solution.

Using 3.3.6, we calculate that

$$\begin{aligned} \mathbf{r}'(t) &= (e^{2t})' \mathbf{i} + (\ln(1 + t))' \mathbf{j} - (\cos t)' \mathbf{k} \\ &= 2e^{2t} \mathbf{i} + \frac{1}{1 + t} \mathbf{j} + \sin t \mathbf{k} \end{aligned}$$

If $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^n$, then $\mathbf{r}' : \mathbb{R} \rightarrow \mathbb{R}^n$ is also a vector-valued function of the same type, so we can consider the second derivatives $\mathbf{r}''$, i.e., the derivative of the function $\mathbf{r}'$.

Activity

Find $\mathbf{r}''(t)$, where $\mathbf{r}$ is as in 3.3.1.

Solution.

In 3.3.1 we calculated that

$$\mathbf{r}'(t) = 2e^{2t} \mathbf{i} + \left(\frac{1}{1+t} \right) \mathbf{j} + \sin t \mathbf{k}$$

. Applying 3.3.6, we find that

$$\begin{aligned} \mathbf{r}''(t) &= (2e^{2t})' \mathbf{i} + \left(\frac{1}{1+t} \right)' \mathbf{j} + (\sin t)' \mathbf{k} \\ &= 4e^{2t} \mathbf{i} - \frac{1}{(1+t)^2} \mathbf{j} + \cos t \mathbf{k} \end{aligned}$$

Differentiation rules

Theorem 3.3.7.

Suppose $\mathbf{u}, \mathbf{v} : \mathbb{R} \rightarrow \mathbb{R}^n$ are differentiable, c is a scalar, and $a : \mathbb{R} \rightarrow \mathbb{R}$ is a scalar-valued function. Then

Sum Rule

$$(\mathbf{u} + \mathbf{v})'(t) = \mathbf{u}'(t) + \mathbf{v}'(t)$$

Product Rule

$$(c\mathbf{u})'(t) = c\mathbf{u}'(t)$$

Product Rule

$$(a\mathbf{u})'(t) = a'(t)\mathbf{u}(t) + a(t)\mathbf{u}'(t)$$

Dot Product rule

$$(\mathbf{u} \cdot \mathbf{v})'(t) = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$$

Proof.

We prove the dot product rule, in the two dimensional case. Let

$$\mathbf{u}(t) = u_1(t)\mathbf{i} + u_2(t)\mathbf{j}$$

and let

$$\mathbf{v}(t) = v_1(t)\mathbf{i} + v_2(t)\mathbf{j}$$

. Then

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2$$

. Applying the sum and product rules for functions from $\mathbb{R}$ to $\mathbb{R}$, we find that

$$\begin{aligned} (\mathbf{u} \cdot \mathbf{v})' &= (u_1 v_1)' + (u_2 v_2)' \\ &= (u_1' v_1 + u_1 v_1') + (u_2' v_2 + u_2 v_2') \\ &= (u_1' v_1 + u_2' v_2) + (u_1 v_1' + u_2 v_2') \\ &= \mathbf{u}' \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{v}' \end{aligned}$$

Activity

Find $(\mathbf{u} \cdot \mathbf{v})'$, where

$$\mathbf{u}(t) = \cos t \mathbf{i} + \sin t \mathbf{j} + t \mathbf{k}$$

and

$$\mathbf{v}(t) = t \mathbf{i} + \ln t \mathbf{j} + \mathbf{k}$$

Solution.

We calculate that

$$\begin{aligned} \mathbf{u}' &= (\cos t)' \mathbf{i} + (\sin t)' \mathbf{j} + (t)' \mathbf{k} \\ &= -\sin t \mathbf{i} + \cos t \mathbf{j} + \mathbf{k} \end{aligned}$$

and

$$\begin{aligned} \mathbf{v}' &= (t)' \mathbf{i} + (\ln t)' \mathbf{j} + (1)' \mathbf{k} \\ &= \mathbf{i} + (1/t) \mathbf{j} + 0 \mathbf{k} \end{aligned}$$

. Thus

$$\mathbf{u} \cdot \mathbf{v}' = \begin{bmatrix} \cos t \\ \sin t \\ t \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1/t \\ 0 \end{bmatrix} = \cos t + \frac{\sin t}{t}$$

and

$$\mathbf{u}' \cdot \mathbf{v} = \begin{bmatrix} -\sin t \\ \cos t \\ 1 \end{bmatrix} \cdot \begin{bmatrix} t \\ \ln t \\ 1 \end{bmatrix} = -t \sin t + \ln t \cos t + 1$$

. So

$$\begin{aligned}(\mathbf{u} \cdot \mathbf{v})' &= \mathbf{u}\mathbf{v}' + \mathbf{u}'\mathbf{v} \\&= \left(\cos t + \frac{\sin t}{t}\right) + (-t \sin t + \ln t \cos t + 1) \\&= \cos t(1 + \ln t) + \sin t(1/t - t) + 1\end{aligned}$$

Remark

Note that in 3.3.1, since $\mathbf{v}$ is only well-defined for $t > 0$, the function $\mathbf{u} \cdot \mathbf{v}$ is also only well-defined for $t > 0$, as is its derivative.

3.4 Partial derivatives, gradients, Hessians, and Jacobian matrices

3.4.1 Partial derivatives

Definition and basic concepts

A partial derivative measures change in one input coordinate while the other input coordinates are held fixed.

Definition 3.4.1 (Partial Derivatives).

Let $f(x, y)$ be a scalar-valued function in two variables. The *partial derivative of f with respect to x* at (a, b) is the scalar quantity

$$f_x(a, b) = \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h}$$

, provided this limit exists. Similarly, the *partial derivative of f with respect to y* at (a, b) is the quantity

$$f_y(a, b) = \lim_{h \rightarrow 0} \frac{f(a, b + h) - f(a, b)}{h}$$

provided this limit exists.

We compute the partial derivative in x of an expression precisely by treating the y -variable as a constant, and differentiating in x . Similarly, we compute the partial derivative in y by treating the x -variable as a constant.

Note

There are several notations used for the partial derivative of a function $f(x, y)$, namely, we might write $f_x(a, b)$ as

$$\partial_x f(a, b) \quad \frac{\partial f}{\partial x}(a, b) \quad (D_x f)(a, b) \quad \text{or} \quad (D_1 f)(a, b)$$

. Similarly, we write $f_y(a, b)$ as

$$\partial_y f(a, b) = \frac{\partial f}{\partial y}(a, b) = (D_y f)(a, b) \quad \text{or} \quad (D_2 f)(a, b)$$

Activity

Let

$$f(x, y) = x^2 - 3xy + 2y^2 - 4x + 5y - 12$$

Task

Calculate $f_x(x, y)$ directly using the definition of the derivative.

Solution.

We must evaluate the limit

$$\lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}$$

. We begin by expanding and simplifying the expression $f(x+h, y)$, that

$$\begin{aligned} f(x+h, y) &= (x+h)^2 - 3(x+h)y + 2y^2 - 4(x+h) + 5y - 12 \\ &= x^2 + 2xh + h^2 - 3xy - 3hy + 2y^2 - 4x - 4h + 5y - 12 \\ &= x^2 - 3xy + 2y^2 - 4x + 5y - 12 + h(2x - 3y - 4 + h) \\ &= f(x, y) + h(2x - 3y - 4 + h) \end{aligned}$$

. So for $h \neq 0$,

$$\lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} = \lim_{h \rightarrow 0} 2x - 3y - 4 + h = 2x - 3y - 4$$

. So

$$f_x(x, y) = 2x - 3y - 4$$

Task

Calculate $f_y(x, y)$ directly using the definition of the derivative.

Solution.

We first calculate that

$$\begin{aligned} f(x, y+h) &= x^2 - 3x(y+h) + 2(y+h)^2 - 4x + 5(y+h) - 12 \\ &= x^2 - 3xy - 3xh + 2y^2 + 4yh + 2h^2 - 4x + 5y + 5h - 12 \\ &= f(x, y) + h(-3x + 4y + 2h + 5) \end{aligned}$$

. So

$$\lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h} = \lim_{h \rightarrow 0} -3x + 4y + 2h + 5 = -3x + 4y + 5$$

. So

$$f_y(x, y) = -3x + 4y + 5$$

.

Activity

For each function $f(x, y)$, find $f_x(x, y)$ and $f_y(x, y)$.

Task

$$f(x, y) = 3x^2y + 2x - 3y$$

Solution.

To find $f_x(x, y)$, treat y as a constant:

$$\begin{aligned} f_x(x, y) &= \frac{\partial}{\partial x}(3x^2y + 2x - 3y) \\ &= 3 \cdot 2xy + 2 - 0 \\ &= 6xy + 2 \end{aligned}$$

.

To find $f_y(x, y)$, treat x as a constant:

$$\begin{aligned} f_y(x, y) &= \frac{\partial}{\partial y}(3x^2y + 2x - 3y) \\ &= 3x^2 \cdot 1 + 0 - 3 \\ &= 3x^2 - 3 \end{aligned}$$

.

Task

$$f(x, y) = x \sin y + y \sin x$$

Solution.

Treating y as a constant, we find that

$$\begin{aligned} f_x(x, y) &= \frac{\partial}{\partial x}(x \sin y + y \sin x) \\ &= 1 \cdot \sin y + y \cdot \cos x \\ &= \sin y + y \cos x \end{aligned}$$

. Treating x as a constant, we find that

$$\begin{aligned} f_y(x, y) &= \frac{\partial}{\partial y}(x \sin y + y \sin x) \\ &= x \cdot \cos y + 1 \cdot \sin x \\ &= x \cos y + \sin x \end{aligned}$$

.

Task

$$f(x, y) = x^2 e^{xy}.$$

Solution.

To find $f_x(x, y)$, treat y as a constant and use the product and chain rules:

$$\begin{aligned} f_x(x, y) &= \frac{\partial}{\partial x}(x^2 e^{xy}) \\ &= \frac{\partial}{\partial x}(x^2) \cdot e^{xy} + x^2 \cdot \frac{\partial}{\partial x}(e^{xy}) \\ &= 2x \cdot e^{xy} + x^2 \cdot e^{xy} \cdot \frac{\partial}{\partial x}(xy) \\ &= 2x \cdot e^{xy} + x^2 \cdot e^{xy} \cdot y \\ &= 2xe^{xy} + x^2 y e^{xy} \\ &= xe^{xy}(2 + xy) \end{aligned}$$

. To find $f_y(x, y)$, treat x as a constant:

$$\begin{aligned} f_y(x, y) &= \frac{\partial}{\partial y}(x^2 e^{xy}) \\ &= x^2 \cdot \frac{\partial}{\partial y}(e^{xy}) \\ &= x^2 \cdot e^{xy} \cdot \frac{\partial}{\partial y}(xy) \\ &= x^2 \cdot e^{xy} \cdot x \\ &= x^3 e^{xy} \end{aligned}$$

.

Geometric interpretation

Definition 3.4.2 (The Slice of a Function).

Given a function $f(x, y)$ and $b \in \mathbb{R}$, the *slice* of the function f in the plane $y = b$ is the function

$g : \mathbb{R} \rightarrow \mathbb{R}$ given by $g(x) = f(x, b)$. Similarly, for $x \in \mathbb{R}$ the slice of f in the plane $x = a$ is the function $g : \mathbb{R} \rightarrow \mathbb{R}$ given by $g(y) = f(a, y)$.

Geometrically, the slice of a function f in the plane $y = b$ corresponding to intersecting the surface given by the equation $z = f(x, y)$ with the plane $y = b$, creating a curve in this plane.

Activity

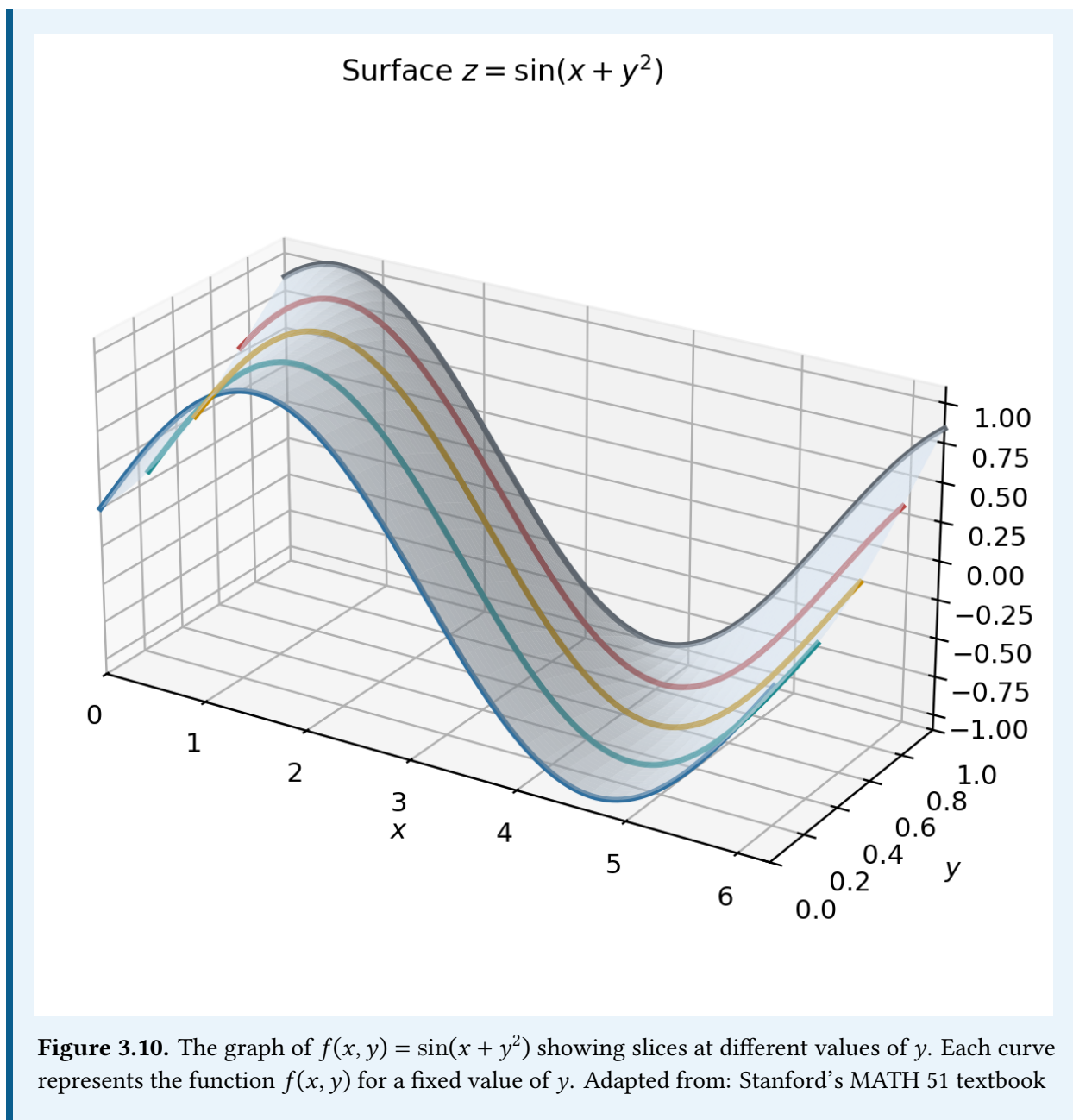
Sketch the slices of the function $f(x, y) = \sin(x + y^2)$ in the plane $y = b$.

Solution.

The slice in the plane $y = b$ is the function

$$g(x) = f(x, b) = \sin(x + b^2)$$

. This slice is a sine function with a phase shift of b^2 . See 3.10 for a sketch of these slices.



Theorem 3.4.3 (Geometric Interpretation).

If we consider the surface in $\mathbb{R}^3$ described by the equation $z = f(x, y)$, then

- $f_x(a, b)$ is the *slope of the tangent line* to the curve formed by intersecting the surface $z = f(x, y)$ with the vertical plane $y = b$.
- $f_y(a, b)$ is the *slope of the tangent line* to the curve formed by intersecting the surface $z = f(x, y)$ with the vertical plane $x = a$.

Activity

Let $f(x, y) = 16 - x^2 - y^2$. Find $f_x(1, 2)$ and $f_y(1, 2)$ and interpret these numbers as slopes of a certain curve.

Solution.

We calculate that

$$\begin{aligned}f_x(x, y) &= \frac{\partial}{\partial x}(16 - x^2 - y^2) \\&= 0 - 2x - 0 \\&= -2x\end{aligned}$$

. So

$$f_x(1, 2) = -2(1) = -2$$

. This is the slope of the curve in the xz plane obtained by intersecting the surface $z = 16 - x^2 - y^2$ with the plane defined by the equation $y = 2$, at the point $(1, 2, 11)$. That is, the tangent line of the curve at this point is given by $z - 11 = -2(x - 1)$.

We calculate that

$$\begin{aligned}f_y(x, y) &= \frac{\partial}{\partial y}(16 - x^2 - y^2) \\&= 0 - 0 - 2y \\&= -2y\end{aligned}$$

. So

$$f_y(1, 2) = -2(2) = -4$$

. This is the slope of the curve in the yz plane obtained by intersecting the surface $z = 16 - x^2 - y^2$ with the plane defined by the equation $x = 1$, at the point $(1, 2, 11)$. That is, the tangent line of the curve at this point is given by $z - 11 = -4(y - 2)$.

Activity

Consider the contour plot shown in 3.11 for a function F with contour lines at increments of 0.2. In particular, consider the curve obtained from the slice $y = 0$, indicated by the red line in 3.11. By interpreting the contour plot, what can we say about the slopes of this curve, or equivalently, the values of F_x on this line?

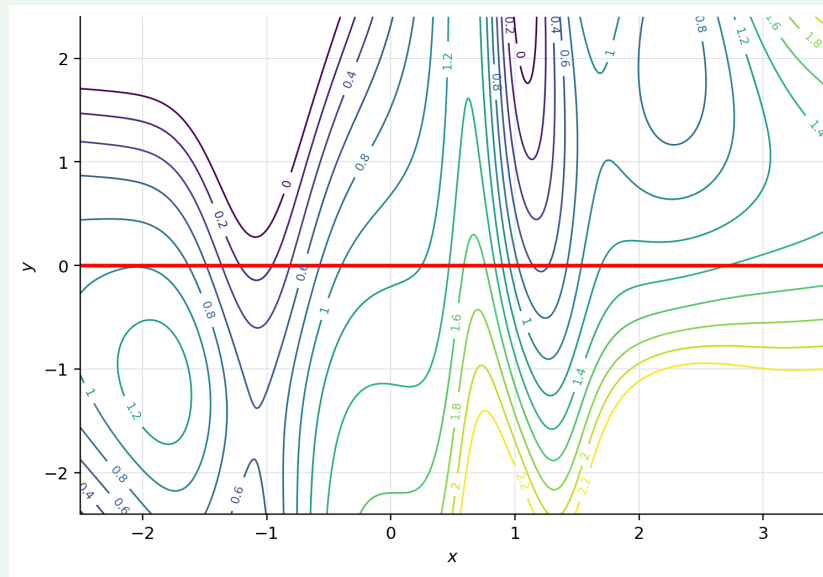


Figure 3.11. A contour plot for a function with increments of 0.2. The line $y = 0$ is indicated in red. Adapted from: Stanford's MATH 51 textbook

Solution.

Here is some of the information we can read from the contour plot:

- Between $x = -1$ and $x = 0$, the contour values increase (from about 0.2 to 1), indicating $F_x > 0$ in this region (the function F is increasing).
- Near $x = 1$, the contour lines are very close together, and the contour values decrease, indicating the $F_x < 0$, and that the magnitude of F_x is large (the curve slopes steeply downward).
- Between $x = 1.5$ and $x = 3$, the contour values increase, indicating $F_x > 0$. (the function F is increasing).

See 3.12 for a graph depicting the curve, to check what we have deduced is correct.

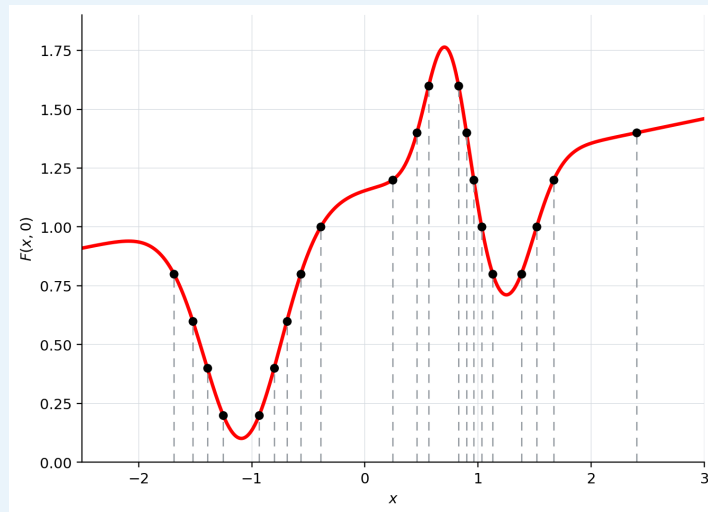
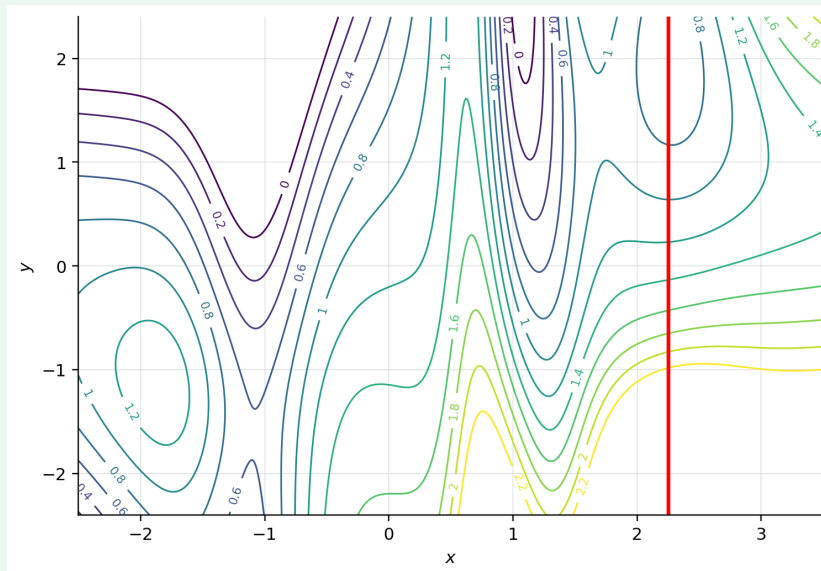


Figure 3.12. The slice of the function $F(x, y)$ where $y = 0$. The dashed lines correspond to increments of 0.2 in the x -axis. Adapted from: Stanford's MATH 51 textbook

Activity

Consider 3.13, which is the same contour plot as in 3.11, but with the vertical line $x = 2.25$ indicated instead. As in 3.4.1, read off the behaviour of the partial derivatives F_y from the contour plot.



Solution.

Here is some information that can be deduced from the contour plot:

- The contour values are always decreasing as we increase y along the red line, so that $F_y < 0$ on this line.
- The contour lines are closest together near $y = -1$, so $F_y(2.25, y)$ will have largest magnitude on the line near $y = -1$.

Generally, the spacing of contour lines corresponds to the magnitude of partial derivatives, and the sign of the partial derivatives corresponds to whether the function increases or decreasing in that direction.

Activity

Consider the function $f(x, y) = x(y^2 + 1)$, whose contour plot is indicated in 3.14. Five points are marked on the plot: $P = (0, 0)$, $Q = (3/2, 0)$, $R = (7/2, 0)$, $S = (2, 1)$, and $T = (2, -1)$. For each of these five points, determine from the contour plot whether f_x is positive, negative, or zero, and whether f_y is positive, negative, or zero. Then calculate these derivatives exactly and check whether your observations were correct.

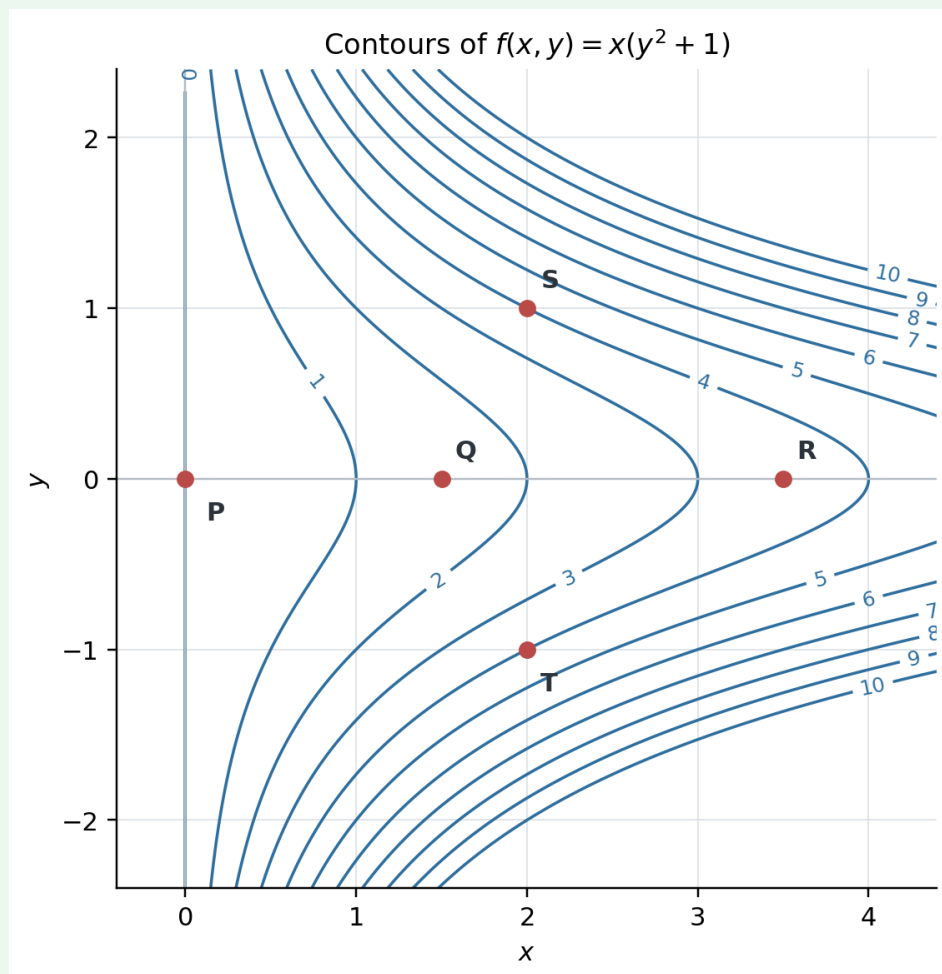


Figure 3.14. Contour plot for $f(x, y) = x(y^2 + 1)$ with increments of 1, and five points P , Q , R , S , and T indicated. Adapted from: Stanford's MATH 51 textbook

Solution.

At all five points, $f_x > 0$ because the contour values increase as increase values of x . The contour lines around points S and T are more tightly packed than around P , Q , and R , indicating that f_x has larger magnitude at S and T than at P , Q , and R .

At the point P , $f_y = 0$ since the contour line passing through P is a straight vertical line. It is not possible to determine the sign of f_y at the points Q and R , but we know the value of f_y must be small since the contour lines are more spread out than anywhere else in the plot. At the point S , $f_y > 0$, since the contour values increase as we increase in y values. At the point T , $f_y < 0$ since the contour values *decrease*.

We now verify these observations by computing partial derivatives. We check that

$$\frac{\partial f}{\partial x} = y^2 + 1 \quad \text{and} \quad \frac{\partial f}{\partial y} = 2xy$$

. So

- $f_x(P) = f_x(0, 0) = 1$ and $f_y(P) = f_y(0, 0) = 0$.
- $f_x(Q) = f_x(3/2, 0) = 1$ and $f_y(Q) = f_y(3/2, 0) = 0$.
- $f_x(R) = f_x(7/2, 0) = 1$ and $f_y(R) = f_y(7/2, 0) = 0$.
- $f_x(S) = f_x(2, 1) = 2$ and $f_y(S) = f_y(2, 1) = 4$.
- $f_x(T) = f_x(2, -1) = 2$ and $f_y(T) = f_y(2, -1) = -4$.

Partial derivatives with more variables

The concept of partial derivatives extends naturally to functions of more than two variables.

Definition 3.4.4.

For a function $f(\mathbf{x}) = f(x_1, \dots, x_n)$ of n variables, the partial derivative with respect to x_i is defined as

$$f_{x_i}(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(x_1, \dots, x_n)}{h}$$

provided this limit exists. Using vector notation, we can write this definition more cleanly, by writing

$$f_{x_i}(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{x} + h\mathbf{e}_i) - f(\mathbf{x})}{h}$$

, where $\mathbf{e}_1, \dots, \mathbf{e}_n$ are the standard basis vectors (recall 1.3.5). This partial derivative is also written as

$$\frac{\partial f}{\partial x_i} \quad \partial_{x_i} f \quad \text{or} \quad (D_i f)(\mathbf{x})$$

. If the variables $x_1, \dots, x_n$ are denoted using different symbols, these symbols might also be used to denote the partial derivatives instead. For instance, in $\mathbb{R}^3$, we might write a function as $f(x, y, z)$, and then the three partial derivatives of f may be denoted using any of the following four notations:

- $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$, and $\frac{\partial f}{\partial z}$.
- f_x , f_y , and f_z .
- $\partial_x f$, $\partial_y f$ and $\partial_z f$.
- $D_1 f$, $D_2 f$, and $D_3 f$.

As for functions of two variables, we can calculate these derivatives by treating all variables but that we are taking the partial derivative of as constants.

Activity

If $f(x, y, z) = 5x^2y^2z^3$, find f_x , f_y , and f_z .

Solution.

To find f_x , treat y and z as constants:

$$\begin{aligned}f_x(x, y, z) &= \frac{\partial}{\partial x}(5x^2y^2z^3) \\&= 5(2x)y^2z^3 \\&= 10xy^2z^3\end{aligned}$$

To find f_y , treat x and z as constants:

$$\begin{aligned}f_y(x, y, z) &= \frac{\partial}{\partial y}(5x^2y^2z^3) \\&= 5x^2(2y)z^3 \\&= 10x^2yz^3\end{aligned}$$

To find f_z , treat x and y as constants:

$$\begin{aligned}f_z(x, y, z) &= \frac{\partial}{\partial z}(5x^2y^2z^3) \\&= 5x^2y^2(3z^2) \\&= 15x^2y^2z^2\end{aligned}$$

Activity

Find f_x , f_y , and f_z where $f(x, y, z) = xy + yz + xz$.

Solution.

$$\begin{aligned}f_x(x, y, z) &= \frac{\partial}{\partial x}(xy + yz + xz) \\&= y + 0 + z \\&= y + z\end{aligned}$$

$$\begin{aligned}
 f_y(x, y, z) &= \frac{\partial}{\partial y}(xy + yz + xz) \\
 &= x + z + 0 \\
 &= x + z
 \end{aligned}$$

$$\begin{aligned}
 f_z(x, y, z) &= \frac{\partial}{\partial z}(xy + yz + xz) \\
 &= 0 + y + x \\
 &= x + y
 \end{aligned}$$

Activity

Find f_x and f_y where $z = f(x, y) = x \ln y + ye^x$.

Solution.

$$\begin{aligned}
 f_x(x, y) &= \frac{\partial}{\partial x}(x \ln y + ye^x) \\
 &= \ln y + ye^x \\
 &= \ln y + ye^x
 \end{aligned}$$

$$\begin{aligned}
 f_y(x, y) &= \frac{\partial}{\partial y}(x \ln y + ye^x) \\
 &= x \cdot \frac{1}{y} + 1 \cdot e^x \\
 &= \frac{x}{y} + e^x
 \end{aligned}$$

Higher-order partial derivatives

Given a scalar-valued function f , the partial derivatives of f (where defined) are also scalar-valued functions, so we can consider partial derivatives of that function as well, which leads to *higher order partial derivatives*.

Definition 3.4.5.

If f is a function of two variables, then the *second-order partial derivatives* of f are defined as:

- The derivative f_{xx} or $\frac{\partial^2 f}{\partial x^2}$, obtained by taking the partial derivative of f twice in the x

variable.

- The derivative f_{yx} or $\frac{\partial^2 f}{\partial y \partial x}$, obtained by *first* differentiating f in the x variable, and *then* differentiating f in the y variable.
- The derivative f_{xy} , or $\frac{\partial^2 f}{\partial x \partial y}$, obtained by *first* differentiating f in the y variable, and *then* differentiating f in the x variable.
- The derivative f_{yy} , or $\frac{\partial^2 f}{\partial y^2}$, obtained by differentiating f twice in the y variable.

Similarly, for a function $f(x_1, \dots, x_n)$, the higher derivatives of f are defined as $f_{x_i x_j} = \frac{\partial^2 f}{\partial x_i \partial x_j}$ for $1 \leq i, j \leq n$, obtained by first differentiating f in the variable x_j , and then differentiating f in the x_i variable.

Theorem 3.4.6 (Clairaut's Theorem).

Suppose a scalar-valued function f is defined in an open ball B that contains a point $\mathbf{a}$. If the functions $f_{x_i x_j}$ and $f_{x_j x_i}$ are well-defined and continuous on the ball B , then $f_{x_i x_j}(\mathbf{a}) = f_{x_j x_i}(\mathbf{a})$.

Activity

In this activity we compute all four partial derivatives of the function $f(x, y) = x \ln y + ye^x$.

Task

First compute f_x and f_y .

Solution.

We calculate that

$$f_x(x, y) = \ln y + ye^x$$

and

$$f_y(x, y) = x/y + e^x$$

.

Task

Next, compute f_{xx} and f_{yy} .

Solution.

We calculate f_{xx} by differentiating the function $f_x(x, y) = \ln y + ye^x$ in the x -variable, i.e.,

$$f_{xx}(x, y) = ye^x$$

. Similarly, we calculate f_{yy} by differentiating the function $f_y(x, y) = x/y + e^x$ in the y -variable, i.e.,

$$f_{yy}(x, y) = -x/y^2$$

.

Task

Now compute f_{xy} and f_{yx} .

Solution.

We compute f_{xy} by differentiating the function $f_y = x/y + e^x$ in the x -variable, i.e., so that

$$f_{xy}(x, y) = 1/y + e^x$$

. We compute f_{yx} by differentiating the function $f_x = \ln y + ye^x$ in the y -variable, i.e., so that

$$f_{yx} = 1/y + e^x$$

. Notice that both functions are continuous, and equal to one another, as guaranteed by 3.4.6.

Note

For most functions f encountered in practice, the second-order partial derivatives are continuous where defined, and so 3.4.6 applies. However, there are functions for which mixed partial derivatives are not equal to one another.

Activity

Find all four second partial derivatives of the function

$$f(x, y) = \ln(x^3 + y^2)$$

.

Solution.

We begin by calculating f_x and f_y using the chain rule, i.e., calculating that

$$f_x(x, y) = \frac{1}{x^3 + y^2}(3x^2) = \frac{3x^2}{x^3 + y^2}$$

and

$$f_y(x, y) = \frac{1}{x^3 + y^2}(2y) = \frac{2y}{x^3 + y^2}$$

. The quotient rule then tells us that

$$\begin{aligned}
 f_{xx} &= (f_x)_x \\
 &= \frac{(x^3 + y^2)(6x) - (3x^2)(3x^2)}{(x^3 + y^2)^2} \\
 &= \frac{6x^4 + 6xy^2 - 9x^4}{(x^3 + y^2)^2}
 \end{aligned}$$

that

$$\begin{aligned}
 f_{yy} &= (f_y)_y \\
 &= \frac{(x^3 + y^2)2 - (2y)(2y)}{(x^3 + y^2)^2} \\
 &= \frac{2x^3 - 2y^2}{(x^3 + y^2)^2}
 \end{aligned}$$

that

$$\begin{aligned}
 f_{xy} &= (f_y)_x \\
 &= \frac{(x^3 + y^2)(0) - (2y)(3x^2)}{(x^3 + y^2)^2} \\
 &= \frac{-6x^2y}{(x^3 + y^2)^2}
 \end{aligned}$$

and that

$$\begin{aligned}
 f_{yx} &= (f_x)_y \\
 &= \frac{(x^3 + y^2)(0) - (3x^2)(2y)}{(x^3 + y^2)^2} \\
 &= \frac{-6x^2y}{(x^3 + y^2)^2}
 \end{aligned}$$

Gradients and Hessians

The gradient collects first partial derivatives of a scalar-valued function.

Definition 3.4.7 (Gradient).

Let f be a scalar-valued function of n variables. At a point $\mathbf{x}$ where all first partial derivatives of f exist, the *gradient* of f is the vector

$$(\nabla f)(\mathbf{x}) = \begin{bmatrix} D_1 f(\mathbf{x}) \\ \vdots \\ D_n f(\mathbf{x}) \end{bmatrix}$$

. When these vectors are defined throughout a domain, they form a vector-valued function ∇f .

Activity

Let $f(x, y) = \sin(x + y^2)$. Find the gradient of f .

Solution.

We begin by computing the partial derivatives of f , that

$$f_x(x, y) = \cos(x + y^2)$$

and

$$f_y(x, y) = \cos(x + y^2)(2y) = 2y \cos(x + y^2)$$

. So

$$\nabla f(x, y) = \begin{bmatrix} \cos(x + y^2) \\ 2y \cos(x + y^2) \end{bmatrix}$$

.

Activity

Let $f(x, y) = x^2 + xy + y^2$. Compute $\nabla f(1, 2)$.

Solution.

We compute that

$$f_x(x, y) = 2x + y$$

and

$$f_y(x, y) = x + 2y$$

. So

$$\nabla f(x, y) = \begin{bmatrix} 2x + y \\ x + 2y \end{bmatrix}$$

and in particular,

$$\nabla f(1, 2) = \begin{bmatrix} 2(1) + (2) \\ (1) + 2(2) \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

An important property of the gradient of a function at a point is that it is a vector that is perpendicular to the level curves of the function.

Theorem 3.4.8.

Let $f(x, y)$ be a function with continuous partial derivatives. If (a, b) is a point where $\nabla f(a, b) \neq 0$, then $\nabla f(a, b)$ is *perpendicular to the level curve* of f passing through the point (a, b) .

See 3.15 for a graphical example of 3.4.8.

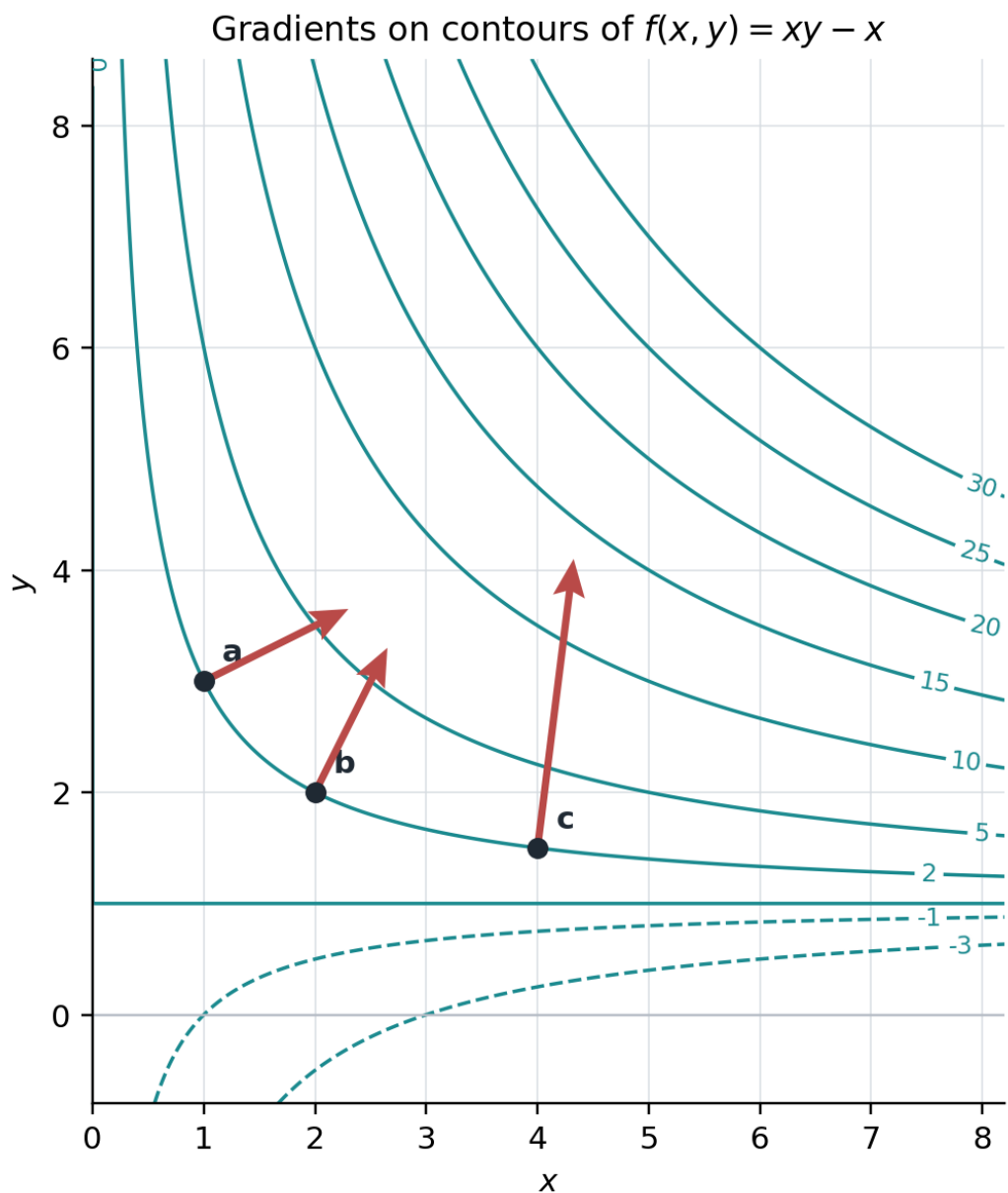


Figure 3.15. A contour plot of the function $f(x, y) = xy - x$, with the gradient vectors at three points $\mathbf{a} = (1, 3)$, $\mathbf{b} = (2, 2)$, and $\mathbf{c} = (4, 3/2)$. Adapted from: Stanford's MATH 51 textbook

The Hessian collects second partial derivatives.

Definition 3.4.9.

Let f be a scalar-valued function of n variables, and suppose the second partial derivatives of f exist at a point $\mathbf{x}$. The *Hessian matrix* of f at $\mathbf{x}$ is the $n \times n$ matrix of second partial derivatives:

$$H_f(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_2^2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(\mathbf{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_n \partial x_2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_n^2}(\mathbf{x}) \end{bmatrix}$$

That is, $H_f(\mathbf{x})$ is the matrix whose (i, j) -entry is the second derivative $\frac{\partial^2 f}{\partial x_i \partial x_j}$ obtained by first differentiating in the x_j variable, and then the x_i variable. When the relevant mixed second partial derivatives are continuous on a ball around $\mathbf{x}$, Clairaut's Theorem implies that the Hessian is symmetric at $\mathbf{x}$.

In particular, the Hessian of a scalar-valued function of two variables can be written

$$H_f = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$$

Activity

Find the Hessian matrix for the function

$$f(x, y) = \sin(x + y^2)$$

at a general point (x, y) .

Solution.

We start by computing the first and second partial derivatives of f . In 3.4.1 we computed that

$$f_x(x, y) = \cos(x + y^2) \quad \text{and} \quad f_y(x, y) = 2y \cos(x + y^2)$$

. We then calculate that

$$\begin{aligned} f_{xx} &= \frac{\partial}{\partial x} \{f_x\} \\ &= -\sin(x + y^2) \end{aligned}$$

, that

$$\begin{aligned}
 f_{yy} &= \frac{\partial}{\partial y}\{f_y\} \\
 &= \frac{\partial}{\partial y}\{2y \cos(x + y^2)\} \\
 &= 2 \cos(x + y^2) - 2y \sin(x + y^2)(2y) \\
 &= 2 \cos(x + y^2) - 4y^2 \sin(x + y^2)
 \end{aligned}$$

, that

$$\begin{aligned}
 f_{xy} &= \frac{\partial}{\partial x}\{f_y\} \\
 &= \frac{\partial}{\partial x}\{2y \cos(x + y^2)\} \\
 &= -2y \sin(x + y^2)
 \end{aligned}$$

, and that

$$\begin{aligned}
 f_{yx} &= \frac{\partial}{\partial y}\{f_x\} \\
 &= \frac{\partial}{\partial y}\{\cos(x + y^2)\} \\
 &= -\sin(x + y^2)(2y) \\
 &= -2y \sin(x + y^2)
 \end{aligned}$$

. Thus

$$\begin{aligned}
 H_f(x, y) &= \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} \\
 &= \begin{bmatrix} -\sin(x + y^2) & -2y \sin(x + y^2) \\ -2y \sin(x + y^2) & 2 \cos(x + y^2) - 4y^2 \sin(x + y^2) \end{bmatrix}
 \end{aligned}$$

Activity

Find the Hessian matrix of the function

$$f(x, y) = x^2 + xy + y^2$$

at a general point (x, y)

Solution.

We compute that

$$f_{xx} = \frac{\partial}{\partial x}\{2x + y\} = 2,$$

that

$$f_{xy} = \frac{\partial}{\partial x}\{x + 2y\} = 1$$

, that

$$f_{yx} = \frac{\partial}{\partial y}\{2x + y\} = 1$$

, and that

$$f_{yy} = \frac{\partial}{\partial y}\{x + 2y\} = 2$$

. So

$$H_f(x, y) = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

.

Activity

Let $f(x, y, z) = xy + yz + xz$. Compute its gradient ∇f and Hessian H_f at a general point (x, y, z) .

Solution.

We begin by computing the gradient. We calculate that

$$f_x(x, y, z) = y + 0 + z$$

, that

$$f_y(x, y, z) = x + z + 0$$

, and that

$$f_z(x, y, z) = 0 + y + x$$

. So

$$\nabla f(x, y, z) = \begin{bmatrix} y + z \\ x + z \\ y + x \end{bmatrix}$$

Next, we compute the second derivatives of f . We compute that $f_{xx} = f_{yy} = f_{zz} = 0$, and that $f_{xy} = f_{xz} = f_{yx} = f_{yz} = f_{zx} = f_{zy} = 1$. So

$$H_f(x, y, z) = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Jacobian matrices

For vector-valued maps, the Jacobian matrix collects the partial derivatives of all component functions.

The rows correspond to component functions. The columns describe how the output changes when one input coordinate changes.

Definition 3.4.10 (Jacobian matrix).

For a vector-valued map $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by

$$F(\mathbf{x}) = \begin{bmatrix} F_1(\mathbf{x}) \\ F_2(\mathbf{x}) \\ \vdots \\ F_m(\mathbf{x}) \end{bmatrix}$$

where $\mathbf{x} = (x_1, x_2, \dots, x_n)$, the *Jacobian matrix* of F is the $m \times n$ matrix:

$$J_F(\mathbf{x}) = \begin{bmatrix} \frac{\partial F_1}{\partial x_1} & \frac{\partial F_1}{\partial x_2} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \frac{\partial F_2}{\partial x_1} & \frac{\partial F_2}{\partial x_2} & \cdots & \frac{\partial F_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1} & \frac{\partial F_m}{\partial x_2} & \cdots & \frac{\partial F_m}{\partial x_n} \end{bmatrix}$$

The Jacobian matrix generalizes the concept of the derivative to a vector-valued map of multiple variables. It provides a compact representation of all first-order partial derivatives of a function.

Activity

Find the Jacobian matrix of $f(x, y, z) = xy + yz + xz$.

Solution.

For a scalar-valued function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the Jacobian matrix is the transpose of the gradient, i.e., $J_f = (\nabla f)^T$. So

$$J_f = (\nabla f)^T = [y + z \quad x + z \quad x + y]$$

Activity

Compute the Jacobian matrix of $F(x, y) = (x^2 - y^2, 2xy, x + y)$.

Solution.

We compute that

$$\begin{aligned} J_F &= \begin{bmatrix} D_1 F_1 & D_2 F_1 \\ D_1 F_2 & D_2 F_2 \\ D_1 F_3 & D_2 F_3 \end{bmatrix} \\ &= \begin{bmatrix} 2x & -2y \\ 2y & 2x \\ 1 & 1 \end{bmatrix} \end{aligned}$$

Activity

Compute the Jacobian matrix of the gradient function $\nabla f(x, y)$ for $f(x, y) = x^2 + xy + y^2$.

Solution.

First, let's compute the gradient of $f(x, y) = x^2 + xy + y^2$:

$$\begin{aligned} \frac{\partial f}{\partial x} &= 2x + y \\ \frac{\partial f}{\partial y} &= x + 2y \end{aligned}$$

So $\nabla f(x, y) = \begin{bmatrix} 2x + y \\ x + 2y \end{bmatrix}$. The Jacobian matrix of the gradient is

$$\begin{aligned} J_{\nabla f} &= \begin{bmatrix} D_1(\nabla f)_1 & D_2(\nabla f)_1 \\ D_1(\nabla f)_2 & D_2(\nabla f)_2 \end{bmatrix} \\ &= \begin{bmatrix} D_1\{2x + y\} & D_2\{2x + y\} \\ D_1\{x + 2y\} & D_2\{x + 2y\} \end{bmatrix} \\ &= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \end{aligned}$$

. Note that this is the *Hessian* of f (recall 3.4.9).

3.5 Local linearization and differentiability

3.5.1 Tangent planes and scalar local linearization

Linear approximations and differentiability

Recall that the linear approximation of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ of one variable at some point a is the affine function $L(x) = f(a) + f'(a)(x - a)$, and this function gives the best local affine approximation of f near a . Similarly, the partial derivatives of a function can be used to define a local approximation to a function of several variables. This leads to 3.5.1.

Definition 3.5.1 (Linear Approximation).

Given a scalar-valued function f , and a point $\mathbf{a}$ in the domain of f where the partial derivatives of f are well-defined, the *linear approximation* of f at the point $\mathbf{a}$ is the affine function

$$L(\mathbf{x}) = f(\mathbf{a}) + (\nabla f)(\mathbf{a})^T(\mathbf{x} - \mathbf{a})$$

, where we can expand the dot product that occurs in the equation, so that

$$L(\mathbf{x}) = f(\mathbf{a}) + \sum_{i=1}^n (D_i f)(\mathbf{a})(x_i - a_i)$$

. The approximation is affine in $\mathbf{x}$; its linear part is the map $\mathbf{h} \mapsto (\nabla f)(\mathbf{a})^T \mathbf{h}$ on small input changes.

In particular, the linear approximation of a scalar-valued function $f(x, y)$ at a point (a, b) is given by

$$L(x, y) = f(a, b) + \frac{\partial f}{\partial x}(a, b)(x - a) + \frac{\partial f}{\partial y}(a, b)(y - b)$$

Activity

Consider the function

$$f(x, y, z) = x^2y + yz^2$$

Task

Find the linear approximation of the function

$$f(x, y, z) = x^2y + yz^2$$

at the point $(1, 2, 0)$.

Solution.

We begin by finding the gradient of f at $(1, 2, 0)$. We compute that

$$\frac{\partial f}{\partial x} = 2xy$$

, that

$$\frac{\partial f}{\partial y} = x^2 + z^2$$

, and that

$$\frac{\partial f}{\partial z} = 2yz$$

. So, evaluating these quantities at $(1, 2, 0)$, we find that

$$\nabla f(1, 2, 0) = \begin{bmatrix} 2(1)(2) \\ (1)^2 + (0)^2 \\ 2(2)(0) \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}$$

. We also calculate that

$$f(1, 2, 0) = (1)^2(2) + (2)(0)^2 = 2$$

. So the linear approximation to f at $(1, 2, 0)$ is the function

$$\begin{aligned} L(x, y, z) &= 2 + \begin{bmatrix} 4 & 1 & 0 \end{bmatrix} \begin{bmatrix} x - 1 \\ y - 2 \\ z - 0 \end{bmatrix} \\ &= 2 + 4(x - 1) + 1(y - 2) \\ &= 4x + y - 4 \end{aligned}$$

.

Task

Use the linear approximation you have found to approximate the value $f(1.04, 1.97, 0.05)$.

Solution.

We calculate that the linear approximation to f is the function

$$L(x, y, z) = 2 + 4(x - 1) + (y - 2)$$

. So

$$\begin{aligned} f(1.04, 1.97, 0.05) &\approx L(1.04, 1.97, 0.05) \\ &= 2 + 4(0.04) + (-0.03) \\ &= 2 + 0.16 - 0.03 \\ &= 2.13 \end{aligned}$$

.

Now suppose a function f of a single variable is differentiable at a point a , i.e., so that the limit

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

exists. We can rearrange this equation to read

$$\lim_{x \rightarrow a} \frac{f(x) - f(a) - f'(a)(x - a)}{x - a} = 0$$

, i.e., so that

$$\lim_{x \rightarrow a} \frac{f(x) - L(x)}{x - a} = 0$$

, where $L(x) = f(a) + f'(a)(x - a)$ is the linear approximation of f at a . A function of several variables is *differentiable* when the same property holds, with the linear approximation given by the partial derivatives of the function.

Definition 3.5.2 (Differentiability).

A scalar-valued function f is *differentiable* at a point $\mathbf{a}$ if the gradient of f is well-defined at $\mathbf{a}$, and if

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} \frac{f(\mathbf{x}) - L(\mathbf{x})}{\|\mathbf{x} - \mathbf{a}\|} = 0$$

, where

$$L(\mathbf{x}) = f(\mathbf{a}) + (\nabla f)(\mathbf{a})^T(\mathbf{x} - \mathbf{a})$$

is the linear approximation of f at $\mathbf{a}$. A vector-valued function is differentiable at a point precisely when each of its component functions is differentiable at that point.

Note

The equation

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} \frac{f(\mathbf{x}) - L(\mathbf{x})}{\|\mathbf{x} - \mathbf{a}\|} = 0$$

can only hold if L is a *good* linear approximation of f near $\mathbf{a}$, so that the magnitude of the numerator is much smaller than the magnitude of the denominator when $\mathbf{x}$ is near $\mathbf{a}$, i.e., when $\|\mathbf{x} - \mathbf{a}\|$ is small.

The following theorem is the main way we will determine if a function is differentiable.

Theorem 3.5.3 (Continuity of First Partial Derivatives Implies Differentiability).

Let f be a scalar-valued function, and suppose $\mathbf{a}$ is an interior point (recall 3.2.6) of the domains of f and its partial derivatives. Moreover, suppose that all of the partial derivatives $D_1 f, \dots, D_n f$ are continuous at $\mathbf{a}$. Then f is differentiable at $\mathbf{a}$.

Tangent planes

Activity

Find an equation for the plane containing the lines L_1 and L_2 , where L_1 is the line parameterized by the function

$$\mathbf{r}(t) = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} + t \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$$

and where L_2 is the line parameterized by the function

$$\mathbf{r}(s) = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} + s \begin{bmatrix} -3 \\ 1 \\ 4 \end{bmatrix}$$

.

Solution.

Recall 2.2.3, which specifies a plane via an equation of the form $\mathbf{n} \cdot (\mathbf{x} - \mathbf{x}_0) = 0$, where $\mathbf{n}$ is a vector normal to the plane, and $\mathbf{x}_0$ is a point on the plane. We can take $\mathbf{x}_0 = (1, 1, -2)$. In order for $\mathbf{n}$ to be the normal vector to the plane containing L_1 and L_2 , it must be perpendicular to the two direction vectors $(2, 3, 1)$ and $(-3, 1, 4)$, i.e., we must have $\mathbf{n} \cdot (2, 3, 1) = \mathbf{n} \cdot (-3, 1, 4) = 0$. If we write $\mathbf{n} = (a, b, c)$, then expanding these dot products gives a homogeneous system

$$\begin{aligned} 2a + 3b + c &= 0 \\ -3a + b + 4c &= 0 \end{aligned}$$

. Let's convert the left hand side to matrix form, i.e., writing the homogeneous system as

$$\begin{bmatrix} 2 & 3 & 1 \\ -3 & 1 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \mathbf{0}$$

. Recalling Homogeneous systems and basic solutions, we solve this homogeneous system by performing row operations to the matrix on the left hand side, from which we obtain that

$$\begin{bmatrix} 2 & 3 & 1 \\ -3 & 1 & 4 \end{bmatrix} \xrightarrow[R_2 \rightarrow R_2 + \frac{3}{2}R_1]{R_1 \rightarrow \frac{1}{2}R_1} \begin{bmatrix} 1 & \frac{3}{2} & \frac{1}{2} \\ 0 & \frac{11}{2} & \frac{11}{2} \end{bmatrix} \xrightarrow[R_1 \rightarrow R_1 - \frac{3}{11}R_2]{R_2 \rightarrow \frac{2}{11}R_2} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}$$

This row reduction tells us that the homogeneous system above is equivalent to the system

$$\begin{aligned} a - c &= 0 \\ b + c &= 0 \end{aligned}$$

. Setting $c = 1$ gives a normal vector $\mathbf{n} = (1, -1, 1)$, and any solution to the homogeneous equation is a scalar multiple of this vector (recalling terminology from 2.2.17, it is a *basic solution*). Thus the equation

$$\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \cdot \left(\mathbf{x} - \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \right) = 0$$

is an equation for the plane, which, if we write $\mathbf{x} = (x, y, z)$, can be expanded as

$$(1)(x - 1) + (-1)(y - 1) + (1)(z + 2) = 0$$

which we simplify to

$$x - y + z = -2$$

Definition 3.5.4 (Tangent Plane).

Let S be a surface, and $\mathbf{x}_0$ a point on S . A *tangent plane* to S at $\mathbf{x}_0$ is a plane that contains the tangent lines of any differentiable curve contained in S as it passes through $\mathbf{x}_0$.

A tangent plane touching a surface along tangent lines through a point.

A curved orange surface S is shown with a translucent blue plane touching it at the labeled point $P_0(x_0, y_0, z_0)$. Several colored curves pass through P_0 on the surface, and straight tangent lines through the same point lie in the plane. The drawing emphasizes that all tangent directions to curves through P_0 sit inside the tangent plane.

The tangent plane to a surface S at a point P_0 contains all the tangent lines to curves in S that pass through P_0 . Figure 4.27 from Edwin “Jed” Herman and Gilbert Strang, *Calculus Volume 3*, OpenStax, © 2018 Rice University, licensed under CC BY-NC-SA 4.0; source: OpenStax Figure 4.27.

Theorem 3.5.5 (Equation of the Tangent Plane).

Suppose S is the graph of a function $f(x, y)$ which is differentiable at a point (x_0, y_0) . Then the tangent plane to S exists, is unique, and is described by the equation

$$z - f(x_0, y_0) = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

. Note that this plane is the graph of the linear approximation L to f at (x_0, y_0) .

Proof.

We verify that the plane described by the equation above is the only possible tangent plane. Recall from 3.4.3 that the curve obtained by the intersection of S with the plane $y = y_0$ has slope $f_x(x_0, y_0)$ at the point (x_0, y_0) , and the tangent line to the curve at this point is by a line with direction vector $(1, 0, f_x(x_0, y_0))$. Similarly, the curve obtained by the intersection of S with the plane $x = x_0$ has slope $f_y(x_0, y_0)$, and so the tangent line to the curve at this point is a line with direction vector $(0, 1, f_y(x_0, y_0))$. But now using the method of 3.5.1, we see that a normal vector to the plane containing these two lines is of the form $\mathbf{n} = (-f_x(x_0, y_0), -f_y(x_0, y_0), 1)$. So the tangent plane to S at $(x_0, y_0, f(x_0, y_0))$ must be described by the equation

$$(-f_x(x_0, y_0), -f_y(x_0, y_0), 1) \cdot (x - x_0, y - y_0, z - f(x_0, y_0)) = 0$$

, an equation we may rearrange to read

$$z - f(x_0, y_0) = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Activity

Find an equation of the tangent plane to the surface

$$z = e^x \cos y$$

at $(0, \frac{\pi}{2}, 0)$.

Solution.

The surface is the graph of the equation $f(x, y) = e^x \cos y$, so we may use 3.5.5 to find the equation of the tangent plane. First, we calculate that

$$f(0, \pi/2) = e^0 \cos(\pi/2) = 0$$

. Next, we calculate that

$$f_x(x, y) = e^x \cos y \quad \text{and} \quad f_y(x, y) = -e^x \sin y$$

so $f_x(0, \pi/2) = e^0 \cos(\pi/2) = 0$ and $f_y(0, \pi/2) = -e^0 \sin(\pi/2) = -1$. This means that an equation describing the tangent plane is given by

$$z - 0 = 0(x - 0) + (-1)(y - \pi/2)$$

which simplifies to

$$z + y = \pi/2$$

3.5.2 Vector-valued local linearization

Linear maps reappear locally

A nonlinear map is usually not linear globally. Near one input, however, differentiability replaces small input changes by a linear map.

$$F(\mathbf{a} + \mathbf{h}) \approx F(\mathbf{a}) + J_F(\mathbf{a})\mathbf{h}.$$

$\mathbf{x} \mapsto F(\mathbf{a}) + J_F(\mathbf{a})(\mathbf{x} - \mathbf{a})$ is affine in $\mathbf{x}$. $\mathbf{h} \mapsto J_F(\mathbf{a})\mathbf{h}$ is the linear map on input changes.

Linear algebra gave us global linear maps $\mathbf{x} \mapsto A\mathbf{x}$. Calculus gives us local linear maps $\mathbf{h} \mapsto J_F(\mathbf{a})\mathbf{h}$. The same matrix language therefore reappears inside nonlinear problems.

Definition 3.5.6 (Linear Approximation of Vector-Valued Functions).

Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a vector-valued function. Provided that the partial derivatives of the components of F all exist at $\mathbf{a}$, so that the Jacobian matrix J_F is well-defined, the *linear approximation* of F at $\mathbf{a}$ is the vector-valued affine function

$$L(\mathbf{x}) = F(\mathbf{a}) + J_F(\mathbf{a})(\mathbf{x} - \mathbf{a})$$

where $J_F(\mathbf{a})(\mathbf{x} - \mathbf{a})$ is the Jacobian matrix of F evaluated at $\mathbf{a}$, and then applied to the vector $\mathbf{x} - \mathbf{a}$. The approximation is affine in $\mathbf{x}$; its linear part is the map $\mathbf{h} \mapsto J_F(\mathbf{a})\mathbf{h}$ on small input changes. The components of L are precisely the linear approximations of the components of F , i.e., $L_i(\mathbf{x}) = F_i(\mathbf{a}) + (\nabla F_i)(\mathbf{a})^T(\mathbf{x} - \mathbf{a})$.

Note

Geometrically, the graph of the linear approximation to a vector-valued function can be visualized as a higher dimensional “tangent plane” of the graph of the function F at $\mathbf{a}$.

Activity

Let $F(x, y) = (x^2y, xe^y)$. Find the linear approximation of F at the point $(1, 0)$, and use it to approximate $F(1.1, -0.2)$.

Solution.

We compute that

$$\begin{aligned} J_F &= \begin{bmatrix} \frac{\partial}{\partial x}\{x^2y\} & \frac{\partial}{\partial y}\{x^2y\} \\ \frac{\partial}{\partial x}\{xe^y\} & \frac{\partial}{\partial y}\{xe^y\} \end{bmatrix} \\ &= \begin{bmatrix} 2xy & x^2 \\ e^y & xe^y \end{bmatrix} \end{aligned}$$

. In particular,

$$J_F(1, 0) = \begin{bmatrix} 2(1)(0) & (1)^2 \\ e^{(0)} & (1)e^{(0)} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

. Since

$$F(1, 0) = \begin{bmatrix} (1)^2(0) \\ (1)e^{(0)} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

. So the linear approximation of F at $(1, 0)$ is the function

$$\begin{aligned}
 L \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x-1 \\ y \end{bmatrix} \\
 &= \begin{bmatrix} 1y \\ 1 + (x-1) + y \end{bmatrix} \\
 &= \begin{bmatrix} y \\ x+y \end{bmatrix}
 \end{aligned}$$

. So

$$F(1.1, -0.2) \approx L(1.1, -0.2) = \begin{bmatrix} -0.2 \\ 1.1 - 0.2 \end{bmatrix} = \begin{bmatrix} -0.2 \\ 0.9 \end{bmatrix}$$

.

Activity: Local square-grid visualization

Return to

$$F \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} x + y^2 \\ y \end{bmatrix}.$$

The Jacobian matrix is

$$J_F(x, y) = \begin{bmatrix} 1 & 2y \\ 0 & 1 \end{bmatrix}.$$

Let

$$\mathbf{a} = \begin{bmatrix} 0 \\ 1/2 \end{bmatrix}.$$

1. Compute $J_F(\mathbf{a})$.
2. Let $\mathbf{h} = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}$. Compute $F(\mathbf{a} + \mathbf{h}) - F(\mathbf{a})$.
3. Compute $J_F(\mathbf{a})\mathbf{h}$.
4. Compute the error

$$F(\mathbf{a} + \mathbf{h}) - F(\mathbf{a}) - J_F(\mathbf{a})\mathbf{h}.$$

5. If $0 \leq h_2 \leq r$, how large can the first coordinate of the error be?
6. What happens to the local square-grid picture as r gets smaller?

Solution.

Since $a_2 = 1/2$,

$$J_F(\mathbf{a}) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

This is the horizontal shear matrix from Unit 1.

Now

$$\mathbf{a} + \mathbf{h} = \begin{bmatrix} h_1 \\ 1/2 + h_2 \end{bmatrix}.$$

Thus

$$F(\mathbf{a} + \mathbf{h}) = \begin{bmatrix} h_1 + (1/2 + h_2)^2 \\ 1/2 + h_2 \end{bmatrix} = \begin{bmatrix} h_1 + 1/4 + h_2 + h_2^2 \\ 1/2 + h_2 \end{bmatrix}.$$

Also,

$$F(\mathbf{a}) = \begin{bmatrix} 1/4 \\ 1/2 \end{bmatrix}.$$

Therefore

$$F(\mathbf{a} + \mathbf{h}) - F(\mathbf{a}) = \begin{bmatrix} h_1 + h_2 + h_2^2 \\ h_2 \end{bmatrix}.$$

The Jacobian matrix prediction is

$$J_F(\mathbf{a})\mathbf{h} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} = \begin{bmatrix} h_1 + h_2 \\ h_2 \end{bmatrix}.$$

So the error is

$$F(\mathbf{a} + \mathbf{h}) - F(\mathbf{a}) - J_F(\mathbf{a})\mathbf{h} = \begin{bmatrix} h_2^2 \\ 0 \end{bmatrix}.$$

If $0 \leq h_2 \leq r$, then $0 \leq h_2^2 \leq r^2$. As the square around $\mathbf{a}$ shrinks, the nonlinear image and the local linear image become closer.

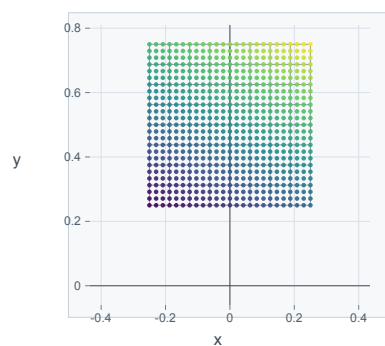
The local linear map acts on the change vector $\mathbf{h}$, not on the original input vector $\mathbf{x}$.

Note

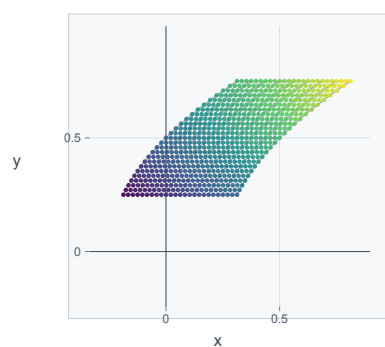
The local square-grid visualization does not produce one matrix for the whole nonlinear map. It shows one matrix approximation near one base point.

Larger square

Input grid, $r=0.25$



Nonlinear image



Local linear image

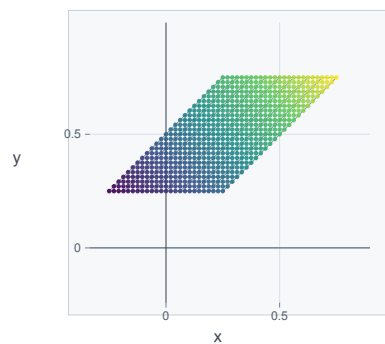


Figure 3.16. Local square-grid visualization near $\mathbf{a} = \begin{bmatrix} 0 \\ 1/2 \end{bmatrix}$ for the larger input square $r = 0.25$. The three panels show the input grid, the nonlinear image, and the local linear image on ticked axes.

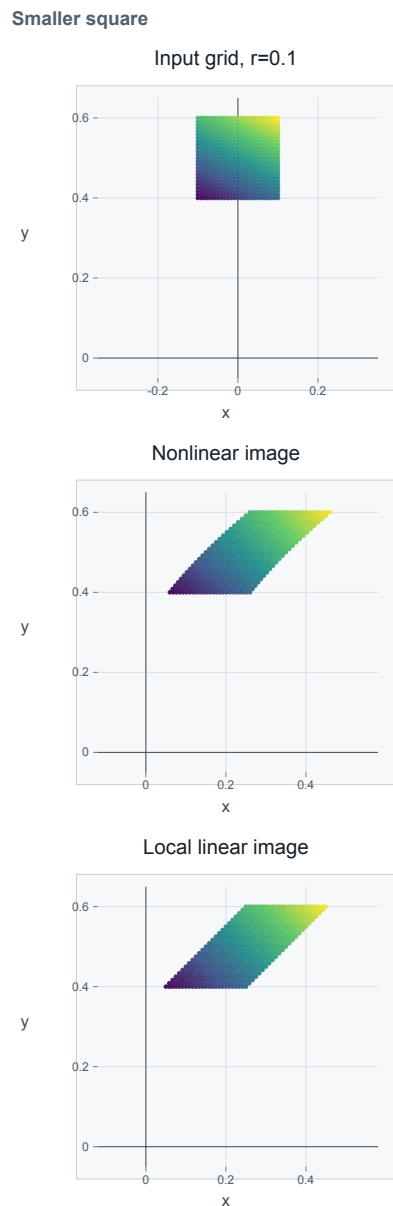


Figure 3.17. Local square-grid visualization near $\mathbf{a} = \begin{bmatrix} 0 \\ 1/2 \end{bmatrix}$ for the smaller input square $r = 0.1$. The nonlinear image is closer to the local linear image than in the larger-square visualization.

The approximation in this unit is local. Near one input $\mathbf{a}$, a differentiable function behaves like

$$F(\mathbf{a} + \mathbf{h}) \approx F(\mathbf{a}) + J_F(\mathbf{a})\mathbf{h}.$$

Unit 4 studies a different kind of approximation:

$$\min_x \|Ax - b\|.$$

That is a global best approximation by a vector in a subspace. Unit 3 asks what a nonlinear function looks like near one point; Unit 4 asks which allowable vector is closest overall.

3.6 Chain rule as matrix multiplication

Unit 1 used matrix multiplication to represent composition of linear maps: applying B first and then A gives $A(B\mathbf{x}) = (AB)\mathbf{x}$. For differentiable nonlinear maps, the same idea reappears locally. The local linear maps compose, so their Jacobian matrices multiply.

3.6.1 The chain rule

The chain rule describes derivatives of composite functions.

Compositions

Definition 3.6.1 (Composite Function).

If $G : \mathbb{R}^n \rightarrow \mathbb{R}^p$ and $F : \mathbb{R}^p \rightarrow \mathbb{R}^m$ are multivariable functions, we can form a new *composite function* as follows. Take an input in $\mathbb{R}^n$, and then apply G to the input to obtain a vector in $\mathbb{R}^p$. Apply F to this vector to obtain a vector in $\mathbb{R}^m$. This is the output of the composite function. In other words, our input is fed through the following diagram

$$\mathbb{R}^n \xrightarrow{G} \mathbb{R}^p \xrightarrow{F} \mathbb{R}^m$$

. We write the composite function as $F \circ G$ (the symbol $\circ$ should be read as “composed with”). In symbols, the composite function is defined as

$$(F \circ G)(\mathbf{x}) = F(G(\mathbf{x}))$$

Note

Note that the composite $F \circ G$ is *only* defined when the output of G is a possible input to F , i.e., the output of G is $\mathbb{R}^p$, which is the space of inputs to F .

Unit 1 used matrix multiplication to represent composition of linear maps: applying B first and then A gives $A(B\mathbf{x}) = (AB)\mathbf{x}$.

Activity

Let $T_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ and $T_2 : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be linear maps defined by their component functions as follows:

$$T_1(x, y) = \begin{bmatrix} 2x - y \\ 3y \\ x + y \end{bmatrix} \quad T_2(u, v, w) = \begin{bmatrix} u + v - w \\ 2u - v \end{bmatrix}$$

Task

Find the matrices A_1 and A_2 associated with these maps, so that $T_1 = T_{A_1}$ and $T_2 = T_{A_2}$ (recall the notation from 1.3.6).

Solution.

We recall the method of 1.3.9. In particular, we obtain the first column of A_1 from the entries of $T_1(\mathbf{e}_1)$, and the second column from $T_1(\mathbf{e}_2)$. We calculate that

$$T_1(\mathbf{e}_1) = T_1(1, 0) = \begin{bmatrix} 2 \cdot 1 - 0 \\ 3 \cdot 0 \\ 1 + 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

and that

$$T_1(\mathbf{e}_2) = T_1(0, 1) = \begin{bmatrix} 2 \cdot 0 - 1 \\ 3 \cdot 1 \\ 0 + 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix}$$

. So the matrix A_1 is equal to

$$A_1 = \begin{bmatrix} 2 & -1 \\ 0 & 3 \\ 1 & 1 \end{bmatrix}$$

Similarly, we calculate that

$$T_2(\mathbf{e}_1) = T_2(1, 0, 0) = \begin{bmatrix} 1 + 0 - 0 \\ 2 \cdot 1 - 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

, that

$$T_2(\mathbf{e}_2) = T_2(0, 1, 0) = \begin{bmatrix} 0 + 1 - 0 \\ 2 \cdot 0 - 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

, and that

$$T_2(\mathbf{e}_3) = T_2(0, 0, 1) = \begin{bmatrix} 0 + 0 - 1 \\ 2 \cdot 0 - 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

. So the matrix A_2 is given by

$$A_2 = \begin{bmatrix} 1 & 1 & -1 \\ 2 & -1 & 0 \end{bmatrix}$$

Task

Calculate the composite map $T_2 \circ T_1$ and express it as a matrix map from $\mathbb{R}^2$ to $\mathbb{R}^2$.

Solution.

Recalling 1.4.2, we have that $T_2 \circ T_1 = T_{A_2 A_1}$. Now

$$\begin{aligned}
A_2 \cdot A_1 &= \begin{bmatrix} 1 & 1 & -1 \\ 2 & -1 & 0 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 0 & 3 \\ 1 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 \cdot 2 + 1 \cdot 0 + (-1) \cdot 1 & 1 \cdot (-1) + 1 \cdot 3 + (-1) \cdot 1 \\ 2 \cdot 2 + (-1) \cdot 0 + 0 \cdot 1 & 2 \cdot (-1) + (-1) \cdot 3 + 0 \cdot 1 \end{bmatrix} \\
&= \begin{bmatrix} 2 + 0 - 1 & -1 + 3 - 1 \\ 4 + 0 + 0 & -2 - 3 + 0 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 1 \\ 4 & -5 \end{bmatrix}
\end{aligned}$$

Therefore, the composite map $T_2 \circ T_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ can be expressed as:

$$(T_2 \circ T_1) \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 4 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + y \\ 4x - 5y \end{bmatrix}$$

Here is a nonlinear example of the composition of two functions.

Activity

Consider $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $G : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$F(u, v) = (uv, u + v) \quad \text{and} \quad G(x, y) = (e^{xy}, x - y)$$

. Find expressions for $(F \circ G)(x, y)$ and $(G \circ F)(u, v)$.

Solution.

We calculate that

$$\begin{aligned}
(F \circ G)(x, y) &= F(G(x, y)) \\
&= F(e^{xy}, x - y) \\
&= (e^{xy}(x - y), e^{xy} + (x - y)) \\
&= (e^{xy}x - e^{xy}y, e^{xy} + x - y)
\end{aligned}$$

, and that

$$\begin{aligned}
(G \circ F)(u, v) &= G(F(u, v)) \\
&= G(uv, u + v) \\
&= (e^{uv(u+v)}, uv - (u + v)) \\
&= (e^{u^2v + uv^2}, uv - u - v)
\end{aligned}$$

Warning

Observe that $F \circ G$ and $G \circ F$ are very different functions. We must therefore be careful that we compose functions in the correct order.

The order matters. The rightmost Jacobian matrix acts first on the input change.

Activity

Consider $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^3$ and $\mathbf{p} : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by

$$\mathbf{r}(t) = (t, \cos(t), \sin(t)) \quad \text{and} \quad \mathbf{p}(x, y, z) = (y, z)$$

Task

Find an expression for $(\mathbf{p} \circ \mathbf{r})(t)$.

Solution.

We calculate that

$$\begin{aligned}(\mathbf{p} \circ \mathbf{r})(t) &= \mathbf{p}(\mathbf{r}(t)) \\ &= \mathbf{p}(t, \cos(t), \sin(t)) \\ &= (\cos(t), \sin(t))\end{aligned}$$

Task

Is the composition $\mathbf{r} \circ \mathbf{p}$ defined? Why or why not?

Solution.

For the composition $\mathbf{r} \circ \mathbf{p}$ to be defined, the outputs of $\mathbf{p}$ must be contained in the inputs of $\mathbf{r}$. But $\mathbf{p}$ outputs points in $\mathbb{R}^2$, whereas $\mathbf{r}$ takes in points in $\mathbb{R}$ as input, so the composition does not make sense.

Chain rule for one-parameter inputs

Recall the chain rule from single-variable calculus, that for two functions $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$, we have

$$(f \circ g)'(t) = (f' \circ g)(t)g'(t)$$

The next theorem is the component form of the chain rule.

Recall that for a function $f(x, y)$, $D_1 f = f_x = \frac{\partial f}{\partial x}$, and that $D_2 f = f_y = \frac{\partial f}{\partial y}$.

Theorem 3.6.2 (Chain Rule for One Independent Variable).

Consider a scalar-valued function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, and a vector-valued function $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^2$ of a single variable. Then the composition $f \circ \mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}$ is a scalar-valued function of a single variable. If $\mathbf{r}$ is differentiable at t_0 , and f is differentiable at $\mathbf{r}(t_0)$, then $f \circ \mathbf{r}$ is differentiable at t_0 and

$$(f \circ \mathbf{r})'(t_0) = (D_1 f \circ \mathbf{r})(t_0)r_1'(t_0) + (D_2 f \circ \mathbf{r})(t_0)r_2'(t_0)$$

, where r_1 and r_2 are the components of $\mathbf{r}$.

Proof.

We note that the right hand side of the equation can be written as $L \circ \mathbf{r} - f(\mathbf{r}(t_0))$, where L is the linear approximation of f at $\mathbf{r}(t_0)$. Because f is differentiable, we know that

$$\lim_{\mathbf{x} \rightarrow \mathbf{r}(t_0)} \frac{f(\mathbf{x}) - L(\mathbf{x})}{\|\mathbf{x} - \mathbf{r}(t_0)\|} = 0$$

. Now

$$\lim_{t \rightarrow t_0} \frac{\|\mathbf{r}(t) - \mathbf{r}(t_0)\|}{t - t_0} = \left\| \lim_{t \rightarrow t_0} \frac{\mathbf{r}(t) - \mathbf{r}(t_0)}{t - t_0} \right\| = \|\mathbf{r}'(t_0)\|$$

. Using these two equations (substituting $\mathbf{r}(t)$ for $\mathbf{x}$ in the first equation), we find that

$$\begin{aligned} \lim_{t \rightarrow t_0} \frac{f(\mathbf{r}(t)) - L(\mathbf{r}(t))}{t - t_0} &= \lim_{t \rightarrow t_0} \frac{f(\mathbf{r}(t)) - L(\mathbf{r}(t))}{\|\mathbf{r}(t) - \mathbf{r}(t_0)\|} \frac{\|\mathbf{r}(t) - \mathbf{r}(t_0)\|}{t - t_0} \\ &= 0 \cdot \|\mathbf{r}'(t_0)\| = 0 \end{aligned}$$

. But recalling that

$$L(x, y) = f(\mathbf{r}(t_0)) + D_1 f(\mathbf{r}(t_0))(x - r_1(t_0)) + D_2 f(\mathbf{r}(t_0))(y - r_2(t_0))$$

, we conclude that

$$\begin{aligned} \lim_{t \rightarrow t_0} \frac{f(\mathbf{r}(t)) - f(\mathbf{r}(t_0))}{t - t_0} &= \lim_{t \rightarrow t_0} \frac{f(\mathbf{r}(t)) - f(\mathbf{r}(t_0))}{t - t_0} - \frac{f(\mathbf{r}(t)) - L(\mathbf{r}(t))}{t - t_0} \\ &= \lim_{t \rightarrow t_0} \frac{L(\mathbf{r}(t)) - f(\mathbf{r}(t_0))}{t - t_0} \\ &= \lim_{t \rightarrow t_0} D_1 f(\mathbf{r}(t_0)) \frac{r_1(t) - r_1(t_0)}{t - t_0} \\ &\quad + D_2 f(\mathbf{r}(t_0)) \frac{r_2(t) - r_2(t_0)}{t - t_0} \\ &= f_x(\mathbf{r}(t_0))r_1'(t_0) + D_2 f(\mathbf{r}(t_0))r_2'(t_0) \end{aligned}$$

.

Note

There are several ways to aid in remembering the formula for the chain rule. The first is to simplify the equation, writing it as

$$(f \circ \mathbf{r})'(t) = (\nabla f \circ \mathbf{r})(t) \cdot \mathbf{r}'(t),$$

so that the equation looks more like the chain rule from single variable calculus. The second is the following mnemonic: if we write z for $f \circ \mathbf{r}$ and for f , and write the components of $\mathbf{r}$ as x and y , then the chain rule reads that

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

, so that the symbols dx and dy in the numerators on the right hand side, and the symbols ∂x and ∂y that occur in the denominators appear to ‘cancel’ like when multiplying fractions.

Activity

Suppose $z = x^2y - y^2x$, where $x = \sin t$ and $y = e^t$. Compute dz/dt .

Solution.

First, we need to calculate

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}(x^2y - y^2x) = 2xy - y^2$$

$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}(x^2y - y^2x) = x^2 - 2yx$$

$$\frac{dx}{dt} = \frac{d}{dt}(\sin t) = \cos t$$

$$\frac{dy}{dt} = \frac{d}{dt}(e^t) = e^t$$

Now we substitute these into the chain rule formula:

$$\begin{aligned} \frac{dz}{dt} &= (2xy - y^2) \cdot \cos t + (x^2 - 2yx) \cdot e^t \\ &= (2 \sin t \cdot e^t - e^{2t}) \cdot \cos t + (\sin^2 t - 2e^t \cdot \sin t) \cdot e^t \\ &= 2 \sin t \cdot e^t \cdot \cos t - e^{2t} \cdot \cos t + e^t \sin^2 t - 2e^{2t} \sin t \end{aligned}$$

We can simplify this further if we want:

$$\begin{aligned} \frac{dz}{dt} &= 2e^t \sin t \cos t - e^{2t} \cos t + e^t \sin^2 t - 2e^{2t} \sin t \\ &= e^t(\sin^2 t + 2 \sin t \cos t) - e^{2t}(\cos t + 2 \sin t) \\ &= e^t(\sin^2 t + \sin 2t) - e^{2t}(\cos t + 2 \sin t) \end{aligned}$$

Therefore,

$$\frac{dz}{dt} = e^t(\sin^2 t + \sin 2t) - e^{2t}(\cos t + 2 \sin t)$$

Activity

If $f(x, y) = x^2 + y^2$, and $\mathbf{r}(t) = (\sin t, \cos(2t))$, find $(f \circ \mathbf{r})'(t)$.

Solution.

We calculate that

$$\nabla f(x, y) = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

and that

$$\mathbf{r}'(t) = \begin{bmatrix} \cos t \\ -2 \sin(2t) \end{bmatrix}$$

. Now

$$\nabla f \circ \mathbf{r}(t) = \nabla f(\sin t, \cos(2t)) = \begin{bmatrix} 2 \sin t \\ 2 \cos(2t) \end{bmatrix}$$

Thus

$$\begin{aligned} (f \circ \mathbf{r})'(t) &= \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \\ &= \begin{bmatrix} 2 \sin t \\ 2 \cos(2t) \end{bmatrix} \cdot \begin{bmatrix} \cos t \\ -2 \sin(2t) \end{bmatrix} \\ &= 2 \sin t \cos t - 4 \sin(2t) \cos(2t) \end{aligned}$$

.

Chain rule for two independent variables

Theorem 3.6.3 (Chain Rule for Two Independent Variables).

Suppose $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a scalar-valued function, and $G : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a vector-valued function. Then $f \circ G : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a scalar-valued function, and the two equations

$$D_1(f \circ G) = (D_1 f \circ G)D_1 G_1 + (D_2 f \circ G)D_1 G_2$$

and

$$D_2(f \circ G) = (D_1 f \circ G)D_2 G_1 + (D_2 f \circ G)D_2 G_2$$

hold.

Proof.

Fix the second variable, and apply 3.6.2 to the first to obtain the formula for $(f \circ G)_x$. Then fix the first variable and apply 3.6.2 to the second variable to obtain the formula for $(f \circ G)_y$.

Remark

Recall from Section 3.4 that the Jacobian matrix collects the partial derivatives of a vector-valued map. For G , this gives

$$J_G = \begin{bmatrix} D_1 G_1 & D_2 G_1 \\ D_1 G_2 & D_2 G_2 \end{bmatrix}$$

. Then we can write the pair of equations in 3.6.3 as the single vector equation

$$\nabla(f \circ G) = J_G^T (\nabla f \circ G)$$

, where we interpret ∇f as a column vector.

The second method is mnemonic. If we let x and y be the variables of f , and u and v be the variables of G , and we write z for $f \circ G$ and for f , and write the components of $G(u, v)$ as x and y , then the chain rule reads that

$$\frac{\partial f}{\partial u} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u}$$

and that

$$\frac{\partial f}{\partial v} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial v}$$

, so that the partials in the numerator and denominator again appear to ‘cancel’.

Activity

Suppose that

$$z = x^2 + xy - y^2$$

that $x = e^{2s+t}$, and that $y = \ln(t/s)$. Compute

$$\frac{\partial z}{\partial s} \quad \text{and} \quad \frac{\partial z}{\partial t}$$

.

Task

First compute the partial derivatives of z in x and y , and the partial derivatives of x and y in t and s .

Solution.

We calculate that

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}(x^2 + xy - y^2) = 2x + y$$

, that

$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}(x^2 + xy - y^2) = x - 2y$$

, that

$$\frac{\partial x}{\partial s} = \frac{\partial}{\partial s}(e^{2s+t}) = 2e^{2s+t}$$

, that

$$\frac{\partial x}{\partial t} = \frac{\partial}{\partial t}(e^{2s+t}) = e^{2s+t}$$

, that

$$\frac{\partial y}{\partial s} = \frac{\partial}{\partial s}\left(\ln\left(\frac{t}{s}\right)\right) = \frac{\partial}{\partial s}(\ln(t) - \ln(s)) = -\frac{1}{s}$$

, and that

$$\frac{\partial y}{\partial t} = \frac{\partial}{\partial t}\left(\ln\left(\frac{t}{s}\right)\right) = \frac{1}{t}$$

Task

Now use the chain rule to compute the partial derivatives of z in s and t .

Solution.

Now we apply the chain rule, from which we conclude that

$$\begin{aligned}\frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} \\ &= (2x + y) \cdot 2e^{2s+t} + (x - 2y) \cdot (-1/s) \\ &= 4xe^{2s+t} + 2ye^{2s+t} - (x/s) + (2y/s)\end{aligned}$$

Substituting $x = e^{2s+t}$ and $y = \ln(t/s)$:

$$\begin{aligned}\frac{\partial z}{\partial s} &= 4e^{2s+t} \cdot e^{2s+t} + 2\ln(t/s) \cdot e^{2s+t} - \frac{e^{2s+t}}{s} + \frac{2\ln(t/s)}{s} \\ &= 4e^{4s+2t} + 2e^{2s+t} \ln(t/s) - \frac{e^{2s+t}}{s} + \frac{2\ln(t/s)}{s}\end{aligned}$$

Similarly for $\frac{\partial z}{\partial t}$:

$$\begin{aligned}
\frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t} \\
&= (2x + y) \cdot e^{2s+t} + (x - 2y) \cdot \frac{1}{t} \\
&= 2xe^{2s+t} + ye^{2s+t} + \frac{x}{t} - \frac{2y}{t}
\end{aligned}$$

Substituting $x = e^{2s+t}$ and $y = \ln(t/s)$:

$$\begin{aligned}
\frac{\partial z}{\partial t} &= 2e^{2s+t} \cdot e^{2s+t} + \ln(t/s) \cdot e^{2s+t} + \frac{e^{2s+t}}{t} - \frac{2 \ln(t/s)}{t} \\
&= 2e^{4s+2t} + e^{2s+t} \ln(t/s) + \frac{e^{2s+t}}{t} - \frac{2 \ln(t/s)}{t}
\end{aligned}$$

Activity

Use the Jacobian matrix version of the chain rule to calculate the partial derivatives of the function $f \circ G$, where $f(x, y) = 3x^2 - 2xy + y^2$, and $G(u, v) = (3u + 2v, 4u - v)$.

Solution.

We compute that

$$\nabla f(x, y) = \begin{bmatrix} 6x - 2y \\ -2x + 2y \end{bmatrix}$$

and that

$$J_G = \begin{bmatrix} D_1\{3u + 2v\} & D_2\{3u + 2v\} \\ D_1\{4u - v\} & D_2\{4u - v\} \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & -1 \end{bmatrix}$$

Since

$$\begin{aligned}
\nabla f(3u + 2v, 4u - v) &= \begin{bmatrix} 6(3u + 2v) - 2(4u - v) \\ -2(3u + 2v) + 2(4u - v) \end{bmatrix} \\
&= \begin{bmatrix} 10u + 14v \\ 2u - 6v \end{bmatrix}
\end{aligned}$$

, we find that

$$\begin{aligned}
\nabla(f \circ G) &= \begin{bmatrix} 3 & 2 \\ 4 & -1 \end{bmatrix}^T \begin{bmatrix} 10u + 14v \\ 2u - 6v \end{bmatrix} \\
&= \begin{bmatrix} 3 & 4 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 10u + 14v \\ 2u - 6v \end{bmatrix} \\
&= \begin{bmatrix} (3)(10u + 14v) + (4)(2u - 6v) \\ (2)(10u + 14v) + (-1)(2u - 6v) \end{bmatrix} \\
&= \begin{bmatrix} 38u + 18v \\ 18u + 34v \end{bmatrix}
\end{aligned}$$

Generalized chain rule

Before multiplying Jacobian matrices, check the input and output dimensions of each map.

Theorem 3.6.4 (Generalized Chain Rule).

Suppose $F : \mathbb{R}^p \rightarrow \mathbb{R}^m$ and $G : \mathbb{R}^n \rightarrow \mathbb{R}^p$ are differentiable functions. Then $F \circ G : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable, and

$$J_{F \circ G}(\mathbf{x}) = J_F(G(\mathbf{x}))J_G(\mathbf{x})$$

, where the right hand side is defined by matrix multiplications. Equivalently, for $1 \leq i \leq m$,

$$D_j\{F_i \circ G\}(\mathbf{x}) = \sum_{k=1}^p (D_k F_i)(G(\mathbf{x}))(D_j G_k)(\mathbf{x})$$

The order matters. The rightmost Jacobian matrix acts first on the input change.

Note

If f is a scalar-valued function, then the generalized chain rule tells us that

$$\nabla(f \circ G) = J_G^T(\nabla f \circ G)$$

as in the two variable case.

Activity

Suppose $f(x, y, z) = 3x^2 - 2xy + 4z^2$, and that $G(u, v) = (e^u \sin v, e^u \cos v, e^u)$. Compute the partial derivatives of $f \circ G$.

Solution.

We calculate that

$$\nabla f(x, y, z) = \begin{bmatrix} 6x - 2y \\ -2x \\ 8z \end{bmatrix}$$

so that

$$\nabla f(e^u \sin v, e^u \cos v, e^u) = e^u \begin{bmatrix} 6 \sin v - 2 \cos v \\ -2 \sin v \\ 8 \end{bmatrix}$$

. Now

$$J_G(u, v) = \begin{bmatrix} e^u \sin v & e^u \cos v \\ e^u \cos v & -e^u \sin v \\ e^u & 0 \end{bmatrix} = e^u \begin{bmatrix} \sin v & \cos v \\ \cos v & -\sin v \\ 1 & 0 \end{bmatrix}$$

. Thus

$$\begin{aligned} \nabla(f \circ G)(u, v) &= J_G(u, v)^T (\nabla f \circ G)(u, v) \\ &= (e^u)^2 \begin{bmatrix} \sin v & \cos v & 1 \\ \cos v & -\sin v & 0 \end{bmatrix} \begin{bmatrix} 6 \sin v - 2 \cos v \\ -2 \sin v \\ 8 \end{bmatrix} \\ &= e^{2u} \begin{bmatrix} 6(\sin v)^2 - 2 \sin v \cos v - 2 \sin v \cos v + 8 \\ 6 \sin v \cos v - 2(\cos v)^2 + 2(\sin v)^2 \end{bmatrix} \\ &= e^{2u} \begin{bmatrix} 6(\sin v)^2 - 4 \sin v \cos v + 8 \\ 4(\sin v)^2 + 6 \sin v \cos v - 2 \end{bmatrix} \end{aligned}$$

. From this calculation, we can read that the partial derivative of $f \circ G$ in the first variable is equal to

$$e^{2u}(6(\sin v)^2 - 4 \sin v \cos v + 8)$$

and the derivative in the second variable is

$$e^{2u}(4(\sin v)^2 + 6 \sin v \cos v - 2)$$

.

Note

The generalized chain rule says that local linear maps compose. The matrix formulation records that composition by multiplying Jacobian matrices in the correct order.

3.6.2 A tiny sigmoid block

A small neural-network block can be read as a composition of three steps: an affine map, a coordinatewise nonlinear rule, and another affine map.

Note: Sigmoid and its derivative

Let

$$\sigma(t) = \frac{1}{1 + e^{-t}}.$$

The derivative is

$$\sigma'(t) = \sigma(t)(1 - \sigma(t)).$$

When σ is applied to a vector, it is applied coordinate by coordinate.

Activity: A tiny sigmoid block

Let

$$W_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad W_2 = \begin{bmatrix} 1 & 0 & -1/2 \\ 0 & 1 & 1/2 \end{bmatrix},$$

and let $\mathbf{b}_1 = \mathbf{0}$ and $\mathbf{b}_2 = \mathbf{0}$.

Define

$$\mathbf{s} = W_1 \mathbf{x} + \mathbf{b}_1, \quad \mathbf{u} = \sigma(\mathbf{s}), \quad N(\mathbf{x}) = W_2 \mathbf{u} + \mathbf{b}_2,$$

where σ is applied coordinatewise.

At the base point

$$\mathbf{a} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

do the following.

1. Compute the pre-activation vector $\mathbf{s}_\mathbf{a} = W_1 \mathbf{a} + \mathbf{b}_1$.
2. Compute the hidden vector $\mathbf{u}_\mathbf{a} = \sigma(\mathbf{s}_\mathbf{a})$.
3. Compute $N(\mathbf{a})$.
4. Use $\sigma'(0) = 1/4$ to compute

$$D_\mathbf{a} = \text{diag}(\sigma'((\mathbf{s}_\mathbf{a})_1), \sigma'((\mathbf{s}_\mathbf{a})_2), \sigma'((\mathbf{s}_\mathbf{a})_3)).$$

5. Use the chain rule to compute

$$J_N(\mathbf{a}) = W_2 D_\mathbf{a} W_1.$$

6. Use

$$N(\mathbf{a} + \mathbf{h}) \approx N(\mathbf{a}) + J_N(\mathbf{a})\mathbf{h}$$

to predict the output change caused by a small input change $\mathbf{h}$.

Learning outcomes. U3-LO5, U3-LO7.

Solution.

Since $\mathbf{a} = \mathbf{0}$ and $\mathbf{b}_1 = \mathbf{0}$,

$$\mathbf{s}_\mathbf{a} = W_1\mathbf{a} + \mathbf{b}_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Therefore

$$\mathbf{u}_\mathbf{a} = \sigma(\mathbf{s}_\mathbf{a}) = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}.$$

Since $\mathbf{b}_2 = \mathbf{0}$,

$$N(\mathbf{a}) = W_2\mathbf{u}_\mathbf{a} = \begin{bmatrix} 1 & 0 & -1/2 \\ 0 & 1 & 1/2 \end{bmatrix} \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 1/4 \\ 3/4 \end{bmatrix}.$$

Since $\sigma'(0) = 1/4$,

$$D_\mathbf{a} = \begin{bmatrix} 1/4 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1/4 \end{bmatrix}.$$

Therefore

$$J_N(\mathbf{a}) = W_2 D_\mathbf{a} W_1 = \begin{bmatrix} 1 & 0 & -1/2 \\ 0 & 1 & 1/2 \end{bmatrix} \begin{bmatrix} 1/4 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1/4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1/8 & -1/8 \\ 1/8 & 3/8 \end{bmatrix}.$$

$$N(\mathbf{a} + \mathbf{h}) \approx \begin{bmatrix} 1/4 \\ 3/4 \end{bmatrix} + \begin{bmatrix} 1/8 & -1/8 \\ 1/8 & 3/8 \end{bmatrix} \mathbf{h}.$$

This gives the intended local prediction: the Jacobian matrix of the composition is a product of local matrices.

Note: Why sigmoid here

The sigmoid function is smooth, so the Jacobian matrix of this block is defined at every input.

Some neural-network models use nonsmooth activation rules. Those require extra case analysis. Here the point is the chain rule for Jacobian matrices.

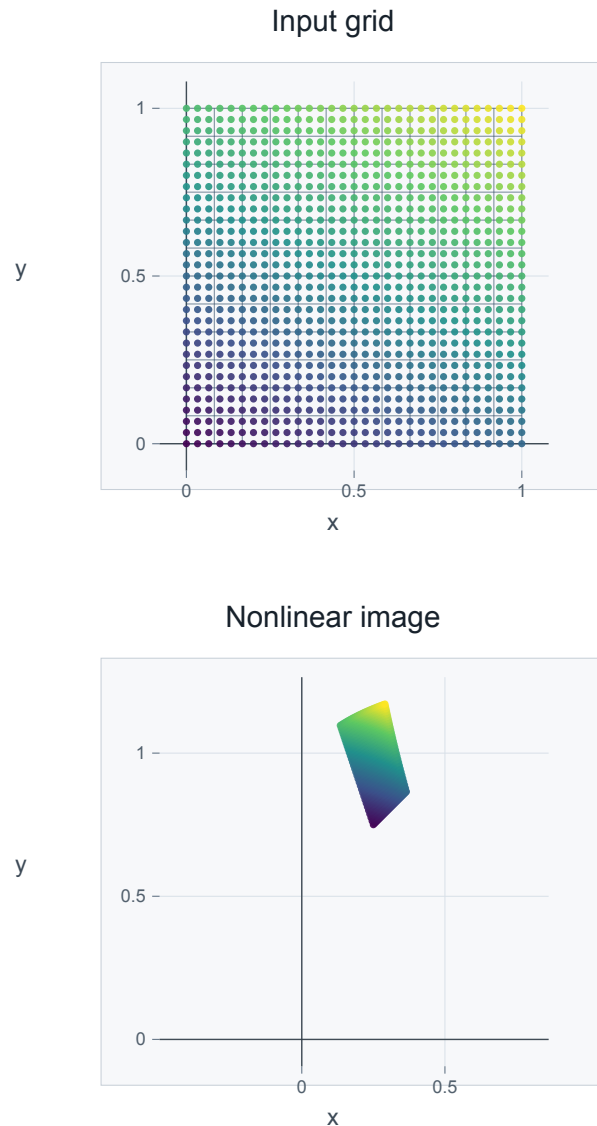


Figure 3.18. Square-grid visualization for a two-output sigmoid block. The map is smooth. The grid bends and compresses because the sigmoid derivative changes with the input. Near one base point, the Jacobian matrix gives the local linear approximation.

Activity: A one-query softmax weighting map

A single-query attention-style map can be viewed as a nonlinear vector-valued map. Let

$$K = \begin{bmatrix} 3 & 0 \\ 0 & 3 \\ -3 & -3 \end{bmatrix}, \quad V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

For a query vector

$$\mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix},$$

define the score vector

$$\mathbf{s} = K\mathbf{q},$$

the softmax weight vector

$$\alpha(\mathbf{q}) = \text{softmax}(\mathbf{s}),$$

and the output

$$F(\mathbf{q}) = V^T \alpha(\mathbf{q}).$$

1. Compute $\mathbf{s} = K\mathbf{q}$ in terms of q_1 and q_2 .
2. What are the rows of V ?
3. Explain why $F(\mathbf{q})$ is a weighted average of the rows of V .
4. Compute $\alpha(\mathbf{0})$ and $F(\mathbf{0})$.
5. Explain why F is not a linear map.
6. What operation makes F nonlinear?

Learning outcomes. U3-LO5, U3-LO7, U1-LO8.

Solution.

The score vector is

$$\mathbf{s} = K\mathbf{q} = \begin{bmatrix} 3q_1 \\ 3q_2 \\ -3q_1 - 3q_2 \end{bmatrix}.$$

The rows of V are the value vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The softmax vector

$$\alpha(\mathbf{q}) = \begin{bmatrix} \alpha_1(\mathbf{q}) \\ \alpha_2(\mathbf{q}) \\ \alpha_3(\mathbf{q}) \end{bmatrix}$$

has positive entries that add to 1. Therefore

$$F(\mathbf{q}) = V^T \alpha(\mathbf{q}) = \alpha_1(\mathbf{q})\mathbf{v}_1 + \alpha_2(\mathbf{q})\mathbf{v}_2 + \alpha_3(\mathbf{q})\mathbf{v}_3$$

is a weighted average of the value vectors.

At $\mathbf{q} = \mathbf{0}$, the score vector is

$$\mathbf{s} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

All three scores are equal, so

$$\alpha(\mathbf{0}) = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}.$$

Thus

$$F(\mathbf{0}) = V^T \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1/3 \end{bmatrix}.$$

A linear map must send $\mathbf{0}$ to $\mathbf{0}$. Since $F(\mathbf{0}) \neq \mathbf{0}$, the map F is not linear.

The matrix multiplication $K\mathbf{q}$ is linear in $\mathbf{q}$. The final weighted average is linear in the weights if the weights are already fixed. The nonlinear step is the softmax rule that turns scores into weights.

Note: Softmax outputs stay in the value triangle

The output always lies in the triangle spanned by the three value vectors. The query vector does not choose fixed weights. It changes the scores, and the softmax rule changes all weights together.



Figure 3.19. Square-grid visualization for an attention-style map $\mathbf{q} \mapsto V^T \text{softmax}(K\mathbf{q})$. The output lies in the triangle spanned by the value vectors. The grid bends and compresses because the softmax weights depend nonlinearly on the query.

Note: An attention-style nonlinear block

Unit 1 treated attention as three visible operations:

dot-product scores $\longrightarrow$ weights $\longrightarrow$ weighted averages.

For a sequence of token vectors stored as rows of a matrix X , fixed weight matrices produce

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V.$$

The score matrix is

$$S = QK^T.$$

The entry S_{ij} compares the query for token i with the key for token j . A weighting rule, such as row-wise softmax, converts each row of S into weights. Write the resulting attention-weight matrix as A_{att} . The attention output is

$$H = A_{\text{att}}V.$$

If A_{att} were fixed, then $V \mapsto A_{\text{att}}V$ would be a linear map: each row of H would be a weighted average of rows of V . In attention, A_{att} is not fixed. It is computed from X through Q , K , scores, and softmax. Thus the full rule $X \mapsto H$ is generally nonlinear.

The Unit 3 question is local. Near one fixed input X_0 , a nonlinear rule can be approximated by a linear map. After the entries of X and H are arranged as vectors, that local linear map is represented by a Jacobian matrix.

Activity: What depends on the input?

Assume the weight matrices W_Q , W_K , and W_V are fixed. The input is the token matrix X . Define

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V,$$

$$S = QK^T, \quad A_{\text{att}} = \text{softmax}_{\text{rows}}(S), \quad H = A_{\text{att}}V.$$

1. Which objects are fixed before the input X is chosen?
2. Which objects are computed from X ?
3. Why is $X \mapsto XW_Q$ a linear map when W_Q is fixed?
4. Why does $S = QK^T$ depend on X in two places?
5. If A_{att} were fixed, what would $H = A_{\text{att}}V$ do to the rows of V ?
6. In actual attention, why is the full rule $X \mapsto H$ not just one linear map?
7. What Unit 3 object describes the local linear approximation near a fixed input X_0 ?

Learning outcomes. U3-LO5, U3-LO6, U3-LO7.

Solution.

The matrices W_Q , W_K , and W_V are fixed. The matrices Q , K , V , S , A_{att} , and H are computed from X .

The rule $X \mapsto XW_Q$ is linear because W_Q is fixed and right multiplication by a fixed matrix distributes over sums and scalar multiples.

The score matrix $S = QK^T$ depends on X through Q and through K . The entries of S are dot products between query rows and key rows, and both sets of rows were computed from X . If A_{att} were fixed, then $H = A_{\text{att}}V$ would form weighted averages of the rows of V . This is linear in V .

In attention, A_{att} is not fixed. The weights are computed from the input through scores and softmax. The values V also depend on X . Therefore the full rule $X \mapsto H$ is generally nonlinear.

Near a fixed input X_0 , the local linear approximation is represented by a Jacobian matrix after the entries of the input and output matrices are arranged as vectors.

Warning: Specify the input

Before asking whether a rule is linear, first say what is variable and what is fixed. The rule $V \mapsto A_{\text{att}}V$ is linear when A_{att} is fixed. The attention rule $X \mapsto A_{\text{att}}(X)V(X)$ is generally nonlinear because the weights and values both depend on X .

Activity: Comparing square-grid visualizations

This unit used three square-grid visualizations: the nonlinear shear from 3.1.1, the sigmoid block from 3.6.2, and the attention-style softmax map from 3.18.

Map A:

$$F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y^2 \\ y \end{bmatrix}.$$

Map B: a two-output sigmoid block

$$\mathbf{s} = W_1\mathbf{x} + \mathbf{b}_1, \quad \mathbf{u} = \sigma(\mathbf{s}), \quad N(\mathbf{x}) = W_2\mathbf{u} + \mathbf{b}_2.$$

Map C: an attention-style softmax map

$$F_{\text{att}}(\mathbf{q}) = V^T \text{softmax}(K\mathbf{q}).$$

1. Which map agrees with a linear shear on the four corners but not at an edge midpoint?
2. Which map is built from affine maps with a coordinatewise sigmoid between them?
3. For which map is the local matrix computed by $W_2 D_a W_1$?
4. Which map produces outputs inside the triangle spanned by value vectors?
5. For which map does the nonlinear step come from softmax weights that depend on the input?
6. In all three examples, what does the square-grid visualization show that four corners alone may miss?

Learning outcomes. U3-LO1, U3-LO5, U3-LO7, U1-LO8.

Solution.

Map A is the nonlinear shear. It can agree with a linear shear on selected points but differ between them.

Map B is the sigmoid block. Its Jacobian matrix near a base point has the form $W_2 D_a W_1$, where D_a records the sigmoid derivatives at that base point. The local matrix can change from point to point because the sigmoid derivatives change.

Map C is the attention-style softmax map. Its outputs are weighted averages of the rows of V , so they stay inside the triangle spanned by the value vectors. The softmax weights depend on the input.

A square-grid visualization can show bending, compression, and changing local behavior that four corners alone may miss.

3.7 Directional derivatives

By 3.5.1, the first-order change in a scalar-valued function near $\mathbf{a}$ is

$$f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) \approx \nabla f(\mathbf{a})^T \mathbf{h}.$$

To study change along a unit direction $\mathbf{v}$, restrict the input to the line

$$\mathbf{c}(t) = \mathbf{a} + t\mathbf{v}.$$

The resulting one-variable rate of change is the directional derivative.

The chain rule for one-parameter inputs in 3.6.2 gives the gradient formula below.

Definition 3.7.1.

Let f be a scalar-valued function. Then if $\mathbf{v}$ is a unit vector, the *directional derivative* of f in the direction of the vector $\mathbf{v}$, denoted $D_{\mathbf{v}}f$, is defined as

$$D_{\mathbf{v}}f(\mathbf{a}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{a} + h\mathbf{v}) - f(\mathbf{a})}{h}$$

, provided the limit exists.

The directional derivative represents the rate of change of the function f at the point $\mathbf{a}$ in the direction of the unit vector $\mathbf{v}$. For a function $f(x, y)$ of two variables, the directional derivative $D_{\mathbf{v}}f(\mathbf{a})$ is the slope of the tangent line to the curve obtained by intersecting the surface $z = f(x, y)$ with the plane containing both the line passing through $\mathbf{a}$ with direction vector $\mathbf{v}$, and the z axis.

Remark

Partial derivatives are a special case of directional derivatives, i.e., we have

$$\frac{\partial f}{\partial x_i} = D_{\mathbf{e}_i}f$$

Despite partial derivatives being a special case of directional derivatives, in most circumstances we can compute directional derivatives using partial derivatives, or equivalently, the gradient.

Theorem 3.7.2.

Suppose f is a function differentiable at a point $\mathbf{a}$. Then for each unit vector $\mathbf{v}$, $D_{\mathbf{v}}f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v}$.

Proof.

Define the path $\mathbf{c}(t) = \mathbf{a} + t\mathbf{v}$ and the single-variable function $g(t) = f(\mathbf{c}(t)) = f(\mathbf{a} + t\mathbf{v})$. Since $\mathbf{c}(0) = \mathbf{a}$, we have $g(0) = f(\mathbf{a})$. Therefore, the definition of the directional derivative gives

$$\begin{aligned} D_{\mathbf{v}}f(\mathbf{a}) &= \lim_{h \rightarrow 0} \frac{f(\mathbf{a} + h\mathbf{v}) - f(\mathbf{a})}{h} \\ &= \lim_{h \rightarrow 0} \frac{g(h) - g(0)}{h} \\ &= g'(0). \end{aligned}$$

Write $\mathbf{a} = (a_1, \dots, a_n)$ and $\mathbf{v} = (v_1, \dots, v_n)$. The i th coordinate of $\mathbf{c}(t)$ is $c_i(t) = a_i + tv_i$, so $c'_i(0) = v_i$. Because $g = f \circ \mathbf{c}$, the chain rule (3.6.2) gives

$$\begin{aligned} g'(0) &= \sum_{i=1}^n \frac{\partial f}{\partial x_i}(\mathbf{c}(0))c'_i(0) \\ &= \sum_{i=1}^n \frac{\partial f}{\partial x_i}(\mathbf{a})v_i \\ &= \nabla f(\mathbf{a}) \cdot \mathbf{v}. \end{aligned}$$

Combining the two calculations proves that $D_{\mathbf{v}}f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v}$.

3.7.2 greatly simplifies the computations of directional derivatives.

Activity: Alignment with the gradient

Suppose

$$\nabla f(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

Compute $D_{\mathbf{v}}f(\mathbf{a})$ for each unit vector below:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \mathbf{v}_4 = -\frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

Which listed direction gives the greatest increase? Which gives the greatest decrease?

Learning outcomes. U3-LO8, U1-LO1.

Solution.

By 3.7.2,

$$D_{\mathbf{v}_1}f(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 3,$$

and

$$D_{\mathbf{v}_2}f(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 4.$$

Also,

$$D_{\mathbf{v}_3}f(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \cdot \frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 5,$$

while

$$D_{\mathbf{v}_4}f(\mathbf{a}) = -5.$$

The greatest listed increase occurs in direction $\mathbf{v}_3$, which is aligned with the gradient. The greatest listed decrease occurs in direction $\mathbf{v}_4$, which points opposite the gradient.

Activity

Let $f(x, y) = x^2y + y^2$.

Task

Compute the directional derivative $D_{\mathbf{v}}f(1, 2)$, where $\mathbf{v} = (1/\sqrt{2}, 1/\sqrt{2})$, and $f(x, y) = x^2y + y^2$.

Solution.

We begin by computing the gradient of f at $(1, 2)$. We have

$$f_x(x, y) = 2xy \quad \text{and} \quad f_y(x, y) = x^2 + 2y$$

. So $\nabla f(x, y) = (2xy, x^2 + 2y)$, and in particular

$$\nabla f(1, 2) = \begin{bmatrix} 2(1)(2) \\ (1)^2 + 2(2) \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

. The given vector $\mathbf{v}$ is already a unit vector because

$$\|\mathbf{v}\| = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} = 1.$$

Therefore, we may apply the directional-derivative formula directly:

$$D_{\mathbf{v}}f(1, 2) = \nabla f(1, 2) \cdot \mathbf{v} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = 9/\sqrt{2}$$

.

Task

Find the directional derivative of f at the point $(-1, 3)$ in the direction of the vector $(-1, \sqrt{3})$. That is, compute $D_{\mathbf{v}}f(-1, 3)$, where $\mathbf{v}$ is the unit vector in the direction of $(-1, \sqrt{3})$.

Solution.

The vector

$$\begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix}$$

has norm

$$\sqrt{(-1)^2 + (\sqrt{3})^2} = 2$$

. So the unit vector in this direction is

$$\mathbf{v} = \frac{1}{2} \begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix} = \begin{bmatrix} -1/2 \\ \sqrt{3}/2 \end{bmatrix}$$

. For

$$f(x, y) = x^2y + y^2,$$

we have

$$\nabla f(x, y) = \begin{bmatrix} 2xy \\ x^2 + 2y \end{bmatrix}.$$

Thus

$$\nabla f(-1, 3) = \begin{bmatrix} -6 \\ 7 \end{bmatrix}.$$

Therefore

$$D_{\mathbf{v}}f(-1, 3) = \nabla f(-1, 3) \cdot \mathbf{v} = \begin{bmatrix} -6 \\ 7 \end{bmatrix} \cdot \begin{bmatrix} -1/2 \\ \sqrt{3}/2 \end{bmatrix} = 3 + \frac{7\sqrt{3}}{2} = \frac{6 + 7\sqrt{3}}{2}$$

.

Activity

Find the directional derivative of f at $(2, 1)$ in the direction from $(2, 1)$ to $(3, 5)$.

Solution.

The unit vector pointing in the direction from $(2, 1)$ to $(3, 5)$ is the unit vector pointing in the same direction as the vector $(3 - 2, 5 - 1) = (1, 4)$. This vector has magnitude $\sqrt{1^2 + 4^2} = \sqrt{17}$,

so the unit vector pointing in this direction is equal to

$$\mathbf{v} = \begin{bmatrix} 1/\sqrt{17} \\ 4/\sqrt{17} \end{bmatrix}$$

. Now

$$\nabla f(x, y) = \begin{bmatrix} 3x^2y \\ x^3 \end{bmatrix}$$

, and in particular,

$$\nabla f(2, 1) = \begin{bmatrix} 3(2)^2(1) \\ (2)^3 \end{bmatrix} = \begin{bmatrix} 12 \\ 8 \end{bmatrix}$$

. So

$$D_{\mathbf{v}}f(2, 1) = \nabla f(2, 1) \cdot \mathbf{v} = \begin{bmatrix} 12 \\ 8 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{17} \\ 4/\sqrt{17} \end{bmatrix} = \frac{12 + 32}{\sqrt{17}} = \frac{44}{\sqrt{17}}$$

.

Activity

Let $f(x, y) = x^2e^y$. Find the directional derivative of f at $(2, 0)$ in the direction pointing from $(2, 0)$ to $(3, 0)$.

Solution.

A vector pointing from $(2, 0)$ to $(3, 0)$ is

$$\mathbf{d} = \begin{bmatrix} 3 - 2 \\ 0 - 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Its norm is

$$\|\mathbf{d}\| = \sqrt{1^2 + 0^2} = 1.$$

Thus this vector is already a unit vector. Therefore the required direction is

$$\mathbf{v} = \mathbf{d} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \mathbf{e}_1.$$

Next,

$$\nabla f(x, y) = \begin{bmatrix} 2xe^y \\ x^2e^y \end{bmatrix}, \quad \nabla f(2, 0) = \begin{bmatrix} 4 \\ 4 \end{bmatrix}.$$

Thus

$$\begin{aligned}
D_{\mathbf{v}}f(2, 0) &= \nabla f(2, 0) \cdot \mathbf{v} \\
&= \begin{bmatrix} 4 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\
&= 4 \cdot 1 + 4 \cdot 0 = 4.
\end{aligned}$$

Theorem 3.7.3.

Suppose that a function f is differentiable at a point $\mathbf{a}$.

- If $\nabla f(\mathbf{a}) = \mathbf{0}$, then $D_{\mathbf{v}}f(\mathbf{a}) = 0$ for any unit vector $\mathbf{v}$.
- If $\nabla f(\mathbf{a}) \neq \mathbf{0}$, then $D_{\mathbf{v}}f(\mathbf{a})$ is maximized, among unit vectors $\mathbf{v}$, when

$$\mathbf{v} = \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}.$$

It is minimized when

$$\mathbf{v} = -\frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}.$$

The maximum value is $\|\nabla f(\mathbf{a})\|$, and the minimum value is $-\|\nabla f(\mathbf{a})\|$.

Proof.

If $\nabla f(\mathbf{a}) = \mathbf{0}$, then 3.7.2 gives

$$D_{\mathbf{v}}f(\mathbf{a}) = 0$$

for every unit vector $\mathbf{v}$.

Now suppose $\nabla f(\mathbf{a}) \neq \mathbf{0}$, and let θ be the angle between $\nabla f(\mathbf{a})$ and a unit vector $\mathbf{v}$. By 3.7.2 and 1.1.7,

$$D_{\mathbf{v}}f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v} = \|\nabla f(\mathbf{a})\| \cos \theta.$$

The largest possible value of $\cos \theta$ is 1, attained when $\mathbf{v}$ points in the direction of the gradient. The smallest possible value is -1 , attained when $\mathbf{v}$ points in the opposite direction. Therefore the maximum and minimum directional derivatives are

$$\|\nabla f(\mathbf{a})\| \quad \text{and} \quad -\|\nabla f(\mathbf{a})\|,$$

attained at

$$\mathbf{v} = \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|} \quad \text{and} \quad \mathbf{v} = -\frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|},$$

respectively.

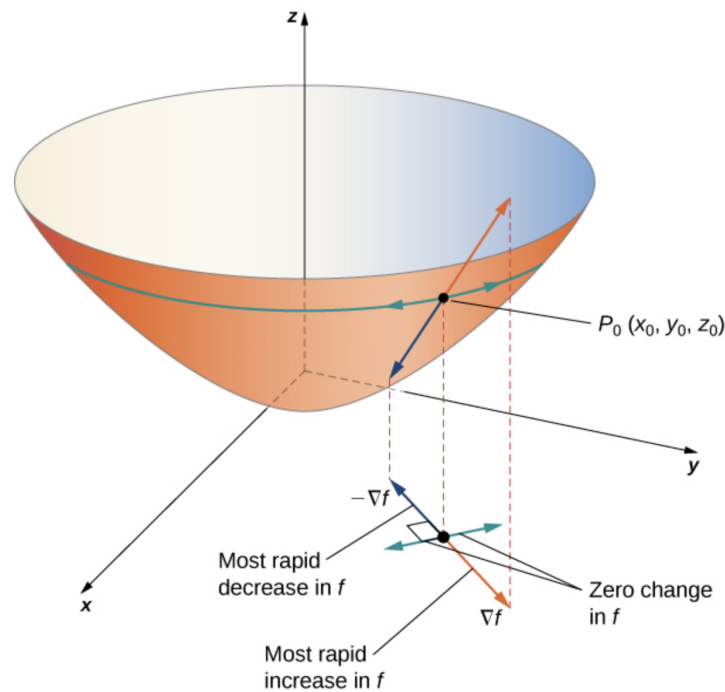


Figure 3.20. The gradient indicates the maximum and minimum values of the directional derivative at a point. Figure 4.41 from Edwin “Jed” Herman and Gilbert Strang, *Calculus Volume 3*, OpenStax, © 2018 Rice University, licensed under CC BY-NC-SA 4.0; source: OpenStax Figure 4.41.

Activity

Find the direction for which the directional derivative of $f(x, y) = 3x^2 - 4xy + 2y^2$ at $(-2, 3)$ is a maximum. What is the maximum value of the directional derivative?

Solution.

First, we calculate that

$$\nabla f(x, y) = \begin{bmatrix} 6x - 4y \\ -4x + 4y \end{bmatrix}$$

. So

$$\nabla f(-2, 3) = \begin{bmatrix} 6(-2) - 4(3) \\ -4(-2) + 4(3) \end{bmatrix} = \begin{bmatrix} -24 \\ 20 \end{bmatrix}$$

. 3.7.3 tells us that the directional derivative $D_{\mathbf{v}}f$ is maximized when $\mathbf{v}$ is the unit vector pointing in the direction of $(-24, 20)$. Since $(-24, 20)$ has magnitude $\sqrt{(-24)^2 + (20)^2} = \sqrt{976}$, we see the direction which maximizes the directional derivative is

$$\mathbf{v} = \begin{bmatrix} -24/\sqrt{976} \\ 20/\sqrt{976} \end{bmatrix}$$

, and the value of the directional derivative in this direction is equal to the magnitude of the gradient, which is

$$\sqrt{976}$$

Activity

Calculate $D_{\mathbf{u}}f(1, -2, 3)$ in the direction of $\mathbf{v} = -\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ for the function

$$f(x, y, z) = 5x^2 - 2xy + y^2 - 4yz + z^2 + 3xz.$$

Solution.

We compute that

$$\|\mathbf{v}\| = \sqrt{(-1)^2 + (2)^2 + (2)^2} = \sqrt{9} = 3$$

So the unit vector in the direction of $\mathbf{v}$ is the vector

$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = -\frac{1}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} + \frac{2}{3}\mathbf{k}$$

Using the formula for the directional derivative, we find that

$$\begin{aligned} D_{\mathbf{u}}f(x, y, z) &= \nabla f(x, y, z) \cdot \mathbf{u} \\ &= \begin{bmatrix} 10x - 2y \\ -2x + 2y - 4z \\ -4y + 2z + 3x \end{bmatrix} \cdot \mathbf{u} \\ &= -\frac{10x - 2y}{3} + 2\frac{-2x + 2y - 4z}{3} + 2\frac{-4y + 2z + 3x}{3} \end{aligned}$$

. At the point $(1, -2, 3)$, we evaluate that

$$D_{\mathbf{u}}f(1, -2, 3) = (23)\left(-\frac{1}{3}\right) + (-18)\left(\frac{2}{3}\right) + (17)\left(\frac{2}{3}\right) = \frac{-25}{3}$$

In 4.2, the same dot product will be interpreted as a signed projection coefficient of the gradient onto the chosen direction.

3.8 Critical points and gradient descent

Directional derivatives describe first-order change along a chosen direction. We now use them to identify candidates for local extrema and to motivate an iterative method for decreasing a function.

3.8.1 Optimization problems and critical points

Optimization problems

In this unit, we apply multivariate calculus to solve *optimization problems*.

Definition 3.8.1 (Optimization problems).

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a scalar-valued function, and consider a set $U \subseteq \mathbb{R}^n$. Then we can consider two optimization problems, in which we call f the *objective function* of the problem, and U the *feasible region* of the optimization problem:

- The *minimization problem* of f on U is the problem of finding $\mathbf{x}^* \in U$ such that the value $f(\mathbf{x}^*)$ is *smallest* among all elements of U (so the function is *minimized* on U by $\mathbf{x}^*$).
- The *maximization problem* of f on U is the problem of finding $\mathbf{x}^* \in U$ such that the value $f(\mathbf{x}^*)$ is *largest* among all elements of U (so the function is *maximized* on U by $\mathbf{x}^*$).

Critical points

Definition 3.8.2 (Minima and maxima).

Suppose f is a scalar-valued function. A point $\mathbf{a}$ in the domain of f is a

- *local minimum* if there is a ball centered at $\mathbf{a}$ such that $f(\mathbf{a}) \leq f(\mathbf{x})$ whenever $\mathbf{x}$ is in that ball and also in the domain of f ; in other words, whenever $\mathbf{x}$ is close enough to $\mathbf{a}$ and lies in the domain of f . In this case, we say $f(\mathbf{a})$ is a *local minimum value* for the function f .
- *local maximum* if there is a ball centered at $\mathbf{a}$ such that $f(\mathbf{a}) \geq f(\mathbf{x})$ whenever $\mathbf{x}$ is in that ball and also in the domain of f ; in other words, whenever $\mathbf{x}$ is close enough to $\mathbf{a}$ and lies in the domain of f . In this case, we say $f(\mathbf{a})$ is a *local maximum value* for the function f .
- *global minimum* if $f(\mathbf{a}) \leq f(\mathbf{x})$ for all $\mathbf{x}$ in the domain of f . In this case, $f(\mathbf{a})$ is the *global minimum value*, or *absolute minimum value*, of the function f .
- *global maximum* if $f(\mathbf{a}) \geq f(\mathbf{x})$ for all $\mathbf{x}$ in the domain of f . In this case, $f(\mathbf{a})$ is the *global maximum value*, or *absolute maximum value*, of the function f .

A local minimum or local maximum is also called a *local extremum*.

Remark

The graph of the function $-f$ is obtained by reflecting the graph of the function f in the output axis. This operation turns local and global maxima into local and global minima, and vice versa. So methods for finding minima for a general function often give methods for finding maxima, which makes the two problems often equivalent.

Definition 3.8.3 (Critical points).

Suppose f is a scalar-valued function that is differentiable at a point $\mathbf{a}$. Then we say $\mathbf{a}$ is a *critical point* if $\nabla f(\mathbf{a}) = \mathbf{0}$.

For a scalar-valued function of two variables, for example, $f(x, y)$, a critical point is a solution to the two equations $f_x(x, y) = 0$ and $f_y(x, y) = 0$.

Just as the derivative of a scalar-valued function of a single variable helps identify local minima and maxima, partial derivatives provide information about possible local minima/maxima.

Theorem 3.8.4.

Suppose f is differentiable at an interior point $\mathbf{a}$ of its domain. If $\mathbf{a}$ is a local extremum of f , then

$$\nabla f(\mathbf{a}) = \mathbf{0}.$$

Proof.

Let $\mathbf{v}$ be a unit vector, and define

$$g(t) = f(\mathbf{a} + t\mathbf{v}).$$

Because $\mathbf{a}$ is an interior local extremum of f , the function g has a local extremum at $t = 0$. Therefore

$$g'(0) = 0.$$

By 3.7.2,

$$g'(0) = D_{\mathbf{v}}f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v}.$$

This holds for every unit vector $\mathbf{v}$. If $\nabla f(\mathbf{a}) \neq \mathbf{0}$, choose

$$\mathbf{v} = \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}.$$

Then

$$\nabla f(\mathbf{a}) \cdot \mathbf{v} = \|\nabla f(\mathbf{a})\| > 0,$$

a contradiction. Hence $\nabla f(\mathbf{a}) = \mathbf{0}$.

Activity

Find the critical points of the function

$$f(x, y) = x^2 + y^2 - 2x + 4y$$

Solution.

To find the critical points, we calculate the partial derivatives and set them equal to zero. We find that

$$f_x(x, y) = 2x - 2 \quad \text{and} \quad f_y(x, y) = 2y + 4$$

. The critical points are the pairs (x, y) that simultaneously solve the equations $2x - 2 = 0$ and $2y + 4 = 0$. So the only critical point is $(1, -2)$.

Activity

Find the critical points of $f(x, y) = x^2 - y^2$.

Solution.

We compute that $f_x(x, y) = 2x$ and $f_y(x, y) = -2y$. The only (x, y) such that $f_x(x, y) = 2x = 0$ and $f_y(x, y) = -2y = 0$ is $(0, 0)$. This is the only critical point.

Warning: Critical points are candidates

The equation $\nabla f = 0$ finds candidates for local extrema. It does not by itself find boundary extrema, constrained extrema, or absolute extrema. Always combine critical-point information with the domain and boundary conditions.

3.8.2 Gradient descent

Activity

Find the system of equations satisfied by the critical points of the function

$$f(x, y, z) = x^2 e^{yz} + y \sin(xz) + z \cos(xy)$$

.

Solution.

We calculate that

$$f_x(x, y, z) = 2xe^{yz} + yz \cos(xz) - yz \sin(xy)$$

, that

$$f_y(x, y, z) = x^2 z e^{yz} + \sin(xz) - xz \sin(xy)$$

and that

$$f_z(x, y, z) = x^2 y e^{yz} + xy \cos(xz) + \cos(xy)$$

. So the critical points of f are the solutions to the system of equations

$$\begin{cases} 2xe^{yz} + yz \cos(xz) - yz \sin(xy) = 0, \\ x^2 z e^{yz} + \sin(xz) - xz \sin(xy) = 0, \\ x^2 y e^{yz} + xy \cos(xz) + \cos(xy) = 0. \end{cases}$$

Using the calculus we have developed, it is usually straightforward to write the equations that critical points must satisfy. 3.8.2 also shows the problem: those equations may be nonlinear and difficult or impossible to solve exactly. This is different from the linear systems in Unit 2, where Gaussian elimination gave a systematic exact method.

When the critical-point equations cannot be solved directly, one often uses a numerical method. By 3.7.3, the negative gradient gives the direction of steepest local decrease.

At a current point $\mathbf{x}_k$, choose

$$\mathbf{h} = -\alpha \nabla f(\mathbf{x}_k), \quad \alpha > 0.$$

By 3.5.1,

$$\begin{aligned} f(\mathbf{x}_k + \mathbf{h}) - f(\mathbf{x}_k) &\approx \nabla f(\mathbf{x}_k)^T \mathbf{h} \\ &= -\alpha \|\nabla f(\mathbf{x}_k)\|^2. \end{aligned}$$

Unless the gradient is zero, the local model predicts a decrease. This is a local prediction, not a convergence theorem. It motivates the gradient-descent update

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k).$$

The gradient-descent algorithm repeats this local step.

1. Start at an initial point $\mathbf{x}_0$.
2. Compute the gradient $\nabla f(\mathbf{x}_0)$.
3. Move in the direction of the negative gradient: set $\mathbf{x}_1 = \mathbf{x}_0 - \alpha_0 \nabla f(\mathbf{x}_0)$, where $\alpha_0 > 0$ is a step size (or learning rate) parameter.
4. Repeat steps 2-3 with step sizes $\alpha_1, \alpha_2, \dots$ to successively define points $\mathbf{x}_2 = \mathbf{x}_1 - \alpha_1 \nabla f(\mathbf{x}_1)$, $\mathbf{x}_3 = \mathbf{x}_2 - \alpha_2 \nabla f(\mathbf{x}_2)$, and so on, with the hope that these points converge to a local minimum.

In data and modeling problems, an objective function being minimized is often called a loss.

Intuitively, gradient descent works because each successive value $\mathbf{x}_k$ is obtained by shifting the previous value $\mathbf{x}_{k-1}$ in the direction in which the function f decreases the fastest. Gradient descent may converge to a local minimum under additional assumptions on the function and the learning rates. The examples below illustrate possible behavior; they do not prove convergence.

Activity: Reading one gradient descent update

Assume $\text{grad_f}(\mathbf{x})$ computes $\nabla f(\mathbf{x})$. Consider the line of code

```
x = x - alpha * grad_f(x)
```

1. What mathematical update rule does this line represent?
2. Which quantity is the learning rate?
3. Why must $\text{grad_f}(\mathbf{x})$ have the same shape as $\mathbf{x}$?
4. What would change if the minus sign were a plus sign?

Learning outcomes. U3-LO8, U3-LO9.

Solution.

The line represents the update

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k).$$

The learning rate is α . The vector $\text{grad_f}(\mathbf{x})$ must have the same shape as $\mathbf{x}$, because the update subtracts one vector from another. If the minus sign were a plus sign, the update would move in the direction of steepest increase rather than steepest decrease.

Activity

Find an approximate minimum of the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) = x^2 - 3xy + 3y^2 + 5y + 2x$$

using gradient descent. Use the starting point $\mathbf{x}_0 = (0, 0)$, a constant step size of 0.1, and stop after computing $\mathbf{x}_3$. You may use a calculator to perform these computations.

Solution.

First, we calculate the gradient:

$$\nabla f(x, y) = \begin{bmatrix} 2x - 3y + 2 \\ -3x + 6y + 5 \end{bmatrix}$$

Starting at $\mathbf{x}_0 = (0, 0)$, we successively compute that

$$\nabla f(0, 0) = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

, so

$$\mathbf{x}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} -0.2 \\ -0.5 \end{bmatrix}$$

. We then compute that

$$\nabla f(-0.2, -0.5) = \begin{bmatrix} -0.4 - (-1.5) + 2 \\ -3(-0.2) + 6(-0.5) + 5 \end{bmatrix} = \begin{bmatrix} 3.1 \\ 2.6 \end{bmatrix}$$

, so

$$\mathbf{x}_2 = \begin{bmatrix} -0.2 \\ -0.5 \end{bmatrix} - 0.1 \begin{bmatrix} 3.1 \\ 2.6 \end{bmatrix} = \begin{bmatrix} -0.51 \\ -0.76 \end{bmatrix}$$

. Finally, we compute that

$$\nabla f(-0.51, -0.76) = \begin{bmatrix} 2(-0.51) - 3(-0.76) + 2 \\ -3(-0.51) + 6(-0.76) + 5 \end{bmatrix} = \begin{bmatrix} 3.26 \\ 1.97 \end{bmatrix}$$

, so

$$\mathbf{x}_3 = \begin{bmatrix} -0.51 \\ -0.76 \end{bmatrix} - 0.1 \begin{bmatrix} 3.26 \\ 1.97 \end{bmatrix} = \begin{bmatrix} -0.836 \\ -0.957 \end{bmatrix}$$

. In 5.5.3, we compute exactly that the unique critical point is

$$\left(-9, -\frac{16}{3}\right)$$

and classify it after developing second-order tools. Thus $\mathbf{x}_3$ is not yet a good approximation. If we continue the iteration, the points $\mathbf{x}_k$ for, say, $k \geq 1000$ are very close to that critical point.

Remark

The iteration is cumbersome by hand but easy to repeat on a computer when the gradient is available.

First gradient descent steps

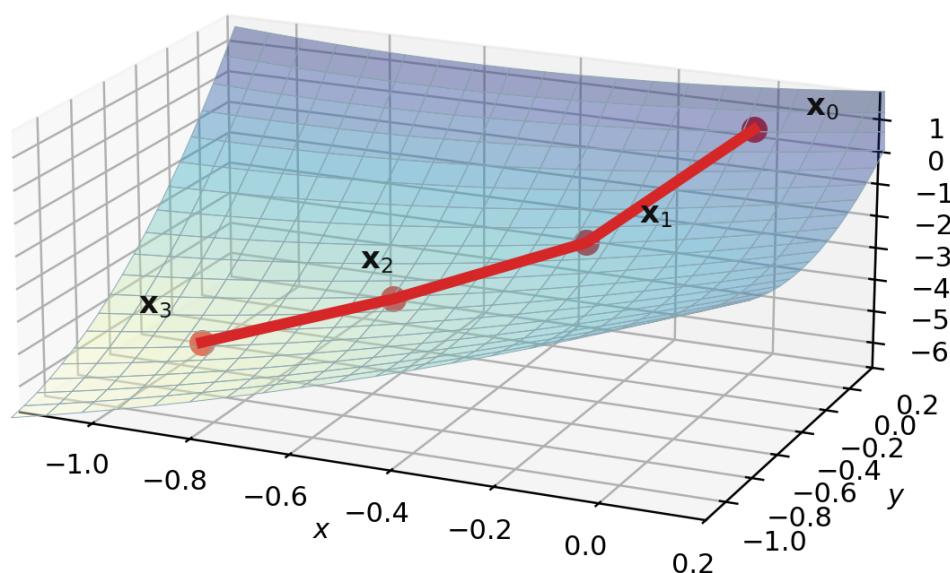


Figure 3.21. Initial steps of gradient descent for $f(x, y) = x^2 - 3xy + 3y^2 + 5y + 2x$ with starting point $\mathbf{x}_0 = (0, 0)$. For $k = 1, 2, 3$, the point $\mathbf{x}_k$ is determined by the rule $\mathbf{x}_k = \mathbf{x}_{k-1} - (0.1)\nabla f(\mathbf{x}_{k-1})$. Adapted from Stanford’s MATH 51 textbook.

3.9 Coding recap

3.9.1 Linked notebook

Run Lab U3: Jacobian matrices and local linearization. The main sequence practices symbolic Jacobian matrices, evaluating a Jacobian matrix at a point, local linear prediction, square-grid visualizations, checking Jacobian matrix shapes in the chain rule, and short nonlinear model-block computations.

Directional derivatives and gradient descent are revisited computationally at the beginning of Lab U5.

For a quick reference on array shapes, matrix products, numerical checks, and SymPy symbolic commands, see the programming appendix sections A.2, A.4, A.9, and A.11.

3.9.2 Review activities

Activity: Reading a symbolic Jacobian matrix

```
x, y = sp.symbols("x y")
F = sp.Matrix([x*y, x**2 + y])
J = F.jacobian([x, y])
J
```

Output:

$$J = \begin{bmatrix} y & x \\ 2x & 1 \end{bmatrix}.$$

1. What are the component functions of F ?
2. What is the shape of J ?
3. What does the first column of J measure?

Solution.

The component functions are $F_1(x, y) = xy$ and $F_2(x, y) = x^2 + y$. The Jacobian matrix is $\begin{bmatrix} y & x \\ 2x & 1 \end{bmatrix}$, so it is 2×2 . Its first column measures how the two output components change with respect to the first input variable x .

Activity: Actual output versus local prediction

```
def F(v):
    x, y = v
    return np.array([x**2 + y, x - y**2])

a = np.array([1.0, 2.0])
h = np.array([0.1, -0.2])
J_at_a = np.array([[2.0, 1.0],
                   [1.0, -4.0]])

actual = F(a + h)
linear = F(a) + J_at_a @ h
actual, linear, actual - linear
```

Output:

$$\text{actual} = \begin{bmatrix} 3.01 \\ -2.14 \end{bmatrix},$$

$$\text{linear} = \begin{bmatrix} 3.00 \\ -2.10 \end{bmatrix},$$

$$\text{actual} - \text{linear} = \begin{bmatrix} 0.01 \\ -0.04 \end{bmatrix}.$$

1. Which output is the true nonlinear value?
2. Which output is the local linear prediction?
3. What does $\text{actual} - \text{linear}$ measure?

Solution.

The true value is actual , which is $[3.01, -2.14]$. The local linear prediction is linear , which is $[3.0, -2.1]$. The difference $\text{actual} - \text{linear}$ is $[0.01, -0.04]$, the local approximation error for this input change.

Activity: Reading Jacobian matrix columns

```
J_at_a = np.array([[2.0, -1.0, 0.0],
                   [0.0, 3.0, 4.0]])

e1_change = np.array([0.1, 0.0, 0.0])
J_at_a[:, 0], J_at_a @ e1_change
```

Output of $J_at_a[:, 0]$:

$$\begin{bmatrix} 2.0 \\ 0.0 \end{bmatrix}.$$

Output of $J_at_a @ e1_change$:

$$\begin{bmatrix} 0.2 \\ 0.0 \end{bmatrix}.$$

1. What are the input and output dimensions?
2. What does $J_at_a[:, 0]$ select?
3. What output change is predicted for $e1_change$?

Solution.

The input dimension is 3 and the output dimension is 2. The slice `J_at_a[:, 0]` selects the first column of the Jacobian matrix, $[2.0, 0.0]$. For the input change $[0.1, 0.0, 0.0]$, the predicted output change is $[0.2, 0.0]$.

Activity: Chain-rule shape check

```
Jg = np.ones((3, 2))
Jf_at_g = np.ones((4, 3))

Jf_at_g @ Jg
```

Output of `Jf_at_g @ Jg`:

$$\begin{bmatrix} 3.0 & 3.0 \\ 3.0 & 3.0 \\ 3.0 & 3.0 \\ 3.0 & 3.0 \end{bmatrix}.$$

1. Which map is applied first in the composition?
2. What are the input and output dimensions of the composition?
3. What is the shape of the product?

Solution.

The map whose Jacobian matrix is `Jg` is applied first. Since `Jg` is 3×2 , it takes two input directions to three intermediate directions; `Jf_at_g` then takes those three directions to four output directions. The product has shape 4×2 , and the displayed output is a 4×2 array whose entries are all 3.

Activity: Reading a tiny sigmoid block

```
def sigmoid(t):
    return 1 / (1 + np.exp(-t))

W1 = np.array([
    [1.0, 0.0],
    [0.0, 1.0],
    [1.0, 1.0],
```

```

])

W2 = np.array([
    [1.0, 0.0, -0.5],
    [0.0, 1.0, 0.5],
])

b1 = np.zeros(3)
b2 = np.zeros(2)

x = np.array([0.0, 0.0])

s = W1 @ x + b1
u = sigmoid(s)
D = np.diag(u * (1 - u))
y = W2 @ u + b2
J_at_x = W2 @ D @ W1

s, u, y, J_at_x

```

Output:

$$\begin{aligned}
 \mathbf{s} &= \begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \end{bmatrix}, \\
 \mathbf{u} &= \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}, \\
 \mathbf{y} &= \begin{bmatrix} 0.25 \\ 0.75 \end{bmatrix}, \\
 J_{\mathbf{x}} &= \begin{bmatrix} 0.125 & -0.125 \\ 0.125 & 0.375 \end{bmatrix}.
 \end{aligned}$$

1. Which lines contain affine maps?
2. Which line contains the coordinatewise nonlinear step?
3. What do $\mathbf{s}$, $\mathbf{u}$, and $\mathbf{y}$ represent?
4. Which line forms the diagonal derivative matrix?
5. Which line computes the Jacobian matrix of the block at $\mathbf{x}$?
6. What is the shape of $J_{\mathbf{at_x}}$?

Solution.

The lines $s = W1 @ x + b1$ and $y = W2 @ u + b2$ are affine maps. The line $u = \text{sigmoid}(s)$ is the coordinatewise nonlinear step.

The vector s is the pre-activation vector. The vector u is the hidden vector after applying sigmoid. The vector y is the output vector.

The line $D = \text{np.diag}(u * (1 - u))$ forms the diagonal matrix of sigmoid derivatives at this input, since $\sigma'(t) = \sigma(t)(1 - \sigma(t))$.

The line $J_at_x = W2 @ D @ W1$ computes the Jacobian matrix of the block at x by the chain rule for Jacobian matrices. The shape of J_at_x is 2×2 : the input has two coordinates and the output has two coordinates.

Activity: Reading a gradient descent loop

Assume $\text{grad_f}(x)$ computes $\nabla f(x)$. Consider the code:

```
x = x0
for k in range(num_steps):
    x = x - alpha * grad_f(x)
```

1. What mathematical update rule is represented?
2. Which quantity is the learning rate?
3. What shape must $\text{grad_f}(x)$ have?
4. What would change if the minus sign were a plus sign?
5. Does this code prove that a minimum has been found?

Learning outcomes. U3-LO8, U3-LO9.

Solution.

The loop represents the update

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k).$$

The learning rate is α . The vector $\text{grad_f}(x)$ must have the same shape as x , since the update subtracts one vector from another. If the minus sign were replaced by a plus sign, the update would move in the direction of steepest increase instead of steepest decrease. The loop does not prove that a minimum has been found. It only describes the repeated update; convergence depends on the function, starting point, learning rate, and stopping rule.

Activity: Diagnosing learning rates

For $f(x) = x^2$, start at $x_0 = 4$ and use the update

$$x_{k+1} = x_k - \alpha f'(x_k).$$

The table shows the loss values $f(x_k)$ for three learning rates.

- $k: 0$
 $\alpha = 0.05: 16.000$
 $\alpha = 0.20: 16.000$
 $\alpha = 1.05: 16.000$
- $k: 1$
 $\alpha = 0.05: 12.960$
 $\alpha = 0.20: 5.760$
 $\alpha = 1.05: 19.360$
- $k: 2$
 $\alpha = 0.05: 10.498$
 $\alpha = 0.20: 2.074$
 $\alpha = 1.05: 23.426$
- $k: 3$
 $\alpha = 0.05: 8.503$
 $\alpha = 0.20: 0.746$
 $\alpha = 1.05: 28.345$

1. Which learning rate is making slow but steady progress?
2. Which learning rate is making faster useful progress?
3. Which learning rate appears unstable?
4. Does a decreasing loss table prove that the global minimum has been found?

Learning outcomes. U3-LO9.

Solution.

The learning rate $\alpha = 0.05$ is making slow but steady progress. The learning rate $\alpha = 0.20$ is making faster useful progress. The learning rate $\alpha = 1.05$ appears unstable because the loss is increasing. A decreasing loss table does not prove that the global minimum has been found. It only shows what happened for the displayed iterates.

Activity: Reading a next-token loss

A language model produces scores for possible next tokens, converts those scores into probabilities, and uses a loss to measure the probability assigned to the observed next token. A simplified training objective for a sequence $t_1, \dots, t_L$ can be written

$$\mathcal{L}(\theta) = - \sum_{i=1}^{L-1} \log p_{\theta}(t_{i+1} \mid t_1, \dots, t_i).$$

The parameter vector θ may collect many matrices and bias vectors. A gradient descent step has the form

$$\theta_{k+1} = \theta_k - \alpha \nabla \mathcal{L}(\theta_k).$$

1. What are the parameters?
2. What quantity is being minimized?
3. Which symbol is the learning rate?
4. What does the gradient point toward?
5. Why is this an optimization problem, even though we are not studying full model training?
6. Why is this not automatically a least-squares problem?

Learning outcomes. U3-LO9.

Solution.

The parameters are collected in θ . The objective $\mathcal{L}(\theta)$ is the loss. The learning rate is α . The gradient $\nabla \mathcal{L}(\theta_k)$ gives the local direction of steepest increase, so the negative gradient is used for descent. This is an optimization problem because training means adjusting parameters to reduce a loss.

Nothing in the displayed objective has the form

$$\|Ac - \mathbf{y}\|^2$$

with a fixed matrix A and a single trained coefficient vector $\mathbf{c}$. It is a general differentiable loss optimized by gradient descent.

Activity: Gradient of a final linear layer

Suppose a fixed hidden representation is

$$\mathbf{h} \in \mathbb{R}^d,$$

and a final linear layer computes

$$\ell = W\mathbf{h}, \quad W \in \mathbb{R}^{m \times d}.$$

Let

$$\mathbf{g} = \nabla_{\ell} \mathcal{L} \in \mathbb{R}^m$$

be the loss gradient with respect to the output scores.

1. For

$$\ell_i = \sum_{j=1}^d W_{ij} h_j,$$

use the chain rule to show that

$$\frac{\partial \mathcal{L}}{\partial W_{ij}} = g_i h_j.$$

2. Conclude that

$$\nabla_W \mathcal{L} = \mathbf{g} \mathbf{h}^T.$$

3. What is the shape of this matrix?

4. Why does it have rank at most one?

5. Interpret the update

$$W_{\text{new}} = W - \alpha \mathbf{g} \mathbf{h}^T.$$

Learning outcomes. U3-LO6, U3-LO9, U2-LO3.

Solution.

Since

$$\ell_k = \sum_{j=1}^d W_{kj} h_j,$$

we have

$$\frac{\partial \ell_k}{\partial W_{ij}} = \begin{cases} h_j, & k = i, \\ 0, & k \neq i. \end{cases}$$

The chain rule gives

$$\frac{\partial \mathcal{L}}{\partial W_{ij}} = \sum_{k=1}^m \frac{\partial \mathcal{L}}{\partial \ell_k} \frac{\partial \ell_k}{\partial W_{ij}} = g_i h_j.$$

Therefore

$$\nabla_W \mathcal{L} = \mathbf{g} \mathbf{h}^T.$$

If $\mathbf{g} \in \mathbb{R}^m$ and $\mathbf{h} \in \mathbb{R}^d$, then this outer product has shape $m \times d$. Every column is a scalar multiple of $\mathbf{g}$, so its rank is at most one. The update changes W in the negative-gradient direction for this training example.

Activity: The same final-layer update in code

```
G = np.outer(g, h)
W_new = W - alpha * G
```

1. Which earlier activity gives the formula represented by `G`?
2. What is the shape of `G`?
3. Why is `np.outer(g, h)` used here?
4. What mathematical update is represented by the second line?
5. Why is this connected to rank?

Learning outcomes. U3-LO6, U3-LO9, U2-LO3.

Solution.

By Gradient of a final linear layer,

$$G = \mathbf{g} \mathbf{h}^T = \nabla_W \mathcal{L}.$$

If $\mathbf{g}$ has length m and $\mathbf{h}$ has length d , then G has shape $m \times d$. The command `np.outer(g, h)` forms the outer product even when the two vectors are stored as one-dimensional NumPy arrays. The second line represents

$$W_{\text{new}} = W - \alpha \nabla_W \mathcal{L}.$$

The gradient matrix has rank at most one for this single training example. This does not imply that W or W_{new} has rank one.

3.10 Exercises

Learning outcomes. The labels below identify the learning outcomes for each exercise group. Individual problems may also involve earlier outcomes.

3.10.1 General functions and domains

Learning outcomes. U3-LO1.

- OpenStax 4.1 #7
- OpenStax 4.1 #9
- OpenStax 4.1 #17
- OpenStax 4.1 #19

3.10.2 One-parameter maps and changes

Learning outcomes. U3-LO1, U3-LO2.

- OpenStax 3.2 #43
- OpenStax 3.2 #49
- OpenStax 3.2 #61

3.10.3 Partial derivatives, gradients, Hessians, and Jacobian matrices

Learning outcomes. U3-LO2, U3-LO3.

- OpenStax 4.3 #119
- OpenStax 4.3 #121
- OpenStax 4.3 #135
- OpenStax 4.3 #137

3.10.4 Local linearization and tangent planes

Learning outcomes. U3-LO4, U3-LO5.

- OpenStax 4.4 #171
- OpenStax 4.4 #173

3.10.5 Chain rule as matrix multiplication

Learning outcomes. U3-LO6.

- OpenStax 4.5 #215
- OpenStax 4.5 #217
- OpenStax 4.5 #219
- OpenStax 4.5 #227

3.10.6 Directional derivatives

Learning outcomes. U3-LO8.

- OpenStax 4.6 #281
- OpenStax 4.6 #283
- OpenStax 4.6 #285
- OpenStax 4.6 #297

3.10.7 Additional exercises

These exercises cover Unit 3 ideas not fully represented in the OpenStax practice: scalar-valued and vector-valued functions, domains, local language, derivatives, gradients, Hessians, Jacobian matrices, tangent planes, local linear prediction, the chain rule, directional derivatives, critical points, gradient descent, and code interpretation. Solutions are collected in Appendix A.3.

Additional exercise: Directional derivative as a dot product

Suppose

$$\nabla f(\mathbf{a}) = \begin{bmatrix} 6 \\ 8 \end{bmatrix}.$$

Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \mathbf{v}_4 = -\frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

1. Compute $D_{\mathbf{v}_i} f(\mathbf{a})$ for $i = 1, 2, 3, 4$.
2. Which listed direction gives the largest directional derivative?
3. Which listed direction gives the smallest directional derivative?
4. Explain how this uses the Unit 1 dot product.

Learning outcomes. U3-LO8.

Additional exercise: Critical points are candidates

Let

$$f(x, y) = x^2 - y^2.$$

1. Find all critical points of f .
2. Evaluate $f(t, 0)$ and $f(0, t)$.
3. Explain why the critical point is not a local minimum or a local maximum.

Learning outcomes. U3-LO9.

Additional exercise: Reading a gradient descent loop

Assume $\text{grad_f}(x)$ computes $\nabla f(x)$. Consider the code:

```
x = x0
for k in range(num_steps):
    x = x - alpha * grad_f(x)
```

1. What mathematical update rule is represented?
2. Which quantity is the learning rate?
3. What shape must $\text{grad_f}(x)$ have?
4. What would change if the minus sign were a plus sign?
5. Does this code prove that a minimum has been found?

Learning outcomes. U3-LO8, U3-LO9.

Additional exercise: Learning-rate diagnosis

For $f(x) = x^2$, start at $x_0 = 4$ and use

$$x_{k+1} = x_k - \alpha f'(x_k).$$

The table shows the loss values $f(x_k)$ for three learning rates.

k	$\alpha = 0.05$	$\alpha = 0.20$	$\alpha = 1.05$
0	16.000	16.000	16.000
1	12.960	5.760	19.360
2	10.498	2.074	23.426
3	8.503	0.746	28.345

1. Which learning rate is making slow but steady progress?
2. Which learning rate is making faster useful progress?
3. Which learning rate appears unstable?
4. Does a decreasing loss table prove that the global minimum has been found?

Learning outcomes. U3-LO9.

Additional exercise: Function type and component functions

Let

$$F(x, y) = \begin{bmatrix} x^2 + y \\ e^{x-y} \\ x - y \end{bmatrix}.$$

1. Is F scalar-valued or vector-valued?
2. What are the domain and codomain?
3. Write the component functions F_1, F_2, F_3 .
4. Compute $F(1, 0)$.
5. What should be the shape of $J_F(1, 0)$?

Learning outcomes. U3-LO1, U3-LO3.

Additional exercise: Domain, boundary, closed, and bounded

Let

$$D = \{(x, y) \in \mathbb{R}^2 : x \geq 0, y > x^2\}.$$

1. Explain why this is the domain of $f(x, y) = \ln(y - x^2) + \sqrt{x}$.
2. Give one interior point of D .

3. Give one boundary point of D .
4. Is D closed?
5. Is D bounded?

Learning outcomes. U3-LO1, U3-LO2.

Additional exercise: One-parameter map and tangent vector

Let

$$\mathbf{r}(t) = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + t \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}.$$

1. Compute $\mathbf{r}(2)$.
2. Compute $\mathbf{r}'(t)$.
3. What does $\mathbf{r}'(t)$ say about the curve?
4. How does this connect to the Unit 2 parametric equation of a line?

Learning outcomes. U3-LO2, U2-LO1.

Additional exercise: Gradient and Hessian

Let

$$f(x, y) = x^2 + xy + 2y^2.$$

1. Compute f_x and f_y .
2. Compute $\nabla f(1, -1)$.
3. Compute $H_f(x, y)$.
4. Is the Hessian symmetric?
5. What does each entry of $\nabla f(1, -1)$ measure?

Learning outcomes. U3-LO3.

Additional exercise: Jacobian matrix rows and columns

Let

$$F(x, y) = \begin{bmatrix} x^2 y \\ x e^y \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

1. Compute $J_F(x, y)$.
2. Compute $J_F(\mathbf{a})$.
3. What are the input and output dimensions?
4. Which row corresponds to the second component of F ?
5. What do the two columns of $J_F(\mathbf{a})$ measure?

Learning outcomes. U3-LO3, U3-LO5, U1-LO4.

Additional exercise: Tangent plane as a Unit 2 plane

Let

$$f(x, y) = x^2 + xy.$$

1. Compute $f(1, 2)$, $f_x(1, 2)$, and $f_y(1, 2)$.
2. Write the tangent plane to $z = f(x, y)$ at $(1, 2, f(1, 2))$.
3. Rewrite the tangent plane in the form

$$\mathbf{n} \cdot \begin{bmatrix} x - 1 \\ y - 2 \\ z - f(1, 2) \end{bmatrix} = 0.$$

4. Identify a normal vector $\mathbf{n}$.
5. Which Unit 2 idea is being reused?

Learning outcomes. U3-LO4, U2-LO1.

Additional exercise: Scalar local linear prediction

Use the same function $f(x, y) = x^2 + xy$ at

$$\mathbf{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{h} = \begin{bmatrix} 0.1 \\ -0.2 \end{bmatrix}.$$

1. Compute $f(\mathbf{a})$.
2. Compute $\nabla f(\mathbf{a})$.
3. Compute the local linear prediction $f(\mathbf{a}) + \nabla f(\mathbf{a})^T \mathbf{h}$.
4. Compute the actual value $f(\mathbf{a} + \mathbf{h})$.
5. Compare the actual value with the prediction.

Learning outcomes. U3-LO4, U3-LO5.

Additional exercise: Vector-valued local linear prediction

Use

$$F(x, y) = \begin{bmatrix} x^2 y \\ x e^y \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{h} = \begin{bmatrix} 0.1 \\ -0.2 \end{bmatrix}.$$

1. Compute $F(\mathbf{a})$.
2. Use $J_F(\mathbf{a})$ from Jacobian matrix rows and columns.
3. Compute $F(\mathbf{a}) + J_F(\mathbf{a})\mathbf{h}$.
4. Compute $F(\mathbf{a} + \mathbf{h})$.
5. What does the error vector measure?

Learning outcomes. U3-LO5.

Additional exercise: Locally forgotten direction

Suppose

$$J_F(\mathbf{a}) = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

1. Find a nonzero vector $\mathbf{h}$ such that $J_F(\mathbf{a})\mathbf{h} = \mathbf{0}$.
2. What does this say about the local linear map?
3. How does this connect to Unit 2 null spaces?

4. Does this prove that $F(\mathbf{a} + \mathbf{h}) = F(\mathbf{a})$ exactly? Explain.

Learning outcomes. U3-LO5, U2-LO3, U2-LO6.

Additional exercise: Chain-rule shape check

Suppose

$$G : \mathbb{R}^2 \rightarrow \mathbb{R}^3, \quad F : \mathbb{R}^3 \rightarrow \mathbb{R}^2.$$

1. What is the shape of $J_G(\mathbf{a})$?
2. What is the shape of $J_F(G(\mathbf{a}))$?
3. What is the shape of $J_{F \circ G}(\mathbf{a})$?
4. Which product gives $J_{F \circ G}(\mathbf{a})$?
5. Why is the reverse order not correct?

Learning outcomes. U3-LO6, U1-LO6.

Additional exercise: Reading a tiny sigmoid block

Consider the code:

```
def sigmoid(t):  
    return 1 / (1 + np.exp(-t))  
  
s = W1 @ x + b1  
u = sigmoid(s)  
D = np.diag(u * (1 - u))  
y = W2 @ u + b2  
J_at_x = W2 @ D @ W1
```

1. Which lines are affine maps?
2. Which line contains the coordinatewise nonlinear step?
3. Which variables represent vectors?
4. What does D store?
5. Which line computes the Jacobian matrix of the block at $\mathbf{x}$?

6. Why is the whole block usually not one linear map?

Learning outcomes. U3-LO7, U1-LO7.

Additional exercise: Corners are not enough

Let

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y \\ y \end{bmatrix}, \quad F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y^2 \\ y \end{bmatrix}.$$

1. Compute the images of the four corners of the unit square under T .
2. Compute the images of the four corners of the unit square under F .
3. Compute the images of $\begin{bmatrix} 0 \\ 1/2 \end{bmatrix}$ under both maps.
4. Explain why the four corners do not prove that two maps are the same.
5. Explain why a grid gives more information than the four corners.

Learning outcomes. U3-LO1, U1-LO5.

Additional exercise: Local square-grid error

Let

$$F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y^2 \\ y \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} 0 \\ 1/2 \end{bmatrix}.$$

1. Compute $J_F(x, y)$.
2. Compute $J_F(\mathbf{a})$.
3. Let $\mathbf{h} = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}$. Compute $F(\mathbf{a} + \mathbf{h}) - F(\mathbf{a})$.
4. Compute $J_F(\mathbf{a})\mathbf{h}$.
5. Compute the error

$$F(\mathbf{a} + \mathbf{h}) - F(\mathbf{a}) - J_F(\mathbf{a})\mathbf{h}.$$

6. If $|h_2| \leq r$, explain why the error goes to zero as $r \rightarrow 0$.

Learning outcomes. U3-LO3, U3-LO5.

Additional exercise: Code interpretation: square-grid arrays

Consider the code:

```
import numpy as np

def F(P):
    x = P[0, :]
    y = P[1, :]
    return np.vstack([x + y**2, y])

P = np.array([
    [0.0, 1.0, 1.0, 0.0, 0.0],
    [0.0, 0.0, 1.0, 1.0, 0.5],
])

Q = F(P)
Q
```

1. What is the shape of P ?
2. Are input points stored as rows or columns?
3. Which column stores $\begin{bmatrix} 0 \\ 1/2 \end{bmatrix}$?
4. What is the corresponding output column of Q ?
5. What does the expression `np.vstack([x + y**2, y])` construct?
6. Why is this computation not a matrix multiplication?

Learning outcomes. U3-LO1, U3-LO5, U1-LO8.

Additional exercise: Sigmoid derivative and local matrix

Let

$$\sigma(t) = \frac{1}{1 + e^{-t}}, \quad \sigma'(t) = \sigma(t)(1 - \sigma(t)),$$

applied coordinatewise. Define

$$W_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{b}_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

$$W_2 = \begin{bmatrix} 1 & 0 & -1/2 \\ 0 & 1 & 1/2 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

For

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix},$$

set

$$N(\mathbf{x}) = W_2 \sigma(W_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2.$$

Let

$$\mathbf{a} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

1. Compute $\mathbf{s}_a = W_1 \mathbf{a} + \mathbf{b}_1$.
2. Compute $\mathbf{u}_a = \sigma(\mathbf{s}_a)$.
3. Compute $D_a = \text{diag}(\sigma'((\mathbf{s}_a)_1), \sigma'((\mathbf{s}_a)_2), \sigma'((\mathbf{s}_a)_3))$.
4. Compute the local matrix $W_2 D_a W_1$.
5. Is this local matrix invertible?
6. Let $\mathbf{c} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Which entries of D_c differ from entries of D_a ?
7. Why can the local matrix change from one base point to another?

Learning outcomes. U3-LO3, U3-LO5, U3-LO7.

Additional exercise: Code interpretation: sigmoid block arrays

Consider the code:

```
import numpy as np

W1 = np.array([
    [1.0, 0.0],
    [0.0, 1.0],
    [1.0, 1.0],
])

b1 = np.zeros((3, 1))
```

```

W2 = np.array([
    [1.0, 0.0, -0.5],
    [0.0, 1.0, 0.5],
])

b2 = np.zeros((2, 1))

def sigmoid(Z):
    return 1 / (1 + np.exp(-Z))

def N(P):
    S = W1 @ P + b1
    U = sigmoid(S)
    return W2 @ U + b2

P = np.array([
    [0.0, 1.0, 0.0],
    [0.0, 0.0, 1.0],
])

Q = N(P)
Q

```

1. What is the shape of P ?
2. Are input points stored as rows or columns?
3. What is the shape of $S = W_1 P + b_1$?
4. What is the shape of $U = \sigma(S)$?
5. Which line applies the nonlinear operation?
6. Compute the first column of Q .
7. Why is the full computation not one matrix multiplication AP ?

Learning outcomes. U3-LO5, U3-LO7, U1-LO8.

Additional exercise: Softmax weights are data-dependent

Let

$$K = \begin{bmatrix} 3 & 0 \\ 0 & 3 \\ -3 & -3 \end{bmatrix}, \quad V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

For

$$\mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix},$$

define

$$\mathbf{s} = K\mathbf{q}, \quad \alpha(\mathbf{q}) = \text{softmax}(\mathbf{s}), \quad F(\mathbf{q}) = V^T \alpha(\mathbf{q}).$$

1. Compute $\mathbf{s}$ in terms of q_1 and q_2 .
2. Compute $\alpha(\mathbf{0})$.
3. Compute $F(\mathbf{0})$.
4. Explain why F is not linear.
5. Explain why $F(\mathbf{q})$ is always a weighted average of the rows of V .
6. Which step is nonlinear?

Learning outcomes. U3-LO1, U3-LO5, U3-LO7, U1-LO8.

Additional exercise: Code interpretation: softmax square-grid arrays

Consider the code:

```
import numpy as np

K = np.array([
    [3.0, 0.0],
    [0.0, 3.0],
    [-3.0, -3.0],
])

V = np.array([
    [1.0, 0.0],
    [0.0, 1.0],
    [0.0, 0.0],
])

def softmax_columns(S):
    S_shifted = S - S.max(axis=0, keepdims=True)
```

```

    E = np.exp(S_shifted)
    return E / E.sum(axis=0, keepdims=True)

def F(P):
    S = K @ P
    Alpha = softmax_columns(S)
    return V.T @ Alpha

P = np.array([
    [0.0, 1.0, 0.0, -1.0],
    [0.0, 0.0, 1.0, -1.0],
])

Q = F(P)
Q

```

1. What is the shape of P ?
2. Are query vectors stored as rows or columns?
3. What is the shape of $S = KP$?
4. What is the shape of Alpha ?
5. What is the shape of Q ?
6. Which line applies the nonlinear operation?
7. Why is the full computation not one matrix multiplication AP ?

Learning outcomes. U3-LO5, U3-LO7, U1-LO8.

Additional exercise: Reading square-grid behavior

A square-grid diagram shows an input grid and its image under a map. Match each behavior with the most likely map type.

Behaviors:

1. Grid lines bend smoothly because of a y^2 term.
2. Grid bends and compresses smoothly because sigmoid derivatives change with input.
3. The output points stay inside a triangle formed by three value vectors.
4. The four corners match a linear shear, but an edge midpoint does not.

5. The local matrix has form $W_2 D_a W_1$.
6. The nonlinear step comes from softmax weights depending on input.

Map types:

1. Nonlinear shear

$$F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y^2 \\ y \end{bmatrix}.$$

2. Two-output sigmoid block

$$N(\mathbf{x}) = W_2 \sigma(W_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2.$$

3. Attention-style softmax map

$$F_{\text{att}}(\mathbf{q}) = V^T \text{softmax}(K\mathbf{q}).$$

1. Match each behavior to Map A, Map B, or Map C.
2. Which map best shows why four corners alone can miss nonlinear behavior?
3. Which map best shows the chain rule for Jacobian matrices?
4. Which map best shows attention-style weighted averages?
5. What can a square-grid visualization show that four corners alone may miss?

Learning outcomes. U3-LO1, U3-LO5, U3-LO7, U1-LO8.

Additional exercise: Final-layer rank-one update

Let

$$\mathbf{g} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \quad \mathbf{h} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}.$$

1. Compute $\mathbf{g}\mathbf{h}^T$.
2. What is the shape of $\mathbf{g}\mathbf{h}^T$?
3. Explain why $\mathbf{g}\mathbf{h}^T$ has rank at most one.

4. Interpret the update $W_{\text{new}} = W - \alpha g h^T$.
5. In NumPy, why is `np.outer(g, h)` appropriate when `g` and `h` are stored as one-dimensional arrays?

Learning outcomes. U3-LO6, U3-LO9, U2-LO3.

3.11 Unit 3 highlights

3.11.1 Mathematical quick reference

Functions, domains, and local language

Function types. A scalar-valued function has the form $f : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$. A vector-valued function $F : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ has scalar component functions $F_1, \dots, F_m$. The domain consists of allowed inputs; the range consists of attained outputs.

Graph and level set. The graph of f is $\{(\mathbf{x}, f(\mathbf{x})) : \mathbf{x} \in D\}$; the level set at c is $\{\mathbf{x} \in D : f(\mathbf{x}) = c\}$.

Open ball. $B_r(\mathbf{a}) = \{\mathbf{x} : \|\mathbf{x} - \mathbf{a}\| < r\}$, with $r > 0$. A point is interior to D if some ball about it lies in D ; it is a boundary point if every ball meets both D and its complement. A set is closed if it contains all its boundary points, and bounded if it lies in some ball.

Limit. At an accumulation point of the domain, $F(\mathbf{x}) \rightarrow \mathbf{L}$ as $\mathbf{x} \rightarrow \mathbf{a}$ means that for every $\epsilon > 0$ there is $\delta > 0$ such that

$$\mathbf{x} \in D, \quad 0 < \|\mathbf{x} - \mathbf{a}\| < \delta \quad \implies \quad \|F(\mathbf{x}) - \mathbf{L}\| < \epsilon.$$

Limits of vector-valued functions are computed componentwise. Two paths giving different limits disprove existence; agreement along a few paths does not prove existence.

Continuity. At an accumulation point $\mathbf{a} \in D$, continuity means $\lim_{\mathbf{x} \rightarrow \mathbf{a}} F(\mathbf{x}) = F(\mathbf{a})$, with inputs restricted to D . Continuity is automatic at isolated domain points.

Derivatives and their dimensions

One-parameter derivative. For $\mathbf{r}(t) = (r_1(t), \dots, r_m(t))$,

$$\mathbf{r}'(t) = (r'_1(t), \dots, r'_m(t)).$$

When $\mathbf{r}'(a) \neq \mathbf{0}$, the tangent line is $\mathbf{r}(a) + s\mathbf{r}'(a)$.

Partial derivative. Differentiate in one coordinate while fixing the others:

$$f_{x_j}(\mathbf{a}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{a} + h\mathbf{e}_j) - f(\mathbf{a})}{h}.$$

Gradient and Hessian. For a scalar-valued function of n variables,

$$\nabla f = \begin{bmatrix} f_{x_1} \\ \vdots \\ f_{x_n} \end{bmatrix} \in \mathbb{R}^n, \quad H_f = [f_{x_i x_j}] \in \mathbb{R}^{n \times n}.$$

Continuous second partial derivatives ensure equality of mixed partials and symmetry of the Hessian.

Jacobian matrix. For $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$,

$$J_F = \left[\frac{\partial F_i}{\partial x_j} \right] \in \mathbb{R}^{m \times n}, \quad J_f = (\nabla f)^T.$$

Row i differentiates component F_i ; column j describes change in input direction $\mathbf{e}_j$.

Differentiability and local approximation

Differentiability. At an interior point $\mathbf{a}$, the derivative is a linear map whose error is small relative to the input change:

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|F(\mathbf{a} + \mathbf{h}) - F(\mathbf{a}) - J_F(\mathbf{a})\mathbf{h}\|}{\|\mathbf{h}\|} = 0.$$

Continuous first partial derivatives near $\mathbf{a}$ are sufficient for differentiability. Existence of partial derivatives alone is not sufficient.

Local prediction.

$$F(\mathbf{a} + \mathbf{h}) \approx F(\mathbf{a}) + J_F(\mathbf{a})\mathbf{h}, \quad f(\mathbf{a} + \mathbf{h}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h}.$$

Tangent plane. For a differentiable $z = f(x, y)$ at (a, b) ,

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

A normal vector is $(f_x(a, b), f_y(a, b), -1)$.

First-order zero change. If $\mathbf{h} \in \text{null}(J_F(\mathbf{a}))$, the predicted first-order output change is zero; the actual nonlinear change need not be zero.

Chain rule

Composition. If $G : \mathbb{R}^n \rightarrow \mathbb{R}^p$ is differentiable at $\mathbf{a}$ and $F : \mathbb{R}^p \rightarrow \mathbb{R}^m$ is differentiable at $G(\mathbf{a})$, then

$$J_{F \circ G}(\mathbf{a}) = J_F(G(\mathbf{a}))J_G(\mathbf{a}).$$

The dimensions are $(m \times p)(p \times n) = m \times n$; the right factor acts first.

Scalar function along a curve.

$$\frac{d}{dt} f(\mathbf{r}(t)) = \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t).$$

Directional change and first-order optimization

Directional derivative. For differentiable f and a unit direction $\mathbf{u}$,

$$D_{\mathbf{u}}f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{u}.$$

Normalize a proposed nonzero direction before using the formula.

Steepest directions. If $\nabla f(\mathbf{a}) \neq \mathbf{0}$, the greatest and least directional derivatives are $\|\nabla f(\mathbf{a})\|$ and $-\|\nabla f(\mathbf{a})\|$, attained in directions $\pm \nabla f(\mathbf{a}) / \|\nabla f(\mathbf{a})\|$. If the gradient is zero, every directional derivative is zero.

Local and global extrema. A local minimum compares values in a neighborhood; a global minimum compares all values on the domain. Reverse the inequalities for maxima.

Critical point. In this book, a differentiable point $\mathbf{a}$ with $\nabla f(\mathbf{a}) = \mathbf{0}$. An interior differentiable local extremum must be critical. The converse fails; boundary points and points where differentiability fails require separate checks.

Gradient descent. Starting from $\mathbf{x}_0$, with step size $\alpha > 0$,

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k).$$

For a differentiable function with a nonzero gradient, a sufficiently small step in the negative-gradient direction decreases the function. Arbitrary step sizes need not decrease it or yield convergence.

3.11.2 Common mistakes

Confusing domain with range; using a few paths to prove a multivariable limit; assuming partial derivatives guarantee differentiability; treating a local approximation as exact; confusing $\mathbf{h}$ with $\mathbf{a} + \mathbf{h}$; reversing chain-rule factors or evaluating them at the wrong point; using a nonunit direction; treating every critical point as an extremum; ignoring boundaries; or assuming every gradient-descent step decreases the function.

3.11.3 Connections

Unit 1 supplies linear maps, affine maps, and composition. Unit 2 supplies planes, normal vectors, systems, and null spaces. Unit 4 interprets directional derivatives through projection. Unit 5 uses Hessian eigenvalues to classify critical points.

Chapter 4

Orthogonality and least squares

Orthogonality turns geometry into computation. In this unit, projection becomes the mechanism for best approximation, and regression becomes a linear algebra problem.

Big question. When $A\mathbf{x} = \mathbf{b}$ has no solution, what is the best approximate solution?

Learning outcomes. By the end of this unit, students should be able to:

- *U4-LO1.* Use orthogonality and orthonormal bases to decompose vectors and compute coefficients.
- *U4-LO2.* Construct orthogonal and orthonormal bases using Gram-Schmidt, and compute orthogonal complements.
- *U4-LO3.* Compute orthogonal projections onto subspaces and solve closest-point problems.
- *U4-LO4.* Formulate least squares as projection onto a column space, identify the residual, and explain residual orthogonality.
- *U4-LO5.* Derive and solve the normal equations $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$, and use them in small regression problems.
- *U4-LO6.* Compute and interpret QR factorizations, and use QR to solve linear systems and least-squares problems.
- *U4-LO7.* Interpret computational least-squares checks such as $\|A^T \mathbf{r}\|$ and explain what they do and do not prove.

Toolbox skills. Compute dot products, test orthogonality, normalize vectors, apply Gram-Schmidt, compute projections, identify residuals, derive and solve normal equations, fit a line by least squares, check $A^T \mathbf{r} \approx \mathbf{0}$, and use QR in small least-squares problems.

4.1 Orthogonality and orthogonal bases

4.1.1 Basic definitions

Definition 4.1.1.

Recall the definition of the dot product of two vectors, introduced in 1.1.6, and the length of a vector as introduced in 1.1.2. The *distance* between two points $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_n)$ is the length of the vector $\mathbf{x} - \mathbf{y}$, i.e.,

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}$$

. If $\mathbf{x}$ is any non-zero vector, then the vector $\mathbf{u} = \mathbf{x}/\|\mathbf{x}\|$ is the *unit vector in the direction of $\mathbf{x}$* (a unit vector is a vector of length one). Two vectors $\mathbf{x}$ and $\mathbf{y}$ are *orthogonal* or *perpendicular* if $\mathbf{x} \cdot \mathbf{y} = 0$.

Theorem 4.1.2 (The Cauchy-Schwarz inequality).

If $\mathbf{x}, \mathbf{y}$ are vectors in $\mathbb{R}^n$,

$$|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|$$

. Moreover, $|\mathbf{x} \cdot \mathbf{y}| = \|\mathbf{x}\| \|\mathbf{y}\|$ only when $\mathbf{x}$ and $\mathbf{y}$ are scalar multiples of each other.

Activity: Cosine similarity revisited

Let $\mathbf{u}$ and $\mathbf{v}$ be nonzero vectors in $\mathbb{R}^n$. The cosine similarity from Unit 1 was

$$\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

1. Use the Cauchy-Schwarz inequality to explain why this number is between -1 and 1 .
2. What condition makes the cosine similarity equal to 0 ?
3. In Unit 1, this number measured cosine similarity. What does a value of 0 say geometrically?

Solution.

By Cauchy-Schwarz, $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$. Since both norms are nonzero, division gives $-1 \leq (\mathbf{u} \cdot \mathbf{v})/(\|\mathbf{u}\| \|\mathbf{v}\|) \leq 1$. The value is 0 exactly when $\mathbf{u} \cdot \mathbf{v} = 0$, which means the vectors are orthogonal.

Theorem 4.1.3 (The triangle inequality).

If $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, then

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$$

.

Remark

The triangle inequality for vectors implies the triangle inequality for the distance function, i.e., that

$$d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z})$$

. Indeed, we calculate that

$$\begin{aligned}
 d(\mathbf{x}, \mathbf{z}) &= \|\mathbf{x} - \mathbf{z}\| \\
 &= \|\mathbf{x} - \mathbf{y} + \mathbf{y} - \mathbf{z}\| \\
 &\leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y} - \mathbf{z}\|
 \end{aligned}$$

Theorem 4.1.4.

Let $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$ be vectors in $\mathbb{R}^n$, and let a be a scalar in $\mathbb{R}$. Then the following properties hold:

- $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$.
- $\mathbf{x} \cdot (\mathbf{y} + \mathbf{z}) = \mathbf{x} \cdot \mathbf{y} + \mathbf{x} \cdot \mathbf{z}$.
- $(a\mathbf{x}) \cdot \mathbf{y} = a(\mathbf{x} \cdot \mathbf{y}) = \mathbf{x} \cdot (a\mathbf{y})$.
- $\|\mathbf{x}\|^2 = \mathbf{x} \cdot \mathbf{x}$.
- $\mathbf{x} \cdot \mathbf{x} \geq 0$, and $\mathbf{x} \cdot \mathbf{x} = 0$ if and only if $\mathbf{x} = \mathbf{0}$.
- $\|a\mathbf{x}\| = |a|\|\mathbf{x}\|$.

Activity

Show that $\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + 2(\mathbf{x} \cdot \mathbf{y}) + \|\mathbf{y}\|^2$ for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

Solution.

We calculate using the properties in 4.1.4 that

$$\begin{aligned}
 \|\mathbf{x} + \mathbf{y}\|^2 &= (\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) \\
 &= \mathbf{x} \cdot \mathbf{x} + \mathbf{y} \cdot \mathbf{x} + \mathbf{x} \cdot \mathbf{y} + \mathbf{y} \cdot \mathbf{y} \\
 &= \|\mathbf{x}\|^2 + 2\mathbf{x} \cdot \mathbf{y} + \|\mathbf{y}\|^2
 \end{aligned}$$

Activity

Suppose that $\mathbb{R}^n = \text{span}\{\mathbf{f}_1, \dots, \mathbf{f}_k\}$ for some vectors $\mathbf{f}_i$. If $\mathbf{x} \in \mathbb{R}^n$, and $\mathbf{x} \cdot \mathbf{f}_i = 0$ for each i , then show $\mathbf{x} = \mathbf{0}$.

Solution.

Since $\mathbb{R}^n$ is spanned by the vectors $\{\mathbf{f}_1, \dots, \mathbf{f}_k\}$, we can write $\mathbf{x} = t_1\mathbf{f}_1 + \dots + t_k\mathbf{f}_k$ for some scalars $t_1, \dots, t_k \in \mathbb{R}$. But then

$$\begin{aligned}
 \|\mathbf{x}\|^2 &= \mathbf{x} \cdot \mathbf{x} \\
 &= \mathbf{x} \cdot (t_1\mathbf{f}_1 + \dots + t_k\mathbf{f}_k) \\
 &= t_1(\mathbf{x} \cdot \mathbf{f}_1) + \dots + t_k(\mathbf{x} \cdot \mathbf{f}_k) \\
 &= t_1(0) + \dots + t_k(0) \\
 &= 0
 \end{aligned}$$

. Thus $\|\mathbf{x}\| = 0$. But this can only occur if $\mathbf{x} = \mathbf{0}$ by 4.1.4.

4.1.2 Orthogonal sets and the expansion theorem**Definition 4.1.5 (Orthogonal and orthonormal sets).**

A set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is called an *orthogonal set* if the set does not contain the zero vector, and $\mathbf{x}_i \cdot \mathbf{x}_j = 0$ for $i \neq j$. A set $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is called *orthonormal* if it is orthogonal and if in addition, $\|\mathbf{x}_i\| = 1$ for each i .

Note that $\{\mathbf{x}\}$ is an orthogonal set if $\mathbf{x} \neq \mathbf{0}$. The standard basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ (see 1.3.5) is an important orthonormal set in $\mathbb{R}^n$.

Activity

Consider the following vectors in $\mathbb{R}^4$, written as column vectors:

$$\mathbf{f}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} \quad \mathbf{f}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix} \quad \mathbf{f}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{f}_4 = \begin{bmatrix} -1 \\ 3 \\ -1 \\ 1 \end{bmatrix}$$

. Verify that these vectors form an orthogonal set in $\mathbb{R}^4$. Then find the orthonormal set obtained by ‘normalizing’ these vectors, i.e., multiplying each vector by an appropriate constant to obtain a unit vector.

Solution.

To verify that $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \mathbf{f}_4\}$ form an orthogonal set, we need to check that $\mathbf{f}_i \cdot \mathbf{f}_j = 0$ for all $i \neq j$. We calculate that

$$\begin{aligned}
\mathbf{f}_1 \cdot \mathbf{f}_2 &= 1 \cdot 1 + 1 \cdot 0 + 1 \cdot 1 + (-1) \cdot 2 = 1 + 0 + 1 - 2 = 0 \\
\mathbf{f}_1 \cdot \mathbf{f}_3 &= 1 \cdot (-1) + 1 \cdot 0 + 1 \cdot 1 + (-1) \cdot 0 = -1 + 0 + 1 + 0 = 0 \\
\mathbf{f}_1 \cdot \mathbf{f}_4 &= 1 \cdot (-1) + 1 \cdot 3 + 1 \cdot (-1) + (-1) \cdot 1 = -1 + 3 - 1 - 1 = 0 \\
\mathbf{f}_2 \cdot \mathbf{f}_3 &= 1 \cdot (-1) + 0 \cdot 0 + 1 \cdot 1 + 2 \cdot 0 = -1 + 0 + 1 + 0 = 0 \\
\mathbf{f}_2 \cdot \mathbf{f}_4 &= 1 \cdot (-1) + 0 \cdot 3 + 1 \cdot (-1) + 2 \cdot 1 = -1 + 0 - 1 + 2 = 0 \\
\mathbf{f}_3 \cdot \mathbf{f}_4 &= (-1) \cdot (-1) + 0 \cdot 3 + 1 \cdot (-1) + 0 \cdot 1 = 1 + 0 - 1 + 0 = 0
\end{aligned}$$

. Since all these dot products are zero, the set is orthogonal.

To normalize these vectors, we need to calculate their lengths and divide each vector by its length. We find that

$$\begin{aligned}
\|\mathbf{f}_1\| &= \sqrt{1^2 + 1^2 + 1^2 + (-1)^2} = \sqrt{4} = 2 \\
\|\mathbf{f}_2\| &= \sqrt{1^2 + 0^2 + 1^2 + 2^2} = \sqrt{6} \\
\|\mathbf{f}_3\| &= \sqrt{(-1)^2 + 0^2 + 1^2 + 0^2} = \sqrt{2} \\
\|\mathbf{f}_4\| &= \sqrt{(-1)^2 + 3^2 + (-1)^2 + 1^2} = \sqrt{12} = 2\sqrt{3}
\end{aligned}$$

. Therefore the normalized vectors are given by:

$$\begin{aligned}
\frac{1}{\|\mathbf{f}_1\|} \mathbf{f}_1 &= \frac{1}{2} \mathbf{f}_1 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \\
\frac{1}{\|\mathbf{f}_2\|} \mathbf{f}_2 &= \frac{1}{\sqrt{6}} \mathbf{f}_2 = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ 0 \\ \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{bmatrix} \\
\frac{1}{\|\mathbf{f}_3\|} \mathbf{f}_3 &= \frac{1}{\sqrt{2}} \mathbf{f}_3 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} \\
\frac{1}{\|\mathbf{f}_4\|} \mathbf{f}_4 &= \frac{1}{2\sqrt{3}} \mathbf{f}_4 = \begin{bmatrix} -\frac{1}{2\sqrt{3}} \\ \frac{3}{2\sqrt{3}} \\ -\frac{1}{2\sqrt{3}} \\ \frac{1}{2\sqrt{3}} \end{bmatrix}
\end{aligned}$$

.

Theorem 4.1.6 (Pythagoras' theorem).

If $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_k\}$ is an orthogonal set in $\mathbb{R}^n$, then

$$\|\mathbf{f}_1 + \mathbf{f}_2 + \dots + \mathbf{f}_k\|^2 = \|\mathbf{f}_1\|^2 + \|\mathbf{f}_2\|^2 + \dots + \|\mathbf{f}_k\|^2.$$

Theorem 4.1.7.

Every orthogonal set in $\mathbb{R}^n$ is linearly independent.

Proof.

Let $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_k\}$ be an orthogonal set in $\mathbb{R}^n$. To show the set is linearly independent (recalling 2.4.1), we must show the only solution to the linear equation

$$t_1 \mathbf{f}_1 + t_2 \mathbf{f}_2 + \dots + t_k \mathbf{f}_k = \mathbf{0}$$

occurs when $t_1 = \dots = t_k = 0$.

If we take the dot product of both sides of the equation with $\mathbf{f}_1$, we find that

$$\mathbf{f}_1 \cdot (t_1 \mathbf{f}_1 + t_2 \mathbf{f}_2 + \dots + t_k \mathbf{f}_k) = \mathbf{f}_1 \cdot \mathbf{0}$$

and therefore that

$$t_1(\mathbf{f}_1 \cdot \mathbf{f}_1) + t_2(\mathbf{f}_1 \cdot \mathbf{f}_2) + \dots + t_k(\mathbf{f}_1 \cdot \mathbf{f}_k) = 0$$

Since the set of vectors is orthogonal, $\mathbf{f}_1 \cdot \mathbf{f}_j = 0$ for all $j \neq 1$, and $\mathbf{f}_1 \cdot \mathbf{f}_1 = \|\mathbf{f}_1\|^2$, so we conclude that

$$t_1 \|\mathbf{f}_1\|^2 = 0.$$

Since $\mathbf{f}_1 \neq \mathbf{0}$ (by definition, an orthogonal set cannot contain the zero vector), $\|\mathbf{f}_1\|^2 > 0$, and so it must be true that $t_1 = 0$. But now repeating this argument with 1 replaced by any index i , we conclude that $t_i = 0$. Therefore the set $\{\mathbf{f}_1, \dots, \mathbf{f}_k\}$ is linearly independent.

Theorem 4.1.8 (Expansion theorem).

Let $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$ be an orthogonal basis of a subspace U of $\mathbb{R}^n$. If $\mathbf{v}$ is any vector in U , we have

$$\mathbf{v} = \left(\frac{\mathbf{v} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \right) \mathbf{f}_1 + \left(\frac{\mathbf{v} \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \right) \mathbf{f}_2 + \dots + \left(\frac{\mathbf{v} \cdot \mathbf{f}_m}{\|\mathbf{f}_m\|^2} \right) \mathbf{f}_m$$

Proof.

Since $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$ spans U , any $\mathbf{v} \in U$ can be written uniquely as

$$\mathbf{v} = t_1 \mathbf{f}_1 + t_2 \mathbf{f}_2 + \dots + t_m \mathbf{f}_m$$

for some $t_1, \dots, t_m$. To find coefficient t_1 , take the dot product of both sides with $\mathbf{f}_1$, obtaining that

$$\begin{aligned} \mathbf{v} \cdot \mathbf{f}_1 &= (t_1 \mathbf{f}_1 + t_2 \mathbf{f}_2 + \dots + t_m \mathbf{f}_m) \cdot \mathbf{f}_1 \\ &= t_1(\mathbf{f}_1 \cdot \mathbf{f}_1) + t_2(\mathbf{f}_2 \cdot \mathbf{f}_1) + \dots + t_m(\mathbf{f}_m \cdot \mathbf{f}_1) \end{aligned}$$

. Since the basis is orthogonal, $\mathbf{f}_i \cdot \mathbf{f}_j = 0$ for $i \neq j$, and so the equation simplifies to

$$\mathbf{v} \cdot \mathbf{f}_1 = t_1 \|\mathbf{f}_1\|^2$$

. Rearranging, we conclude that

$$t_1 = \frac{\mathbf{v} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2}$$

A similar argument shows that

$$t_i = \frac{\mathbf{v} \cdot \mathbf{f}_i}{\|\mathbf{f}_i\|^2}$$

for each i , concluding the proof.

Definition 4.1.9 (The Fourier expansion in an orthogonal basis).

Given an orthogonal basis $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$ for some subspace V of $\mathbb{R}^n$, the expansion of $\mathbf{v} \in V$ as a linear combination of this basis is called the *Fourier expansion* of the vector $\mathbf{v}$, and the coefficients

$$t_i = \frac{\mathbf{v} \cdot \mathbf{f}_i}{\|\mathbf{f}_i\|^2}$$

are called the *Fourier coefficients* of $\mathbf{v}$.

Activity

Let $\mathbf{v} = (a, b, c, d)$ be a vector in $\mathbb{R}^4$. Find the Fourier expansion of $\mathbf{v}$ as a linear combination of the orthogonal basis $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \mathbf{f}_4\}$ given in 4.1.2.

Solution.

We need to find the Fourier coefficients

$$t_i = \frac{\mathbf{v} \cdot \mathbf{f}_i}{\|\mathbf{f}_i\|^2}$$

for $i = 1, 2, 3, 4$. In 4.1.2 we already computed the lengths of the elements of the basis, namely, that

$$\|\mathbf{f}_1\| = 2, \quad \|\mathbf{f}_2\| = \sqrt{6}, \quad \|\mathbf{f}_3\| = \sqrt{2} \quad \text{and} \quad \|\mathbf{f}_4\| = 2\sqrt{3}$$

. We now calculate the dot products:

$$\mathbf{v} \cdot \mathbf{f}_1 = a \cdot 1 + b \cdot 1 + c \cdot 1 + d \cdot (-1) = a + b + c - d$$

$$\mathbf{v} \cdot \mathbf{f}_2 = a \cdot 1 + b \cdot 0 + c \cdot 1 + d \cdot 2 = a + c + 2d$$

$$\mathbf{v} \cdot \mathbf{f}_3 = a \cdot (-1) + b \cdot 0 + c \cdot 1 + d \cdot 0 = -a + c$$

$$\mathbf{v} \cdot \mathbf{f}_4 = a \cdot (-1) + b \cdot 3 + c \cdot (-1) + d \cdot 1 = -a + 3b - c + d$$

Therefore, the Fourier coefficients are:

$$\begin{aligned} t_1 &= \frac{\mathbf{v} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} = \frac{a + b + c - d}{4} \\ t_2 &= \frac{\mathbf{v} \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} = \frac{a + c + 2d}{6} \\ t_3 &= \frac{\mathbf{v} \cdot \mathbf{f}_3}{\|\mathbf{f}_3\|^2} = \frac{-a + c}{2} \\ t_4 &= \frac{\mathbf{v} \cdot \mathbf{f}_4}{\|\mathbf{f}_4\|^2} = \frac{-a + 3b - c + d}{12} \end{aligned}$$

The Fourier expansion of $\mathbf{v}$ in the basis $\{\mathbf{f}_1, \dots, \mathbf{f}_4\}$ is therefore

$$\begin{aligned} \mathbf{v} &= \left(\frac{a + b + c - d}{4} \right) \mathbf{f}_1 + \left(\frac{a + c + 2d}{6} \right) \mathbf{f}_2 \\ &\quad + \left(\frac{-a + c}{2} \right) \mathbf{f}_3 + \left(\frac{-a + 3b - c + d}{12} \right) \mathbf{f}_4 \end{aligned}$$

Dot products and orthogonality now become tools for finding closest vectors.

4.2 Orthogonal projection

4.2.1 Orthogonal complements and projections

Definition 4.2.1 (The orthogonal complement of a subspace).

If U is a subspace of $\mathbb{R}^n$, the *orthogonal complement* of U is the subset of $\mathbb{R}^n$ given by

$$U^\perp = \{\mathbf{x} \text{ in } \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{y} = 0 \text{ for all } \mathbf{y} \text{ in } U\}$$

Theorem 4.2.2.

The following properties of the orthogonal complement hold:

1. For any subspace U of $\mathbb{R}^n$, $U^\perp$ is a subspace of $\mathbb{R}^n$
2. $\{0\}^\perp = \mathbb{R}^n$ and $(\mathbb{R}^n)^\perp = \{0\}$
3. If $U = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$, then

$$U^\perp = \{\mathbf{x} \text{ in } \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{x}_i = 0 \text{ for } i = 1, 2, \dots, k\}$$

Activity

Find $U^\perp$ if $U = \text{span}\{(1, 1, 0), (0, 1, 1)\}$ in $\mathbb{R}^3$.

Solution.

The third property of 4.2.2 implies that a vector $\mathbf{w} = (x, y, z)$ lies in $U^\perp$ if and only if it is orthogonal to both $(1, 1, 0)$ and $(0, 1, 1)$. Taking these two dot products gives

$$\mathbf{w} \cdot (1, 1, 0) = x + y = 0$$

$$\mathbf{w} \cdot (0, 1, 1) = y + z = 0.$$

From $x + y = 0$, we obtain $x = -y$. From $y + z = 0$, we obtain $z = -y$. The variable y is free, so every solution can be written as

$$\mathbf{w} = (-y, y, -y) = y(-1, 1, -1), \quad y \in \mathbb{R}.$$

Thus every vector in $U^\perp$ is a multiple of $(-1, 1, -1)$, and

$$U^\perp = \text{span} \left\{ \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} \right\} = \left\{ \begin{bmatrix} -y \\ y \\ -y \end{bmatrix} : y \in \mathbb{R} \right\}$$

Theorem 4.2.3 (Expanding an orthogonal set).

Let $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$ be an orthogonal set in $\mathbb{R}^n$. Given $\mathbf{x}$ in $\mathbb{R}^n$, write

$$\mathbf{f}_{m+1} = \mathbf{x} - \frac{\mathbf{x} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 - \frac{\mathbf{x} \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 - \dots - \frac{\mathbf{x} \cdot \mathbf{f}_m}{\|\mathbf{f}_m\|^2} \mathbf{f}_m$$

Then:

- $\mathbf{f}_{m+1} \cdot \mathbf{f}_k = 0$ for $k = 1, 2, \dots, m$
- If $\mathbf{x} \notin \text{span}\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$, then $\mathbf{f}_{m+1} \neq \mathbf{0}$ and $\{\mathbf{f}_1, \dots, \mathbf{f}_m, \mathbf{f}_{m+1}\}$ is an orthogonal set.

Proof.

For convenience, let $t_i = \frac{\mathbf{x} \cdot \mathbf{f}_i}{\|\mathbf{f}_i\|^2}$ for each i . Given $1 \leq k \leq m$, we calculate that

$$\begin{aligned} \mathbf{f}_{m+1} \cdot \mathbf{f}_k &= (\mathbf{x} - t_1 \mathbf{f}_1 - t_2 \mathbf{f}_2 - \dots - t_m \mathbf{f}_m) \cdot \mathbf{f}_k \\ &= \mathbf{x} \cdot \mathbf{f}_k - t_1 (\mathbf{f}_1 \cdot \mathbf{f}_k) - \dots - t_k (\mathbf{f}_k \cdot \mathbf{f}_k) - \dots - t_m (\mathbf{f}_m \cdot \mathbf{f}_k) \end{aligned}$$

Since $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$ is an orthogonal set, $\mathbf{f}_i \cdot \mathbf{f}_k = 0$ for all $i \neq k$. So

$$\begin{aligned}
\mathbf{f}_{m+1} \cdot \mathbf{f}_k &= \mathbf{x} \cdot \mathbf{f}_k - t_k(\mathbf{f}_k \cdot \mathbf{f}_k) \\
&= \mathbf{x} \cdot \mathbf{f}_k - \frac{\mathbf{x} \cdot \mathbf{f}_k}{\|\mathbf{f}_k\|^2} \cdot \|\mathbf{f}_k\|^2 \\
&= \mathbf{x} \cdot \mathbf{f}_k - \mathbf{x} \cdot \mathbf{f}_k = 0
\end{aligned}$$

This proves the first property of the lemma. To prove the second, we note that if $\mathbf{f}_{m+1}$ was zero, then $\mathbf{x}$ *would* be in the span of the vectors $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$. So if $\mathbf{x}$ is not in the span, then $\mathbf{f}_{m+1}$ can not be equal to zero. And we have proved $\mathbf{f}_{m+1}$ is orthogonal to the vectors $\mathbf{f}_1, \dots, \mathbf{f}_m$, so that $\{\mathbf{f}_1, \dots, \mathbf{f}_{m+1}\}$ is an orthogonal set.

Activity

Use 4.2.3 to extend the orthogonal set $\{\mathbf{f}_1, \mathbf{f}_2\}$ in $\mathbb{R}^3$ to an orthogonal basis $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3\}$, where

$$\mathbf{f}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{f}_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

, using the vector $\mathbf{x} = (1, 1, 2)$ as in that lemma.

Solution.

We calculate that

$$\begin{aligned}
\mathbf{x} \cdot \mathbf{f}_1 &= (1, 1, 2) \cdot (1, 1, 0) = 1 + 1 + 0 = 2 \\
\|\mathbf{f}_1\|^2 &= 1^2 + 1^2 + 0^2 = 2 \\
\mathbf{x} \cdot \mathbf{f}_2 &= (1, 1, 2) \cdot (1, -1, 0) = 1 - 1 + 0 = 0 \\
\|\mathbf{f}_2\|^2 &= 1^2 + (-1)^2 + 0^2 = 2
\end{aligned}$$

Using 4.2.3, we define

$$\begin{aligned}
\mathbf{f}_3 &= \mathbf{x} - \frac{\mathbf{x} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 - \frac{\mathbf{x} \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 \\
&= \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - \frac{2}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \frac{0}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}
\end{aligned}$$

Definition 4.2.4 (The orthogonal projection onto a subspace).

Let U be a subspace of $\mathbb{R}^n$. Then the *orthogonal projection* of the vector $\mathbf{x} \in \mathbb{R}^n$ onto U is the

vector $\mathbf{y} = \text{proj}_U(\mathbf{x}) \in U$ which is *closest* to $\mathbf{x}$, in the sense that for any $\mathbf{z} \in U$ with $\mathbf{z} \neq \mathbf{y}$, $\|\mathbf{x} - \mathbf{y}\| < \|\mathbf{x} - \mathbf{z}\|$. The *residual* is the difference between the original vector and its projection:

$$\mathbf{r} = \mathbf{x} - \mathbf{y} = \mathbf{x} - \text{proj}_U(\mathbf{x}).$$

Orthogonal projection decomposes the original vector into a vector in the subspace and a residual perpendicular to the subspace.

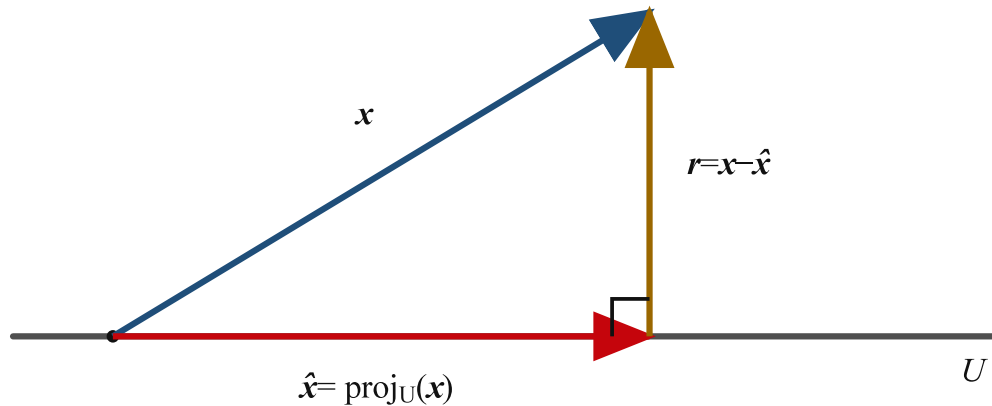


Figure 4.1. Projection and residual. Orthogonal projection writes $\mathbf{x} = \hat{\mathbf{x}} + \mathbf{r}$, where $\hat{\mathbf{x}} \in U$ and $\mathbf{r} \in U^\perp$.

The following theorem gives us a practical way to calculate orthogonal projections.

Theorem 4.2.5 (Projection theorem).

Let U be a subspace of $\mathbb{R}^n$ with orthogonal basis $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$. For any vector $\mathbf{x} \in \mathbb{R}^n$, define

$$\mathbf{y} = \frac{\mathbf{x} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 + \frac{\mathbf{x} \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 + \dots + \frac{\mathbf{x} \cdot \mathbf{f}_m}{\|\mathbf{f}_m\|^2} \mathbf{f}_m.$$

Then $\mathbf{y} = \text{proj}_U(\mathbf{x})$. The residual

$$\mathbf{r} = \mathbf{x} - \mathbf{y} = \mathbf{x} - \text{proj}_U(\mathbf{x})$$

lies in $U^\perp$. In particular, if $\mathbf{x}$ is already in U , then $\mathbf{x}$ equals the displayed sum.

Proof.

If $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$ is an orthogonal basis for U , and we define

$$\mathbf{y} = \frac{\mathbf{x} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 + \frac{\mathbf{x} \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 + \dots + \frac{\mathbf{x} \cdot \mathbf{f}_m}{\|\mathbf{f}_m\|^2} \mathbf{f}_m$$

, then $\mathbf{y}$ is evidently a linear combination of the basis $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$, so $\mathbf{y} \in U$. And 4.2.3 states that $\mathbf{x} - \mathbf{y} \in U^\perp$. Thus if $\mathbf{z} \in U$ and $\mathbf{z} \neq \mathbf{y}$, the Pythagorean theorem (4.1.6) tells us that

$$\begin{aligned}\|\mathbf{x} - \mathbf{z}\|^2 &= \|(\mathbf{x} - \mathbf{y}) + (\mathbf{y} - \mathbf{z})\|^2 \\ &= \|\mathbf{x} - \mathbf{y}\|^2 + \|\mathbf{y} - \mathbf{z}\|^2\end{aligned}$$

. Since $\mathbf{y} \neq \mathbf{z}$, we find that

$$\|\mathbf{x} - \mathbf{z}\|^2 > \|\mathbf{x} - \mathbf{y}\|^2$$

, and taking square roots gives that

$$\|\mathbf{x} - \mathbf{z}\| > \|\mathbf{x} - \mathbf{y}\|$$

. Thus the vector $\mathbf{y}$ we have defined is equal to $\text{proj}_U(\mathbf{x})$. Since we have already verified that $\mathbf{x} - \mathbf{y} = \mathbf{x} - \text{proj}_U(\mathbf{x})$ is in $U^\perp$, the theorem follows.

Activity

Find $\text{proj}_U \mathbf{x}$ where $U = \text{span}\{(1, 1, 0), (1, -1, 0)\}$ and $\mathbf{x} = (2, 0, 1)$ in $\mathbb{R}^3$.

Solution.

The vectors $\mathbf{f}_1 = (1, 1, 0)$ and $\mathbf{f}_2 = (1, -1, 0)$ are already orthogonal, as

$$\mathbf{f}_1 \cdot \mathbf{f}_2 = 1 \cdot 1 + 1 \cdot (-1) + 0 \cdot 0 = 0$$

. They thus form an orthogonal basis for U . We calculate the following values:

$$\mathbf{x} \cdot \mathbf{f}_1 = (2, 0, 1) \cdot (1, 1, 0) = 2 \cdot 1 + 0 \cdot 1 + 1 \cdot 0 = 2$$

$$\|\mathbf{f}_1\|^2 = 1^2 + 1^2 + 0^2 = 2$$

$$\mathbf{x} \cdot \mathbf{f}_2 = (2, 0, 1) \cdot (1, -1, 0) = 2 \cdot 1 + 0 \cdot (-1) + 1 \cdot 0 = 2$$

$$\|\mathbf{f}_2\|^2 = 1^2 + (-1)^2 + 0^2 = 2$$

Using 4.2.5, we calculate that

$$\begin{aligned}\text{proj}_U \mathbf{x} &= \frac{\mathbf{x} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 + \frac{\mathbf{x} \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 \\ &= \frac{2}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \frac{2}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}\end{aligned}$$

.

Activity: Projection onto a line

Let

$$\mathbf{u} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \quad L = \text{span}\{\mathbf{u}\}.$$

1. Compute $\text{proj}_L(\mathbf{x})$.
2. Compute the residual $\mathbf{r} = \mathbf{x} - \text{proj}_L(\mathbf{x})$.
3. Check that $\mathbf{r} \cdot \mathbf{u} = 0$.

Solution.

We have $\mathbf{x} \cdot \mathbf{u} = 26$ and $\mathbf{u} \cdot \mathbf{u} = 25$. Therefore

$$\text{proj}_L(\mathbf{x}) = \frac{\mathbf{x} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u} = \frac{26}{25} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 78/25 \\ 104/25 \end{bmatrix}.$$

The residual is

$$\mathbf{r} = \begin{bmatrix} 2 \\ 5 \end{bmatrix} - \begin{bmatrix} 78/25 \\ 104/25 \end{bmatrix} = \begin{bmatrix} -28/25 \\ 21/25 \end{bmatrix}.$$

Finally,

$$\mathbf{r} \cdot \mathbf{u} = \left(-\frac{28}{25}\right) 3 + \left(\frac{21}{25}\right) 4 = 0.$$

Activity: Projection onto a line in code

The following code computes the same projection and residual as the previous activity.

```
import numpy as np

u = np.array([3.0, 4.0])
x = np.array([2.0, 5.0])

y = (x @ u) / (u @ u) * u
r = x - y

y, r, r @ u
```

1. Which variable stores the projection?
2. Which variable stores the residual?

3. Why should $r \cdot u$ be close to zero?
4. What does $(x \cdot u) / (u \cdot u) \cdot u$ compute?

Solution.

The variable y stores the projection and r stores the residual. The value $r \cdot u$ checks whether the residual is orthogonal to the direction of the line, so it should be close to zero. The expression $(x \cdot u) / (u \cdot u) \cdot u$ computes the orthogonal projection of x onto $L = \text{span}\{u\}$:

$$\text{proj}_L(x) = \frac{x \cdot u}{u \cdot u} u.$$

Directional derivatives as projection coefficients

Let v be a unit vector. By 4.2.5,

$$\text{proj}_{\text{span}\{v\}}(\nabla f(a)) = (\nabla f(a) \cdot v)v.$$

By 3.7.2,

$$\text{proj}_{\text{span}\{v\}}(\nabla f(a)) = D_v f(a)v.$$

Thus $D_v f(a)$ is the signed projection coefficient of the gradient in the oriented direction v . Its absolute value is the length of the projection. The Cauchy–Schwarz inequality in 4.1.2 gives

$$|D_v f(a)| \leq \|\nabla f(a)\|.$$

Activity: Directional derivative as a projection coefficient

Reopen 3.7. For each listed unit vector v_i , compute

$$\text{proj}_{\text{span}\{v_i\}}(\nabla f(a)) = D_{v_i} f(a)v_i.$$

1. Which projection coefficients are positive, and which are negative?
2. Why do v_3 and v_4 give the same projection vector but opposite directional derivatives?
3. Reopen 3.20. How does the figure distinguish greatest increase, greatest decrease, and zero change?

Learning outcomes. U3-LO8, U4-LO3.

Solution.

From 3.7,

$$D_{\mathbf{v}_1}f(\mathbf{a}) = 3, \quad D_{\mathbf{v}_2}f(\mathbf{a}) = 4, \quad D_{\mathbf{v}_3}f(\mathbf{a}) = 5, \quad D_{\mathbf{v}_4}f(\mathbf{a}) = -5.$$

Therefore,

$$\text{proj}_{\text{span}\{\mathbf{v}_1\}}(\nabla f(\mathbf{a})) = 3\mathbf{v}_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix},$$

$$\text{proj}_{\text{span}\{\mathbf{v}_2\}}(\nabla f(\mathbf{a})) = 4\mathbf{v}_2 = \begin{bmatrix} 0 \\ 4 \end{bmatrix},$$

and

$$\text{proj}_{\text{span}\{\mathbf{v}_3\}}(\nabla f(\mathbf{a})) = 5\mathbf{v}_3 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

Since $\mathbf{v}_4 = -\mathbf{v}_3$,

$$\text{proj}_{\text{span}\{\mathbf{v}_4\}}(\nabla f(\mathbf{a})) = (-5)\mathbf{v}_4 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

The vectors $\mathbf{v}_3$ and $\mathbf{v}_4$ span the same line, so projection onto that line gives the same vector. The directional derivatives have opposite signs because the two unit vectors give opposite orientations of the line. Figure 3.7.5 shows greatest increase in the gradient direction, greatest decrease in the opposite direction, and zero change in directions orthogonal to the gradient.

Projection as an optimization problem

Orthogonal projection can also be described by minimizing a distance function.

Example

A common optimization problem that we have already encountered in 4.2.4 involves taking a subspace $U \subseteq \mathbb{R}^n$ and a given vector $\mathbf{x}_0 \in \mathbb{R}^n$, and finding the vector $\mathbf{x}^* \in U$ that is *closest* to $\mathbf{x}_0$, i.e., finding the vector that minimizes the scalar-valued function $g : \mathbb{R}^n \rightarrow \mathbb{R}$ on U , defined by setting $g(\mathbf{x}) = \|\mathbf{x}_0 - \mathbf{x}\|$. Then $\mathbf{x}^*$ is the orthogonal projection of $\mathbf{x}_0$ onto U , and we have seen in 4.2.5 that if U has an orthogonal basis $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$, then

$$\mathbf{x}^* = \frac{\mathbf{x}_0 \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 + \frac{\mathbf{x}_0 \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 + \dots + \frac{\mathbf{x}_0 \cdot \mathbf{f}_m}{\|\mathbf{f}_m\|^2} \mathbf{f}_m.$$

In 5.6, we return to this viewpoint and derive the normal equations by setting a gradient equal to zero.

Activity: Projection as optimization

Let

$$W = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\} \subseteq \mathbb{R}^2, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

A general point of W has the form

$$\mathbf{w}(t) = \begin{bmatrix} t \\ 0 \end{bmatrix}.$$

1. Write the squared distance $g(t) = \|\mathbf{b} - \mathbf{w}(t)\|^2$.
2. Compute $g'(t)$ and find the critical point of g .
3. Use $g''(t)$ to classify the critical point.
4. What point of W is closest to $\mathbf{b}$?
5. How does this agree with 4.1 and 4.2.5?

Learning outcomes. U3-LO9, U4-LO4.

Solution.

We have

$$\mathbf{b} - \mathbf{w}(t) = \begin{bmatrix} 2 - t \\ 3 \end{bmatrix},$$

so

$$g(t) = \|\mathbf{b} - \mathbf{w}(t)\|^2 = (2 - t)^2 + 9.$$

Differentiating gives

$$g'(t) = 2(t - 2).$$

The unique critical point is $t = 2$. Since

$$g''(t) = 2 > 0,$$

this critical point is a local minimum. Because g is an upward-opening quadratic, it is also the unique global minimum. Therefore the closest point in W is

$$\hat{\mathbf{b}} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

This agrees with 4.1 and 4.2.5: the residual is

$$\mathbf{b} - \hat{\mathbf{b}} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

For every $\mathbf{w}(t) \in W$,

$$(\mathbf{b} - \hat{\mathbf{b}}) \cdot \mathbf{w}(t) = \begin{bmatrix} 0 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} t \\ 0 \end{bmatrix} = 0 \cdot t + 3 \cdot 0 = 0.$$

Therefore the residual is orthogonal to W .

Connection: score, then combine

Projection and attention both use dot products to score directions and then use the resulting numbers in a linear combination.

Let

$$Q = [\mathbf{q}_1 \quad \cdots \quad \mathbf{q}_k]$$

have orthonormal columns. Then

$$Q^T \mathbf{x} = \begin{bmatrix} \mathbf{q}_1^T \mathbf{x} \\ \vdots \\ \mathbf{q}_k^T \mathbf{x} \end{bmatrix}$$

collects the directional coefficients of $\mathbf{x}$. Because each $\mathbf{q}_i$ has length 1, these dot products are the coordinates used in the projection. If $\mathbf{x}$ also has length 1, they are cosine similarities with the directions $\mathbf{q}_i$.

Multiplying by Q combines the directions using those coefficients:

$$Q(Q^T \mathbf{x}) = \sum_{i=1}^k (\mathbf{q}_i^T \mathbf{x}) \mathbf{q}_i = \text{proj}_{\text{col}(Q)}(\mathbf{x}).$$

Thus projection follows the pattern

$$\text{score with } Q^T \longrightarrow \text{combine with } Q.$$

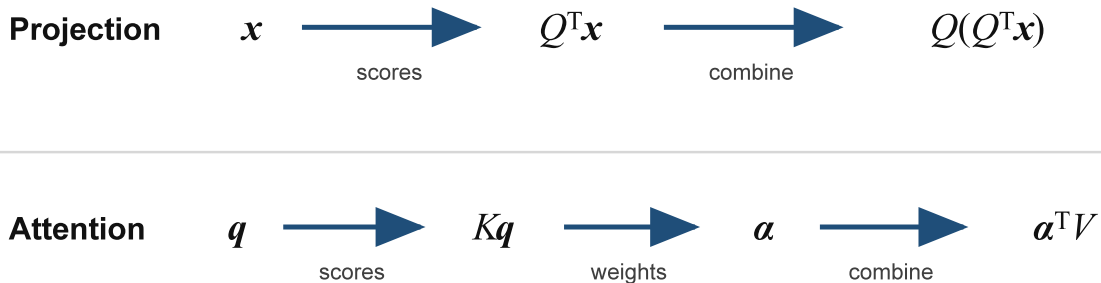


Figure 4.2. Score, then combine. Projection uses $Q^T \mathbf{x}$ to score orthonormal directions and $Q(Q^T \mathbf{x})$ to combine them. Attention uses query-key scores, converts them into weights, and combines value vectors.

Attention has the same broad pattern. Query-key dot products produce scores, a weighting rule converts the scores into weights, and the weights combine value vectors. Projection scores and combines the same orthonormal directions. Attention scores key vectors and then combines value vectors, which may be different vectors. Projection also chooses its coefficients so that the residual is orthogonal to the subspace. Attention is therefore not generally an orthogonal projection, but both computations are built from dot products and linear combinations.

Activity: Dot-product scores as projection coordinates

Let

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{q}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix},$$

and let

$$Q = [\mathbf{q}_1 \quad \mathbf{q}_2].$$

1. Verify that the columns of Q are orthonormal.
2. Compute the score vector

$$\mathbf{c} = Q^T \mathbf{x}.$$

3. Compute

$$\hat{\mathbf{x}} = Q\mathbf{c}.$$

4. Compute the residual

$$\mathbf{r} = \mathbf{x} - \hat{\mathbf{x}},$$

and verify that $Q^T \mathbf{r} = \mathbf{0}$.

5. Explain why the entries of $\mathbf{c}$ can be read both as dot-product scores and as coordinates of the projection.
6. Explain how the calculation follows the pattern “score, then combine.” What extra condition identifies $\hat{\mathbf{x}}$ as an orthogonal projection?

Solution.

The two columns have length 1, and

$$\mathbf{q}_1^T \mathbf{q}_2 = \frac{1}{2}(1 - 1) = 0.$$

Thus the columns of Q are orthonormal.

The score vector is

$$\mathbf{c} = Q^T \mathbf{x} = \begin{bmatrix} \mathbf{q}_1^T \mathbf{x} \\ \mathbf{q}_2^T \mathbf{x} \end{bmatrix} = \begin{bmatrix} \sqrt{2} \\ \sqrt{2} \end{bmatrix}.$$

Combining the columns of Q using these scores gives

$$\hat{\mathbf{x}} = Q\mathbf{c} = \sqrt{2}\mathbf{q}_1 + \sqrt{2}\mathbf{q}_2 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}.$$

The residual is

$$\mathbf{r} = \mathbf{x} - \hat{\mathbf{x}} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore

$$Q^T \mathbf{r} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Each entry of $\mathbf{c}$ is a dot product with one of the unit directions, so it is both a directional score and a coordinate used to reconstruct the projection. The calculation first scores $\mathbf{x}$ with Q^T , then combines the columns of Q using those scores. The condition $Q^T \mathbf{r} = \mathbf{0}$ says that the residual is orthogonal to every column of Q . This is the extra condition that identifies $\hat{\mathbf{x}}$ as the orthogonal projection.

4.2.2 Gram-Schmidt orthogonalization

Gram-Schmidt is a sequence of projection, subtraction, measurement, and normalization steps. Use *Perfectly Normal* to practice the same sequence geometrically. The game uses an original column $\mathbf{a}_j$, its projection $\mathbf{p}_j$, the residual $\mathbf{v}_j$, the new unit direction $\mathbf{q}_j$, and coefficients r_{ij} .

Algorithm: Gram-Schmidt algorithm

Input: A matrix

$$A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n]$$

with linearly independent columns.

For $j = 1, \dots, n$:

1. Current column. Start with $\mathbf{a}_j$.

2. Project. For each $i < j$, compute

$$r_{ij} = \mathbf{q}_i^T \mathbf{a}_j.$$

Then assemble

$$\mathbf{p}_j = \sum_{i < j} r_{ij} \mathbf{q}_i.$$

For the first column, $\mathbf{p}_1 = \mathbf{0}$.

3. Subtract.

$$\mathbf{v}_j = \mathbf{a}_j - \mathbf{p}_j.$$

4. Measure.

$$r_{jj} = \|\mathbf{v}_j\|.$$

5. Normalize.

$$\mathbf{q}_j = \frac{\mathbf{v}_j}{r_{jj}}.$$

Output: The vectors $\mathbf{q}_1, \dots, \mathbf{q}_n$ are orthonormal, and

$$\text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_j\} = \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_j\}$$

at every stage.

Warning

If $r_{jj} = 0$, then $\mathbf{a}_j$ is already in the span of the earlier columns. The input columns are not linearly independent, and no new unit direction is produced.

The same algorithm in orthogonal form. Repeatedly applying 4.2.3 gives the theorem below. The theorem first constructs orthogonal residuals $\mathbf{f}_j$, while the algorithm above normalizes each residual to obtain $\mathbf{q}_j$. The two forms agree because

$$\mathbf{q}_i = \frac{\mathbf{f}_i}{\|\mathbf{f}_i\|}$$

and

$$\frac{\mathbf{a}_j^T \mathbf{f}_i}{\|\mathbf{f}_i\|^2} \mathbf{f}_i = (\mathbf{q}_i^T \mathbf{a}_j) \mathbf{q}_i.$$

Theorem 4.2.6 (The Gram-Schmidt orthogonalization algorithm).

Let U be a subspace of $\mathbb{R}^n$. Then U has an orthogonal basis.

Given a basis $\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ of U , we can iteratively define an orthogonal basis $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$ of U by defining

$$\begin{aligned}\mathbf{f}_1 &= \mathbf{x}_1 \\ \mathbf{f}_2 &= \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 \\ \mathbf{f}_3 &= \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2\end{aligned}$$

and so on, i.e., for $2 \leq k \leq m$ defining

$$\mathbf{f}_k = \mathbf{x}_k - \frac{\mathbf{x}_k \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 - \frac{\mathbf{x}_k \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 - \dots - \frac{\mathbf{x}_k \cdot \mathbf{f}_{k-1}}{\|\mathbf{f}_{k-1}\|^2} \mathbf{f}_{k-1}$$

. Then

- $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$ is an orthogonal basis of U .
- $\text{span}\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_k\} = \text{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ for each $k = 1, 2, \dots, m$.

Activity

Apply the Gram-Schmidt algorithm to the columns of A^T , where

$$A = \begin{bmatrix} 1 & 1 & -1 & -1 \\ 3 & 2 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

. These columns are the rows of A , written as column vectors. At each stage, record the projection coefficients, projection, residual, residual length, and normalized direction. Give an orthogonal basis and an orthonormal basis for $\text{row}(A)$ (recall 2.4.11).

1. Compute $\mathbf{v}_1$, r_{11} , and $\mathbf{q}_1$.
2. Compute r_{12} , $\mathbf{p}_2$, $\mathbf{v}_2$, r_{22} , and $\mathbf{q}_2$.
3. Compute r_{13} , r_{23} , $\mathbf{p}_3$, $\mathbf{v}_3$, r_{33} , and $\mathbf{q}_3$.
4. Give an orthogonal basis from the residuals.
5. Give an orthonormal basis from the unit vectors.
6. Explain why both bases span $\text{row}(A)$.

Solution.

Let the columns of A^T be

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix},$$

$$\mathbf{a}_2 = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 1 \end{bmatrix},$$

$$\mathbf{a}_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}.$$

For the first column,

$$\mathbf{v}_1 = \mathbf{a}_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix},$$

$$r_{11} = \|\mathbf{v}_1\| = 2,$$

$$\mathbf{q}_1 = \frac{\mathbf{v}_1}{r_{11}} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}.$$

For the second column,

$$r_{12} = \mathbf{q}_1^T \mathbf{a}_2 = 2,$$

$$\mathbf{p}_2 = r_{12} \mathbf{q}_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix},$$

$$\mathbf{v}_2 = \mathbf{a}_2 - \mathbf{p}_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix},$$

$$r_{22} = \|\mathbf{v}_2\| = \sqrt{10},$$

$$\mathbf{q}_2 = \frac{\mathbf{v}_2}{r_{22}} = \frac{1}{\sqrt{10}} \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix}.$$

For the third column,

$$r_{13} = \mathbf{q}_1^T \mathbf{a}_3 = 0,$$

$$r_{23} = \mathbf{q}_2^T \mathbf{a}_3 = \frac{3}{\sqrt{10}},$$

$$\mathbf{p}_3 = r_{13} \mathbf{q}_1 + r_{23} \mathbf{q}_2 = \frac{3}{10} \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix},$$

$$\mathbf{v}_3 = \mathbf{a}_3 - \mathbf{p}_3 = \frac{1}{10} \begin{bmatrix} 4 \\ -3 \\ 7 \\ -6 \end{bmatrix},$$

$$r_{33} = \|\mathbf{v}_3\| = \frac{\sqrt{110}}{10},$$

$$\mathbf{q}_3 = \frac{\mathbf{v}_3}{r_{33}} = \frac{1}{\sqrt{110}} \begin{bmatrix} 4 \\ -3 \\ 7 \\ -6 \end{bmatrix}.$$

An orthogonal basis from the residuals is

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix}, \frac{1}{10} \begin{bmatrix} 4 \\ -3 \\ 7 \\ -6 \end{bmatrix} \right\}.$$

The corresponding orthonormal basis is

$$\left\{ \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \frac{1}{\sqrt{10}} \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix}, \frac{1}{\sqrt{110}} \begin{bmatrix} 4 \\ -3 \\ 7 \\ -6 \end{bmatrix} \right\}.$$

The columns of A^T are the rows of A , and Gram-Schmidt preserves each successive span. Therefore the residuals and their normalized unit vectors both span $\text{row}(A)$.

Activity

Find the point in the plane defined by the equation $2x + y - z = 0$, which is closest to the point $(1, 2, -1)$.

Solution.

Let U be the plane. Solving its equation for z gives $z = 2x + y$. If we set $x = s$ and $y = t$, then every vector in U has the form

$$\begin{bmatrix} s \\ t \\ 2s + t \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + t \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

Therefore

$$U = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

For a convenient ordering of this basis, let

$$\mathbf{u}_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{u}_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}.$$

Applying the Gram-Schmidt process from 4.2.6 gives

$$\begin{aligned} \mathbf{f}_1 &= \mathbf{u}_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \\ \mathbf{f}_2 &= \mathbf{u}_2 - \frac{\mathbf{u}_2 \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 \\ &= \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - \frac{2}{2} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}. \end{aligned}$$

Thus $\{f_1, f_2\}$ is an orthogonal basis of U .

For the given point $x = (1, 2, -1)$,

$$x \cdot f_1 = 1, \quad \|f_1\|^2 = 2, \quad x \cdot f_2 = -2, \quad \|f_2\|^2 = 3.$$

The projection theorem now gives

$$\begin{aligned} \text{proj}_U(x) &= \frac{x \cdot f_1}{\|f_1\|^2} f_1 + \frac{x \cdot f_2}{\|f_2\|^2} f_2 \\ &= \frac{1}{2} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \frac{2}{3} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2/3 \\ 7/6 \\ -1/6 \end{bmatrix}. \end{aligned}$$

Hence the point in the plane closest to $(1, 2, -1)$ is $(-2/3, 7/6, -1/6)$.

4.3 Least squares as projection

2.2.1 asked whether a target vector b is reachable as Ax . When $b \notin \text{col}(A)$, no input reaches the target exactly. Least squares replaces the unreachable target by the closest reachable output.

Recall the Unit 2 matrix and target

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

For

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

we have

$$Ax = \begin{bmatrix} x_1 \\ x_2 \\ x_1 + x_2 \end{bmatrix}.$$

Every reachable output therefore has third coordinate equal to the sum of its first two coordinates. The target b is not reachable because $4 \neq 1 + 2$.

Definition 4.3.1 (Least-squares solution, fitted vector, and residual).

Let A be an $m \times n$ matrix and let $b \in \mathbb{R}^m$. A least-squares solution is a vector $\hat{x} \in \mathbb{R}^n$ that minimizes

$$\|Ax - b\|^2$$

over all $x \in \mathbb{R}^n$.

The vector

$$\hat{\mathbf{b}} = A\hat{\mathbf{x}}$$

is the fitted vector. The vector

$$\mathbf{r} = \mathbf{b} - \hat{\mathbf{b}} = \mathbf{b} - A\hat{\mathbf{x}}$$

is the residual.

The quantity

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=1}^m ((A\mathbf{x})_i - b_i)^2$$

is the sum of the squares of the coordinate errors. This explains the name least squares. Minimizing the norm or its square gives the same minimizers.

If $\mathbf{b} \in \text{col}(A)$, the minimum error is 0, and least squares agrees with solving $A\mathbf{x} = \mathbf{b}$ exactly. If the target is not reachable, the fitted vector $\hat{\mathbf{b}}$ is the closest reachable output.

In the projection-and-residual diagram from Section 4.2, take the subspace to be $\text{col}(A)$, relabel the original vector as $\mathbf{b}$, and relabel its projection as

$$\hat{\mathbf{b}} = A\hat{\mathbf{x}}.$$

The residual is

$$\mathbf{r} = \mathbf{b} - \hat{\mathbf{b}}.$$

The projection theorem guarantees a closest vector $\hat{\mathbf{b}} \in \text{col}(A)$. Since $\hat{\mathbf{b}}$ lies in the column space, at least one coefficient vector $\hat{\mathbf{x}}$ satisfies

$$A\hat{\mathbf{x}} = \hat{\mathbf{b}}.$$

Fact: A^T checks column orthogonality

Write

$$A = [\mathbf{a}_1 \quad \cdots \quad \mathbf{a}_n].$$

For any vector $\mathbf{r} \in \mathbb{R}^m$,

$$A^T \mathbf{r} = \begin{bmatrix} \mathbf{a}_1^T \mathbf{r} \\ \vdots \\ \mathbf{a}_n^T \mathbf{r} \end{bmatrix}.$$

Therefore

$$A^T \mathbf{r} = \mathbf{0}$$

exactly when $\mathbf{r}$ is orthogonal to every column of A . Equivalently,

$$\text{col}(A)^\perp = \text{null}(A^T).$$

Every fitted vector $A\mathbf{x}$ lies in $\text{col}(A)$. Therefore the least-squares fitted vector is

$$\hat{\mathbf{b}} = A\hat{\mathbf{x}} = \text{proj}_{\text{col}(A)}(\mathbf{b}).$$

Its residual

$$\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$$

is orthogonal to $\text{col}(A)$.

Theorem 4.3.2 (Least squares and the normal equations).

A vector $\hat{\mathbf{x}}$ is a least-squares solution exactly when the residual

$$\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$$

is orthogonal to $\text{col}(A)$. Equivalently,

$$A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}.$$

Thus the least-squares solutions are exactly the solutions of the normal equations

$$A^T A\hat{\mathbf{x}} = A^T \mathbf{b}.$$

Proof.

Every candidate fitted vector $A\mathbf{x}$ lies in $\text{col}(A)$. By the 4.2.5, $A\hat{\mathbf{x}}$ is closest to $\mathbf{b}$ exactly when

$$\mathbf{b} - A\hat{\mathbf{x}} \in \text{col}(A)^\perp.$$

By the fact above, this is equivalent to

$$A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}.$$

Expanding and rearranging gives

$$A^T \mathbf{b} - A^T A\hat{\mathbf{x}} = \mathbf{0},$$

or

$$A^T A\hat{\mathbf{x}} = A^T \mathbf{b}.$$

The fitted vector $\hat{\mathbf{b}}$ is unique because orthogonal projection onto a subspace is unique. The coefficient vector $\hat{\mathbf{x}}$ need not be unique. If $\mathbf{z} \in \text{null}(A)$, then

$$A(\hat{\mathbf{x}} + \mathbf{z}) = A\hat{\mathbf{x}}.$$

When the columns of A are linearly independent, $\text{null}(A) = \{\mathbf{0}\}$, so the least-squares coefficient vector is unique.

Activity: Independent columns and $A^T A$

Let A be an $m \times n$ matrix. Use the identity

$$\mathbf{x}^T A^T A \mathbf{x} = \|A\mathbf{x}\|^2$$

to explain why $A^T A$ is invertible exactly when the columns of A are linearly independent.

1. Suppose

$$A^T A \mathbf{x} = \mathbf{0}.$$

What does the displayed identity say about $\|A\mathbf{x}\|^2$?

2. If the columns of A are linearly independent, what must $\mathbf{x}$ be?
3. Conversely, suppose $A^T A$ is invertible and

$$A\mathbf{x} = \mathbf{0}.$$

What equation results after multiplying by A^T ?

4. Explain why the two conditions are equivalent.

Solution.

If

$$A^T A \mathbf{x} = \mathbf{0},$$

then

$$0 = \mathbf{x}^T A^T A \mathbf{x} = \|A\mathbf{x}\|^2.$$

Therefore $A\mathbf{x} = \mathbf{0}$. If the columns of A are linearly independent, then

$$\text{null}(A) = \{\mathbf{0}\},$$

so $\mathbf{x} = \mathbf{0}$. Thus $\text{null}(A^T A) = \{\mathbf{0}\}$, and the square matrix $A^T A$ is invertible. Conversely, suppose $A^T A$ is invertible and

$$A\mathbf{x} = \mathbf{0}.$$

Multiplying by A^T gives

$$A^T A \mathbf{x} = \mathbf{0}.$$

Since $A^T A$ is invertible, $\mathbf{x} = \mathbf{0}$. Therefore

$$\text{null}(A) = \{\mathbf{0}\},$$

so the columns of A are linearly independent.

Activity: Unit 2 target revisited

Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

1. Recall why $A\mathbf{x} = \mathbf{b}$ has no exact solution.
2. Form $A^T A$ and $A^T \mathbf{b}$.
3. Solve the normal equations for $\hat{\mathbf{x}}$.
4. Compute the fitted vector

$$\hat{\mathbf{b}} = A\hat{\mathbf{x}}$$

and the residual

$$\mathbf{r} = \mathbf{b} - \hat{\mathbf{b}}.$$

5. Check that

$$A^T \mathbf{r} = \mathbf{0}.$$

6. Compute $\|\mathbf{r}\|^2$.
7. Which vector is the closest reachable output: $\hat{\mathbf{x}}$ or $\hat{\mathbf{b}}$?
8. The residual is not zero. Explain why this does not mean least squares failed.

Solution.

For

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

we have

$$A\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_1 + x_2 \end{bmatrix}.$$

The first two coordinates of $A\mathbf{x} = \mathbf{b}$ would force $x_1 = 1$ and $x_2 = 2$, but then the third coordinate would be 3, not 4. Thus the target is not reachable exactly.

The normal equations use

$$A^T A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad A^T \mathbf{b} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

Solving

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \hat{\mathbf{x}} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

gives

$$\hat{\mathbf{x}} = \begin{bmatrix} 4/3 \\ 7/3 \end{bmatrix}.$$

The fitted vector is

$$\hat{\mathbf{b}} = A\hat{\mathbf{x}} = \begin{bmatrix} 4/3 \\ 7/3 \\ 11/3 \end{bmatrix},$$

and the residual is

$$\mathbf{r} = \mathbf{b} - \hat{\mathbf{b}} = \begin{bmatrix} -1/3 \\ -1/3 \\ 1/3 \end{bmatrix}.$$

The orthogonality check gives

$$A^T \mathbf{r} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1/3 \\ -1/3 \\ 1/3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Also,

$$\|\mathbf{r}\|^2 = \frac{1}{9} + \frac{1}{9} + \frac{1}{9} = \frac{1}{3}.$$

The coefficient vector is $\hat{\mathbf{x}}$. The closest reachable output is the fitted vector

$$\hat{\mathbf{b}} = A\hat{\mathbf{x}}.$$

The residual is not zero because the original target is not reachable. Least squares did not fail: it found the reachable output closest to $\mathbf{b}$, and the residual is orthogonal to the column space.

Linear regression

Suppose data are given as pairs

$$(t_1, y_1), \dots, (t_m, y_m).$$

A line model predicts

$$\hat{y} = c_0 + c_1 t.$$

The coefficient c_0 is the intercept, and c_1 is the slope. At input t_i , the predicted value is

$$\hat{y}_i = c_0 + c_1 t_i.$$

The residual for the i -th data point is

$$r_i = y_i - \hat{y}_i.$$

Linear regression chooses c_0 and c_1 to make

$$r_1^2 + \dots + r_m^2$$

as small as possible.

Linear regression is the same closest-reachable-output problem as “4.3,” but the columns of the matrix now represent model features.

Activity: A line model as a matrix product

Consider the data points

$$(0, 1), \quad (1, 2), \quad (2, 2),$$

and the line model

$$\hat{y} = c_0 + c_1 t.$$

1. Write the three predicted values in terms of c_0 and c_1 .
2. Define

$$\mathbf{c} = \begin{bmatrix} c_0 \\ c_1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

Build a matrix X such that $X\mathbf{c}$ is the vector of predicted values.

3. What does the first column of X represent?
4. What does the second column of X represent?

5. Write the residual vector

$$\mathbf{r} = \mathbf{y} - X\mathbf{c}.$$

6. Explain why

$$\|\mathbf{r}\|^2 = \sum_{i=1}^3 (y_i - \hat{y}_i)^2$$

is the sum of the squared vertical errors in the data plot.

Learning outcomes. U4-LO4, U4-LO5.

Solution.

The three predictions are

$$\hat{y}_1 = c_0, \quad \hat{y}_2 = c_0 + c_1, \quad \hat{y}_3 = c_0 + 2c_1.$$

Therefore

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix},$$

because

$$X\mathbf{c} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} c_0 \\ c_0 + c_1 \\ c_0 + 2c_1 \end{bmatrix}.$$

The first column is the constant feature: it records the coefficient of the intercept c_0 . The second column is the t -feature: it records the input values 0, 1, 2.

The residual vector is

$$\mathbf{r} = \mathbf{y} - X\mathbf{c} = \begin{bmatrix} 1 - c_0 \\ 2 - (c_0 + c_1) \\ 2 - (c_0 + 2c_1) \end{bmatrix}.$$

Its squared norm is

$$\|\mathbf{r}\|^2 = (1 - c_0)^2 + (2 - (c_0 + c_1))^2 + (2 - (c_0 + 2c_1))^2.$$

Each entry is an observed output minus the predicted output at the same input, so these are vertical errors in the data plot.

For general data (t_i, y_i) , define the design matrix

$$X = \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} c_0 \\ c_1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}.$$

Then $X\mathbf{c}$ is the vector of predicted values. Linear regression is the least-squares problem

$$\min_{\mathbf{c}} \|X\mathbf{c} - \mathbf{y}\|^2.$$

The matrix X is called the design matrix. Each column represents one feature used by the model.

Warning: Two different geometric pictures

The projection occurs in $\mathbb{R}^m$:

$$\hat{\mathbf{y}} = X\hat{\mathbf{c}} = \text{proj}_{\text{col}(X)}(\mathbf{y}).$$

In the usual t - y data plot, the residuals are vertical differences

$$r_i = y_i - \hat{y}_i.$$

They are not generally perpendicular distances from the plotted points to the fitted line.

Activity: Fit a line to three data points

Continue with the design matrix and data vector from “4.3.”

1. Use “4.3.2” to form

$$X^T X \hat{\mathbf{c}} = X^T \mathbf{y}.$$

2. Solve for

$$\hat{\mathbf{c}} = \begin{bmatrix} \hat{c}_0 \\ \hat{c}_1 \end{bmatrix}.$$

3. Write the fitted line.
4. Compute the fitted-value vector

$$\hat{\mathbf{y}} = X\hat{\mathbf{c}}$$

and the residual vector

$$\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}}.$$

5. Check that

$$X^T \mathbf{r} = \mathbf{0}.$$

6. Write the two scalar equations contained in $X^T \mathbf{r} = \mathbf{0}$.

7. Compute

$$\|\mathbf{r}\|^2.$$

8. Which object is the fitted vector: $\hat{\mathbf{c}}$ or $\hat{\mathbf{y}}$?

9. In the t - y plot, are the residual segments vertical or perpendicular to the fitted line?

Learning outcomes. U4-LO4, U4-LO5.

Solution.

We have

$$X^T X = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, \quad X^T \mathbf{y} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

Thus the normal equations are

$$\begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \hat{\mathbf{c}} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

Solving gives

$$\hat{\mathbf{c}} = \begin{bmatrix} 7/6 \\ 1/2 \end{bmatrix}.$$

The fitted line is

$$\hat{y} = \frac{7}{6} + \frac{1}{2}t.$$

The fitted-value vector is

$$\hat{\mathbf{y}} = X\hat{\mathbf{c}} = \begin{bmatrix} 7/6 \\ 5/3 \\ 13/6 \end{bmatrix},$$

and the residual vector is

$$\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} -1/6 \\ 1/3 \\ -1/6 \end{bmatrix}.$$

The residual-orthogonality check is

$$X^T \mathbf{r} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} -1/6 \\ 1/3 \\ -1/6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The two scalar equations are

$$r_1 + r_2 + r_3 = 0$$

and

$$0r_1 + 1r_2 + 2r_3 = 0.$$

The first says that the residuals add to zero. The second says that the residuals weighted by the input values add to zero.

The squared residual norm is

$$\|\mathbf{r}\|^2 = \frac{1}{36} + \frac{1}{9} + \frac{1}{36} = \frac{1}{6}.$$

The coefficient vector is $\hat{\mathbf{c}}$. The fitted vector is

$$\hat{\mathbf{y}} = X\hat{\mathbf{c}}.$$

In the data plot, the residual segments are vertical. They are not generally perpendicular to the fitted line.

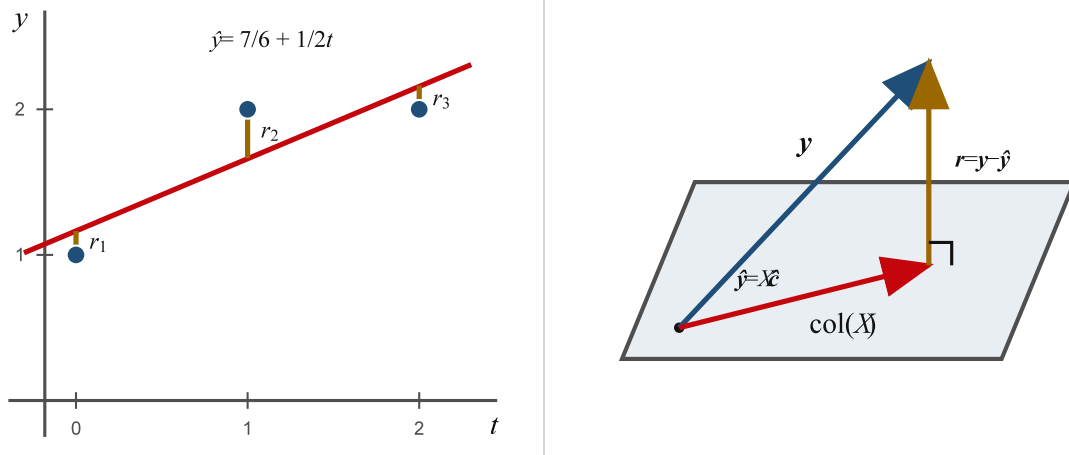


Figure 4.3. Two views of the same line fit. In the data plot, the residuals are vertical differences $r_i = y_i - \hat{y}_i$. In $\mathbb{R}^3$, the fitted vector $\hat{\mathbf{y}} = X\hat{\mathbf{c}}$ is the projection of $\mathbf{y}$ onto $\text{col}(X)$, and $\mathbf{r} \perp \text{col}(X)$.

The two panels in 4.3 show different spaces: the left panel is the t - y data plot, while the right panel is the output space $\mathbb{R}^3$.

Activity: Reading regression code

The following code repeats the line fit from “4.3.”

```
import numpy as np

t = np.array([0.0, 1.0, 2.0])
y = np.array([1.0, 2.0, 2.0])

X = np.column_stack([np.ones_like(t), t])

c = np.linalg.lstsq(X, y, rcond=None)[0]
yhat = X @ c
r = y - yhat

X, c, yhat, r, X.T @ r, np.linalg.norm(r)**2
```

1. What array is produced by `np.ones_like(t)`?
2. What does `np.column_stack([np.ones_like(t), t])` construct?
3. What is the shape of `X`?
4. What does `[0]` select from the value returned by `np.linalg.lstsq`?
5. Why does `c` have two entries?
6. What mathematical objects are stored in `yhat` and `r`?
7. Why is the residual check `X.T @ r` rather than `X @ r`?
8. What should `X.T @ r` be close to?
9. Does a nonzero residual vector mean the code failed?
10. What quantity is computed by `np.linalg.norm(r)**2`?

Learning outcomes. U4-LO5, U4-LO7.

Solution.

The expression

```
np.ones_like(t)
```

produces

```
array([1., 1., 1.])
```

The call

```
np.column_stack([np.ones_like(t), t])
```

uses those two arrays as columns, so

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

Thus X has shape $(3, 2)$.

The function `np.linalg.lstsq` returns several objects. The index `[0]` selects the least-squares coefficient vector. The vector $\hat{c}$ has two entries because the design matrix has two columns: one for the constant feature and one for the t -feature.

The variable `yhat` stores the fitted vector

$$\hat{\mathbf{y}} = X\hat{\mathbf{c}},$$

and `r` stores the residual vector

$$\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}}.$$

The product `X.T @ r` computes dot products with the columns of X . It checks that the residual is orthogonal to the column space. The product `X @ r` is not the correct operation and does not have compatible shapes in this example.

The output `X.T @ r` should be close to

```
array([0., 0.])
```

up to roundoff.

A nonzero residual does not mean the code failed. The data vector is not exactly in the column space, so the closest fit has a nonzero residual.

Finally,

```
np.linalg.norm(r)**2
```

computes the sum of squared residuals. Its value is

$$\frac{1}{6}$$

up to roundoff.

The normal equations describe the geometry of least squares. The next section uses 4.2.2 and 4.4 to compute the same solution in a different form.

Explore least squares interactively with the Residual Squares mini-game.

4.4 *QR* factorization and least squares

4.4.1 From Gram–Schmidt to *QR*

The 4.2.2 processes the independent columns of a matrix from left to right. It constructs orthonormal directions $\mathbf{q}_1, \dots, \mathbf{q}_n$ and records coefficients r_{ij} . These same quantities form a matrix factorization.

Let

$$A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n].$$

At column j , the 4.2.2 computes

$$\begin{aligned} \mathbf{p}_j &= \sum_{i < j} r_{ij} \mathbf{q}_i, & \mathbf{v}_j &= \mathbf{a}_j - \mathbf{p}_j, \\ r_{jj} &= \|\mathbf{v}_j\|, & \mathbf{q}_j &= \frac{\mathbf{v}_j}{r_{jj}}. \end{aligned}$$

Since

$$\mathbf{v}_j = r_{jj} \mathbf{q}_j,$$

we can reconstruct the original column:

$$\mathbf{a}_j = \mathbf{p}_j + \mathbf{v}_j = \sum_{i < j} r_{ij} \mathbf{q}_i + r_{jj} \mathbf{q}_j = \sum_{i \leq j} r_{ij} \mathbf{q}_i.$$

Thus column j uses only $\mathbf{q}_1, \dots, \mathbf{q}_j$.

Collect the orthonormal directions in

$$Q = [\mathbf{q}_1 \quad \mathbf{q}_2 \quad \cdots \quad \mathbf{q}_n]$$

and the coefficients in

$$R = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ 0 & r_{22} & \cdots & r_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & r_{nn} \end{bmatrix}.$$

The column equations above become the single matrix equation

$$A = QR.$$

The zeros below the diagonal of R record the order of the algorithm: column j uses only directions constructed by stage j .

Definition 4.4.1 (Reduced QR factorization).

Let A be an $m \times n$ matrix with linearly independent columns. A reduced QR factorization of A is a factorization

$$A = QR,$$

where

- Q is an $m \times n$ matrix with orthonormal columns;
- R is an invertible upper-triangular $n \times n$ matrix.

For the Gram–Schmidt convention used in this unit, the diagonal entries of R are positive:

$$r_{jj} = \|\mathbf{v}_j\| > 0.$$

Activity: Reading $Q^T Q$

Let

$$Q = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{6} \\ 1/\sqrt{3} & -2/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{6} \end{bmatrix} = [\mathbf{q}_1 \quad \mathbf{q}_2].$$

1. Compute $\|\mathbf{q}_1\|^2$ and $\|\mathbf{q}_2\|^2$.
2. Compute $\mathbf{q}_1^T \mathbf{q}_2$.
3. Compute $Q^T Q$.
4. What does the (i, j) -entry of $Q^T Q$ measure?
5. Explain why

$$Q^T Q = I_2$$

is the matrix check that the columns of Q are orthonormal.

Solution.

We have

$$\|\mathbf{q}_1\|^2 = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$$

and

$$\|\mathbf{q}_2\|^2 = \frac{1}{6} + \frac{4}{6} + \frac{1}{6} = 1.$$

Also,

$$\mathbf{q}_1^T \mathbf{q}_2 = \frac{1}{\sqrt{18}} - \frac{2}{\sqrt{18}} + \frac{1}{\sqrt{18}} = 0.$$

Therefore

$$Q^T Q = \begin{bmatrix} \mathbf{q}_1^T \mathbf{q}_1 & \mathbf{q}_1^T \mathbf{q}_2 \\ \mathbf{q}_2^T \mathbf{q}_1 & \mathbf{q}_2^T \mathbf{q}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2.$$

The (i, j) -entry of $Q^T Q$ is the dot product $\mathbf{q}_i^T \mathbf{q}_j$. The diagonal entries check unit length, and the off-diagonal entries check orthogonality.

Theorem 4.4.2 (Gram–Schmidt produces QR).

Every $m \times n$ matrix A with linearly independent columns has a reduced QR factorization

$$A = QR$$

in which the columns of Q are orthonormal and R is upper triangular with positive diagonal entries.

Proof.

Apply the 4.2.2 to the columns

$$\mathbf{a}_1, \dots, \mathbf{a}_n.$$

At stage j , the algorithm gives

$$\mathbf{a}_j = \sum_{i \leq j} r_{ij} \mathbf{q}_i.$$

Collecting these equations column by column gives

$$A = QR.$$

The columns of Q are orthonormal. The condition $i \leq j$ makes R upper triangular, and

$$r_{jj} = \|\mathbf{v}_j\| > 0$$

because the input columns are linearly independent.

Theorem 4.4.3 (Square case).

Every invertible square matrix has a QR factorization

$$A = QR,$$

where Q is an orthogonal matrix and R is upper triangular with positive diagonal entries.

Proof.

An invertible square matrix has linearly independent columns. Apply “4.4.2.” Since Q is square and has orthonormal columns,

$$Q^T Q = Q Q^T = I.$$

Thus Q is an orthogonal matrix.

Activity: Build Q and R from the Gram–Schmidt ledger

Let

$$A = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix} = [\mathbf{a}_1 \quad \mathbf{a}_2].$$

1. Compute

$$r_{11} = \|\mathbf{a}_1\|$$

and

$$\mathbf{q}_1 = \frac{\mathbf{a}_1}{r_{11}}.$$

2. Compute the projection coefficient

$$r_{12} = \mathbf{q}_1^T \mathbf{a}_2.$$

3. Compute

$$\mathbf{p}_2 = r_{12} \mathbf{q}_1$$

and

$$\mathbf{v}_2 = \mathbf{a}_2 - \mathbf{p}_2.$$

4. Compute

$$r_{22} = \|\mathbf{v}_2\|$$

and

$$\mathbf{q}_2 = \frac{\mathbf{v}_2}{r_{22}}.$$

5. Assemble

$$Q = [\mathbf{q}_1 \quad \mathbf{q}_2]$$

and

$$R = \begin{bmatrix} r_{11} & r_{12} \\ 0 & r_{22} \end{bmatrix}.$$

6. Check

$$Q^T Q = I_2$$

and

$$QR = A.$$

Solution.

The first column gives

$$r_{11} = \|\mathbf{a}_1\| = 5, \quad \mathbf{q}_1 = \frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

The projection coefficient is

$$r_{12} = \mathbf{q}_1^T \mathbf{a}_2 = \frac{3}{5}(1) + \frac{4}{5}(2) = \frac{11}{5}.$$

Therefore

$$\mathbf{p}_2 = r_{12} \mathbf{q}_1 = \frac{11}{5} \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} = \begin{bmatrix} 33/25 \\ 44/25 \end{bmatrix},$$

and

$$\mathbf{v}_2 = \mathbf{a}_2 - \mathbf{p}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 33/25 \\ 44/25 \end{bmatrix} = \begin{bmatrix} -8/25 \\ 6/25 \end{bmatrix}.$$

Its length is

$$r_{22} = \|\mathbf{v}_2\| = \frac{2}{5},$$

so

$$\mathbf{q}_2 = \frac{\mathbf{v}_2}{r_{22}} = \begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix}.$$

Thus

$$Q = \begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix}, \quad R = \begin{bmatrix} 5 & 11/5 \\ 0 & 2/5 \end{bmatrix}.$$

The orthonormality check gives

$$Q^T Q = I_2,$$

and direct multiplication gives

$$QR = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix} = A.$$

Activity: An upper-triangular matrix already has a QR factorization

Let

$$A = \begin{bmatrix} 6 & 2 \\ 0 & 3 \end{bmatrix}.$$

Explain why

$$Q = I_2, \quad R = A$$

is a QR factorization. Check each condition in the definition.

Solution.

The columns of I_2 are orthonormal, so

$$Q^T Q = I_2.$$

The matrix

$$R = \begin{bmatrix} 6 & 2 \\ 0 & 3 \end{bmatrix}$$

is upper triangular and has positive diagonal entries. Finally,

$$QR = I_2 A = A.$$

Thus $A = QR$ is a QR factorization.

Warning: Rectangular Q

In a reduced QR factorization of an $m \times n$ matrix with $m > n$, the matrix Q has orthonormal columns, so

$$Q^T Q = I_n.$$

However, Q is not square, and generally

$$Q Q^T \neq I_m.$$

Instead,

$$Q Q^T \mathbf{b} = \text{proj}_{\text{col}(Q)}(\mathbf{b}) = \text{proj}_{\text{col}(A)}(\mathbf{b}).$$

Warning: Signs in a QR factorization

The positive-diagonal convention fixes the signs of the Gram–Schmidt factorization. Without that convention, one may replace a column $\mathbf{q}_j$ by $-\mathbf{q}_j$ and replace row j of R by its negative. The product QR does not change.

Numerical software may therefore return Q and R with signs different from a hand calculation. Check

$$Q^T Q = I \quad \text{and} \quad QR = A$$

rather than requiring every entry to have the same sign.

4.4.2 Least squares with QR

The 4.3.2 give a direct route from a QR factorization to a least-squares solution.

Suppose

$$A = QR$$

is a reduced QR factorization. The 4.3.2 are

$$A^T A \hat{\mathbf{x}} = A^T \mathbf{b}.$$

Substituting $A = QR$ gives

$$R^T Q^T Q R \hat{\mathbf{x}} = R^T Q^T \mathbf{b}.$$

Since

$$Q^T Q = I,$$

this becomes

$$R^T R \hat{\mathbf{x}} = R^T Q^T \mathbf{b}.$$

The matrix R is invertible, so R^T is invertible. Canceling R^T gives

$$R \hat{\mathbf{x}} = Q^T \mathbf{b}.$$

Algorithm: Least squares with QR

Input. An $m \times n$ matrix A with linearly independent columns and a vector $\mathbf{b} \in \mathbb{R}^m$.

1. Compute a reduced QR factorization

$$A = QR.$$

2. Compute

$$\mathbf{y} = Q^T \mathbf{b}.$$

3. Solve the upper-triangular system

$$R\hat{\mathbf{x}} = \mathbf{y}$$

by back-substitution.

Output. The vector $\hat{\mathbf{x}}$ is the least-squares coefficient vector. The fitted vector is

$$A\hat{\mathbf{x}} = QR\hat{\mathbf{x}} = QQ^T \mathbf{b}.$$

Activity: Back-substitution

Solve

$$R\mathbf{x} = \mathbf{y},$$

where

$$R = \begin{bmatrix} 4 & 2 & 7 \\ 0 & 3 & 5 \\ 0 & 0 & 2 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 29 \\ 21 \\ 6 \end{bmatrix}.$$

Work from the last equation upward.

Solution.

The last equation gives

$$2x_3 = 6, \quad x_3 = 3.$$

The second equation gives

$$3x_2 + 5x_3 = 21,$$

so

$$3x_2 + 15 = 21, \quad x_2 = 2.$$

The first equation gives

$$4x_1 + 2x_2 + 7x_3 = 29,$$

so

$$4x_1 + 4 + 21 = 29, \quad x_1 = 1.$$

Therefore

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Activity: Solving least squares with QR in code

The following code solves the regression problem from “4.3” using *QR*.

```
import numpy as np

t = np.array([0.0, 1.0, 2.0])
y = np.array([1.0, 2.0, 2.0])

X = np.column_stack([np.ones_like(t), t])

Q, R = np.linalg.qr(X, mode="reduced")

c_qr = np.linalg.solve(R, Q.T @ y)
c_lstsq = np.linalg.lstsq(X, y, rcond=None)[0]

(
    Q.shape,
    R.shape,
    np.allclose(Q.T @ Q, np.eye(Q.shape[1])),
    np.allclose(Q @ R, X),
    c_qr,
    c_lstsq,
    np.allclose(c_qr, c_lstsq),
)
```

1. What does the assignment `Q, R = np.linalg.qr(X, mode="reduced")`

store in Q and R?

2. Why does the code use `mode="reduced"`?
3. What should `Q.shape` and `R.shape` be?
4. What does `Q.T @ Q` check?
5. What does `Q @ R` check?
6. What array is constructed by `np.eye(Q.shape[1])` in this example?
7. What mathematical vector is computed by `Q.T @ y`?
8. Why does the code solve a system with R?
9. Why is `np.linalg.solve(R, Q.T @ y)` preferable to explicitly computing an inverse of R?
10. Why should `c_qr` and `c_lstsq` agree?
11. Does their agreement mean that

$$Xc = y$$

has an exact solution?

12. Why may the entries of Q and R have signs different from a hand Gram–Schmidt calculation?

Learning outcomes. U4-LO5, U4-LO6, U4-LO7.

Solution.

The call

```
Q, R = np.linalg.qr(X, mode="reduced")
```

computes a reduced QR factorization and stores the two returned arrays in Q and R. Since X has shape (3, 2), reduced QR gives

```
Q.shape == (3, 2)
R.shape == (2, 2)
```

The expression

```
Q.T @ Q
```

checks whether the columns of Q are orthonormal. The expression

```
Q @ R
```

checks whether the factors reconstruct X.
Here,

```
np.eye(Q.shape[1])
```

constructs the 2×2 identity matrix.
The vector

```
Q.T @ y
```

is the right-hand side $Q^T y$ in the reduced system

$$R\hat{c} = Q^T y.$$

The matrix R is square, invertible, and upper triangular, so the code solves this system with `np.linalg.solve`. It does not need to construct an inverse matrix.
Both methods compute the same least-squares coefficient vector, so

```
c_qr
```

and

```
c_lstsq
```

should both be close to

```
array([1.16666667, 0.5])
```

and the final `np.allclose` check should return `True`.
This agreement does not mean that

$$Xc = y$$

has an exact solution. The residual from the 4.3 is nonzero.

A valid QR factorization can use simultaneous sign changes in one column of Q and the corresponding row of R . Therefore numerical output may have different signs from a hand calculation while still satisfying

$$Q^T Q = I \quad \text{and} \quad QR = X.$$

The 4.3.2 and QR compute the same least-squares solution. The normal equations expose residual orthogonality directly. QR reuses the orthonormal directions produced by 4.2.2 and reduces the computation to an upper-triangular solve.

4.5 Coding recap

4.5.1 Linked notebook

Run Lab U4: Regression as projection. Focus on the design matrix, fitted coefficients, residual, and the computation `A.T @ r`.

For a quick reference on least-squares commands, transposes, matrix products, and numerical checks such as “close to zero,” see the programming appendix sections A.10, A.4, and A.9.

4.5.2 Review activities

These activities interpret Python code and output for least squares, residual orthogonality, regression, and QR . Reopen the 4.3.2, the 4.3, or the 4.4.2 as needed.

Activity: Reading least-squares objects

The following code repeats “4.3.”

```
A = np.array([
    [1.0, 0.0],
    [0.0, 1.0],
    [1.0, 1.0],
])

b = np.array([1.0, 2.0, 4.0])

xhat = np.linalg.lstsq(A, b, rcond=None)[0]
bhat = A @ xhat
```

```
r = b - bhat
```

```
xhat.shape, bhat.shape, r.shape, A.T @ r
```

1. What is the shape of xhat?
2. What is the shape of bhat?
3. What is the shape of r?
4. What mathematical object is stored in xhat?
5. What mathematical object is stored in bhat?
6. Which variable stores the closest reachable output?
7. What does $A.T @ r$ check?
8. Does $A.T @ r$ being close to zero imply that r is close to the zero vector?

Learning outcomes. U4-LO4, U4-LO5, U4-LO7.

Solution.

The matrix A has two columns and three rows. Therefore xhat has shape $(2,)$, while bhat and r both have shape $(3,)$.

The vector xhat stores the least-squares coefficient vector

$$\hat{\mathbf{x}} = \begin{bmatrix} 4/3 \\ 7/3 \end{bmatrix}.$$

The vector bhat stores the fitted vector

$$\hat{\mathbf{b}} = A\hat{\mathbf{x}} = \begin{bmatrix} 4/3 \\ 7/3 \\ 11/3 \end{bmatrix}.$$

The vector r stores the residual

$$\mathbf{r} = \mathbf{b} - \hat{\mathbf{b}} = \begin{bmatrix} -1/3 \\ -1/3 \\ 1/3 \end{bmatrix}.$$

The closest reachable output is bhat, not xhat.

The product

```
A.T @ r
```

computes the dot products of the residual with the columns of A . It checks that

$$\mathbf{r} \perp \text{col}(A).$$

This does not mean that the residual is zero. Here,

$$\|\mathbf{r}\|^2 = \frac{1}{3},$$

so the residual is nonzero.

Activity: Roundoff and the wrong residual check

For the same problem as “4.5.2,” suppose NumPy gives

```
A.T @ r
```

with output

```
array([2.2e-16, -1.1e-16])
```

and

```
np.linalg.norm(r)
```

with output

```
0.5773502691896257
```

1. What geometric condition is checked by the first output?
2. Why are the entries tiny rather than exactly zero?
3. Does the first output say that the residual vector is zero?
4. What does the second output show?
5. Why is
`np.allclose(A.T @ r, np.zeros(A.shape[1]))`
a more appropriate numerical check than `A.T @ r == 0`?
6. A student instead writes
`A @ r`
to check residual orthogonality. What is wrong with this expression?

Learning outcomes. U4-LO4, U4-LO7.

Solution.

The first output checks whether the residual is orthogonal to every column of A . In exact arithmetic,

$$A^T \mathbf{r} = \mathbf{0}.$$

Floating-point arithmetic may produce tiny roundoff errors instead of exact zeros. The first output does not say that $\mathbf{r} = \mathbf{0}$. The second output shows that

$$\|\mathbf{r}\| \approx 0.57735 = \frac{1}{\sqrt{3}},$$

so the residual is nonzero.

The function `np.allclose` checks whether the entries are numerically close to zero within a tolerance. An entry-by-entry equality check is usually too strict for floating-point output. The expression `A @ r` is not the correct check. Here A has shape 3×2 , while $\mathbf{r}$ has length 3, so the product is not even defined. More importantly, residual orthogonality requires dot products with the columns of A , which are collected by

$$A^T \mathbf{r}.$$

Activity: Reading a regression design matrix

The following code repeats “4.3.”

```
t = np.array([0.0, 1.0, 2.0])
y = np.array([1.0, 2.0, 2.0])

X = np.column_stack([np.ones_like(t), t])

c = np.linalg.lstsq(X, y, rcond=None)[0]
yhat = X @ c
r = y - yhat
```

1. What do the two columns of X represent?
2. Why does $\mathbf{c}$ have two entries?
3. Why does $\mathbf{yhat}$ have three entries?
4. Which variable is the fitted vector?
5. Which variable is the coefficient vector?
6. Which vector lies in $\text{col}(X)$?

7. In “4.3,” which panel shows the projection in $\mathbb{R}^3$?
8. In the t - y data plot, are the residual segments vertical or perpendicular to the fitted line?
9. What quantity is computed by
`np.linalg.norm(r)**2`
 ?

Learning outcomes. U4-LO5, U4-LO7.

Solution.

The first column of X is the constant feature. The second column contains the input values t . The vector c has two entries because the model has two coefficients: the intercept and slope. The vector $\hat{y}$ has three entries because the model makes one prediction at each of the three input values.

The coefficient vector is

$$\hat{c} = \begin{bmatrix} 7/6 \\ 1/2 \end{bmatrix}.$$

The fitted vector is

$$\hat{y} = X\hat{c} = \begin{bmatrix} 7/6 \\ 5/3 \\ 13/6 \end{bmatrix}.$$

Thus c is the coefficient vector, while $\hat{y}$ is the fitted vector. The vector $\hat{y}$ lies in $\text{col}(X)$.

The right panel of “4.3” shows the projection in $\mathbb{R}^3$. In the t - y data plot, the residual segments are vertical. They are not generally perpendicular to the fitted line.

Finally,

```
np.linalg.norm(r)**2
```

computes the sum of squared residuals. Its value is

$$\frac{1}{6}$$

up to roundoff.

Activity: Reading QR least-squares code

The following code applies the 4.4.2 to the 4.3.

```
Q, R = np.linalg.qr(X, mode="reduced")

c_qr = np.linalg.solve(R, Q.T @ y)

np.allclose(Q.T @ Q, np.eye(Q.shape[1]))
np.allclose(Q @ R, X)
c_qr
```

1. What should the shapes of Q and R be?
2. What does the first `np.allclose` line check?
3. What does the second `np.allclose` line check?
4. What mathematical system is solved by
`np.linalg.solve(R, Q.T @ y)`
?
5. Why is a system with R easier to solve than the original least-squares problem?
6. What should `c_qr` be close to?
7. Does obtaining `c_qr` mean that

$$Xc = y$$

has an exact solution?

8. Why might NumPy return signs in Q and R that differ from a hand Gram–Schmidt calculation?

Learning outcomes. U4-LO6, U4-LO7.

Solution.

The design matrix X has shape 3×2 . Reduced QR therefore gives

```
Q.shape == (3, 2)
R.shape == (2, 2)
```

The first check verifies

$$Q^T Q = I_2,$$

so the columns of Q are orthonormal. The second check verifies

$$QR = X.$$

The call

```
np.linalg.solve(R, Q.T @ y)
```

solves

$$R\hat{c} = Q^T y.$$

The matrix R is square and upper triangular, so this is a standard linear system rather than a rectangular least-squares problem.

The result should be close to

```
array([1.16666667, 0.5])
```

which represents

$$\hat{c} = \begin{bmatrix} 7/6 \\ 1/2 \end{bmatrix}.$$

This does not mean that $Xc = y$ has an exact solution. The fitted vector has a nonzero residual.

A QR factorization allows simultaneous sign changes in a column of Q and the corresponding row of R . Therefore numerical software may return signs different from a hand calculation while still satisfying

$$Q^T Q = I \quad \text{and} \quad QR = X.$$

4.6 Exercises

Learning outcomes. The labels below identify the learning outcomes for each exercise group. Individual problems may also involve earlier outcomes.

4.6.1 Orthogonality and orthogonal bases

Learning outcomes. U4-LO1, U4-LO2.

- Nicholson 5.3.3(b)
- Nicholson 5.3.6(b)

- Nicholson 5.3.9

4.6.2 Orthogonal projection and Gram-Schmidt

Learning outcomes. U4-LO2, U4-LO3.

- Nicholson 8.1.1(d)
- Nicholson 8.1.2(b,d)
- Nicholson 8.1.4(b)

4.6.3 QR factorization and least squares

Learning outcomes. U4-LO6.

- Nicholson 8.4.1(b,d)

4.6.4 Additional exercises

These exercises connect the main Unit 4 ideas: orthogonality, orthogonal bases, Gram–Schmidt, projections, residuals, least squares, normal equations, regression, QR factorization, and code interpretation. Solutions are collected in Appendix A.4.

Additional exercise: Orthogonal coordinates and projection

Let

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{q}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix},$$

$$Q = [\mathbf{q}_1 \quad \mathbf{q}_2], \quad \mathbf{x} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}.$$

Use “4.1” as a geometric guide.

1. Check that

$$Q^T Q = I_2.$$

2. Compute the coordinate vector

$$\mathbf{c} = Q^T \mathbf{x}.$$

3. Compute

$$\hat{\mathbf{x}} = Q\mathbf{c}.$$

4. Compute the residual

$$\mathbf{r} = \mathbf{x} - \hat{\mathbf{x}}.$$

5. Check that

$$Q^T \mathbf{r} = \mathbf{0}.$$

6. Explain how the calculation follows the pattern “score, then combine.”

7. Which vector is the projection of $\mathbf{x}$: $\mathbf{c}$ or $\hat{\mathbf{x}}$?

Learning outcomes. U4-LO1, U4-LO3.

Additional exercise: One Gram–Schmidt column step

Suppose the first completed unit direction is

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix},$$

and the next original column is

$$\mathbf{a}_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

Use the 4.2.2.

1. Compute

$$r_{12} = \mathbf{q}_1^T \mathbf{a}_2.$$

2. Compute

$$\mathbf{p}_2 = r_{12} \mathbf{q}_1.$$

3. Compute

$$\mathbf{v}_2 = \mathbf{a}_2 - \mathbf{p}_2.$$

4. Compute

$$r_{22} = \|\mathbf{v}_2\|.$$

5. Compute

$$\mathbf{q}_2 = \frac{\mathbf{v}_2}{r_{22}}.$$

6. Check that

$$\mathbf{q}_1^T \mathbf{q}_2 = 0.$$

7. What would $r_{22} = 0$ say about $\mathbf{a}_2$?

Learning outcomes. U4-LO2, U4-LO6.

Additional exercise: Orthogonal-basis coefficients

Let

$$\mathbf{f}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{f}_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{f}_3 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix},$$

and let

$$\mathbf{x} = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}.$$

1. Verify that $\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3$ are pairwise orthogonal.
2. Use dot products to write $\mathbf{x}$ as a linear combination of $\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3$.
3. Explain why each dot product isolates one coefficient.

Learning outcomes. U4-LO1, U4-LO2.

Additional exercise: Projection onto a line

Let

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 3 \\ 1 \end{bmatrix},$$

and let $L = \text{span}\{\mathbf{u}\}$.

1. Compute $\text{proj}_L(\mathbf{x})$.
2. Compute $\mathbf{r} = \mathbf{x} - \text{proj}_L(\mathbf{x})$.
3. Check that $\mathbf{r} \cdot \mathbf{u} = 0$.
4. Explain what the residual measures.

Learning outcomes. U4-LO3.

Additional exercise: Column-space orthogonal complement

Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

1. Compute $A^T \mathbf{r}$ for $\mathbf{r} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$.
2. Explain why $\mathbf{r}$ is orthogonal to $\text{col}(A)$.
3. Describe $\text{col}(A)^\perp$.
4. Explain the identity $\text{col}(A)^\perp = \text{null}(A^T)$ in this example.

Learning outcomes. U4-LO3, U4-LO4, U2-LO3.

Additional exercise: Projection onto a plane

Let U be the plane $x + y + z = 0$, and let

$$\mathbf{p} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

1. Give a normal vector $\mathbf{n}$ for U .

2. Compute $\text{proj}_U(\mathbf{p})$ by subtracting the component of $\mathbf{p}$ in the normal direction.
3. Check that the projected point lies in U .

Learning outcomes. U4-LO3.

Additional exercise: Unreachable target, closest output

Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}.$$

1. Explain why $A\mathbf{x} = \mathbf{b}$ has no exact solution.
2. Form $A^T A$ and $A^T \mathbf{b}$.
3. Solve the normal equations.
4. Compute $A\hat{\mathbf{x}}$ and $\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$.
5. Check $A^T \mathbf{r} = \mathbf{0}$.

Learning outcomes. U4-LO4, U4-LO5, U2-LO3.

Additional exercise: Regression design matrix and residual

Fit a line $y = c_0 + c_1 t$ to the data

$$(0, 1), \quad (1, 2), \quad (2, 2).$$

1. Build the design matrix X .
2. Form $X^T X$ and $X^T \mathbf{y}$.
3. Solve the normal equations.
4. Compute the fitted values and residual.
5. Check $X^T \mathbf{r} = \mathbf{0}$.

Learning outcomes. U4-LO5.

Additional exercise: Code interpretation: residual orthogonality

Use “Least squares and the normal equations” to interpret the following computation.

```
import numpy as np

A = np.array([
    [1.0, 0.0],
    [0.0, 1.0],
    [1.0, 1.0],
])
b = np.array([2.0, 1.0, 5.0])

xhat = np.linalg.lstsq(A, b, rcond=None)[0]
bhat = A @ xhat
r = b - bhat
orthogonality = A.T @ r

xhat, bhat, r, orthogonality, np.linalg.norm(r)
```

Suppose the last two outputs are approximately

```
array([1.3e-15, -2.2e-16])
```

and

```
1.1547005383792515
```

1. What does `[0]` select from the value returned by `np.linalg.lstsq`?
2. What mathematical object is stored in `xhat`?
3. What mathematical object is stored in `bhat`?
4. What mathematical object is stored in `r`?
5. What geometric condition does `A.T @ r` check?
6. Why are the entries of `orthogonality` tiny rather than exactly zero?
7. Does the nonzero residual norm mean that least squares failed?

Learning outcomes. U4-LO4, U4-LO5, U4-LO7.

Additional exercise: Debug: wrong residual check

Use “ A^T checks column orthogonality” to diagnose the following mistake.
A student tries to check residual orthogonality with

```
A @ r
```

instead of

```
A.T @ r
```

Assume that A is an $m \times n$ matrix and $\mathbf{r} \in \mathbb{R}^m$.

1. What are the shapes of A , A^T , and $\mathbf{r}$?
2. Why is $A @ \mathbf{r}$ usually not defined?
3. Suppose the dimensions happened to make $A @ \mathbf{r}$ defined. Why would it still be the wrong geometric check?
4. State the correct mathematical residual-orthogonality condition.
5. Write a numerical Boolean check using `np.allclose`, `A.shape[1]`, and `np.zeros`.

Learning outcomes. U4-LO4, U4-LO7.

Additional exercise: Rank and nonunique coefficients

Let

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix},$$

and let $\mathbf{z} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$.

1. Compute $A\mathbf{z}$.
2. Suppose $\hat{\mathbf{x}}$ is a least-squares solution. Compare $A\hat{\mathbf{x}}$ and $A(\hat{\mathbf{x}} + 5\mathbf{z})$.
3. Which can be nonunique: the fitted vector or the coefficient vector?
4. How does this connect to dependent columns?

Learning outcomes. U4-LO4, U2-LO6.

Additional exercise: QR least-squares code reading

Use “Least squares with QR ” to interpret the following code.

```
import numpy as np

A = np.array([
    [1.0, 0.0],
    [0.0, 1.0],
    [1.0, 1.0],
])
b = np.array([2.0, 1.0, 5.0])

Q, R = np.linalg.qr(A, mode="reduced")

x_qr = np.linalg.solve(R, Q.T @ b)
x_lstsq = np.linalg.lstsq(A, b, rcond=None)[0]

(
    Q.shape,
    R.shape,
    np.allclose(Q.T @ Q, np.eye(Q.shape[1])),
    np.allclose(Q @ R, A),
    x_qr,
    x_lstsq,
    np.allclose(x_qr, x_lstsq),
)
```

1. Why does the code request `mode="reduced"`?
2. What should the shapes of Q and R be?
3. What does
`np.allclose(Q.T @ Q, np.eye(Q.shape[1]))`
check?
4. What does
`np.allclose(Q @ R, A)`
check?
5. What mathematical system is solved by
`np.linalg.solve(R, Q.T @ b)`
?

6. Why should x_{qr} and x_{lstsq} agree?

7. Does their agreement mean that

$$Ax = b$$

has an exact solution?

8. Why may the entries of Q and R have signs different from a hand Gram–Schmidt calculation?

Learning outcomes. U4-LO5, U4-LO6, U4-LO7.

Additional exercise: Projection and attention: score, then combine

Use “Connection: score, then combine” and “Dot-product scores as projection coordinates” as guides.

Projection pipeline. Let

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}.$$

1. Compute the score vector

$$\mathbf{c} = Q^T \mathbf{x}.$$

2. Compute

$$\hat{\mathbf{x}} = Q\mathbf{c}.$$

3. Compute

$$\mathbf{r} = \mathbf{x} - \hat{\mathbf{x}},$$

and check that

$$Q^T \mathbf{r} = \mathbf{0}.$$

Attention pipeline. Let

$$K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{q} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

The score vector is

$$\mathbf{s} = K\mathbf{q}.$$

A weighting rule divides the positive scores by their sum:

$$\alpha = \frac{s}{s_1 + s_2}.$$

Let the value vectors be the rows of

$$V = \begin{bmatrix} 3 & 0 \\ 0 & 6 \end{bmatrix}.$$

4. Compute s and α .
5. Compute the attention output

$$\mathbf{h} = \alpha^T V.$$

6. Identify the score step and the combine step in each pipeline.
7. What mathematical operations do both pipelines use?
8. What extra condition identifies $\hat{\mathbf{x}}$ as an orthogonal projection?
9. In the attention pipeline, which vectors are scored and which vectors are combined?

Learning outcomes. U4-LO1, U4-LO3, U1-LO8.

4.7 Unit 4 highlights

4.7.1 Mathematical quick reference

Orthogonal coordinates

Orthogonal and orthonormal lists. Distinct vectors in an orthogonal list have dot product zero. An orthonormal list also has every norm equal to 1. A list of nonzero pairwise orthogonal vectors is independent.

Coordinates in an orthogonal basis. If $\mathbf{f}_1, \dots, \mathbf{f}_k$ is an orthogonal basis of U and $\mathbf{x} \in U$, then

$$\mathbf{x} = \sum_{i=1}^k c_i \mathbf{f}_i, \quad c_i = \frac{\mathbf{x} \cdot \mathbf{f}_i}{\mathbf{f}_i \cdot \mathbf{f}_i}.$$

For an orthonormal basis $\mathbf{q}_i$, this simplifies to $c_i = \mathbf{x} \cdot \mathbf{q}_i$.

Orthonormal columns. For $Q \in \mathbb{R}^{m \times k}$, $Q^T Q = I_k$ exactly when its columns are orthonormal. If Q is square, it is an orthogonal matrix and $Q^{-1} = Q^T$.

Orthogonal complements and projection

Orthogonal complement.

$$U^\perp = \{\mathbf{r} : \mathbf{r} \cdot \mathbf{u} = 0 \text{ for every } \mathbf{u} \in U\}.$$

For $A \in \mathbb{R}^{m \times n}$, $\text{col}(A)^\perp = \text{null}(A^T)$ and $\text{row}(A)^\perp = \text{null}(A)$.

Orthogonal decomposition. For a subspace $U \subseteq \mathbb{R}^m$, every $\mathbf{b}$ has a unique decomposition

$$\mathbf{b} = \mathbf{p} + \mathbf{r}, \quad \mathbf{p} \in U, \quad \mathbf{r} \in U^\perp.$$

Here $\mathbf{p} = \text{proj}_U \mathbf{b}$, and $\dim U + \dim U^\perp = m$.

Projection formulas. For a nonzero $\mathbf{v}$, an orthogonal basis $\mathbf{f}_1, \dots, \mathbf{f}_k$ of U , or a matrix Q with orthonormal columns spanning U , respectively,

$$\text{proj}_{\text{span}\{\mathbf{v}\}} \mathbf{b} = \frac{\mathbf{b} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}, \quad \text{proj}_U \mathbf{b} = \sum_{i=1}^k \frac{\mathbf{b} \cdot \mathbf{f}_i}{\mathbf{f}_i \cdot \mathbf{f}_i} \mathbf{f}_i = QQ^T \mathbf{b}.$$

Best approximation. If $\mathbf{p} = \text{proj}_U \mathbf{b}$, then for every $\mathbf{u} \in U$,

$$\|\mathbf{b} - \mathbf{u}\|^2 = \|\mathbf{b} - \mathbf{p}\|^2 + \|\mathbf{p} - \mathbf{u}\|^2.$$

Thus $\mathbf{p}$ is the unique closest vector in U , and $\text{dist}(\mathbf{b}, U) = \|\mathbf{b} - \mathbf{p}\|$.

Gram–Schmidt and QR factorization

Gram–Schmidt. For independent columns $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^m$, process $j = 1, \dots, n$:

$$r_{ij} = \mathbf{q}_i^T \mathbf{a}_j \quad (i < j), \quad \mathbf{v}_j = \mathbf{a}_j - \sum_{i < j} r_{ij} \mathbf{q}_i, \\ r_{jj} = \|\mathbf{v}_j\|, \quad \mathbf{q}_j = \frac{\mathbf{v}_j}{r_{jj}}.$$

Independence guarantees $r_{jj} > 0$. Each initial span is preserved.

Reduced QR factorization. For $A \in \mathbb{R}^{m \times n}$ with independent columns,

$$A = QR, \quad Q \in \mathbb{R}^{m \times n}, \quad Q^T Q = I_n, \quad R \in \mathbb{R}^{n \times n}.$$

The matrix R is upper triangular, with the coefficients r_{ij} above and on its diagonal, and is invertible.

Least squares and normal equations

Least-squares problem. Given $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$, minimize $\|A\mathbf{x} - \mathbf{b}\|^2$. For a minimizer $\hat{\mathbf{x}}$, distinguish the coefficient vector, fitted vector, and residual:

$$\hat{\mathbf{x}} \in \mathbb{R}^n, \quad \mathbf{p} = A\hat{\mathbf{x}} \in \mathbb{R}^m, \quad \mathbf{r} = \mathbf{b} - \mathbf{p} \in \mathbb{R}^m.$$

Equivalent tests.

$$A\hat{\mathbf{x}} = \text{proj}_{\text{col}(A)} \mathbf{b} \iff A^T \mathbf{r} = \mathbf{0} \iff A^T A\hat{\mathbf{x}} = A^T \mathbf{b}.$$

Least-squares solutions always exist. The fitted vector and residual are unique; the coefficients are unique exactly when the columns of A are independent.

Full column rank. In this case $A^T A$ is invertible, so

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}, \quad \mathbf{p} = A(A^T A)^{-1} A^T \mathbf{b}.$$

With $A = QR$, solve the upper-triangular system $R\hat{\mathbf{x}} = Q^T \mathbf{b}$ instead.

Design matrix. For prescribed functions ϕ_j and sample points t_i , set $A_{ij} = \phi_j(t_i)$. Fitting the linear combination $\sum_j c_j \phi_j$ to values b_i gives $A^T A \hat{\mathbf{c}} = A^T \mathbf{b}$.

4.7.2 Common mistakes

Confusing orthogonal with orthonormal; omitting the squared norm in a projection coefficient; assuming $QQ^T = I_m$ for a rectangular Q ; normalizing a zero Gram–Schmidt residual; confusing coefficients with fitted values; assuming $A^T \mathbf{r} = \mathbf{0}$ means $\mathbf{r} = \mathbf{0}$; or using $(A^T A)^{-1}$ without checking column independence.

4.7.3 Connections

Unit 1 supplies dot products and linear combinations. Unit 2 supplies subspaces, rank, and null spaces. Unit 3 uses the projection of a gradient onto a unit direction for directional derivatives. Unit 5 derives the normal equations using gradients. Unit 7 extends projection and Gram–Schmidt to polynomial and inner product spaces.

Chapter 5

Eigenvalues and second-order optimization

Eigenvectors identify directions that a square matrix maps back to the same line. Diagonalization collects enough of these directions into a coordinate system in which the matrix acts by separate scalings. Symmetric matrices admit orthonormal eigenvector coordinates, which also simplify quadratic forms. We then apply these tools to Hessians and revisit least squares through projection, gradients, and curvature.

Big questions. Which directions make a matrix simple? How do those directions describe quadratic forms and the local shape of a function? How do projection, zero gradient, and curvature describe the same least-squares problem?

Learning outcomes. By the end of this unit, students should be able to:

- *U5-LO1.* Compute eigenvalues, eigenvectors, and eigenspaces, and interpret eigenvectors as directions whose lines are preserved by a square matrix.
- *U5-LO2.* Determine whether a matrix is diagonalizable, construct a diagonalization

$$A = PDP^{-1}$$

when possible, and use it to simplify matrix actions and powers.

- *U5-LO3.* Orthogonally diagonalize symmetric matrices and interpret the spectral theorem as an orthonormal change of coordinates.
- *U5-LO4.* Analyze quadratic forms in principal-axis coordinates and classify symmetric matrices as positive definite, negative definite, or indefinite.
- *U5-LO5.* Use quadratic approximations, Hessian eigenvalues, contour plots, and the two-variable discriminant shortcut to classify critical points.
- *U5-LO6.* Derive and interpret least squares through residual orthogonality, zero gradient, Hessian curvature, global minimality, and uniqueness.
- *U5-LO7.* Construct a design matrix from fixed nonlinear features, write the squared-error loss

$$L(\mathbf{c}) = \|\mathbf{A}\mathbf{c} - \mathbf{y}\|^2,$$

and explain why fitting only $\mathbf{c}$ is a least-squares problem.

Toolbox skills. Compute eigenvalues, eigenvectors, eigenspaces, diagonalizations, matrix powers, orthogonal diagonalizations, quadratic forms, definiteness, Hessian eigenvalue classifications, normal equations, residual checks, and fixed-feature design matrices.

5.1 Eigenvalues and eigenvectors

Let A be a square matrix. An eigenvector is a nonzero vector whose line is preserved by A : applying A may stretch, reverse, or collapse the vector, but it does not turn the vector away from that line. Eigenvectors are general tools for understanding matrix actions, independently of any particular application.

5.1.1 Basic definitions

Definition 5.1.1.

Let A be an $n \times n$ matrix. Suppose $\mathbf{v}$ is a *nonzero* vector in $\mathbb{R}^n$ and λ is a scalar (which could be 0) such that

$$A\mathbf{v} = \lambda\mathbf{v}$$

. Then λ is an *eigenvalue* of A , and $\mathbf{v}$ an *eigenvector* of A associated with the *eigenvalue* λ .

Activity

Check that

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$$

are eigenvectors for the matrix

$$A = \begin{bmatrix} 3 & -1 \\ 6 & -4 \end{bmatrix}$$

, and find the corresponding eigenvalues.

Solution.

We plug in $\mathbf{v}_1$ into the matrix A , and calculate that

$$\begin{aligned}
 A\mathbf{v}_1 &= \begin{bmatrix} 3 & -1 \\ 6 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\
 &= \begin{bmatrix} 3(1) + (-1)(1) \\ 6(1) + (-4)(1) \end{bmatrix} \\
 &= \begin{bmatrix} 2 \\ 2 \end{bmatrix} \\
 &= 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 2\mathbf{v}_1
 \end{aligned}$$

. Since $A\mathbf{v}_1 = 2\mathbf{v}_1$, the vector $\mathbf{v}_1$ is an eigenvector of A associated with the eigenvalue 2. Next, we plug in $\mathbf{v}_2$, and check that

$$\begin{aligned}
 A\mathbf{v}_2 &= \begin{bmatrix} 3 & -1 \\ 6 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 6 \end{bmatrix} \\
 &= \begin{bmatrix} 3(1) + (-1)(6) \\ 6(1) + (-4)(6) \end{bmatrix} \\
 &= \begin{bmatrix} -3 \\ -18 \end{bmatrix} \\
 &= -3 \begin{bmatrix} 1 \\ 6 \end{bmatrix} = -3\mathbf{v}_2
 \end{aligned}$$

. Since $A\mathbf{v}_2 = -3\mathbf{v}_2$, $\mathbf{v}_2$ is an eigenvector of A associated with the eigenvalue -3 .

Activity

Let

$$A = \begin{bmatrix} 3 & 0 \\ -1 & -4 \end{bmatrix}$$

be a matrix. Find the eigenvalues and eigenvectors of A .

Solution.

We need to solve the equation $A\mathbf{v} = \lambda\mathbf{v}$, where both λ and $\mathbf{v} \neq \mathbf{0}$ are unknown.

The equation is equivalent to the equation $(\lambda I - A)\mathbf{v} = \mathbf{0}$. For each λ , in order for this equation to have a nontrivial solution $\mathbf{v}$, the matrix $\lambda I - A$ must be singular, i.e., $\det(\lambda I - A) = 0$. This allows us to temporarily eliminate $\mathbf{v}$ from the problem, determining the values of λ that have solutions $\mathbf{v}$.

We calculate that

$$\begin{aligned}\det(\lambda I - A) &= \det \begin{bmatrix} \lambda - 3 & 0 \\ 1 & 4 + \lambda \end{bmatrix} \\ &= (\lambda - 3)(\lambda + 4)\end{aligned}$$

Thus $\det(\lambda I - A) = 0$ holds precisely when $(\lambda - 3)(\lambda + 4) = 0$, i.e., when $\lambda = 3$ or $\lambda = -4$, and so these two values are the two eigenvalues of A . It remains to find the *eigenvectors* for each of the eigenvalues.

We now find the nontrivial solutions to the equation $(A - \lambda I)\mathbf{v} = \mathbf{0}$ for $\lambda = 3$ and $\lambda = -4$. Each of these is a linear system that can be solved, either using the methods of 2, or more ad hoc methods, which may be faster since the matrices we are dealing with are small.

For $\lambda = 3$, we calculate that

$$A - 3I = \begin{bmatrix} 0 & 0 \\ -1 & -7 \end{bmatrix}$$

. Thus the solutions to $(A - 3I)\mathbf{v} = \mathbf{0}$ are given by $-v_1 - 7v_2 = 0$, or $v_1 = -7v_2$. Thus the solutions to this equation are given by nonzero multiples of the basic solution

$$\begin{bmatrix} -7 \\ 1 \end{bmatrix}$$

. These are all eigenvectors of A with eigenvalue 3.

For $\lambda = -4$, we calculate that

$$A + 4I = \begin{bmatrix} 7 & 0 \\ -1 & 0 \end{bmatrix}$$

. Thus the solutions to $(A + 4I)\mathbf{v} = \mathbf{0}$ are those vectors $\mathbf{v}$ such that $v_1 = 0$, and so the eigenvectors for A with eigenvalue -4 are given by nonzero multiples of the basic solution

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

. To conclude, the two eigenvalues of the matrix A are $\lambda = 3$ and $\lambda = -4$. The set of all eigenvectors of A associated with the eigenvalue $\lambda = 3$ is given by the expression

$$\left\{ \begin{bmatrix} -7t \\ t \end{bmatrix} : t \neq 0 \right\}$$

, and the set of eigenvectors of A associated with the eigenvalue $\lambda = -4$ is given by

$$\left\{ \begin{bmatrix} 0 \\ t \end{bmatrix} : t \neq 0 \right\}$$

.

Definition 5.1.2.

Let $A = [a_{ij}]$ be an $n \times n$ matrix. Then

$$\det(\lambda I_n - A) = \det \begin{bmatrix} \lambda - a_{11} & -a_{12} & \cdots & -a_{1n} \\ -a_{21} & \lambda - a_{22} & \cdots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & -a_{n2} & \cdots & \lambda - a_{nn} \end{bmatrix}$$

is called the *characteristic polynomial* of A and is sometimes denoted $c_A(\lambda)$. The equation

$$\det(\lambda I_n - A) = 0$$

is called the *characteristic equation* of A .

Theorem 5.1.3.

Let A be an $n \times n$ matrix. The eigenvalues of A are the *roots* of the characteristic polynomial of A .

Remark

Some authors define the characteristic polynomial to be $\det(A - \lambda I_n)$ rather than $\det(\lambda I_n - A)$. The advantage of the definition chosen in these notes is that the characteristic polynomial is always *monic*, i.e., the coefficient associated with the highest power of λ is always equal to one. The zeros of the two polynomials are both the same, and as we saw in 5.1.1, they are the eigenvalues of A .

Activity

Find the characteristic polynomial of A if

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{bmatrix}$$

Solution.

We calculate that

$$\begin{aligned} \lambda I_3 - A &= \begin{bmatrix} \lambda - 1 & -2 & 1 \\ -1 & \lambda - 0 & -1 \\ -4 & 4 & \lambda - 5 \end{bmatrix} \\ &= \begin{bmatrix} \lambda - 1 & -2 & 1 \\ -1 & \lambda & -1 \\ -4 & 4 & \lambda - 5 \end{bmatrix} \end{aligned}$$

The characteristic polynomial is therefore

$$\begin{aligned}
\det(\lambda I_3 - A) &= \det \begin{bmatrix} \lambda - 1 & -2 & 1 \\ -1 & \lambda & -1 \\ -4 & 4 & \lambda - 5 \end{bmatrix} \\
&= (\lambda - 1) \det \begin{bmatrix} \lambda & -1 \\ 4 & \lambda - 5 \end{bmatrix} \\
&\quad - (-2) \det \begin{bmatrix} -1 & -1 \\ -4 & \lambda - 5 \end{bmatrix} \\
&\quad + (1) \det \begin{bmatrix} -1 & \lambda \\ -4 & 4 \end{bmatrix} \\
&= (\lambda - 1)[\lambda(\lambda - 5) - (-1)(4)] \\
&\quad - (-2)[(-1)(\lambda - 5) - (-1)(-4)] \\
&\quad + (1)[(-1)(4) - (\lambda)(-4)] \\
&= (\lambda - 1)[\lambda^2 - 5\lambda + 4] \\
&\quad - (-2)[- \lambda + 1] + (1)[4\lambda - 4] \\
&= (\lambda - 1)[\lambda^2 - 5\lambda + 4] \\
&\quad + (\lambda - 1)[-2] + (\lambda - 1)[4] \\
&= (\lambda - 1)[\lambda^2 - 5\lambda + 6] \\
&= \lambda^3 - 6\lambda^2 + 11\lambda - 6
\end{aligned}$$

Note: Finding eigenvalues and eigenvectors of a matrix A

1. Determine the roots of the characteristic polynomial $c_A(\lambda) = \det(\lambda I_n - A)$. These are the *eigenvalues* of A .
2. For each eigenvalue λ , find all nontrivial solutions to the homogeneous system $(\lambda I_n - A)\mathbf{x} = \mathbf{0}$. These are the *eigenvectors* of A associated with λ .

Definition 5.1.4.

A basic solution (recall 2.2.17) to the homogeneous equation $(\lambda I_n - A)\mathbf{x} = \mathbf{0}$ will be called a *basic eigenvector* associated with the eigenvalue λ .

Warning

Do not make the common mistake of first row-reducing A to RREF and then finding the eigenvalues and eigenvectors of the RREF matrix. Row operations on the matrix A *do not* preserve the set of basic solutions. They only preserve solutions to the homogeneous equation $A\mathbf{v} = \mathbf{0}$.

If $\mathbf{v}$ is an eigenvector of a matrix A with eigenvalue λ , then any nonzero scalar multiple of $\mathbf{v}$ is also an eigenvector with the same eigenvalue λ . Indeed, if $\mathbf{w} = c\mathbf{v}$ for some scalar $c \neq 0$, then:

$$\begin{aligned}
A\mathbf{w} &= A(c\mathbf{v}) \\
&= c(A\mathbf{v}) \quad (\text{by linearity of matrix multiplication}) \\
&= c(\lambda\mathbf{v}) \quad (\text{since } \mathbf{v} \text{ is an eigenvector with eigenvalue } \lambda) \\
&= \lambda(c\mathbf{v}) \\
&= \lambda\mathbf{w}
\end{aligned}$$

Put differently, the entire line $\text{span}(\mathbf{v})$ is “invariant” under the map defined by A , with every vector in this line being stretched (or compressed) by the factor λ .

Fact

Every nonzero linear combination of eigenvectors associated with the same eigenvalue is again an eigenvector for that eigenvalue. Conversely, every eigenvector in the eigenspace has a unique expression in a chosen basis of that eigenspace.

Activity

Find the eigenvalues and basic eigenvectors of the matrix

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Solution.

We begin by calculating the characteristic polynomial of A , i.e.,

$$\det(\lambda I_3 - A) = \det \begin{bmatrix} \lambda - 2 & -1 & 0 \\ 0 & \lambda - 2 & 0 \\ 0 & 0 & \lambda - 2 \end{bmatrix}$$

Since this matrix is upper triangular, the determinant is the product of the diagonal entries, and so the characteristic polynomial is $c_A(\lambda) = (\lambda - 2)^3$. The polynomial has only a single zero, i.e., $\lambda = 2$, and so $\lambda = 2$ is the only eigenvalue of A .

To find the basic eigenvectors of A with eigenvalue 2, we solve the system $(2I - A)\mathbf{x} = \mathbf{0}$, i.e.,

$$\begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Thus the solutions are given by $x_2 = 0$, with x_1 and x_3 as free variables. The set of basic eigenvectors is thus given by

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

5.2 Diagonalization

Eigenvalues describe individual special directions. Diagonalization asks whether there are enough independent eigenvectors to form a basis. In eigenvector coordinates, the matrix acts by scaling each coordinate separately.

Definition 5.2.1.

An $n \times n$ matrix A is called *diagonalizable* if there exists an invertible $n \times n$ matrix P so that $P^{-1}AP$ is a diagonal matrix (recall 2.5.11). The matrix P is called a *diagonalizing matrix* for A .

Activity

Suppose

$$A = \begin{bmatrix} 3 & 1 \\ -2 & 0 \end{bmatrix}$$

.

Task

Compute $P^{-1}AP$ where

$$P = \begin{bmatrix} -1/2 & -1 \\ 1 & 1 \end{bmatrix}$$

. You may use the fact that

$$P^{-1} = \begin{bmatrix} 2 & 2 \\ -2 & -1 \end{bmatrix}$$

. What does this computation say about the matrix A ?

Solution.

We start by calculating that

$$\begin{aligned} AP &= \begin{bmatrix} (3)(-1/2) + (1)(1) & (3)(-1) + (1)(1) \\ (-2)(-1/2) + (0)(1) & (-2)(-1) + (0)(1) \end{bmatrix} \\ &= \begin{bmatrix} -1/2 & -2 \\ 1 & 2 \end{bmatrix} \end{aligned}$$

and thus that

$$\begin{aligned}
P^{-1}AP &= \begin{bmatrix} 2 & 2 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} -1/2 & -2 \\ 1 & 2 \end{bmatrix} \\
&= \begin{bmatrix} (2)(-1/2) + (2)(1) & (2)(-2) + (2)(2) \\ (-2)(-1/2) + (-1)(1) & (-2)(-2) + (-1)(2) \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}
\end{aligned}$$

Since $P^{-1}AP$ is a diagonal matrix, the matrix A is diagonalizable.

Task

Use the previous computation to calculate A^5 .

Solution.

Let's start by computing smaller powers of A . We find that

$$\begin{aligned}
A^2 &= A \cdot A = (PDP^{-1})(PDP^{-1}) \\
&= PD(P^{-1}P)DP^{-1} = PD^2P^{-1}
\end{aligned}$$

and

$$\begin{aligned}
A^3 &= A \cdot A^2 = (PDP^{-1})(PD^2P^{-1}) \\
&= PD(P^{-1}P)D^2P^{-1} = PD^3P^{-1}
\end{aligned}$$

Continuing this calculation, noticing the pattern, we conclude that $A^5 = PD^5P^{-1}$. Multiplying diagonal matrices is easy, i.e.,

$$\begin{bmatrix} a_1 & 0 \\ 0 & b_1 \end{bmatrix} \begin{bmatrix} a_2 & 0 \\ 0 & b_2 \end{bmatrix} \cdots \begin{bmatrix} a_m & 0 \\ 0 & b_m \end{bmatrix} = \begin{bmatrix} a_1 \cdots a_m & 0 \\ 0 & b_1 \cdots b_m \end{bmatrix}$$

. Thus

$$D^5 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}^5 = \begin{bmatrix} 1^5 & 0 \\ 0 & 2^5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 32 \end{bmatrix}$$

. We use this computation to compute A^5 , i.e., we calculate that

$$PD^5 = \begin{bmatrix} -1/2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 32 \end{bmatrix} = \begin{bmatrix} -1/2 & -32 \\ 1 & 32 \end{bmatrix}$$

and thus that

$$\begin{aligned}
PD^5P^{-1} &= \begin{bmatrix} -1/2 & -32 \\ 1 & 32 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ -2 & -1 \end{bmatrix} \\
&= \begin{bmatrix} 63 & 31 \\ -62 & -30 \end{bmatrix}
\end{aligned}$$

.

The previous activity shows why diagonalizing a matrix may be useful. But given a matrix A , how do we potentially find P such that $P^{-1}AP$ is a diagonal matrix?

Activity

Suppose $P^{-1}AP = D$ for some diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$. Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be the columns of P . Note that the equation $P^{-1}AP = D$ holds if and only if $AP = PD$.

Task

What happens when I take PD ?

Solution.

$$PD = [\mathbf{v}_1 \mathbf{v}_2 \cdots \mathbf{v}_n] \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

Matrix multiplication gives us:

$$PD = [\lambda_1 \mathbf{v}_1 \quad \lambda_2 \mathbf{v}_2 \quad \cdots \quad \lambda_n \mathbf{v}_n]$$

So PD is the matrix whose columns are the vectors $\mathbf{v}_i$ scaled by their corresponding λ_i values.

Task

Calculate AP and PD in terms of the vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$, the scalars $\lambda_1, \dots, \lambda_n$ and the matrix A .

Solution.

When we take AP , we compute that

$$AP = A[\mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_n] = [A\mathbf{v}_1 \quad A\mathbf{v}_2 \quad \cdots \quad A\mathbf{v}_n]$$

.

When we calculate PD , we compute that

$$\begin{aligned} PD &= [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_n] \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} \\ &= [\lambda_1 \mathbf{v}_1 \quad \cdots \quad \lambda_n \mathbf{v}_n] \end{aligned}$$

.

Task

So what conditions on P and D are necessary and sufficient for the equation $AP = PD$ to hold?

Solution.

For the equation $AP = PD$ to hold, we need $A\mathbf{v}_i = \lambda_i\mathbf{v}_i$ for each i . Thus we need each column of P to be an eigenvector of A , whose eigenvalue is the corresponding entry of the diagonal matrix D .

To see this differently, suppose $P^{-1}AP$ is a diagonal matrix with diagonal entries $\lambda_1, \dots, \lambda_n$. Then for each standard basis vector (recall 1.3.5) $P^{-1}AP\mathbf{e}_i = \lambda_i\mathbf{e}_i$. Multiplying this equation by P on the left-hand side and setting $\mathbf{v}_i = P\mathbf{e}_i$, we obtain

$$A\mathbf{v}_i = P(P^{-1}AP\mathbf{e}_i) = P(\lambda_i\mathbf{e}_i) = \lambda_i\mathbf{v}_i$$

. Thus $\mathbf{v}_i$ is an eigenvector of the matrix A associated with the eigenvalue λ_i .

Theorem 5.2.2.

An $n \times n$ matrix A is *diagonalizable* if and only if A has n eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ such that the matrix P with columns $\mathbf{v}_1, \dots, \mathbf{v}_n$ is invertible. In this case, $P^{-1}AP = D$, where D is a diagonal matrix such that the elements of D on the main diagonal are the *eigenvalues* of A .

Remark

Recalling 2.5.1, the matrix P in 5.2.2 is invertible if and only if the vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ form a basis for $\mathbb{R}^n$.

If $\mathbb{R}^n$ has a basis of eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$, then every vector $\mathbf{v} \in \mathbb{R}^n$ can be written as a linear combination of these eigenvectors, i.e., there exists constants $a_1, \dots, a_n$ such that

$$\mathbf{v} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n$$

. Thus

$$\begin{aligned} T_A(\mathbf{v}) &= A\mathbf{v} \\ &= A(a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n) \\ &= a_1A\mathbf{v}_1 + a_2A\mathbf{v}_2 + \dots + a_nA\mathbf{v}_n \\ &= a_1\lambda_1\mathbf{v}_1 + a_2\lambda_2\mathbf{v}_2 + \dots + a_n\lambda_n\mathbf{v}_n \\ &= \lambda_1(a_1\mathbf{v}_1) + \lambda_2(a_2\mathbf{v}_2) + \dots + \lambda_n(a_n\mathbf{v}_n) \end{aligned}$$

In other words, T_A ‘stretches’ the components of $\mathbf{v}$ along each eigenvector by the corresponding eigenvalue.

Activity

Consider the matrix

$$M = \begin{bmatrix} 5/3 & -2/3 \\ -1/3 & 4/3 \end{bmatrix}$$

. Verify that

$$\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}$$

are eigenvectors of M corresponding to different eigenvalues. What are the eigenvalues?

Solution.

We compute that

$$\begin{aligned} M\mathbf{u} &= \begin{bmatrix} 5/3 & -2/3 \\ -1/3 & 4/3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 5/3 - 2/3 \\ -1/3 + 4/3 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= 1 \cdot \mathbf{u} \end{aligned}$$

and

$$\begin{aligned} M\mathbf{v} &= \begin{bmatrix} 5/3 & -2/3 \\ -1/3 & 4/3 \end{bmatrix} \begin{bmatrix} 1 \\ -1/2 \end{bmatrix} \\ &= \begin{bmatrix} 5/3 - 2/3 \cdot (-1/2) \\ -1/3 + 4/3 \cdot (-1/2) \end{bmatrix} \\ &= \begin{bmatrix} 5/3 + 1/3 \\ -1/3 - 2/3 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ -1 \end{bmatrix} \\ &= 2 \cdot \begin{bmatrix} 1 \\ -1/2 \end{bmatrix} \\ &= 2 \cdot \mathbf{v} \end{aligned}$$

. Therefore, $\mathbf{u}$ is an eigenvector with eigenvalue $\lambda_1 = 1$ and $\mathbf{v}$ is an eigenvector with eigenvalue $\lambda_2 = 2$.

5.1 illustrates the effect of the matrix M from 5.2 on a grid in $\mathbb{R}^2$. In particular, it highlights an important geometric interpretation of eigenvectors and eigenvalues; when we apply the linear map defined by the matrix M to any vector in $\mathbb{R}^2$, the matrix map stretches eigenvector directions by their eigenvalues.

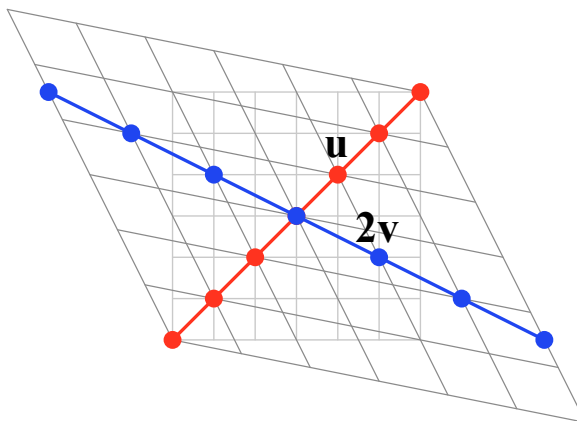
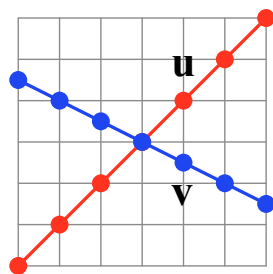


Figure 5.1. Visualization of the action of the matrix M from 5.2 on a grid. The red line along the eigenvector $\mathbf{u}$ is unchanged, because $\mathbf{u}$ has eigenvalue 1. The blue line along the eigenvector $\mathbf{v}$ is stretched by a factor of 2, because $\mathbf{v}$ has eigenvalue 2. Adapted from: Stanford's MATH 51 textbook.

Remark: Non-diagonalizable matrices

Consider the rotation matrix

$$A_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

where θ is not a multiple of 180° . For any nonzero vector $\mathbf{v} \in \mathbb{R}^2$, the vector $A_\theta \mathbf{v}$ is obtained by rotating $\mathbf{v}$ counterclockwise by an angle θ , so $A_\theta \mathbf{v}$ is never on the line spanned by $\mathbf{v}$. Thus $A_\theta \mathbf{v} = \lambda \mathbf{v}$ cannot hold for any real scalar λ and any nonzero $\mathbf{v}$.

Thus the matrix A_θ has no real eigenvalues or real eigenvectors. It is therefore not diagonalizable over $\mathbb{R}$. This makes geometric sense: we cannot find special directions in $\mathbb{R}^2$ so that rotating vectors in the plane is obtained by stretching in those directions.

5.1 illustrates why not every matrix is diagonalizable. A matrix is diagonalizable precisely when it has a basis of eigenvectors, but this is not always possible.

Activity

Let

$$A = \begin{bmatrix} 3 & 0 \\ -1 & -4 \end{bmatrix}$$

. Is A diagonalizable?

Solution.

We need to check if A has a basis of eigenvectors. In 5.1.1, we showed that A has eigenvectors

$$\mathbf{v}_1 = \begin{bmatrix} -7 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

. The two vectors are linearly independent as they are not parallel, and thus form a basis for $\mathbb{R}^2$. Therefore, A is diagonalizable.

Activity

Let

$$A = \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}$$

. Is A diagonalizable?

Solution.

We start by calculating the characteristic polynomial of A , that

$$\det(\lambda I - A) = \det \begin{bmatrix} \lambda - 4 & -1 \\ 0 & \lambda - 4 \end{bmatrix} = (\lambda - 4)^2$$

. Thus $\lambda = 4$ is the only eigenvalue of A .

To find the eigenvectors of A , we solve the equation $(4I - A)\mathbf{v} = \mathbf{0}$, which we expand as

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

. The equation has only a single basic solution, i.e.,

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

. Recalling 5.1.1, every eigenvector of A is a scalar multiple of this vector. So A cannot have a basis of eigenvectors, and so A is not diagonalizable.

5.3 Symmetric matrices and orthogonal diagonalization

A symmetric matrix, as defined in 1.2.9, is especially well behaved. Its eigenvectors can be chosen to form an orthonormal basis; recall 4.1.1. Thus diagonalization uses a rigid change of coordinates that preserves dot products and lengths. This section develops that structure before applying it to quadratic forms.

5.3.1 Orthogonality of eigenvectors, orthogonal matrices, and orthogonal diagonalization

A key property of symmetric matrices is that for any two vectors $\mathbf{x}$ and $\mathbf{y}$, $(A\mathbf{x}) \cdot \mathbf{y} = \mathbf{x} \cdot (A\mathbf{y})$. Indeed, we calculate that

$$\begin{aligned}
(A\mathbf{x}) \cdot \mathbf{y} &= (A\mathbf{x})^T \mathbf{y} \\
&= \mathbf{x}^T A^T \mathbf{y} \\
&= \mathbf{x}^T A \mathbf{y} \quad (\text{since } A^T = A) \\
&= \mathbf{x} \cdot (A\mathbf{y})
\end{aligned}$$

This property leads to a fundamental result about the eigenvectors of symmetric matrices.

Theorem 5.3.1 (Orthogonality of Eigenvectors).

If A is a *symmetric matrix*, then the *eigenvectors* of A corresponding to *distinct* eigenvalues are *orthogonal*.

Proof.

Let $A\mathbf{x} = \lambda\mathbf{x}$ and $A\mathbf{y} = \mu\mathbf{y}$, where $\lambda \neq \mu$. Using the property above:

$$\begin{aligned}
\lambda(\mathbf{x} \cdot \mathbf{y}) &= (\lambda\mathbf{x}) \cdot \mathbf{y} \\
&= (A\mathbf{x}) \cdot \mathbf{y} \\
&= \mathbf{x} \cdot (A\mathbf{y}) \\
&= \mathbf{x} \cdot (\mu\mathbf{y}) \\
&= \mu(\mathbf{x} \cdot \mathbf{y})
\end{aligned}$$

This gives us $(\lambda - \mu)(\mathbf{x} \cdot \mathbf{y}) = 0$. Since $\lambda \neq \mu$, we must have $\mathbf{x} \cdot \mathbf{y} = 0$, which means the eigenvectors are orthogonal.

As we will see, it turns out that for any symmetric matrix, there exists a pairwise orthogonal set of eigenvectors which form a basis of $\mathbb{R}^n$. In particular, every symmetric matrix is diagonalizable.

Recall from 2.5.22 that a square matrix Q is *orthogonal* if $Q^T = Q^{-1}$.

Theorem 5.3.2 (Equivalent Conditions for Orthogonal Matrices).

Let Q be an $n \times n$ matrix. Then the following conditions are equivalent to one another:

1. Q is invertible and $Q^{-1} = Q^T$.
2. The rows of Q are orthonormal.
3. The columns of Q are orthonormal.

Proof.

Let us start by showing the first property is equivalent to the third. Suppose Q is invertible, and $Q^{-1} = Q^T$. Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be the columns of Q . Recall from 1.4.2 that the (i, j) -entry of $Q^T Q$ is equal to the dot product $\mathbf{v}_i \cdot \mathbf{v}_j$. Since $Q^{-1} = Q^T$, $Q^T Q = I$, and so for $i \neq j$, $\mathbf{v}_i \cdot \mathbf{v}_j = 0$, and for each i , $\mathbf{v}_i \cdot \mathbf{v}_i = 1$. Thus the column vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ are orthonormal. Conversely, if the column vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ are orthonormal, then we immediately see that $Q^T Q = I$.

The proof that the first property is equivalent to the second is very similar, using the rows of Q rather than the columns and that if $Q^T Q = I$, then $Q Q^T = I$, and vice versa.

Theorem 5.3.3.

Let Q be an $n \times n$ orthogonal matrix. Then:

1. Q preserves dot products, i.e., for any vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$, $(Q\mathbf{v}) \cdot (Q\mathbf{w}) = \mathbf{v} \cdot \mathbf{w}$.
2. Q preserves lengths, i.e., for any vector $\mathbf{v} \in \mathbb{R}^n$, $\|Q\mathbf{v}\| = \|\mathbf{v}\|$.

Proof.

Let $\mathbf{v}$ and $\mathbf{w}$ be column vectors. Recalling the fifth property of 1.4.3, we calculate that

$$\begin{aligned} (Q\mathbf{v}) \cdot (Q\mathbf{w}) &= (Q\mathbf{v})^T (Q\mathbf{w}) \\ &= \mathbf{v}^T Q^T Q \mathbf{w} \\ &= \mathbf{v}^T I_n \mathbf{w} \\ &= \mathbf{v}^T \mathbf{w} \\ &= \mathbf{v} \cdot \mathbf{w} \end{aligned}$$

. Thus Q preserves dot products.

Next, using that Q preserves dot products, and also 1.5, we find

$$\|Q\mathbf{v}\|^2 = (Q\mathbf{v}) \cdot (Q\mathbf{v}) = \mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$$

. Taking square roots on both sides of this equation gives $\|Q\mathbf{v}\| = \|\mathbf{v}\|$.

5.2 provides a visualization of an orthogonal change of coordinates.

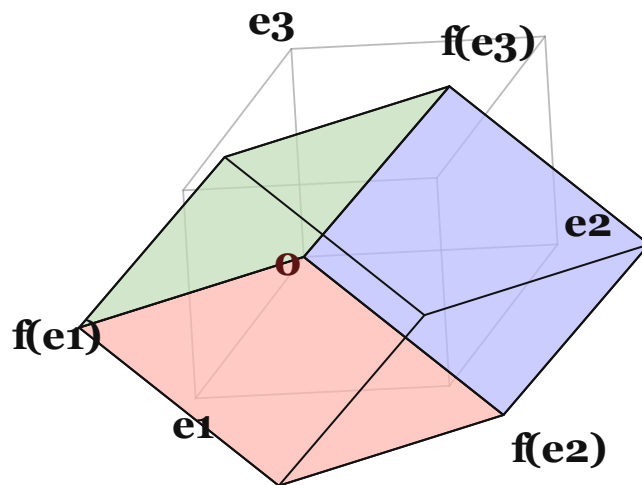


Figure 5.2. The effect of applying the linear map $f(\mathbf{x}) = Q\mathbf{x}$ to a cube, where Q is orthogonal. Notice that the map is “rigid”, i.e., applying the map to the cube results in a cube with the same dimensions. Adapted from: Stanford’s MATH 51 textbook.

Definition 5.3.4 (Orthogonal diagonalizability).

A matrix A is *orthogonally diagonalizable* when an *orthogonal matrix* Q can be found such that $Q^{-1}AQ$ is a diagonal matrix. Since $Q^{-1} = Q^T$, this is equivalent to $Q^T A Q$ being a diagonal matrix.

Activity

Consider the symmetric matrix

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

Check that the vectors

$$\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

are eigenvectors of A which are orthogonal to each other. What are the corresponding eigenvalues?

Solution.

The vectors $\mathbf{u}$ and $\mathbf{v}$ are eigenvectors of A with eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 3$ respectively, since

$$A\mathbf{u} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1\mathbf{u}$$

and

$$A\mathbf{v} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \end{bmatrix} = 3\mathbf{v}$$

We also calculate that

$$\mathbf{u} \cdot \mathbf{v} = 1 \cdot 1 + 1 \cdot (-1) = 0$$

. Thus the two vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.

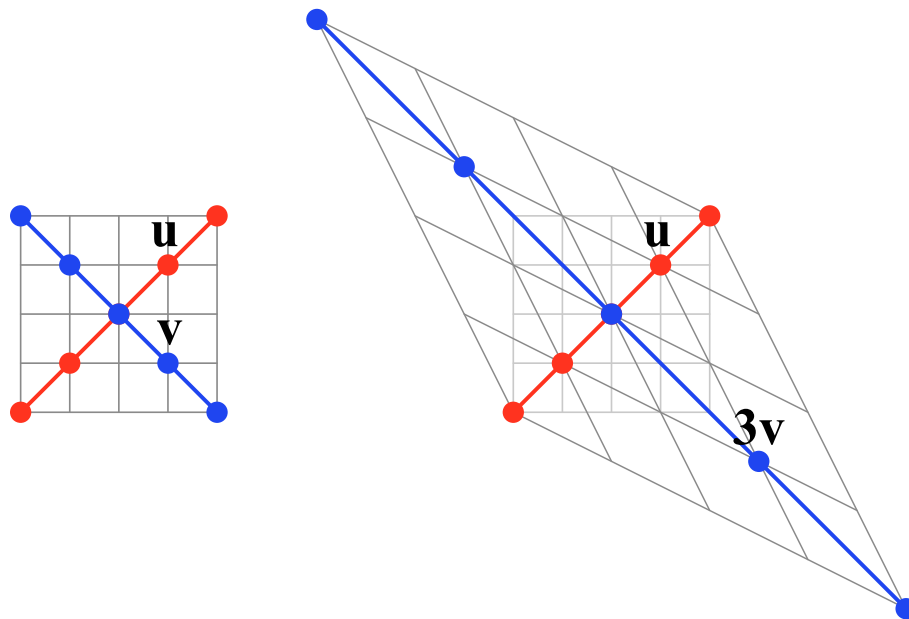


Figure 5.3. Visualization of the action of matrix A from 5.2 on a grid. The red line along eigenvector $\mathbf{u}$ is unchanged (since $\mathbf{u}$ is an eigenvector with eigenvalue 1), while the blue line along eigenvector $\mathbf{v}$ is stretched by a factor of 3 (since $\mathbf{v}$ is an eigenvector with eigenvalue 3). Adapted from: Stanford's MATH 51 textbook.

5.3 should be compared to 5.1. Notice that since the matrix A in 5.2 is symmetric, the red and blue line in 5.3 meet at right angles, since the vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, whereas the red and blue lines in 5.1 do not meet at right angles, since the eigenvectors for that example are not orthogonal to one another.

Activity

Consider the matrix

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

. Find an orthogonal matrix Q such that $Q^{-1}AQ$ is diagonal.

Solution.

We proceed as in 5.2, i.e., finding a basis of eigenvectors of A , and defining Q by taking those columns as eigenvectors. By 5.3.2, as long as the eigenvectors are orthonormal, the matrix Q will be orthogonal. We start by finding the eigenvalues of A by computing its characteristic polynomial. We calculate that

$$\begin{aligned}
c_A(\lambda) &= \det \begin{bmatrix} 1-\lambda & 1 \\ 1 & -1-\lambda \end{bmatrix} \\
&= (1-\lambda)(-1-\lambda) - 1 \\
&= \lambda^2 - 2 \\
&= (\lambda - \sqrt{2})(\lambda + \sqrt{2})
\end{aligned}$$

. The solutions to $c_A(\lambda) = 0$ are thus given by $\lambda = \pm\sqrt{2}$, and so $\lambda_1 = \sqrt{2}$ and $\lambda_2 = -\sqrt{2}$ are the two eigenvalues of A .

To calculate an eigenvector with eigenvalue $\sqrt{2}$, we solve the system

$$\begin{bmatrix} 1-\sqrt{2} & 1 \\ 1 & -1-\sqrt{2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

. Expanding the left-hand side and reading off the second entry of the resulting vector gives

$$v_1 + (-1 - \sqrt{2})v_2 = 0$$

. Because $\sqrt{2}$ is an eigenvalue, the matrix $A - \sqrt{2}I$ is singular. The displayed matrix is nonzero, so its rank is 1. Therefore there is one pivot variable and one free variable. Setting $v_2 = 1$ and solving for v_1 , we obtain the basic eigenvector

$$\begin{bmatrix} 1 + \sqrt{2} \\ 1 \end{bmatrix}$$

. We are looking for an orthonormal basis of eigenvectors, and so in particular, the eigenvectors we choose as a basis must all have length 1. We thus consider a multiple of the basic eigenvector which has length 1. Since the basic eigenvector has length

$$\sqrt{(1 + \sqrt{2})^2 + (1)^2} = \sqrt{4 + 2\sqrt{2}}$$

The vector

$$\mathbf{v} = \frac{1}{\sqrt{4 + 2\sqrt{2}}} \begin{bmatrix} 1 + \sqrt{2} \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1+\sqrt{2}}{\sqrt{4+2\sqrt{2}}} \\ \frac{1}{\sqrt{4+2\sqrt{2}}} \end{bmatrix}$$

has length 1. This will be the first vector in our basis.

With a similar technique, one may find the basic eigenvector with eigenvalue $-\sqrt{2}$. It is the vector

$$\begin{bmatrix} 1 - \sqrt{2} \\ 1 \end{bmatrix}$$

. Normalizing gives the vector

$$\mathbf{w} = \frac{1}{\sqrt{4-2\sqrt{2}}} \begin{bmatrix} 1-\sqrt{2} \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1-\sqrt{2}}{\sqrt{4-2\sqrt{2}}} \\ \frac{1}{\sqrt{4-2\sqrt{2}}} \end{bmatrix}$$

, which has length 1. This is the second vector in our basis.

Using 5.3.1, since A is symmetric, $\mathbf{v}$ and $\mathbf{w}$ are orthogonal, and thus, by 4.1.7, are linearly independent, thus forming a basis. But this means that the matrix

$$Q = \begin{bmatrix} \frac{1+\sqrt{2}}{\sqrt{4+2\sqrt{2}}} & \frac{1-\sqrt{2}}{\sqrt{4-2\sqrt{2}}} \\ \frac{1}{\sqrt{4+2\sqrt{2}}} & \frac{1}{\sqrt{4-2\sqrt{2}}} \end{bmatrix}$$

is orthogonal, and

$$Q^{-1}AQ = \begin{bmatrix} \sqrt{2} & 0 \\ 0 & -\sqrt{2} \end{bmatrix}$$

is diagonal.

Activity

Suppose that A is orthogonally diagonalizable, i.e., we can find an orthogonal matrix Q so that $Q^T A Q = D$, where D is diagonal. Explain why $A = Q D Q^T$.

Solution.

We calculate that

$$Q D Q^T = Q(Q^T A Q)Q^T = (Q Q^T) A (Q Q^T) = I A I = A$$

Remark

The factorization $A = Q D Q^T$ writes A as a product of three matrices: an orthogonal matrix, a diagonal matrix, and the transpose of the orthogonal matrix. This product is sometimes called the *spectral decomposition* of a symmetric matrix A .

5.3.2 Principal axes theorem

Recall 5.2.2. That theorem tells us that if A is orthogonally diagonalizable, where $Q^{-1}AQ$ is diagonal, then the columns of Q must be linearly independent eigenvectors of A . In addition, in order for Q to be orthogonal, 5.3.2 tells us that the eigenvectors which form the columns of Q must be orthonormal.

Remarkably, *all* symmetric matrices are orthogonally diagonalizable.

Theorem 5.3.5 (Principal Axes Theorem).

Let A be any $n \times n$ matrix. Then the following conditions are equivalent:

1. A has an orthonormal set of n eigenvectors
2. A is orthogonally diagonalizable
3. A is symmetric

A set of orthonormal eigenvectors for a symmetric matrix A is called a set of *principal axes* for A .

Proof.

We do not prove the “hard” part of the theorem (that the third property implies the second). But the proof that the second property implies the third is more simple (see 5.3.2).

Remark

5.3.5 is also called *The Spectral Theorem* for symmetric matrices (the set of eigenvalues of a matrix is often called the *spectrum* of the matrix).

We do not prove the “hard” part of this theorem, that is, that (3) implies (2). Let’s however check that (2) implies (3).

Activity

Show that if A is orthogonally diagonalizable, then A is symmetric.

Solution.

Suppose A is orthogonally diagonalizable. Then there exists an orthogonal matrix Q and a diagonal matrix D such that $Q^T A Q = D$. 5.3 tells us that $A = Q D Q^T$. But now we check that

$$\begin{aligned} A^T &= (Q D Q^T)^T \\ &= (Q^T)^T D^T Q^T \\ &= Q D Q^T = A \end{aligned}$$

. The equation $A^T = A$ means precisely that A is symmetric.

Activity

Let

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ -1 & 2 & 5 \end{bmatrix}$$

Find an orthogonal matrix Q such that $Q^{-1} A Q$ is diagonal.

Solution.

First, compute the eigenvalues. We calculate that

$$\begin{aligned}
 \det(A - \lambda I) &= \det \begin{bmatrix} 1 - \lambda & 0 & -1 \\ 0 & 1 - \lambda & 2 \\ -1 & 2 & 5 - \lambda \end{bmatrix} \\
 &= (1 - \lambda)[(1 - \lambda)(5 - \lambda) - 4] + (-1)[0 - (1 - \lambda)(-1)] \\
 &= (1 - \lambda)[(1 - \lambda)(5 - \lambda) - 4] - (1 - \lambda) \\
 &= (1 - \lambda)[(1 - \lambda)(5 - \lambda) - 4 - 1] \\
 &= (1 - \lambda)[-6\lambda + \lambda^2] \\
 &= -\lambda(\lambda - 1)(\lambda - 6)
 \end{aligned}$$

This calculation verifies that the eigenvalues of A are $\lambda_1 = 0$, $\lambda_2 = 1$, and $\lambda_3 = 6$. Now we proceed to find an eigenvector for each of these eigenvalues.

For the eigenvalue $\lambda_1 = 0$, we solve the equation $A\mathbf{v}_1 = \mathbf{0}$, which we expand as

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ -1 & 2 & 5 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

By observation, we see that the first and second columns will contain pivots in the reduced row echelon form of the matrix on the left-hand side, so setting $v_3 = 1$ gives the basic eigenvector

$$\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

. Normalizing gives the normalized eigenvector

$$\mathbf{v}_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{6} \\ -2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}$$

, which will be the first vector in our basis.

For the eigenvalue $\lambda_2 = 1$, we solve the equation $(A - I)\mathbf{v}_2 = \mathbf{0}$, which we expand as

$$\begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 2 \\ -1 & 2 & 4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

We now see that first and third columns will contain pivots when we row reduce this matrix, so setting $v_2 = 1$ and solving gives the basic eigenvector

$$\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

. Normalizing gives the eigenvector

$$\mathbf{v}_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \\ 0 \end{bmatrix}$$

. This will be the second vector in our basis.

For $\lambda_3 = 6$, we solve $(A - 6I)\mathbf{v}_3 = \mathbf{0}$, i.e.,

$$\begin{bmatrix} -5 & 0 & -1 \\ 0 & -5 & 2 \\ -1 & 2 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The first two equations give $v_1 = -\frac{1}{5}v_3$ and $v_2 = \frac{2}{5}v_3$. Setting $v_3 = 1$ gives the basic eigenvector

$$\begin{bmatrix} -1/5 \\ 2/5 \\ 1 \end{bmatrix}$$

. Multiplying by 5 (to prevent working with fractions) gives the integer eigenvector

$$\begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix}$$

Normalizing gives the eigenvector

$$\mathbf{v}_3 = \frac{1}{\sqrt{30}} \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix} = \begin{bmatrix} -1/\sqrt{30} \\ 2/\sqrt{30} \\ 5/\sqrt{30} \end{bmatrix}$$

. This will be the third and final vector in our basis.

We now set

$$Q = \begin{bmatrix} 1/\sqrt{6} & 2/\sqrt{5} & -1/\sqrt{30} \\ -2/\sqrt{6} & 1/\sqrt{5} & 2/\sqrt{30} \\ 1/\sqrt{6} & 0 & 5/\sqrt{30} \end{bmatrix}$$

. Then

$$Q^{-1}AQ = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

. We have thus found an orthogonal diagonalization of the matrix A .

In 5.3.2, the eigenvalues of A were distinct. To come up with a general strategy to obtain an orthogonal diagonalization, we need to make one further observation.

Definition 5.3.6 (Eigenspaces of a matrix).

Suppose A is a square matrix. Then for any scalar λ , the set

$$E_\lambda(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \lambda\mathbf{x}\}$$

is called the *eigenspace* of A corresponding to λ .

Theorem 5.3.7.

Suppose A is a square matrix and $\lambda \in \mathbb{R}$. Then $E_\lambda(A)$ is a subspace of $\mathbb{R}^n$.

Proof.

Note that $\mathbf{x} \in E_\lambda(A)$ precisely when $\mathbf{x}$ solves the equation $(A - \lambda I)\mathbf{x} = \mathbf{0}$. Thus $E_\lambda(A)$ is precisely the null space of the matrix $A - \lambda I$, and the theorem follows from 2.3.4.

Note: Procedure for orthogonally diagonalizing a symmetric matrix A

1. Find the eigenvalues of A .
2. For each eigenvalue λ of A , find a basis for $E_\lambda(A)$, and convert it to an orthonormal basis using the Gram-Schmidt orthogonalization process (see 4.2.6) if necessary.
3. Collecting the orthonormal bases for each of the eigenspaces gives an orthonormal basis of eigenvectors for A . Taking these eigenvectors as the columns of a matrix Q gives an orthogonal matrix such that $Q^{-1}AQ$ is diagonal.

Activity

If

$$A = \begin{bmatrix} 8 & -2 & 2 \\ -2 & 5 & 4 \\ 2 & 4 & 5 \end{bmatrix},$$

compute an orthonormal diagonalization of A . You may use the fact that the characteristic polynomial of A is $\lambda(\lambda - 9)^2$, and the null space of A is spanned by the vector

$$\begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}$$

Solution.

The eigenvalues are $\lambda_1 = 0$ and $\lambda_2 = 9$. Since the null space of A is spanned by the vector

$$\begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}$$

, the eigenvectors of A with eigenvalue 0 are all multiples of this vector. Thus an orthonormal basis for $E_0(A)$ is given by

$$\left\{ \begin{bmatrix} 1/3 \\ 2/3 \\ -2/3 \end{bmatrix} \right\}$$

. On the other hand, to find an orthonormal basis for $E_9(A)$ we form the matrix

$$9I - A = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} - \begin{bmatrix} 8 & -2 & 2 \\ -2 & 5 & 4 \\ 2 & 4 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -2 \\ 2 & 4 & -4 \\ -2 & -4 & 4 \end{bmatrix}$$

. Using the row operations $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 + 2R_1$, we obtain the reduced row echelon form for the matrix $9I - A$: the matrix

$$\begin{bmatrix} 1 & 2 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

. We thus obtain the two basic solutions

$$\mathbf{v}_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

. We now apply Gram-Schmidt to this pair of vectors, which form a basis of $E_9(A)$, in order to obtain an orthogonal basis. We define

$$\mathbf{w}_1 = \mathbf{v}_1 \quad \text{and} \quad \mathbf{w}_2 = \mathbf{v}_2 - \frac{\mathbf{v}_2 \cdot \mathbf{w}_1}{\mathbf{w}_1 \cdot \mathbf{w}_1} \mathbf{w}_1.$$

We calculate that

$$\frac{\mathbf{v}_2 \cdot \mathbf{w}_1}{\mathbf{w}_1 \cdot \mathbf{w}_1} = \frac{-4}{5}$$

and thus that

$$\mathbf{w}_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} + \frac{4}{5} \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2/5 \\ 4/5 \\ 1 \end{bmatrix}$$

. Normalizing gives the orthonormal basis

$$\left\{ \frac{\mathbf{w}_1}{\|\mathbf{w}_1\|}, \frac{\mathbf{w}_2}{\|\mathbf{w}_2\|} \right\} = \left\{ \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \\ 0 \end{bmatrix}, \begin{bmatrix} 2/\sqrt{45} \\ 4/\sqrt{45} \\ 5/\sqrt{45} \end{bmatrix} \right\}$$

for $E_9(A)$.

Combining the two orthonormal bases we have calculated gives an orthonormal basis

$$\left\{ \begin{bmatrix} 1/3 \\ 2/3 \\ -2/3 \end{bmatrix}, \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \\ 0 \end{bmatrix}, \begin{bmatrix} 2/\sqrt{45} \\ 4/\sqrt{45} \\ 5/\sqrt{45} \end{bmatrix} \right\}$$

of $\mathbb{R}^n$ consisting of eigenvectors, and so if we define

$$Q = \begin{bmatrix} 1/3 & -2/\sqrt{5} & 2/\sqrt{45} \\ 2/3 & 1/\sqrt{5} & 4/\sqrt{45} \\ -2/3 & 0 & 5/\sqrt{45} \end{bmatrix}$$

, then

$$Q^{-1}AQ = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

gives an orthogonal diagonalization of A .

5.4 Quadratic forms and definiteness

So far, a matrix A has defined the matrix map

$$\mathbf{x} \mapsto A\mathbf{x},$$

whose output is a vector. The same matrix also defines the scalar-valued rule

$$q_A(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}.$$

This rule is quadratic rather than linear. Although the input now appears twice, eigenvectors and the orthogonal diagonalization theorem still make the rule simple.

Definition 5.4.1.

The *quadratic form* associated with an $n \times n$ matrix A is the scalar-valued function q_A on $\mathbb{R}^n$ defined by $q_A(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$.

Activity: Linear and quadratic scaling

Let A be an $n \times n$ matrix, let $\mathbf{x} \in \mathbb{R}^n$, and let $t \in \mathbb{R}$.

1. Compare $A(t\mathbf{x})$ with $tA\mathbf{x}$.
2. Compare $q_A(t\mathbf{x})$ with $tq_A(\mathbf{x})$.
3. Which rule is linear in $\mathbf{x}$? Which rule is quadratic?

Learning outcomes. U5-LO4, U1-LO7.

Solution.

Matrix multiplication is linear, so

$$A(t\mathbf{x}) = tA\mathbf{x}.$$

For the quadratic form,

$$\begin{aligned} q_A(t\mathbf{x}) &= (t\mathbf{x})^T A(t\mathbf{x}) \\ &= t^2 \mathbf{x}^T A\mathbf{x} \\ &= t^2 q_A(\mathbf{x}). \end{aligned}$$

Thus $\mathbf{x} \mapsto A\mathbf{x}$ is linear, while $\mathbf{x} \mapsto q_A(\mathbf{x})$ is quadratic.

The remaining results use an orthogonal eigenvector basis to remove cross terms and expose the sign of the quadratic form.

The next definition records the eigenvalue sign patterns that will control the sign of the quadratic form.

Definition 5.4.2 (Positive definite, negative definite, and indefinite matrices).

A symmetric matrix is called *positive definite* if all of its eigenvalues are strictly positive. It is called *negative definite* if all of its eigenvalues are strictly negative. It is called *indefinite* if it has at least one strictly positive eigenvalue and at least one strictly negative eigenvalue.

Activity: A quadratic form as curvature

Let

$$H = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}.$$

1. Compute $\mathbf{h}^T H \mathbf{h}$ for

$$\mathbf{h} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{h} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{h} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

2. Write $\mathbf{h}^T H \mathbf{h}$ for a general vector

$$\mathbf{h} = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}.$$

3. In which coordinate direction is the quadratic form larger?

Learning outcomes. U5-LO4, U5-LO5.

Solution.

We compute

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}^T H \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 4,$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}^T H \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1,$$

and

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}^T H \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 5.$$

For a general vector,

$$\mathbf{h}^T H \mathbf{h} = 4h_1^2 + h_2^2.$$

The quadratic form is larger in the first coordinate direction than in the second coordinate direction. For Hessians, this kind of comparison measures curvature in different directions.

If we can understand quadratic forms, we can understand the quadratic approximations of functions at critical points.

Activity

For a symmetric 3×3 matrix $A = [A_{ij}]$, write out the quadratic form $q_A(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ explicitly in terms of the entries of A and the components of $\mathbf{x} = (x_1, x_2, x_3)$.

Solution.

$$\begin{aligned} q_A(\mathbf{x}) &= \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ &= \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} A_{11}x_1 + A_{12}x_2 + A_{13}x_3 \\ A_{21}x_1 + A_{22}x_2 + A_{23}x_3 \\ A_{31}x_1 + A_{32}x_2 + A_{33}x_3 \end{bmatrix} \\ &= x_1(A_{11}x_1 + A_{12}x_2 + A_{13}x_3) \\ &\quad + x_2(A_{21}x_1 + A_{22}x_2 + A_{23}x_3) \\ &\quad + x_3(A_{31}x_1 + A_{32}x_2 + A_{33}x_3) \\ &= A_{11}x_1^2 + A_{22}x_2^2 + A_{33}x_3^2 \\ &\quad + A_{12}x_1x_2 + A_{13}x_1x_3 + A_{21}x_2x_1 \\ &\quad + A_{23}x_2x_3 + A_{31}x_3x_1 + A_{32}x_3x_2 \end{aligned}$$

Since A is symmetric, we have $A_{ij} = A_{ji}$ for all i, j . Using this property:

$$\begin{aligned} q_A(\mathbf{x}) &= A_{11}x_1^2 + A_{22}x_2^2 + A_{33}x_3^2 + (A_{12} + A_{21})x_1x_2 \\ &\quad + (A_{13} + A_{31})x_1x_3 + (A_{23} + A_{32})x_2x_3 \\ &= A_{11}x_1^2 + A_{22}x_2^2 + A_{33}x_3^2 \\ &\quad + 2A_{12}x_1x_2 + 2A_{13}x_1x_3 + 2A_{23}x_2x_3 \end{aligned}$$

Remark

More generally, for a symmetric $n \times n$ matrix $A = [A_{ij}]$, the quadratic form $q_A(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ can be written as:

$$q_A(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} = \sum_{i=1}^n \sum_{j=1}^n A_{ij}x_i x_j = \sum_{i=1}^n A_{ii}x_i^2 + 2 \sum_{1 \leq i < j \leq n} A_{ij}x_i x_j$$

In particular, note that if A is a diagonal matrix, then

$$q_A(\mathbf{x}) = A_{11}x_1^2 + \cdots + A_{nn}x_n^2$$

which is simple to understand. We will use spectral theory to show that a rotation of a coordinate system reduces a quadratic form to a quadratic form defined by a diagonal matrix. The signs of the diagonal entries will then tell us the behavior of the critical point.

Recall the orthogonal diagonalization theorem, which says that there is an orthogonal matrix Q so that $Q^T A Q$ is a diagonal matrix D .

Theorem 5.4.3 (Diagonalization theorem).

Consider a quadratic form q_A , defined in terms of a symmetric matrix A . Suppose Q is an orthogonal matrix such that

$$Q^T A Q = D = \text{diag}(\lambda_1, \dots, \lambda_n).$$

If we set $\mathbf{x} = Q\mathbf{y}$, then

$$q_A(\mathbf{x}) = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2.$$

Proof.

Since $\mathbf{x} = Q\mathbf{y}$, we have

$$\begin{aligned} q_A(\mathbf{x}) &= \mathbf{x}^T A \mathbf{x} \\ &= (Q\mathbf{y})^T A (Q\mathbf{y}) \\ &= \mathbf{y}^T Q^T A Q \mathbf{y} \\ &= \mathbf{y}^T D \mathbf{y} \\ &= \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2. \end{aligned}$$

Definition 5.4.4 (The principal axes).

If A is a symmetric, $n \times n$ matrix, then an orthonormal basis of eigenvectors of A is called a set of *principal axes* for the quadratic form q_A .

Activity

Consider the quadratic form $q(x, y) = 13x^2 - 6xy + 5y^2$.

Task

Find the symmetric matrix A such that $q = q_A$, i.e., such that

$$q(x, y) = \begin{bmatrix} x & y \end{bmatrix} A \begin{bmatrix} x \\ y \end{bmatrix}$$

.

Solution.

If

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

, then

$$q_A(x, y) = ax^2 + 2bxy + cy^2.$$

So $a = 13$, $b = -3$, and $c = 5$, i.e.,

$$A = \begin{bmatrix} 13 & -3 \\ -3 & 5 \end{bmatrix}$$

.

Task

Find the eigenvalues and a basis of eigenvectors for A .

Solution.

The *characteristic polynomial* of A (recall 5.1.2) is given by

$$\begin{aligned} c_A(\lambda) &= \det(\lambda I_2 - A) \\ &= \det \begin{bmatrix} \lambda - 13 & 3 \\ 3 & \lambda - 5 \end{bmatrix} \\ &= (\lambda - 13)(\lambda - 5) - 9 \\ &= \lambda^2 - 18\lambda + 56 \\ &= (\lambda - 14)(\lambda - 4) \end{aligned}$$

. So $\lambda = 4$ and $\lambda = 14$ are the two eigenvalues of A .

For the eigenvalue $\lambda_1 = 14$, we solve $(A - \lambda_1 I)\mathbf{v}_1 = \mathbf{0}$:

$$\begin{bmatrix} -1 & -3 \\ -3 & -9 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

That is, $v_1 = -3v_2$. Taking $v_2 = 1$, we get $v_1 = -3$. So an eigenvector for λ_1 is

$$\mathbf{v}_1 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

. For the eigenvalue $\lambda_2 = 4$, we solve $(A - \lambda_2 I)\mathbf{v}_2 = \mathbf{0}$:

$$\begin{bmatrix} 9 & -3 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

That is, $v_2 = 3v_1$. Taking $v_1 = 1$, we get $v_2 = 3$. So an eigenvector for λ_2 is

$$\mathbf{v}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

Task

Apply the previous theorem to express $q(x_1, x_2)$ in terms of new coordinates y_1 and y_2 that eliminate the cross term.

Solution.

We first normalize the eigenvectors to get:

$$\begin{aligned} \mathbf{v}'_1 &= \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{\begin{bmatrix} -3 \\ 1 \end{bmatrix}}{\sqrt{(-3)^2 + 1^2}} = \frac{1}{\sqrt{10}} \begin{bmatrix} -3 \\ 1 \end{bmatrix} \\ \mathbf{v}'_2 &= \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \frac{\begin{bmatrix} 1 \\ 3 \end{bmatrix}}{\sqrt{1^2 + 3^2}} = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \end{aligned}$$

The orthogonal matrix Q with these unit eigenvectors as columns diagonalizes A . When we express a vector $\mathbf{x} = Q\mathbf{y} = y_1\mathbf{v}'_1 + y_2\mathbf{v}'_2$, the quadratic form becomes:

$$\begin{aligned} q(\mathbf{x}) &= \mathbf{x}^T A \mathbf{x} = \mathbf{y}^T Q^T A Q \mathbf{y} \\ &= \mathbf{y}^T \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \mathbf{y} \\ &= \lambda_1 y_1^2 + \lambda_2 y_2^2 \\ &= 14y_1^2 + 4y_2^2. \end{aligned}$$

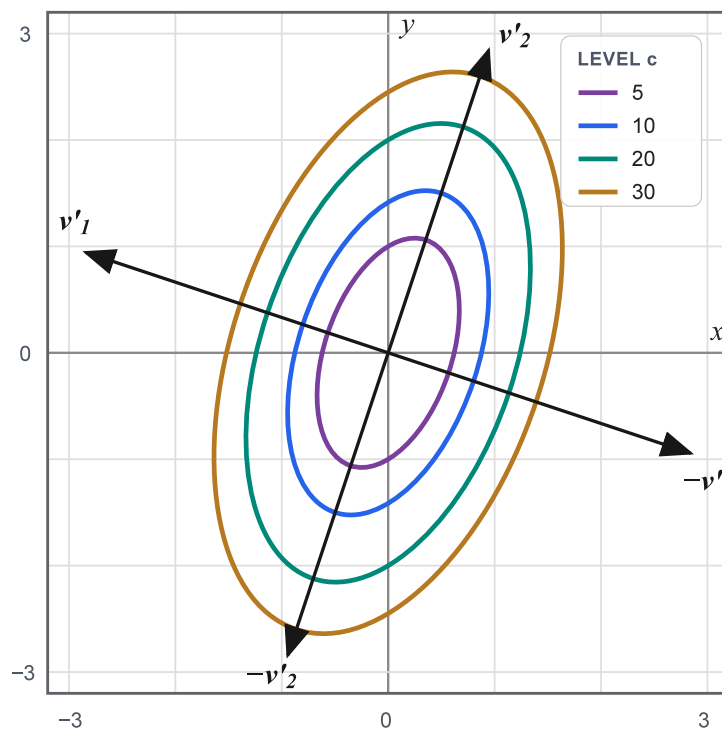


Figure 5.4. Level curves of $q(x, y) = 13x^2 - 6xy + 5y^2$. The principal-axis directions $\mathbf{v}'_1$ and $\mathbf{v}'_2$ are orthonormal eigenvectors of A with eigenvalues 14 and 4, respectively.

Activity

Consider the quadratic form

$$q(x, y) = x^2 + 4xy - 2y^2$$

.

Task

Find the symmetric matrix A such that $q = q_A$.

Solution.

The matrix is

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$$

.

Task

Find the eigenvalues and corresponding eigenvectors of A .

Solution.

The characteristic polynomial is

$$\begin{aligned}\det(A - \lambda I) &= \det \begin{bmatrix} 1 - \lambda & 2 \\ 2 & -2 - \lambda \end{bmatrix} \\ &= (1 - \lambda)(-2 - \lambda) - 4 \\ &= \lambda^2 + \lambda - 6 \\ &= (\lambda - 2)(\lambda + 3).\end{aligned}$$

Thus the eigenvalues are 2 and -3 . For eigenvalue $\lambda_1 = 2$, we solve $(A - \lambda_1 I)\mathbf{v}_1 = \mathbf{0}$:

$$\begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

That is, $v_1 = 2v_2$. Taking $v_2 = 1$, we get $v_1 = 2$. So

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

is an eigenvector for λ_1 . For the eigenvalue $\lambda_2 = -3$, we solve $(A - \lambda_2 I)\mathbf{v}_2 = \mathbf{0}$:

$$\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

That is, $v_2 = -2v_1$. Taking $v_1 = 1$, we get $v_2 = -2$. So an eigenvector for λ_2 is $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$.

Task

Apply the previous theorem to express $q(x_1, x_2)$ in terms of new coordinates y_1 and y_2 that eliminate the cross term, and classify the level curves $q(x_1, x_2) = c$ for different values of c .

Solution.

We first normalize the eigenvectors to get:

$$\begin{aligned}\mathbf{v}'_1 &= \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{\begin{bmatrix} 2 \\ 1 \end{bmatrix}}{\sqrt{2^2 + 1^2}} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ \mathbf{v}'_2 &= \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \frac{\begin{bmatrix} 1 \\ -2 \end{bmatrix}}{\sqrt{1^2 + (-2)^2}} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ -2 \end{bmatrix}\end{aligned}$$

The orthogonal matrix Q with these unit eigenvectors as columns diagonalizes A . When we express a vector $\mathbf{x} = Q\mathbf{y} = y_1\mathbf{v}'_1 + y_2\mathbf{v}'_2$, the quadratic form becomes:

$$\begin{aligned}
 q(\mathbf{x}) &= \mathbf{x}^T A \mathbf{x} = \mathbf{y}^T Q^T A Q \mathbf{y} \\
 &= \mathbf{y}^T \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \mathbf{y} \\
 &= \lambda_1 y_1^2 + \lambda_2 y_2^2 \\
 &= 2y_1^2 - 3y_2^2
 \end{aligned}$$

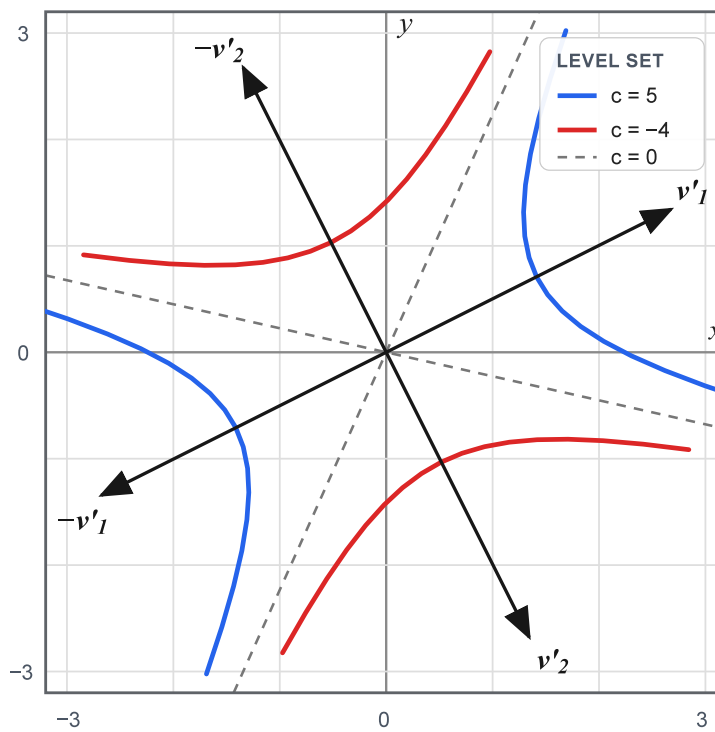


Figure 5.5. Level curves of $q(x, y) = x^2 + 4xy - 2y^2$. In principal-axis coordinates, $q = 2y_1^2 - 3y_2^2$.

Theorem 5.4.5.

A symmetric matrix A is positive definite if and only if $\mathbf{x}^T A \mathbf{x} > 0$ for every vector $\mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$. It is negative definite if and only if $\mathbf{x}^T A \mathbf{x} < 0$ for every vector $\mathbf{x} \neq \mathbf{0}$.

Proof.

By 5.4.3, there is an orthogonal matrix Q such that

$$Q^T A Q = D = \text{diag}(\lambda_1, \dots, \lambda_n)$$

, where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A . Set $\mathbf{y} = Q^T \mathbf{x}$, so $\mathbf{x} = Q\mathbf{y}$. Because Q is invertible, $\mathbf{x} \neq \mathbf{0}$ if and only if $\mathbf{y} \neq \mathbf{0}$. The diagonalization theorem gives

$$\mathbf{x}^T A \mathbf{x} = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2.$$

Suppose first that A is positive definite. Then every λ_i is strictly positive. If $\mathbf{x} \neq \mathbf{0}$, then $\mathbf{y} \neq \mathbf{0}$, so at least one coordinate y_i is nonzero. Therefore $\lambda_1 y_1^2 + \cdots + \lambda_n y_n^2 > 0$, and hence $\mathbf{x}^T A \mathbf{x} > 0$.

Conversely, suppose $\mathbf{x}^T A \mathbf{x} > 0$ for every $\mathbf{x} \neq \mathbf{0}$. For each i , take $\mathbf{y} = \mathbf{e}_i$ and $\mathbf{x} = Q\mathbf{e}_i$. Then $\mathbf{x} \neq \mathbf{0}$ and

$$\mathbf{x}^T A \mathbf{x} = \lambda_i > 0$$

. Thus every eigenvalue of A is strictly positive, so A is positive definite.

The negative-definite case is analogous. If every $\lambda_i < 0$, then $\lambda_1 y_1^2 + \cdots + \lambda_n y_n^2 < 0$ for every $\mathbf{y} \neq \mathbf{0}$, so $\mathbf{x}^T A \mathbf{x} < 0$ for every $\mathbf{x} \neq \mathbf{0}$. Conversely, if this quadratic form is strictly negative for every nonzero $\mathbf{x}$, taking $\mathbf{x} = Q\mathbf{e}_i$ gives $\lambda_i < 0$ for each i . Hence A is negative definite.

Note: Second-order optimization

At a critical point, the linear part of the local approximation disappears. The next term is quadratic:

$$f(\mathbf{a} + \mathbf{h}) \approx f(\mathbf{a}) + \frac{1}{2} \mathbf{h}^T H_f(\mathbf{a}) \mathbf{h}.$$

Thus the second derivative test is a linear algebra test applied to the Hessian matrix.

5.5 Hessians and the second derivative test

In 3.8, critical points are found by solving

$$\nabla f(\mathbf{a}) = \mathbf{0}.$$

This condition removes the linear term from the local approximation, but it does not determine whether $\mathbf{a}$ is a local minimum, a local maximum, or a saddle point. The next term is quadratic. In 5.4, eigenvalues determined the sign of a quadratic form. We now apply that linear algebra to the Hessian.

5.5.1 Local behavior and quadratic approximation

A critical point need not be a local extremum. We first name the third possibility, then review the one-variable Taylor polynomial before writing its multivariable analogue.

Definition 5.5.1.

A critical point of a function which is not a local minimum or local maximum is called a *saddle point*.

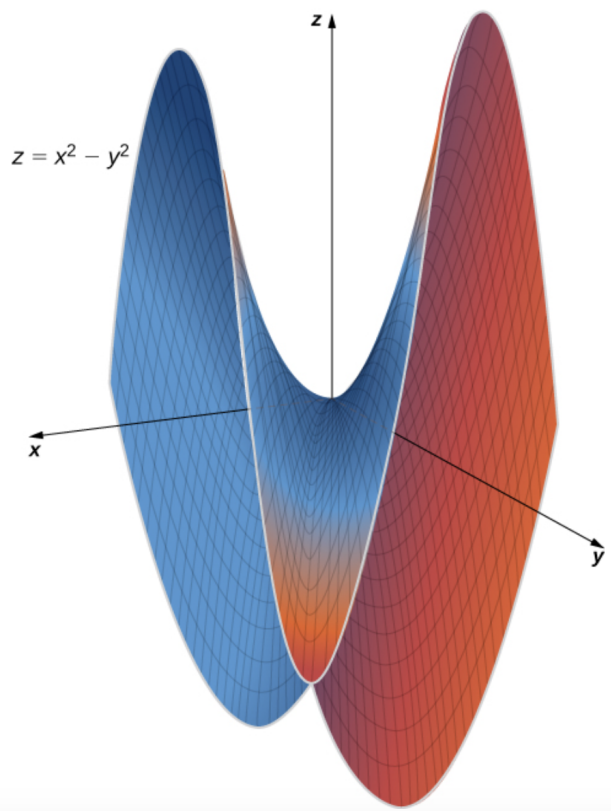


Figure 5.6. A saddle point at the origin of the function $f(x, y) = y^2 - x^2$. Note that along the x -axis ($y = 0$), the function has a global maximum at $x = 0$, while along the y -axis ($x = 0$), the function has a global minimum at $y = 0$. This combination of maximum in one direction and minimum in another is characteristic of saddle points. Figure 4.48 from Edwin “Jed” Herman and Gilbert Strang, *Calculus Volume 3*, OpenStax, © 2018 Rice University, licensed under CC BY-NC-SA 4.0; source: OpenStax Figure 4.48.

The second-order approximation explains this local behavior. We begin with the one-variable case.

For a review, see Section 6.3: Taylor and Maclaurin Series in Strang and Herman’s *Calculus Volume 2*.

Recall that if a single-variable function f has n derivatives at $x = a$, then the n th Taylor polynomial for f at a is the function

$$\begin{aligned} p_n(x) &= f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n \\ &= \sum_{k=0}^n \frac{f^{(k)}(a)}{k!}(x - a)^k \end{aligned}$$

. In particular, for $n = 1$ we obtain the *linear approximation*

$$p_1(x) = f(a) + f'(a)(x - a)$$

of f , and for $n = 2$ we obtain the *quadratic approximation*

$$p_2(x) = f(a) + f'(a)(x - a) + f''(a)\frac{(x - a)^2}{2}$$

. The Taylor polynomial $p_n(x)$ approximates the function $f(x)$ near $x = a$, with the approximation improving as n increases.

Activity

Consider the function $f(x) = \sqrt[3]{x}$.

Find the first and second Taylor polynomials for f at $x = 8$.

Solution.

For $f(x) = \sqrt[3]{x}$, the values of the function and its first two derivatives at $x = 8$ are as follows:

$$\begin{array}{ll} f(x) = \sqrt[3]{x} & f(8) = 2 \\ f'(x) = \frac{1}{3x^{2/3}} & f'(8) = \frac{1}{12} \\ f''(x) = \frac{-2}{9x^{5/3}} & f''(8) = -\frac{1}{144} \end{array}$$

Thus, the first and second Taylor polynomials at $x = 8$ are given by:

$$\begin{aligned} p_1(x) &= f(8) + f'(8)(x - 8) \\ &= 2 + \frac{1}{12}(x - 8) \\ p_2(x) &= f(8) + f'(8)(x - 8) + \frac{f''(8)}{2!}(x - 8)^2 \\ &= 2 + \frac{1}{12}(x - 8) - \frac{1}{288}(x - 8)^2 \end{aligned}$$

Activity: Using Taylor polynomials

Use these two polynomials to estimate $\sqrt[3]{11}$.

Solution.

Using the first Taylor polynomial at $x = 8$, we can estimate:

$$\begin{aligned} \sqrt[3]{11} &\approx p_1(11) = 2 + \frac{1}{12}(11 - 8) \\ &= 2.25 \end{aligned}$$

Using the second Taylor polynomial at $x = 8$, we obtain:

$$\begin{aligned}\sqrt[3]{11} \approx p_2(11) &= 2 + \frac{1}{12}(11 - 8) - \frac{1}{288}(11 - 8)^2 \\ &= 2 + \frac{1}{4} - \frac{1}{32} \\ &= 2.21875\end{aligned}$$

If a is a critical point of f , then the quadratic approximation of f is given by the expression

$$f(a) + f''(a)\frac{(x - a)^2}{2}$$

If $f''(a) > 0$, then the quadratic approximation of f is an upward pointing parabola with a vertex at $(a, f(a))$. This approximation has a minimum at this vertex, explaining why a is a local minimum of the function f (since f is closely approximated by its quadratic approximation). Similarly, if $f''(a) < 0$, then the quadratic approximation is a downward pointing parabola with a maximum at $(a, f(a))$, explaining why a is a local maximum of the function f .

In several variables, the gradient replaces the first derivative and the Hessian replaces the second derivative.

Definition 5.5.2 (Quadratic approximation).

The *quadratic approximation* $Q(\mathbf{x})$ of a scalar-valued function f near a point $\mathbf{a}$ is the function defined by

$$Q(\mathbf{x}) = f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a}) + \frac{1}{2}(\mathbf{x} - \mathbf{a})^T H_f(\mathbf{a})(\mathbf{x} - \mathbf{a})$$

Remark

The quadratic approximation of a function f refines the linear approximation via the gradient that we previously encountered. In particular, if the second derivatives of f are continuous, then

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} \frac{f(\mathbf{x}) - Q(\mathbf{x})}{\|\mathbf{x} - \mathbf{a}\|^2} = 0$$

. If $\mathbf{x}$ is close to $\mathbf{a}$, then $\|\mathbf{x} - \mathbf{a}\|^2$ is much smaller than $\|\mathbf{x} - \mathbf{a}\|$, so that the quadratic approximation $Q(\mathbf{x})$ to f is in general a much better approximation than the linear approximation $L(\mathbf{x})$.

Activity

Compute the gradient and the Hessian of the function $f(x, y) = \ln(xy - 1)$ at the point $(x, y) = (1, 2)$, use these values to determine the quadratic approximation to f at the point $(1, 2)$, and then use the quadratic approximation to estimate the value of f at $(1.2, 1.8)$.

Solution.

We calculate that

$$f_x = \frac{y}{xy-1} \quad \text{and} \quad f_y = \frac{x}{xy-1}$$

. So $(\nabla f)(1, 2) = (2, 1)$. Next, we compute that

$$f_{xx} = \frac{-y^2}{(xy-1)^2} \quad f_{yy} = \frac{-x^2}{(xy-1)^2} \quad f_{xy} = f_{yx} = \frac{-1}{(xy-1)^2}$$

. Evaluating at $(1, 2)$, we find that

$$H_f(1, 2) = \begin{bmatrix} \frac{-4}{1} & \frac{-1}{1} \\ \frac{-1}{1} & \frac{-1}{1} \end{bmatrix} = \begin{bmatrix} -4 & -1 \\ -1 & -1 \end{bmatrix}$$

. Since

$$f(1, 2) = \ln(1) = 0$$

, using the Hessian and gradient, we find that the quadratic approximation of f near $(1, 2)$ is given by

$$\begin{aligned} Q(x, y) &= (2, 1) \cdot (x-1, y-2) + \frac{1}{2} \begin{bmatrix} x-1 & y-2 \end{bmatrix} \begin{bmatrix} -4 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x-1 \\ y-2 \end{bmatrix} \\ &= 2(x-1) + (y-2) - 2(x-1)^2 - (x-1)(y-2) - (1/2)(y-2)^2 \end{aligned}$$

. So

$$\begin{aligned} Q(1.2, 1.8) &= 2(0.2) + (-0.2) - 2(0.2)^2 - (0.2)(-0.2) - (1/2)(-0.2)^2 \\ &= 0.4 - 0.2 - 0.08 + 0.04 - 0.02 \\ &= 0.14 \end{aligned}$$

. The actual value is $f(1.2, 1.8) \approx 0.1484$, while the quadratic approximation is 0.14. Thus the quadratic approximation is quite accurate. For comparison, the linear approximation L at $(1, 2)$ is given by

$$L(x, y) = 2(0.2) + (-0.2) = 0.2$$

, which is far less accurate.

Activity

Let $f(x, y) = \cos(x/y)$. Use the gradient and Hessian to obtain a quadratic approximation of the function $f(x, y)$ near $\mathbf{a} = (0, 1)$.

Solution.

We calculate that

$$\nabla f(x, y) = \left(-\frac{\sin(x/y)}{y}, \frac{x \sin(x/y)}{y^2} \right)$$

In particular, $\nabla f(0, 1) = (0, 0)$. Next, we calculate that

$$\begin{aligned} f_{xx} &= -\frac{1}{y^2} \cos\left(\frac{x}{y}\right), & f_{yy} &= \frac{x}{y^4} \left(2y \sin\left(\frac{x}{y}\right) + x \cos\left(\frac{x}{y}\right) \right) \\ f_{xy} &= f_{yx} = \frac{1}{y^3} \left(y \sin\left(\frac{x}{y}\right) + x \cos\left(\frac{x}{y}\right) \right) \end{aligned}$$

So

$$H_f(0, 1) = \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}$$

Since $f(0, 1) = \cos(0) = 1$, the quadratic approximation of f at $(0, 1)$ is thus given by

$$Q(x, y) = 1 - x^2/2$$

5.5.2 Hessian eigenvalues and the general second derivative test

Suppose $\mathbf{a}$ is a critical point. Then

$$\nabla f(\mathbf{a}) = \mathbf{0},$$

so the quadratic approximation becomes

$$f(\mathbf{a} + \mathbf{h}) \approx f(\mathbf{a}) + \frac{1}{2} \mathbf{h}^T H_f(\mathbf{a}) \mathbf{h}.$$

Thus the local behavior is controlled by the quadratic form associated with the Hessian. Because the Hessian is symmetric, the spectral theorem gives an orthonormal basis of eigenvectors. The signs of the corresponding eigenvalues determine the sign of the quadratic term.

Theorem 5.5.3 (Second derivative test).

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice continuously differentiable near a critical point $\mathbf{a}$, and let

$$H = H_f(\mathbf{a}).$$

1. If every eigenvalue of H is strictly positive, then $\mathbf{a}$ is a strict local minimum.
2. If every eigenvalue of H is strictly negative, then $\mathbf{a}$ is a strict local maximum.

3. If H has at least one strictly positive eigenvalue and at least one strictly negative eigenvalue, then $\mathbf{a}$ is a saddle point.

In all other cases, the test is inconclusive.

Proof.

Since $\mathbf{a}$ is a critical point,

$$f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) = \frac{1}{2}\mathbf{h}^T H \mathbf{h} + r(\mathbf{h}),$$

where

$$\frac{r(\mathbf{h})}{\|\mathbf{h}\|^2} \longrightarrow 0 \quad \text{as } \mathbf{h} \longrightarrow \mathbf{0}.$$

Suppose first that every eigenvalue of H is positive, and let $\mu > 0$ be the smallest eigenvalue. By the quadratic-form diagonalization theorem,

$$\mathbf{h}^T H \mathbf{h} \geq \mu \|\mathbf{h}\|^2.$$

For sufficiently small nonzero $\mathbf{h}$,

$$|r(\mathbf{h})| \leq \frac{\mu}{4} \|\mathbf{h}\|^2.$$

Therefore

$$f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) \geq \frac{\mu}{2} \|\mathbf{h}\|^2 - \frac{\mu}{4} \|\mathbf{h}\|^2 > 0.$$

Hence $\mathbf{a}$ is a strict local minimum. The negative-eigenvalue case follows by applying the same argument to $-f$.

Finally, suppose H has a positive eigenvalue λ_+ with unit eigenvector $\mathbf{v}_+$, and a negative eigenvalue λ_- with unit eigenvector $\mathbf{v}_-$. Along the two lines through $\mathbf{a}$,

$$f(\mathbf{a} + t\mathbf{v}_+) - f(\mathbf{a}) = \frac{1}{2}\lambda_+ t^2 + o(t^2) > 0$$

and

$$f(\mathbf{a} + t\mathbf{v}_-) - f(\mathbf{a}) = \frac{1}{2}\lambda_- t^2 + o(t^2) < 0$$

for all sufficiently small nonzero t . Nearby values therefore occur on both sides of $f(\mathbf{a})$, so $\mathbf{a}$ is a saddle point.

Warning

A zero Hessian eigenvalue does not classify the critical point. It means the quadratic approximation has a flat direction, so higher-order terms may matter.

Activity: Predicting from Hessian eigenvalues

Suppose $\mathbf{a}$ is a critical point of a scalar-valued function f . For each list of eigenvalues of $H_f(\mathbf{a})$, classify the critical point as a local minimum, local maximum, saddle point, or inconclusive.

1. 3, 5
2. $-2, -7$
3. $4, -1$
4. $0, 2$

Learning outcomes. U5-LO5.

Solution.

If the eigenvalues are 3, 5, then the Hessian is positive definite, so the critical point is a local minimum.

If the eigenvalues are $-2, -7$, then the Hessian is negative definite, so the critical point is a local maximum.

If the eigenvalues are $4, -1$, then the Hessian is indefinite, so the critical point is a saddle point.

If the eigenvalues are $0, 2$, the test is inconclusive. The zero eigenvalue means the second-order approximation has a flat direction, so the second derivative test alone does not decide the local behavior.

Activity

Consider the function $f(x, y, z) = x^2 + y^2 - xz$. Find all critical points of f and determine the behavior of f near each critical point.

Task

First find the critical points.

Solution.

We compute that $\nabla f(x, y, z) = (2x - z, 2y, -x)$, which vanishes only at $(0, 0, 0)$, which is the only critical point.

Task

Now classify each critical point.

Solution.

The Hessian is the constant matrix

$$H_f = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

. The associated quadratic form is

$$\mathbf{h}^T H_f \mathbf{h} = 2h_1^2 + 2h_2^2 - 2h_1h_3.$$

This quadratic form takes both positive and negative values. For example,

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}^T H_f \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = 2 > 0,$$

but

$$\begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}^T H_f \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} = 2 - 6 = -4 < 0.$$

Thus H_f is indefinite. By the second derivative test, the critical point $(0, 0, 0)$ is a saddle point.

Activity

Classify the critical points of the function $f(x, y, z) = xy + yz + xz + xyz$ using the second derivative test.

Solution.

We compute that

$$\nabla f(x, y, z) = \begin{bmatrix} y + z + yz \\ x + z + xz \\ x + y + xy \end{bmatrix}$$

. Notice that $\nabla f(x, -1, z) = (-1, x + z + xz, -1)$ is non-vanishing, so that there are no critical points on the plane $y = -1$. Now suppose that $\nabla f(x, y, z) = 0$ and $y \neq -1$. The third entry tells us that $x = -y/(1 + y)$. Substituting this into the second equation gives that $-y/(1 + y) + z - yz/(1 + y) = 0$, i.e., that $z = y$. Substituting this into the first equation gives that $y^2 + 2y = 0$, i.e., that $y = 0$ or $y = -2$. Substituting back gives the two critical points of f , namely $(0, 0, 0)$ and $(-2, -2, -2)$.

Next, we classify these critical points. We calculate that the Hessian of f is

$$H_f(x, y, z) = \begin{bmatrix} 0 & 1+z & 1+y \\ 1+z & 0 & 1+x \\ 1+y & 1+x & 0 \end{bmatrix}$$

. In particular,

$$H_f(0, 0, 0) = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

, which has characteristic polynomial

$$\begin{aligned} \det \begin{bmatrix} \lambda & -1 & -1 \\ -1 & \lambda & -1 \\ -1 & -1 & \lambda \end{bmatrix} &= \lambda(\lambda^2 - 1) - (-1)(-\lambda - 1) + (-1)(1 + \lambda) \\ &= \lambda^3 - 3\lambda - 2 \\ &= (\lambda + 1)(\lambda^2 - \lambda - 2) \\ &= (\lambda + 1)^2(\lambda - 2) \end{aligned}$$

. So $H_f(0, 0, 0)$ has eigenvalues -1 and 2 , and so is indefinite, so that f has a saddle point at $(0, 0, 0)$. We have

$$H_f(-2, -2, -2) = \begin{bmatrix} 0 & -1 & -1 \\ -1 & 0 & -1 \\ -1 & -1 & 0 \end{bmatrix}$$

, which has characteristic polynomial

$$\begin{aligned} \det \begin{bmatrix} \lambda & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{bmatrix} &= \lambda(\lambda^2 - 1) - (\lambda - 1) + (1 - \lambda) \\ &= \lambda^3 - 3\lambda + 2 \\ &= (\lambda - 1)(\lambda^2 + \lambda - 2) \\ &= (\lambda - 1)^2(\lambda + 2) \end{aligned}$$

, so $H_f(-2, -2, -2)$ has eigenvalues 1 and -2 , and is thus an indefinite matrix. So $(-2, -2, -2)$ is also a saddle point of f .

5.5.3 The two-variable second derivative test

For a function of two variables, the Hessian at a critical point has the form

$$H_f(a, b) = \begin{bmatrix} f_{xx}(a, b) & f_{xy}(a, b) \\ f_{xy}(a, b) & f_{yy}(a, b) \end{bmatrix}.$$

Its determinant is

$$D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - f_{xy}(a, b)^2.$$

Since the determinant is the product of the two eigenvalues, D detects whether their signs agree or disagree. The standard two-variable test is therefore a shortcut for the general Hessian test.

Activity: The 2x2 definiteness criterion

Show that a matrix

$$A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$$

is positive definite if and only if $ad - b^2$ is positive and $a > 0$, negative definite if and only if $ad - b^2$ is positive and $a < 0$, and indefinite if and only if $ad - b^2$ is negative. It follows from this that the two-variable test is a special case of the general second derivative test.

Solution.

Let λ_1 and λ_2 be the eigenvalues of A . A direct calculation gives

$$\begin{aligned} c_A(\lambda) &= \det(\lambda I_2 - A) \\ &= (\lambda - a)(\lambda - d) - b^2 \\ &= \lambda^2 - (a + d)\lambda + (ad - b^2). \end{aligned}$$

On the other hand, the characteristic polynomial is monic and has roots λ_1 and λ_2 , so

$$\begin{aligned} c_A(\lambda) &= (\lambda - \lambda_1)(\lambda - \lambda_2) \\ &= \lambda^2 - (\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2. \end{aligned}$$

Comparing coefficients gives

$$a + d = \lambda_1 + \lambda_2, \quad ad - b^2 = \lambda_1\lambda_2.$$

Suppose first that A is positive definite. Then $\lambda_1, \lambda_2 > 0$, so $ad - b^2 = \lambda_1\lambda_2 > 0$. Also, by 5.4.5,

$$a = \mathbf{e}_1^T A \mathbf{e}_1 > 0.$$

Conversely, suppose $ad - b^2 > 0$ and $a > 0$. Then $ad > b^2 \geq 0$, so $d > 0$ as well. Hence

$$\lambda_1 + \lambda_2 = a + d > 0, \quad \lambda_1 \lambda_2 = ad - b^2 > 0.$$

The positive product means that the two eigenvalues have the same sign, and the positive sum means that this sign is positive. Thus both eigenvalues are positive, so A is positive definite.

The negative-definite case is analogous. If A is negative definite, then $\lambda_1, \lambda_2 < 0$, so $ad - b^2 > 0$, and 5.4.5 gives $a = \mathbf{e}_1^T A \mathbf{e}_1 < 0$. Conversely, if $ad - b^2 > 0$ and $a < 0$, then $d < 0$. Thus the eigenvalues have a positive product and a negative sum, so both eigenvalues are negative and A is negative definite.

Finally, A is indefinite if and only if its two eigenvalues have opposite signs. This holds if and only if $\lambda_1 \lambda_2 < 0$, which, by the coefficient comparison above, holds if and only if $ad - b^2 < 0$.

Theorem 5.5.4 (Two-variable second derivative test).

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be twice continuously differentiable near a critical point (a, b) . Define

$$D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - f_{xy}(a, b)^2.$$

- If $D(a, b) > 0$ and $f_{xx}(a, b) > 0$, then (a, b) is a local minimum.
- If $D(a, b) > 0$ and $f_{xx}(a, b) < 0$, then (a, b) is a local maximum.
- If $D(a, b) < 0$, then (a, b) is a saddle point.
- If $D(a, b) = 0$, the test is inconclusive.

Proof.

Apply 5.5.3 to the Hessian matrix $H_f(a, b)$, then apply 5.5.3.

Activity

Use the two-variable second derivative test to determine if the critical point $(1, -2)$ of $f(x, y) = x^2 + y^2 - 2x + 4y$ found in the earlier activity is a local maximum, a local minimum, or a saddle point.

Solution.

We apply the second derivative test. We compute the second-order partial derivatives of f . We calculate that

$$f_{xx}(x, y) = 2 \quad f_{yy} = 2 \quad \text{and} \quad f_{xy}(x, y) = 0$$

. So the discriminant is the constant function

$$D = f_{xx}f_{yy} - f_{xy}^2 = (2)(2) - (0)^2 = 4$$

. Since the discriminant is positive, by the second derivative test, the critical point $(1, -2)$ is a local minimum.

Activity

Use the two-variable second derivative test to analyze the critical point $(0, 0)$ of $f(x, y) = x^2 - y^2$ found in the earlier activity.

Solution.

We need to compute the second-order partial derivatives:

$$f_{xx} = 2 \quad f_{yy} = -2 \quad \text{and} \quad f_{xy} = 0$$

. So the discriminant is also the constant function

$$D = f_{xx}f_{yy} - f_{xy}^2 = (2)(-2) - (0)^2 = -4$$

. Since the discriminant is negative, the second derivative test tells us that $(0, 0)$ is a *saddle point* of f .

Activity

Find all local maxima, local minima, and saddle points for the function

$$f(x, y) = x^2 + xy + y^2 - 6x + 6$$

.

Solution.

We begin by finding the critical points of f . We calculate that

$$f_x(x, y) = 2x + y - 6 \quad \text{and} \quad f_y(x, y) = x + 2y$$

. So the critical points satisfy

$$2x + y - 6 = 0 \quad \text{and} \quad x + 2y = 0.$$

The second equation gives $x = -2y$. Substituting into the first equation gives

$$2(-2y) + y - 6 = 0,$$

so $-3y = 6$, and therefore $y = -2$. Thus $x = 4$, and the only critical point is $(4, -2)$. To classify this critical point, we apply the second derivative test. We calculate that

$$f_{xx}(x, y) = 2, \quad f_{yy}(x, y) = 2, \quad f_{xy}(x, y) = 1$$

. So the discriminant is the constant

$$D = f_{xx}f_{yy} - f_{xy}^2 = (2)(2) - (1)^2 = 3$$

. Since $D > 0$ and $f_{xx} > 0$, the critical point $(4, -2)$ is a local minimum.

Activity

Consider the function

$$f(x, y) = x^2 - 3xy + 3y^2 + 5y + 2x.$$

Find all critical points of f and classify each as a local minimum, a local maximum, or a saddle point.

Solution.

We find the critical points by solving $\nabla f(x, y) = 0$. Since

$$f_x(x, y) = 2x - 3y + 2 \quad \text{and} \quad f_y(x, y) = -3x + 6y + 5,$$

the critical points satisfy

$$2x - 3y + 2 = 0 \quad \text{and} \quad -3x + 6y + 5 = 0.$$

The first equation gives $x = (3y - 2)/2$. Substituting this expression into the second equation gives

$$\begin{aligned} -3\left(\frac{3y - 2}{2}\right) + 6y + 5 &= 0, \\ \frac{3y}{2} &= -8, \\ y &= -\frac{16}{3}. \end{aligned}$$

It follows that

$$x = \frac{3(-16/3) - 2}{2} = -9,$$

so the only critical point is $(-9, -16/3)$.

The Hessian matrix is the constant matrix

$$H_f(x, y) = \begin{bmatrix} 2 & -3 \\ -3 & 6 \end{bmatrix}.$$

Thus the discriminant is

$$D = f_{xx}f_{yy} - f_{xy}^2 = (2)(6) - (-3)^2 = 3 > 0.$$

Since $D > 0$ and $f_{xx} = 2 > 0$, the second derivative test tells us that $(-9, -16/3)$ is a local minimum of f .

In the previous examples, the discriminant was constant because f was a quadratic function. Let us consider an example where the discriminant is nonconstant.

Activity

Find all local maxima, local minima, and saddle points for the function $f(x, y) = x^3 + y^2 + 2xy - 4x - 3y + 5$.

Solution.

We begin by finding the critical point. We calculate that

$$f_x(x, y) = 3x^2 + 2y - 4 \quad \text{and} \quad f_y(x, y) = 2y + 2x - 3$$

. Setting both equal to zero gives the pair of equations

$$3x^2 + 2y - 4 = 0 \quad \text{and} \quad 2y + 2x - 3 = 0$$

. We solve the second equation for y , i.e., that $y = 3/2 - x$, and then substitute this into the first equation, which leads to the equation

$$3x^2 + 2(3/2 - x) - 4 = 0$$

, which simplifies to the quadratic equation

$$3x^2 - 2x - 1 = 0$$

. The quadratic equation tells us that the roots of this equation are given by the formula

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(3)(-1)}}{2(3)} = 1/3 \pm \frac{\sqrt{16}}{6} = 1/3 \pm 2/3$$

. So $x = 1$ or $x = -1/3$. For each such x , setting $y = 3/2 - x$ gives us a critical point for f , so f has two critical points, i.e., $(1, 1/2)$ and $(-1/3, 11/6)$.

To apply the second derivative test to each critical point, we calculate the discriminant of f . We find that

$$f_{xx} = 6x \quad f_{yy} = 2 \quad \text{and} \quad f_{xy} = 2$$

. So the discriminant is the function

$$D(x, y) = f_{xx}(x, y)f_{yy}(x, y) - f_{xy}(x, y)^2 = (6x)(2) - (2)^2 = 12x - 4$$

. So

$$D(1, 1/2) = 12(1) - 4 = 8 \quad \text{and} \quad D(-1/3, 11/6) = 12(-1/3) - 4 = -8$$

. Since $D(1, 1/2)$ is positive, and $f_{xx}(1, 1/2) = 6$ is positive, the critical point $(1, 1/2)$ is a local minimum of the function f . Since $D(-1/3, 11/6)$ is negative, this tells us that $(-1/3, 11/6)$ is a saddle point of the function f .

Remark

The following activity explains why we only look at f_{xx} in the first two cases of the second derivative test.

Activity

True or False: If $D > 0$, then f_{xx} and f_{yy} have the same sign.

Solution.

True. If $D = f_{xx}f_{yy} - (f_{xy})^2 > 0$, then $f_{xx}f_{yy} > (f_{xy})^2 \geq 0$, so $f_{xx}f_{yy} > 0$. Since $f_{xx}f_{yy}$ is positive, this implies either both f_{xx} and f_{yy} are positive, or both f_{xx} and f_{yy} are negative. So f_{xx} and f_{yy} always have the same sign when the discriminant is positive.

5.5.4 Contour plots and saddle points

The Hessian tests classify local behavior algebraically. Contour plots and restrictions to lines show the same behavior geometrically.

Note

A common way to detect a saddle point is to find two directions through the point such that the function increases in one direction and decreases in another. This is not a complete definition of every possible saddle point; it is a useful test in many examples.

For example, the function

$$f(x, y) = x^2 - y^2$$

has a saddle point at $(0, 0)$. Along the line $y = 0$,

$$f(x, 0) = x^2,$$

which has a local minimum at $x = 0$. Along the line $x = 0$,

$$f(0, y) = -y^2,$$

which has a local maximum at $y = 0$. The contour plot below shows the same behavior.

In this example, the surface increases in one coordinate direction and decreases in the other. This is the behavior suggested by the saddle shape.

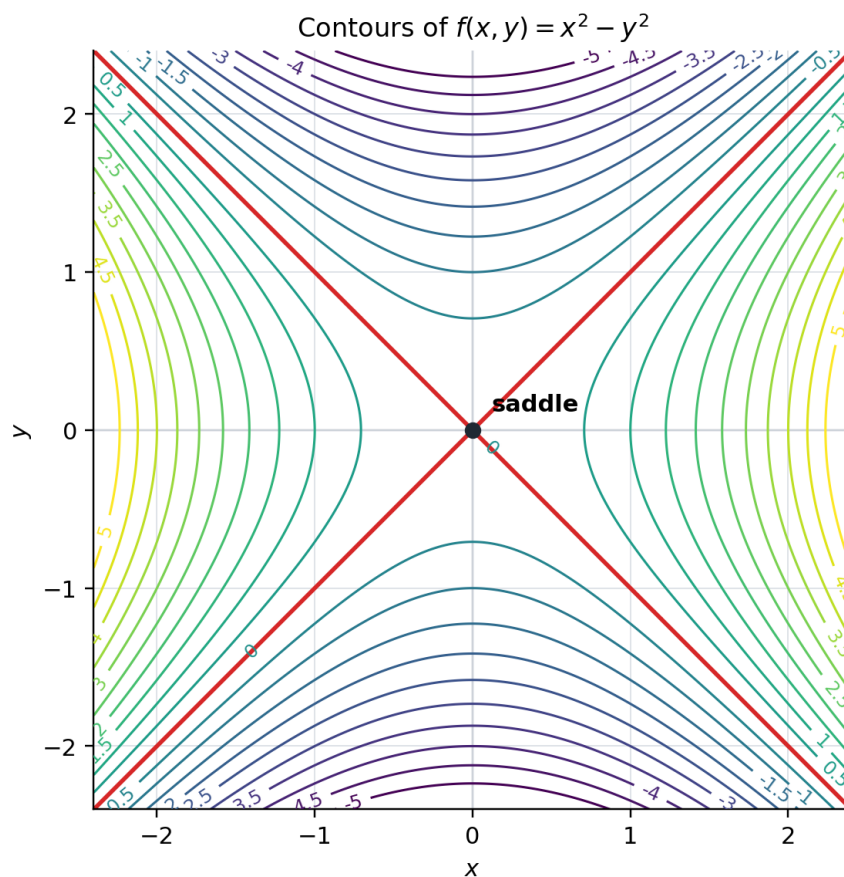


Figure 5.7. A contour plot of $f(x, y) = x^2 - y^2$. Adapted from: Stanford's MATH 51 textbook.

Remark

More formally, if the lines are given in parametric form by

$$\{\mathbf{a} + t\mathbf{d}_1\} \quad \text{and} \quad \{\mathbf{a} + t\mathbf{d}_2\},$$

then the one-variable restrictions

$$g(t) = f(\mathbf{a} + t\mathbf{d}_1) \quad \text{and} \quad h(t) = f(\mathbf{a} + t\mathbf{d}_2)$$

can show opposite local behavior at $t = 0$. In the example above, this verifies that nearby points occur on both sides of $f(\mathbf{a})$.

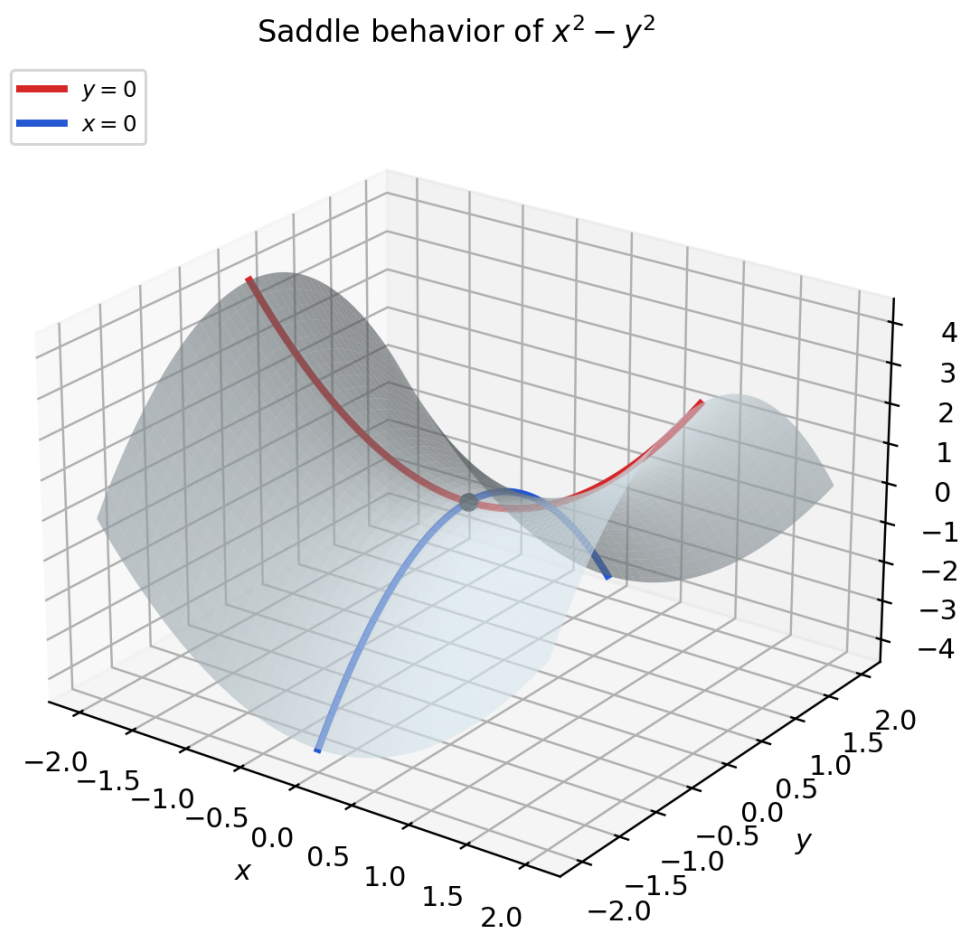


Figure 5.8. At a saddle point for $f(x, y) = x^2 - y^2$, there is a local minimum along $y = 0$ and a local maximum along $x = 0$.

Activity

For each point in the figure below, indicate whether it is a local maximum, a local minimum, or a saddle point.

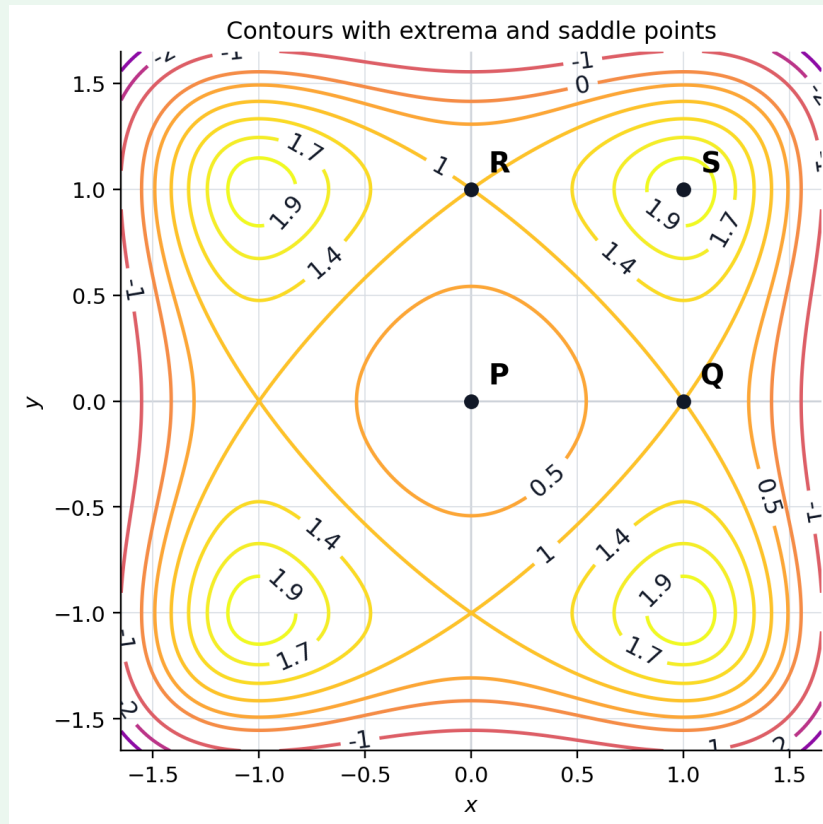


Figure 5.9. A contour plot with a variety of types of critical points.

Solution.

If we look at the collection of nested ovals centered on S , and inspect the numerical values labeling the level sets, we see those numbers are strictly increasing. These ovals seem to be honing in on a point S that is a local maximum. Likewise, the collection of nested ovals near P have function values going in a decreasing direction, suggesting P is a local minimum. We also see saddle points at R and Q . As we move away from R in the north-east direction, the function value increases, whereas moving away in the south-east direction causes the function value to decrease. Near saddle points, the contour plot forms an “X” shape dividing the space into 4 regions. Function values increase in one pair of opposite regions and decrease in the other pair. Near local extrema, level sets form oval shapes resembling ellipses.

5.6 Least squares: three viewpoints

The squared-residual loss

In the Unit 4 least-squares section, least squares was a projection problem. Given a matrix A and a target vector $\mathbf{b}$, a least-squares coefficient vector $\hat{\mathbf{x}}$ makes $A\hat{\mathbf{x}}$ as close as possible to $\mathbf{b}$.

We now study the same problem through the squared-residual loss

$$L(\mathbf{x}) = \|\mathbf{Ax} - \mathbf{b}\|^2.$$

Squaring the residual norm does not change its minimizers, and the squared expression is easier to differentiate. This section connects three viewpoints:

$$\text{residual orthogonality} \longleftrightarrow \text{zero gradient} \longleftrightarrow \text{curvature from } A^T A.$$

Activity: Derivative identities for least squares

Compute the following derivatives.

1. For a fixed vector $\mathbf{d}$, let

$$g(\mathbf{x}) = \mathbf{d} \cdot \mathbf{x}.$$

Compute $\nabla g(\mathbf{x})$.

2. For a fixed $n \times n$ matrix A , let

$$q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}.$$

Compute $\nabla q(\mathbf{x})$.

3. For a fixed $m \times n$ matrix C , let

$$F(\mathbf{x}) = C\mathbf{x}.$$

Compute $J_F(\mathbf{x})$.

Solution.

Expanding the first function gives

$$g(\mathbf{x}) = d_1 x_1 + \cdots + d_n x_n.$$

Therefore

$$\frac{\partial g}{\partial x_i} = d_i, \quad \nabla g(\mathbf{x}) = \mathbf{d}.$$

For the quadratic function,

$$q(\mathbf{x}) = \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j.$$

For $1 \leq k \leq n$,

$$\frac{\partial q}{\partial x_k} = \sum_{j=1}^n A_{kj}x_j + \sum_{i=1}^n A_{ik}x_i.$$

This is the k -th entry of

$$(A + A^T)\mathbf{x}.$$

Hence

$$\nabla q(\mathbf{x}) = (A + A^T)\mathbf{x}.$$

In particular, if A is symmetric, then

$$\nabla q(\mathbf{x}) = 2A\mathbf{x}.$$

Finally, the i -th component of F is

$$F_i(\mathbf{x}) = C_{i1}x_1 + \cdots + C_{in}x_n.$$

Thus

$$\frac{\partial F_i}{\partial x_j} = C_{ij},$$

and therefore

$$J_F(\mathbf{x}) = C.$$

Activity: Gradient zero gives the normal equations

Let A be an $m \times n$ matrix and let $\mathbf{b} \in \mathbb{R}^m$. Define

$$L(\mathbf{x}) = \|A\mathbf{x} - \mathbf{b}\|^2.$$

1. Expand $L(\mathbf{x})$ as a quadratic function of $\mathbf{x}$.
2. Compute $\nabla L(\mathbf{x})$.
3. Show that every critical point satisfies

$$A^T A \mathbf{x} = A^T \mathbf{b}.$$

4. If

$$\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}},$$

show that the normal equations are equivalent to

$$A^T \mathbf{r} = \mathbf{0}.$$

How was this condition interpreted in the Unit 4 normal-equations theorem?

Solution.

We expand

$$\begin{aligned} L(\mathbf{x}) &= (\mathbf{Ax} - \mathbf{b})^T (\mathbf{Ax} - \mathbf{b}) \\ &= \mathbf{x}^T A^T A \mathbf{x} - 2(A^T \mathbf{b})^T \mathbf{x} + \mathbf{b}^T \mathbf{b}. \end{aligned}$$

Since $A^T A$ is symmetric, the preceding activity gives

$$\nabla L(\mathbf{x}) = 2A^T A \mathbf{x} - 2A^T \mathbf{b} = 2A^T (\mathbf{Ax} - \mathbf{b}).$$

Therefore

$$\nabla L(\mathbf{x}) = \mathbf{0}$$

exactly when

$$A^T A \mathbf{x} = A^T \mathbf{b}.$$

These are the normal equations.

If

$$\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}},$$

then

$$A^T \mathbf{r} = A^T \mathbf{b} - A^T A \hat{\mathbf{x}}.$$

Thus the normal equations are equivalent to

$$A^T \mathbf{r} = \mathbf{0}.$$

In Unit 4, this says that the residual is orthogonal to every column of A , or equivalently,

$$\mathbf{r} \perp \text{col}(A).$$

Activity: The same condition in code

Consider the code

```
xhat = np.linalg.lstsq(A, b, rcond=None)[0]
r = b - A @ xhat
A.T @ r
```

1. What does xhat represent?
2. What does r represent?
3. What condition is checked by $A.T @ r$?
4. How do the projection and gradient viewpoints interpret the same output?

Learning outcomes. U5-LO6, U4-LO4.

Solution.

The vector xhat is a least-squares coefficient vector. The vector r is the residual

$$\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}.$$

The expression $A.T @ r$ checks numerically whether

$$A^T \mathbf{r} = \mathbf{0}.$$

Unit 4 interprets this condition as residual orthogonality. In the current section,

$$\nabla L(\hat{\mathbf{x}}) = 2A^T(A\hat{\mathbf{x}} - \mathbf{b}) = -2A^T \mathbf{r},$$

so the same calculation also checks whether the gradient is zero.

Activity: Curvature, global minimality, and uniqueness

Suppose $\hat{\mathbf{x}}$ satisfies the normal equations, and define

$$\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}.$$

Let $\mathbf{z} \in \mathbb{R}^n$.

1. Compute the Hessian $H_L(\mathbf{x})$.
2. Show that

$$\mathbf{z}^T A^T A \mathbf{z} = \|A\mathbf{z}\|^2.$$

What does this say about the curvature of L ?

3. Use $A^T \mathbf{r} = \mathbf{0}$ to prove the exact identity

$$L(\hat{\mathbf{x}} + \mathbf{z}) = L(\hat{\mathbf{x}}) + \|A\mathbf{z}\|^2.$$

4. Explain why $\hat{\mathbf{x}}$ is a global minimizer.
5. Suppose the columns of A are linearly independent. Explain why the minimizer is unique.
6. Suppose $\mathbf{z} \neq \mathbf{0}$ lies in $\text{null}(A)$. Compare

$$A(\hat{\mathbf{x}} + \mathbf{z}) \quad \text{and} \quad A\hat{\mathbf{x}},$$

and compare their loss values.

Solution.

From

$$\nabla L(\mathbf{x}) = 2A^T(A\mathbf{x} - \mathbf{b}),$$

we obtain

$$H_L(\mathbf{x}) = 2A^T A.$$

For every $\mathbf{z}$,

$$\mathbf{z}^T A^T A \mathbf{z} = (A\mathbf{z})^T (A\mathbf{z}) = \|A\mathbf{z}\|^2 \geq 0.$$

Thus the loss has nonnegative quadratic curvature in every direction. A direction is flat exactly when

$$A\mathbf{z} = \mathbf{0}.$$

Since

$$A(\hat{\mathbf{x}} + \mathbf{z}) - \mathbf{b} = A\mathbf{z} - \mathbf{r},$$

we have

$$\begin{aligned} L(\hat{\mathbf{x}} + \mathbf{z}) &= \|A\mathbf{z} - \mathbf{r}\|^2 \\ &= \|A\mathbf{z}\|^2 - 2\mathbf{r}^T A\mathbf{z} + \|\mathbf{r}\|^2. \end{aligned}$$

But

$$\mathbf{r}^T A\mathbf{z} = (A^T \mathbf{r})^T \mathbf{z} = 0,$$

so

$$L(\hat{\mathbf{x}} + \mathbf{z}) = \|\mathbf{r}\|^2 + \|A\mathbf{z}\|^2 = L(\hat{\mathbf{x}}) + \|A\mathbf{z}\|^2.$$

Since $\|A\mathbf{z}\|^2 \geq 0$, no other coefficient vector has smaller loss. Therefore $\hat{\mathbf{x}}$ is a global minimizer. If the columns of A are linearly independent, then

$$A\mathbf{z} = \mathbf{0}$$

implies $\mathbf{z} = \mathbf{0}$. The displayed identity is then strict for every nonzero $\mathbf{z}$, so the minimizer is unique. Equivalently, $A^T A$ is invertible and

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}.$$

If $\mathbf{z} \neq \mathbf{0}$ lies in $\text{null}(A)$, then

$$A(\hat{\mathbf{x}} + \mathbf{z}) = A\hat{\mathbf{x}},$$

and

$$L(\hat{\mathbf{x}} + \mathbf{z}) = L(\hat{\mathbf{x}}).$$

Thus the fitted vector remains the same, but the coefficient vector is not unique.

Warning

The second derivative test is a local test. The exact identity

$$L(\hat{\mathbf{x}} + \mathbf{z}) = L(\hat{\mathbf{x}}) + \|A\mathbf{z}\|^2$$

gives the stronger global conclusion. Linear independence is needed for uniqueness of the coefficient vector, not for the least-squares problem to have a minimizer.

One equation, three viewpoints

- **Viewpoint:** Projection
Key calculation: $A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$
Interpretation: The residual is orthogonal to $\text{col}(A)$.
- **Viewpoint:** Gradient
Key calculation: $\nabla L(\hat{\mathbf{x}}) = \mathbf{0}$
Interpretation: The least-squares coefficient vector is a critical point of the squared-residual loss.
- **Viewpoint:** Curvature
Key calculation: $H_L = 2A^T A$ and $L(\hat{\mathbf{x}} + \mathbf{z}) - L(\hat{\mathbf{x}}) = \|A\mathbf{z}\|^2$
Interpretation: The critical point is a global minimizer; it is unique when $\text{null}(A) = \{\mathbf{0}\}$.

The normal equations are therefore one condition with three readings. Unit 4 supplied the geometry, the gradient gives the critical-point equation, and $A^T A$ describes curvature and uniqueness.

In the fixed-feature section, the columns of a design matrix will be fixed nonlinear features. Training only their coefficient vector will then be exactly this least-squares problem.

5.7 Least squares with fixed nonlinear features

Fixed features and trained coefficients

A model can be nonlinear in its input and still be linear in the parameters being fitted. Let

$$\phi_0, \phi_1, \dots, \phi_p$$

be fixed scalar-valued feature functions. For a coefficient vector

$$\mathbf{c} = \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_p \end{bmatrix},$$

define

$$N_{\mathbf{c}}(t) = c_0\phi_0(t) + c_1\phi_1(t) + \dots + c_p\phi_p(t).$$

Equivalently, if

$$\phi(t) = \begin{bmatrix} \phi_0(t) \\ \phi_1(t) \\ \vdots \\ \phi_p(t) \end{bmatrix},$$

then

$$N_{\mathbf{c}}(t) = \phi(t)^T \mathbf{c}.$$

Given data

$$(t_1, y_1), \dots, (t_m, y_m),$$

form the design matrix

$$A = \begin{bmatrix} \phi_0(t_1) & \phi_1(t_1) & \cdots & \phi_p(t_1) \\ \phi_0(t_2) & \phi_1(t_2) & \cdots & \phi_p(t_2) \\ \vdots & \vdots & & \vdots \\ \phi_0(t_m) & \phi_1(t_m) & \cdots & \phi_p(t_m) \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}.$$

The prediction vector is

$$A\mathbf{c} = \begin{bmatrix} N_{\mathbf{c}}(t_1) \\ N_{\mathbf{c}}(t_2) \\ \vdots \\ N_{\mathbf{c}}(t_m) \end{bmatrix}.$$

With squared-error loss,

$$L(\mathbf{c}) = \sum_{i=1}^m (N_{\mathbf{c}}(t_i) - y_i)^2 = \|A\mathbf{c} - \mathbf{y}\|^2.$$

Thus minimizing L is exactly the least-squares problem from Least squares: three viewpoints.

The feature functions determine the columns of A . They are fixed before the coefficients are fitted. Only $\mathbf{c}$ is trained.

Activity: Fixed nonlinear features, trained coefficients

Let

$$\sigma(t) = \tanh(t),$$

and define the fixed feature functions

$$\phi_0(t) = 1, \quad \phi_1(t) = \sigma(t), \quad \phi_2(t) = \sigma(t-1), \quad \phi_3(t) = \sigma(t+1).$$

The model is

$$N_{\mathbf{c}}(t) = c_0 + c_1\sigma(t) + c_2\sigma(t-1) + c_3\sigma(t+1).$$

For data (t_i, y_i) , define row i of A to be $\phi(t_i)^T$.

1. Write the feature vector $\phi(t)$.
2. Write the design matrix A .
3. Explain why the prediction vector is $A\mathbf{c}$.
4. Write the squared-error loss $L(\mathbf{c})$.
5. Which quantities are fixed, and which vector is trained?
6. Is the model linear as a function of t ?
7. Is the model linear as a function of $\mathbf{c}$?
8. Why is fitting $\mathbf{c}$ a least-squares problem?

Learning outcomes. U5-LO6, U5-LO7.

Solution.

The feature vector is

$$\phi(t) = \begin{bmatrix} 1 \\ \tanh(t) \\ \tanh(t-1) \\ \tanh(t+1) \end{bmatrix}.$$

Therefore

$$A = \begin{bmatrix} 1 & \tanh(t_1) & \tanh(t_1-1) & \tanh(t_1+1) \\ 1 & \tanh(t_2) & \tanh(t_2-1) & \tanh(t_2+1) \\ \vdots & \vdots & \vdots & \vdots \\ 1 & \tanh(t_m) & \tanh(t_m-1) & \tanh(t_m+1) \end{bmatrix}.$$

Row i of A is $\phi(t_i)^T$. Hence the i -th entry of $A\mathbf{c}$ is

$$\phi(t_i)^T \mathbf{c} = N_{\mathbf{c}}(t_i).$$

The squared-error loss is

$$L(\mathbf{c}) = \|A\mathbf{c} - \mathbf{y}\|^2.$$

The inputs t_i , targets y_i , feature functions, and entries of A are fixed. The trained vector is

$$\mathbf{c} = \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix}.$$

The model is nonlinear as a function of t , because the feature functions contain $\tanh$. It is linear as a function of $\mathbf{c}$, because the predictions have the form $A\mathbf{c}$. Therefore fitting $\mathbf{c}$ by squared error is a least-squares problem.

For

$$\mathbf{t} = \begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 4 \\ 1 \\ 0 \\ 1 \\ 4 \end{bmatrix},$$

the design matrix has shape 5×4 . The linked lab constructs this matrix, solves for $\mathbf{c}$, and compares the fitted values with the data.

What “fixed hidden layer” means

In machine-learning language, the feature vector $\phi(t)$ may be produced by a hidden layer. Calling the hidden layer fixed means that its internal weights, shifts, and nonlinear feature functions are

not trained. Training only the final coefficients still gives the model

$$N_{\mathbf{c}}(t) = \phi(t)^T \mathbf{c},$$

so squared-error training is ordinary least squares.

Activity: The same fixed-feature fit in code

```
A = np.column_stack([
    np.ones_like(t),
    np.tanh(t),
    np.tanh(t - 1),
    np.tanh(t + 1)
])

c = np.linalg.lstsq(A, y, rcond=None)[0]
yhat = A @ c
loss = np.linalg.norm(yhat - y)**2
```

1. What feature is stored in each column of A?
2. Which vector is trained?
3. What does yhat contain?
4. What quantity is stored in loss?
5. Why can `np.linalg.lstsq` be used?
6. Is this code fitting the locations of the shifts inside `tanh`?

Learning outcomes. U5-LO6, U5-LO7.

Solution.

The columns of A are the fixed feature vectors

$$\mathbf{1}, \quad \left[\tanh(t_i) \right]_{i=1}^m, \quad \left[\tanh(t_i - 1) \right]_{i=1}^m, \quad \left[\tanh(t_i + 1) \right]_{i=1}^m,$$

where each bracketed vector is evaluated entrywise at the data inputs. The trained vector is $\mathbf{c}$. The vector `yhat` contains the fitted values

$$A\mathbf{c}.$$

The variable `loss` stores

$$\|A\mathbf{c} - \mathbf{y}\|^2.$$

The command `np.linalg.lstsq` applies because A is fixed and the prediction vector is linear in $\mathbf{c}$. The shifts 0, 1, -1 inside the feature functions are not being fitted.

Warning

The least-squares reduction depends on the features being fixed. If the shifts β_j in features such as

$$\tanh(t - \beta_j)$$

are also trained, then the design matrix depends on the parameters:

$$A = A(\beta).$$

The loss

$$\|A(\beta)\mathbf{c} - \mathbf{y}\|^2$$

is no longer a quadratic function of all trained parameters. One least-squares solve does not train the whole model; gradient descent and the chain rule from Unit 3 are then needed.

5.8 Coding recap

5.8.1 Linked notebook

Run Lab U5: Hessian eigenvalues, least squares, and fixed nonlinear features. The opening block reviews gradient descent and a final-layer update from Unit 3. The Unit 5 material interprets Hessian eigenvalues, checks least squares through residual and gradient conditions, distinguishes global minimality from coefficient uniqueness, and fits fixed nonlinear features.

For a quick reference on entrywise functions, design matrices, eigenvalue and least-squares commands, transposes, and numerical checks such as “close to zero”, see the programming appendix sections A.4, A.8, A.10, and A.9.

5.8.2 Review activities

Activity: Reading Hessian eigenvalues

Suppose a critical point has Hessian matrix

```
H = np.array([[4.0, 0.0],
              [0.0, -1.0]])

evals, evecs = np.linalg.eig(H)
evals
```

Output:

```
array([ 4., -1.])
```

1. What matrix is being studied?
2. What do the signs of the eigenvalues say about the quadratic form $\mathbf{h}^T H \mathbf{h}$?
3. If H is the Hessian at a critical point, what does the second derivative test conclude?
4. Why is the phrase “at a critical point” important?

Learning outcomes. U5-LO4, U5-LO5.

Solution.

The matrix being studied is the Hessian matrix. Its eigenvalues have opposite signs, so the quadratic form $\mathbf{h}^T H \mathbf{h}$ takes both positive and negative values. If H is the Hessian at a critical point, the second derivative test classifies the critical point as a saddle point. The phrase “at a critical point” is important because the Hessian test classifies local behavior after the first-order term has disappeared.

Activity: Reading a least-squares residual check

Consider the code:

```
xhat = np.linalg.lstsq(A, b, rcond=None)[0]
r = b - A @ xhat
A.T @ r
```

1. What is `xhat`?
2. What is `r`?
3. What condition is checked by `A.T @ r`?
4. Does `A.T @ r` being “close to zero” mean that `r` is “close to zero”?
5. Which earlier unit used this same condition?

Learning outcomes. U5-LO6, U4-LO4.

Solution.

The vector $\hat{\mathbf{x}}$ is a least-squares coefficient vector. The vector $\mathbf{r}$ is the residual

$$\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}.$$

The expression $A^T @ \mathbf{r}$ checks whether the residual is orthogonal to the columns of A . It can be close to zero even when $\mathbf{r}$ is not close to zero. Unit 4 used this same residual-orthogonality condition to derive the normal equations. Unit 5 obtains the same equations by setting the gradient of $\|A\mathbf{x} - \mathbf{b}\|^2$ equal to zero.

Activity: Reading spectral tools

Consider the code:

```
A = np.array([[2.0, 1.0],
              [1.0, 2.0]])

evals, Q = np.linalg.eigh(A)
D = Q.T @ A @ Q
x = np.array([1.0, -1.0])
q = x @ (A @ x)
```

One possible output is

$$\begin{aligned}\text{evals} &= \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \\ Q &= \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}, \\ D = Q^T A Q &= \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}, \\ q = \mathbf{x}^T A \mathbf{x} &= 2.\end{aligned}$$

1. What do the entries of `evals` and the columns of `Q` represent?
2. Why does `Q.T @ A @ Q` give a diagonal matrix?
3. What does `x @ (A @ x)` compute?
4. Use the eigenvalues to classify the quadratic form $q_A(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$.

Learning outcomes. U5-LO1, U5-LO2, U5-LO3, U5-LO4.

Solution.

The entries of `eigs` are the eigenvalues 1 and 3. The corresponding columns of Q are orthonormal eigenvectors. NumPy may return either sign for either eigenvector, so the signs of the displayed columns may change without changing the mathematics.

Because A is symmetric, its orthonormal eigenvectors form the columns of an orthogonal matrix Q , and $Q^T A Q$ is the diagonal matrix of eigenvalues. The expression `x @ (A @ x)` computes the quadratic form $q_A(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$, which equals 2 for the displayed vector. Both eigenvalues are positive, so q_A is positive definite.

Reopen The same fixed-feature fit in code for the fixed-feature design-matrix calculation.

5.9 Exercises

Learning outcomes. The labels below identify the learning outcomes for each exercise group. Individual problems may also involve earlier outcomes.

5.9.1 Symmetric matrices and orthogonal diagonalization

Learning outcomes. U5-LO3.

- Nicholson 8.2.5(b,d,f)

5.9.2 Hessians and critical-point classification

Learning outcomes. U3-LO9, U5-LO5.

- OpenStax 4.7 #319
- OpenStax 4.7 #323

5.9.3 Additional exercises

These exercises connect eigenvectors, quadratic forms, Hessian eigenvalues, least squares, residual checks, fixed nonlinear features, and code interpretation. Solutions are collected in Appendix A.5.

Additional exercise: Eigenvectors as special directions

Let

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}, \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

1. Compute $A\mathbf{v}_1$.
2. Compute $A\mathbf{v}_2$.

3. Identify the eigenvalue for each eigenvector.
4. Explain what the matrix map does to these two special directions.

Learning outcomes. U5-LO1.

Additional exercise: Diagonalization and matrix powers

Let

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}, \quad P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

1. Verify that the columns of P are eigenvectors of A , and identify their eigenvalues.
2. Compute P^{-1} and verify that $A = PDP^{-1}$.
3. Use $A^k = PD^kP^{-1}$ to derive a formula for A^k .
4. Compute A^4 .
5. Explain why diagonalization makes repeated matrix actions easier to compute.

Learning outcomes. U5-LO1, U5-LO2.

Additional exercise: Quadratic form and definiteness

Let

$$H = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}.$$

1. Compute $\mathbf{h}^T H \mathbf{h}$ for

$$\mathbf{h} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{h} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{h} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

2. Write $\mathbf{h}^T H \mathbf{h}$ for a general vector

$$\mathbf{h} = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}.$$

3. Is H positive definite, negative definite, or indefinite?
4. If H is the Hessian at a critical point, what does the second derivative test conclude?

Learning outcomes. U5-LO4, U5-LO5.

Additional exercise: Hessian eigenvalue classification

Suppose $\mathbf{a}$ is a critical point of a scalar-valued function f . For each possible list of Hessian eigenvalues, classify the critical point as a local minimum, local maximum, saddle point, or inconclusive.

1. 3, 5
2. $-2, -7$
3. $4, -1$
4. $0, 2$
5. Explain why the zero eigenvalue case is different.

Learning outcomes. U5-LO5.

Additional exercise: Least squares from gradients

Let

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

Define

$$h(\mathbf{x}) = \|\mathbf{Ax} - \mathbf{b}\|^2.$$

1. Compute $A^T A$ and $A^T \mathbf{b}$.
2. Use $\nabla h(\mathbf{x}) = 2A^T(\mathbf{Ax} - \mathbf{b})$ to write the normal equations.
3. Solve the normal equations.
4. Compute the residual $\mathbf{r} = \mathbf{b} - \mathbf{Ax}$.
5. Check that $A^T \mathbf{r} = \mathbf{0}$.

Learning outcomes. U5-LO6, U4-LO4.

Additional exercise: Residual orthogonality in code

Consider the code:

```
xhat = np.linalg.lstsq(A, b, rcond=None)[0]
r = b - A @ xhat
```

A.T @ r

1. What is $\hat{x}$?
2. What is r ?
3. What condition is checked by A.T @ r?
4. Does A.T @ r being close to zero mean that r is close to zero?
5. Which earlier unit used this same condition?

Learning outcomes. U5-LO6, U4-LO4.

Additional exercise: Fixed-feature design matrix

Let $\sigma(t) = \tanh(t)$, and define

$$N_{\mathbf{c}}(t) = c_0 + c_1\sigma(t) + c_2\sigma(t-1) + c_3\sigma(t+1).$$

Use the input values $t = -1, 0, 1$.

1. Write the design matrix A for the model $A\mathbf{c} \approx \mathbf{y}$.
2. Which vector is trained?
3. Why is this a least-squares problem?
4. Is the model linear as a function of t ?
5. Is the model linear as a function of $\mathbf{c}$?

Learning outcomes. U5-LO6, U5-LO7.

5.10 Unit 5 highlights

5.10.1 Mathematical quick reference

Eigenvalues and eigenspaces

Eigenpair. For a square matrix A , an eigenvalue λ and eigenvector $\mathbf{v} \neq \mathbf{0}$ satisfy

$$A\mathbf{v} = \lambda\mathbf{v}.$$

The eigenspace is $E_\lambda = \text{null}(A - \lambda I)$; it includes the zero vector, although an eigenvector cannot be zero.

Calculation. Solve $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$ for each eigenvalue. Eigenvectors for distinct eigenvalues are independent.

Multiplicities. Algebraic multiplicity is multiplicity as a root of the characteristic polynomial; geometric multiplicity is $\dim E_\lambda$, and is at least 1 and at most the algebraic multiplicity.

Diagonalization

Eigenvector basis. A real $n \times n$ matrix has a real diagonalization exactly when it has a basis of real eigenvectors. With eigenvectors as columns of P and their eigenvalues in the same order on D ,

$$AP = PD, \quad A = PDP^{-1}, \quad A^k = PD^kP^{-1} \quad (k = 0, 1, 2, \dots).$$

In particular, n distinct real eigenvalues suffice. Repeated eigenvalues require checking eigenspace dimensions.

Spectral theorem. A real symmetric matrix has an orthonormal eigenvector basis:

$$A = QDQ^T, \quad Q^TQ = I.$$

Find orthonormal bases in the eigenspaces and assemble Q ; eigenvectors for distinct eigenvalues are automatically orthogonal.

Quadratic forms and definiteness

Quadratic form. $q_A(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$. For symmetric $A = QDQ^T$, use principal coordinates $\mathbf{y} = Q^T \mathbf{x}$:

$$q_A(\mathbf{x}) = \mathbf{y}^T D \mathbf{y} = \sum_{i=1}^n \lambda_i y_i^2.$$

Sign tests for symmetric matrices. Positive definite means every eigenvalue is positive, equivalently $q_A(\mathbf{x}) > 0$ for all $\mathbf{x} \neq \mathbf{0}$. Negative definite means every eigenvalue is negative, equivalently $q_A(\mathbf{x}) < 0$ for all nonzero $\mathbf{x}$. Indefinite means at least one positive and one negative eigenvalue, so the quadratic form takes both signs.

Two-by-two shortcut. For $A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$ and $\Delta = ad - b^2$: positive definite if $\Delta > 0$ and $a > 0$; negative definite if $\Delta > 0$ and $a < 0$; indefinite if $\Delta < 0$.

Hessians and the second derivative test

Quadratic local approximation. If f has continuous second partial derivatives near $\mathbf{a}$, then

$$f(\mathbf{a} + \mathbf{h}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a})^T \mathbf{h} + \frac{1}{2} \mathbf{h}^T H_f(\mathbf{a}) \mathbf{h}.$$

Classify an interior critical point. First check $\nabla f(\mathbf{a}) = \mathbf{0}$. For a function with continuous second partials near that point, all positive Hessian eigenvalues give a strict local minimum; all negative give a strict local maximum; a positive and a negative eigenvalue give a saddle point. All remaining sign patterns are inconclusive.

Two-variable test. At a critical point (a, b) , set

$$D = f_{xx}(a, b)f_{yy}(a, b) - f_{xy}(a, b)^2.$$

If $D > 0$, use the sign of $f_{xx}(a, b)$: positive gives a strict local minimum, negative a strict local maximum. If $D < 0$, there is a saddle point. If $D = 0$, the test is inconclusive.

Least squares: projection, gradient, and curvature

Squared-error function. For $L(\mathbf{x}) = \|\mathbf{Ax} - \mathbf{b}\|^2$,

$$\nabla L(\mathbf{x}) = 2A^T(\mathbf{Ax} - \mathbf{b}), \quad H_L = 2A^T A, \quad \mathbf{z}^T H_L \mathbf{z} = 2\|A\mathbf{z}\|^2 \geq 0.$$

Minimizer conditions. With $\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$,

$$\nabla L(\hat{\mathbf{x}}) = \mathbf{0} \iff A^T \mathbf{r} = \mathbf{0} \iff A^T A \hat{\mathbf{x}} = A^T \mathbf{b}.$$

Every solution of these equations is a global minimizer, since

$$L(\hat{\mathbf{x}} + \mathbf{z}) = L(\hat{\mathbf{x}}) + \|A\mathbf{z}\|^2.$$

Uniqueness. The minimizer is unique exactly when $\text{null}(A) = \{\mathbf{0}\}$. Otherwise the complete minimizer set is $\hat{\mathbf{x}} + \text{null}(A)$. Singularity of the Hessian does not prevent global minimality here.

Fixed functions, variable coefficients. If $q(t) = \sum_j c_j \phi_j(t)$ and the functions ϕ_j are fixed, set $A_{ij} = \phi_j(t_i)$. Minimizing $\sum_i (q(t_i) - b_i)^2$ is the same least-squares problem in $\mathbf{c}$, even when the ϕ_j are nonlinear functions of t .

5.10.2 Common mistakes

Using the zero vector as an eigenvector; assuming every square matrix is diagonalizable; ordering the eigenvalues differently from the columns of P ; replacing P^{-1} by P^T without orthonormal columns; judging definiteness only from diagonal entries; applying the Hessian test away from a critical point; or treating a zero eigenvalue as conclusive. Mixed positive and negative eigenvalues still imply a saddle even if other eigenvalues are zero.

Confusing a local extremum with a global one; assuming residual orthogonality means a zero residual; assuming least-squares coefficients are unique without a rank check; or overlooking linearity in the coefficients when the fixed functions are nonlinear.

5.10.3 Connections

Unit 1 supplies matrix maps, products, and transposes. Unit 2 supplies eigenspace calculations through null spaces and explains nonuniqueness through rank. Unit 3 supplies gradients, Hessians, and critical points. Unit 4 supplies projection and residual orthogonality. Unit 6 relates constrained quadratic-form extrema to eigenvectors, principal directions, and singular values.

Chapter 6

Constrained optimization and Lagrange multipliers

Recall from 3.8.1 that an optimization problem has an objective function and a feasible region. In a constrained optimization problem, one or more equations or inequalities restrict the feasible region.

The section “Optimization problems and critical points” found interior critical points by solving

$$\nabla f(\mathbf{x}) = \mathbf{0}.$$

On a closed bounded region, absolute extrema may also occur on the boundary. Lagrange multipliers give equations for critical points on boundaries and feasible sets described by one or more equality constraints.

The same method connects constrained optimization to linear algebra. On the unit sphere, the extrema of a symmetric quadratic form occur in eigenvector directions. This leads to the section “Principal component analysis” for centered data and the section “Singular value decomposition” for general matrix maps.

Big questions. How do we find absolute extrema when the allowable inputs are restricted? Why do eigenvectors identify extremal directions of quadratic forms, data sets, and matrix maps?

Learning outcomes. By the end of this unit, students should be able to:

- *U6-LO1.* Find absolute extrema on closed bounded regions by comparing interior and boundary candidates.
- *U6-LO2.* Formulate and solve Lagrange-multiplier problems with one equality constraint, and explain the parallel-gradient and Lagrangian formulations.
- *U6-LO3.* Formulate and solve Lagrange-multiplier problems with several equality constraints, and interpret the objective gradient as a linear combination of constraint gradients.
- *U6-LO4.* Use Lagrange multipliers and eigenvalues to find extrema of symmetric quadratic forms on the unit sphere, and apply this result to maximum stretch and singular values.
- *U6-LO5.* Center a small data set, form its covariance matrix, compute and interpret first and second principal directions, scores, reconstructions, and captured variation, and distinguish PCA from linear regression.

- U6-LO6. Construct and interpret a small singular value decomposition

$$A = U\Sigma V^T,$$

including its orthonormal input and output directions and singular-value stretch factors.

- U6-LO7. Use an SVD to identify rank and orthonormal bases for the four fundamental subspaces, and connect the SVD of centered data with PCA directions, scores, and reconstructions.

Toolbox skills. Compare interior and boundary candidates; restrict a function to simple boundary pieces; form and solve one- and several-constraint Lagrange systems; write and differentiate Lagrangians; evaluate and diagonalize symmetric quadratic forms; center data and form covariance matrices; compute principal scores, projections, and reconstruction residuals; form $A^T A$; compute singular values and singular vectors; assemble and read small SVDs; and identify rank and bases for the four fundamental subspaces.

6.1 Absolute extrema on closed bounded regions

Recall from 3.2.6 that a point $\mathbf{x}_0 \in \mathbb{R}^n$ is a *boundary point* of a set S if every ball centered at $\mathbf{x}_0$ contains at least one point of S and at least one point outside S . The *boundary* of S is the set of all its boundary points, and 3.2.7 says that S is *closed* if it contains its boundary.

Activity

Determine whether each of the given regions is closed.

Task

The region $A = \{(x, y) : x^2 + y^2 < 1\}$.

Solution.

The set A is the interior of the unit disk, which doesn't include any of the points on the circle $x^2 + y^2 = 1$, which are the boundary points of the set.

Task

The region $B = \{(x, y) : x^2 + y^2 = 1\}$

Solution.

The set B is closed, because the set of boundary points of B is equal to B itself.

Task

$C = \{(x, y) : 1 \leq x \leq 2, 1 \leq y \leq 2\}$

Solution.

The set C is closed. The boundary points are the set of points (x, y) such that either

- $x = 1$ and $1 \leq y \leq 2$.

- $x = 2$ and $1 \leq y \leq 2$.
- $1 \leq x \leq 2$ and $y = 1$.
- $1 \leq x \leq 2$ and $y = 2$.

The set C contains all such points, and so is closed.

Task

The set $D = \{(x, y) : x > 1\}$

Solution.

The set D is not closed. The set of boundary points of D is the set $\{(x, y) : x = 1\}$, which is not contained in D .

Theorem 6.1.1.

If f is a scalar-valued function which is continuous on a closed and bounded set D , then f has an absolute maximum and an absolute minimum value on D .

The second derivative test informs us whether a critical point is a local maximum, a local minimum, or a saddle point. However, it doesn't identify global (absolute) extrema. For functions defined on closed, bounded regions, we can use the following procedure to find absolute extrema, provided that the boundary points of the region are of a suitably simple nature.

Algorithm

Given a function f , continuous on a closed bounded set D , this algorithm finds the absolute extrema of f on D .

1. Find all critical points of f in the interior of D .
2. Find all boundary candidates. On each boundary piece, reduce to a one-variable problem when possible, and include endpoints and corners.
3. Evaluate f at every candidate from the first two steps.
4. The largest value is the absolute maximum value, and the smallest value is the absolute minimum value.

If the boundary of D is made from one or more curves, we can usually parameterize each curve and find boundary candidates using one-variable calculus. Endpoints and corners must still be checked separately.

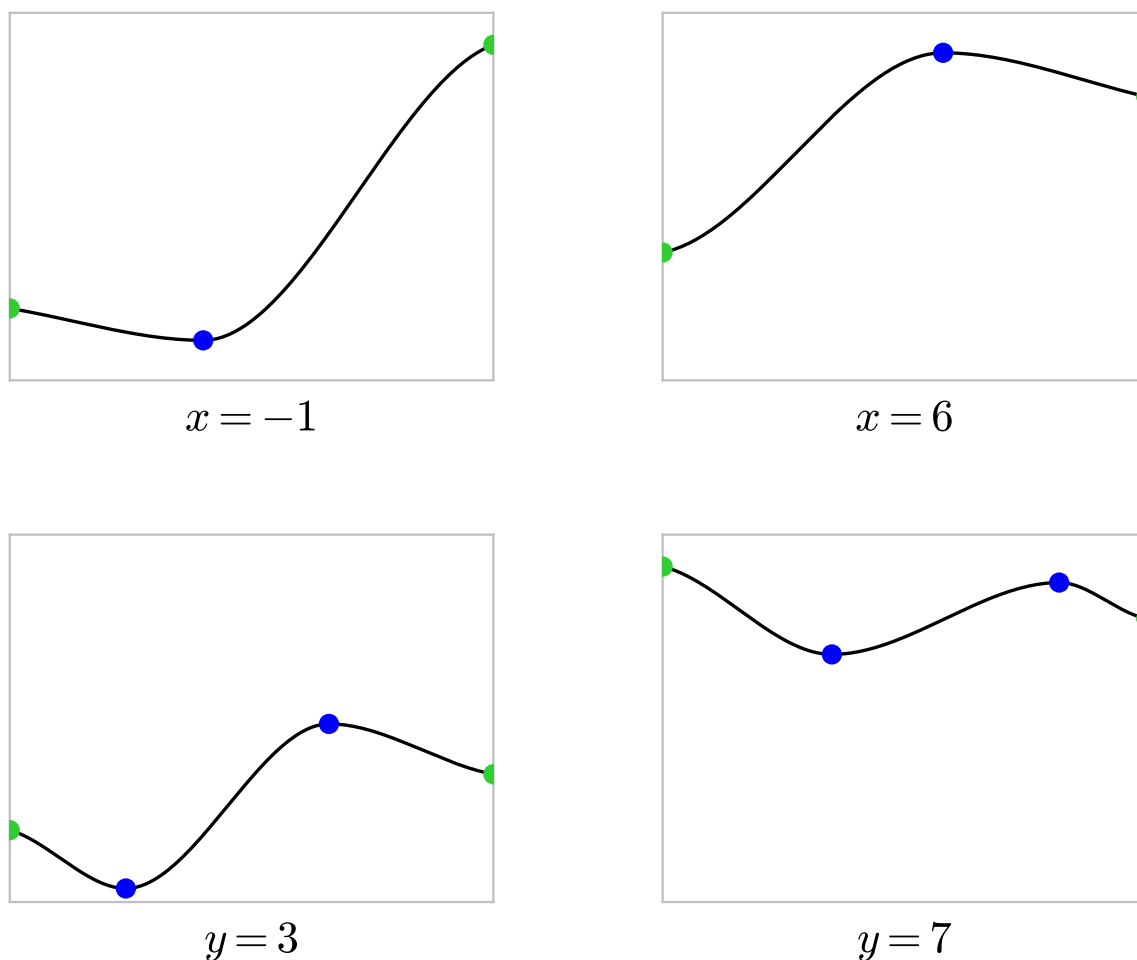


Figure 6.1. A schematic for finding absolute extrema on a rectangular domain. The top panel shows a surface over a rectangle with interior candidates in red, boundary candidates in blue, and corners in green. The lower four panels show the one-variable boundary slices, with endpoints in green and boundary critical points in blue. Adapted from: Stanford's MATH 51 textbook.

Activity

Consider the function

$$f(x, y) = 3x^2 - 2y^2 + 2x + 2y - 1$$

on the unit square

$$D = \{(x, y) \in \mathbb{R}^2 : -1 \leq x \leq 1, -1 \leq y \leq 1\}.$$

Find the absolute maximum and minimum values of f on D .

Solution.

The set D is closed and bounded, so we can apply 6.1.

We start by finding the critical points of f in the interior of D , i.e., on the set

$$\text{int}(D) = \{(x, y) \in \mathbb{R}^2 : -1 < x < 1, -1 < y < 1\}$$

. We compute that

$$f_x = 6x + 2 \quad \text{and} \quad f_y = -4y + 2$$

. These two quantities vanish simultaneously at the point $(-1/3, 1/2)$, which is therefore the only critical point of f . At this point

$$\begin{aligned} f(-1/3, 1/2) &= 3(-1/3)^2 - 2(1/2)^2 + 2(-1/3) + 2(1/2) - 1 \\ &= -5/6 \end{aligned}$$

. Now we check the boundary of D . The boundary is the union of four curves, and so we can parameterize these curves, and find the critical points of the parameterization, to determine potential absolute minima and maxima on these curves (treating the corners of D separately). On the bottom edge $y = -1$, the function takes the form

$$f(x, -1) = 3x^2 - 2(-1)^2 + 2x + 2(-1) - 1 = 3x^2 + 2x - 5$$

. The only critical point of the expression $3x^2 + 2x - 5$ occurs at $x = -1/3$, and

$$f(-1/3, -1) = 3(-1/3)^2 + 2(-1/3) - 5 = -16/3$$

. Similarly, on the top edge $y = 1$, the function takes the form

$$f(x, 1) = 3x^2 - 2(1)^2 + 2x + 2(1) - 1 = 3x^2 + 2x - 1$$

. As for the bottom edge, the only critical point of the expression $3x^2 + 2x - 1$ occurs at $x = -1/3$, and

$$f(-1/3, 1) = 3(-1/3)^2 + 2(-1/3) - 1 = -4/3$$

. On the left edge $x = -1$, the function takes the form

$$f(-1, y) = 3(-1)^2 - 2y^2 + 2(-1) + 2y - 1 = -2y^2 + 2y$$

. The only critical point of the expression $-2y^2 + 2y$ occurs when $y = 1/2$, and

$$f(-1, 1/2) = 2(1/2) - 2(1/2)^2 = 1/2$$

. Similarly, on the right edge $x = 1$, the function takes the form

$$f(1, y) = 3(1)^2 - 2y^2 + 2(1) + 2y - 1 = -2y^2 + 2y + 4$$

. The only critical point of this expression occurs when $y = 1/2$, and

$$f(1, 1/2) = -2(1/2)^2 + 2(1/2) + 4 = 9/2$$

. Among the boundary critical points, the smallest value occurs at $(-1/3, -1)$, where f has value $-16/3$, and the largest value occurs at $(1, 1/2)$, where f has value $9/2$.

Finally, we must check the value of the function at the four corners $(-1, -1)$, $(-1, 1)$, $(1, -1)$, and $(1, 1)$. We calculate that $f(-1, -1) = -4$, $f(-1, 1) = 0$, $f(1, -1) = 0$, and $f(1, 1) = 4$.

Comparing all values, we see that the absolute maximum of f on D occurs at $(1, 1/2)$, with maximum value $9/2$, and the absolute minimum occurs at $(-1/3, -1)$, with minimum value $-16/3$.

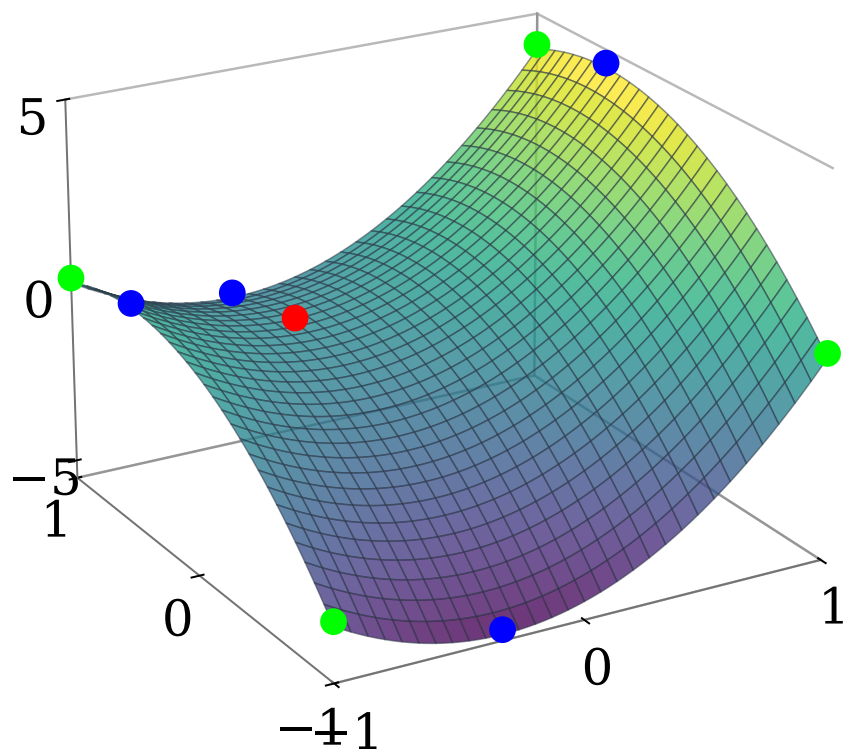


Figure 6.2. Critical points on D for $f(x, y) = 3x^2 - 2y^2 + 2x + 2y - 1$: interior point (red), edge points (blue), corners (green). Adapted from: Stanford's MATH 51 textbook.

Activity

In this exercise, we compute the absolute extrema of the function $f(x, y) = 2x - 2xy + y^2$ for $0 \leq x \leq 4$ and $0 \leq y \leq 3$.

Task

Start by finding all critical points on the interior of the domain upon which we are trying to optimize f .

Solution.

The interior region is defined by the equations $0 < x < 4$ and $0 < y < 3$. We calculate that

$$f_x = 2 - 2y \quad \text{and} \quad f_y = -2x + 2y$$

, and so the only critical point occurs at the point $(1, 1)$. Here we have $f(1, 1) = 2(1) - 2(1)(1) + (1)^2 = 1$.

Task

Next, find the maximum and minimum values on the interior of the boundary curves of the region, and on the corners of the region.

Solution.

The boundary curves consist of the following four regions:

- Bottom edge: $y = 0, 0 \leq x \leq 4$
- Right edge: $x = 4, 0 \leq y \leq 3$
- Top edge: $y = 3, 0 \leq x \leq 4$
- Left edge: $x = 0, 0 \leq y \leq 3$

We begin by computing critical points on the interior of the curves, once parameterized.

On the bottom edge $y = 0$, we have $f(x, y) = 2x$, an expression with no critical points. On the top edge $y = 3$, we have $f(x, 3) = 2x - 2x(3) + 3^2 = 9 - 4x$, which also has no critical points. On the left edge $x = 0$, we have $f(0, y) = y^2$, which has a critical point at $y = 0$, but this is not in the interior of the curve. On the right edge $x = 4$, we have $f(4, y) = 8 - 8y + y^2$, which has a critical point when $y = 4$, which is not in the region D . So there are no critical points on the interior of the boundary curves.

Nonetheless, we must also compute the values of the function directly on the four corners. We have $f(0, 0) = 0$, $f(0, 3) = 9$, $f(4, 0) = 8$, and $f(4, 3) = 8 - 24 + 9 = -7$.

Task

Finally, compare all values.

Solution.

Comparing all values of the possible minima and maxima we have collected, we see that the absolute maximum of the function f on D occurs at $(0, 3)$, where $f(0, 3) = 9$, and the absolute minimum of the function occurs at $(4, 3)$, where $f(4, 3) = -7$.

On a rectangle, each boundary piece is given by a simple equation such as $x = a$ or $y = b$, so Step 2 of 6.1 reduces to one-variable calculus. A curved boundary may instead be given implicitly

by an equation

$$g(\mathbf{x}) = k.$$

Parameterizing such a boundary may be inconvenient. Lagrange multipliers give a direct way to find candidates on it.

More generally, many optimization problems include a constraint from the beginning, such as a fixed budget, total, area, or vector length. Thus Lagrange multipliers are both a boundary method for closed regions and a method for naturally constrained problems.

6.2 Lagrange multipliers

Recall from 3.8.1 that an optimization problem has an objective function and a feasible region. A constrained optimization problem restricts the feasible region. We begin with one equality constraint.

Definition 6.2.1 (Constrained optimization with one equality constraint).

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be an objective function, let $g : \mathbb{R}^n \rightarrow \mathbb{R}$, and let $k \in \mathbb{R}$. To maximize or minimize $f(\mathbf{x})$ subject to

$$g(\mathbf{x}) = k$$

means to optimize f on the feasible region

$$S = \{\mathbf{x} \in \mathbb{R}^n : g(\mathbf{x}) = k\}.$$

The equation $g(\mathbf{x}) = k$ is the constraint. A point of S is feasible.

Suppose $\mathbf{a} \in S$ is a local extremum of f on S . At $\mathbf{a}$, allowed motion follows tangent directions to S . The first-order change of f must vanish along those directions, so $\nabla f(\mathbf{a})$ is normal to S . The vector $\nabla g(\mathbf{a})$ is also normal to the level set $g(\mathbf{x}) = k$. If

$$\nabla g(\mathbf{a}) \neq \mathbf{0},$$

this suggests that the two gradients are parallel.

Definition 6.2.2 (Constrained critical point).

Let $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable, and let

$$S = \{\mathbf{x} \in \mathbb{R}^n : g(\mathbf{x}) = k\}.$$

Assume

$$\nabla g(\mathbf{x}) \neq \mathbf{0}$$

for every $\mathbf{x} \in S$. A point $\mathbf{a} \in S$ in the interior of the domain of f is a constrained critical point of f on S if there exists $\lambda \in \mathbb{R}$ such that

$$\nabla f(\mathbf{a}) = \lambda \nabla g(\mathbf{a}).$$

The scalar λ is a Lagrange multiplier.

Lagrangian formulation

The same equations can be packaged in one function. Define the Lagrangian

$$L(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda(g(\mathbf{x}) - k).$$

Then

$$\nabla_{\mathbf{x}} L(\mathbf{x}, \lambda) = \nabla f(\mathbf{x}) - \lambda \nabla g(\mathbf{x}),$$

and

$$\frac{\partial L}{\partial \lambda}(\mathbf{x}, \lambda) = -(g(\mathbf{x}) - k).$$

Therefore the equations

$$\nabla_{\mathbf{x}} L(\mathbf{x}, \lambda) = \mathbf{0}, \quad \frac{\partial L}{\partial \lambda}(\mathbf{x}, \lambda) = 0$$

are exactly the equations

$$\nabla f(\mathbf{x}) = \lambda \nabla g(\mathbf{x}), \quad g(\mathbf{x}) = k.$$

The Lagrangian does not replace the constraint. It packages the parallel-gradient equation and the constraint equation into one critical-point system.

The method of Lagrange multipliers solves this system to find constrained critical points. The next theorem explains why constrained extrema occur among these candidates.

Theorem 6.2.3 (Lagrange multiplier theorem).

Consider a constraint set

$$S = \{\mathbf{x} \in \mathbb{R}^n : g(\mathbf{x}) = k\},$$

and assume ∇g is nonzero on S . If a point $\mathbf{a} \in S$ is a local extremum of a differentiable function f on S , and $\mathbf{a}$ is an interior point of the domain of f , then $\mathbf{a}$ is a constrained critical point of f on S . In particular, if S is closed and bounded and f is continuous on S , then absolute extrema of f on S exist. Under the same differentiability and nonzero-gradient hypotheses, every absolute extremum of f on S is among the constrained critical points.

Proof.

Suppose $\mathbf{a}$ is a local extremum of f on S . The directional derivative

$$D_{\mathbf{v}} f(\mathbf{a})$$

must be 0 for every unit vector $\mathbf{v}$ tangent to S at $\mathbf{a}$. Otherwise, moving along S in the direction $\mathbf{v}$ or $-\mathbf{v}$ would increase or decrease f .

By 3.7.2,

$$D_{\mathbf{v}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v}.$$

Thus $\nabla f(\mathbf{a})$ is perpendicular to every tangent direction to S . Since S is the level set

$$g(\mathbf{x}) = k,$$

the vector $\nabla g(\mathbf{a})$ is also perpendicular to every tangent direction. Therefore the two gradients are parallel:

$$\nabla f(\mathbf{a}) = \lambda \nabla g(\mathbf{a})$$

for some $\lambda \in \mathbb{R}$.

The final statement follows from 6.1.1. If S is closed and bounded and f is continuous, absolute extrema exist. Any absolute extremum is also a local extremum on S , so it must be a constrained critical point.

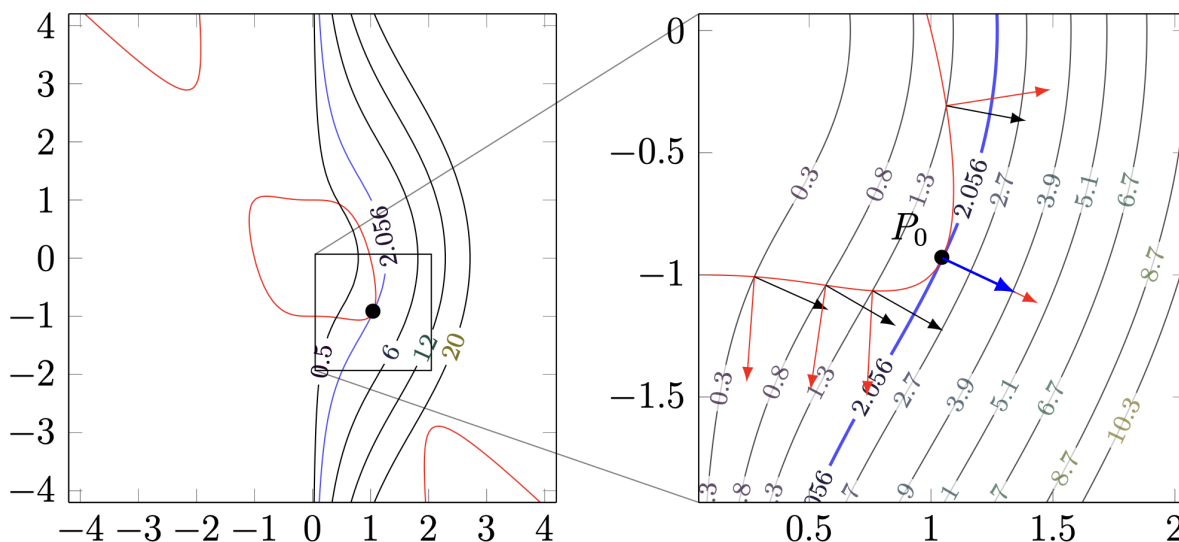


Figure 6.3. Suppose we optimize a function whose contours are the red curves, subject to the constraint shown by the blue curve. Constrained critical points occur where an objective contour is tangent to the constraint curve, so their normal vectors are parallel. From: Stanford's MATH 51 textbook

Algorithm: Lagrange multipliers with one constraint

Given differentiable scalar-valued functions f and g , to find candidates for extrema of $f(\mathbf{x})$ subject to

$$g(\mathbf{x}) = k :$$

1. Compute ∇f and ∇g .
2. Solve

$$\nabla f(\mathbf{x}) = \lambda \nabla g(\mathbf{x})$$

together with

$$g(\mathbf{x}) = k.$$

3. Evaluate f at every candidate.
4. Compare the values when absolute extrema are requested.

If $\nabla g(\mathbf{x}) = \mathbf{0}$ at a feasible point, 6.2.3 does not apply there, so check that point separately. When the constraint describes one boundary piece of a larger region, combine these candidates with the other candidates required by 6.1.

Activity

Find the absolute extrema of $f(x, y) = 3x + y$ subject to the constraint $x^2 + y^2 = 10$.

Solution.

Write $g(x, y) = x^2 + y^2$. Then we are finding extrema of f on the surface

$$S = \{(x, y) : g(x, y) = 10\}$$

. We calculate that

$$\nabla f(x, y) = (3, 1) \quad \text{and} \quad \nabla g(x, y) = (2x, 2y)$$

. So the constrained critical points of f on S are the points $(x, y) \in S$ for which there exists λ such that

$$(3, 1) = \lambda(2x, 2y)$$

, i.e., the points that satisfy the three equations

$$x^2 + y^2 = 10$$

$$2\lambda x = 3$$

$$2\lambda y = 1$$

. If the second and third equations hold, then λ cannot be zero, and we can divide both sides of the equations by λ to obtain that $x = \frac{3}{2\lambda}$ and $y = \frac{1}{2\lambda}$. Substituting this into the first equation, we find that

$$\left(\frac{3}{2\lambda}\right)^2 + \left(\frac{1}{2\lambda}\right)^2 = 10$$

, which simplifies to $1/4 = \lambda^2$, so either $\lambda = 1/2$ or $\lambda = -1/2$. If $\lambda = 1/2$, then $x = 3$ and $y = 1$, and if $\lambda = -1/2$, then $x = -3$ and $y = -1$. So the two points $(3, 1)$ and $(-3, -1)$ are

the two constrained critical points of f on S . Since S is a circle and is closed and bounded, the absolute maximum and minimum occur among these two points. We have

$$f(3, 1) = 10 \quad \text{and} \quad f(-3, -1) = -10$$

, so $(3, 1)$ is the absolute maximum of f on S , and $(-3, -1)$ is the absolute minimum of f on S .

Activity

Find the absolute extrema of the function $f(x, y, z) = x^2 + 2y^2 + 3z^2$, subject to the constraint $x^2 + y^2 + z^2 = 1$.

Solution.

Let $g(x, y, z) = x^2 + y^2 + z^2$. Then $\nabla f(x, y, z) = (2x, 4y, 6z)$ and $\nabla g(x, y, z) = (2x, 2y, 2z)$. A point (x, y, z) is a constrained critical point of f on the sphere $S = \{(x, y, z) : x^2 + y^2 + z^2 = 1\}$ only when there exists $\lambda \in \mathbb{R}$ so that the two equations

$$\nabla f(x, y, z) = \lambda \nabla g(x, y, z) \quad \text{and} \quad x^2 + y^2 + z^2 = 1$$

both hold, which expands to the four equations

$$2x = 2\lambda x$$

$$4y = 2\lambda y$$

$$6z = 2\lambda z$$

$$x^2 + y^2 + z^2 = 1$$

. Since $x^2 + y^2 + z^2 = 1$, either $x \neq 0$ (and dividing the first equation by x we obtain $\lambda = 1$), $y \neq 0$ (and dividing the second equation we obtain $\lambda = 2$), or $z \neq 0$ (and dividing the third equation by z gives $\lambda = 3$). When $\lambda = 1$, the first equation automatically holds, and the second and third equations can only hold when $y = z = 0$. Thus the fourth equation becomes $x^2 = 1$, so $x = \pm 1$. Similarly, when $\lambda = 2$ we have $x = z = 0$ and $y = \pm 1$, and when $\lambda = 3$ we have $x = y = 0$ and $z = \pm 1$. So the function f has six constrained critical points on S , at the points $(\pm 1, 0, 0)$, $(0, \pm 1, 0)$, and $(0, 0, \pm 1)$. We have

$$f(\pm 1, 0, 0) = 1 \quad f(0, \pm 1, 0) = 2 \quad f(0, 0, \pm 1) = 3$$

. So the absolute minima of f occur at $(\pm 1, 0, 0)$, and the absolute maxima of f occur at $(0, 0, \pm 1)$.

Activity: Absolute extrema on a disk

Let

$$f(x, y) = x^2 + y^2 + 4x - 6y$$

on the closed disk

$$D = \{(x, y) : x^2 + y^2 \leq 16\}.$$

Find the absolute extrema of f on D , using Lagrange multipliers to analyze the circular boundary.

Task

Find the critical points of f in the interior of D and evaluate f at each one.

Solution.

The interior critical points satisfy

$$\nabla f(x, y) = (2x + 4, 2y - 6) = (0, 0).$$

Thus $(x, y) = (-2, 3)$. Since $(-2)^2 + 3^2 = 13 < 16$, this point lies in the interior of D . Its function value is

$$f(-2, 3) = 4 + 9 - 8 - 18 = -13.$$

Task

Use Lagrange multipliers to find and evaluate all candidates on the boundary $x^2 + y^2 = 16$.

Solution.

Let $g(x, y) = x^2 + y^2$. On the boundary, the Lagrange multiplier equations are

$$\nabla f(x, y) = \lambda \nabla g(x, y), \quad x^2 + y^2 = 16.$$

Since $\nabla g(x, y) = (2x, 2y)$, the gradient equation becomes

$$\begin{aligned} 2x + 4 &= 2\lambda x, \\ 2y - 6 &= 2\lambda y. \end{aligned}$$

Equivalently,

$$(1 - \lambda)x = -2, \quad (1 - \lambda)y = 3.$$

The value $\lambda = 1$ cannot satisfy these equations. Set $\mu = 1 - \lambda$. Then

$$x = -\frac{2}{\mu}, \quad y = \frac{3}{\mu}.$$

Substitution into the constraint gives

$$\frac{13}{\mu^2} = 16,$$

so $\mu = \pm\sqrt{13}/4$. Therefore the boundary candidates are

$$\left(-\frac{8}{\sqrt{13}}, \frac{12}{\sqrt{13}}\right) \quad \text{and} \quad \left(\frac{8}{\sqrt{13}}, -\frac{12}{\sqrt{13}}\right).$$

On the boundary, $x^2 + y^2 = 16$, so $f(x, y) = 16 + 4x - 6y$. Hence the corresponding values are

$$\begin{aligned} f\left(-\frac{8}{\sqrt{13}}, \frac{12}{\sqrt{13}}\right) &= 16 - 8\sqrt{13}, \\ f\left(\frac{8}{\sqrt{13}}, -\frac{12}{\sqrt{13}}\right) &= 16 + 8\sqrt{13}. \end{aligned}$$

Task

Compare the interior and boundary candidates to identify the absolute maximum and absolute minimum of f on D .

Solution.

The disk D is closed and bounded, and f is continuous, so the extreme value theorem guarantees that absolute extrema exist. The candidate values are

$$-13, \quad 16 - 8\sqrt{13}, \quad 16 + 8\sqrt{13}.$$

Since $16 - 8\sqrt{13} \approx -12.84$, the absolute minimum is -13 at $(-2, 3)$. The absolute maximum is $16 + 8\sqrt{13}$ at

$$\left(\frac{8}{\sqrt{13}}, -\frac{12}{\sqrt{13}}\right).$$

Several equality constraints

Some feasible sets are intersections of several level sets. Let

$$f, g_1, \dots, g_m : \mathbb{R}^n \rightarrow \mathbb{R}$$

be differentiable scalar-valued functions. To optimize $f(\mathbf{x})$ subject to

$$g_1(\mathbf{x}) = k_1, \quad \dots, \quad g_m(\mathbf{x}) = k_m$$

means to optimize f on the feasible set

$$S = \{\mathbf{x} \in \mathbb{R}^n : g_1(\mathbf{x}) = k_1, \dots, g_m(\mathbf{x}) = k_m\}.$$

Definition 6.2.4 (Constrained critical point with several constraints).

This extends 6.2.2. Let

$$S = \{\mathbf{x} \in \mathbb{R}^n : g_1(\mathbf{x}) = k_1, \dots, g_m(\mathbf{x}) = k_m\}.$$

A feasible point $\mathbf{a} \in S$ in the interior of the domain of f is a constrained critical point of f on S if there are scalars

$$\lambda_1, \dots, \lambda_m$$

such that

$$\nabla f(\mathbf{a}) = \lambda_1 \nabla g_1(\mathbf{a}) + \dots + \lambda_m \nabla g_m(\mathbf{a}).$$

The scalars $\lambda_1, \dots, \lambda_m$ are Lagrange multipliers.

By 2.3.5, the condition says that

$$\nabla f(\mathbf{a}) \in \text{span}\{\nabla g_1(\mathbf{a}), \dots, \nabla g_m(\mathbf{a})\}.$$

With one constraint, this span is a line, so the condition reduces to parallel gradients.

Lagrangian formulation

Define

$$L(\mathbf{x}, \lambda_1, \dots, \lambda_m) = f(\mathbf{x}) - \sum_{j=1}^m \lambda_j (g_j(\mathbf{x}) - k_j).$$

Then

$$\nabla_{\mathbf{x}} L = \nabla f - \sum_{j=1}^m \lambda_j \nabla g_j,$$

while

$$\frac{\partial L}{\partial \lambda_j} = -(g_j(\mathbf{x}) - k_j)$$

for $j = 1, \dots, m$. Therefore the critical-point equations for L are exactly

$$\nabla f = \lambda_1 \nabla g_1 + \dots + \lambda_m \nabla g_m$$

together with

$$g_1(\mathbf{x}) = k_1, \quad \dots, \quad g_m(\mathbf{x}) = k_m.$$

As in the one-constraint case, the Lagrangian packages the gradient equation and all constraint equations into one system.

Theorem 6.2.5 (Lagrange multiplier theorem with several constraints).

Let

$$S = \{\mathbf{x} \in \mathbb{R}^n : g_1(\mathbf{x}) = k_1, \dots, g_m(\mathbf{x}) = k_m\}.$$

Suppose $\mathbf{a} \in S$ is a local extremum of f on S , and suppose the constraint gradients

$$\nabla g_1(\mathbf{a}), \dots, \nabla g_m(\mathbf{a})$$

are linearly independent in the sense of 2.4.1. Then there are scalars

$$\lambda_1, \dots, \lambda_m$$

such that

$$\nabla f(\mathbf{a}) = \lambda_1 \nabla g_1(\mathbf{a}) + \dots + \lambda_m \nabla g_m(\mathbf{a}).$$

Proof.

A unit vector tangent to S at $\mathbf{a}$ must be tangent to every constraint level set. It is therefore perpendicular to each of

$$\nabla g_1(\mathbf{a}), \dots, \nabla g_m(\mathbf{a}).$$

At a constrained local extremum, the directional derivative of f must vanish in every allowed tangent direction. By 3.7.2,

$$D_{\mathbf{v}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v},$$

so $\nabla f(\mathbf{a})$ is also normal to the feasible set.

When the constraint gradients are linearly independent, they span the normal directions to the intersection. Therefore $\nabla f(\mathbf{a})$ is a linear combination of the constraint gradients.

Algorithm: Lagrange multipliers with several constraints
To find candidates for extrema of $f(\mathbf{x})$ subject to $g_1(\mathbf{x}) = k_1, \quad \dots, \quad g_m(\mathbf{x}) = k_m :$ 1. Compute $\nabla f, \quad \nabla g_1, \quad \dots, \quad \nabla g_m.$ 2. Solve

$$\nabla f = \lambda_1 \nabla g_1 + \cdots + \lambda_m \nabla g_m$$

together with all m constraint equations.

3. Evaluate f at every candidate.
4. Compare the values when absolute extrema are requested.

If the constraint gradients are linearly dependent at a feasible point, the theorem does not apply there. Check such points separately.

Activity: A sphere and a plane

Find the absolute maximum and minimum values of

$$f(x, y, z) = x$$

subject to

$$x^2 + y^2 + z^2 = 1$$

and

$$x + y + z = 0.$$

1. Describe the feasible set geometrically. Why do absolute extrema exist?
2. Show that the two constraint gradients are linearly independent at every feasible point.
3. Form the Lagrangian using multipliers λ and μ .
4. Solve the Lagrange equations.
5. Evaluate f at the candidates and identify the absolute extrema.

Learning outcomes. U6-LO3, U2-LO4, U4-LO1.

Solution.

The feasible set is the intersection of the unit sphere and the plane

$$x + y + z = 0.$$

It is a circle. In particular, it is closed and bounded, so absolute extrema exist by 6.1.1.
Let

$$g_1(x, y, z) = x^2 + y^2 + z^2, \quad g_2(x, y, z) = x + y + z.$$

Then

$$\nabla g_1 = 2 \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad \nabla g_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

At a feasible point,

$$\nabla g_1 \cdot \nabla g_2 = 2(x + y + z) = 0.$$

Both gradients are nonzero on the feasible set. They are therefore an orthogonal set and hence linearly independent by 4.1.7.

Define

$$L(x, y, z, \lambda, \mu) = x - \lambda(x^2 + y^2 + z^2 - 1) - \mu(x + y + z).$$

The critical-point equations are

$$1 = 2\lambda x + \mu,$$

$$0 = 2\lambda y + \mu,$$

$$0 = 2\lambda z + \mu,$$

together with

$$x^2 + y^2 + z^2 = 1, \quad x + y + z = 0.$$

We cannot have $\lambda = 0$, since the second and third equations would give $\mu = 0$, contradicting the first equation. Thus $\lambda \neq 0$. Subtracting the second and third equations gives

$$y = z.$$

Write $y = z = t$. The plane constraint gives

$$x = -2t.$$

The sphere constraint becomes

$$4t^2 + t^2 + t^2 = 1,$$

so

$$t = \pm \frac{1}{\sqrt{6}}.$$

The candidates are

$$\frac{1}{\sqrt{6}} \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix} \quad \text{and} \quad \frac{1}{\sqrt{6}} \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}.$$

Their f -values are

$$\frac{2}{\sqrt{6}} \quad \text{and} \quad -\frac{2}{\sqrt{6}}.$$

Therefore the absolute maximum is

$$\frac{2}{\sqrt{6}}$$

at

$$\frac{1}{\sqrt{6}} \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix},$$

and the absolute minimum is

$$-\frac{2}{\sqrt{6}}$$

at

$$\frac{1}{\sqrt{6}} \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}.$$

6.3 Quadratic forms and maximum stretch

Recall from 5.4.1 that an $n \times n$ matrix B determines the scalar-valued function

$$q_B(\mathbf{x}) = \mathbf{x}^T B \mathbf{x}.$$

In this section, B will be symmetric. By 5.3.5, B has an orthonormal basis of eigenvectors. Lagrange multipliers give these eigenvectors an optimization interpretation.

Fact: Extrema of a symmetric quadratic form on the unit sphere

Let B be a symmetric $n \times n$ matrix, and let

$$q_B(\mathbf{x}) = \mathbf{x}^T B \mathbf{x}.$$

If $\lambda_{\min}$ and $\lambda_{\max}$ are the smallest and largest eigenvalues of B , then

$$\min_{\|\mathbf{x}\|=1} q_B(\mathbf{x}) = \lambda_{\min}$$

and

$$\max_{\|\mathbf{x}\|=1} q_B(\mathbf{x}) = \lambda_{\max}.$$

These values are achieved at unit eigenvectors for $\lambda_{\min}$ and $\lambda_{\max}$. More generally, every constrained critical point is a unit eigenvector of B .

Proof.

Use the constraint

$$g(\mathbf{x}) = \mathbf{x}^T \mathbf{x} = 1.$$

Since B is symmetric,

$$\nabla q_B(\mathbf{x}) = 2B\mathbf{x}, \quad \nabla g(\mathbf{x}) = 2\mathbf{x}.$$

On the unit sphere, $\nabla g(\mathbf{x}) \neq \mathbf{0}$, so 6.2.3 applies. Its equations are

$$2B\mathbf{x} = 2\mu\mathbf{x},$$

or

$$B\mathbf{x} = \mu\mathbf{x}.$$

Thus the constrained critical points are unit eigenvectors of B . In this calculation, the Lagrange multiplier μ is the corresponding eigenvalue.

If $B\mathbf{x} = \lambda\mathbf{x}$ and $\|\mathbf{x}\| = 1$, then

$$\begin{aligned} q_B(\mathbf{x}) &= \mathbf{x}^T B\mathbf{x} \\ &= \mathbf{x}^T (\lambda\mathbf{x}) \\ &= \lambda \|\mathbf{x}\|^2 \\ &= \lambda. \end{aligned}$$

The unit sphere is closed and bounded, so absolute extrema exist by 6.1.1. Comparing the eigenvalues gives the smallest and largest values.

Principal-coordinate view

The Lagrange calculation identifies the critical directions. Orthogonal diagonalization explains all possible values of the quadratic form.

By 5.3.5, write

$$B = QDQ^T, \quad D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix},$$

where the columns of Q are orthonormal eigenvectors of B . Set

$$\mathbf{c} = Q^T \mathbf{x}.$$

Since Q is orthogonal, 5.3.3 gives

$$\|\mathbf{c}\| = \|\mathbf{x}\|.$$

Also,

$$\begin{aligned} q_B(\mathbf{x}) &= \mathbf{x}^T B \mathbf{x} \\ &= \mathbf{c}^T D \mathbf{c} \\ &= \lambda_1 c_1^2 + \cdots + \lambda_n c_n^2. \end{aligned}$$

On the unit sphere,

$$c_1^2 + \cdots + c_n^2 = 1.$$

Thus $q_B(\mathbf{x})$ is a weighted average of the eigenvalues, with nonnegative weights c_i^2 . It must lie between the smallest and largest eigenvalues.

Activity: A non-diagonal quadratic form on the unit circle

Return to the symmetric matrix from 5.2:

$$B = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

Let

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad q_B(\mathbf{x}) = \mathbf{x}^T B \mathbf{x}.$$

1. Write $q_B(x, y)$ in scalar form.
2. Set up the Lagrange equations for finding the extrema of q_B subject to

$$x^2 + y^2 = 1.$$

3. Find all constrained critical points and their q_B -values.
4. Find an orthogonal matrix Q and diagonal matrix D such that

$$B = QDQ^T,$$

with the larger eigenvalue first.

5. If

$$\mathbf{c} = Q^T \mathbf{x},$$

rewrite q_B in principal coordinates and use this form to identify its absolute maximum and minimum.

Learning outcomes. U6-LO4, U5-LO4.

Solution.

We have

$$\begin{aligned} q_B(x, y) &= \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \\ &= 2x^2 - 2xy + 2y^2. \end{aligned}$$

Its gradient is

$$\nabla q_B(x, y) = \begin{bmatrix} 4x - 2y \\ -2x + 4y \end{bmatrix}.$$

For

$$g(x, y) = x^2 + y^2,$$

we have

$$\nabla g(x, y) = \begin{bmatrix} 2x \\ 2y \end{bmatrix}.$$

The Lagrange equations are therefore

$$\begin{bmatrix} 4x - 2y \\ -2x + 4y \end{bmatrix} = \mu \begin{bmatrix} 2x \\ 2y \end{bmatrix}, \quad x^2 + y^2 = 1.$$

Dividing the gradient equation by 2 gives

$$B\mathbf{x} = \mu\mathbf{x}.$$

The characteristic polynomial is

$$\begin{aligned} \det(B - \mu I) &= (2 - \mu)^2 - 1 \\ &= (\mu - 1)(\mu - 3). \end{aligned}$$

For $\mu = 1$, the eigenvectors satisfy $x = y$. The corresponding unit vectors are

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad -\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

At either point,

$$q_B(\mathbf{x}) = 1.$$

For $\mu = 3$, the eigenvectors satisfy $x = -y$. The corresponding unit vectors are

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \text{and} \quad -\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

At either point,

$$q_B(\mathbf{x}) = 3.$$

Choose

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

The columns of Q are orthonormal eigenvectors of B , so

$$B = QDQ^T.$$

If

$$\mathbf{c} = Q^T \mathbf{x},$$

then

$$q_B(\mathbf{x}) = 3c_1^2 + c_2^2.$$

On the unit circle,

$$c_1^2 + c_2^2 = 1.$$

Therefore the absolute maximum is 3, attained when $c_1^2 = 1$, and the absolute minimum is 1, attained when $c_2^2 = 1$. These are the same values and directions found from the Lagrange equations.

Maximum stretch of a matrix

Now let A be an $m \times n$ matrix. The matrix A need not be square or symmetric. One way to study the matrix map

$$\mathbf{x} \mapsto A\mathbf{x}$$

is to ask which unit inputs produce the largest output lengths.

Because lengths are nonnegative, maximizing $\|A\mathbf{x}\|$ is equivalent to maximizing its square. We have

$$\begin{aligned}\|A\mathbf{x}\|^2 &= (A\mathbf{x})^T(A\mathbf{x}) \\ &= \mathbf{x}^T A^T A \mathbf{x}.\end{aligned}$$

Thus maximum stretch is a quadratic-form problem for the symmetric matrix $A^T A$. The eigenvalues of $A^T A$ are nonnegative. Indeed, if

$$A^T A \mathbf{v} = \lambda \mathbf{v}$$

for a nonzero vector $\mathbf{v}$, then

$$\begin{aligned}\lambda \|\mathbf{v}\|^2 &= \mathbf{v}^T A^T A \mathbf{v} \\ &= \|A\mathbf{v}\|^2 \\ &\geq 0.\end{aligned}$$

Therefore $\lambda \geq 0$.

Order the eigenvalues of $A^T A$ from largest to smallest and take their nonnegative square roots:

$$\sigma_i = \sqrt{\lambda_i}.$$

These numbers are the singular values of A . A unit eigenvector $\mathbf{v}_i$ of $A^T A$ is a right singular vector of A . Since

$$\begin{aligned}\|A\mathbf{v}_i\|^2 &= \mathbf{v}_i^T A^T A \mathbf{v}_i \\ &= \lambda_i,\end{aligned}$$

we have

$$\|A\mathbf{v}_i\| = \sigma_i.$$

Thus the right singular vectors are special input directions, and the singular values are their stretch factors. In particular,

$$\max_{\|\mathbf{x}\|=1} \|A\mathbf{x}\| = \sigma_1.$$

Activity: A small stretch computation

Let

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

Let

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

1. Compute $A\mathbf{e}_1$ and $A\mathbf{e}_2$.
2. Compute $\|A\mathbf{e}_1\|$ and $\|A\mathbf{e}_2\|$.
3. Compute $A^T A$.
4. Find the eigenvalues of $A^T A$.
5. What are the singular values of A ?
6. Which unit direction is stretched most?

Solution.

We have

$$A\mathbf{e}_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix}, \quad A\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Thus

$$\|A\mathbf{e}_1\| = 3, \quad \|A\mathbf{e}_2\| = 1.$$

Also

$$A^T A = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}^T \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}.$$

The eigenvalues of $A^T A$ are 9 and 1. Therefore the singular values of A are

$$\sigma_1 = 3, \quad \sigma_2 = 1.$$

The direction $\mathbf{e}_1$ is stretched most. This is the simplest version of the question

$$\max_{\|\mathbf{x}\|=1} \|A\mathbf{x}\|.$$

Activity: Reading the same stretch in code

The following code repeats the previous activity using $A^T A$.

```
import numpy as np

A = np.array([[3., 0.],
              [0., 1.]])

B = A.T @ A
eigvals, eigvecs = np.linalg.eigh(B)

singular_values = np.sqrt(eigvals[::-1])
dominant_direction = eigvecs[:, -1]

B, eigvals, singular_values, dominant_direction
```

Output:

```
(array([[9., 0.],
        [0., 1.]]),
 array([1., 9.]),
 array([3., 1.]),
 array([1., 0.]])
```

1. What mathematical matrix is stored in B?
2. Why are the entries of `eigvals` squared stretch factors?
3. Why does the code take a square root?
4. Why does `eigvecs[:, -1]` give the most-stretched input direction?
5. What value of $\max_{\|x\|=1} \|Ax\|$ does the code predict?

Solution.

The matrix B is $A^T A$. The entries of `eigvals` are eigenvalues of $A^T A$, so they are values of $\|Ax\|^2$ for special unit input directions. The code takes a square root because singular values measure $\|Ax\|$, not $\|Ax\|^2$. The command `np.linalg.eigh` lists eigenvalues in increasing order, so the largest eigenvalue is last, and `eigvecs[:, -1]` is its eigenvector. The largest singular value is 3, so

$$\max_{\|\mathbf{x}\|=1} \|\mathbf{A}\mathbf{x}\| = 3.$$

The same quadratic-form problem appears when a matrix stores centered data. 6.4 uses it to choose directions that capture variation in a data set.

6.4 Principal component analysis

A data matrix can have many feature columns. Principal component analysis, or PCA, looks for orthogonal directions that capture as much centered variation as possible. The first direction is found with one constraint: it must have unit length. Each later direction has additional orthogonality constraints.

Definition 6.4.1 (Centered data and covariance matrix).

Let

$$\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$$

be data vectors. Their mean is

$$\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i.$$

The centered data vectors are

$$\mathbf{z}_i = \mathbf{x}_i - \bar{\mathbf{x}}.$$

Let Z be the $n \times d$ matrix whose i -th row is $\mathbf{z}_i^T$. The covariance matrix is

$$\begin{aligned} C &= \frac{1}{n} Z^T Z \\ &= \frac{1}{n} \sum_{i=1}^n \mathbf{z}_i \mathbf{z}_i^T. \end{aligned}$$

Some conventions divide by $n - 1$ instead of n . This multiplies every covariance eigenvalue by the same positive constant and does not change the eigenvectors.

Scores and captured variation

Let $\mathbf{v} \in \mathbb{R}^d$ be a unit vector. The score of the i -th centered data point in direction $\mathbf{v}$ is

$$t_i = \mathbf{v}^T \mathbf{z}_i.$$

The score vector is

$$\mathbf{t} = Z\mathbf{v}.$$

Its average squared score is

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n t_i^2 &= \frac{1}{n} \|Z\mathbf{v}\|^2 \\ &= \mathbf{v}^T C \mathbf{v}. \end{aligned}$$

Thus captured variation is a quadratic form.

The matrix C is symmetric. Also,

$$\begin{aligned} \mathbf{v}^T C \mathbf{v} &= \frac{1}{n} \|Z\mathbf{v}\|^2 \\ &\geq 0 \end{aligned}$$

for every $\mathbf{v}$, so the eigenvalues of C are nonnegative.

Definition 6.4.2 (First principal direction).

A first principal direction is a unit vector $\mathbf{v}_1$ that maximizes

$$\mathbf{v}^T C \mathbf{v}$$

subject to

$$\|\mathbf{v}\| = 1.$$

The entries of

$$\mathbf{t}_1 = Z\mathbf{v}_1$$

are the first principal-component scores.

By 6.3, $\mathbf{v}_1$ may be chosen as a unit eigenvector of C corresponding to its largest eigenvalue λ_1 . The maximum captured variation is λ_1 .

Reconstruction view

For a unit direction $\mathbf{v}$, the orthogonal projection of $\mathbf{z}_i$ onto

$$\text{span}\{\mathbf{v}\}$$

is

$$\hat{\mathbf{z}}_i = t_i \mathbf{v}.$$

The corresponding reconstructed data point is

$$\hat{\mathbf{x}}_i = \bar{\mathbf{x}} + \hat{\mathbf{z}}_i.$$

Define the reconstruction residual by

$$\mathbf{r}_i = \mathbf{z}_i - \hat{\mathbf{z}}_i.$$

By 4.2.4 and 4.2.5,

$$\mathbf{r}_i \perp \mathbf{v}.$$

Therefore

$$\|\mathbf{z}_i\|^2 = t_i^2 + \|\mathbf{r}_i\|^2.$$

Summing over the data gives

$$\sum_{i=1}^n \|\mathbf{z}_i\|^2 = \sum_{i=1}^n t_i^2 + \sum_{i=1}^n \|\mathbf{r}_i\|^2.$$

The quantity on the left does not depend on $\mathbf{v}$. Thus maximizing the total squared scores is equivalent to minimizing the total squared reconstruction residuals.

Activity: Principal directions of a small data set

The rows of

$$X = \begin{bmatrix} 4 & 3 \\ 3 & 4 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

are four data points in $\mathbb{R}^2$.

1. Compute the mean $\bar{\mathbf{x}}$ and the centered data matrix Z .
2. Compute

$$C = \frac{1}{4} Z^T Z.$$

3. Find the eigenvalues of C and an orthonormal basis of eigenvectors. Choose $\mathbf{v}_1$ for the larger eigenvalue and $\mathbf{v}_2$ for the smaller eigenvalue.
4. Compute the first principal-component score vector

$$\mathbf{t}_1 = Z \mathbf{v}_1.$$

5. For the first centered point $\mathbf{z}_1$, compute

$$\hat{\mathbf{z}}_1 = (\mathbf{v}_1^T \mathbf{z}_1) \mathbf{v}_1$$

and

$$\mathbf{r}_1 = \mathbf{z}_1 - \hat{\mathbf{z}}_1.$$

Check that

$$\mathbf{r}_1 \cdot \mathbf{v}_1 = 0$$

and

$$\|\mathbf{z}_1\|^2 = (\mathbf{v}_1^T \mathbf{z}_1)^2 + \|\mathbf{r}_1\|^2.$$

6. Form

$$V = [\mathbf{v}_1 \quad \mathbf{v}_2]$$

and compute $V^T C V$.

7. What fraction of the total centered variation is captured by the first principal direction?

8. Describe the affine line

$$\bar{\mathbf{x}} + \text{span}\{\mathbf{v}_1\}.$$

Learning outcomes. U6-LO5, U5-LO4, U4-LO3.

Solution.

The mean is

$$\bar{\mathbf{x}} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Therefore

$$Z = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ -2 & -1 \\ -1 & -2 \end{bmatrix}.$$

We calculate

$$\begin{aligned} C &= \frac{1}{4} Z^T Z \\ &= \frac{1}{4} \begin{bmatrix} 10 & 8 \\ 8 & 10 \end{bmatrix} \\ &= \begin{bmatrix} 5/2 & 2 \\ 2 & 5/2 \end{bmatrix}. \end{aligned}$$

The eigenvalues are

$$\lambda_1 = \frac{9}{2}, \quad \lambda_2 = \frac{1}{2}.$$

Corresponding unit eigenvectors are

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

The first score vector is

$$\mathbf{t}_1 = Z\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 3 \\ 3 \\ -3 \\ -3 \end{bmatrix}.$$

For

$$\mathbf{z}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix},$$

the score is

$$\mathbf{v}_1^T \mathbf{z}_1 = \frac{3}{\sqrt{2}}.$$

Thus

$$\hat{\mathbf{z}}_1 = \frac{3}{\sqrt{2}} \mathbf{v}_1 = \begin{bmatrix} 3/2 \\ 3/2 \end{bmatrix},$$

and

$$\mathbf{r}_1 = \begin{bmatrix} 1/2 \\ -1/2 \end{bmatrix}.$$

We have

$$\mathbf{r}_1 \cdot \mathbf{v}_1 = 0.$$

Also,

$$\|\mathbf{z}_1\|^2 = 5, \quad (\mathbf{v}_1^T \mathbf{z}_1)^2 = \frac{9}{2}, \quad \|\mathbf{r}_1\|^2 = \frac{1}{2},$$

so

$$5 = \frac{9}{2} + \frac{1}{2}.$$

With

$$V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix},$$

we obtain

$$V^T C V = \begin{bmatrix} 9/2 & 0 \\ 0 & 1/2 \end{bmatrix}.$$

The first direction captures the fraction

$$\begin{aligned} \frac{\lambda_1}{\lambda_1 + \lambda_2} &= \frac{9/2}{9/2 + 1/2} \\ &= \frac{9}{10} \end{aligned}$$

of the total centered variation.

Finally,

$$\bar{\mathbf{x}} + \text{span}\{\mathbf{v}_1\} = \left\{ \begin{bmatrix} 2 \\ 2 \end{bmatrix} + s \begin{bmatrix} 1 \\ 1 \end{bmatrix} : s \in \mathbb{R} \right\}.$$

This is the line $y = x$.

Definition 6.4.3 (Second principal direction).

Suppose $\mathbf{v}_1$ is a first principal direction. A second principal direction is a unit vector $\mathbf{v}_2$ that maximizes

$$\mathbf{v}^T C \mathbf{v}$$

subject to the two constraints

$$\mathbf{v}^T \mathbf{v} = 1$$

and

$$\mathbf{v}^T \mathbf{v}_1 = 0.$$

In $\mathbb{R}^2$, the second constraint leaves only one axis, up to sign. In higher dimensions, it restricts the optimization to the unit vectors orthogonal to $\mathbf{v}_1$.

Fact: The second principal direction is another eigenvector

Let the eigenvalues of C be ordered so that

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d.$$

If $\mathbf{v}_1$ is a unit eigenvector for λ_1 , then a second principal direction may be chosen as a unit eigenvector $\mathbf{v}_2$ for λ_2 that is orthogonal to $\mathbf{v}_1$.

Proof.

On the feasible set, the gradients of the two constraints are

$$2\mathbf{v} \quad \text{and} \quad \mathbf{v}_1.$$

They are nonzero and orthogonal, so they are linearly independent by 4.1.7. Thus 6.2.5 applies. Define

$$L(\mathbf{v}, \mu, \nu) = \mathbf{v}^T C \mathbf{v} - \mu(\mathbf{v}^T \mathbf{v} - 1) - \nu \mathbf{v}^T \mathbf{v}_1.$$

The stationarity equation is

$$2C\mathbf{v} = 2\mu\mathbf{v} + \nu\mathbf{v}_1.$$

Take the dot product with $\mathbf{v}_1$. Since C is symmetric,

$$\begin{aligned} \mathbf{v}_1^T C \mathbf{v} &= (C\mathbf{v}_1)^T \mathbf{v} \\ &= \lambda_1 \mathbf{v}_1^T \mathbf{v} \\ &= 0. \end{aligned}$$

The first constraint term also has zero dot product with $\mathbf{v}_1$. Since

$$\|\mathbf{v}_1\| = 1,$$

it follows that $\nu = 0$. Hence

$$C\mathbf{v} = \mu\mathbf{v}.$$

Thus a constrained critical direction is an eigenvector of C . Among the eigenvectors orthogonal to $\mathbf{v}_1$, the largest remaining value of the quadratic form is λ_2 .

More generally, the j -th principal direction maximizes

$$\mathbf{v}^T C \mathbf{v}$$

subject to unit length and orthogonality to

$$\mathbf{v}_1, \dots, \mathbf{v}_{j-1}.$$

The same argument gives a unit eigenvector for λ_j . If an eigenvalue is repeated, the individual principal directions inside its eigenspace are not unique, but their span is fixed.

Principal coordinates

Collect an orthonormal set of principal directions as the columns of

$$V = [\mathbf{v}_1 \quad \dots \quad \mathbf{v}_d].$$

By 5.3.5,

$$V^T C V = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_d \end{bmatrix}.$$

The principal-coordinate matrix is

$$T = ZV.$$

Its j -th column contains the scores in direction $\mathbf{v}_j$, and

$$\frac{1}{n} T^T T = V^T C V.$$

Thus λ_j is the average squared score in the j -th principal direction, while the off-diagonal entries are zero.

The total centered variation is

$$\lambda_1 + \dots + \lambda_d.$$

When the centered data are not all zero, the fraction captured by the first k principal directions is

$$\frac{\lambda_1 + \dots + \lambda_k}{\lambda_1 + \dots + \lambda_d}.$$

PCA and linear regression

PCA and linear regression both use projection and sums of squared residuals, but they solve different problems.

PCA treats the feature coordinates symmetrically. It chooses an affine subspace through the data mean, and its reconstruction residuals are perpendicular to that subspace.

Linear regression designates one coordinate as the response and minimizes vertical prediction residuals in the data plot. Algebraically, it projects the response vector onto the column space of a design matrix, as in 4.3.

Thus PCA and linear regression can choose different best-fit lines for the same data.

For PCA, the principal directions come from

$$C = \frac{1}{n} Z^T Z.$$

6.5 packages the corresponding input directions, output directions, and stretch factors directly from Z , and extends the same picture to an arbitrary rectangular matrix.

6.5 Singular value decomposition

6.3 found singular values from the symmetric matrix $A^T A$. 6.4 used the same construction for a centered data matrix Z , since

$$C = \frac{1}{n} Z^T Z.$$

This section packages input directions, output directions, and stretch factors into one factorization.

The singular value decomposition is a rectangular analogue of orthogonal diagonalization. Orthogonal diagonalization uses an orthonormal eigenvector basis for a symmetric square matrix. SVD uses an orthonormal basis in the input space and an orthonormal basis in the output space, and it applies to every real matrix.

Definition 6.5.1 (Singular value decomposition).

Let A be an $m \times n$ matrix, and let $p = \min(m, n)$. A singular value decomposition, or SVD, of A is a factorization

$$A = U \Sigma V^T,$$

where U is an $m \times m$ orthogonal matrix, V is an $n \times n$ orthogonal matrix, and Σ is an $m \times n$ diagonal matrix whose diagonal entries

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_p \geq 0$$

are nonnegative.

The numbers σ_i are the singular values of A . The columns

$$\mathbf{u}_1, \dots, \mathbf{u}_m$$

of U are left singular vectors. The columns

$$\mathbf{v}_1, \dots, \mathbf{v}_n$$

of V are right singular vectors.

If r is the number of positive singular values, then

$$A \mathbf{v}_i = \sigma_i \mathbf{u}_i \quad \text{for } 1 \leq i \leq r,$$

and

$$A \mathbf{v}_i = \mathbf{0} \quad \text{for } i > r.$$

Note: Notation for SVD

Some references write the same factorization as

$$A = P\Sigma Q^T.$$

In this book we use

$$A = U\Sigma V^T$$

because this matches the NumPy output names `U`, `s`, `Vt` = `np.linalg.svd(A)`. The roles are the same: U stores left singular vectors, V stores right singular vectors, and Σ stores singular values. The definition above uses a full SVD; NumPy with `full_matrices=False` returns the reduced factors needed for the matrix product.

Input directions, stretch, and output directions

Write

$$U = [\mathbf{u}_1 \quad \cdots \quad \mathbf{u}_m], \quad V = [\mathbf{v}_1 \quad \cdots \quad \mathbf{v}_n].$$

Read

$$A = U\Sigma V^T$$

from right to left.

1. The vector $V^T \mathbf{x}$ gives the coordinates of $\mathbf{x}$ in the orthonormal basis of right singular vectors.
2. The matrix Σ multiplies the i -th coordinate by σ_i . A zero singular value erases that direction.
3. The matrix U combines the resulting coordinates using the left singular vectors.

By 4.1.8, if r singular values are positive, then

$$A\mathbf{x} = \sum_{i=1}^r \sigma_i (\mathbf{v}_i \cdot \mathbf{x}) \mathbf{u}_i.$$

The dot product $\mathbf{v}_i \cdot \mathbf{x}$ measures how much of the input lies in the i -th right singular direction. The singular value σ_i stretches that coordinate, and $\mathbf{u}_i$ gives the corresponding output direction.

Since U and V are orthogonal, they preserve lengths by 5.3.3. The changes in length occur in Σ . Geometrically, the right singular vectors give principal input directions, the singular values give the corresponding stretch factors, and the left singular vectors give principal output directions. The following figure illustrates this picture for a rectangular matrix.

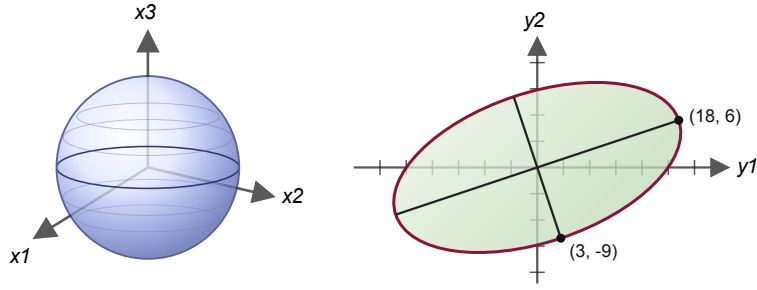


Figure 6.4. A matrix map from $\mathbb{R}^3$ to $\mathbb{R}^2$ corresponding to the matrix $A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$. The left panel shows the unit sphere in $\mathbb{R}^3$. The right panel shows its image under A in $\mathbb{R}^2$, an ellipse with principal semiaxis endpoints $(18, 6)$ and $(3, -9)$. Adapted from David C. Lay, Steven R. Lay, and Judi J. McDonald, *Linear Algebra and Its Applications*, 6th edition, Pearson, © 2021, Pearson.

Theorem 6.5.2 (From right singular vectors to left singular vectors).

Let

$$\mathbf{v}_1, \dots, \mathbf{v}_n$$

be orthonormal eigenvectors of $A^T A$, with corresponding eigenvalues

$$\lambda_1, \dots, \lambda_n \geq 0.$$

Set

$$\sigma_i = \sqrt{\lambda_i}.$$

For each positive eigenvalue, define

$$\mathbf{u}_i = \frac{1}{\sigma_i} A \mathbf{v}_i.$$

Then the vectors $\mathbf{u}_i$ obtained from the positive eigenvalues are orthonormal, and

$$A \mathbf{v}_i = \sigma_i \mathbf{u}_i.$$

If $\lambda_i = 0$, then

$$A \mathbf{v}_i = \mathbf{0}.$$

Proof.

Since $\mathbf{v}_i$ is a unit eigenvector,

$$\begin{aligned} \|A \mathbf{v}_i\|^2 &= \mathbf{v}_i^T A^T A \mathbf{v}_i \\ &= \lambda_i. \end{aligned}$$

Thus, when $\lambda_i > 0$,

$$\left\| \frac{1}{\sigma_i} A \mathbf{v}_i \right\| = 1.$$

If $i \neq j$, then

$$\begin{aligned} (A \mathbf{v}_i) \cdot (A \mathbf{v}_j) &= \mathbf{v}_i^T A^T A \mathbf{v}_j \\ &= \lambda_j (\mathbf{v}_i \cdot \mathbf{v}_j) \\ &= 0. \end{aligned}$$

Therefore the resulting left singular vectors are orthonormal. If $\lambda_i = 0$, then

$$\|A \mathbf{v}_i\|^2 = 0,$$

so $A \mathbf{v}_i = \mathbf{0}$.

Theorem 6.5.3 (Every real matrix has an SVD).

Let A be a real $m \times n$ matrix. Then A has a singular value decomposition

$$A = U \Sigma V^T.$$

One such factorization can be constructed as follows.

1. Find an orthonormal eigenbasis

$$\mathbf{v}_1, \dots, \mathbf{v}_n$$

for $A^T A$. Order the vectors so that

$$\lambda_1 \geq \dots \geq \lambda_r > 0$$

and $\lambda_i = 0$ for $i > r$.

2. Set

$$\sigma_i = \sqrt{\lambda_i}$$

for $1 \leq i \leq r$.

3. Define

$$\mathbf{u}_i = \frac{1}{\sigma_i} A \mathbf{v}_i$$

for $1 \leq i \leq r$.

4. Extend

$$\mathbf{u}_1, \dots, \mathbf{u}_r$$

to an orthonormal basis

$$\mathbf{u}_1, \dots, \mathbf{u}_m$$

of $\mathbb{R}^m$.

5. Use these vectors as the columns of V and U , and place

$$\sigma_1, \dots, \sigma_r$$

on the diagonal of Σ , followed by zeros.

Proof.

The matrix $A^T A$ is symmetric, so 5.3.5 gives an orthonormal eigenbasis. Its eigenvalues are nonnegative because

$$\mathbf{x}^T A^T A \mathbf{x} = \|A\mathbf{x}\|^2 \geq 0.$$

The previous lemma shows that the vectors

$$\mathbf{u}_1, \dots, \mathbf{u}_r$$

are orthonormal and satisfy

$$A\mathbf{v}_i = \sigma_i \mathbf{u}_i.$$

Use 4.2.3 to complete them to an orthonormal basis of $\mathbb{R}^m$.

Comparing columns gives

$$AV = [\sigma_1 \mathbf{u}_1 \ \cdots \ \sigma_r \mathbf{u}_r \ \mathbf{0} \ \cdots \ \mathbf{0}] = U\Sigma.$$

Since V is orthogonal, multiplying by V^T gives

$$A = U\Sigma V^T.$$

Note: Non-uniqueness of the SVD

The SVD is not unique. A matched pair of singular vectors can be multiplied by -1 without changing the product:

$$\sigma_i \mathbf{u}_i \mathbf{v}_i^T = \sigma_i (-\mathbf{u}_i)(-\mathbf{v}_i)^T.$$

Repeated positive singular values also allow orthogonal changes of basis within the corresponding singular subspaces. When a singular value is zero, there can also be many valid choices for the corresponding orthonormal basis vectors. The roles are more important than the specific signs.

Here is one complete SVD computation, broken into the same steps as the theorem.

Activity: Computing an SVD, part 1: right singular vectors

Let

$$A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}.$$

1. Compute $A^T A$.
2. Find the eigenvalues of $A^T A$.
3. Find an orthonormal eigenbasis

$$\mathbf{v}_1, \mathbf{v}_2$$

for $A^T A$, with the positive eigenvalue first.

4. Find the singular values of A .

Solution.

We compute

$$\begin{aligned} A^T A &= \begin{bmatrix} 1 & -2 & 2 \\ -1 & 2 & -2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 9 & -9 \\ -9 & 9 \end{bmatrix}. \end{aligned}$$

The characteristic polynomial is

$$\begin{aligned} \det(A^T A - \lambda I) &= \det \begin{bmatrix} 9 - \lambda & -9 \\ -9 & 9 - \lambda \end{bmatrix} \\ &= \lambda(\lambda - 18). \end{aligned}$$

Thus the eigenvalues are

$$\lambda_1 = 18, \quad \lambda_2 = 0.$$

For $\lambda_1 = 18$, one eigenvector is

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

so we choose the unit vector

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

For $\lambda_2 = 0$, one eigenvector is

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

so we choose

$$\mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

The singular values are the square roots of the eigenvalues of $A^T A$:

$$\sigma_1 = \sqrt{18} = 3\sqrt{2}, \quad \sigma_2 = 0.$$

Activity: Computing an SVD, part 2: output directions

Use the vectors from the previous activity.

1. Compute $A\mathbf{v}_1$.
2. Compute $A\mathbf{v}_2$.
3. Compute

$$\mathbf{u}_1 = \frac{1}{\sigma_1} A\mathbf{v}_1.$$

4. What does $A\mathbf{v}_2 = \mathbf{0}$ say about the matrix map?

Solution.

First,

$$\begin{aligned}
 A\mathbf{v}_1 &= \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\
 &= \begin{bmatrix} \sqrt{2} \\ -2\sqrt{2} \\ 2\sqrt{2} \end{bmatrix}.
 \end{aligned}$$

Since $\sigma_1 = 3\sqrt{2}$,

$$\begin{aligned}
 \mathbf{u}_1 &= \frac{1}{3\sqrt{2}} \begin{bmatrix} \sqrt{2} \\ -2\sqrt{2} \\ 2\sqrt{2} \end{bmatrix} \\
 &= \begin{bmatrix} 1/3 \\ -2/3 \\ 2/3 \end{bmatrix}.
 \end{aligned}$$

Also,

$$\begin{aligned}
 A\mathbf{v}_2 &= \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\
 &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.
 \end{aligned}$$

Thus $\mathbf{v}_2$ is a right null direction. The matrix map forgets that input direction. Since only one singular value is positive, the rank of A is 1.

Activity: Checking a completed SVD

Continue with the matrix and singular vectors from the previous two activities. One orthonormal completion of

$$\mathbf{u}_1 = \begin{bmatrix} 1/3 \\ -2/3 \\ 2/3 \end{bmatrix}.$$

gives

$$U = \begin{bmatrix} 1/3 & 2\sqrt{2}/3 & 0 \\ -2/3 & \sqrt{2}/6 & \sqrt{2}/2 \\ 2/3 & -\sqrt{2}/6 & \sqrt{2}/2 \end{bmatrix}.$$

Also set

$$\Sigma = \begin{bmatrix} 3\sqrt{2} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

1. Identify the shapes of U , Σ , and V .
2. Check that

$$U^T U = I_3, \quad V^T V = I_2.$$

3. Use

$$A \mathbf{v}_1 = 3\sqrt{2} \mathbf{u}_1, \quad A \mathbf{v}_2 = \mathbf{0}$$

to explain why

$$AV = U\Sigma$$

without multiplying all the matrices entry by entry.

4. Conclude that

$$A = U\Sigma V^T.$$

5. Which column of U comes from a positive singular value? Which columns arise from an orthonormal completion?

Solution.

The shapes are

$$U \in \mathbb{R}^{3 \times 3}, \quad \Sigma \in \mathbb{R}^{3 \times 2}, \quad V \in \mathbb{R}^{2 \times 2}.$$

Direct multiplication gives

$$U^T U = I_3, \quad V^T V = I_2,$$

so U and V are orthogonal.
The columns of AV are

$$Av_1 \quad \text{and} \quad Av_2.$$

From the previous activities,

$$AV = [3\sqrt{2} \mathbf{u}_1 \quad \mathbf{0}] = U\Sigma.$$

Since V is orthogonal,

$$V^{-1} = V^T.$$

Multiplying $AV = U\Sigma$ on the right by V^T gives

$$A = U\Sigma V^T.$$

The first column $\mathbf{u}_1$ comes from the positive singular value. The remaining columns complete $\mathbf{u}_1$ to an orthonormal basis of $\mathbb{R}^3$. That completion is not unique.

Unit 2 introduced row spaces, column spaces, and null spaces in 2.4.11 and 2.3.3. The null space of A^T is called the *left null space* of A . Together, these are the four fundamental subspaces of a matrix.

Definition 6.5.4 (Four fundamental subspaces).

Let A be an $m \times n$ matrix. Its four fundamental subspaces are

$$\text{row}(A) \subseteq \mathbb{R}^n, \quad \text{col}(A) \subseteq \mathbb{R}^m,$$

$$\text{null}(A) \subseteq \mathbb{R}^n, \quad \text{null}(A^T) \subseteq \mathbb{R}^m.$$

More explicitly,

$$\text{row}(A) = \text{span}\{\text{rows of } A\},$$

$$\text{col}(A) = \text{span}\{\text{columns of } A\},$$

$$\text{null}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\},$$

and

$$\text{null}(A^T) = \{\mathbf{y} \in \mathbb{R}^m : A^T\mathbf{y} = \mathbf{0}\}.$$

Theorem 6.5.5 (Fundamental subspaces from an SVD).

Let

$$A = U\Sigma V^T$$

be an SVD of an $m \times n$ matrix, and suppose exactly r singular values are positive. Write

$$U = [\mathbf{u}_1 \quad \cdots \quad \mathbf{u}_m], \quad V = [\mathbf{v}_1 \quad \cdots \quad \mathbf{v}_n].$$

Then:

- $\{\mathbf{u}_1, \dots, \mathbf{u}_r\}$ is an orthonormal basis for $\text{col}(A)$.
- $\{\mathbf{u}_{r+1}, \dots, \mathbf{u}_m\}$ is an orthonormal basis for $\text{null}(A^T)$.
- $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is an orthonormal basis for $\text{row}(A)$.
- $\{\mathbf{v}_{r+1}, \dots, \mathbf{v}_n\}$ is an orthonormal basis for $\text{null}(A)$.

In particular,

$$\text{rank}(A) = r,$$

$$\dim \text{null}(A) = n - r, \quad \dim \text{null}(A^T) = m - r.$$

Proof.

By the discussion “Input directions, stretch, and output directions”,

$$A\mathbf{x} = \sum_{i=1}^r \sigma_i (\mathbf{v}_i \cdot \mathbf{x}) \mathbf{u}_i.$$

Thus every output lies in

$$\text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_r\}.$$

Conversely,

$$\mathbf{u}_i = \frac{1}{\sigma_i} A\mathbf{v}_i$$

belongs to $\text{col}(A)$ for $1 \leq i \leq r$. Therefore the first r left singular vectors form a basis for the column space.

Write an input in the right singular basis:

$$\mathbf{x} = c_1 \mathbf{v}_1 + \cdots + c_n \mathbf{v}_n.$$

Then

$$A\mathbf{x} = \sigma_1 c_1 \mathbf{u}_1 + \cdots + \sigma_r c_r \mathbf{u}_r.$$

Since the displayed left singular vectors are independent, $A\mathbf{x} = \mathbf{0}$ exactly when

$$c_1 = \cdots = c_r = 0.$$

Hence the remaining right singular vectors form a basis for $\text{null}(A)$.
Apply the same argument to

$$A^T = V\Sigma^T U^T.$$

This gives the row space and left null space statements. Finally, the column space has r basis vectors, so

$$\text{rank}(A) = r.$$

Activity: Reading the computed SVD

Use the SVD computed above:

$$A = U\Sigma V^T,$$

where

$$V = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix},$$

$$U = \begin{bmatrix} 1/3 & (2/3)\sqrt{2} & 0 \\ -2/3 & (1/6)\sqrt{2} & (1/2)\sqrt{2} \\ 2/3 & -(1/6)\sqrt{2} & (1/2)\sqrt{2} \end{bmatrix},$$

and

$$\Sigma = \begin{bmatrix} 3\sqrt{2} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

1. What is $\text{rank}(A)$?
2. Give an orthonormal basis for $\text{row}(A)$.
3. Give an orthonormal basis for $\text{null}(A)$.
4. Give an orthonormal basis for $\text{col}(A)$.
5. Give an orthonormal basis for $\text{null}(A^T)$.
6. Which input direction is forgotten?
7. Which input direction is transmitted, and by what stretch factor?

Solution.

There is one positive singular value, so

$$\text{rank}(A) = 1.$$

The first right singular vector gives a basis for the row space:

$$\text{row}(A) = \text{span} \left\{ \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix} \right\}.$$

The zero singular value gives the null-space direction:

$$\text{null}(A) = \text{span} \left\{ \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}.$$

The first left singular vector gives a basis for the column space:

$$\text{col}(A) = \text{span} \left\{ \begin{bmatrix} 1/3 \\ -2/3 \\ 2/3 \end{bmatrix} \right\}.$$

The remaining left singular vectors give a basis for $\text{null}(A^T)$:

$$\text{null}(A^T) = \text{span} \left\{ \begin{bmatrix} (2/3)\sqrt{2} \\ (1/6)\sqrt{2} \\ -(1/6)\sqrt{2} \end{bmatrix}, \begin{bmatrix} 0 \\ (1/2)\sqrt{2} \\ (1/2)\sqrt{2} \end{bmatrix} \right\}.$$

The forgotten input direction is

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

The transmitted input direction is

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

and its stretch factor is

$$3\sqrt{2}.$$

Note: SVD and orthogonal diagonalization

The SVD also gives orthogonal diagonalizations of two symmetric matrices:

$$A^T A = V \Sigma^T \Sigma V^T$$

and

$$A A^T = U \Sigma \Sigma^T U^T.$$

Thus the right singular vectors are eigenvectors of $A^T A$, the left singular vectors are eigenvectors of $A A^T$, and the positive eigenvalues of both symmetric matrices are

$$\sigma_1^2, \dots, \sigma_r^2.$$

Orthogonal diagonalization writes a symmetric square matrix as

$$B = Q D Q^T,$$

using the same orthonormal basis in its domain and codomain; see 5.3.4 and 5.3.5. The diagonal entries of D are signed eigenvalues.

The SVD

$$A = U \Sigma V^T$$

uses one orthonormal basis in the input space and another in the output space. Its diagonal entries are nonnegative stretch factors. This two-basis version applies to every real matrix, including rectangular and nonsymmetric matrices.

It gives a second description of the matrix map

$$\mathbf{x} \mapsto A\mathbf{x} :$$

measure the input in special directions, stretch or erase those coordinates, and express the result in special output directions. This is a precise version of the Unit 2 question: what information does a matrix map keep, weaken, or forget?

6.6 PCA from the SVD

6.4 found principal directions by diagonalizing the covariance matrix

$$C = \frac{1}{n} Z^T Z.$$

The SVD of the centered data matrix Z contains the same directions and the same captured-variation values. It also connects principal-component scores with left singular vectors.

Fact: PCA from a centered-data SVD

Let

$$Z \in \mathbb{R}^{n \times d}$$

be a centered data matrix, and let

$$Z = U \Sigma V^T$$

be an SVD. Write

$$V = [\mathbf{v}_1 \quad \cdots \quad \mathbf{v}_d],$$

and include zero singular values when needed. Then

$$C = \frac{1}{n}Z^T Z = V \left(\frac{1}{n}\Sigma^T \Sigma \right) V^T.$$

Therefore:

- the right singular vectors $\mathbf{v}_j$ are principal directions;
- the covariance eigenvalues are

$$\lambda_j = \frac{\sigma_j^2}{n};$$

- the principal-coordinate matrix is

$$T = ZV = U\Sigma.$$

In particular, the j -th score vector is

$$\mathbf{t}_j = \sigma_j \mathbf{u}_j$$

when $\sigma_j > 0$.

Proof.

Since U is orthogonal,

$$U^T U = I.$$

Therefore

$$\begin{aligned} Z^T Z &= (U\Sigma V^T)^T (U\Sigma V^T) \\ &= V\Sigma^T U^T U \Sigma V^T \\ &= V\Sigma^T \Sigma V^T. \end{aligned}$$

Dividing by n gives the displayed orthogonal diagonalization of C . The columns of V are therefore covariance eigenvectors, with eigenvalues σ_j^2/n .

Also,

$$\begin{aligned} ZV &= U\Sigma V^T V \\ &= U\Sigma. \end{aligned}$$

Thus the columns of $U\Sigma$ are the principal-component score vectors.

Keeping the first k principal directions

Let $p = \min(n, d)$ and $1 \leq k \leq p$. Set

$$V_k = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_k], \quad T_k = ZV_k,$$

where $V_k \in \mathbb{R}^{d \times k}$ and $T_k \in \mathbb{R}^{n \times k}$.

The centered-data reconstruction using the first k principal directions is

$$\hat{Z}_k = T_k V_k^T = ZV_k V_k^T.$$

Since the columns of V_k are orthonormal, each row of $\hat{Z}_k$ is the orthogonal projection of the corresponding centered data vector onto

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\};$$

see 4.2.5.

If $U_k \in \mathbb{R}^{n \times k}$ contains the first k left singular vectors and

$$\Sigma_k = \text{diag}(\sigma_1, \dots, \sigma_k) \in \mathbb{R}^{k \times k},$$

then

$$\hat{Z}_k = U_k \Sigma_k V_k^T.$$

The reconstruction has rank at most k . Each d -dimensional centered data point is represented by its k scores and reconstructed from those scores.

The fraction of total centered variation captured is

$$\frac{\lambda_1 + \cdots + \lambda_k}{\lambda_1 + \cdots + \lambda_d} = \frac{\sigma_1^2 + \cdots + \sigma_k^2}{\sigma_1^2 + \cdots + \sigma_p^2},$$

provided the centered data are not all zero.

Activity: The small PCA data set through the SVD

Return to the activity “Principal directions of a small data set”. There,

$$Z = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ -2 & -1 \\ -1 & -2 \end{bmatrix}, \quad V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix},$$

and the covariance eigenvalues are

$$\lambda_1 = \frac{9}{2}, \quad \lambda_2 = \frac{1}{2}.$$

1. Use

$$\lambda_j = \frac{\sigma_j^2}{4}$$

to find the singular values of Z .

2. Compute the principal-coordinate matrix

$$T = ZV.$$

3. Let $\mathbf{t}_1, \mathbf{t}_2$ be the columns of T . Compute

$$\mathbf{u}_1 = \frac{\mathbf{t}_1}{\sigma_1}, \quad \mathbf{u}_2 = \frac{\mathbf{t}_2}{\sigma_2},$$

and verify that $\mathbf{u}_1, \mathbf{u}_2$ are orthonormal.

4. Set

$$U_2 = [\mathbf{u}_1 \quad \mathbf{u}_2], \quad \Sigma_2 = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}.$$

Explain why

$$Z = U_2 \Sigma_2 V^T.$$

5. Compute the reconstruction using only the first principal direction:

$$\hat{Z}_1 = \mathbf{t}_1 \mathbf{v}_1^T.$$

6. Interpret the rows of $\hat{Z}_1$. How does its first row compare with the projection computed in the earlier PCA activity?
7. Use the singular values to compute the fraction of total centered variation captured by the first principal direction.

Learning outcomes. U6-LO5, U6-LO7, U4-LO3.

Solution.

Since $n = 4$,

$$\sigma_1 = \sqrt{4\lambda_1} = \sqrt{18} = 3\sqrt{2}, \quad \sigma_2 = \sqrt{4\lambda_2} = \sqrt{2}.$$

We compute

$$T = ZV = \frac{1}{\sqrt{2}} \begin{bmatrix} 3 & 1 \\ 3 & -1 \\ -3 & -1 \\ -3 & 1 \end{bmatrix}.$$

Thus

$$\mathbf{t}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 3 \\ 3 \\ -3 \\ -3 \end{bmatrix}, \quad \mathbf{t}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}.$$

Dividing by the singular values gives

$$\mathbf{u}_1 = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \quad \mathbf{u}_2 = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}.$$

Both vectors have norm 1, and

$$\mathbf{u}_1 \cdot \mathbf{u}_2 = 0.$$

Therefore they are orthonormal.

We have

$$T = U_2 \Sigma_2.$$

Since

$$T = ZV$$

and V is orthogonal,

$$Z = TV^T = U_2 \Sigma_2 V^T.$$

This factorization uses the two left singular vectors needed to reconstruct Z .

Keeping only the first principal direction gives

$$\hat{Z}_1 = \mathbf{t}_1 \mathbf{v}_1^T = \begin{bmatrix} 3/2 & 3/2 \\ 3/2 & 3/2 \\ -3/2 & -3/2 \\ -3/2 & -3/2 \end{bmatrix}.$$

Each row is the projection of the corresponding centered data vector onto

$$\text{span}\{\mathbf{v}_1\}.$$

In particular, the first row is

$$\begin{bmatrix} 3/2 & 3/2 \end{bmatrix},$$

which is the transpose of the projected centered vector

$$\hat{\mathbf{z}}_1 = \begin{bmatrix} 3/2 \\ 3/2 \end{bmatrix}$$

found in the earlier PCA activity.

Finally,

$$\frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} = \frac{18}{18 + 2} = \frac{9}{10}.$$

Thus the first principal direction captures 90% of the total centered variation.

6.7 Coding recap

6.7.1 Linked notebook

Run Lab U6: Constraints, PCA, and SVD. The lab checks one- and several-constraint Lagrange conditions, connects quadratic forms with maximum stretch, computes PCA from covariance and from the SVD, compares PCA with linear regression, and reads rank and the four fundamental subspaces from an SVD.

For a quick reference on arrays and shapes, matrix products, numerical checks, eigenvalue and SVD commands, and plotting, see the programming appendix sections A.2, A.4, A.9, A.10, and A.14.

6.7.2 Review activities

These activities interpret Python code and output for PCA and SVD: centering data, covariance matrices, principal directions, reconstruction, reduced-SVD shapes, and singular-vector identities. Reopen the definition “First principal direction” or the definition “Singular value decomposition” as needed.

Activity: Reading a PCA computation

The following code repeats part of the activity “Principal directions of a small data set”.

```
import numpy as np

np.set_printoptions(precision=3, suppress=True)

X = np.array([
    [4.0, 3.0],
    [3.0, 4.0],
    [0.0, 1.0],
```

```

    [1.0, 0.0],
])

xbar = X.mean(axis=0)
Z = X - xbar
C = Z.T @ Z / len(X)

eigvals, V = np.linalg.eigh(C)
order = np.argsort(eigvals)[::-1]
eigvals = eigvals[order]
V = V[:, order]

v1 = V[:, 0]
scores1 = Z @ v1
Zhat1 = np.outer(scores1, v1)
R1 = Z - Zhat1

(
    xbar,
    C,
    eigvals,
    V,
    Zhat1,
    np.allclose(R1 @ v1, np.zeros(len(X))),
    V.T @ C @ V,
)

```

Output:

```

(array([2., 2.]),
 array([[2.5, 2. ],
        [2. , 2.5]]),
 array([4.5, 0.5]),
 array([[ 0.707, -0.707],
        [ 0.707,  0.707]]),
 array([[ 1.5,  1.5],
        [ 1.5,  1.5],
        [-1.5, -1.5],
        [-1.5, -1.5]]),
 True,
 array([[4.5, 0. ],
        [0. , 0.5]]))

```

1. Give the shapes of X, xbar, Z, C, V, scores1, and Zhat1.

2. What mathematical objects are stored in `xbar` and `C`?
3. Why does the code reverse the order returned by `np.linalg.eigh`?
4. What do the columns of `V` represent?
5. Interpret `scores1`, `Zhat1`, and `R1`.
6. What geometric condition is checked by `np.allclose(R1 @ v1, np.zeros(len(X)))`?
7. Another valid numerical output could replace a column of `V` by its negative. What would happen to the corresponding scores and reconstructions?
8. Interpret the final diagonal matrix. What fraction of the total centered variation is captured by the first principal direction?

Learning outcomes. U6-LO5.

Solution.

The shapes are

$$X \in \mathbb{R}^{4 \times 2}, \quad \bar{\mathbf{x}} \in \mathbb{R}^2, \quad Z \in \mathbb{R}^{4 \times 2},$$

$$C \in \mathbb{R}^{2 \times 2}, \quad V \in \mathbb{R}^{2 \times 2}, \quad \mathbf{t}_1 \in \mathbb{R}^4, \quad \hat{Z}_1 \in \mathbb{R}^{4 \times 2}.$$

The array `xbar` stores the sample mean

$$\bar{\mathbf{x}} = \begin{bmatrix} 2 \\ 2 \end{bmatrix},$$

while `C` stores the covariance matrix

$$C = \begin{bmatrix} 5/2 & 2 \\ 2 & 5/2 \end{bmatrix}.$$

The command `np.linalg.eigh` returns the eigenvalues in increasing order. The reversal places the largest eigenvalue first. The first column of `V` is therefore a first principal direction, and the second column is a second principal direction.

The vector `scores1` stores

$$\mathbf{t}_1 = Z\mathbf{v}_1.$$

The rows of `Zhat1` are the projections of the centered data vectors onto

$$\text{span}\{\mathbf{v}_1\}.$$

The matrix $R1$ stores the corresponding reconstruction residuals.

The product $R1 @ v1$ computes the dot product of each residual row with $v1$. The value `True` checks that every reconstruction residual is numerically perpendicular to the first principal direction.

Replacing v_1 by $-v_1$ reverses the signs of its scores. However,

$$(-t_i)(-v_1) = t_i v_1,$$

so the projected points and reconstruction residuals do not change. The principal axis is the same.

The final output says that covariance in the principal-coordinate system is

$$V^T C V = \begin{bmatrix} 4.5 & 0 \\ 0 & 0.5 \end{bmatrix}.$$

The diagonal entries are the captured variations in the two principal directions. The first direction captures

$$\frac{4.5}{4.5 + 0.5} = \frac{9}{10}$$

of the total centered variation.

Activity: Reading SVD shapes

The following code computes a reduced SVD.

```
import numpy as np

A = np.array([[ 1., -1.],
              [-2.,  2.],
              [ 2., -2.]])

U, s, Vt = np.linalg.svd(A, full_matrices=False)

A.shape, U.shape, s.shape, Vt.shape
```

Output:

```
((3, 2), (3, 2), (2,), (2, 2))
```

1. What is the shape of A ?
2. How many singular values are returned?
3. Why is $s.shape$ equal to $(2,)$ rather than $(2, 2)$?

4. What matrix does `Vt` represent?
5. What does `full_matrices=False` do in this example?

Solution.

The matrix A is 3×2 . The reduced SVD returns two singular values because

$$\min(3, 2) = 2.$$

The array `s` stores only the diagonal entries of Σ , not the full diagonal matrix, so its shape is $(2,)$. The array `Vt` represents V^T . With `full_matrices=False`, NumPy returns the reduced SVD shapes needed for reconstruction:

$$U \in \mathbb{R}^{3 \times 2}, \quad s \in \mathbb{R}^2, \quad V^T \in \mathbb{R}^{2 \times 2}.$$

Activity: Checking a singular-vector identity in code

For an SVD

$$A = U\Sigma V^T,$$

the identity

$$A\mathbf{v}_i = \sigma_i \mathbf{u}_i$$

is one of the main ways to read the factors. The following code checks this identity for a diagonal stretch.

```
A = np.array([[3., 0.],
               [0., 1.]])

U, s, Vt = np.linalg.svd(A, full_matrices=False)

i = 0
v = Vt.T[:, i]
lhs = A @ v
rhs = s[i] * U[:, i]

lhs, rhs, np.allclose(lhs, rhs)
```

Output:

```
(array([3., 0.]), array([3., 0.]), True)
```

1. Which vector is stored in v ?
2. What does `lhs` compute?
3. What does `rhs` compute?
4. Why does `np.allclose(lhs, rhs)` return `True`?
5. Which stretch factor appears in this computation?

Solution.

The vector v is the first right singular vector v_1 . The array `lhs` computes Av_1 . The array `rhs` computes $\sigma_1 u_1$. The value `True` means that the numerical computation agrees with

$$Av_1 = \sigma_1 u_1.$$

The stretch factor is

$$\sigma_1 = 3.$$

6.8 Exercises

Learning outcomes. The labels below identify the learning outcomes for each exercise group. Individual problems may also involve earlier outcomes.

6.8.1 Lagrange multipliers

Learning outcomes. U6-LO1, U6-LO2.

- OpenStax 4.8 #358
- OpenStax 4.8 #360
- OpenStax 4.8 #362
- OpenStax 4.8 #364
- OpenStax 4.8 #366
- OpenStax 4.8 #368
- OpenStax 4.8 #370
- OpenStax 4.8 #372
- OpenStax 4.8 #374

- OpenStax 4.8 #376
- OpenStax 4.8 #378
- OpenStax 4.8 #380
- OpenStax 4.8 #382
- OpenStax 4.8 #384
- OpenStax 4.8 #386
- OpenStax 4.8 #388
- OpenStax 4.8 #390
- OpenStax 4.8 #392

6.8.2 Singular value decomposition

Learning outcomes. U6-LO4, U6-LO6, U6-LO7.

- Nicholson 8.6.9(b)

6.8.3 Additional exercises

These exercises connect absolute extrema, Lagrange multipliers with one and several constraints, quadratic forms, maximum stretch, PCA, SVD, rank, and the four fundamental subspaces. Solution sketches are collected in Appendix A.6.

Additional exercise: Choosing candidates and equations

For each problem below, state the equations used to find candidates and say which candidates must be compared. Do not solve the equations.

1. Find the absolute extrema of a differentiable function $f(x, y)$ on the rectangle $D = [a, b] \times [c, d]$.
2. Find extrema of $f(\mathbf{x})$ subject to

$$g(\mathbf{x}) = k,$$

assuming $\nabla g \neq \mathbf{0}$ on the feasible set.

3. Find extrema of $f(\mathbf{x})$ subject to

$$g_1(\mathbf{x}) = k_1, \quad g_2(\mathbf{x}) = k_2,$$

assuming the two constraint gradients are linearly independent on the feasible set.

4. Find the absolute extrema of $f(\mathbf{x})$ on the closed bounded region

$$D = \{\mathbf{x} : g(\mathbf{x}) \leq k\},$$

assuming $\nabla g \neq \mathbf{0}$ on its boundary.

Learning outcomes. U6-LO1, U6-LO2, U6-LO3.

Additional exercise: One quadratic form, three readings

Let

$$B = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \quad C = \frac{1}{2}B.$$

1. Find the eigenvalues of B and an orthonormal basis of eigenvectors.
2. Find the absolute maximum and minimum values of

$$q_B(\mathbf{x}) = \mathbf{x}^T B \mathbf{x}$$

subject to

$$\|\mathbf{x}\| = 1.$$

3. Check that

$$A^T A = B.$$

Find the singular values and right singular directions of A . What is

$$\max_{\|\mathbf{x}\|=1} \|A\mathbf{x}\|?$$

4. Suppose C is the covariance matrix from the activity “Principal directions of a small data set”. Find the first and second principal directions and their captured variations.
5. What fraction of the total centered variation is captured by the first principal direction?
6. Why do the same two directions appear in all three readings?

Learning outcomes. U6-LO4, U6-LO5, U6-LO6, U5-LO4.

Additional exercise: Candidate table on a rectangle

Let

$$f(x, y) = x^2 + y^2 - 2x + 4y$$

on the rectangle

$$D = \{(x, y) : 0 \leq x \leq 3, -3 \leq y \leq 1\}.$$

1. Find the critical point of f in the interior of D .
2. Find the boundary candidates on each of the four edges.
3. Include the four corner points.
4. Make a candidate table with the source of each candidate and the value of f .
5. Find the absolute maximum and absolute minimum of f on D .

Learning outcomes. U6-LO1.

Additional exercise: Parallel gradients on a circle

Let

$$f(x, y) = 3x + 4y$$

and constrain the input to the circle

$$x^2 + y^2 = 25.$$

1. Write the constraint as $g(x, y) = 25$.
2. Solve

$$\nabla f = \lambda \nabla g.$$

3. Find the constrained critical points.
4. Evaluate f at the constrained critical points.
5. Which point gives the absolute maximum? Which point gives the absolute minimum?
6. Explain geometrically why ∇f is parallel to ∇g at those points.

Learning outcomes. U6-LO2.

Additional exercise: Quadratic form on the unit circle

Let

$$B = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$$

and

$$q(\mathbf{x}) = \mathbf{x}^T B \mathbf{x}.$$

Write

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

1. Write $q(\mathbf{x})$ in terms of x and y .
2. Maximize $q(\mathbf{x})$ subject to

$$\|\mathbf{x}\| = 1.$$

3. Minimize $q(\mathbf{x})$ subject to

$$\|\mathbf{x}\| = 1.$$

4. Which unit vectors give the extreme values?
5. How do the answers relate to eigenvalues and eigenvectors of B ?

Learning outcomes. U6-LO4.

Additional exercise: Maximum stretch of a matrix

Let

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

1. Compute $A^T A$.
2. Find the eigenvalues of $A^T A$.
3. Find the singular values of A .
4. Find a unit right singular vector corresponding to the largest singular value.

5. What is

$$\max_{\|\mathbf{x}\|=1} \|A\mathbf{x}\|?$$

6. Which input direction is stretched most?

Learning outcomes. U6-LO4.

Additional exercise: Reading an SVD

Suppose

$$A = U\Sigma V^T,$$

where

$$U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix},$$

and

$$V = \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \end{bmatrix}.$$

Thus the right singular vectors are

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}.$$

1. What are the singular values of A ?
2. What is $\text{rank}(A)$?
3. Compute $A\mathbf{v}_1$, $A\mathbf{v}_2$, and $A\mathbf{v}_3$ using the SVD.
4. Which input direction is forgotten?
5. Which input directions are transmitted?

Learning outcomes. U6-LO6, U6-LO7.

Additional exercise: Fundamental subspaces from an SVD

Use the SVD data from the previous checkpoint.

1. Give an orthonormal basis for $\text{row}(A)$.
2. Give an orthonormal basis for $\text{null}(A)$.
3. Give an orthonormal basis for $\text{col}(A)$.
4. Give an orthonormal basis for $\text{null}(A^T)$.
5. Check that the dimensions agree with rank-nullity.

Learning outcomes. U6-LO7.

Additional exercise: Two constraints on a sphere

Let

$$f(x, y, z) = x - y.$$

Find the absolute extrema of f subject to

$$x^2 + y^2 + z^2 = 1$$

and

$$x + y + z = 0.$$

1. Describe the feasible set geometrically. Why do absolute extrema exist?
2. Show that the gradients of the two constraints are linearly independent at every feasible point.
3. Form the Lagrangian using multipliers λ and μ .
4. Write and solve the Lagrange equations.
5. Evaluate f at the candidates and identify the absolute maximum and minimum.

Learning outcomes. U6-LO3.

Additional exercise: Principal directions of a centered data set

The rows of

$$X = \begin{bmatrix} 4 & 2 \\ 2 & 4 \\ -2 & 0 \\ 0 & -2 \end{bmatrix}$$

are four data points in $\mathbb{R}^2$.

1. Compute the mean $\bar{\mathbf{x}}$ and the centered data matrix Z .
2. Compute the covariance matrix

$$C = \frac{1}{4} Z^T Z.$$

3. Find the eigenvalues of C and corresponding orthonormal eigenvectors. Choose $\mathbf{v}_1$ for the larger eigenvalue and $\mathbf{v}_2$ for the smaller eigenvalue.
4. Form

$$V = [\mathbf{v}_1 \quad \mathbf{v}_2]$$

and compute the principal-coordinate matrix

$$T = ZV.$$

5. Compute

$$V^T C V.$$

6. What fraction of the total centered variation is captured by the first principal direction?

Learning outcomes. U6-LO5.

Additional exercise: PCA reconstruction and regression

Use the data and principal directions from the checkpoint “Principal directions of a centered data set”. The first centered data vector is

$$\mathbf{z}_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix},$$

and

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

1. Compute the first principal-component score

$$t_1 = \mathbf{v}_1^T \mathbf{z}_1.$$

2. Compute the projected centered vector

$$\hat{\mathbf{z}}_1 = t_1 \mathbf{v}_1$$

and the reconstruction residual

$$\mathbf{r}_1 = \mathbf{z}_1 - \hat{\mathbf{z}}_1.$$

3. Check that

$$\mathbf{r}_1 \cdot \mathbf{v}_1 = 0.$$

4. Find the reconstructed original data point

$$\hat{\mathbf{x}}_1 = \bar{\mathbf{x}} + \hat{\mathbf{z}}_1.$$

5. Give an equation for the PCA line

$$\bar{\mathbf{x}} + \text{span}\{\mathbf{v}_1\}.$$

6. For the regression of y on x , use

$$m = \frac{C_{12}}{C_{11}}, \quad b = \bar{y} - m\bar{x}$$

to find the regression line.

7. Why are the PCA and regression lines different? Describe the residual direction minimized by each method.

Learning outcomes. U6-LO5, U4-LO5.

Additional exercise: Redundant features revisited by singular values

Return to the Unit 2 redundant-feature matrix

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}.$$

The third column is the sum of the first two columns.

1. Verify that

$$X \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \mathbf{0}.$$

2. What does this say about $\text{null}(X)$?
3. What does this say about the rank of X ?
4. What singular value should appear because of this redundancy?
5. If $X\mathbf{c} = \mathbf{y}$, what is

$$X \left(\mathbf{c} + t \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right)?$$

6. Explain how this revisits the Unit 2 idea of nonunique coefficient vectors.

Learning outcomes. U2-LO6, U6-LO7.

Additional exercise: PCA from an SVD

Return to the centered data matrix from the checkpoint “Principal directions of a centered data set”. Suppose

$$Z = U_2 \Sigma_2 V^T,$$

where

$$\mathbf{u}_1 = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \quad \mathbf{u}_2 = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix},$$

$$U_2 = [\mathbf{u}_1 \quad \mathbf{u}_2], \quad \Sigma_2 = \begin{bmatrix} 4\sqrt{2} & 0 \\ 0 & 2\sqrt{2} \end{bmatrix},$$

and

$$V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

1. Use

$$\lambda_j = \frac{\sigma_j^2}{4}$$

to recover the two covariance eigenvalues.

2. Which columns of V are the first and second principal directions?
3. Compute the principal-coordinate matrix

$$T = U_2 \Sigma_2.$$

4. Compute the reconstruction using only the first principal direction:

$$\hat{Z}_1 = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T.$$

5. What fraction of the total centered variation is captured by the first principal direction?
6. What do the two columns of T represent?

Learning outcomes. U6-LO5, U6-LO7.

6.9 Unit 6 highlights

6.9.1 Mathematical quick reference

Absolute extrema

Extreme value theorem. A continuous real-valued function on a nonempty closed bounded subset of $\mathbb{R}^n$ attains an absolute maximum and an absolute minimum.

Candidate procedure. Find interior critical points, examine interior points where differentiability fails, and optimize on every boundary piece. Include endpoints, corners, and intersections of boundary pieces. Compare the function values at all candidates. When a boundary is parametrized by $\mathbf{r}(t)$, optimize $f(\mathbf{r}(t))$ over its parameter domain.

Lagrange multipliers

One equality constraint. At a constrained local extremum of f on $g(\mathbf{x}) = k$, with f, g continuously differentiable nearby and $\nabla g \neq \mathbf{0}$, there is a scalar λ such that

$$\nabla f(\mathbf{x}) = \lambda \nabla g(\mathbf{x}), \quad g(\mathbf{x}) = k.$$

Several equality constraints. For $g_i(\mathbf{x}) = k_i$, $i = 1, \dots, s$, with continuously differentiable functions and independent constraint gradients at the extremum,

$$\nabla f(\mathbf{x}) = \sum_{i=1}^s \lambda_i \nabla g_i(\mathbf{x}), \quad g_i(\mathbf{x}) = k_i \quad (i = 1, \dots, s).$$

Lagrangian. Package both the gradient equation and all constraints as

$$L(\mathbf{x}, \lambda) = f(\mathbf{x}) - \sum_{i=1}^s \lambda_i (g_i(\mathbf{x}) - k_i), \quad \nabla_{\mathbf{x}} L = \mathbf{0}, \quad L_{\lambda_i} = 0.$$

Use the conditions. Solve the system for feasible candidates and compare objective values. Examine separately any points where constraint gradients fail the hypotheses, and any additional boundary restrictions. The multiplier equations are necessary under these hypotheses, not sufficient for an extremum.

Quadratic forms and maximum stretch

Unit-sphere extrema. For a real symmetric B ,

$$\min_{\|\mathbf{x}\|=1} \mathbf{x}^T B \mathbf{x} = \lambda_{\min}(B), \quad \max_{\|\mathbf{x}\|=1} \mathbf{x}^T B \mathbf{x} = \lambda_{\max}(B).$$

Unit eigenvectors for the extreme eigenvalues attain the extrema.

Matrix stretch. For $A \in \mathbb{R}^{m \times n}$,

$$\|A\mathbf{x}\|^2 = \mathbf{x}^T A^T A \mathbf{x}, \quad \max_{\|\mathbf{x}\|=1} \|A\mathbf{x}\| = \sqrt{\lambda_{\max}(A^T A)} = \sigma_1.$$

The positive singular values are square roots of the positive eigenvalues of $A^T A$.

Principal directions and orthogonal reconstruction

Centering and covariance. For $X \in \mathbb{R}^{n \times d}$ with data rows $\mathbf{x}_i^T$, let $\mu = \frac{1}{n} \sum_i \mathbf{x}_i$ and let Z have rows $(\mathbf{x}_i - \mu)^T$. This book uses

$$C = \frac{1}{n} Z^T Z.$$

The symmetric matrix C has nonnegative eigenvalues.

Principal directions. Choose orthonormal eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_d$ of C , ordered by $\lambda_1 \geq \dots \geq \lambda_d \geq 0$. The first direction maximizes $\mathbf{v}^T C \mathbf{v}$ over unit vectors; each later direction maximizes it subject to orthogonality to earlier directions.

Scores and reconstruction. Put $V_k = [\mathbf{v}_1 \ \dots \ \mathbf{v}_k]$. Then

$$T_k = ZV_k, \quad \hat{Z} = T_k V_k^T = ZV_k V_k^T, \quad R = Z - \hat{Z}, \quad RV_k = 0.$$

Rows of $\hat{Z}$ are projections onto the chosen subspace. Add the mean vector to recover approximations in the original coordinates.

Captured variation. When total variation is positive, the fraction retained by the first k directions is

$$\frac{\lambda_1 + \cdots + \lambda_k}{\lambda_1 + \cdots + \lambda_d}.$$

Maximizing the sum of squared scores is equivalent to minimizing the sum of squared reconstruction residuals. A repeated eigenvalue permits different orthonormal choices within its eigenspace.

Singular value decomposition

Full SVD. Every real $m \times n$ matrix has a factorization

$$A = U\Sigma V^T, \quad U \in \mathbb{R}^{m \times m}, \quad \Sigma \in \mathbb{R}^{m \times n}, \quad V \in \mathbb{R}^{n \times n},$$

with U, V orthogonal and rectangular diagonal Σ . For $p = \min(m, n)$, its diagonal entries satisfy $\sigma_1 \geq \cdots \geq \sigma_p \geq 0$.

Construction and action. Choose an orthonormal eigenvector basis of $A^T A$ for the columns of V . For each positive singular value, set $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$; complete these to an orthonormal basis of $\mathbb{R}^m$. If $r = \text{rank}(A)$, then

$$A\mathbf{v}_i = \sigma_i \mathbf{u}_i \quad (1 \leq i \leq r), \quad A\mathbf{v}_i = \mathbf{0} \quad (r < i \leq n),$$

$$A = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^T.$$

Do not divide by a zero singular value.

Fundamental subspaces. In a full SVD,

$$\text{col}(A) = \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_r\}, \quad \text{null}(A^T) = \text{span}\{\mathbf{u}_{r+1}, \dots, \mathbf{u}_m\},$$

$$\text{row}(A) = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_r\}, \quad \text{null}(A) = \text{span}\{\mathbf{v}_{r+1}, \dots, \mathbf{v}_n\}.$$

Empty lists give the zero subspace. The number of positive singular values is the rank.

PCA from an SVD. For centered $Z = U\Sigma V^T$,

$$C = V \left(\frac{1}{n} \Sigma^T \Sigma \right) V^T, \quad T = ZV = U\Sigma.$$

Thus principal directions are right singular vectors, and corresponding covariance eigenvalues are σ_i^2/n , with additional zeros when needed.

6.9.2 Common mistakes

Ignoring boundaries or constraint equations; treating every multiplier solution as an extremum; failing to compare objective values; applying the multiplier theorem when constraint gradients are dependent; or requiring the objective gradient to be parallel to every constraint gradient separately.

Dropping the unit-length condition in a quadratic-form extremum; forgetting to center before forming covariance; selecting the smallest eigenvalue for the first principal direction; confusing a direction, a score, and a reconstructed vector; dividing by a zero singular value; interchanging left and right singular vectors; or using an incomplete set of singular vectors to describe a null space.

6.9.3 Connections

Unit 1 supplies vectors and matrix actions. Unit 2 supplies rank and fundamental spaces. Unit 3 supplies gradients, level sets, and closed bounded regions. Unit 4 supplies orthogonal projection. Unit 5 supplies symmetric diagonalization and quadratic forms. Unit 7 extends orthogonality and projection to polynomial spaces.

Chapter 7

Abstract vector spaces and polynomial approximation

Abstract vector spaces

Earlier units worked mostly with vectors in $\mathbb{R}^n$. This unit introduces vector spaces and inner product spaces as abstract settings where the same algebraic and geometric ideas still work. Matrices and polynomials are the main examples. Polynomial spaces then serve as a synthesis example: after an inner product is chosen, projection connects least squares, Taylor approximation, and Gram-Schmidt.

Big question. What remains true when vectors are matrices or polynomials, and how does an inner product determine projection and best approximation?

Learning outcomes. By the end of this unit, students should be able to:

- U7-LO1. Verify vector-space properties and produce counterexamples when a property fails.
- U7-LO2. Work in nonstandard vector spaces, especially matrix spaces and polynomial spaces, using subspaces, span, independence, bases, dimension, and coordinates.
- U7-LO3. Verify inner-product properties, compute norms and distances, and explain how changing the inner product changes geometry.
- U7-LO4. Derive projection equations in finite-dimensional inner product spaces, including Gram-matrix equations $Gc = b$.
- U7-LO5. Compare Taylor, continuous least-squares, and sampled least-squares approximations by their residual conditions and error measures.
- U7-LO6. Apply Gram-Schmidt in polynomial spaces and interpret the resulting orthogonal polynomial basis.

Toolbox skills. Verify vector-space axioms; test subspace properties; write bases, dimensions, and coordinate vectors in matrix and polynomial spaces; test inner-product properties; compute norms and distances from inner products; form Gram matrices; solve projection equations; compare residual conditions; and run Gram-Schmidt.

7.1 Abstract vector spaces

Earlier units worked mostly with vectors in $\mathbb{R}^n$. The same algebraic rules also make sense for matrices and polynomials. This section gives the abstract definition so that later courses can use the same language in many settings: vectors may be columns, matrices, polynomials, or other objects.

Definition 7.1.1.

A real *vector space* consists of a nonempty set V of objects (which we will call *vectors*) that can be added together, and multiplied by a real number (called a *scalar* in this context) and for which the ten axioms below hold. If $\mathbf{v}$ and $\mathbf{w}$ are two vectors in V , their sum is denoted $\mathbf{v} + \mathbf{w}$, and the scalar product of $\mathbf{v}$ by a real number a is denoted by $a\mathbf{v}$. The axioms of a vector space are described below:

(A1) Closure Under Addition: If $\mathbf{u}$ and $\mathbf{v}$ are any elements in V , then $\mathbf{u} + \mathbf{v}$ is in V .

(A2) Commutativity of Addition: For any $\mathbf{u}, \mathbf{v} \in V$, $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.

(A3) Associativity of Addition: For any $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$, $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$.

(A4) Existence of a Zero Vector: There exists a *zero vector* $\mathbf{0}$ in V , such that $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$ for any $\mathbf{u} \in V$.

(A5) Existence of Additive Inverses For each $\mathbf{v} \in V$ there exists $-\mathbf{v}$ in V such that

$$\mathbf{v} + (-\mathbf{v}) = -\mathbf{v} + \mathbf{v} = \mathbf{0}$$

. The vector $-\mathbf{v}$ is the *negative* or *additive inverse* of $\mathbf{v}$.

(S1) Closure Under Scalar Multiplication If $\mathbf{v}$ is any element of V and a is any real number, then $a\mathbf{v}$ is in V .

(S2) Distributive Law 1 For any $\mathbf{u}, \mathbf{v} \in V$ and any $a \in \mathbb{R}$, $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$.

(S3) Distributive Law 2 For any $\mathbf{v} \in V$ and any $a, b \in \mathbb{R}$, $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$.

(S4) Associativity of Scalar Multiplication For any $\mathbf{v} \in V$ and $a, b \in \mathbb{R}$, $a(b\mathbf{v}) = (ab)\mathbf{v}$.

(S5) $1\mathbf{v} = \mathbf{v}$ for any $\mathbf{v} \in V$.

The operation $\mathbf{v} + \mathbf{w}$ is called *vector addition* and the operation $a\mathbf{v}$ is called *scalar multiplication*.

Note

In order to be clear about the operations used, we will often denote newly defined addition and scalar multiplication operations using the symbols $\oplus$ and $\odot$, respectively.

Note

We often use the phrase “vector space”, dropping the “real” and assuming our scalars are real numbers in this course. However, it is possible to define complex vector spaces, rational vector spaces, and more, using different systems of scalars than the real numbers.

Note

From now on, a “vector” will mean an element of a vector space, not necessarily a column vector or line segment.

The main examples in this unit are

$$\mathbb{R}^n, \quad M_{m,n}, \quad \mathcal{P}_{\leq d}.$$

Here $M_{m,n}$ is the set of $m \times n$ matrices, and

$$\mathcal{P}_{\leq d} = \{a_0 + a_1x + \cdots + a_dx^d : a_0, \dots, a_d \in \mathbb{R}\}.$$

In $\mathbb{R}^n$, addition and scalar multiplication are entrywise. In $M_{m,n}$, they are entrywise. In $\mathcal{P}_{\leq d}$, they are polynomial addition and scalar multiplication. The set $\mathcal{P}$ of all polynomials is also a vector space, but it is infinite-dimensional and will not be our main example.

Activity

Consider the vector space M_{22} of all 2×2 matrices.

Task

Prove that M_{22} is closed under addition and scalar multiplication. What is the zero vector in this vector space?

Solution.

For two matrices

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

in M_{22} , their sum is

$$\begin{bmatrix} a+e & b+f \\ c+g & d+h \end{bmatrix}$$

, which is an element of M_{22} . The product of a scalar k with a matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is

$$k \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$$

, which is an element of M_{22} .

Task

What is the zero vector in M_{22} ?

Solution.

The zero vector in M_{22} is the zero matrix

$$\mathbf{0} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

, because $X + \mathbf{0} = \mathbf{0} + X$ for any $X \in M_{22}$.

Activity

Let's walk through the proof that $\mathcal{P}_{\leq 2}$ is a vector space.

Solution.

We go through each axiom one by one:

Let us start by verifying Axiom (A1). We must verify the sum of two polynomials with degree at most two has degree at most two. Let $u(x) = a_0 + a_1x + a_2x^2$ and $v(x) = b_0 + b_1x + b_2x^2$ be two arbitrary polynomials in $\mathcal{P}_{\leq 2}$. Then

$$(u + v)(x) = (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2$$

has degree at most 2, so lies in $\mathcal{P}_{\leq 2}$. Thus Axiom (A1) holds.

Next, we verify Axiom (A2). We must verify that for any two polynomials, $u, v \in \mathcal{P}_{\leq 2}$, $u + v = v + u$. But using the commutativity of real numbers, for any real number x , if $u(x) = a_0 + a_1x + a_2x^2$ and $v(x) = b_0 + b_1x + b_2x^2$, then

$$\begin{aligned}(u + v)(x) &= (a_0 + a_1x + a_2x^2) + (b_0 + b_1x + b_2x^2) \\ &= (b_0 + b_1x + b_2x^2) + (a_0 + a_1x + a_2x^2) \\ &= (v + u)(x)\end{aligned}$$

. Thus Axiom (A2) holds.

Next, we verify Axiom (A3). We must verify that for any two polynomials $u, v, w \in \mathcal{P}_{\leq 2}$, $u + (v + w) = (u + v) + w$. But this follows, as above, by using the associativity of real numbers, i.e., that if $u(x) = a_0 + a_1x + a_2x^2$, $v(x) = b_0 + b_1x + b_2x^2$, and $w(x) = c_0 + c_1x + c_2x^2$, then for any $x \in \mathbb{R}$,

$$\begin{aligned}(u + (v + w))(x) &= u(x) + (v(x) + w(x)) \\ &= (a_0 + a_1x + a_2x^2) \\ &\quad + ((b_0 + b_1x + b_2x^2) + (c_0 + c_1x + c_2x^2)) \\ &= (a_0 + b_0 + c_0) + (a_1 + b_1 + c_1)x \\ &\quad + (a_2 + b_2 + c_2)x^2 \\ &= ((a_0 + a_1x + a_2x^2) + (b_0 + b_1x + b_2x^2)) \\ &\quad + (c_0 + c_1x + c_2x^2) \\ &= ((u + v) + w)(x)\end{aligned}$$

. Thus Axiom (A3) holds.

Next, we verify Axiom (A4). We must check that if $\mathbf{0}$ is the zero polynomial, then $\mathbf{0} + v = v$ for any $v \in \mathcal{P}_{\leq 2}$. Now if $v = a_0 + a_1x + a_2x^2$, then for any $x \in \mathbb{R}$,

$$\begin{aligned}
0(x) + v(x) &= (0 + 0x + 0x^2) + (a_0 + a_1x + a_2x^2) \\
&= (0 + a_0) + (0 + a_1)x + (0 + a_2)x^2 \\
&= a_0 + a_1x + a_2x^2 \\
&= v(x)
\end{aligned}$$

Thus Axiom (A4) holds.

Next, we verify Axiom (A5). For each polynomial $u \in \mathcal{P}_{\leq 2}$ given by $u(x) = a_0 + a_1x + a_2x^2$, if we define $(-u)(x) = (-a_0) + (-a_1)x + (-a_2)x^2$, then $u + (-u) = 0$. Indeed

$$\begin{aligned}
(u + (-u))(x) &= (a_0 + a_1x + a_2x^2) + ((-a_0) + (-a_1)x + (-a_2)x^2) \\
&= ((a_0 - a_0) + (a_1 - a_1)x + (a_2 - a_2)x^2) \\
&= 0
\end{aligned}$$

. Thus $u + (-u) = 0$, verifying Axiom (A5).

Next, we verify Axiom (S1). We must verify that for $v \in \mathcal{P}_{\leq 2}$ and $r \in \mathbb{R}$, $rv \in \mathcal{P}_{\leq 2}$. If $u(x) = a_0 + a_1x + a_2x^2$, and a scalar r ,

$$\begin{aligned}
(ru)(x) &= r(a_0 + a_1x + a_2x^2) \\
&= (ra_0) + (ra_1)x + (ra_2)x^2
\end{aligned}$$

. The polynomial ru still has degree at most 2, so $ru \in \mathcal{P}_{\leq 2}$. Thus Axiom (S1) holds.

Next, we verify Axiom (S2). We must verify that for any $u, v \in \mathcal{P}_{\leq 2}$ and $r \in \mathbb{R}$, then $r(u + v) = ru + rv$. If $u(x) = a_0 + a_1x + a_2x^2$ and $v(x) = b_0 + b_1x + b_2x^2$, then

$$\begin{aligned}
(r(u + v))(x) &= r((a_0 + a_1x + a_2x^2) + (b_0 + b_1x + b_2x^2)) \\
&= (ra_0 + rb_0) + (ra_1 + rb_1)x + (ra_2 + rb_2)x^2 \\
&= (ru + rv)(x)
\end{aligned}$$

. Thus Axiom (S2) holds.

Next, we verify Axiom (S3). We must verify that for any $u \in \mathcal{P}_{\leq 2}$ and $r, s \in \mathbb{R}$, $(r+s)u = ru + su$. If $u(x) = a_0 + a_1x + a_2x^2$, then

$$\begin{aligned}
((r + s)u)(x) &= (r + s)(a_0 + a_1x + a_2x^2) \\
&= ((ra_0 + sa_0) + (ra_1 + sa_1)x + (ra_2 + sa_2)x^2) \\
&= (ru + su)(x)
\end{aligned}$$

. Thus Axiom (S3) holds.

Next, we verify Axiom (S4). We must verify that for any $r, s \in \mathbb{R}$ and $u \in \mathcal{P}_{\leq 2}$, $r(su) = (rs)u$. If $u(x) = a_0 + a_1x + a_2x^2$, then

$$\begin{aligned}(r(su))(x) &= r(s(a_0 + a_1x + a_2x^2)) \\ &= (rs)(a_0 + a_1x + a_2x^2) \\ &= ((rs)u)(x)\end{aligned}$$

. Thus Axiom (S4) holds.

Finally, we verify Axiom (S5). We must verify that for any $u \in \mathcal{P}_{\leq 2}$, $1u = u$. If $u(x) = a_0 + a_1x + a_2x^2$, then

$$(1u)(x) = 1(a_0 + a_1x + a_2x^2) = u(x)$$

. Thus Axiom (S5) holds.

Activity

Let V be the set of all ordered pairs of real numbers (x, y) with the operations

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1, 0) \quad \text{and} \quad c \odot (x, y) = (cx, cy)$$

. Show that V is *not* a vector space.

Solution.

The set V , and the operations defined upon it, fail to satisfy Axiom (A4), i.e., no zero vector exists in V . Since $(x_1, y_1) \oplus (x_2, y_2) = (x_1, 0)$, it is impossible for there to exist a vector (x_0, y_0) such that $(x_0, y_0) \oplus (x, y) = (x, y)$ for all ordered pairs (x, y) .

Note that this is not the only axiom that V fails to satisfy. V also fails to satisfy axioms (A2), (A5), and (S3). Verifying that any one of these axioms fail to hold would also constitute a valid argument that V is not a vector space. But a valid argument only needs to check that *one* of the axioms fails to hold.

Activity

Let V be the set of all ordered pairs of real numbers (x, y) with the operations

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2, y_1 + y_2) \quad \text{and} \quad c \odot (x, y) = (cx, y)$$

. Show that V is not a vector space.

Solution.

Under this definition of addition and multiplication, V fails to satisfy Axiom (S3). We calculate that

$$(a + b) \odot (x, y) = ((a + b)x, y)$$

, and that

$$(a \odot (x, y)) \oplus (b \odot (x, y)) = (ax, y) \oplus (bx, y) = (ax + bx, 2y)$$

. If $y \neq 0$, then the two tuples will not be equal to one another. Thus a *particular counterexample* that shows that Axiom (S3) fails is obtained by taking $a = b = 1$, and $(x, y) = (0, 1)$. Then

$$(1 + 1) \odot (0, 1) = (0, 1)$$

whereas

$$(1 \odot (0, 1)) \oplus (1 \odot (0, 1)) = (0, 2)$$

. These two quantities are evidently not equal to one another, which shows Axiom (S3) fails, and so V is not a vector space.

Activity

Let V be the set of all non-negative real numbers, and define vector addition as

$$\mathbf{v} \oplus \mathbf{u} = |\mathbf{v} - \mathbf{u}|$$

, and scalar multiplication as

$$a \odot \mathbf{v} = |a|\mathbf{v}$$

. Does this structure form a vector space?

Solution.

With this definition, *addition is not associative*, i.e., Axiom (A3) fails. To find a counterexample, we must find three elements of V such that

$$|x - |y - z|| \neq ||x - y| - z|$$

. We might as well start simply by guessing some values for which this equation fails to hold, because often if an algebraic identity fails to hold, it fails to hold for *most* inputs. Choose $x = 3$, $y = 1$, and $z = 2$. Then

$$(x \oplus y) \oplus z = ||3 - 1| - 2| = |2 - 2| = 0$$

, whereas

$$x \oplus (y \oplus z) = |3 - |1 - 2|| = |3 - 1| = 2$$

. Thus we have found a particular counterexample that shows that addition is not associative.

Below we list several facts about the interaction of addition and multiplication that hold in any vector space.

Theorem 7.1.2.

Let V be a vector space. For any $\mathbf{v} \in V$ and $a \in \mathbb{R}$,

- $0\mathbf{u} = \mathbf{0}$.
- $a\mathbf{0} = \mathbf{0}$.
- If $a\mathbf{v} = \mathbf{0}$, then $a = 0$ or $\mathbf{v} = \mathbf{0}$.
- $(-1)\mathbf{v} = -\mathbf{v}$.
- $(-a)\mathbf{v} = -(a\mathbf{v}) = a(-\mathbf{v})$.

The point is not that every vector looks like an arrow. The point is that the same algebraic rules let us reuse span, independence, bases, coordinates, dimension, and linear maps in settings where the objects are matrices or polynomials.

7.2 Inner product spaces

Unit 4 used the dot product on $\mathbb{R}^n$ to define length, distance, orthogonality, projection, and least squares. An inner product space keeps exactly the rules needed for those ideas. This section introduces the abstract definition and keeps three examples in view: $\mathbb{R}^n$, polynomial spaces, and matrix spaces.

First, we define inner products.

Definition 7.2.1 (Inner Products).

Let V be a vector space. An *inner product* on V is a function $\langle \cdot, \cdot \rangle$ that assigns to each *ordered pair* of vectors $\mathbf{v}, \mathbf{w}$ in V a *real number* $\langle \mathbf{v}, \mathbf{w} \rangle$ satisfying the following properties:

1. $\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{w}, \mathbf{v} \rangle$ for all $\mathbf{v}, \mathbf{w}$ in V .
2. $\langle \mathbf{v} + \mathbf{w}, \mathbf{u} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{w}, \mathbf{u} \rangle$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in V .
3. $\langle r\mathbf{v}, \mathbf{w} \rangle = r\langle \mathbf{v}, \mathbf{w} \rangle$ for all $\mathbf{v}$ and $\mathbf{w}$ in V and r in $\mathbb{R}$.
4. $\langle \mathbf{v}, \mathbf{v} \rangle > 0$ for all $\mathbf{v} \neq \mathbf{0}$ in V .

The vector space V together with $\langle \cdot, \cdot \rangle$ is called an *inner product space*.

Theorem 7.2.2.

For an inner product $\langle \cdot, \cdot \rangle$ on a vector space V , the following additional properties automatically hold:

1. $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$ for any $\mathbf{u}, \mathbf{v}$ in V .
2. $\langle \mathbf{u}, r\mathbf{v} \rangle = r\langle \mathbf{u}, \mathbf{v} \rangle$ for any $\mathbf{u}, \mathbf{v}$ in V , and r in $\mathbb{R}$.
3. $\langle \mathbf{v}, \mathbf{0} \rangle = 0 = \langle \mathbf{0}, \mathbf{v} \rangle$
4. $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

The dot product on $\mathbb{R}^n$ is an inner product. We will often call the dot product the *standard inner product*.

Activity

Let

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

be vectors in $\mathbb{R}^2$, and define

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 - u_2 v_1 - u_1 v_2 + 3u_2 v_2$$

Show this definition satisfies the 4th property of 7.2.2, i.e., that $\langle \mathbf{u}, \mathbf{u} \rangle > 0$ for $\mathbf{u} \neq \mathbf{0}$.

Solution.

We calculate that

$$\begin{aligned} \langle \mathbf{u}, \mathbf{u} \rangle &= u_1 u_1 - u_2 u_1 - u_1 u_2 + 3u_2 u_2 \\ &= u_1^2 - 2u_1 u_2 + 3u_2^2 \\ &= u_1^2 - 2u_1 u_2 + u_2^2 + 2u_2^2 \\ &= (u_1 - u_2)^2 + 2u_2^2 \end{aligned}$$

This quantity can only be zero if $u_1 - u_2 = 0$ and $u_2 = 0$. But this is only possible if $u_1 = 0$ and $u_2 = 0$, i.e., $\mathbf{u} = \mathbf{0}$.

Theorem 7.2.3.

Let $E = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ be the standard basis for $\mathbb{R}^n$, and let $\langle \cdot, \cdot \rangle$ be any inner product on $\mathbb{R}^n$. Let $c_{ij} = \langle \mathbf{e}_i, \mathbf{e}_j \rangle$ and consider the matrix $C = [c_{ij}]$. Then

1. C is a *symmetric matrix*.
2. $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T C \mathbf{v}$ for all $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^n$.

Definition 7.2.4.

Given an inner product $\langle \cdot, \cdot \rangle$ on $\mathbb{R}^n$, the symmetric matrix C analyzed in 7.2.3 is called the *Gram matrix* associated with the inner product.

Remark

If a symmetric $n \times n$ matrix C satisfies $\mathbf{x}^T C \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$, then C defines an inner product on $\mathbb{R}^n$. Moreover, these are the *only* symmetric matrices that define inner products. They are called *positive definite matrices*.

Definition 7.2.5 (Positive Definite Matrices).

An $n \times n$ matrix A is *positive definite* if it is symmetric, and if $\mathbf{x}^T A \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

Activity

Consider the inner product on $\mathbb{R}^2$ defined by

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 - u_2 v_1 - u_1 v_2 + 3u_2 v_2$$

. Find C such that $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T C \mathbf{v}$.

Solution.

We employ 7.2.3. We compute that if $C = [c_{ij}]$, then

$$c_{11} = \langle \mathbf{e}_1, \mathbf{e}_1 \rangle = (1)(1) - (0)(1) - (1)(0) + 3(0)(0) = 1$$

$$c_{12} = \langle \mathbf{e}_1, \mathbf{e}_2 \rangle = (1)(0) - (0)(0) - (1)(1) + 3(0)(1) = -1$$

$$c_{21} = \langle \mathbf{e}_2, \mathbf{e}_1 \rangle = (0)(1) - (1)(1) - (0)(0) + 3(1)(0) = -1$$

$$c_{22} = \langle \mathbf{e}_2, \mathbf{e}_2 \rangle = (0)(0) - (1)(0) - (0)(1) + 3(1)(1) = 3$$

Therefore,

$$C = \begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix}$$

Definition 7.2.6 (The Norm of a Vector).

In an inner product space, the *norm* of a vector $\mathbf{v}$ is

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}.$$

The *distance* between two vectors $\mathbf{u}$ and $\mathbf{v}$ in an inner product space is

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

Definition 7.2.7 (Unit Vectors).

A vector $\mathbf{v}$ in an inner product space V is called a *unit vector* if $\|\mathbf{v}\| = 1$. This is equivalent to the equation $\langle \mathbf{v}, \mathbf{v} \rangle = 1$. The set of all unit vectors in V is called the *unit sphere* in V .

Theorem 7.2.8.

If $\mathbf{v}$ is a nonzero vector of an inner product space, then

$$\mathbf{u} = \frac{1}{\|\mathbf{v}\|} \mathbf{v}$$

is the unique unit vector that is a positive multiple of $\mathbf{v}$. We often say that $\mathbf{u}$ is the unit vector ‘in the same direction as $\mathbf{v}$ ’.

Definition 7.2.9 (Orthogonal Vectors).

Two vectors $\mathbf{u}$ and $\mathbf{v}$ in an inner product space are *orthogonal* if

$$\langle \mathbf{u}, \mathbf{v} \rangle = 0$$

The next examples show that inner products are not limited to ordinary coordinate vectors. The objects can be polynomials or matrices, as long as the rule for $\langle \cdot, \cdot \rangle$ satisfies the inner-product properties.

7.2.1 Spaces of polynomials

Activity

Let $V = \mathbf{P}_1$ be the vector space of all polynomials of degree at most 1. Define

$$\langle f, g \rangle = \int_0^1 f(x)g(x) dx$$

. Does this definition define an inner product on V ?

Solution.

Let

$$f(x) = a_0 + a_1x \quad g(x) = b_0 + b_1x \quad h(x) = c_0 + c_1x$$

be arbitrary polynomials in $\mathbf{P}_1$.

We begin by checking that property (a) of the definition of an inner product holds, i.e., that $\langle f, g \rangle = \langle g, f \rangle$. Indeed,

$$\langle f, g \rangle = \int_0^1 f(x)g(x) dx = \int_0^1 g(x)f(x) dx = \langle g, f \rangle$$

. To verify that property (b) holds, we check that $\langle f + g, h \rangle = \langle f, h \rangle + \langle g, h \rangle$. We calculate that

$$\begin{aligned} \langle f + g, h \rangle &= \int_0^1 (f(x) + g(x))h(x) dx \\ &= \int_0^1 f(x)h(x) dx + \int_0^1 g(x)h(x) dx \\ &= \langle f, h \rangle + \langle g, h \rangle \end{aligned}$$

To verify that property (c) holds, we must check that for any scalar r , $\langle rf, g \rangle = r\langle f, g \rangle$. We calculate that

$$\langle rf, g \rangle = \int_0^1 (rf(x))g(x) dx = r \int_0^1 f(x)g(x) dx = r\langle f, g \rangle$$

. To verify that property (d) holds, we must check that $\langle f, f \rangle > 0$ if $f \neq 0$. But

$$\begin{aligned} \langle f, f \rangle &= \int_0^1 (a_0 + a_1x)^2 dx \\ &= \int_0^1 (a_0^2 + 2a_0a_1x + a_1^2x^2) dx \\ &= a_0^2 + a_0a_1 + a_1^2/3 \\ &= (a_0 + a_1/2)^2 + a_1^2/12 \end{aligned}$$

. This quantity is positive unless $a_0 + a_1/2 = 0$ and $a_1 = 0$, and these equations only hold simultaneously when $a_0 = 0$ and $a_1 = 0$, i.e., when $f = 0$. Thus property (d) holds. Therefore the definition of the map $\langle, \rangle$ defined above is an inner product.

Activity

Consider P_1 with the inner product

$$\langle p, q \rangle = \int_0^1 p(x)q(x) dx$$

.

Task

Calculate the norm of the polynomial $p(x) = x$.

Solution.

We calculate that

$$\begin{aligned}\|p\|^2 &= \langle p, p \rangle \\ &= \int_0^1 p(x)p(x) \, dx \\ &= \int_0^1 x^2 \, dx \\ &= \left[\frac{x^3}{3} \right]_0^1 \\ &= 1/3\end{aligned}$$

. Therefore, $\|p\| = \sqrt{\frac{1}{3}} = \frac{1}{\sqrt{3}}$.

Task

Calculate the distance between $q_1(x) = 2x + 1$ and $q_2(x) = 3x$.

Solution.

We calculate that $q_1(x) - q_2(x) = (2x + 1) - 3x = -x + 1$, so

$$\begin{aligned}\|q_1 - q_2\|^2 &= \langle q_1 - q_2, q_1 - q_2 \rangle = \int_0^1 (1 - x)^2 \, dx \\ &= (-1/3)(1 - x)^3 \Big|_0^1 \\ &= 1/3\end{aligned}$$

. Therefore, $d(q_1, q_2) = \|q_1 - q_2\| = \sqrt{\frac{1}{3}} = \frac{1}{\sqrt{3}}$.

Task

Find the unit vector in the same direction as $p(x) = x$.

Solution.

We calculated $\|p\| = \frac{1}{\sqrt{3}}$ in 7.2.1. Therefore, the unit vector in the same direction as $p(x)$ is the polynomial

$$\begin{aligned}\frac{p(x)}{\|p\|} &= \frac{x}{\frac{1}{\sqrt{3}}} \\ &= \sqrt{3} \cdot x\end{aligned}$$

Task

Are x and $x - 2/3$ orthogonal polynomials in $\mathbf{P}_1$?

Solution.

To determine if the two polynomials are orthogonal, we must check whether $\langle x, x - 2/3 \rangle = 0$. We calculate that

$$\begin{aligned}\langle x, x - \frac{2}{3} \rangle &= \int_0^1 x \left(x - \frac{2}{3} \right) dx \\ &= \int_0^1 \left(x^2 - \frac{2x}{3} \right) dx \\ &= \frac{x^3}{3} - \frac{x^2}{3} \Big|_0^1 \\ &= 0\end{aligned}$$

So the polynomials x and $x - 2/3$ are orthogonal to one another.

7.2.2 Spaces of matrices

Definition 7.2.10 (The Trace of a Matrix).

The trace of a square $n \times n$ matrix A , denoted by $\text{Tr}(A)$, is the sum of its diagonal entries. For $A = [a_{ij}]$, we have

$$\text{Tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}.$$

Theorem 7.2.11.

For any $n \times n$ matrices A and B ,

$$\text{Tr}(A^T) = \text{Tr}(A) \quad \text{and} \quad \text{Tr}(A + B) = \text{Tr}(A) + \text{Tr}(B)$$

Activity

Consider $\mathbf{M}_{22}$, and let A, B be matrices in $\mathbf{M}_{22}$. Define

$$\langle A, B \rangle = \text{Tr}(A^T B)$$

Task

Find $\langle A, B \rangle$ for $A = \begin{bmatrix} 1 & 2 \\ 2 & -3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 2 \\ -1 & 1 \end{bmatrix}$.

Solution.

We calculate that

$$A^T B = \begin{bmatrix} 1 & 2 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ 3 & 1 \end{bmatrix}$$

Thus

$$\langle A, B \rangle = \text{Tr}(A^T B) = -2 + 1 = -1$$

Task

Show that the map $\langle, \rangle$ defined above is an inner product.

Solution.

We verify each of the four properties that define an inner product. Let A, B, C be arbitrary 2×2 matrices:

(a) We calculate that

$$\text{Tr}(A^T B) = \text{Tr}((A^T B)^T) = \text{Tr}(B^T (A^T)^T) = \text{Tr}(B^T A)$$

So $\langle A, B \rangle = \langle B, A \rangle$.

(b) We calculate that

$$\text{Tr}((A + C)^T B) = \text{Tr}(A^T B + C^T B) = \text{Tr}(A^T B) + \text{Tr}(C^T B)$$

So $\langle A + C, B \rangle = \langle A, B \rangle + \langle C, B \rangle$.

(c) We calculate that

$$\text{Tr}(rA^T B) = r \text{Tr}(A^T B)$$

. So $\langle rA, B \rangle = r\langle A, B \rangle$.

(d) For $A \neq 0$ with $A = [a_{ij}]$,

$$A^T A = [c_{ij}] \text{ where } c_{ij} = \sum_k a_{ki} a_{kj}$$

The diagonal elements are: $c_{ii} = \sum_k a_{ki}^2 \geq 0$, and so

$$\text{Tr}(A^T A) = \sum_i \sum_k a_{ki}^2$$

is precisely the sum of the squares of the entries of A , which is always non-negative, and positive when at least one entry is non-zero. Thus $\langle A, A \rangle > 0$ for $A \neq 0$.

Task

Find the distance between the two matrices

$$A = \begin{bmatrix} 3 & 1 \\ 0 & -1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -1 & 0 \\ 1 & 2 \end{bmatrix}$$

Solution.

We recall that

$$d(A, B) = \|A - B\| = \sqrt{\langle A - B, A - B \rangle}$$

. We begin by calculating

$$A - B = \begin{bmatrix} 3 & 1 \\ 0 & -1 \end{bmatrix} - \begin{bmatrix} -1 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ -1 & -3 \end{bmatrix}$$

. Next, we calculate that

$$(A - B)^T(A - B) = \begin{bmatrix} 4 & -1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ -1 & -3 \end{bmatrix} = \begin{bmatrix} 17 & 7 \\ 7 & 10 \end{bmatrix}$$

. Thus

$$\langle A - B, A - B \rangle = \text{Tr}((A - B)^T(A - B)) = 17 + 10 = 27$$

. Therefore, $d(A, B) = \sqrt{27} = 3\sqrt{3}$.

Task

Find a value for x so that the two matrices

$$A = \begin{bmatrix} 1 & 0 \\ -2 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -3 & 2 \\ x & 0 \end{bmatrix}$$

are orthogonal.

Solution.

We calculate that

$$A^T B = \begin{bmatrix} 1 & -2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ x & 0 \end{bmatrix} = \begin{bmatrix} -3 - 2x & 2 \\ 2x & 0 \end{bmatrix}$$

. So

$$\langle A, B \rangle = \text{Tr}(A^T B) = (-3 - 2x) + 0 = -3 - 2x = 0$$

. Now $\langle A, B \rangle = 0$ precisely when $x = -3/2$, and so this is precisely the value of x which makes the matrices A and B orthogonal.

7.3 Polynomial spaces and course connections

Sections 7.1 and 7.2 introduced the abstract language. We now use bounded-degree polynomial spaces as a finite-dimensional test world for that language. A polynomial has coefficient coordinates, but the same polynomial space can carry several useful inner products. Changing the inner

product changes the projection, so it changes what “best approximation” means.

The goal is not a full theory of polynomial approximation. The goal is to see how vector spaces, inner products, projections, least squares, Taylor approximation, and Gram-Schmidt fit together in one concrete setting.

7.3.1 Polynomials as vectors

A bounded-degree polynomial space is finite-dimensional:

$$\mathcal{P}_{\leq d} = \{a_0 + a_1x + \cdots + a_dx^d : a_0, \dots, a_d \in \mathbb{R}\}.$$

The standard ordered basis is

$$B = (1, x, x^2, \dots, x^d).$$

If

$$p(x) = a_0 + a_1x + \cdots + a_dx^d,$$

then the coordinate vector of p in the basis B is

$$\mathbf{c}_p = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_d \end{bmatrix}.$$

The polynomial p is the vector in $\mathcal{P}_{\leq d}$. The column $\mathbf{c}_p$ is its coordinate vector after the basis has been chosen.

Activity: Polynomial coordinates

Work in $\mathcal{P}_{\leq 3}$ with ordered basis

$$B = (1, x, x^2, x^3).$$

Find the coordinate vectors of

$$p(x) = 2 - 3x + x^3, \quad q(x) = x^2 - 4x, \quad r(x) = 5.$$

Solution.

Match coefficients in the order $1, x, x^2, x^3$:

$$\mathbf{c}_p = \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{c}_q = \begin{bmatrix} 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{c}_r = \begin{bmatrix} 5 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

The degree-at-most-one polynomial space

$$\mathcal{P}_{\leq 1} = \text{span}\{1, x\}$$

is a subspace of $\mathcal{P}_{\leq d}$ when $d \geq 1$. Its elements have the form

$$c_0 + c_1x.$$

These are the affine polynomials in one variable. Although the word affine often signals a shifted object, $\mathcal{P}_{\leq 1}$ is a vector subspace: it contains the zero polynomial and is closed under addition and scalar multiplication.

Activity: A root condition as a subspace

Let

$$U = \{p \in \mathcal{P}_{\leq 4} : p(-2) = 0\}.$$

Explain why U is a subspace of $\mathcal{P}_{\leq 4}$, and find a basis for U .

Solution.

The zero polynomial satisfies $0(-2) = 0$, so $0 \in U$. If $p(-2) = 0$ and $q(-2) = 0$, then

$$(p + q)(-2) = p(-2) + q(-2) = 0.$$

If $c \in \mathbb{R}$, then

$$(cp)(-2) = c p(-2) = 0.$$

Thus U is a subspace.

A polynomial satisfies $p(-2) = 0$ exactly when $x + 2$ is a factor. Since $p \in \mathcal{P}_{\leq 4}$,

$$p(x) = (x + 2)(a_0 + a_1x + a_2x^2 + a_3x^3).$$

Therefore a basis is

$$x + 2, \quad x(x + 2), \quad x^2(x + 2), \quad x^3(x + 2).$$

Activity: Vanishing value and slope

Let

$$W = \{p \in \mathcal{P}_{\leq 3} : p(0) = 0, p'(0) = 0\}.$$

Find a basis for W .

Solution.

Write

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3.$$

Then

$$p(0) = a_0, \quad p'(0) = a_1.$$

The conditions $p(0) = 0$ and $p'(0) = 0$ force $a_0 = 0$ and $a_1 = 0$. Thus

$$p(x) = a_2x^2 + a_3x^3,$$

so

$$W = \text{span}\{x^2, x^3\}.$$

A basis is x^2, x^3 .

The next question is not whether $\mathcal{P}_{\leq 1}$ is a subspace. It is: which element of $\mathcal{P}_{\leq 1}$ is closest to a given polynomial or function? That question needs an inner product.

7.3.2 Inner products and projection equations

The same vector space can carry different inner products. Different inner products give different lengths, distances, orthogonality conditions, and projections.

Let

$$p(x) = a_0 + a_1x + \cdots + a_dx^d, \quad q(x) = b_0 + b_1x + \cdots + b_dx^d.$$

The coefficient inner product is

$$\langle p, q \rangle_{\text{coef}} = a_0b_0 + a_1b_1 + \cdots + a_db_d.$$

This is the ordinary dot product of the coordinate vectors in the basis $1, x, \dots, x^d$.

Another useful inner product is the $L^2[-1, 1]$ inner product:

$$\langle p, q \rangle_{L^2} = \int_{-1}^1 p(x)q(x) dx.$$

This inner product measures agreement over the whole interval $[-1, 1]$.

Given sample points $x_1, \dots, x_N$, the sampled inner product is

$$\langle p, q \rangle_{\text{sample}} = \sum_{i=1}^N p(x_i)q(x_i).$$

This is the inner product behind sampled least squares and polynomial regression.

Warning

A sampled rule can fail to be an inner product on all of $\mathcal{P}_{\leq d}$ if a nonzero polynomial in $\mathcal{P}_{\leq d}$ vanishes at every sample point. In the computations below, the sample points are chosen so the feature columns in the approximation subspace are independent. That is enough for the least-squares projection used here.

Suppose

$$U = \text{span}\{\phi_0, \phi_1, \dots, \phi_m\}$$

is a polynomial subspace. We want the best approximation to f inside U . Write

$$q = c_0\phi_0 + c_1\phi_1 + \dots + c_m\phi_m.$$

The residual is

$$r = f - q.$$

The projection condition is

$$\langle r, \phi_i \rangle = 0, \quad i = 0, \dots, m.$$

Substituting the formula for q gives

$$\sum_{j=0}^m c_j \langle \phi_j, \phi_i \rangle = \langle f, \phi_i \rangle, \quad i = 0, \dots, m.$$

This is the Gram-matrix system

$$G\mathbf{c} = \mathbf{b},$$

where

$$G_{ij} = \langle \phi_j, \phi_i \rangle, \quad b_i = \langle f, \phi_i \rangle,$$

and

$$\mathbf{c} = \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_m \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_m \end{bmatrix}.$$

Activity: The Gram matrix for affine polynomials

Let

$$U = \mathcal{P}_{\leq 1} = \text{span}\{1, x\}$$

with the $L^2[-1, 1]$ inner product. Compute the Gram matrix for the ordered basis

$$\phi_0 = 1, \quad \phi_1 = x.$$

Solution.

We compute

$$\langle 1, 1 \rangle = \int_{-1}^1 1 \, dx = 2,$$

$$\langle 1, x \rangle = \int_{-1}^1 x \, dx = 0,$$

$$\langle x, 1 \rangle = \int_{-1}^1 x \, dx = 0,$$

and

$$\langle x, x \rangle = \int_{-1}^1 x^2 \, dx = \frac{2}{3}.$$

Thus

$$G = \begin{bmatrix} 2 & 0 \\ 0 & \frac{2}{3} \end{bmatrix}.$$

If $q = c_0 + c_1 x$ is the $L^2[-1, 1]$ -projection of f onto $\mathcal{P}_{\leq 1}$, then

$$\begin{bmatrix} 2 & 0 \\ 0 & \frac{2}{3} \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} \int_{-1}^1 f(x) \, dx \\ \int_{-1}^1 x f(x) \, dx \end{bmatrix}.$$

Activity: Sampled projection and the design matrix

Let $x_1, \dots, x_N$ be sample points and suppose we approximate sampled values of f by

$$q(x) = c_0 + c_1 x.$$

Define

$$A = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_N \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} c_0 \\ c_1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_N) \end{bmatrix}.$$

Explain why the sampled residual condition is

$$A^T \mathbf{r} = \mathbf{0}, \quad \mathbf{r} = \mathbf{y} - A\mathbf{c}.$$

Solution.

The vector Ac stores the fitted values

$$q(x_1), q(x_2), \dots, q(x_N).$$

Thus

$$\mathbf{r} = \mathbf{y} - Ac$$

stores the sampled residuals

$$r_i = f(x_i) - q(x_i).$$

The sampled projection condition says the residual is orthogonal to the sampled constant feature and the sampled x -feature:

$$\sum_i r_i = 0, \quad \sum_i x_i r_i = 0.$$

These are exactly the two entries of

$$A^T \mathbf{r} = \begin{bmatrix} \sum_i r_i \\ \sum_i x_i r_i \end{bmatrix}.$$

So $A^T \mathbf{r} = \mathbf{0}$ is the sampled residual-orthogonality condition.

Note

Polynomial regression is linear regression with polynomial feature columns. For example, fitting

$$y \approx c_0 + c_1 x + c_2 x^2$$

uses the design matrix

$$A = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_N & x_N^2 \end{bmatrix}.$$

The model is nonlinear in x , but it is linear in the coefficient vector

$$\mathbf{c} = \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix}.$$

The next subsection uses one polynomial and one approximation subspace, but three different error measures: local Taylor matching, continuous L^2 -least squares, and sampled least squares.

7.3.3 Taylor line versus least-squares line

We now use one polynomial and one approximation subspace, but three different meanings of "best".

Let

$$p(x) = 1 + 2x + x^2 + x^3,$$

and let

$$U = \mathcal{P}_{\leq 1} = \text{span}\{1, x\}.$$

Every element of U has the form

$$q(x) = \alpha + \beta x.$$

The Taylor line, the continuous least-squares line, and the sampled least-squares line all live in U . They differ because they measure error differently.

Activity: Taylor line at the origin

Find the first-order Taylor approximation to p at 0. Then verify the residual conditions.

Solution.

We compute

$$p(0) = 1.$$

Also

$$p'(x) = 2 + 2x + 3x^2,$$

so

$$p'(0) = 2.$$

Thus the first-order Taylor approximation at 0 is

$$T(x) = p(0) + p'(0)x = 1 + 2x.$$

The residual is

$$r_T(x) = p(x) - T(x) = x^2 + x^3.$$

Therefore

$$r_T(0) = 0,$$

and

$$r'_T(x) = 2x + 3x^2,$$

so

$$r'_T(0) = 0.$$

The Taylor line matches the value and slope of p at the origin.

Activity: Continuous least-squares line

Use the $L^2[-1, 1]$ inner product to find the best approximation to p in $U = \text{span}\{1, x\}$.

Solution.

Write

$$q_{L^2}(x) = \alpha + \beta x.$$

The residual is

$$r_{L^2}(x) = p(x) - q_{L^2}(x).$$

The projection condition says

$$\langle r_{L^2}, 1 \rangle_{L^2} = 0, \quad \langle r_{L^2}, x \rangle_{L^2} = 0.$$

Equivalently,

$$\int_{-1}^1 r_{L^2}(x) dx = 0, \quad \int_{-1}^1 x r_{L^2}(x) dx = 0.$$

Now

$$r_{L^2}(x) = (1 - \alpha) + (2 - \beta)x + x^2 + x^3.$$

The first residual equation gives

$$\int_{-1}^1 ((1 - \alpha) + (2 - \beta)x + x^2 + x^3) dx = 0.$$

The odd terms integrate to zero on $[-1, 1]$, so

$$2(1 - \alpha) + \frac{2}{3} = 0.$$

Thus

$$\alpha = \frac{4}{3}.$$

The second residual equation gives

$$\int_{-1}^1 x((1 - \alpha) + (2 - \beta)x + x^2 + x^3) dx = 0.$$

Again using symmetry,

$$(2 - \beta)\frac{2}{3} + \frac{2}{5} = 0.$$

Thus

$$\beta = \frac{13}{5}.$$

So the continuous least-squares line is

$$q_{L^2}(x) = \frac{4}{3} + \frac{13}{5}x.$$

Activity: Sampled least-squares line

Use the sample points

$$-1, -\frac{1}{2}, 0, \frac{1}{2}, 1$$

to find the sampled least-squares approximation to p in $U = \text{span}\{1, x\}$.

Solution.

Let the sample points be $x_1, \dots, x_5$, and write

$$q_{\text{sample}}(x) = \alpha + \beta x.$$

The sampled residuals are

$$r_i = p(x_i) - q_{\text{sample}}(x_i).$$

The sampled projection condition is

$$\sum_i r_i = 0, \quad \sum_i x_i r_i = 0.$$

Equivalently, if

$$A = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_5 \end{bmatrix}, \quad \mathbf{r} = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_5 \end{bmatrix},$$

then

$$A^T \mathbf{r} = \mathbf{0}.$$

For the symmetric sample points,

$$\sum_i x_i = 0, \quad \sum_i x_i^2 = \frac{5}{2}.$$

Also

$$\sum_i p(x_i) = \sum_i (1 + 2x_i + x_i^2 + x_i^3) = 5 + \frac{5}{2} = \frac{15}{2},$$

so

$$5\alpha = \frac{15}{2}, \quad \alpha = \frac{3}{2}.$$

Next,

$$\sum_i x_i p(x_i) = \sum_i (x_i + 2x_i^2 + x_i^3 + x_i^4) = 2 \sum_i x_i^2 + \sum_i x_i^4.$$

Here

$$\sum_i x_i^4 = 1 + \frac{1}{16} + 0 + \frac{1}{16} + 1 = \frac{17}{8},$$

so

$$\sum_i x_i p(x_i) = 5 + \frac{17}{8} = \frac{57}{8}.$$

The second residual equation gives

$$\beta \sum_i x_i^2 = \sum_i x_i p(x_i),$$

so

$$\beta \cdot \frac{5}{2} = \frac{57}{8}, \quad \beta = \frac{57}{20}.$$

Thus the sampled least-squares line is

$$q_{\text{sample}}(x) = \frac{3}{2} + \frac{57}{20}x.$$

The three approximations are

$$T(x) = 1 + 2x,$$

$$q_{L^2}(x) = \frac{4}{3} + \frac{13}{5}x,$$

$$q_{\text{sample}}(x) = \frac{3}{2} + \frac{57}{20}x.$$

They are different because they solve different projection problems.

- **Approximation:** Taylor line
How it is chosen: match value and slope at 0
Residual condition: $r_T(0) = 0, r'_T(0) = 0$
- **Approximation:** Continuous least squares
How it is chosen: minimize squared error over $[-1, 1]$
Residual condition: $\int_{-1}^1 r(x) dx = 0, \int_{-1}^1 x r(x) dx = 0$
- **Approximation:** Sampled least squares
How it is chosen: minimize squared error at sample points
Residual condition: $\sum_i r_i = 0, \sum_i x_i r_i = 0$

One cubic, three notions of best affine approximation

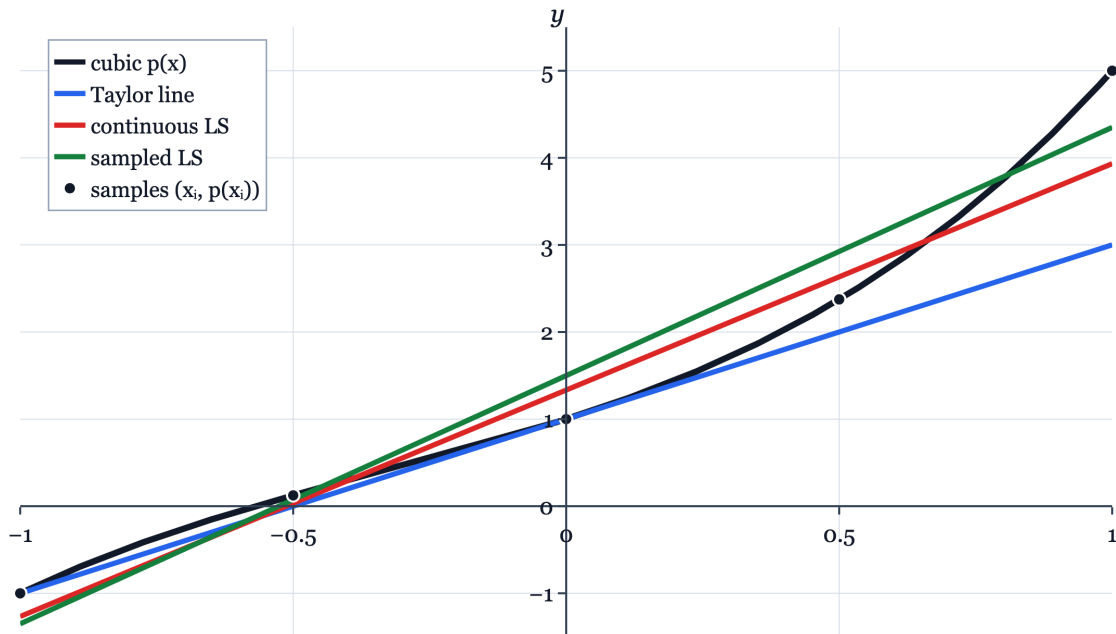


Figure 7.1. One curve, three lines: Taylor, continuous least squares, and sampled least squares measure best approximation differently.

Warning

The phrase “best affine approximation” is incomplete until the error measure is specified. Taylor matching, continuous least squares, and sampled least squares give different answers because they use different residual conditions.

The next subsection explains how Taylor approximation itself can be viewed as projection, but with a local derivative-based inner product.

7.3.4 Taylor approximation as local projection

The previous subsection showed that Taylor and least squares can give different lines. This is not a contradiction. Taylor approximation is local. Ordinary L^2 -least squares over a fixed interval is global.

There are two useful ways to connect Taylor approximation with projection.

First, Taylor approximation is exactly projection for a derivative-based inner product at the expansion point.

Second, ordinary L^2 -least-squares projection over a shrinking interval converges to Taylor approximation as the interval shrinks to the expansion point.

Derivative-based inner product. Fix a point a , and work in $\mathcal{P}_{\leq d}$. Define

$$\langle p, q \rangle_{\text{jet}, a} = \sum_{k=0}^d w_k \frac{p^{(k)}(a)}{k!} \frac{q^{(k)}(a)}{k!}, \quad w_k > 0.$$

This is an inner product on $\mathcal{P}_{\leq d}$, because a polynomial of degree at most d is determined by its derivatives through order d at a .

Activity: Taylor line as exact projection

Suppose

$$p(x) = c_0 + c_1(x - a) + c_2(x - a)^2 + c_3(x - a)^3$$

and let

$$q(x) = \alpha + \beta(x - a).$$

Use the derivative-based inner product above to find the closest affine polynomial q to p .

Solution.

In the centered basis

$$1, \quad (x - a), \quad (x - a)^2, \quad (x - a)^3,$$

the coefficients of p are c_0, c_1, c_2, c_3 , while the coefficients of q are $\alpha, \beta, 0, 0$. Therefore

$$p - q = (c_0 - \alpha) + (c_1 - \beta)(x - a) + c_2(x - a)^2 + c_3(x - a)^3.$$

The squared norm of the residual has the form

$$\|p - q\|_{\text{jet},a}^2 = w_0(c_0 - \alpha)^2 + w_1(c_1 - \beta)^2 + w_2c_2^2 + w_3c_3^2.$$

The last two terms do not depend on α or β . The expression is minimized when

$$\alpha = c_0, \quad \beta = c_1.$$

Thus the closest affine polynomial is

$$q(x) = c_0 + c_1(x - a).$$

Since

$$c_0 = p(a), \quad c_1 = p'(a),$$

we get

$$q(x) = p(a) + p'(a)(x - a).$$

This is the first-order Taylor approximation at a .

More generally, projection onto $\mathcal{P}_{\leq m}$, written in powers of $x - a$, keeps the Taylor terms through degree m and discards the higher-degree terms.

Warning

The inner product $\langle \cdot, \cdot \rangle_{\text{jet},a}$ depends on the expansion point a . One fixed projection cannot produce the Taylor line at every point, because the Taylor line of the same polynomial changes when the expansion point changes.

Activity: Fixed-interval least squares is not Taylor approximation

Project

$$p(x) = x^2$$

onto $\mathcal{P}_{\leq 1} = \text{span}\{1, x\}$ using the $L^2[-1, 1]$ inner product. Compare the answer with the Taylor line at 0.

Solution.

The Taylor line of x^2 at 0 is

$$T(x) = 0.$$

Now compute the $L^2[-1, 1]$ -projection. Write

$$q(x) = \alpha + \beta x.$$

Since x^2 is even and x is odd, the best line has no x -term:

$$\beta = 0.$$

The constant term is the average value of x^2 over $[-1, 1]$:

$$\alpha = \frac{\int_{-1}^1 x^2 dx}{\int_{-1}^1 1 dx} = \frac{2/3}{2} = \frac{1}{3}.$$

Thus the $L^2[-1, 1]$ -best affine approximation is

$$q(x) = \frac{1}{3},$$

not 0. Taylor approximation is local, while least squares over a fixed interval is global.

Shrinking intervals. For $t > 0$, let $P_t f$ be the $L^2[-t, t]$ -projection of f onto $\text{span}\{1, x\}$. Write

$$P_t f(x) = a_t + b_t x.$$

The residual conditions are

$$\int_{-t}^t (f(x) - a_t - b_t x) dx = 0,$$

and

$$\int_{-t}^t x(f(x) - a_t - b_t x) dx = 0.$$

Because the interval is symmetric,

$$\int_{-t}^t x dx = 0.$$

Thus the formulas simplify to

$$a_t = \frac{1}{2t} \int_{-t}^t f(x) dx,$$

and

$$b_t = \frac{\int_{-t}^t x f(x) dx}{\int_{-t}^t x^2 dx} = \frac{3}{2t^3} \int_{-t}^t x f(x) dx.$$

Activity: Shrinking least squares for x^2 and x^3

Compute the $L^2[-t, t]$ -projection of x^2 and x^3 onto $\text{span}\{1, x\}$. What happens as $t \rightarrow 0$?

Solution.

For $f(x) = x^2$, symmetry gives $b_t = 0$, and

$$a_t = \frac{1}{2t} \int_{-t}^t x^2 dx = \frac{1}{2t} \cdot \frac{2t^3}{3} = \frac{t^2}{3}.$$

Thus

$$P_t(x^2) = \frac{t^2}{3}.$$

As $t \rightarrow 0$,

$$P_t(x^2) \rightarrow 0,$$

which is the Taylor line of x^2 at 0.

For $f(x) = x^3$, symmetry gives $a_t = 0$, and

$$b_t = \frac{\int_{-t}^t x^4 dx}{\int_{-t}^t x^2 dx} = \frac{2t^5/5}{2t^3/3} = \frac{3t^2}{5}.$$

Thus

$$P_t(x^3) = \frac{3t^2}{5}x.$$

As $t \rightarrow 0$,

$$P_t(x^3) \rightarrow 0,$$

which is again the Taylor line at 0.

Note

If

$$p(x) = c_0 + c_1x + c_2x^2 + \cdots + c_dx^d,$$

then the $L^2[-t, t]$ -projection of p onto $\text{span}\{1, x\}$ has the form

$$P_tp(x) = a_t + b_tx,$$

where

$$a_t = c_0 + \frac{c_2t^2}{3} + \frac{c_4t^4}{5} + \frac{c_6t^6}{7} + \cdots$$

and

$$b_t = c_1 + \frac{3c_3t^2}{5} + \frac{3c_5t^4}{7} + \frac{3c_7t^6}{9} + \cdots.$$

Therefore

$$a_t \rightarrow c_0 = p(0), \quad b_t \rightarrow c_1 = p'(0).$$

So

$$P_t p(x) \rightarrow p(0) + p'(0)x.$$

This is the first-order Taylor polynomial at 0.

Warning

The projections $P_t p$ can converge to Taylor approximation even though the inner products collapse as $t \rightarrow 0$. In fact, for

$$\langle p, q \rangle_t = \int_{-t}^t p(x)q(x) dx,$$

we have

$$\langle p, q \rangle_t \rightarrow 0$$

for fixed polynomials p, q . If we normalize by interval length, the limit only remembers the value at the origin:

$$\frac{1}{2t} \int_{-t}^t p(x)q(x) dx \rightarrow p(0)q(0).$$

That is not a nondegenerate inner product on $\mathcal{P}_{\leq d}$. The Taylor-coefficient inner product appears only after separating degrees by scale. For this course, the important point is: shrinking L^2 -projections recover Taylor polynomials, but the ordinary $L^2[-t, t]$ inner products themselves do not converge to the derivative-based Taylor inner product.

The single-variable picture is now complete:

fixed-interval least squares is global,
shrinking-interval least squares becomes local,
the derivative-based inner product makes Taylor projection exact.

7.3.5 Gram-Schmidt and Legendre polynomials

Gram-Schmidt is not only an algorithm for column vectors in $\mathbb{R}^n$. It works in any finite-dimensional inner product space. Here the vectors are polynomials.

Use the inner product

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx$$

on $\mathcal{P}_{\leq 3}$. Start with the standard ordered basis

$$1, \quad x, \quad x^2, \quad x^3.$$

This basis spans $\mathcal{P}_{\leq 3}$, but it is not an orthogonal basis. Gram-Schmidt replaces it with an orthogonal basis for the same polynomial space.

Note: Symmetry on $[-1, 1]$

If $h(x)$ is an odd function, then

$$\int_{-1}^1 h(x) dx = 0.$$

This explains many zeros below. Products with odd total power integrate to zero on the symmetric interval.

Activity: The first two orthogonal polynomials

Show that 1 and x are orthogonal using the $L^2[-1, 1]$ inner product.

Solution.

We compute

$$\langle 1, x \rangle = \int_{-1}^1 x dx = 0.$$

Thus 1 and x are already orthogonal.

Activity: Orthogonalizing x^2

Let

$$\phi_0(x) = 1, \quad \phi_1(x) = x.$$

Use Gram-Schmidt to subtract from x^2 its projections onto ϕ_0 and ϕ_1 .

Solution.

First,

$$\langle x^2, 1 \rangle = \int_{-1}^1 x^2 dx = \frac{2}{3},$$

and

$$\langle 1, 1 \rangle = \int_{-1}^1 1 dx = 2.$$

Therefore

$$\text{proj}_{\text{span}\{1\}}(x^2) = \frac{\langle x^2, 1 \rangle}{\langle 1, 1 \rangle} 1 = \frac{2/3}{2} 1 = \frac{1}{3}.$$

Also,

$$\langle x^2, x \rangle = \int_{-1}^1 x^3 dx = 0,$$

so there is no x -component to subtract. Thus the next orthogonal vector is

$$\phi_2(x) = x^2 - \frac{1}{3}.$$

Activity: Orthogonalizing x^3

Let

$$\phi_0(x) = 1, \quad \phi_1(x) = x, \quad \phi_2(x) = x^2 - \frac{1}{3}.$$

Use Gram-Schmidt to subtract from x^3 its projections onto ϕ_0, ϕ_1, ϕ_2 .

Solution.

By symmetry,

$$\langle x^3, 1 \rangle = \int_{-1}^1 x^3 dx = 0.$$

Also,

$$\langle x^3, \phi_2 \rangle = \int_{-1}^1 x^3 \left(x^2 - \frac{1}{3} \right) dx = \int_{-1}^1 \left(x^5 - \frac{1}{3} x^3 \right) dx = 0,$$

again by symmetry.

The only projection to subtract is the projection onto x . We compute

$$\langle x^3, x \rangle = \int_{-1}^1 x^4 dx = \frac{2}{5},$$

and

$$\langle x, x \rangle = \int_{-1}^1 x^2 dx = \frac{2}{3}.$$

Thus

$$\text{proj}_{\text{span}\{x\}}(x^3) = \frac{\langle x^3, x \rangle}{\langle x, x \rangle} x = \frac{2/5}{2/3} x = \frac{3}{5} x.$$

Therefore the next orthogonal vector is

$$\phi_3(x) = x^3 - \frac{3}{5}x.$$

Gram-Schmidt has produced the orthogonal basis

$$1, \quad x, \quad x^2 - \frac{1}{3}, \quad x^3 - \frac{3}{5}x$$

for $\mathcal{P}_{\leq 3}$, using the $L^2[-1, 1]$ inner product.

Warning: Orthogonal does not mean normalized

The vectors above are orthogonal, but their norms are not all equal to 1. Gram-Schmidt can produce an orthogonal basis first; normalizing each vector is a separate step.

Note: Scaled Legendre polynomials

The conventional first Legendre polynomials begin

$$P_0(x) = 1, \quad P_1(x) = x,$$

$$P_2(x) = \frac{1}{2}(3x^2 - 1), \quad P_3(x) = \frac{1}{2}(5x^3 - 3x).$$

These are nonzero scalar multiples of

$$1, \quad x, \quad x^2 - \frac{1}{3}, \quad x^3 - \frac{3}{5}x.$$

We do not develop the theory of Legendre polynomials here. The point is that Gram-Schmidt creates coordinates adapted to an inner product.

When $\phi_0, \dots, \phi_m$ are nonzero and orthogonal, projection onto

$$U = \text{span}\{\phi_0, \dots, \phi_m\}$$

has the simple form

$$\text{proj}_U(f) = \sum_{j=0}^m \frac{\langle f, \phi_j \rangle}{\langle \phi_j, \phi_j \rangle} \phi_j.$$

The Gram matrix is diagonal in an orthogonal basis, so the projection equations decouple.

Activity: Revisiting the continuous least-squares line

Let

$$p(x) = 1 + 2x + x^2 + x^3.$$

Use the orthogonal basis

$$\phi_0 = 1, \quad \phi_1 = x, \quad \phi_2 = x^2 - \frac{1}{3}, \quad \phi_3 = x^3 - \frac{3}{5}x$$

to express p . Then project p onto

$$\mathcal{P}_{\leq 1} = \text{span}\{1, x\}.$$

Solution.

Since

$$x^2 = \phi_2 + \frac{1}{3}\phi_0$$

and

$$x^3 = \phi_3 + \frac{3}{5}\phi_1,$$

we have

$$\begin{aligned} p(x) &= 1 + 2x + x^2 + x^3 \\ &= 1 + 2\phi_1 + \left(\phi_2 + \frac{1}{3}\phi_0\right) + \left(\phi_3 + \frac{3}{5}\phi_1\right). \end{aligned}$$

Thus

$$p = \frac{4}{3}\phi_0 + \frac{13}{5}\phi_1 + \phi_2 + \phi_3.$$

Projection onto $\mathcal{P}_{\leq 1} = \text{span}\{\phi_0, \phi_1\}$ keeps only the ϕ_0 and ϕ_1 components:

$$\text{proj}_{\mathcal{P}_{\leq 1}}(p) = \frac{4}{3}\phi_0 + \frac{13}{5}\phi_1.$$

Since $\phi_0 = 1$ and $\phi_1 = x$,

$$\text{proj}_{\mathcal{P}_{\leq 1}}(p) = \frac{4}{3} + \frac{13}{5}x.$$

This is the continuous least-squares line from the earlier subsection.

The same idea explains why orthogonal bases and QR are useful in least squares: changing to an orthogonal basis does not change the approximation space, but it makes projection computations easier.

The next subsection moves from one-variable polynomials to two-variable polynomial spaces, where affine approximations are planes.

7.3.6 Two-variable polynomials and tangent planes

The same ideas work for polynomials in more than one input variable. This is the bridge back to multivariable calculus: affine approximations become planes, and sampled least squares becomes plane fitting.

The space of two-variable polynomials of total degree at most 2 is

$$\mathcal{P}_{\leq 2}^{(2)} = \text{span}\{1, x, y, x^2, xy, y^2\}.$$

A standard ordered basis is

$$B = (1, x, y, x^2, xy, y^2).$$

An element of $\mathcal{P}_{\leq 2}^{(2)}$ has the form

$$a_0 + a_1x + a_2y + a_3x^2 + a_4xy + a_5y^2.$$

Its coordinate vector in the basis B is

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix}.$$

Activity: Coordinates in a two-variable polynomial space

Work in $\mathcal{P}_{\leq 2}^{(2)}$ with ordered basis

$$B = (1, x, y, x^2, xy, y^2).$$

Find the coordinate vector of

$$p(x, y) = 3 - 2x + y + 5xy.$$

Solution.

Match coefficients in the order $1, x, y, x^2, xy, y^2$. We get

$$\mathbf{c}_p = \begin{bmatrix} 3 \\ -2 \\ 1 \\ 0 \\ 5 \\ 0 \end{bmatrix}.$$

The affine-polynomial subspace is

$$\mathcal{A} = \text{span}\{1, x, y\}.$$

Its elements have the form

$$q(x, y) = \alpha + \beta x + \gamma y.$$

Geometrically, the graph of q is a plane.

Now consider

$$f(x, y) = 1 + 2x - y + x^2 + xy + 2y^2 + x^3.$$

At $(0, 0)$, the tangent plane uses only the value and first partial derivatives of f .

Activity: Tangent plane at the origin

Compute the tangent plane to

$$f(x, y) = 1 + 2x - y + x^2 + xy + 2y^2 + x^3$$

at $(0, 0)$. Then state the residual conditions it satisfies.

Solution.

We compute

$$f(0, 0) = 1.$$

Also

$$f_x(x, y) = 2 + 2x + y + 3x^2,$$

so

$$f_x(0, 0) = 2.$$

And

$$f_y(x, y) = -1 + x + 4y,$$

so

$$f_y(0, 0) = -1.$$

Therefore the tangent plane is

$$T(x, y) = 1 + 2x - y.$$

The residual is

$$r_T(x, y) = f(x, y) - T(x, y) = x^2 + xy + 2y^2 + x^3.$$

Thus

$$r_T(0, 0) = 0,$$

and

$$\nabla r_T(0, 0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The tangent plane matches the value and first partial derivatives at the origin.

A best-fit plane uses a different idea. Instead of matching at one point, it minimizes error over many points.

Let

$$(x_1, y_1), \dots, (x_N, y_N)$$

be sampled points. Define the sampled inner product by

$$\langle u, v \rangle_{\text{sample}} = \sum_{i=1}^N u(x_i, y_i) v(x_i, y_i).$$

For a plane

$$q(x, y) = \alpha + \beta x + \gamma y,$$

the sampled residuals are

$$r_i = f(x_i, y_i) - q(x_i, y_i).$$

The sampled projection conditions are

$$\sum_i r_i = 0, \quad \sum_i x_i r_i = 0, \quad \sum_i y_i r_i = 0.$$

Activity: Design matrix for a sampled best-fit plane

Let

$$q(x, y) = \alpha + \beta x + \gamma y.$$

Construct the design matrix for fitting q to sampled values of f at points

$$(x_1, y_1), \dots, (x_N, y_N).$$

Explain the residual condition in matrix form.

Solution.

The design matrix is

$$A = \begin{bmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ \vdots & \vdots & \vdots \\ 1 & x_N & y_N \end{bmatrix}.$$

The coefficient vector is

$$\mathbf{c} = \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix},$$

and the sampled-value vector is

$$\mathbf{f} = \begin{bmatrix} f(x_1, y_1) \\ f(x_2, y_2) \\ \vdots \\ f(x_N, y_N) \end{bmatrix}.$$

The fitted values are $A\mathbf{c}$, so the residual vector is

$$\mathbf{r} = \mathbf{f} - A\mathbf{c}.$$

The sampled residual-orthogonality condition is

$$A^T \mathbf{r} = \mathbf{0}.$$

Written entry by entry, this says

$$\sum_i r_i = 0, \quad \sum_i x_i r_i = 0, \quad \sum_i y_i r_i = 0.$$

These are the same projection conditions as before, now for the sampled constant feature, sampled x -feature, and sampled y -feature.

Activity: A sampled best-fit plane on a symmetric grid

Use the grid

$$x, y \in \left\{ -1, -\frac{1}{2}, 0, \frac{1}{2}, 1 \right\}$$

and the function

$$f(x, y) = 1 + 2x - y + x^2 + xy + 2y^2 + x^3.$$

Find the sampled least-squares plane

$$q_{\text{sample}}(x, y) = \alpha + \beta x + \gamma y.$$

Solution.

The grid has 25 points. By symmetry,

$$\sum_i x_i = 0, \quad \sum_i y_i = 0, \quad \sum_i x_i y_i = 0.$$

Also

$$\sum_i x_i^2 = \sum_i y_i^2 = \frac{25}{2},$$

and

$$\sum_i x_i^4 = \frac{85}{8}.$$

Because the grid is symmetric, the normal equations decouple. For the constant term,

$$25\alpha = \sum_i f(x_i, y_i).$$

All odd terms have zero sum on the symmetric grid, and the xy term also has zero sum. Thus

$$\sum_i f(x_i, y_i) = 25 + \sum_i x_i^2 + 2 \sum_i y_i^2 = 25 + \frac{25}{2} + 25 = \frac{125}{2}.$$

Therefore

$$\alpha = \frac{5}{2}.$$

For the x -coefficient,

$$\beta \sum_i x_i^2 = \sum_i x_i f(x_i, y_i).$$

By symmetry,

$$\sum_i x_i f(x_i, y_i) = 2 \sum_i x_i^2 + \sum_i x_i^4 = 25 + \frac{85}{8} = \frac{285}{8}.$$

Thus

$$\beta \cdot \frac{25}{2} = \frac{285}{8},$$

so

$$\beta = \frac{57}{20}.$$

For the y -coefficient,

$$\gamma \sum_i y_i^2 = \sum_i y_i f(x_i, y_i).$$

By symmetry, the only surviving term is $-y$, so

$$\sum_i y_i f(x_i, y_i) = - \sum_i y_i^2.$$

Therefore

$$\gamma = -1.$$

The sampled best-fit plane is

$$q_{\text{sample}}(x, y) = \frac{5}{2} + \frac{57}{20}x - y.$$

The tangent plane and sampled best-fit plane are different:

$$T(x, y) = 1 + 2x - y,$$

while

$$q_{\text{sample}}(x, y) = \frac{5}{2} + \frac{57}{20}x - y.$$

They are different because they solve different approximation problems.

- **Plane:** Tangent plane

How it is chosen: match value and first partial derivatives at $(0, 0)$

Residual condition: $r(0, 0) = 0, \nabla r(0, 0) = \mathbf{0}$

- **Plane:** Sampled best-fit plane

How it is chosen: minimize squared error on sampled points

Residual condition: $\sum_i r_i = 0, \sum_i x_i r_i = 0, \sum_i y_i r_i = 0$

- **Plane:** Continuous best-fit plane

How it is chosen: minimize squared error over a region

Residual condition: $\iint_R r \, dA = 0, \iint_R x r \, dA = 0, \iint_R y r \, dA = 0$

Warning

A tangent plane is local. A sampled best-fit plane depends on the chosen sample points. A continuous best-fit plane depends on the chosen region and uses double integrals.

Note: Continuous best fit as a preview

In one variable, the continuous inner product

$$\langle u, v \rangle = \int_{-1}^1 u(x)v(x) dx$$

measures agreement over an interval. In two variables, the analogous continuous inner product over a region R is

$$\langle u, v \rangle_R = \iint_R u(x, y)v(x, y) dA.$$

MATH 345 does not develop double integrals, so the course-level computation uses sampled sums:

$$\langle u, v \rangle_{\text{sample}} = \sum_i u(x_i, y_i)v(x_i, y_i).$$

With many well-spread sample points, a scaled sampled sum can approximate a continuous integral. This is also a bridge to probability: averages over sampled points approximate expected values.

For the square

$$R = [-1, 1] \times [-1, 1],$$

the continuous $L^2(R)$ -best-fit plane for the same function is

$$q_{L^2}(x, y) = 2 + \frac{13}{5}x - y.$$

This is shown only as a preview. The sampled plane above is the computation used in this course.

Note: Looking ahead

Students who want to study probability together with the multivariate-integration tools behind continuous averaging can look at MATH 331. MATH 331 is an introductory probability course for students who have not taken multivariable calculus and introduces the needed tools from multivariate integration alongside probability.

To summarize: bounded-degree polynomial spaces are vector spaces. Inner products define projection and least squares. Taylor approximation is local projection. Gram-Schmidt builds orthogonal polynomial coordinates. In two variables, tangent planes and best-fit planes are affine polynomial approximations.

7.4 Coding recap

7.4.1 Linked notebook

Run Lab U7: Polynomial Approximation. Use it to compare Taylor, continuous least-squares, and sampled least-squares approximations, and to practice reading feature matrices, least-squares coefficients, fitted values, and residual checks.

For a quick reference on arrays, sample grids, design matrices, matrix products, transposes, least squares, and numerical checks, see the programming appendix sections A.2, A.4, A.8, A.9, and A.10.

7.4.2 Review activities

Activity: Feature matrix for an affine fit

Consider the code:

```
import numpy as np

xs = np.array([-1.0, -0.5, 0.0, 0.5, 1.0])
A = np.column_stack([np.ones_like(xs), xs])

A.shape, A
```

1. How many rows does A have?
2. How many columns does A have?
3. What does the first column represent?
4. What does the second column represent?
5. What polynomial subspace is represented by the columns?

Solution.

The matrix A has five rows, one for each sample point. It has two columns. The first column is the sampled constant feature 1. The second column is the sampled x -feature. Together the columns represent the approximation subspace

$$\mathcal{P}_{\leq 1} = \text{span}\{1, x\}.$$

A coefficient vector

$$\mathbf{c} = \begin{bmatrix} c_0 \\ c_1 \end{bmatrix}$$

produces fitted values for

$$q(x) = c_0 + c_1 x.$$

Activity: Sampled least squares and residual orthogonality

Consider the code:

```
y = 1 + 2*xs + xs**2 + xs**3
```

```
c = np.linalg.lstsq(A, y, rcond=None)[0]
r = y - A @ c

c, A.T @ r
```

Suppose the output is approximately

```
(array([1.5 , 2.85]), array([0., 0.]))
```

1. What line does the coefficient vector represent?
2. What does Ac store?
3. What does r store?
4. What does $A.T @ r$ check?
5. Does $A.T @ r$ being close to zero mean $r = 0$?

Solution.

The coefficient vector

$$\mathbf{c} \approx \begin{bmatrix} 1.5 \\ 2.85 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ \frac{57}{20} \end{bmatrix}$$

represents the sampled least-squares line

$$q_{\text{sample}}(x) = \frac{3}{2} + \frac{57}{20}x.$$

The vector Ac stores the fitted values

$$q_{\text{sample}}(x_i)$$

at the sampled points. The residual vector

$$\mathbf{r} = \mathbf{y} - A\mathbf{c}$$

stores the sampled errors

$$r_i = p(x_i) - q_{\text{sample}}(x_i).$$

The computation $A.T @ r$ checks whether the residual is orthogonal to the sampled constant feature and the sampled x -feature:

$$\sum_i r_i = 0, \quad \sum_i x_i r_i = 0.$$

It does not mean $\mathbf{r} = \mathbf{0}$. It means the residual is orthogonal to the approximation subspace.

Activity: Three coefficient vectors

Consider the code:

```
c_taylor = np.array([1.0, 2.0])
c_continuous = np.array([4/3, 13/5])
c_sampled = np.array([3/2, 57/20])

c_taylor, c_continuous, c_sampled
```

1. Which vector represents the Taylor line at 0?
2. Which vector represents the continuous $L^2[-1, 1]$ -least-squares line?
3. Which vector represents the sampled least-squares line from the five sample points?
4. Which approximation is local?
5. Which approximations depend on an interval or sampled points?

Solution.

The vector

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

represents the Taylor line

$$T(x) = 1 + 2x.$$

The vector

$$\begin{bmatrix} 4/3 \\ 13/5 \end{bmatrix}$$

represents the continuous least-squares line

$$q_{L^2}(x) = \frac{4}{3} + \frac{13}{5}x.$$

The vector

$$\begin{bmatrix} 3/2 \\ 57/20 \end{bmatrix}$$

represents the sampled least-squares line

$$q_{\text{sample}}(x) = \frac{3}{2} + \frac{57}{20}x.$$

The Taylor line is local because it uses value and derivative data at one point. The continuous least-squares line depends on the interval $[-1, 1]$. The sampled least-squares line depends on the chosen sample points.

Activity: Shrinking interval coefficients

Consider the code:

```
ts = np.array([1.0, 0.5, 0.25, 0.125])

x2_const = ts**2 / 3
x3_slope = 3*ts**2 / 5

x2_const, x3_slope
```

1. What polynomial does `x2_const` come from?
2. What polynomial does `x3_slope` come from?
3. What happens to both arrays as $t \rightarrow 0$?
4. How does this connect to Taylor approximation?

Solution.

The array `x2_const` contains the constants in the $L^2[-t, t]$ -projection

$$P_t(x^2) = \frac{t^2}{3}$$

onto $\text{span}\{1, x\}$. The array `x3_slope` contains the slopes in the projection

$$P_t(x^3) = \frac{3t^2}{5}x.$$

Both arrays tend to zero as $t \rightarrow 0$. This matches the Taylor linear approximations of x^2 and x^3 at 0, both of which are zero. The code is reading the limiting local least-squares idea from the previous section.

Activity: Quadratic feature matrix

Consider the code:

```
A2 = np.column_stack([np.ones_like(xs), xs, xs**2])
c2 = np.linalg.lstsq(A2, y, rcond=None)[0]
fit2 = A2 @ c2
r2 = y - fit2

A2.shape, c2, A2.T @ r2
```

1. What polynomial subspace is represented by the columns of $A2$?
2. What does $c2$ store?
3. What does $fit2$ store?
4. What does $A2.T @ r2$ check?
5. Why are there three entries in $A2.T @ r2$?

Solution.

The columns of $A2$ represent

$$\text{span}\{1, x, x^2\} = \mathcal{P}_{\leq 2}$$

sampled at the points in xs . The vector $c2$ stores the coefficients of the sampled least-squares quadratic

$$c_0 + c_1 x + c_2 x^2.$$

The vector $fit2$ stores the fitted values at the sample points. The vector $r2$ stores sampled residuals. The computation $A2.T @ r2$ checks that the residual is orthogonal to each sampled feature column:

$$\sum_i r_i = 0, \quad \sum_i x_i r_i = 0, \quad \sum_i x_i^2 r_i = 0.$$

There are three entries because the approximation subspace has three feature columns.

Activity: Sampled best-fit plane in code

Consider the code:

```
grid = np.linspace(-1, 1, 5)
X, Y = np.meshgrid(grid, grid)
```

```

xx = X.ravel()
yy = Y.ravel()

z = 1 + 2*xx - yy + xx**2 + xx*yy + 2*yy**2 + xx**3

A = np.column_stack([np.ones_like(xx), xx, yy])
c = np.linalg.lstsq(A, z, rcond=None)[0]
r = z - A @ c

c, A.T @ r

```

Suppose the output is approximately

```
(array([ 2.5 ,  2.85, -1.  ]), array([0., 0., 0.]))
```

1. How many sampled points are used?
2. What do the three columns of A represent?
3. What plane does the coefficient vector represent?
4. What does $A^T \mathbf{r} \approx \mathbf{0}$ check?
5. How is this the two-variable version of sampled line fitting?

Solution.

The grid has $5 \cdot 5 = 25$ sampled points. The three columns of A are the sampled constant feature, the sampled x -feature, and the sampled y -feature. The coefficient vector represents the sampled best-fit plane

$$q_{\text{sample}}(x, y) = \frac{5}{2} + \frac{57}{20}x - y.$$

The check $A^T \mathbf{r} \approx \mathbf{0}$ means

$$\sum_i r_i = 0, \quad \sum_i x_i r_i = 0, \quad \sum_i y_i r_i = 0.$$

This is the same residual-orthogonality condition as sampled line fitting, but now the approximation subspace is

$$\text{span}\{1, x, y\}.$$

7.5 Exercises

Learning outcomes. The labels below identify the learning outcomes for each exercise group. Individual problems may also involve earlier outcomes.

7.5.1 Abstract vector spaces

Learning outcomes. U7-LO1, U7-LO2.

- Nicholson 6.1.1(b)
- Nicholson 6.1.2(f)
- Nicholson 6.1.6(b)

7.5.2 Subspaces, bases, and coordinates in nonstandard spaces

Learning outcomes. U7-LO1, U7-LO2.

- Nicholson 6.2.2(d)
- Nicholson 6.2.2(f)
- Nicholson 6.3.2(d)

7.5.3 Inner product spaces

Learning outcomes. U7-LO3, U7-LO4.

- Nicholson 10.1.1(b)
- Nicholson 10.1.1(d)
- Nicholson 10.1.3(d)

7.5.4 Additional exercises

These exercises connect the main Unit 7 ideas: abstract vector spaces, matrix and polynomial examples, inner products, projection equations, Taylor and least-squares approximations, Gram-Schmidt in polynomial spaces, sampled polynomial regression, two-variable planes, and code interpretation. Solutions are collected in Appendix A.7.

Additional exercise: Matrix-space coordinates

Let

$$E_{11} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad E_{12} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad E_{21} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad E_{22} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Let

$$A = \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix}.$$

1. Write A as a linear combination of $E_{11}, E_{12}, E_{21}, E_{22}$.
2. Write the coordinate vector of A in the ordered basis $(E_{11}, E_{12}, E_{21}, E_{22})$.
3. What is $\dim M_{22}$?
4. What is the zero vector in M_{22} ?

Learning outcomes. U7-LO2.

Additional exercise: Polynomial coordinates and subspaces

Let $p(x) = 2 - 3x + x^3$.

1. Write the coordinate vector of p in the ordered basis $(1, x, x^2, x^3)$.
2. Find a basis for $U = \{p \in \mathcal{P}_{\leq 3} : p(0) = 0\}$.
3. Find a basis for $W = \{p \in \mathcal{P}_{\leq 3} : p(0) = p'(0) = 0\}$.
4. Explain why U and W are subspaces.

Learning outcomes. U7-LO1, U7-LO2.

Additional exercise: Derivative and evaluation as linear maps

Define

$$D : \mathcal{P}_{\leq 3} \rightarrow \mathcal{P}_{\leq 2}, \quad D(p) = p',$$

and

$$E_0 : \mathcal{P}_{\leq 3} \rightarrow \mathbb{R}, \quad E_0(p) = p(0).$$

Use the ordered bases $(1, x, x^2, x^3)$ for $\mathcal{P}_{\leq 3}$ and $(1, x, x^2)$ for $\mathcal{P}_{\leq 2}$.

1. Explain why D is linear.
2. Find the matrix of D in these bases.
3. Explain why E_0 is linear.
4. Find a basis for the kernel of E_0 .

Learning outcomes. U7-LO1, U7-LO2.

Additional exercise: Matrix inner product

Use the matrix inner product $\langle A, B \rangle = \text{tr}(A^T B)$. Let

$$A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 0 \\ 4 & 1 \end{bmatrix}.$$

1. Compute $\langle A, B \rangle$.
2. Compute $\|A\|$.
3. Compute $d(A, B)$.
4. Are A and B orthogonal?

Learning outcomes. U7-LO3.

Additional exercise: Same polynomial space, different inner products

In $\mathcal{P}_{\leq 1}$, let $p(x) = 1 + x$ and $q(x) = 1 - x$.

1. Compute $\langle p, q \rangle_{\text{coef}}$.
2. Compute $\langle p, q \rangle_{L^2}$, where $\langle p, q \rangle_{L^2} = \int_{-1}^1 p(x)q(x) dx$.
3. Are p and q orthogonal for the coefficient inner product?
4. Are p and q orthogonal for the $L^2[-1, 1]$ inner product?
5. What does this show about the phrase “orthogonal polynomials”?

Learning outcomes. U7-LO3.

Additional exercise: Projection equations for the continuous least-squares line

Let $p(x) = 1 + 2x + x^2 + x^3$, and $U = \text{span}\{1, x\}$. Use the $L^2[-1, 1]$ inner product.

1. Compute the Gram matrix for the ordered basis $(1, x)$.
2. Compute the right-hand side vector $\mathbf{b}$ for projecting p onto U .
3. Solve $G\mathbf{c} = \mathbf{b}$.
4. What is the continuous least-squares line?

Learning outcomes. U7-LO4, U7-LO5.

Additional exercise: Taylor line versus fixed-interval least squares

Let $f(x) = x^2$.

1. Find the Taylor line of f at 0.
2. Find the $L^2[-1, 1]$ -projection of f onto $\mathcal{P}_{\leq 1} = \text{span}\{1, x\}$.
3. Explain why the answers differ.

Learning outcomes. U7-LO5.

Additional exercise: Taylor projection from a local inner product

On $\mathcal{P}_{\leq 3}$, define

$$\langle p, q \rangle_0 = \sum_{k=0}^3 \frac{p^{(k)}(0)}{k!} \frac{q^{(k)}(0)}{k!}.$$

Let $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$. Show that the orthogonal projection of p onto $\mathcal{P}_{\leq 1}$ is $a_0 + a_1x$. Then rewrite the answer using $p(0)$ and $p'(0)$.

Learning outcomes. U7-LO3, U7-LO5.

Additional exercise: Shrinking least squares

Let $P_t f$ denote the $L^2[-t, t]$ -projection of f onto $\text{span}\{1, x\}$.

1. Show that $P_t(x^2) = \frac{t^2}{3}$.
2. Show that $P_t(x^3) = \frac{3t^2}{5}x$.
3. What happens to both projections as $t \rightarrow 0$?
4. How does this connect to Taylor approximation?

Learning outcomes. U7-LO5.

Additional exercise: Gram-Schmidt in mathcal P_le3

Use the inner product $\langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx$. Apply Gram-Schmidt to $1, x, x^2, x^3$ through the x^3 step.

1. Explain why 1 and x are orthogonal.
2. Derive $x^2 - \frac{1}{3}$.

- Derive $x^3 - \frac{3}{5}x$.
- Why does an orthogonal basis make projection easier?

Learning outcomes. U7-LO6.

Additional exercise: Code interpretation: polynomial regression

Consider the code:

```
A = np.column_stack([np.ones_like(xs), xs, xs**2])
c = np.linalg.lstsq(A, y, rcond=None)[0]
r = y - A @ c
A.T @ r
```

- What polynomial subspace is represented by the columns of A?
- What does the vector c store?
- What does A @ c store?
- What does r store?
- What does A.T @ r check?
- Why does $A^T r \approx 0$ not mean $r \approx 0$?

Learning outcomes. U7-LO4, U7-LO5.

Additional exercise: Two-variable tangent plane and sampled design matrix

Let $f(x, y) = 1 + 2x - y + x^2 + xy + 2y^2 + x^3$.

- Compute the tangent plane to f at $(0, 0)$.
- Write the residual $r_T = f - T$.
- What conditions do $r_T(0, 0)$ and $\nabla r_T(0, 0)$ satisfy?
- For sampled points (x_i, y_i) , write the design matrix for fitting a plane $q(x, y) = \alpha + \beta x + \gamma y$.
- Write the sampled residual-orthogonality equations.

Learning outcomes. U7-LO5.

7.6 Unit 7 highlights

7.6.1 Mathematical quick reference

Vector spaces, bases, and coordinates

Real vector space. A set with addition and real scalar multiplication satisfying closure, associativity and commutativity of addition, a zero vector and additive inverses, both distributive laws, $a(b\mathbf{v}) = (ab)\mathbf{v}$, and $1\mathbf{v} = \mathbf{v}$. A vector need not be a numerical column.

Subspace test. A subset $U \subseteq V$ is a subspace if it contains the zero vector and is closed under addition and scalar multiplication.

Basis and coordinates. A basis is an independent spanning list. For an ordered basis $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_k)$,

$$\mathbf{v} = \sum_{j=1}^k c_j \mathbf{v}_j, \quad [\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_k \end{bmatrix}, \quad \dim V = k.$$

Independence means the zero vector has only the all-zero coefficient representation.

Matrix and polynomial spaces. $M_{m,n}$ has dimension mn , with basis matrices having one entry 1 and all others zero. The space $\mathcal{P}_{\leq d}$ has basis $(1, x, \dots, x^d)$ and dimension $d + 1$. The coordinates of $p(x) = \sum_{j=0}^d c_j x^j$ are its coefficients in that order.

Inner products and geometry

Real inner product. The rule $\langle \mathbf{u}, \mathbf{v} \rangle$ is symmetric, linear in each argument, and positive definite:

$$\langle \mathbf{v}, \mathbf{v} \rangle \geq 0, \quad \langle \mathbf{v}, \mathbf{v} \rangle = 0 \iff \mathbf{v} = \mathbf{0}.$$

Norm, distance, and orthogonality.

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}, \quad d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|, \quad \mathbf{u} \perp \mathbf{v} \iff \langle \mathbf{u}, \mathbf{v} \rangle = 0.$$

Normalize a nonzero vector by dividing by its norm.

Matrix inner product. For matrices of the same size,

$$\langle A, B \rangle = \text{Tr}(A^T B) = \sum_{i,j} a_{ij} b_{ij}.$$

The trace $\text{Tr}(C)$ of a square matrix is the sum of its diagonal entries.

Polynomial inner products. For coefficient lists a_j, b_j ,

$$\langle p, q \rangle_{\text{coef}} = \sum_{j=0}^d a_j b_j, \quad \langle p, q \rangle_{L^2} = \int_{-1}^1 p(x) q(x) dx.$$

The sampled rule $\sum_i p(x_i) q(x_i)$ is an inner product on a chosen polynomial space exactly when no nonzero polynomial in that space vanishes at every sample point. Otherwise it fails positive definiteness.

Projection and Gram matrices

Best approximation. In a real inner product space, the projection $\mathbf{q}$ of $\mathbf{f}$ onto a finite-dimensional subspace U is the unique vector satisfying

$$\mathbf{q} \in U, \quad \mathbf{f} - \mathbf{q} \perp U.$$

It uniquely minimizes $\|\mathbf{f} - \mathbf{u}\|$ over $\mathbf{u} \in U$. Changing the inner product can change the projection.

Gram-matrix equations. For a basis $(\phi_0, \dots, \phi_m)$ of U , write $q = \sum_{j=0}^m c_j \phi_j$ and solve

$$G\mathbf{c} = \mathbf{b}, \quad G_{ij} = \langle \phi_j, \phi_i \rangle, \quad b_i = \langle f, \phi_i \rangle.$$

A genuine inner product and an independent basis make G positive definite and invertible.

Orthogonal basis shortcut. If the nonzero ϕ_j are pairwise orthogonal, then

$$q = \sum_{j=0}^m \frac{\langle f, \phi_j \rangle}{\langle \phi_j, \phi_j \rangle} \phi_j.$$

Three approximation conditions

Taylor matching. For a polynomial p , the degree- m Taylor polynomial at a is

$$T_m p(x) = \sum_{j=0}^m \frac{p^{(j)}(a)}{j!} (x - a)^j.$$

Its residual has zero derivatives through order m at a .

Continuous least squares. Over an interval $[a, b]$ with $a < b$, minimize $\int_a^b (f - q)^2 dx$ over $q \in U$. The residual satisfies $\int_a^b (f - q) \phi_j dx = 0$ for every basis function.

Sampled least squares. With $A_{ij} = \phi_j(x_i)$ and $y_i = f(x_i)$, minimize $\|A\mathbf{c} - \mathbf{y}\|^2$. The equations are

$$A^T A \hat{\mathbf{c}} = A^T \mathbf{y}, \quad \mathbf{r} = \mathbf{y} - A \hat{\mathbf{c}}, \quad A^T \mathbf{r} = \mathbf{0}.$$

Coefficients are unique when A has independent columns. Neither continuous nor sampled least squares generally agrees with Taylor matching.

Local projection and shrinking intervals

Derivative-based inner product. On $\mathcal{P}_{\leq d}$, fix a and positive weights w_j :

$$\langle p, q \rangle_{\text{jet}, a} = \sum_{j=0}^d w_j \frac{p^{(j)}(a)}{j!} \frac{q^{(j)}(a)}{j!}.$$

For $0 \leq m \leq d$, projection onto $\mathcal{P}_{\leq m}$ is exactly $T_m p$: retain the first $m + 1$ coefficients in powers of $x - a$.

Affine projection on a shrinking interval. For a polynomial p and $t > 0$, its $L^2[-t, t]$ -projection onto $\text{span}\{1, x\}$ is

$$P_t p(x) = a_t + b_t x, \quad a_t = \frac{1}{2t} \int_{-t}^t p(x) dx, \quad b_t = \frac{3}{2t^3} \int_{-t}^t x p(x) dx.$$

As $t \rightarrow 0$, $a_t \rightarrow p(0)$ and $b_t \rightarrow p'(0)$, so the coefficients approach those of the Taylor line.

Gram–Schmidt in an inner product space

Orthogonalize an independent list. Starting with $v_0, \dots, v_m$, set

$$u_0 = v_0, \quad u_j = v_j - \sum_{i=0}^{j-1} \frac{\langle v_j, u_i \rangle}{\langle u_i, u_i \rangle} u_i.$$

The vectors u_j are nonzero and pairwise orthogonal, with the same successive spans as the input list. Normalize separately by $q_j = u_j / \|u_j\|$ for an orthonormal basis. Use the specified inner product in every coefficient and norm.

7.6.2 Common mistakes

Assuming vectors must be numerical columns; forgetting that the zero vector in a polynomial space is the zero polynomial; confusing degree d with dimension $d + 1$; treating an affine set as a subspace; changing a basis without changing coordinates; or using a sampled rule as an inner product without checking positive definiteness.

Using “best approximation” without specifying an inner product or error measure; confusing local Taylor matching with fixed-interval or sampled least squares; interpreting residual orthogonality as a zero residual; applying ordinary dot products where a different inner product was specified; or calling an orthogonal basis orthonormal before normalizing it.

7.6.3 Connections

Unit 2 supplies subspaces, bases, coordinates, and dimension. Unit 3 supplies local approximation and tangent planes. Unit 4 supplies projection, residual orthogonality, least squares, and Gram–Schmidt, which extend here to abstract inner product spaces.

Appendix A

Transformer Wrap-Up Worksheet

This optional worksheet asks you to read a simplified transformer block using tools from the course: vectors, dot products, matrix multiplication, affine maps, null spaces, nonlinear functions, Jacobians, gradients, optimization, rank, and low-rank structure.

The goal is not to teach the full engineering of large language models. The goal is to recognize the mathematics and explain how information moves through a simplified model.

Big question. How do the tools from Units 1-6 help us read a transformer-style computation?

How to use this worksheet. Work through the activities in order. Each activity asks you to identify which course tool is being used, rather than to learn a new transformer topic from scratch.

Note: Scope of This Worksheet

This worksheet does not cover tokenization, real training datasets, hardware, backpropagation through attention, model evaluation, alignment, fine-tuning pipelines, or prompt engineering. Those topics belong to other courses. Here we focus on reading a simplified computation.

Note: Linked notebook

For a small computation attached to this worksheet, use Transformer Wrap-Up Lab: A Tiny Attention Block. The notebook is for interpreting shapes, dot products, masks, weighted averages, logits, and one small loss; it is not a real model-training assignment.

For attention-block code involving row-wise softmax, masks, `np.triu`, `.astype(bool)`, `.copy()`, `np.outer`, and plotting, see the programming quick-reference sections A.13 and A.14.

Worksheet goals. After this appendix, you should be able to:

1. Read a token matrix X and check the shapes of $Q = XW_Q$, $K = XW_K$, $V = XW_V$, $S = QK^T$, and $H = A_{\text{att}}V$.
2. Explain attention as dot-product scores, attention weights, and weighted averages of value vectors.
3. Distinguish linear, affine, nonlinear, and data-dependent operations inside a transformer-style block.
4. Interpret an output layer $\ell = Wh + b$ using null-space language.
5. Read a training loss and gradient descent update as an optimization problem.
6. Interpret a low-rank update $W_{\text{new}} = W + BC$ using rank and storage.

Notation card: one simplified attention head. Suppose a sequence has L tokens, each represented by a vector in $\mathbb{R}^d$. Store the token vectors as rows of

$$X \in \mathbb{R}^{L \times d}.$$

Let

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V.$$

The score matrix is

$$S = QK^T.$$

A nonlinear weighting rule converts each row of S into attention weights. We write the resulting matrix as

$$A_{\text{att}}.$$

The attention output is

$$H = A_{\text{att}}V.$$

In transformer attention, the weighting rule is usually row-wise softmax, possibly after scaling and masking. Unit 3 treats row-wise softmax as a nonlinear function.

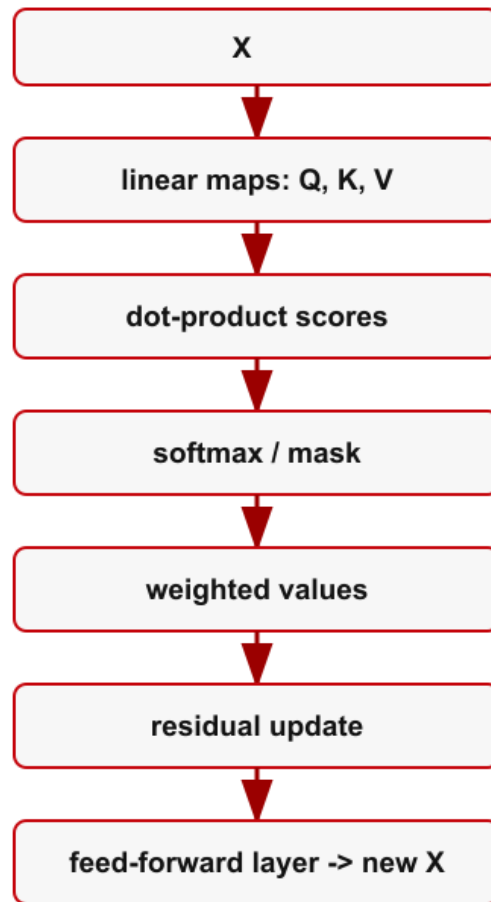


Figure A.1. A transformer block is a nonlinear information processor built from linear maps, dot products, weighted averages, and nonlinear operations.

Activity: Transformer data card

Suppose

$$X \in \mathbb{R}^{4 \times 3}, \quad W_Q, W_K, W_V \in \mathbb{R}^{3 \times 2}.$$

Define

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \quad S = QK^T.$$

A row-wise weighting rule produces

$$A_{\text{att}} \in \mathbb{R}^{4 \times 4},$$

and the attention output is

$$H = A_{\text{att}}V.$$

1. What are the shapes of $Q, K, V, S, A_{\text{att}}, H$?
2. What does S_{ij} compare?
3. What does row i of A_{att} control?
4. Which product forms weighted averages of value vectors?
5. Which step is nonlinear?
6. Which units explain these steps?

Learning outcomes. U1-LO3, U1-LO6, U1-LO8, U3-LO7.

Solution.

Q, K, V are 4×2 . The score matrix $S = QK^T$ is 4×4 . The attention-weight matrix A_{att} is 4×4 , so $H = A_{\text{att}}V$ is 4×2 . The entry S_{ij} compares the query for token i with the key for token j . Row i of A_{att} gives the weights used to combine rows of V . The product $A_{\text{att}}V$ forms weighted averages. The weighting rule that turns scores into attention weights is nonlinear. Units 1, 2, 3, 5, and 6 supply the main tools.

Activity: Linearity audit

Classify each operation as linear, affine, nonlinear, or data-dependent in the relevant input.

1. $X \mapsto XW_Q$
2. $x \mapsto W_1x + b_1$
3. $x \mapsto \sigma(x)$, where σ is applied coordinatewise
4. $S \mapsto \text{softmax}_{\text{rows}}(S)$
5. $(A_{\text{att}}, V) \mapsto A_{\text{att}}V$
6. $(x, \Delta x) \mapsto x + \Delta x$
7. $x \mapsto \text{normalize}(x)$

Learning outcomes. U1-LO7, U3-LO5, U3-LO7.

Solution.

1 is linear in X . 2 is affine in x . 3 is nonlinear. 4 is nonlinear. 5 is bilinear in the pair (A_{att}, V) , and linear in either one if the other is held fixed. 6 is linear as a map of the pair $(x, \Delta x)$, and represents vector addition. 7 is generally nonlinear. Thus a transformer-style block can

contain many matrix multiplications without being a linear map.

Activity: Attention or projection?

Both attention and orthogonal projection use dot products. Match each description with attention, projection, or both.

1. Forms dot-product scores.
2. Produces weights used in a weighted average.
3. Finds the closest vector in a subspace.
4. Has an orthogonal residual condition.
5. May use a nonlinear normalization rule on scores.
6. Is checked by equations such as $A^T r = 0$.

Learning outcomes. U1-LO1, U4-LO4, U4-LO5.

Solution.

1 can appear in both settings. 2 is attention-like. 3 is projection. 4 is projection. 5 is attention-like. 6 is projection. The main distinction is that attention uses scores to mix value vectors, while orthogonal projection solves a closest-vector problem with an orthogonal residual.

Activity: Output layer and forgotten hidden directions

A final hidden vector h is converted into scores by

$$\ell = Wh + b.$$

Suppose $Wz = 0$.

1. Compare $W(h + z) + b$ with $Wh + b$.
2. What subspace contains z ?
3. What does this say about whether the scores determine the hidden vector uniquely?
4. Does the bias vector b change which directions are forgotten?

Learning outcomes. U2-LO3, U2-LO6, U1-LO7.

Solution.

$$W(h + z) + b = Wh + Wz + b = Wh + b.$$

The vector z is in $\text{null}(W)$. Hidden vectors that differ by z give the same scores. The bias shifts the scores but does not change the condition $Wz = 0$.

Activity: Reading a training objective

A simplified next-token training objective is

$$L(\theta) = - \sum_{i=1}^{L-1} \log p_{\theta}(t_{i+1} \mid t_1, \dots, t_i).$$

A gradient descent step has the form

$$\theta_{k+1} = \theta_k - \alpha \nabla L(\theta_k).$$

1. What are the parameters?
2. What quantity is being minimized?
3. Which symbol is the learning rate?
4. What does $\nabla L(\theta_k)$ describe locally?
5. Why is this an optimization problem?

Learning outcomes. U3-LO9.

Solution.

The parameters are collected in θ . The loss $L(\theta)$ is being minimized. The learning rate is α . The gradient $\nabla L(\theta_k)$ gives the local direction of steepest increase of the loss, so the negative gradient is used for descent. This is an optimization problem because training means adjusting parameters to reduce a loss.

Activity: Reading a low-rank update

A model matrix is updated by

$$W_{\text{new}} = W + BC,$$

where

$$W \in \mathbb{R}^{m \times n}, \quad B \in \mathbb{R}^{m \times r}, \quad C \in \mathbb{R}^{r \times n}, \quad r \ll \min(m, n).$$

1. What is the shape of BC ?
2. Why is $\text{rank}(BC) \leq r$?
3. How many entries are stored in B and C , compared with W ?
4. Does this imply that W_{new} has low rank?

Learning outcomes. U2-LO3.

Solution.

The product BC has shape $m \times n$, matching W . Since BC factors through an r -dimensional middle space, its rank is at most r . The two factors store $mr + rn = r(m+n)$ entries, compared with mn entries for W . This does not imply that W_{new} has low rank. The update BC has low rank; the whole matrix $W + BC$ may still have large rank.

Activity: Course concept map

Complete the table.

- **Course idea:** Vectors
Where it appears in the worksheet: ____
Course source: ____
- **Course idea:** Dot products
Where it appears in the worksheet: ____
Course source: ____
- **Course idea:** Matrix multiplication
Where it appears in the worksheet: ____
Course source: ____
- **Course idea:** Affine maps
Where it appears in the worksheet: ____
Course source: ____
- **Course idea:** Null space
Where it appears in the worksheet: ____
Course source: ____
- **Course idea:** Nonlinear functions
Where it appears in the worksheet: ____
Course source: ____
- **Course idea:** Jacobians
Where it appears in the worksheet: ____
Course source: ____

- **Course idea:** Projection and least squares
Where it appears in the worksheet: ____
Course source: ____
- **Course idea:** Gradients
Where it appears in the worksheet: ____
Course source: ____
- **Course idea:** Eigenvalues, quadratic forms, and Hessian curvature
Where it appears in the worksheet: ____
Course source: ____
- **Course idea:** Rank and SVD
Where it appears in the worksheet: ____
Course source: ____

Learning outcomes. U1-LO8, U2-LO3, U3-LO7, U3-LO9, U4-LO4, U6-LO7.

Solution.

- Vectors: token representations; Unit 1.
- Dot products: attention scores; Unit 1.
- Matrix multiplication: $Q = XW_Q$, $K = XW_K$, $V = XW_V$, $S = QK^T$, $H = A_{\text{att}}V$; Unit 1.
- Affine maps: layers such as $Wh + b$; Unit 1.
- Null space: hidden directions an output layer cannot detect; Unit 2.
- Nonlinear functions: softmax, activations, normalization; Unit 3.
- Jacobians: local linear behavior of nonlinear blocks; Unit 3.
- Projection and least squares: closest-vector problems and residual checks in Unit 4, with least-squares applications in Unit 5.
- Gradients, critical points, and gradient descent: training as first-order optimization; Unit 3.
- Eigenvalues, quadratic forms, and Hessian curvature: second-order structure and classification; Unit 5.
- Rank and SVD: transmitted and forgotten directions, singular values, and fundamental subspaces; Unit 6.

Note: Wrap-up highlights

Big question. How do course tools help us read a transformer-style computation?

Notation. $X, Q = XW_Q, K = XW_K, V = XW_V, S = QK^T, A_{\text{att}}, H = A_{\text{att}}V, \ell = Wh + b, L(\theta), W_{\text{new}} = W + BC.$

Main idea. A transformer-style block is not one linear map. It is a nonlinear function built from linear maps, dot products, weighted averages, affine maps, nonlinear weighting rules, residual additions, normalization, and optimization.

Course synthesis. Unit 1 explains vectors, dot products, matrix products, weighted averages, and affine maps. Unit 2 explains forgotten directions. Unit 3 explains nonlinear functions, local linearization, gradients, critical points, and gradient descent. Unit 4 separates attention from projection. Unit 5 supplies eigenvalues, quadratic forms, Hessian curvature, and least-squares applications. Unit 6 explains constrained optimization, principal directions, and SVD. The low-rank-update activity combines matrix shapes from Unit 1 with rank from Unit 2.

The worksheet is successful if you can read the formulas and explain which course tool is being used at each step.

Further reading

For transformer, GPT-style language model, and low-rank adaptation readings connected to this worksheet, see References.

Appendix A

NumPy and SymPy Quick Reference for the Labs

This appendix explains the computational notation used in the Applications and Computation sections and linked notebooks. It assumes familiarity with basic Python variables, arithmetic, functions, and loops, but it does not assume prior experience with NumPy arrays, array slicing, matrix products, numerical linear algebra, or SymPy symbolic matrices.

The goal is not to teach all of Python. The goal is to help you read short computational snapshots as mathematical notation.

Where to look.

Arrays, shapes, and slicing Use A.1, A.2, and A.3 when code imports libraries, creates arrays, checks shapes, or selects rows and columns.

Products and vector comparisons Use A.4, A.5, and A.6 when code uses $*$, $@$, norms, distances, cosine similarity, or rankings.

Row-wise computations and constructors Use A.7 and A.8 when code uses axes, sums, maxima, design matrices, or common arrays.

Numerical checks Use A.9 when code checks approximate equality, residual orthogonality, or rounded output.

Linear algebra commands Use A.10 for rank, determinants, solves, least squares, factorizations, eigenvalue commands, and outer products.

Symbolic computation Use A.11 for SymPy matrices, row reduction, null spaces, Jacobians, substitution, and conversion to NumPy.

Small Python patterns Use A.12 for helper functions, loops, comprehensions, and assertions.

Optional code Use A.13 and A.14 for optional attention masks and plotting.

A.1 How to Read Computational Cells

In this course, code is usually included to express a mathematical idea, not to teach software engineering.

- *Read imports as setup.* Commands such as `import numpy as np`, `import sympy as sp`, and `import matplotlib.pyplot as plt` make a library available under a short name; they have no mathematical output.

- *Read comments as notes.* A line beginning with # is ignored by Python and is included only to help a reader.
- *Read display commands as output choices.* The last line of many notebook cells is displayed automatically, while `print(...)` explicitly displays a value.
- *Read assertions as sanity checks.* A command such as `assert A.shape == (2, 3)` checks that a condition is true.
- *Read shape requests as dimension checks.* A command such as `A.shape` returns a tuple describing the dimensions of an array or matrix.
- *Read display settings as display only.* A command such as `np.set_printoptions(...)` changes how arrays are printed, not the underlying numerical values.

```
import numpy as np

A = np.array([[1, 2, 3],
              [4, 5, 6]])

print(A.shape)
assert A.shape == (2, 3)
```

```
np.set_printoptions(precision=3, suppress=True)
```

The command `np.set_printoptions(precision=3, suppress=True)` only changes how arrays are displayed. It does not change the underlying numerical values.

Warning: Read code as mathematical notation

In the labs, read the code as compressed mathematical notation. The important question is usually what object is represented, what operation is being performed, and what the output means.

Most sections below follow the same pattern. First, a short code fragment shows the kind of computation used in the labs. Then a reading guide translates code patterns into a mathematical reading, expected output or shape, and where the pattern is used. When a command has a common trap, a short warning explains what to watch for.

A.2 NumPy Arrays, Vectors, Matrices, and Shapes

This section helps you read NumPy arrays as vectors, matrices, or data tables, and helps you predict their shapes.

```
import numpy as np

u = np.array([3.0, 4.0])
A = np.array([[1.0, 2.0],
              [3.0, 4.0]])

u.shape
A.shape
```

Here `u.shape` is `(2,)`, a one-dimensional array of length 2. The shape `A.shape` is `(2, 2)`, a two-dimensional array with 2 rows and 2 columns.

```
A = np.array([[1, 2],
              [3, 4]], dtype=float)
```

Use `dtype=float` when converting integer or symbolic data for numerical linear algebra.

```
u = np.asarray([3, 4], dtype=float)
```

The command `np.asarray` treats existing array-like data as a NumPy array. In these notes it is used only to make short helper functions accept either lists or arrays.

```
v = np.asarray([1, 2, 3, 4], dtype=float).reshape(2, 2)
```

The method `.reshape(2, 2)` changes how entries are arranged into rows and columns when the total number of entries matches.

One-dimensional array

```
np.array([3.0, 4.0])
```

Read as. A vector-like one-dimensional array.

Shape/return. Shape `(2,)`.

Used for. Vectors in computations.

Matrix array

```
np.array([[1.0, 2.0],
          [3.0, 4.0]])
```

Read as. A matrix.

Shape/return. Shape `(2, 2)`.

Used for. Linear maps and data matrices.

Shape tuple

`A.shape`

Read as. The dimensions of `A`.

Shape/return. A tuple.

Used for. Shape checks.

Floating-point entries

`dtype=float`

Read as. Store entries as floating-point numerical values.

Shape/return. Numerical entries.

Used for. Numerical linear algebra.

Array conversion

`np.asarray(...)`

Read as. Convert array-like input to a NumPy array.

Shape/return. A NumPy array.

Used for. Helper functions.

Reshape

`.reshape(m, n)`

Read as. Arrange the same entries in a new shape.

Shape/return. Shape (m, n) if the number of entries matches.

Used for. Converting flat numerical output into a matrix shape.

Warning: One-dimensional arrays are not column matrices

The array `np.array([3.0, 4.0])` is not a 2×1 matrix. It is a one-dimensional array of length 2. NumPy still allows it in expressions such as `A @ u` when the dimensions match.

A.3 Indexing and Slicing

This section helps you read Python indexing and slicing as requests for entries, rows, columns, or blocks of a vector or matrix.

```
x = np.array([10, 20, 30, 40])

x[0]      # first entry
x[1]      # second entry
x[-1]     # last entry
x[:2]     # first two entries
x[2:]     # entries from index 2 onward
x[::-1]   # reversed order
```

```
A = np.array([[1, 2, 3],
              [4, 5, 6]])

A[0, 1]    # row 0, column 1
A[0, :]    # first row
A[:, 0]    # first column
A[:, 1]    # second column
A[:, :2]   # all rows, first two columns
```

The first k principal directions use the same slicing language:

```
V_k = V[:, :k]
T_k = Z @ V_k
Zhat_k = T_k @ V_k.T
```

First entry

`x[0]`

Read as. The first entry of x .

Shape/return. A scalar.

Used for. Indexing.

Last entry

`x[-1]`

Read as. The last entry of x .

Shape/return. A scalar.

Used for. Indexing.

Column extraction

`A[:, 0]`

Read as. The first column of A .

Shape/return. A one-dimensional array.

Used for. Column extraction.

First columns

`A[:, :k]`

Read as. All rows and the first k columns.

Shape/return. An array block.

Used for. Principal-direction, basis-column, and leading singular-vector blocks.

First singular values

`s[:k]`

Read as. The first k singular values.

Shape/return. A one-dimensional array.

Used for. Leading singular values.

First rows

`Vt[:k, :]`

Read as. The first k rows and all columns.

Shape/return. An array block.

Used for. Leading right singular directions stored as rows.

Reverse order

`[::-1]`

Read as. Reverse the order.

Shape/return. A reordered array.

Used for. Rankings.

Boolean mask indexing selects entries where a Boolean array is true. In the optional attention notebook, `S_masked[mask] = -1e9` changes only the entries where `mask` is true.

Activity: Check Yourself: Slices

```
A = np.array([[1, 2, 3],
              [4, 5, 6],
              [7, 8, 9]])
```

1. What is `A[:, 0]`?
2. What is `A[:2, :]`?
3. What is `A[:, :2]`?
4. What is the shape of each result?

A.4 Elementwise Arithmetic Versus Linear Algebra

This section helps you read the difference between entry-by-entry arithmetic and linear-algebra operations.

```
u = np.array([1.0, 2.0])
v = np.array([3.0, 4.0])
```



```

u + v
2 * u
u * v      # entry-by-entry product
u**2       # entry-by-entry square
u @ v      # dot product

```

```

A = np.array([[1.0, 2.0],
              [3.0, 4.0]])

A @ u      # matrix-vector product
A.T        # transpose
A.T @ u

```

- **Code pattern:** `u * v`
Read as: entry-by-entry product
- **Code pattern:** `u @ v`
Read as: dot product
- **Code pattern:** `A @ u`
Read as: matrix-vector product
- **Code pattern:** `A @ B`
Read as: matrix product

Vector arithmetic

```
u + v, 2 * u, u**2
```

Read as. Vector sum, scalar multiple, and entry-by-entry square.

Shape/return. Vectors with matching shape.

Used for. Vector arithmetic and coordinatewise operations.

Transpose

```
A.T
```

Read as. The transpose of A.

Shape/return. A matrix with rows and columns switched.

Used for. Residual checks and shape changes.

Quadratic form

`h @ H @ h`

Read as. The scalar $\mathbf{h}^T \mathbf{H} \mathbf{h}$.

Shape/return. A scalar.

Used for. Hessian quadratic forms and the quadratic part of a Taylor polynomial.

Watch for. NumPy evaluates this left to right, but for one-dimensional $\mathbf{h}$ this reads as $\mathbf{h}^T \mathbf{H} \mathbf{h}$.

Sampled polynomial features

`xs**2, xs**3, xx*yy`

Read as. Entry-by-entry powers or products of sampled coordinate arrays.

Shape/return. Arrays with the same shape as the sampled input arrays.

Used for. Polynomial feature columns in design matrices.

Warning: Entrywise multiplication is not matrix multiplication

The symbol `*` is entry-by-entry multiplication. The symbol `@` is the dot product, matrix-vector product, or matrix-matrix product.

Unit 1 uses $\mathbf{u} @ \mathbf{v}$, $\mathbf{K} @ \mathbf{q}$, $\alpha @ \mathbf{V}$, and $\mathbf{Y} = \mathbf{A} @ \mathbf{X}$. These are linear-algebra operations, not entry-by-entry multiplication.

Additional exercise: Check Yourself: `*` versus `@`

Let $\mathbf{u} = \text{np.array}([1, 2])$ and $\mathbf{v} = \text{np.array}([3, 4])$. Which expression computes the dot product, $\mathbf{u} * \mathbf{v}$ or $\mathbf{u} @ \mathbf{v}$? What does the other expression compute?

A.5 Dot Products, Norms, Distances, and Cosine Similarity

This section helps you read short NumPy expressions for dot products, lengths, distances, and direction-based comparisons.

```
u = np.array([3.0, 4.0])
v = np.array([4.0, 3.0])

dot = u @ v
norm_u = np.linalg.norm(u)
distance = np.linalg.norm(u - v)
cosine = (u @ v) / (np.linalg.norm(u) * np.linalg.norm(v))
```

Dot product

`u @ v`

Read as. The dot product $\mathbf{u} \cdot \mathbf{v}$.

Shape/return. A scalar.

Used for. Alignment or score.

Norm

```
np.linalg.norm(u)
```

Read as. The norm $\|u\|$.

Shape/return. A scalar.

Used for. Length.

Row-wise norms

```
np.linalg.norm(X, axis=1)
```

Read as. One norm for each row of X .

Shape/return. A one-dimensional array with one entry per row.

Used for. Row-wise cosine similarities, such as document-ranking scores.

Euclidean distance

```
np.linalg.norm(u - v)
```

Read as. The Euclidean distance between u and v .

Shape/return. A scalar.

Used for. Distance between vectors.

Cosine similarity

```
(u @ v) / (...)
```

Read as. The cosine similarity $\frac{u \cdot v}{\|u\| \|v\|}$.

Shape/return. A scalar.

Used for. Direction-based comparison.

Warning: Cosine similarity is not distance

Cosine similarity compares direction after normalization. It is not the same as Euclidean distance.

A.6 Sorting and Rankings

This section helps you read sorting commands as requests for positions and rankings.

```
scores = np.array([0.2, 0.9, 0.5])
doc_names = np.array(["D0", "D1", "D2"])

np.argsort(scores)          # indices from smallest score to largest
np.argsort(scores)[::-1]    # indices from largest score to smallest
np.argmax(scores)           # index of the largest score

ranking = doc_names[np.argsort(scores)[::-1]]
```

Sorting positions

`np.argsort(scores)`

Read as. Indices from smallest score to largest score.

Shape/return. An index array.

Used for. Sorting.

Largest-value position

`np.argmax(scores)`

Read as. The index of the largest entry of scores.

Shape/return. An integer index.

Used for. Locating a sampled maximum.

Watch for. This returns a position, not the largest value itself. If several entries tie for largest, NumPy returns the first such position.

Reverse ranking

`np.argsort(scores)[::-1]`

Read as. Indices from largest score to smallest score.

Shape/return. An index array.

Used for. Rankings.

Reordered names

`doc_names[...]`

Read as. Select or reorder names by indices.

Shape/return. A reordered array.

Used for. Document ranking.

Warning: Sorted positions are not sorted scores

`np.argsort(scores)` returns positions, not sorted scores.

A.7 Reductions, Axes, and Row-Wise Computations

This section helps you read reductions, axes, and entry-by-entry scalar functions in array computations.

```
A = np.array([[1, 2, 3],
               [4, 5, 6]])

A.sum()
A.sum(axis=0)    # one sum per column
A.sum(axis=1)    # one sum per row
np.sum(A)
```

```
S = np.array([[1.0, 2.0, 3.0],
              [2.0, 4.0, 1.0]])

S_shifted = S - np.max(S, axis=1, keepdims=True)
e = np.exp(S_shifted)
A = e / e.sum(axis=1, keepdims=True)
```

Axes and reductions. These commands collapse all entries, one column at a time, or one row at a time.

All entries

`A.sum()`
Read as. Sum all entries.
Shape/return. A scalar.
Used for. Totals.

Column sums

`A.sum(axis=0)`
Read as. One sum per column.
Shape/return. A one-dimensional array.
Used for. Column summaries.

Row sums

`A.sum(axis=1)`
Read as. One sum per row.
Shape/return. A one-dimensional array.
Used for. Row summaries.

Row maximum

`np.max(S, axis=1, keepdims=True)`
Read as. Row maxima with the reduced dimension preserved.
Shape/return. A column-shaped array.
Used for. Stable softmax-style code.

Row sum with shape

`e.sum(axis=1, keepdims=True)`
Read as. One row sum, with the reduced dimension preserved.
Shape/return. A column-shaped array.
Used for. Row normalization.

Keeping the reduced dimension makes the result line up with the original array for row-by-row subtraction or division. This is why the softmax-style example uses `keepdims=True`.

NumPy also applies many familiar scalar functions entry-by-entry to arrays.

```
x = np.array([-1.0, 0.0, 1.0])

np.exp(x)      # entry-by-entry exponential
np.log(x + 2)  # entry-by-entry logarithm
np.sqrt(x + 1) # entry-by-entry square root
np.tanh(x)     # entry-by-entry hyperbolic tangent
np.abs(x)      # entry-by-entry absolute value
np.sin(x)      # entry-by-entry sine
np.cos(x)      # entry-by-entry cosine
```

Entrywise functions. These commands apply the same scalar function to each entry of an array.

Exponential

`np.exp(x)`

Read as. Entry-by-entry exponential.

Shape/return. An array.

Used for. Softmax.

Logarithm

`np.log(x)`

Read as. Entry-by-entry logarithm.

Shape/return. An array.

Used for. Losses and log probabilities.

Square root

`np.sqrt(x)`

Read as. Entry-by-entry square root.

Shape/return. An array.

Used for. Norms and scales.

Hyperbolic tangent

`np.tanh(x)`

Read as. Entry-by-entry hyperbolic tangent.

Shape/return. An array.

Used for. Nonlinear hidden layer.

Absolute value

`np.abs(x)`

Read as. Entry-by-entry absolute value.

Shape/return. An array with the same shape as `x`.

Used for. Magnitudes and sign-independent alignment checks.

Sine and cosine

`np.sin(x)` and `np.cos(x)`

Read as. Entry-by-entry sine and cosine, with inputs measured in radians.

Shape/return. Arrays with the same shape as `x`.

Used for. Sampling points on a circle.

Coordinatewise maximum

`np.maximum(z, 0)`

Read as. Replace negative entries by 0.

Shape/return. Same shape as `z`.

Used for. Coordinatewise ReLU.

Cumulative energy

```
np.cumsum(s**2) / np.sum(s**2)
```

Read as. Cumulative fraction of squared singular-value energy.

Output. A one-dimensional array.

Used for. Choosing how many SVD terms to keep.

These commands do not perform matrix multiplication. They apply the same scalar function to each entry of the array.

The same idea appears in cumulative SVD energy calculations.

Additional exercise: Check Yourself: Axis

For a two-row, three-column array `A`, how many entries are in `A.sum(axis=0)`? How many are in `A.sum(axis=1)`?

A.8 Constructing Common Arrays and Design Matrices

This section helps you read code that creates standard arrays, identity matrices, sample grids, and design matrices.

```
np.ones(4)
np.ones_like(t)
np.full_like(t, 2.0)
```

```
np.zeros((2, 3))  
np.eye(3)
```

```
t = np.array([0.0, 1.0, 2.0])  
X = np.column_stack([np.ones_like(t), t])
```

```
s = np.array([5.0, 2.0, 1.0])  
Sigma = np.diag(s)
```

```
xs = np.linspace(-1, 1, 41)  
xx, yy = np.meshgrid(xs, xs)  
points = np.vstack([xx.ravel(), yy.ravel()]).T
```

```
mask = np.triu(np.ones_like(S), k=1).astype(bool)
```

Ones vector

```
np.ones(4)
```

Read as. A vector of ones.

Shape/return. Shape (4,).

Used for. Simple test data.

Matching ones

```
np.ones_like(t)
```

Read as. Ones with the same shape as t.

Shape/return. Same shape as t.

Used for. Constant feature column.

Matching constant array

```
np.full_like(t, value)
```

Read as. Fill an array with the same shape as t using one value.

Shape/return. Same shape as t.

Used for. Baseline arrays and plotting helpers.

Zero array

```
np.zeros((m, n))
```

Read as. An $m \times n$ zero array.

Shape/return. Shape (m, n).

Used for. Placeholders and test data.

Identity matrix

```
np.eye(n)
```

Read as. The identity matrix.

Shape/return. Shape (n, n).

Used for. Inverses and rank checks.

Column stack

```
np.column_stack(...)
```

Read as. Put several one-dimensional arrays side by side as columns.

Shape/return. A two-dimensional array.

Used for. Design matrices in least-squares models.

Affine polynomial design matrix

```
A = np.column_stack([np.ones_like(xs), xs])
```

Read as. Put the sampled constant feature and sampled x -feature into columns.

Shape/return. A two-column design matrix.

Used for. Sampled least-squares line fitting.

Quadratic polynomial design matrix

```
A2 = np.column_stack([np.ones_like(xs), xs, xs**2])
```

Read as. Put sampled 1, x , and x^2 features into columns.

Shape/return. A three-column design matrix.

Used for. Sampled least-squares quadratic fitting.

Sampled plane design matrix

```
A = np.column_stack([np.ones_like(xx), xx, yy])
```

Read as. Put sampled 1, x , and y features into columns.

Shape/return. A three-column design matrix.

Used for. Sampled best-fit planes.

Diagonal matrix

`np.diag(s)`

Read as. Build a diagonal matrix with entries of `s` on the diagonal.

Shape/return. A square matrix when `s` is one-dimensional.

Used for. SVD factor matrices and PCA score checks.

Sample grid

`np.linspace(a, b, N)`

Read as. Evenly spaced sample points.

Shape/return. A one-dimensional array.

Used for. Plotting and sample grids.

Two-variable sample grid

`np.meshgrid(grid, grid)`

Read as. Build coordinate arrays for all pairs of grid values.

Shape/return. Two arrays representing *x*- and *y*-coordinates on a rectangular grid.

Used for. Sampling a two-variable function on a grid.

Vertical stack

`np.vstack([a, b])`

Read as. Stack arrays as rows.

Shape/return. A two-dimensional array.

Used for. Turning sampled coordinates into a point table.

Flatten a grid

`X.ravel()`

Read as. List all entries of `X` in a one-dimensional array.

Shape/return. A one-dimensional array.

Used for. Turning grid coordinates into sample-point columns.

Attention mask

`np.triu(np.ones_like(S), k=1).astype(bool)`

Read as. The optional attention mask pattern.

Shape/return. A Boolean array.

Used for. Optional attention code.

Design matrices in least squares often use `np.column_stack` to build one column per feature. The attention-mask pattern is optional and is explained in Appendix A.13.

A.9 Numerical Checks and Roundoff

This section helps you read numerical checks that decide whether computed quantities are close enough for interpretation.

```
np.isclose(0.1 + 0.2, 0.3)
np.allclose(A @ x, b)
np.round(x, 3)
```

```
xhat = np.linalg.lstsq(A, b, rcond=None)[0]
r = b - A @ xhat
np.allclose(A.T @ r, 0)
```

Scalar closeness

```
np.isclose(a, b)
```

Read as. Approximate scalar equality.

Output. A Boolean.

Used for. Scalar checks.

Array closeness

```
np.allclose(x, y)
```

Read as. Approximate array equality.

Output. A Boolean.

Used for. Vector and matrix checks.

Rounded display

```
np.round(x, 3)
```

Read as. Round for display.

Output. A number or array.

Used for. Readable output.

Fitted values

```
A @ c
```

Read as. Evaluate the fitted model at the sampled inputs.

Shape/return. A vector of fitted values.

Used for. Sampled polynomial least-squares fits.

Trapezoid approximation

`np.trapezoid(values, grid)`

Read as. Approximate an integral from sampled values.

Output. A scalar.

Used for. Polynomial inner products and residual-size comparisons.

Residual vector

$y - A @ c$

Read as. Observed or sampled values minus fitted values.

Shape/return. A vector of residuals.

Used for. Residual checks such as $A.T @ r$.

Solution check

`np.allclose(A @ x, b)`

Read as. Check an approximate solution.

Output. A Boolean.

Used for. Numerical systems.

Residual orthogonality

`np.allclose(A.T @ r, 0)`

Read as. Check residual orthogonality.

Output. A Boolean.

Used for. Least squares.

- Numerical linear algebra may produce tiny numbers such as $1e-15$.
- Close to zero is often the right numerical interpretation.
- $A.T @ r$ checks residual orthogonality, not whether r itself is zero.

Warning: Numerical checks are not symbolic proof

Rounding or `allclose` checks support numerical interpretation; they are not symbolic proof.

A.10 NumPy Linear Algebra Commands

Use this section when code calls common NumPy linear-algebra commands: rank, determinant, solve, least squares, eigenvalues, QR, or SVD.

Basic computations. These commands return single quantities or solutions for common linear-algebra tasks.

Norm

`np.linalg.norm(x)`

Read as. The norm $\|x\|$.

Shape/return. A scalar.

Used for. Lengths, distances, and residuals.

Numerical rank

`np.linalg.matrix_rank(A)`

Read as. The numerical rank of A .

Shape/return. An integer.

Used for. Independent directions.

Determinant

`np.linalg.det(A)`

Read as. The determinant of square A .

Shape/return. A scalar.

Used for. Invertibility checks.

Linear solve

`np.linalg.solve(A, b)`

Read as. Solve $Ax = b$ numerically.

Shape/return. A vector or matrix.

Used for. Square full-rank systems.

Least-squares coefficients

```
np.linalg.lstsq(A, b, rcond=None)[0]
```

Read as. The least-squares coefficient vector.

Shape/return. A vector.

Used for. Regression, projection, and sampled polynomial approximation.

Eigenvalue commands. These commands read square or symmetric matrices through eigenvalues and eigenvectors.

General eigenvalues

`np.linalg.eig(H)`

Read as. Eigenvalues and eigenvectors of a square matrix.

Shape/return. A pair: eigenvalue array and eigenvector matrix.

Used for. General eigenvalue computations.

Watch for. For symmetric real matrices, `eigvalsh` or `eigh` is usually cleaner.

Symmetric eigenvalues

`np.linalg.eigvalsh(H)`

Read as. Eigenvalues of a symmetric matrix.

Shape/return. A one-dimensional array of real eigenvalues.

Used for. Hessian classification and symmetric quadratic forms.

Symmetric eigenvectors

`np.linalg.eigh(B)`

Read as. Eigenvalues and orthonormal eigenvectors of a symmetric matrix.

Shape/return. A pair: eigenvalue array and eigenvector matrix.

Used for. Symmetric quadratic forms, covariance matrices, and $A^T A$.

Watch for. Eigenvalues are returned in increasing order.

Outer product

`np.outer(g, h)`

Read as. The outer product gh^T .

Shape/return. Shape (m, d) when g has length m and h has length d.

Used for. PCA reconstructions and rank-one matrices.

Watch for. This is different from $g @ h$, which is a dot product when both are one-dimensional arrays of the same length.

Factorizations. These commands split a matrix into pieces that can be reused for interpretation or reconstruction.

QR factorization

`np.linalg.qr(A)`

Read as. QR factorization.

Shape/return. Q and R.

Used for. Least-squares checks.

Compact SVD

`np.linalg.svd(A, full_matrices=False)`

Read as. Compact singular value decomposition.

Shape/return. U, s, and Vt.

Used for. Matrix stretch, PCA, rank, and singular-vector identities.

Full SVD

`np.linalg.svd(A, full_matrices=True)`

Read as. Full singular value decomposition.

Shape/return. U, s, and Vt; the columns of U and the rows of Vt give complete orthonormal bases.

Used for. The four fundamental subspaces, especially the left null space.

Watch for. The length of s is $\min(m, n)$.

```
result = np.linalg.lstsq(A, b, rcond=None)
xhat = result[0]
```

The command `np.linalg.lstsq` returns several pieces of information. The coefficient vector is the first item, so the textbook often writes $[0]$.

```
U, s, Vt = np.linalg.svd(Z, full_matrices=False)
V = Vt.T
T = Z @ V
covariance_eigenvalues = s**2 / len(Z)
```

- V stores the principal directions as columns.
- T stores the principal scores, with

$$T = ZV = U\Sigma.$$

- $s**2 / \text{len}(Z)$ gives the covariance eigenvalues.
- `full_matrices=False` returns compact arrays that are convenient for PCA and reconstruction.

Additional exercise: Check Yourself: Least Squares Output

In the line that defines `xhat` using `np.linalg.lstsq`, what does `xhat` represent? Why is $[0]$ used? What should be close to zero if the residual is orthogonal to the columns of A ?

Additional exercise: Check Yourself: PCA and SVD

If $V = Vt.T$ and $T = Z @ V$, what do the columns of V represent, and what does the matrix T store? Why do the covariance eigenvalues equal $s**2 / \text{len}(Z)$?

A.11 SymPy Symbolic Computation

This section helps you read SymPy commands for exact symbolic formulas, matrices, Jacobians, substitutions, and null spaces.

SymPy is for exact symbolic expressions. NumPy is for numerical arrays and numerical linear algebra.

```

import sympy as sp
import numpy as np

x, y = sp.symbols("x y")

F = sp.Matrix([
    x**2 * y,
    x * sp.exp(y)
])

J = F.jacobian([x, y])
J_at_a = J.subs({x: 1.0, y: 0.0})
J_at_a = np.array(J_at_a, dtype=float)
sp.diff(x**3 + x*y, x)

```

```

B = sp.Matrix([[1, 2, 3],
               [2, 4, 6],
               [0, 1, 1]])

B.rref()
B.nullspace()
sp.eye(3)

A = [[1, 2, 3],
     [2, 4, 6],
     [0, 1, 1]]
sp.Matrix(A).nullspace()

```

Symbols

`sp.symbols("x y")`

Read as. Create symbolic variables.

Shape/return. Symbols.

Used for. Symbolic formulas.

Symbolic matrix

`sp.Matrix([...])`

Read as. A symbolic vector or matrix.

Shape/return. A SymPy Matrix.

Used for. Exact linear algebra.

Jacobian

`F.jacobian([x, y])`

Read as. A symbolic Jacobian.

Shape/return. A SymPy Matrix.

Used for. Local linearization.

Symbolic derivative

`sp.diff(f, x)`

Read as. Differentiate a symbolic expression with respect to x .

Shape/return. A symbolic expression.

Used for. Exact derivative checks.

Substitution

`J.subs({...})`

Read as. Substitute values.

Shape/return. A symbolic or numeric expression.

Used for. Evaluate at a point.

NumPy conversion

`np.array(J_at_a, dtype=float)`

Read as. Convert to NumPy.

Shape/return. A NumPy array.

Used for. Numerical matrix multiplication.

Reduced row echelon form

`M.rref()`

Read as. Compute the reduced row echelon form exactly.

Output. A pair (R, pivots) .

Used for. Pivot columns, free variables, and solution sets.

Watch for. Pivot indices are zero-based; 0 means the first column.

Nullspace

`M.nullspace()`

Read as. Return a basis for the null space.

Shape/return. A list of SymPy column vectors.

Used for. Exact null-space directions.

Symbolic identity matrix

`sp.eye(n)`

Read as. The $n \times n$ identity matrix in SymPy.

Shape/return. A SymPy Matrix.

Used for. Exact matrix examples.

Convert symbolic matrices to NumPy arrays with `np.array(..., dtype=float)` before numerical matrix multiplication.

A.12 Small Python Patterns Used in the Labs

This section helps you read small Python patterns that support repeated mathematical computations in the labs.

```
def cosim(u, v):  
    return (u @ v) / (np.linalg.norm(u) * np.linalg.norm(v))
```

```
losses = []  
  
for k in range(num_steps):  
    losses.append(float(x @ x))  
    x = x - alpha * grad_f(x)
```

```
results = {alpha: run_gd(alpha) for alpha in [0.05, 0.2, 1.05]}
```

Helper function

`def cosim(u, v):`

Read as. Define a helper function.

Shape/return. A function.

Used for. Repeated computations.

Function output

`return ...`

Read as. Output from a function.

Shape/return. A value.

Used for. Helper functions.

Empty list

```
losses = []
```

Read as. An empty list.

Shape/return. A list.

Used for. Recording values.

Repeated loop

```
for k in range(num_steps):
```

Read as. Repeat the indented commands.

Shape/return. Repeated commands.

Used for. Iterative algorithms.

Append value

```
losses.append(...)
```

Read as. Add one recorded value.

Shape/return. A modified list.

Used for. Training or descent history.

Plain number

```
float(...)
```

Read as. Convert to a plain number.

Shape/return. A scalar.

Used for. Cleaner output.

Dictionary comprehension

```
{alpha: run_gd(alpha) for ...}
```

Read as. Build a dictionary by repeating a computation.

Shape/return. A dictionary.

Used for. Comparing learning rates.

List comprehension

```
[f(x) for x in values]
```

Read as. Build a list by applying the same expression to each item.

Shape/return. A list.

Used for. Compact repeated calculations.

The labs also use list comprehensions, `zip`, `dict`, `len`, `round`, `print`, and `assert`. These are ordinary Python patterns. The course emphasis remains the mathematical meaning of the computation.

A.13 Boolean Masks, Copying, and Attention Helpers

This optional section helps you read the array-manipulation code used in the optional attention material.

```
S = np.array([[1.0, 2.0, 3.0],
              [4.0, 5.0, 6.0],
              [7.0, 8.0, 9.0]])

mask = np.triu(np.ones_like(S), k=1).astype(bool)

S_masked = S.copy()
S_masked[mask] = -1e9
```

```
def softmax_rows(S):
    S_shifted = S - np.max(S, axis=1, keepdims=True)
    e = np.exp(S_shifted)
    return e / e.sum(axis=1, keepdims=True)
```

```
grad_W = np.outer(g, h)
```

Matching ones

`np.ones_like(S)`

Read as. Ones with the same shape as S.

Shape/return. An array.

Used for. Mask construction.

Upper triangle

`np.triu(..., k=1)`

Read as. Entries above the diagonal.

Shape/return. An array.

Used for. Causal mask.

Boolean conversion

`.astype(bool)`

Read as. Convert to True/False.

Shape/return. A Boolean array.

Used for. Mask indexing.

Independent copy

`S.copy()`
Read as. Make an independent copy.
Shape/return. An array.
Used for. Safe modification.

Mask assignment

`S_masked[mask] = -1e9`
Read as. Change selected entries.
Shape/return. A modified array.
Used for. Masked attention scores.

Row maximum

`np.max(S, axis=1, keepdims=True)`
Read as. Row maximum.
Shape/return. A column-shaped array.
Used for. Stable softmax.

Exponential

`np.exp(S_shifted)`
Read as. Entry-by-entry exponential.
Shape/return. An array.
Used for. Softmax.

Row normalization

`e / e.sum(axis=1, keepdims=True)`
Read as. Row normalization.
Shape/return. An array with row sums 1.
Used for. Attention weights.

Outer product

`np.outer(g, h)`
Read as. Outer product.
Shape/return. A matrix.
Used for. Rank-one update.

Warning: Large negative mask values are numerical placeholders

The value `-1e9` is a numerical stand-in for a very large negative score in the optional attention code.

A.14 Optional Plotting

This optional section helps you read optional visualization code. It is not part of the main mathematical calculations.

```
import matplotlib.pyplot as plt

plt.figure(figsize=(5, 4))
plt.imshow(A)
plt.colorbar(label="attention weight")
plt.title("Attention weights")
plt.xlabel("Key index")
plt.ylabel("Query index")
plt.xticks(range(len(tokens)), tokens)
plt.yticks(range(len(tokens)), tokens)
plt.show()
```

Some optional geometric transformation pictures use line plots instead of image plots.

```
plt.figure(figsize=(4, 4))
plt.plot(X_square[0], X_square[1], "o-", label="original")
plt.plot(Y_square[0], Y_square[1], "o-", label="transformed")
plt.axis("equal")
plt.grid(True)
plt.legend()
plt.show()
```

```
fig, ax = plt.subplots(figsize=(5, 4))
ax.plot(xs, ys)
plt.show()
```

Figure setup

```
plt.figure(...)
```

Read as. Start a figure.

Output. A figure.

Used for. Plotting setup.

Figure and axes setup

```
plt.subplots(...)
```

Read as. Start a figure together with an axes object.

Output. A figure and axes pair.

Used for. Optional plot customization.

Matrix image

```
plt.imshow(A)
```

Read as. Display a matrix as an image.

Output. An image plot.

Used for. Attention weights and matrices.

Color scale

```
plt.colorbar(...)
```

Read as. Add a scale bar.

Output. A colorbar.

Used for. Matrix image interpretation.

Labels

```
plt.title(...), plt.xlabel(...), plt.ylabel(...)
```

Read as. Add labels.

Output. Text labels.

Used for. Readable plots.

Tick labels

```
plt.xticks(...), plt.yticks(...)
```

Read as. Add tick labels.

Output. Axis labels.

Used for. Token labels.

Line plot

```
plt.plot(...)
```

Read as. Draw connected points or a curve.

Output. A line plot.

Used for. Geometry pictures.

Equal scales

```
plt.axis("equal")
```

Read as. Use equal scales.

Output. A plot setting.

Used for. Geometric accuracy.

Grid

`plt.grid(True)`

Read as. Show a grid.

Output. A plot setting.

Used for. Optional geometry plots.

Legend

`plt.legend()`

Read as. Show labels.

Output. A legend.

Used for. Comparing curves.

Display figure

`plt.show()`

Read as. Display the figure.

Output. A rendered plot.

Used for. Notebooks.

A.15 Quick Reference

Array setup and shapes. See A.1 and A.2.

Array `np.array([...])`. *Read as.* Create an array. *Used for.* All labs.

Array conversion `np.asarray(...)`. *Read as.* Treat array-like input as an array. *Used for.* Helper functions.

Reshape `.reshape(m, n)`. *Read as.* Rearrange entries into a new shape. *Used for.* Matrix-shaped numerical output.

Display options `np.set_printoptions(...)`. *Read as.* Change printed display of arrays. *Used for.* Readability.

Floating-point data `dtype=float`. *Read as.* Request floating-point numerical data. *Used for.* SymPy-to-NumPy conversion.

Shape `A.shape`. *Read as.* Dimensions. *Used for.* Shape checks.

Matrix products and entrywise operations. See A.4 and A.5.

Transpose `A.T`. *Read as.* Transpose. *Used for.* Residual checks and normal equations.

Dot product `u @ v`. *Read as.* Dot product. *Used for.* Cosine similarity.

Matrix-vector product `A @ x`. *Read as.* Matrix-vector product. *Used for.* Linear maps.

Matrix product `A @ B`. *Read as.* Matrix product. *Used for.* Composition and attention.

Outer product `np.outer(g, h)`. *Read as.* Rank-one outer product. *Used for.* Matrix updates such as gradient steps.

Entrywise product `u * v`. *Read as.* Entry-by-entry product. *Used for.* Elementwise operations.

Norm `np.linalg.norm(x)`. *Read as.* Norm. *Used for.* Distances and residuals.

Row-wise norms `np.linalg.norm(X, axis=1)`. *Read as.* One norm per row. *Used for.* Row-wise cosine similarity.

Indexing, reductions, and entrywise functions. See A.3, A.6, and A.7.

Ranking `np.argsort(scores)[::-1]`. *Read as.* Rank scores largest-to-smallest. *Used for.* Document similarity.

Largest-value position `np.argmax(scores)`. *Read as.* Index of the largest entry. *Used for.* Locating a sampled maximum.

First column `A[:, 0]`. *Read as.* All rows, first column. *Used for.* Column extraction.

Column block `A[:, :k]`. *Read as.* All rows, first k columns. *Used for.* Principal directions, basis columns, and leading singular vectors.

Row block `Vt[:k, :]`. *Read as.* First k rows, all columns. *Used for.* Leading right singular directions stored as rows.

Row sums `A.sum(axis=1)`. *Read as.* One sum per row. *Used for.* Attention normalization.

General sum `np.sum(...)`. *Read as.* Sum selected or all entries. *Used for.* Totals and captured-variation fractions.

Row maximum `np.max(..., axis=..., keepdims=True)`. *Read as.* Maximum along an axis with shape preserved. *Used for.* Stable softmax.

Coordinatewise maximum `np.maximum(z, 0)`. *Read as.* Entry-by-entry maximum with zero. *Used for.* ReLU.

Exponential `np.exp(x)`. *Read as.* Entry-by-entry exponential. *Used for.* Softmax.

Logarithm `np.log(x)`. *Read as.* Entry-by-entry logarithm. *Used for.* Loss and log probabilities.

Square root `np.sqrt(x)`. *Read as.* Entry-by-entry square root. *Used for.* Norms and scales.

Hyperbolic tangent `np.tanh(x)`. *Read as.* Entry-by-entry hyperbolic tangent. *Used for.* Fixed hidden layer.

Absolute value `np.abs(x)`. *Read as.* Entry-by-entry absolute value. *Used for.* Magnitudes and alignment checks.

Sine and cosine `np.sin(x)`, `np.cos(x)`. *Read as.* Entry-by-entry trigonometric values in radians. *Used for.* Sampling circles.

Constructors and linear algebra. See A.8, A.9, and A.10.

Ones vector `np.ones(n)`. *Read as.* Vector of ones. *Used for.* Test data and constant columns.

Matching ones `np.ones_like(t)`. *Read as.* Ones with the same shape. *Used for.* Constant feature columns and masks.

Matching constants `np.full_like(t, value)`. *Read as.* Fill the same shape with one value. *Used for.* Baseline arrays and plotting helpers.

Zero array `np.zeros((m, n))`. *Read as.* Make an $m \times n$ zero array. *Used for.* Test data and placeholders.

Identity matrix `np.eye(n)`. *Read as.* Make an $n \times n$ identity matrix. *Used for.* Rank and inverse checks.

Column stack `np.column_stack([...])`. *Read as.* Build a matrix from columns. *Used for.* Design matrices.

Vertical stack `np.vstack([...])`. *Read as.* Stack arrays as rows. *Used for.* Point-table construction.

Diagonal matrix `np.diag(s)`. *Read as.* Put vector entries on the diagonal. *Used for.* SVD factor matrices and PCA score checks.

Sample grid `np.linspace(a, b, N)`. *Read as.* Evenly spaced sample points. *Used for.* Plotting and sampling.

Trapezoid rule `np.trapezoid(values, grid)`. *Read as.* Approximate an integral from samples. *Used for.* Polynomial inner products.

Numerical rank `np.linalg.matrix_rank(A)`. *Read as.* Numerical rank. *Used for.* Independent directions.

Determinant `np.linalg.det(A)`. *Read as.* Determinant of a square matrix. *Used for.* Invertibility checks.

Least squares `np.linalg.lstsq(A, b, rcond=None)[0]`. *Read as.* Least-squares coefficients. *Used for.* Regression, projection, and sampled polynomial approximation.

Residual orthogonality `A.T @ r`. *Read as.* Dot products with the columns of A . *Used for.* Least-squares residual checks.

Linear solve `np.linalg.solve(A, b)`. *Read as.* Solve a square nonsingular system. *Used for.* Linear systems.

General eigenvalues `np.linalg.eig(H)`. *Read as.* Eigenvalues and eigenvectors. *Used for.* General square matrices.

Symmetric eigenvalues `np.linalg.eigvalsh(H)`. *Read as.* Eigenvalues of a symmetric matrix. *Used for.* Hessian classification.

Symmetric eigenvectors `np.linalg.eigh(B)`. *Read as.* Symmetric eigenvalues and eigenvectors. *Used for.* Quadratic forms, covariance matrices, and $A^T A$.

QR factorization `np.linalg.qr(A)`. *Read as.* QR factorization. *Used for.* Least-squares checks.

SVD `np.linalg.svd(A, full_matrices=False)`. *Read as.* Compact SVD. *Used for.* Matrix stretch, PCA, rank, and singular-vector identities.

Polynomial approximation patterns. See A.4, A.8, A.9, and A.10.

Sampled powers `xs**2, xs**3`. *Read as.* Sampled powers of x . *Used for.* Polynomial feature columns.

Sampled product `xx*yy`. *Read as.* Entrywise product of sampled x - and y -coordinates. *Used for.* Two-variable polynomial features.

Two-variable grid `np.meshgrid(grid, grid)`. *Read as.* Coordinate arrays for a rectangular sample grid. *Used for.* Two-variable sampled fits.

Flatten grid `X.ravel()`. *Read as.* Flatten an array into one dimension. *Used for.* Turning grid coordinates into sample lists.

Fitted sampled values `A @ c`. *Read as.* Fitted sampled values. *Used for.* Polynomial least-squares fits.

Residual vector `y - A @ c`. *Read as.* Residual vector. *Used for.* Residual-orthogonality checks.

Quadratic form `h @ H @ h`. *Read as.* Quadratic form $\mathbf{h}^T H \mathbf{h}$. *Used for.* Hessian quadratic forms.

Python helper patterns. See A.12.

Define function `def f(...):`. *Read as.* Define a helper function. *Used for.* Repeated computations.

Return value `return ...`. *Read as.* Output from a function. *Used for.* Helper functions.

Loop `for k in range(n):`. *Read as.* Repeat an indented block. *Used for.* Iterative algorithms.

Append `losses.append(...)`. *Read as.* Add one item to a list. *Used for.* Recording histories.

Plain number `float(...)`. *Read as.* Convert to a scalar. *Used for.* Cleaner output.

List comprehension `[f(x) for x in values]`. *Read as.* Build a list by looping. *Used for.* Compact repeated calculations.

Dictionary comprehension `{k: v for ...}`. *Read as.* Build a dictionary by looping. *Used for.* Comparing parameter choices.

Pairing `zip(a, b)`. *Read as.* Pair entries from two iterables. *Used for.* Labeled loops.

Dictionary `dict(...)`. *Read as.* Build a dictionary. *Used for.* Named results.

Length `len(values)`. *Read as.* Count entries. *Used for.* Sizes and loops.

Round `round(x, 3)`. *Read as.* Round a Python number. *Used for.* Readable output.

Print `print(...)`. *Read as.* Display a value. *Used for.* Notebook output.

Assert `assert condition`. *Read as.* Check that a condition holds. *Used for.* Lab sanity checks.

Optional attention, SymPy, checks, and plotting. See A.11, A.9, A.13, and A.14.

Upper triangle `np.triu(..., k=1)`. *Read as.* Upper-triangular part above the diagonal. *Used for.* Causal mask.

Boolean mask `.astype(bool)`. *Read as.* Convert to Boolean mask. *Used for.* Optional attention.

Copy `.copy()`. *Read as.* Make an independent copy. *Used for.* Optional attention.

Mask assignment `S_masked[mask] = -1e9`. *Read as.* Change selected entries. *Used for.* Optional attention masks.

Row normalization `e / e.sum(axis=1, keepdims=True)`. *Read as.* Divide each row by its row sum. *Used for.* Softmax rows.

Symbols `sp.symbols("x y")`. *Read as.* Create symbolic variables. *Used for.* SymPy formulas.

Symbolic matrix `sp.Matrix([...])`. *Read as.* Symbolic matrix or vector. *Used for.* SymPy.

Symbolic identity `sp.eye(n)`. *Read as.* SymPy identity matrix. *Used for.* Exact examples.

Derivative `sp.diff(f, x)`. *Read as.* Symbolic derivative. *Used for.* Exact derivative checks.

Jacobian `F.jacobian([x, y])`. *Read as.* Symbolic Jacobian. *Used for.* Local linearization.

Substitution `J.subs({...})`. *Read as.* Substitute values. *Used for.* Jacobian at a point.

NumPy conversion `np.array(..., dtype=float)`. *Read as.* Convert symbolic output to NumPy. *Used for.* Numerical multiplication.

Row reduction `M.rref()`. *Read as.* Reduced row echelon form and pivots. *Used for.* Exact pivot analysis.

Nullspace `M.nullspace()`. *Read as.* List of symbolic column-vector basis elements. *Used for.* Exact null spaces.

Scalar closeness `np.isclose(a, b)`. *Read as.* Approximate scalar equality. *Used for.* Numerical checks.

Approximate equality `np.allclose(x, y)`. *Read as.* Approximate equality. *Used for.* Numerical checks.

Rounded display `np.round(x, 3)`. *Read as.* Round an array for display. *Used for.* Readable output.

Figure setup `plt.figure(...)`. *Read as.* Start a figure. *Used for.* Optional plots.

Figure and axes `plt.subplots(...)`. *Read as.* Start a figure and axes pair. *Used for.* Optional plots.

Matrix image `plt.imshow(A)`. *Read as.* Display a matrix as an image. *Used for.* Optional attention plots.

Color scale `plt.colorbar(...)`. *Read as.* Add a scale bar. *Used for.* Matrix image interpretation.

Plot labels `plt.title(...)`, `plt.xlabel(...)`, `plt.ylabel(...)`. *Read as.* Add labels. *Used for.* Readable plots.

Tick labels `plt.xticks(...)`, `plt.yticks(...)`. *Read as.* Add tick labels. *Used for.* Token labels.

Line plot `plt.plot(...)`. *Read as.* Draw connected points or curves. *Used for.* Optional geometry plots.

Equal scales `plt.axis("equal")`. *Read as.* Use equal horizontal and vertical scales. *Used for.* Optional geometry plots.

Grid `plt.grid(True)`. *Read as.* Show a background grid. *Used for.* Optional geometry plots.

Legend `plt.legend()`. *Read as.* Show plot labels. *Used for.* Optional geometry plots.

Display figure `plt.show()`. *Read as.* Display the figure. *Used for.* Notebooks.

Appendix A

Applied exercise solution sketches

A.1 Unit 1 applied, geometric, and computational interpretation

These solution sketches correspond to the applied, geometric, and computational interpretation exercises at the end of Exercises.

A.1.1 Word-count vectors and cosine similarity

We have $\|\mathbf{q}\| = \sqrt{2}$. For $\mathbf{D}_1 = (2, 0, 2)$,

$$\mathbf{q} \cdot \mathbf{D}_1 = 4, \quad \|\mathbf{D}_1\| = \sqrt{8},$$

so

$$\text{cosim}(\mathbf{q}, \mathbf{D}_1) = \frac{4}{\sqrt{2}\sqrt{8}} = 1.$$

For $\mathbf{D}_2 = (1, 1, 0)$,

$$\mathbf{q} \cdot \mathbf{D}_2 = 1, \quad \|\mathbf{D}_2\| = \sqrt{2},$$

so

$$\text{cosim}(\mathbf{q}, \mathbf{D}_2) = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2}.$$

For $\mathbf{D}_3 = (0, 2, 0)$, $\mathbf{q} \cdot \mathbf{D}_3 = 0$, so $\text{cosim}(\mathbf{q}, \mathbf{D}_3) = 0$. The cosine ranking is $\mathbf{D}_1, \mathbf{D}_2, \mathbf{D}_3$.

The vector $\mathbf{D}_1$ has cosine similarity 1 because $\mathbf{D}_1 = 2\mathbf{q}$. The document $\mathbf{D}_3$ is orthogonal to $\mathbf{q}$. In terms of the dictionary words, $\mathbf{D}_3$ contains only “matrix”, while the query uses “linear” and “data”.

The Euclidean distances are

$$\text{dist}(\mathbf{q}, \mathbf{D}_1) = \|\mathbf{q} - \mathbf{D}_1\| = \sqrt{2},$$

$$\text{dist}(\mathbf{q}, \mathbf{D}_2) = \|\mathbf{q} - \mathbf{D}_2\| = \sqrt{2},$$

$$\text{dist}(\mathbf{q}, \mathbf{D}_3) = \|\mathbf{q} - \mathbf{D}_3\| = \sqrt{6}.$$

The distance ranking has $\mathbf{D}_1$ and $\mathbf{D}_2$ tied, followed by $\mathbf{D}_3$. It does not exactly match the cosine ranking $\mathbf{D}_1, \mathbf{D}_2, \mathbf{D}_3$.

A.1.2 Weighted average of value vectors

The weighted average is

$$\frac{1}{4}(10, 0) + \frac{1}{4}(0, 10) + \frac{1}{2}(10, 10) = (2.5, 0) + (0, 2.5) + (5, 5) = (7.5, 7.5).$$

It is a weighted average because all coefficients are nonnegative and $\frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1$. The largest weight is $\frac{1}{2}$, so $\mathbf{v}_3$ receives the largest weight.

A.1.3 Attention bridge: hand scores versus $K\mathbf{q}$

The dot products are

$$\mathbf{q} \cdot \mathbf{k}_{\text{small}} = (1, 1) \cdot (1, 0) = 1,$$

$$\mathbf{q} \cdot \mathbf{k}_{\text{red}} = (1, 1) \cdot (0, 1) = 1,$$

$$\mathbf{q} \cdot \mathbf{k}_{\text{bird}} = (1, 1) \cdot (1, 1) = 2.$$

Since the key vectors are rows of K ,

$$K\mathbf{q} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}.$$

The entries of $K\mathbf{q}$ are the row dot products with $\mathbf{q}$. The sum of the scores is 4, so $\alpha = (\frac{1}{4}, \frac{1}{4}, \frac{1}{2})$.

A.1.4 Rows measure; columns contribute

The row-dot-product view gives

$$A\mathbf{x} = \begin{pmatrix} (-1, 4, -5) \cdot (2, -3, 4) \\ (3, 1, -2) \cdot (2, -3, 4) \end{pmatrix} = \begin{pmatrix} -34 \\ -5 \end{pmatrix}.$$

The column-combination view gives

$$A\mathbf{x} = 2 \begin{pmatrix} -1 \\ 3 \end{pmatrix} - 3 \begin{pmatrix} 4 \\ 1 \end{pmatrix} + 4 \begin{pmatrix} -5 \\ -2 \end{pmatrix} = \begin{pmatrix} -34 \\ -5 \end{pmatrix}.$$

The first row measures the input by the dot product $(-1, 4, -5) \cdot \mathbf{x}$. The coordinates 2, -3, 4 are the coefficients of the column combination.

A.1.5 The unit-square visualization

We have

$$A\mathbf{e}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad A\mathbf{e}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Also,

$$A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = A\mathbf{e}_1 + A\mathbf{e}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

The four corners transform as

$$A(0, 0) = (0, 0), \quad A(1, 0) = (2, 1), \quad A(1, 1) = (1, 2), \quad A(0, 1) = (-1, 1).$$

The image of the unit square is the parallelogram with vertices $(0, 0)$, $(2, 1)$, $(1, 2)$, and $(-1, 1)$. The two transformed edges leaving the origin are $A\mathbf{e}_1 = (2, 1)$ and $A\mathbf{e}_2 = (-1, 1)$, the columns of A .

A.1.6 Transformation matching

The matrix $A_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is a horizontal shear because $A_1(x, y) = (x + y, y)$. The matrix $A_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ is a 90° counterclockwise rotation because $\mathbf{e}_1 \mapsto \mathbf{e}_2$ and $\mathbf{e}_2 \mapsto -\mathbf{e}_1$.

The matrix $A_3 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ is projection onto the x -axis because $(x, y) \mapsto (x, 0)$. The matrix $A_4 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ is reflection across the y -axis because $(x, y) \mapsto (-x, y)$. The matrix $A_5 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ is a horizontal stretch because $(x, y) \mapsto (2x, y)$.

For

$$A_6 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad A_6\mathbf{e}_1 = \mathbf{e}_2, \quad A_6\mathbf{e}_2 = \mathbf{e}_1.$$

Thus A_6 is reflection across the line $y = x$.

Thus the matches are $A_1 \rightarrow 4$, $A_2 \rightarrow 5$, $A_3 \rightarrow 2$, $A_4 \rightarrow 1$, $A_5 \rightarrow 3$, and $A_6 \rightarrow 6$.

A.1.7 Composition order: RS versus SR

We compute

$$RS = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 2 & 0 \end{pmatrix},$$

while

$$SR = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}.$$

The products are not equal. The product RS means apply S first, then R . The two orders differ because horizontal stretching before rotation is not the same as rotation before horizontal stretching. Thus R and S do not commute.

A.1.8 Affine but not linear

We have $T(0, 0) = (1, 0)$. If $T(\mathbf{x}) = A\mathbf{x}$, then $T(\mathbf{0}) = A\mathbf{0} = \mathbf{0}$. But this transformation sends $\mathbf{0}$ to $(1, 0)$, so it cannot be written as $A\mathbf{x}$.

It can be written as

$$T(\mathbf{x}) = I\mathbf{x} + \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

So T is affine but not linear.

A.1.9 Code interpretation: sorting similarity scores

The command `np.argsort(scores)` returns the indices that sort the scores from smallest to largest:

```
np.argsort(scores)
```

gives

```
array([2, 1, 0])
```

because $0.0 < 0.5 < 1.0$. The slice `[::-1]` reverses the order, giving

```
array([0, 1, 2])
```

Therefore ranking is

```
array(['D1', 'D2', 'D3'])
```

The code ranks documents by the given similarity scores. If the scores are cosine similarities, then the ranking is based on direction after normalization, not Euclidean distance or alphabetical order.

A.1.10 Transformer shape check

Since $X \in \mathbb{R}^{6 \times 5}$ and $W_Q, W_K, W_V \in \mathbb{R}^{5 \times 2}$, we have

$$Q = XW_Q \in \mathbb{R}^{6 \times 2}, \quad K = XW_K \in \mathbb{R}^{6 \times 2}, \quad V = XW_V \in \mathbb{R}^{6 \times 2}.$$

Since $K^T \in \mathbb{R}^{2 \times 6}$, we get $QK^T \in \mathbb{R}^{6 \times 6}$. The (i, j) -entry of QK^T is the dot product of query row i with key row j . It is a relevance score.

In the notation of the exercise, $S = QK^T$ has shape 6×6 , so the later attention-weight matrix A_{att} also has shape 6×6 . Then $H = A_{\text{att}}V$ has shape 6×2 . The product $A_{\text{att}}V$ forms the weighted averages of the value vectors. In this convention, token vectors are stored as rows of X .

A.2 Unit 2 applied, geometric, and computational interpretation

These solution sketches correspond to the additional applied and computational interpretation exercises at the end of Unit 2.

A.2.1 Projection and forgotten height in code

The matrix P keeps the first two coordinates and drops the third:

$$P \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

The two inputs $\mathbf{x}$ and $\mathbf{x} + 6\mathbf{z}$ have the same first two coordinates, so their outputs agree. Since $P\mathbf{z} = \mathbf{0}$, the height direction is forgotten. The rank is 2 and the nullity is 1, so two input directions are transmitted and one input direction is forgotten.

A.2.2 Reachable outputs from augmented rref

For $\mathbf{b}_1$, the rref is

$$\left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{array} \right],$$

so the system is consistent and $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is a solution.

For $\mathbf{b}_2$, the rref is

$$\left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right].$$

The last row is $[0 \ 0 \ | \ 1]$, which means $0 = 1$. Thus $\mathbf{b}_2$ is not reachable. A pivot to the right of the vertical line means inconsistency.

A.2.3 Line of intersection from an rref output

The rref gives

$$x + \frac{5}{3}z = 1, \quad y - \frac{2}{3}z = 0.$$

Let $z = t$. Then

$$x = 1 - \frac{5}{3}t, \quad y = \frac{2}{3}t.$$

So the solution set is

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -5/3 \\ 2/3 \\ 1 \end{bmatrix}.$$

There is one free parameter, so the solution set is a line. Geometrically, it is the line of intersection of the two planes.

A.2.4 Plane through three points from a null-space computation

The rows of M are displacement vectors in the plane. A vector in $\text{null}(M)$ is perpendicular to both rows, so it is normal to the plane. The output gives

$$\mathbf{n} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}.$$

Using $P = (1, 1, -2)$, the plane equation is

$$2(x - 1) - (y - 1) + (z + 2) = 0,$$

or

$$2x - y + z = -1.$$

A.2.5 Reading a null-space basis in code

The vector

$$\mathbf{z} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

is an input direction. Direct multiplication gives $A\mathbf{z} = \mathbf{0}$. If $A\mathbf{x} = \mathbf{y}$, then

$$A(\mathbf{x} + t\mathbf{z}) = A\mathbf{x} + tA\mathbf{z} = \mathbf{y}.$$

Thus different inputs can give the same output.

A.2.6 Redundant square-footage features in code

The equation

$$X \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = 0$$

means

$$\text{first-level area} + \text{second-level area} = \text{total area}$$

for every row. The rank is 2, so the three feature columns contain only two independent directions.

The two coefficient vectors differ by the null-space direction:

$$\mathbf{c} - \mathbf{c}_{\text{alt}} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

Since $X\mathbf{z} = 0$, the two coefficient vectors make the same predictions. With redundant features, individual coefficient importance is not uniquely determined by the prediction vector.

A.2.7 Difference matrix: levels versus changes

The product $D\mathbf{x}$ gives consecutive differences:

$$5 - 2 = 3, \quad 9 - 5 = 4, \quad 10 - 9 = 1.$$

Adding 10 to every entry changes the level but not the differences, so the first two outputs agree. Since

$$D\mathbf{1} = 0,$$

the constant direction is forgotten. The rank is 3. Since D has four columns, the nullity is 1.

A.2.8 Debugging a column-space basis from code

The pivot indices (0, 1) mean mathematical columns 1 and 2. Row reduction identifies pivot positions, but row operations do not preserve the original columns. A basis for $\text{col}(A)$ must use columns of the original matrix A , not columns of R .

Thus a basis is

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix} \right\}.$$

A.2.9 Invertibility gallery in code

A 2×2 matrix is invertible exactly when its determinant is nonzero, equivalently when its rank is 2. The invertible matrices are the second, third, and fourth matrices. The first and fifth matrices are singular because their determinants are 0 and their ranks are 1.

The first matrix

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

forgets the x -direction:

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The fifth matrix

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

forgets the y -direction:

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

A.2.10 Reading an inverse-computation output

The left side of the vertical line has reduced to I_3 , so A is invertible. The right side is the inverse:

$$A^{-1} = \begin{bmatrix} 3 & -1 & -10 \\ -2 & 1 & 6 \\ 0 & 0 & 1 \end{bmatrix}.$$

Row-reducing $[A \mid I_3]$ applies the same row operations to A and I_3 . If the left side becomes I_3 , then the right side is the matrix that undoes A , namely A^{-1} .

A.3 Unit 3 conceptual, geometric, and computational checkpoints

These solution sketches correspond to the additional conceptual, geometric, and computational checkpoints at the end of Unit 3.

A.3.1 Directional derivative as a dot product

Use

$$D_{\mathbf{v}}f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v}.$$

Then

$$D_{\mathbf{v}_1}f(\mathbf{a}) = 6, \quad D_{\mathbf{v}_2}f(\mathbf{a}) = 8.$$

Also,

$$D_{\mathbf{v}_3}f(\mathbf{a}) = \begin{bmatrix} 6 \\ 8 \end{bmatrix} \cdot \frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \frac{18 + 32}{5} = 10,$$

and

$$D_{\mathbf{v}_4}f(\mathbf{a}) = -10.$$

The largest listed directional derivative occurs in direction $\mathbf{v}_3$, and the smallest occurs in direction $\mathbf{v}_4$. This is the Unit 1 dot product measuring alignment: the derivative is largest when the direction is aligned with the gradient and smallest when it points opposite the gradient.

A.3.2 Final-layer rank-one update

The outer product is

$$\mathbf{g}\mathbf{h}^T = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} \begin{bmatrix} 4 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ -8 & 0 \\ 12 & 0 \end{bmatrix}.$$

Its shape is 3×2 . Since every column is a scalar multiple of $\mathbf{g}$, the column space has dimension at most one, so the rank is at most one.

The update

$$W_{\text{new}} = W - \alpha \mathbf{g}\mathbf{h}^T$$

changes W by subtracting a scaled rank-one matrix. In NumPy, `np.outer(g, h)` is appropriate when $\mathbf{g}$ and $\mathbf{h}$ are stored as one-dimensional arrays because it explicitly forms the matrix $\mathbf{g}\mathbf{h}^T$.

The update matrix has rank at most one. This does not imply that the updated matrix W_{new} has rank at most one.

A.3.3 Critical points are candidates

We have

$$\nabla f(x, y) = \begin{bmatrix} 2x \\ -2y \end{bmatrix}.$$

Thus the only critical point is $(0, 0)$.

Along the x -axis,

$$f(t, 0) = t^2,$$

which is positive for $t \neq 0$. Along the y -axis,

$$f(0, t) = -t^2,$$

which is negative for $t \neq 0$. Nearby points therefore occur both above and below $f(0, 0) = 0$. The critical point is a saddle point, not a local minimum or local maximum.

A.3.4 Reading a gradient descent loop

The loop represents

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k).$$

The learning rate is α . The vector $\text{grad}_f(\mathbf{x})$ must have the same shape as $\mathbf{x}$. Replacing the minus sign by a plus sign would move in the direction of steepest increase instead of steepest decrease. The code does not prove that a minimum has been found. It only describes the update rule; convergence depends on the function, starting point, learning rate, and stopping rule.

A.3.5 Learning-rate diagnosis

The learning rate $\alpha = 0.05$ is making slow but steady progress. The learning rate $\alpha = 0.20$ is making faster useful progress. The learning rate $\alpha = 1.05$ appears unstable because the loss increases.

A decreasing loss table does not prove that the global minimum has been found. It only reports what happened for the displayed iterates.

A.3.6 Function type and component functions

The function is vector-valued:

$$F : \mathbb{R}^2 \rightarrow \mathbb{R}^3.$$

Its component functions are

$$F_1(x, y) = x^2 + y, \quad F_2(x, y) = e^{x-y}, \quad F_3(x, y) = x - y.$$

We have

$$F(1, 0) = \begin{bmatrix} 1 \\ e \\ 1 \end{bmatrix}.$$

Since the output has three components and the input has two variables, $J_F(1, 0)$ has shape 3×2 .

A.3.7 Domain, boundary, closed, and bounded

The logarithm requires $y - x^2 > 0$, and the square root requires $x \geq 0$. Thus the domain is

$$D = \{(x, y) \in \mathbb{R}^2 : x \geq 0, y > x^2\}.$$

One interior point is $(1, 2)$. Boundary points include points on $y = x^2$ with $x \geq 0$, such as $(1, 1)$, and points on $x = 0$ with $y \geq 0$, such as $(0, 1)$. The set is not closed because it does not include the parabola $y = x^2$. It is not bounded because points in D can have arbitrarily large coordinates.

A.3.8 One-parameter map and tangent vector

We have

$$\mathbf{r}(2) = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 5 \end{bmatrix}.$$

The derivative is constant:

$$\mathbf{r}'(t) = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}.$$

This is the direction vector of the line, matching the Unit 2 parametric-line form point plus parameter times direction.

A.3.9 Gradient and Hessian

The first partial derivatives are

$$f_x = 2x + y, \quad f_y = x + 4y.$$

Hence

$$\nabla f(x, y) = \begin{bmatrix} 2x + y \\ x + 4y \end{bmatrix}, \quad \nabla f(1, -1) = \begin{bmatrix} 1 \\ -3 \end{bmatrix}.$$

The Hessian is

$$H_f = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix},$$

which is symmetric. The entries of $\nabla f(1, -1)$ measure the sensitivity of f to changes in the x - and y -directions at that point.

A.3.10 Jacobian matrix rows and columns

The Jacobian matrix is

$$J_F(x, y) = \begin{bmatrix} 2xy & x^2 \\ e^y & xe^y \end{bmatrix}.$$

At $\mathbf{a} = (1, 0)$,

$$J_F(\mathbf{a}) = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

The input and output dimensions are both 2. The second row corresponds to the second component $F_2(x, y) = xe^y$. The columns measure the predicted output changes from changing the first and second input coordinates.

A.3.11 Tangent plane as a Unit 2 plane

We have $f(1, 2) = 3$, $f_x(1, 2) = 4$, and $f_y(1, 2) = 1$. The tangent plane is

$$z - 3 = 4(x - 1) + (y - 2).$$

Equivalently,

$$\begin{bmatrix} -4 \\ -1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} x - 1 \\ y - 2 \\ z - 3 \end{bmatrix} = 0.$$

A normal vector is $\mathbf{n} = (-4, -1, 1)^T$. The Unit 2 idea being reused is a plane described by a point and a normal vector.

A.3.12 Scalar local linear prediction

From the previous function, $f(\mathbf{a}) = f(1, 2) = 3$ and

$$\nabla f(\mathbf{a}) = \begin{bmatrix} 4 \\ 1 \end{bmatrix}.$$

The local linear prediction is

$$3 + \begin{bmatrix} 4 & 1 \end{bmatrix} \begin{bmatrix} 0.1 \\ -0.2 \end{bmatrix} = 3 + 0.2 = 3.2.$$

The actual value is

$$f(1.1, 1.8) = 1.1^2 + 1.1(1.8) = 3.19.$$

The error actual minus prediction is $3.19 - 3.2 = -0.01$.

A.3.13 Vector-valued local linear prediction

We have

$$F(\mathbf{a}) = F(1, 0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

and, from the previous Jacobian matrix exercise,

$$J_F(\mathbf{a})\mathbf{h} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0.1 \\ -0.2 \end{bmatrix} = \begin{bmatrix} -0.2 \\ -0.1 \end{bmatrix}.$$

Thus the local prediction is

$$F(\mathbf{a}) + J_F(\mathbf{a})\mathbf{h} = \begin{bmatrix} -0.2 \\ 0.9 \end{bmatrix}.$$

The actual value is approximately

$$F(1.1, -0.2) = \begin{bmatrix} -0.242 \\ 0.9006 \end{bmatrix}.$$

The error vector measures actual output minus predicted output.

A.3.14 Locally forgotten direction

One choice is

$$\mathbf{h} = \begin{bmatrix} -2 \\ 1 \end{bmatrix},$$

since

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

This direction is in the null space of the Jacobian matrix, so the local linear map forgets it to first order. This connects directly to Unit 2 null spaces. It does not prove $F(\mathbf{a} + \mathbf{h}) = F(\mathbf{a})$ exactly, because the Jacobian matrix only gives a local linear approximation.

A.3.15 Chain-rule shape check

Since $G : \mathbb{R}^2 \rightarrow \mathbb{R}^3$, the Jacobian matrix $J_G(\mathbf{a})$ has shape 3×2 . Since $F : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, the Jacobian matrix $J_F(G(\mathbf{a}))$ has shape 2×3 .

Therefore

$$J_{F \circ G}(\mathbf{a}) = J_F(G(\mathbf{a}))J_G(\mathbf{a})$$

has shape 2×2 . The reverse order is not correct because the shapes do not match the order of composition, and in general Jacobian matrix products do not commute.

A.3.16 Reading a tiny sigmoid block

The lines

$$\mathbf{s} = W_1 \mathbf{x} + \mathbf{b}_1, \quad \mathbf{y} = W_2 \mathbf{u} + \mathbf{b}_2$$

are affine maps. The line $\mathbf{u} = \text{sigmoid}(\mathbf{s})$ contains the coordinatewise nonlinear step.

The variables $\mathbf{x}$, $\mathbf{s}$, $\mathbf{u}$, $\mathbf{b}_1$, $\mathbf{b}_2$, and $\mathbf{y}$ represent vectors. The matrix D stores the sigmoid derivative at each coordinate of $\mathbf{s}$. The line $J_{\text{at_x}} = W_2 @ D @ W_1$ computes the Jacobian matrix at $\mathbf{x}$. The full block is usually not one linear map because of the bias terms and the coordinatewise sigmoid step.

A.3.17 Corners are not enough

Both maps send the four corners to the same four outputs:

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

But

$$T\left(\begin{bmatrix} 0 \\ 1/2 \end{bmatrix}\right) = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}, \quad F\left(\begin{bmatrix} 0 \\ 1/2 \end{bmatrix}\right) = \begin{bmatrix} 1/4 \\ 1/2 \end{bmatrix}.$$

So the corner test misses the difference. A grid samples more points, including edge and interior points, and can show bending.

A.3.18 Local square-grid error

The Jacobian matrix is

$$J_F(\mathbf{x}, \mathbf{y}) = \begin{bmatrix} 1 & 2y \\ 0 & 1 \end{bmatrix}.$$

At

$$\mathbf{a} = \begin{bmatrix} 0 \\ 1/2 \end{bmatrix},$$

we get

$$J_F(\mathbf{a}) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

For

$$\mathbf{h} = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix},$$

we have

$$F(\mathbf{a} + \mathbf{h}) - F(\mathbf{a}) = \begin{bmatrix} h_1 + h_2 + h_2^2 \\ h_2 \end{bmatrix}$$

and

$$J_F(\mathbf{a})\mathbf{h} = \begin{bmatrix} h_1 + h_2 \\ h_2 \end{bmatrix}.$$

Thus

$$F(\mathbf{a} + \mathbf{h}) - F(\mathbf{a}) - J_F(\mathbf{a})\mathbf{h} = \begin{bmatrix} h_2^2 \\ 0 \end{bmatrix}.$$

If $|h_2| \leq r$, then $h_2^2 \leq r^2$, so the error goes to zero as the square shrinks.

A.3.19 Code interpretation: square-grid arrays

The array P has shape 2×5 . Its columns are input points. The last column is

$$\begin{bmatrix} 0 \\ 1/2 \end{bmatrix}.$$

The corresponding output column is

$$\begin{bmatrix} 1/4 \\ 1/2 \end{bmatrix}.$$

The expression `np.vstack([x + y**2, y])` builds the two output components of F as rows of a new 2×5 array. This is not matrix multiplication because the first output component contains y^2 , a nonlinear operation.

A.3.20 Sigmoid derivative and local matrix

At

$$\mathbf{a} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

we get

$$\mathbf{s}_\mathbf{a} = W_1\mathbf{a} + \mathbf{b}_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Thus

$$\mathbf{u}_\mathbf{a} = \sigma(\mathbf{s}_\mathbf{a}) = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}, \quad D_\mathbf{a} = \text{diag}(1/4, 1/4, 1/4).$$

Thus

$$W_2 D_a W_1 = \begin{bmatrix} 1 & 0 & -1/2 \\ 0 & 1 & 1/2 \end{bmatrix} \begin{bmatrix} 1/4 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1/4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1/8 & -1/8 \\ 1/8 & 3/8 \end{bmatrix}.$$

Its determinant is

$$(1/8)(3/8) - (-1/8)(1/8) = 1/16,$$

so this local matrix is invertible.

For $\mathbf{c} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, we get

$$\mathbf{s}_c = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad D_c = \text{diag}(d, 1/4, d), \quad d = \sigma(1)(1 - \sigma(1)).$$

Since $d \neq 1/4$, the first and third entries of D_c differ from the corresponding entries of D_a . The block is smooth, but the sigmoid derivative changes with the input, so the Jacobian matrix can change with the base point.

A.3.21 Code interpretation: sigmoid block arrays

The array P has shape 2×3 . Its columns are input points. Since W_1 has shape 3×2 , the array $S = W_1 P + \mathbf{b}_1$ has shape 3×3 . The array $U = \sigma(S)$ has the same shape.

The nonlinear operation is

```
U = sigmoid(S)
```

The first column of P is

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

For this input,

$$\mathbf{s} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}.$$

Therefore the first output column is

$$W_2 \mathbf{u} + \mathbf{b}_2 = \begin{bmatrix} 1/4 \\ 3/4 \end{bmatrix}.$$

The full computation is not one matrix multiplication AP because the sigmoid step is nonlinear.

A.3.22 Softmax weights are data-dependent

The score vector is

$$\mathbf{s} = K\mathbf{q} = \begin{bmatrix} 3q_1 \\ 3q_2 \\ -3q_1 - 3q_2 \end{bmatrix}.$$

At $\mathbf{q} = \mathbf{0}$,

$$\mathbf{s} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

All three scores are equal, so the softmax weights are

$$\alpha(\mathbf{0}) = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}.$$

Therefore

$$F(\mathbf{0}) = V^T \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1/3 \end{bmatrix}.$$

Since a linear map must send $\mathbf{0}$ to $\mathbf{0}$, the map F is not linear.

The softmax entries are positive and add to 1, so $F(\mathbf{q})$ is always a weighted average of the rows of V . The nonlinear step is the softmax rule that turns scores into weights.

A.3.23 Code interpretation: softmax square-grid arrays

The array P has shape 2×4 . Its columns are query vectors. Since K has shape 3×2 , the product

$$S = KP$$

has shape 3×4 . The array Alpha also has shape 3×4 , with one softmax weight vector in each column.

Since V^T has shape 2×3 , the product

$$Q = V^T \text{Alpha}$$

has shape 2×4 . Its columns are output vectors.

The nonlinear operation is

```
Alpha = softmax_columns(S)
```

The full computation is not one matrix multiplication AP because the softmax step uses exponentials and column sums. The weights depend nonlinearly on the scores, and the scores depend on P .

A.3.24 Reading square-grid behavior

Map A is the nonlinear shear. It bends grid lines smoothly because of the y^2 term, and it can agree with a linear shear on four corners while differing at edge and interior points.

Map B is the sigmoid block. Its local matrix has the form $W_2 D_a W_1$, where D_a stores sigmoid derivatives. Its grid can bend and compress because those derivatives depend on the input.

Map C is the attention-style softmax map. The output is

$$V^T \alpha,$$

where the entries of α are positive, add to 1, and depend on the input. Therefore the output is a weighted average of the value vectors and stays inside the triangle they form.

Thus behaviors 1 and 4 match Map A, behaviors 2 and 5 match Map B, and behaviors 3 and 6 match Map C. Map A best shows why four corners alone can miss nonlinear behavior. Map B best shows the chain rule for Jacobian matrices. Map C best shows attention-style weighted averages. A square-grid visualization can show bending, compression, and changing local behavior that four corners alone may miss.

A.4 Unit 4 applied, geometric, and computational interpretation

These solution sketches correspond to the additional applied and computational interpretation exercises at the end of Unit 4.

A.4.1 Orthogonal coordinates and projection

Solution to Orthogonal coordinates and projection.

The columns have length 1, and

$$\mathbf{q}_1^T \mathbf{q}_2 = \frac{1}{2}(1 - 1) = 0.$$

Therefore

$$Q^T Q = I_2.$$

The coordinate vector is

$$\mathbf{c} = Q^T \mathbf{x} = \begin{bmatrix} \mathbf{q}_1^T \mathbf{x} \\ \mathbf{q}_2^T \mathbf{x} \end{bmatrix} = \begin{bmatrix} 2\sqrt{2} \\ \sqrt{2} \end{bmatrix}.$$

Combining the columns of Q gives

$$\hat{\mathbf{x}} = Q\mathbf{c} = 2\sqrt{2}\mathbf{q}_1 + \sqrt{2}\mathbf{q}_2 = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}.$$

The residual is

$$\mathbf{r} = \mathbf{x} - \hat{\mathbf{x}} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}.$$

Since both columns of Q have third coordinate 0,

$$Q^T \mathbf{r} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The multiplication $Q^T \mathbf{x}$ scores the orthonormal directions. The multiplication $Q\mathbf{c}$ combines those directions using the resulting coordinates. The vector $\mathbf{c}$ contains coordinates. The projected vector is

$$\hat{\mathbf{x}} = Q\mathbf{c}.$$

A.4.2 One Gram–Schmidt column step

Solution to One Gram–Schmidt column step.

The projection coefficient is

$$r_{12} = \mathbf{q}_1^T \mathbf{a}_2 = \frac{2}{\sqrt{2}} = \sqrt{2}.$$

Therefore

$$\mathbf{p}_2 = r_{12} \mathbf{q}_1 = \sqrt{2} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

The residual is

$$\mathbf{v}_2 = \mathbf{a}_2 - \mathbf{p}_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

Its length is

$$r_{22} = \|\mathbf{v}_2\| = \sqrt{3}.$$

Thus

$$\mathbf{q}_2 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

The orthogonality check gives

$$\mathbf{q}_1^T \mathbf{q}_2 = \frac{1}{\sqrt{6}}(1 - 1 + 0) = 0.$$

If $r_{22} = 0$, then $\mathbf{v}_2 = \mathbf{0}$. The column $\mathbf{a}_2$ would already be in the span of the earlier directions and would not produce a new unit direction.

A.4.3 Orthogonal-basis coefficients

The vectors are pairwise orthogonal. The coefficients are

$$\begin{aligned}\frac{\mathbf{x} \cdot \mathbf{f}_1}{\mathbf{f}_1 \cdot \mathbf{f}_1} &= 2, \\ \frac{\mathbf{x} \cdot \mathbf{f}_2}{\mathbf{f}_2 \cdot \mathbf{f}_2} &= 1, \\ \frac{\mathbf{x} \cdot \mathbf{f}_3}{\mathbf{f}_3 \cdot \mathbf{f}_3} &= 2.\end{aligned}$$

So $\mathbf{x} = 2\mathbf{f}_1 + \mathbf{f}_2 + 2\mathbf{f}_3$. Orthogonality makes the cross terms vanish.

A.4.4 Projection onto a line

$\mathbf{x} \cdot \mathbf{u} = 5$ and $\mathbf{u} \cdot \mathbf{u} = 5$, so $\text{proj}_L(\mathbf{x}) = \mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. The residual is $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$. Its dot product with $\mathbf{u}$ is $2 - 2 = 0$. The residual is the part of $\mathbf{x}$ perpendicular to the line.

A.4.5 Column-space orthogonal complement

$A^T \mathbf{r} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Therefore $\mathbf{r}$ is orthogonal to each column of A . The orthogonal complement is $\text{span}\left\{\begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}\right\}$. This is exactly the null space of A^T .

A.4.6 Projection onto a plane

A normal vector is $\mathbf{n} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Since $\mathbf{p} \cdot \mathbf{n} = 7$ and $\mathbf{n} \cdot \mathbf{n} = 3$,

$$\text{proj}_U(\mathbf{p}) = \mathbf{p} - (7/3)\mathbf{n} = \begin{bmatrix} -4/3 \\ -1/3 \\ 5/3 \end{bmatrix}.$$

The entries add to 0, so the projected point lies in U .

A.4.7 Unreachable target, closest output

The output of A has the form $(x_1, x_2, x_1 + x_2)$. The first two target entries force $x_1 = 2$ and $x_2 = 1$, which would make the third entry 3, not 5.

$$A^T A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, A^T \mathbf{b} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}.$$

Solving gives $\hat{\mathbf{x}} = \begin{bmatrix} 8/3 \\ 5/3 \end{bmatrix}$. Then

$$A\hat{\mathbf{x}} = \begin{bmatrix} 8/3 \\ 5/3 \\ 13/3 \end{bmatrix},$$

$$\mathbf{r} = \begin{bmatrix} -2/3 \\ -2/3 \\ 2/3 \end{bmatrix},$$

$$\text{and } A^T \mathbf{r} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

A.4.8 Regression design matrix and residual

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

$$X^T X = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, X^T \mathbf{y} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

Solving gives $\beta = \begin{bmatrix} 7/6 \\ 1/2 \end{bmatrix}$. The fitted values are $\begin{bmatrix} 7/6 \\ 5/3 \\ 13/6 \end{bmatrix}$. The residual is $\begin{bmatrix} -1/6 \\ 1/3 \\ -1/6 \end{bmatrix}$, and $X^T \mathbf{r} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

A.4.9 Code interpretation: residual orthogonality

The index `[0]` selects the least-squares coefficient vector returned by `np.linalg.lstsq`.

The coefficient vector is

$$\hat{\mathbf{x}} = \begin{bmatrix} 8/3 \\ 5/3 \end{bmatrix}.$$

The fitted vector is

$$\hat{\mathbf{b}} = A\hat{\mathbf{x}} = \begin{bmatrix} 8/3 \\ 5/3 \\ 13/3 \end{bmatrix},$$

and the residual is

$$\mathbf{r} = \mathbf{b} - \hat{\mathbf{b}} = \begin{bmatrix} -2/3 \\ -2/3 \\ 2/3 \end{bmatrix}.$$

The expression `A.T @ r` checks

$$A^T \mathbf{r} = \mathbf{0},$$

so it checks that the residual is orthogonal to every column of A . The tiny entries come from floating-point roundoff.

The residual is not zero:

$$\|\mathbf{r}\| = \frac{2}{\sqrt{3}} \approx 1.15470.$$

This does not mean least squares failed. The target is not reachable, so the closest reachable output has a nonzero residual.

A.4.10 Debug: wrong residual check

The matrix A has shape $m \times n$, the transpose A^T has shape $n \times m$, and $\mathbf{r}$ has length m . Thus

$$A^T \mathbf{r}$$

is defined and has length n . The product $A\mathbf{r}$ is usually not defined because A expects an input of length n , not m .

Even if the dimensions happened to match, $A\mathbf{r}$ would form a linear combination of the columns of A . It would not collect the dot products of $\mathbf{r}$ with those columns.

The correct condition is

$$A^T \mathbf{r} = \mathbf{0}.$$

A numerical Boolean check is

```
np.allclose(A.T @ r, np.zeros(A.shape[1]))
```

A.4.11 Rank and nonunique coefficients

$A\mathbf{z} = \mathbf{0}$, so $\mathbf{z}$ is a null-space direction. Therefore

$$A(\hat{\mathbf{x}} + 5\mathbf{z}) = A\hat{\mathbf{x}} + 5A\mathbf{z} = A\hat{\mathbf{x}}.$$

The coefficient vector can change while the fitted vector stays fixed. This happens because the third column is the sum of the first two columns.

A.4.12 QR least-squares code reading

Since A has shape 3×2 , reduced QR gives

$$Q : 3 \times 2, \quad R : 2 \times 2.$$

The first Boolean check verifies

$$Q^T Q = I_2,$$

and the second verifies

$$QR = A.$$

The solve computes the reduced system

$$R\hat{\mathbf{x}} = Q^T \mathbf{b}.$$

Both methods return

$$\hat{\mathbf{x}} = \begin{bmatrix} 8/3 \\ 5/3 \end{bmatrix}$$

up to roundoff, so the final comparison is `True`.

The agreement does not mean that $A\mathbf{x} = \mathbf{b}$ has an exact solution. The residual is nonzero.

A QR factorization permits a simultaneous sign change in one column of Q and the corresponding row of R . Numerical software may therefore return signs different from a hand calculation while still satisfying

$$Q^T Q = I \quad \text{and} \quad QR = A.$$

A.4.13 Projection and attention: score, then combine

For the projection pipeline,

$$\mathbf{c} = Q^T \mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Therefore

$$\hat{\mathbf{x}} = Q\mathbf{c} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{r} = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}.$$

The residual check gives

$$Q^T \mathbf{r} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

For the attention pipeline,

$$\mathbf{s} = K\mathbf{q} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \alpha = \begin{bmatrix} 2/3 \\ 1/3 \end{bmatrix}.$$

The output is

$$\mathbf{h} = \alpha^T V = \frac{2}{3} \begin{bmatrix} 3 & 0 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 0 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 2 \end{bmatrix}.$$

Both pipelines use dot products or matrix products to produce scores and use linear combinations to combine vectors.

Projection scores the orthonormal directions in Q and combines those same directions. Its residual is orthogonal to their span. Attention scores key vectors and combines value vectors, which may be different vectors.

A.5 Unit 5 applied, geometric, and computational interpretation

These solution sketches correspond to the additional applied and computational interpretation exercises at the end of Unit 5.

A.5.1 Eigenvectors as special directions

Learning outcomes. U5-LO1.

We compute

$$A\mathbf{v}_1 = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix} = 3\mathbf{v}_1.$$

So $\mathbf{v}_1$ is an eigenvector with eigenvalue 3.

Next,

$$A\mathbf{v}_2 = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = 2\mathbf{v}_2.$$

So $\mathbf{v}_2$ is an eigenvector with eigenvalue 2. The matrix map stretches the $\mathbf{v}_1$ -direction by a factor of 3 and the $\mathbf{v}_2$ -direction by a factor of 2.

A.5.2 Diagonalization and matrix powers

Learning outcomes. U5-LO1, U5-LO2.

The columns of P are

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Direct calculation gives

$$A\mathbf{v}_1 = 2\mathbf{v}_1, \quad A\mathbf{v}_2 = 3\mathbf{v}_2.$$

Thus the columns of P are eigenvectors with eigenvalues 2 and 3, respectively.

We have

$$P^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}.$$

Therefore

$$\begin{aligned} PDP^{-1} &= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} = A. \end{aligned}$$

Since $D^k = \text{diag}(2^k, 3^k)$,

$$\begin{aligned}
A^k &= PD^kP^{-1} \\
&= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2^k & 0 \\ 0 & 3^k \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 2^k & 3^k - 2^k \\ 0 & 3^k \end{bmatrix}.
\end{aligned}$$

In particular,

$$A^4 = \begin{bmatrix} 16 & 65 \\ 0 & 81 \end{bmatrix}.$$

Diagonalization reduces repeated matrix actions to taking powers of the diagonal entries of D and then changing coordinates with P and P^{-1} .

A.5.3 Quadratic form and definiteness

Learning outcomes. U5-LO4, U5-LO5.

For the three listed vectors,

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}^T H \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 4,$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}^T H \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1,$$

and

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}^T H \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 5.$$

For a general vector

$$\mathbf{h} = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix},$$

$$\mathbf{h}^T H \mathbf{h} = 4h_1^2 + h_2^2.$$

This is positive for every nonzero $\mathbf{h}$, so H is positive definite. If H is the Hessian at a critical point, the second derivative test gives a local minimum.

A.5.4 Hessian eigenvalue classification

Learning outcomes. U5-LO5.

If the Hessian eigenvalues are 3, 5, the Hessian is positive definite, so the critical point is a local minimum.

If the Hessian eigenvalues are $-2, -7$, the Hessian is negative definite, so the critical point is a local maximum.

If the Hessian eigenvalues are $4, -1$, the Hessian is indefinite, so the critical point is a saddle point.

If the Hessian eigenvalues are $0, 2$, the test is inconclusive. The zero eigenvalue means the quadratic approximation has a flat direction, so the second-order term alone does not decide the local behavior.

A.5.5 Least squares from gradients

Learning outcomes. U5-LO6, U4-LO4.

We compute

$$A^T A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, \quad A^T \mathbf{b} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

The normal equations are

$$A^T A \hat{\mathbf{x}} = A^T \mathbf{b},$$

or

$$\begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \hat{\mathbf{x}} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

Solving gives

$$\hat{\mathbf{x}} = \begin{bmatrix} 7/6 \\ 1/2 \end{bmatrix}.$$

Then

$$A\hat{\mathbf{x}} = \begin{bmatrix} 7/6 \\ 5/3 \\ 13/6 \end{bmatrix},$$

so the residual is

$$\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}} = \begin{bmatrix} -1/6 \\ 1/3 \\ -1/6 \end{bmatrix}.$$

Finally,

$$A^T \mathbf{r} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

This is the residual-orthogonality condition from Unit 4 and the gradient-zero condition from Unit 5.

A.5.6 Residual orthogonality in code

Learning outcomes. U5-LO6, U4-LO4.

The vector $\hat{\mathbf{x}}$ is a least-squares coefficient vector. The vector $\mathbf{r}$ is the residual

$$\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}.$$

The expression $A^T \mathbf{r}$ checks whether the residual is orthogonal to the columns of A . It can be close to zero even when $\mathbf{r}$ is not close to zero. Unit 4 used this residual-orthogonality condition to derive the normal equations. Unit 5 obtains the same equations by setting the gradient of $\|A\mathbf{x} - \mathbf{b}\|^2$ equal to zero.

A.5.7 Fixed-feature design matrix

Learning outcomes. U5-LO6, U5-LO7.

For $t = -1, 0, 1$, the design matrix is

$$A = \begin{bmatrix} 1 & \tanh(-1) & \tanh(-2) & \tanh(0) \\ 1 & \tanh(0) & \tanh(-1) & \tanh(1) \\ 1 & \tanh(1) & \tanh(0) & \tanh(2) \end{bmatrix}.$$

The trained vector is

$$\mathbf{c} = \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix}.$$

This is a least-squares problem because the prediction vector is $A\mathbf{c}$, and training chooses $\mathbf{c}$ so that $A\mathbf{c}$ is close to $\mathbf{y}$. The model is not linear as a function of t , because it contains $\tanh(t)$, $\tanh(t-1)$, and $\tanh(t+1)$. It is linear as a function of $\mathbf{c}$, which is why least squares applies.

A.6 Unit 6 applied, geometric, and computational interpretation

These solution sketches correspond to the additional applied and computational interpretation exercises at the end of Exercises.

A.6.1 Choosing candidates and equations

Solution to Choosing candidates and equations.

For the rectangle, first solve

$$\nabla f(x, y) = \mathbf{0}$$

for critical points in the interior. Restrict f to each of the four edges and find the one-variable critical points on those edges. Include all endpoints and corners. Evaluate f at every candidate and compare the values.

With one equality constraint, solve

$$\nabla f(\mathbf{x}) = \lambda \nabla g(\mathbf{x})$$

together with

$$g(\mathbf{x}) = k.$$

Evaluate f at every resulting candidate and compare the values when absolute extrema are requested.

With two equality constraints, solve

$$\nabla f(\mathbf{x}) = \lambda_1 \nabla g_1(\mathbf{x}) + \lambda_2 \nabla g_2(\mathbf{x})$$

together with

$$g_1(\mathbf{x}) = k_1, \quad g_2(\mathbf{x}) = k_2.$$

Again, evaluate f at all candidates and compare the values.

For the region $g(\mathbf{x}) \leq k$, solve

$$\nabla f(\mathbf{x}) = \mathbf{0}$$

at feasible interior points satisfying $g(\mathbf{x}) < k$. On the boundary, solve

$$\nabla f(\mathbf{x}) = \lambda \nabla g(\mathbf{x}), \quad g(\mathbf{x}) = k.$$

Compare all interior and boundary candidates. Since the region is closed and bounded, absolute extrema exist when f is continuous by the extreme value theorem.

A.6.2 One quadratic form, three readings

Solution to One quadratic form, three readings.

The characteristic polynomial of B is

$$\det(B - \lambda I) = (5 - \lambda)^2 - 16 = (\lambda - 9)(\lambda - 1).$$

Thus the eigenvalues are

$$\lambda_1 = 9, \quad \lambda_2 = 1.$$

Corresponding unit eigenvectors are

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

By the fact “Extrema of a symmetric quadratic form on the unit sphere”,

$$\max_{\|\mathbf{x}\|=1} q_B(\mathbf{x}) = 9$$

in direction $\mathbf{v}_1$, while

$$\min_{\|\mathbf{x}\|=1} q_B(\mathbf{x}) = 1$$

in direction $\mathbf{v}_2$.

We calculate

$$A^T A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} = B.$$

Therefore the singular values of A are

$$\sigma_1 = \sqrt{9} = 3, \quad \sigma_2 = \sqrt{1} = 1,$$

and the right singular directions are $\mathbf{v}_1$ and $\mathbf{v}_2$. In particular,

$$\max_{\|\mathbf{x}\|=1} \|A\mathbf{x}\| = 3.$$

Since

$$C = \frac{1}{2}B,$$

the covariance matrix has the same eigenvectors, with eigenvalues

$$\frac{9}{2} \quad \text{and} \quad \frac{1}{2}.$$

Thus $\mathbf{v}_1$ is a first principal direction and $\mathbf{v}_2$ is a second principal direction. Their captured variations are $9/2$ and $1/2$.

The first principal direction captures the fraction

$$\frac{9/2}{9/2 + 1/2} = \frac{9}{10}$$

of the total centered variation.

The same directions appear because the three symmetric matrices being analyzed are

$$B, \quad A^T A = B, \quad C = \frac{1}{2}B.$$

Multiplying a matrix by a positive scalar changes its eigenvalues but not its eigenvectors.

A.6.3 Candidate table on a rectangle

The gradient is

$$\nabla f(x, y) = \begin{bmatrix} 2x - 2 \\ 2y + 4 \end{bmatrix}.$$

The interior critical point is

$$(1, -2),$$

and

$$f(1, -2) = -5.$$

On $x = 0$,

$$f(0, y) = y^2 + 4y,$$

so the boundary critical point is $y = -2$, giving $(0, -2)$ and value -4 .

On $x = 3$,

$$f(3, y) = y^2 + 4y + 3,$$

so the boundary critical point is $y = -2$, giving $(3, -2)$ and value -1 .

On $y = -3$,

$$f(x, -3) = x^2 - 2x - 3,$$

so the boundary critical point is $x = 1$, giving $(1, -3)$ and value -4 .

On $y = 1$,

$$f(x, 1) = x^2 - 2x + 5,$$

so the boundary critical point is $x = 1$, giving $(1, 1)$ and value 4 .

The corner values are

$$f(0, -3) = -3, \quad f(3, -3) = 0, \quad f(0, 1) = 5, \quad f(3, 1) = 8.$$

Thus the absolute minimum value is

$$-5$$

at

$$(1, -2),$$

and the absolute maximum value is

$$8$$

at

$$(3, 1).$$

A.6.4 Parallel gradients on a circle

Use

$$g(x, y) = x^2 + y^2.$$

Then

$$\nabla f = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \nabla g = \begin{bmatrix} 2x \\ 2y \end{bmatrix}.$$

The Lagrange equations are

$$3 = 2\lambda x, \quad 4 = 2\lambda y, \quad x^2 + y^2 = 25.$$

The first two equations say that

$$\begin{bmatrix} x \\ y \end{bmatrix}$$

is parallel to

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

Since

$$\left\| \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\| = 5,$$

the constrained critical points are

$$(3, 4) \quad \text{and} \quad (-3, -4).$$

The function values are

$$f(3, 4) = 25, \quad f(-3, -4) = -25.$$

So the absolute maximum is 25 at (3, 4), and the absolute minimum is -25 at (-3, -4).

Geometrically, ∇f points in the direction of fastest increase of f , while ∇g is perpendicular to the circle. At the maximum and minimum, these directions are parallel or antiparallel.

A.6.5 Quadratic form on the unit circle

For

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix},$$

we have

$$q(\mathbf{x}) = \mathbf{x}^T \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix} \mathbf{x} = 5x^2 + 2y^2.$$

On the unit circle,

$$x^2 + y^2 = 1.$$

The maximum value is 5, attained at

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{x} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

The minimum value is 2, attained at

$$\mathbf{x} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{x} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

These are the eigenvalues and eigenvectors of B : the eigenvalues are 5 and 2, with eigenvector directions $\mathbf{e}_1$ and $\mathbf{e}_2$.

A.6.6 Maximum stretch of a matrix

Since A is symmetric,

$$A^T A = A^2.$$

Compute

$$A^T A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}.$$

The eigenvectors

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

have eigenvalues 9 and 1, respectively. Therefore the singular values are

$$3 \quad \text{and} \quad 1.$$

The largest singular value is 3, so

$$\max_{\|\mathbf{x}\|=1} \|A\mathbf{x}\| = 3.$$

The most-stretched input direction is

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

A.6.7 Reading an SVD

The nonzero singular values are

$$5 \quad \text{and} \quad 2.$$

The matrix has rank

$$2.$$

Using

$$A\mathbf{v}_i = \sigma_i \mathbf{u}_i$$

for positive singular values, and $A\mathbf{v}_i = \mathbf{0}$ for a null direction,

$$A\mathbf{v}_1 = 5 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix},$$

$$A\mathbf{v}_2 = 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix},$$

and

$$A\mathbf{v}_3 = \mathbf{0}.$$

The forgotten input direction is

$$\mathbf{v}_3 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}.$$

The transmitted input directions are

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

and

$$\mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

A.6.8 Fundamental subspaces from an SVD

Since the rank is 2, the first two right singular vectors form an orthonormal basis for the row space:

$$\text{row}(A) = \text{span} \left\{ \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

The remaining right singular vector gives the null space:

$$\text{null}(A) = \text{span} \left\{ \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \right\}.$$

The first two left singular vectors form an orthonormal basis for the column space:

$$\text{col}(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} = \mathbb{R}^2.$$

Since A has two rows and rank 2,

$$\text{null}(A^T) = \{\mathbf{0}\},$$

whose basis is the empty set.

The dimensions are

$$\dim(\text{row}(A)) = 2, \quad \dim(\text{null}(A)) = 1,$$

and

$$2 + 1 = 3,$$

which agrees with rank-nullity for a 2×3 matrix. Also,

$$\dim(\text{col}(A)) = 2, \quad \dim(\text{null}(A^T)) = 0,$$

and

$$2 + 0 = 2.$$

A.6.9 Two constraints on a sphere

The feasible set is the intersection of the unit sphere and the plane

$$x + y + z = 0.$$

It is a circle, so it is closed and bounded. Since f is continuous, absolute extrema exist by the extreme value theorem.

Let

$$g_1(x, y, z) = x^2 + y^2 + z^2, \quad g_2(x, y, z) = x + y + z.$$

Then

$$\nabla g_1 = 2 \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad \nabla g_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

At a feasible point,

$$\nabla g_1 \cdot \nabla g_2 = 2(x + y + z) = 0.$$

Both gradients are nonzero there. They are therefore linearly independent. Define

$$L(x, y, z, \lambda, \mu) = x - y - \lambda(x^2 + y^2 + z^2 - 1) - \mu(x + y + z).$$

The equations are

$$1 = 2\lambda x + \mu,$$

$$-1 = 2\lambda y + \mu,$$

$$0 = 2\lambda z + \mu,$$

together with the two constraints.

Adding the first three equations and using $x + y + z = 0$ gives

$$\mu = 0.$$

Also, $\lambda \neq 0$, since otherwise the first equation would be impossible. Thus

$$x = \frac{1}{2\lambda}, \quad y = -\frac{1}{2\lambda}, \quad z = 0.$$

The sphere constraint gives

$$\frac{1}{2\lambda^2} = 1,$$

so

$$\lambda = \pm \frac{1}{\sqrt{2}}.$$

The candidates are

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \quad \text{and} \quad \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}.$$

Their f -values are

$$\sqrt{2} \quad \text{and} \quad -\sqrt{2}.$$

Therefore the absolute maximum is $\sqrt{2}$, and the absolute minimum is $-\sqrt{2}$.

A.6.10 Principal directions of a centered data set

The mean is

$$\bar{\mathbf{x}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Therefore

$$Z = \begin{bmatrix} 3 & 1 \\ 1 & 3 \\ -3 & -1 \\ -1 & -3 \end{bmatrix}.$$

The covariance matrix is

$$C = \frac{1}{4} Z^T Z = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}.$$

Its eigenvalues are

$$\lambda_1 = 8, \quad \lambda_2 = 2.$$

Corresponding unit eigenvectors are

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Thus

$$V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

The principal-coordinate matrix is

$$T = ZV = \begin{bmatrix} 2\sqrt{2} & \sqrt{2} \\ 2\sqrt{2} & -\sqrt{2} \\ -2\sqrt{2} & -\sqrt{2} \\ -2\sqrt{2} & \sqrt{2} \end{bmatrix}.$$

Also,

$$V^T C V = \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix}.$$

The first principal direction captures

$$\frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{8}{8 + 2} = \frac{4}{5}$$

of the total centered variation.

A.6.11 PCA reconstruction and regression

The score is

$$t_1 = \mathbf{v}_1^T \mathbf{z}_1 = \frac{4}{\sqrt{2}} = 2\sqrt{2}.$$

Therefore

$$\hat{\mathbf{z}}_1 = t_1 \mathbf{v}_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix},$$

and

$$\mathbf{r}_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

We check that

$$\mathbf{r}_1 \cdot \mathbf{v}_1 = \frac{1}{\sqrt{2}}(1 - 1) = 0.$$

Since

$$\bar{\mathbf{x}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

the reconstructed original data point is

$$\hat{\mathbf{x}}_1 = \begin{bmatrix} 3 \\ 3 \end{bmatrix}.$$

The PCA line passes through (1, 1) in direction (1, 1), so its equation is

$$y = x.$$

For regression,

$$m = \frac{3}{5}, \quad b = 1 - \frac{3}{5} = \frac{2}{5}.$$

Thus the regression line is

$$\hat{y} = \frac{2}{5} + \frac{3}{5}x.$$

PCA treats the two feature coordinates symmetrically and minimizes perpendicular reconstruction residuals. Regression treats x as the predictor and y as the response, and minimizes vertical prediction residuals. The two objectives need not produce the same line.

A.6.12 Redundant features revisited by singular values

Compute

$$X \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1+2-3 \\ 2+4-6 \\ 0+1-1 \\ 1-1-0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Thus

$$\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \in \text{null}(X).$$

The third column is the sum of the first two columns, so the columns are dependent. The first two columns are not multiples of each other, so

$$\text{rank}(X) = 2.$$

Because X has three columns but rank 2, one singular value is zero.

If $X\mathbf{c} = \mathbf{y}$, then

$$X \left(\mathbf{c} + t \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right) = X\mathbf{c} + tX \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \mathbf{y}.$$

So different coefficient vectors can give the same output. This is the same nonuniqueness from Unit 2, now detected by a zero singular value.

A.6.13 PCA from an SVD

The covariance eigenvalues are

$$\lambda_1 = \frac{(4\sqrt{2})^2}{4} = 8, \quad \lambda_2 = \frac{(2\sqrt{2})^2}{4} = 2.$$

Thus

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

is a first principal direction, and

$$\mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

is a second principal direction.

The principal-coordinate matrix is

$$T = U_2 \Sigma_2 = \begin{bmatrix} 2\sqrt{2} & \sqrt{2} \\ 2\sqrt{2} & -\sqrt{2} \\ -2\sqrt{2} & -\sqrt{2} \\ -2\sqrt{2} & \sqrt{2} \end{bmatrix}.$$

Keeping only the first principal direction gives

$$\hat{Z}_1 = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T = \begin{bmatrix} 2 & 2 \\ 2 & 2 \\ -2 & -2 \\ -2 & -2 \end{bmatrix}.$$

The captured fraction is

$$\frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} = \frac{32}{32 + 8} = \frac{4}{5}.$$

The two columns of T are the score vectors for the first and second principal directions.

A.7 Unit 7 applied, geometric, and computational interpretation

These are solution sketches for the additional applied and computational problems in Unit 7.

A.7.1 Matrix-space coordinates

We have

$$A = \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix} = 2E_{11} - E_{12} + 0E_{21} + 3E_{22}.$$

Thus the coordinate vector in the ordered basis $(E_{11}, E_{12}, E_{21}, E_{22})$ is

$$\begin{bmatrix} 2 \\ -1 \\ 0 \\ 3 \end{bmatrix}.$$

The space M_{22} has dimension 4. Its zero vector is the zero matrix

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

A.7.2 Polynomial coordinates and subspaces

In the ordered basis $(1, x, x^2, x^3)$, the coordinate vector of $p(x) = 2 - 3x + x^3$ is

$$\begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \end{bmatrix}.$$

For $U = \{p \in \mathcal{P}_{\leq 3} : p(0) = 0\}$, the constant coefficient must be zero, so

$$U = \text{span}\{x, x^2, x^3\}.$$

For $W = \{p \in \mathcal{P}_{\leq 3} : p(0) = p'(0) = 0\}$, the constant and linear coefficients must be zero, so

$$W = \text{span}\{x^2, x^3\}.$$

Both are subspaces because they are solution sets of homogeneous linear conditions on the coefficients.

A.7.3 Derivative and evaluation as linear maps

The derivative map is linear because

$$D(p + q) = (p + q)' = p' + q'$$

and

$$D(cp) = (cp)' = cp'.$$

If $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$, then

$$D(p) = a_1 + 2a_2x + 3a_3x^2.$$

Thus the matrix of D , using the ordered bases $(1, x, x^2, x^3)$ and $(1, x, x^2)$, is

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

The evaluation map $E_0(p) = p(0)$ is linear because evaluation distributes over addition and scalar multiplication. Its kernel is

$$\{p \in \mathcal{P}_{\leq 3} : p(0) = 0\} = \text{span}\{x, x^2, x^3\}.$$

A.7.4 Matrix inner product

The trace inner product is the sum of entrywise products:

$$\langle A, B \rangle = 1 \cdot 3 + 2 \cdot 0 + 0 \cdot 4 + (-1) \cdot 1 = 2.$$

Also

$$\|A\|^2 = 1^2 + 2^2 + 0^2 + (-1)^2 = 6,$$

so $\|A\| = \sqrt{6}$. Next

$$A - B = \begin{bmatrix} -2 & 2 \\ -4 & -2 \end{bmatrix},$$

so $d(A, B)^2 = 4 + 4 + 16 + 4 = 28$. Therefore $d(A, B) = \sqrt{28} = 2\sqrt{7}$. Since $\langle A, B \rangle = 2 \neq 0$, the matrices are not orthogonal.

A.7.5 Same polynomial space, different inner products

In the coefficient inner product,

$$p(x) = 1 + x \quad \leftrightarrow \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad q(x) = 1 - x \quad \leftrightarrow \quad \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Thus

$$\langle p, q \rangle_{\text{coef}} = 1 \cdot 1 + 1 \cdot (-1) = 0.$$

For the $L^2[-1, 1]$ inner product,

$$\langle p, q \rangle_{L^2} = \int_{-1}^1 (1+x)(1-x) dx = \int_{-1}^1 (1-x^2) dx.$$

This is $2 - \frac{2}{3} = \frac{4}{3}$. Thus p and q are orthogonal for the coefficient inner product but not for the $L^2[-1, 1]$ inner product. Orthogonality depends on the chosen inner product.

A.7.6 Projection equations for the continuous least-squares line

With basis $(1, x)$,

$$G = \begin{bmatrix} \langle 1, 1 \rangle & \langle x, 1 \rangle \\ \langle 1, x \rangle & \langle x, x \rangle \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & \frac{2}{3} \end{bmatrix}.$$

For $p(x) = 1 + 2x + x^2 + x^3$, we get

$$b_0 = \int_{-1}^1 p(x) dx = 2 + \frac{2}{3} = \frac{8}{3},$$

and

$$b_1 = \int_{-1}^1 x p(x) dx = \int_{-1}^1 (x + 2x^2 + x^3 + x^4) dx = \frac{4}{3} + \frac{2}{5} = \frac{26}{15}.$$

Thus

$$\begin{bmatrix} 2 & 0 \\ 0 & \frac{2}{3} \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} \frac{8}{3} \\ \frac{26}{15} \end{bmatrix}.$$

So $c_0 = \frac{4}{3}$ and $c_1 = \frac{13}{5}$. The continuous least-squares line is $q_{L^2}(x) = \frac{4}{3} + \frac{13}{5}x$.

A.7.7 Taylor line versus fixed-interval least squares

The Taylor line of $f(x) = x^2$ at 0 is $T(x) = 0$. For the $L^2[-1, 1]$ -projection onto $\text{span}\{1, x\}$, symmetry gives no x -term. The best constant is

$$\frac{\int_{-1}^1 x^2 dx}{\int_{-1}^1 1 dx} = \frac{2/3}{2} = \frac{1}{3}.$$

Thus the fixed-interval least-squares approximation is $q(x) = \frac{1}{3}$. The answers differ because Taylor approximation is local, while fixed-interval least squares minimizes error over the whole interval.

A.7.8 Taylor projection from a local inner product

If $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$ and $q(x) = c_0 + c_1x$, then

$$p - q = (a_0 - c_0) + (a_1 - c_1)x + a_2x^2 + a_3x^3.$$

The squared norm from the local inner product is

$$(a_0 - c_0)^2 + (a_1 - c_1)^2 + a_2^2 + a_3^2.$$

The last two terms do not depend on c_0, c_1 . The minimum occurs when $c_0 = a_0$ and $c_1 = a_1$. Thus the projection is $a_0 + a_1x$. Since $a_0 = p(0)$ and $a_1 = p'(0)$, the answer is $p(0) + p'(0)x$.

A.7.9 Shrinking least squares

For x^2 , symmetry gives slope 0, and the constant is

$$\frac{1}{2t} \int_{-t}^t x^2 dx = \frac{1}{2t} \cdot \frac{2t^3}{3} = \frac{t^2}{3}.$$

Thus $P_t(x^2) = \frac{t^2}{3}$. For x^3 , symmetry gives constant 0. The slope is

$$\frac{\int_{-t}^t x^4 dx}{\int_{-t}^t x^2 dx} = \frac{2t^5/5}{2t^3/3} = \frac{3t^2}{5}.$$

Thus $P_t(x^3) = \frac{3t^2}{5}x$. Both projections tend to 0 as $t \rightarrow 0$, which is the Taylor linear approximation at the origin for both x^2 and x^3 .

A.7.10 Gram-Schmidt in $\mathcal{P}_{\leq 3}$

We have $\langle 1, x \rangle = \int_{-1}^1 x dx = 0$, so 1 and x are orthogonal. For x^2 ,

$$\text{proj}_{\text{span}\{1\}}(x^2) = \frac{\langle x^2, 1 \rangle}{\langle 1, 1 \rangle} 1 = \frac{2/3}{2} = \frac{1}{3}.$$

Also $\langle x^2, x \rangle = \int_{-1}^1 x^3 dx = 0$. So the next orthogonal polynomial is $x^2 - \frac{1}{3}$. For x^3 , the projections onto 1 and $x^2 - \frac{1}{3}$ vanish by symmetry. The projection onto x is

$$\frac{\langle x^3, x \rangle}{\langle x, x \rangle} x = \frac{2/5}{2/3} x = \frac{3}{5} x.$$

So the next orthogonal polynomial is $x^3 - \frac{3}{5}x$. An orthogonal basis makes projection easier because the Gram matrix is diagonal, so each coefficient can be computed separately.

A.7.11 Code interpretation: polynomial regression

The columns of

```
A = np.column_stack([np.ones_like(xs), xs, xs**2])
```

represent sampled values of 1, x , x^2 . Thus the approximation subspace is $\mathcal{P}_{\leq 2} = \text{span}\{1, x, x^2\}$. The vector c stores coefficients of the fitted quadratic $c_0 + c_1x + c_2x^2$. The product $A @ c$ stores fitted sampled values. The vector $r = y - A @ c$ stores sampled residuals. The computation $A.T @ r$ checks that the residual is orthogonal to each sampled feature column:

$$\sum_i r_i = 0, \quad \sum_i x_i r_i = 0, \quad \sum_i x_i^2 r_i = 0.$$

This does not mean $r \approx 0$. It means r is orthogonal to the approximation subspace.

A.7.12 Two-variable tangent plane and sampled design matrix

We compute $f(0,0) = 1$, $f_x(0,0) = 2$, and $f_y(0,0) = -1$. Thus the tangent plane is $T(x,y) = 1 + 2x - y$. The residual is $r_T(x,y) = f(x,y) - T(x,y) = x^2 + xy + 2y^2 + x^3$. It satisfies

$$r_T(0,0) = 0, \quad \nabla r_T(0,0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

For sampled points (x_i, y_i) , the design matrix for fitting $q(x,y) = \alpha + \beta x + \gamma y$ is

$$A = \begin{bmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ \vdots & \vdots & \vdots \\ 1 & x_N & y_N \end{bmatrix}.$$

The sampled residual equations are

$$\sum_i r_i = 0, \quad \sum_i x_i r_i = 0, \quad \sum_i y_i r_i = 0.$$

References

- W. Keith Nicholson, *Linear Algebra with Applications*, 2023-A open edition.
- Gilbert Strang and Edwin “Jed” Herman, *Calculus Volume 3*, OpenStax.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin, “Attention Is All You Need.”
- Tom B. Brown et al., “Language Models are Few-Shot Learners.”
- Edward J. Hu et al., “LoRA: Low-Rank Adaptation of Large Language Models.”

Lab Index

The notebooks are linked supplements to the textbook. They are not meant to be long take-home programming assignments. The main sequence in each lab is designed for in-class use or guided review.

Lab U1: Vectors, Similarity, Attention, and Matrix Actions. Unit 1. Includes optional warm-up and optional extension sections. Focus: read short NumPy snippets as mathematical notation; interpret norms, dot products, cosine similarity, attention-style weighted averages, matrix-vector products, matrix products, geometric matrix actions, and affine maps. *Learning outcomes.* U1-LO1, U1-LO2, U1-LO3, U1-LO4, U1-LO5, U1-LO6, U1-LO7, U1-LO8.

Lab U2: Understanding a Linear Map. Unit 2. Focus: read short NumPy and SymPy snippets as mathematical notation; interpret reachable outputs, row reduction, plane intersections, homogeneous systems, forgotten directions, rank, null vectors, redundant features, determinants, and inverse checks. *Learning outcomes.* U2-LO1, U2-LO2, U2-LO3, U2-LO5, U2-LO6, U2-LO7.

Lab U3: Jacobian matrices and local linearization. Unit 3. Focus: Jacobian matrices and local linear prediction. *Learning outcomes.* U3-LO3, U3-LO5.

Lab U4: Regression as Projection. Unit 4. Focus: design matrices, least squares, residual orthogonality. *Learning outcomes.* U4-LO4, U4-LO5, U4-LO7.

Lab U5: Hessian eigenvalues, least squares, and fixed nonlinear features. Unit 5. Focus: interpret Hessian eigenvalues as curvature information; connect residual orthogonality, zero gradient, and curvature in least squares; and fit a coefficient vector for fixed nonlinear features. The opening block revisits gradient descent and a final-layer outer-product update from Unit 3. *Learning outcomes.* U3-LO6, U3-LO9, U5-LO5, U5-LO6, U5-LO7.

Lab U6: Constraints, PCA, and SVD. Unit 6. Focus: read short NumPy snippets as mathematical notation; check one- and several-constraint Lagrange conditions; connect quadratic forms with maximum stretch; center data and form covariance; compute PCA scores, reconstructions, residuals, and principal coordinates; compare PCA with regression; connect PCA with SVD; and read rank and the four fundamental subspaces from an SVD. *Learning outcomes.* U6-LO2, U6-LO3, U6-LO4, U6-LO5, U6-LO6, U6-LO7.

Lab U7: Polynomial Approximation. Unit 7. Focus: read polynomial feature matrices, least-squares coefficients, fitted values, residual checks, shrinking-interval coefficients, Gram-Schmidt checks, and sampled plane fits. Compare Taylor, continuous least-squares, and sampled least-squares approximations. *Learning outcomes.* U7-LO3, U7-LO4, U7-LO5, U7-LO6.

Transformer Wrap-Up Lab: A Tiny Attention Block. Appendix: Transformer Wrap-Up Worksheet. Status: optional extension. Focus: token matrices, attention scores, masks, weighted values, logits, and one small loss. *Learning outcomes.* U1-LO3, U1-LO6, U1-LO8, U3-LO7, U3-LO8.

Colophon

This book was authored in PreTeXt.